

**US Army Corps
of Engineers**
Louisville District

Solicitation For

HVAC & Controls Replacement F/20824 Wright-Patterson AFB, OH

P2: 513948

Design-Bid-Build

**Specifications: Vol 1 of 2 (Div00-10)
Certified Final Design**

**27 June 2025
W912QR25B0053**

**19 August 2025
W912QR25B0053-0004**

RRS-A: 200A Disposition: Maintain for 15 yrs after construction.

AMENDMENT OF SOLICITATION/MODIFICATION OF CONTRACT				1. CONTRACT ID CODE		PAGE OF PAGES 1 11	
2. AMENDMENT/MODIFICATION NO. AMDT 0004		3. EFFECTIVE DATE 19-Aug-2025		4. REQUISITION/PURCHASE REQ. NO. W22W9K51353222		5. PROJECT NO.(If applicable)	
6. ISSUED BY U. S. ARMY ENGINEER DISTRICT, LOUISVILLE 600 DR. MARTIN LUTHER KING, JR. PLACE ROOM 821 LOUISVILLE KY 40202-2239		CODE W912QR		7. ADMINISTERED BY (If other than item 6) See Item 6		CODE	
8. NAME AND ADDRESS OF CONTRACTOR (No., Street, County, State and Zip Code)				X		9A. AMENDMENT OF SOLICITATION NO. W912QR25B0053	
				X		9B. DATED (SEE ITEM 11) 27-Jun-2025	
						10A. MOD. OF CONTRACT/ORDER NO.	
						10B. DATED (SEE ITEM 13)	
CODE		FACILITY CODE					
11. THIS ITEM ONLY APPLIES TO AMENDMENTS OF SOLICITATIONS							
☒ The above numbered solicitation is amended as set forth in Item 14. The hour and date specified for receipt of Offer ☒ is extended, ☐ is not extended. Offer must acknowledge receipt of this amendment prior to the hour and date specified in the solicitation or as amended by one of the following methods: (a) By completing Items 8 and 15, and returning <u>1</u> copies of the amendment; (b) By acknowledging receipt of this amendment on each copy of the offer submitted; or (c) By separate letter or telegram which includes a reference to the solicitation and amendment numbers. FAILURE OF YOUR ACKNOWLEDGMENT TO BE RECEIVED AT THE PLACE DESIGNATED FOR THE RECEIPT OF OFFERS PRIOR TO THE HOUR AND DATE SPECIFIED MAY RESULT IN REJECTION OF YOUR OFFER. If by virtue of this amendment you desire to change an offer already submitted, such change may be made by telegram or letter, provided each telegram or letter makes reference to the solicitation and this amendment, and is received prior to the opening hour and date specified.							
12. ACCOUNTING AND APPROPRIATION DATA (If required)							
13. THIS ITEM APPLIES ONLY TO MODIFICATIONS OF CONTRACTS/ORDERS. IT MODIFIES THE CONTRACT/ORDER NO. AS DESCRIBED IN ITEM 14.							
A. THIS CHANGE ORDER IS ISSUED PURSUANT TO: (Specify authority) THE CHANGES SET FORTH IN ITEM 14 ARE MADE IN THE CONTRACT ORDER NO. IN ITEM 10A.							
B. THE ABOVE NUMBERED CONTRACT/ORDER IS MODIFIED TO REFLECT THE ADMINISTRATIVE CHANGES (such as changes in paying office, appropriation date, etc.) SET FORTH IN ITEM 14, PURSUANT TO THE AUTHORITY OF FAR 43.103(B).							
C. THIS SUPPLEMENTAL AGREEMENT IS ENTERED INTO PURSUANT TO AUTHORITY OF:							
D. OTHER (Specify type of modification and authority)							
E. IMPORTANT: Contractor ☐ is not, ☐ is required to sign this document and return _____ copies to the issuing office.							
14. DESCRIPTION OF AMENDMENT/MODIFICATION (Organized by UCF section headings, including solicitation/contract subject matter where feasible.) SEE SUMMARY OF CHANGES							
Except as provided herein, all terms and conditions of the document referenced in Item 9A or 10A, as heretofore changed, remains unchanged and in full force and effect.							
15A. NAME AND TITLE OF SIGNER (Type or print)				16A. NAME AND TITLE OF CONTRACTING OFFICER (Type or print)			
				TEL: _____ EMAIL: _____			
15B. CONTRACTOR/OFFEROR _____ (Signature of person authorized to sign)		15C. DATE SIGNED		16B. UNITED STATES OF AMERICA BY _____ (Signature of Contracting Officer)		16C. DATE SIGNED	

SECTION SF 30 BLOCK 14 CONTINUATION PAGE

SUMMARY OF CHANGES

The following have been added by full text:

SUMMARY OF CHANGES AMDT 0004

P2#: 513948 HVAC & CONTROLS RENOVATION F/20824
WPAFB, OH
Amendment 004

SUMMARY OF CHANGES

The submission date is extended to 27 August 2025 at 10:00 AM EST. Acknowledgment of this amendment and proposals are due at that time

1. The following SPECIFICATION SECTIONS have been revised and replaced in their entirety:

LOW PRESSURE WATER HEATING BOILERS (UNDER 800,000 BTU/HR OUTPUT) - 23 52
43.00 20

2. The following SPECIFICATION SECTIONS have been added in their entirety:

None.

3. The following SPECIFICATION SECTIONS have been deleted in their entirety:

None

4. The following PLAN SHEETS have been revised and replaced in their entirety:

A-002

5. The following PLAN SHEETS have been added new:

None.

6. The following PLAN SHEETS have been deleted in their entirety new:

None.

7. The following DOCUMENTS have been replaced in their entirety:

Specifications have been divided into two Volumes with the Project Number being revised from P2#513984 to P2#513948. The only revisions to the technical specifications are noted in line item 1 above.

Drawings have an updated Project Number being revised from P2#513984 to P2#513948. The only revisions to the drawings are noted in line item 4 above.

Section 00 73 00 – Supplementary Conditions: The Wage Determination OH20250081 has been revised from 07/25/2025 Revision 08 to 08/15/2025 Revision 10 and must be adhered to.

SECTION 00 10 00 - SOLICITATION

The required response date/time has changed from 21-Aug-2025 10:00 AM to 27-Aug-2025 10:00 AM.

SECTION 00 73 00 - SUPPLEMENTARY CONDITIONS

The following have been modified:

The Wage Determination OH20250081 has been revised from 07/25/2025 Revision 08 to 08/15/2025 Revision 10 and must be adhered to.

"General Decision Number: OH20250081 08/15/2025

Superseded General Decision Number: OH20240081

State: Ohio

Construction Type: Building

County: Greene County in Ohio.

BUILDING CONSTRUCTION PROJECTS (does not include single family homes or apartments up to and including 4 stories).

Note: Contracts subject to the Davis-Bacon Act are generally required to pay at least the applicable minimum wage rate required under Executive Order 14026 or Executive Order 13658. Please note that these Executive Orders apply to covered contracts entered into by the federal government that are subject to the Davis-Bacon Act itself, but do not apply to contracts subject only to the Davis-Bacon Related Acts, including those set forth at 29 CFR 5.1(a)(1).

If the contract is entered into on or after January 30, 2022, or the contract is renewed or extended (e.g., an option is exercised) on or after January 30, 2022:	. Executive Order 14026 generally applies to the contract. The contractor must pay all covered workers at least \$17.75 per hour (or the applicable wage rate listed on this wage determination, if it is higher) for all hours spent performing on the contract in 2025.
If the contract was awarded on or between January 1, 2015 and January 29, 2022, and the	. Executive Order 13658 generally applies to the contract.

contract is not renewed or extended on or after January 30, 2022:	. The contractor must pay all covered workers at least \$13.30 per hour (or the applicable wage rate listed on this wage determination, if it is higher) for all hours spent performing on that contract in 2025.
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The applicable Executive Order minimum wage rate will be adjusted annually. If this contract is covered by one of the Executive Orders and a classification considered necessary for performance of work on the contract does not appear on this wage determination, the contractor must still submit a conformance request.

Additional information on contractor requirements and worker protections under the Executive Orders is available at <http://www.dol.gov/whd/govcontracts>.

Modification Number Publication Date

0	01/03/2025
1	02/07/2025
2	02/14/2025
3	03/14/2025
4	04/25/2025
5	05/30/2025
6	06/06/2025
7	07/18/2025
8	07/25/2025
9	08/08/2025
10	08/15/2025

ASBE0008-010 03/01/2025

Rates Fringes

ASBESTOS WORKER/HEAT & FROST
INSULATOR.....\$ 35.23 23.04

BROH0018-003 06/01/2024

Rates Fringes

BRICKLAYER.....\$ 33.30 18.28

BROH0022-010 07/01/2024

Rates Fringes

TILE FINISHER.....\$ 28.28 11.05
TILE SETTER.....\$ 30.98 15.82

CARP0002-017 06/01/2025

	Rates	Fringes
CARPENTER.....	\$ 30.96	18.44

ELEC0082-004 12/02/2024

	Rates	Fringes
ELECTRICIAN.....	\$ 38.00	22.49

ELEV0011-002 01/01/2025

	Rates	Fringes
ELEVATOR MECHANIC.....	\$ 57.41	38.435+a+b

PAID HOLIDAYS:

a. New Year's Day, Memorial Day, Independence Day, Labor Day, Vetern's Day, Thanksgiving Day, the Friday after Thanksgiving, and Christmas Day.

b. Employer contributes 8% of regular hourly rate to vacation pay credit for employee who has worked in business more than 5 years; 6% for less than 5 years' service.

ENGI0018-034 05/01/2024

	Rates	Fringes
POWER EQUIPMENT OPERATOR Crane.....	\$ 44.14	16.41

ENGI0066-045 06/01/2017

	Rates	Fringes
POWER EQUIPMENT OPERATOR Forklift.....	\$ 28.87	19.66
Grader/Blade.....	\$ 32.42	19.66
Mechanic.....	\$ 32.92	19.66

IRON0044-020 06/01/2025

	Rates	Fringes
IRONWORKER, ORNAMENTAL.....	\$ 37.77	23.90

IRON0290-006 06/01/2025

	Rates	Fringes
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IRONWORKER (Reinforcing and
Structural).....\$ 37.39 25.35

* LABO0265-018 06/01/2025

Rates Fringes

LABORER
Mason Tender - Brick.....\$ 27.30 18.75

LABO1410-004 04/01/2024

Rates Fringes

LABORER
Common or General;
Asbestos Abatement
(Removal from Ceilings,
Floors, and Walls).....\$ 31.65 12.95

PAIN0249-005 05/01/2025

Rates Fringes

PAINTER (Brush and Roller).....\$ 29.15 13.97

PAIN0387-002 11/01/2023

Rates Fringes

GLAZIER.....\$ 31.95 18.20

PLAS0132-020 06/01/2025

Rates Fringes

CEMENT MASON/CONCRETE FINISHER...\$ 32.71 19.27

PLUM0050-009 06/30/2025

Rates Fringes

PIPEFITTER (Excludes HVAC
Pipe Installation).....\$ 51.00 32.56

PLUM0162-009 06/01/2024

Rates Fringes

PLUMBER (Includes HVAC Pipe
Installation).....\$ 43.05 27.18

ROOF0042-001 08/01/2024

Rates Fringes

ROOFER.....\$ 33.00 19.42

* SFOH0669-009 04/01/2025

Rates Fringes

SPRINKLER FITTER (Fire
Sprinklers).....\$ 48.28 28.08

* UAVG-OH-0021 01/01/2019

Rates Fringes

OPERATOR: Oiler.....\$ 27.56 16.37

* UAVG-OH-0025 01/01/2018

Rates Fringes

SHEET METAL WORKER, Excludes
HVAC Duct and Unit
Installation.....\$ 28.10 23.41

* UAVG-OH-0027 01/01/2018

Rates Fringes

OPERATOR:
Backhoe/Excavator/Trackhoe.....\$ 34.84 14.76

SUOH2012-083 08/29/2014

Rates Fringes

DRYWALL HANGER AND METAL STUD
INSTALLER.....\$ 21.02 4.21

FORM WORKER.....\$ 22.41 9.01

LABORER: Mason Tender -
Cement/Concrete.....\$ 22.95 8.60

LABORER: Pipelayer.....\$ 23.98 8.58

OPERATOR: Bobcat/Skid
Steer/Skid Loader.....\$ 30.26 12.58

OPERATOR: Bulldozer.....\$ 26.01 4.95

OPERATOR: Loader.....\$ 29.99 12.80

OPERATOR: Paver (Asphalt,
Aggregate, and Concrete).....\$ 30.28 13.29

OPERATOR: Roller.....\$ 28.25 12.61

PAINTER: Spray.....\$ 22.78 12.40

SHEET METAL WORKER (HVAC Duct
and HVAC Unit Installation
Only).....\$ 26.41 17.34

TRUCK DRIVER: Dump (All Types)...\$ 22.08 11.51

WELDERS - Receive rate prescribed for craft performing
operation to which welding is incidental.

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Note: Executive Order (EO) 13706, Establishing Paid Sick Leave for Federal Contractors applies to all contracts subject to the Davis-Bacon Act for which the contract is awarded (and any solicitation was issued) on or after January 1, 2017. If this contract is covered by the EO, the contractor must provide employees with 1 hour of paid sick leave for every 30 hours they work, up to 56 hours of paid sick leave each year. Employees must be permitted to use paid sick leave for their own illness, injury or other health-related needs, including preventive care; to assist a family member (or person who is like family to the employee) who is ill, injured, or has other health-related needs, including preventive care; or for reasons resulting from, or to assist a family member (or person who is like family to the employee) who is a victim of, domestic violence, sexual assault, or stalking. Additional information on contractor requirements and worker protections under the EO is available at <https://www.dol.gov/agencies/whd/government-contracts>.

Unlisted classifications needed for work not included within the scope of the classifications listed may be added after award only as provided in the labor standards contract clauses (29CFR 5.5 (a) (1) (iii)).

The body of each wage determination lists the classifications and wage rates that have been found to be prevailing for the type(s) of construction and geographic area covered by the wage determination. The classifications are listed in alphabetical order under rate identifiers indicating whether the particular rate is a union rate (current union negotiated rate), a survey rate, a weighted union average rate, a state adopted rate, or a supplemental classification rate.

Union Rate Identifiers

A four-letter identifier beginning with characters other than

""SU"", ""UAVG"", ?SA?, or ?SC? denotes that a union rate was prevailing for that classification in the survey. Example: PLUM0198-005 07/01/2024. PLUM is an identifier of the union whose collectively bargained rate prevailed in the survey for this classification, which in this example would be Plumbers. 0198 indicates the local union number or district council number where applicable, i.e., Plumbers Local 0198. The next number, 005 in the example, is an internal number used in processing the wage determination. The date, 07/01/2024 in the example, is the effective date of the most current negotiated rate.

Union prevailing wage rates are updated to reflect all changes over time that are reported to WHD in the rates in the collective bargaining agreement (CBA) governing the classification.

Union Average Rate Identifiers

The UAVG identifier indicates that no single rate prevailed for those classifications, but that 100% of the data reported for the classifications reflected union rates. EXAMPLE: UAVG-OH-0010 01/01/2024. UAVG indicates that the rate is a weighted union average rate. OH indicates the State of Ohio. The next number, 0010 in the example, is an internal number used in producing the wage determination. The date, 01/01/2024 in the example, indicates the date the wage determination was updated to reflect the most current union average rate.

A UAVG rate will be updated once a year, usually in January, to reflect a weighted average of the current rates in the collective bargaining agreements on which the rate is based.

Survey Rate Identifiers

The ""SU"" identifier indicates that either a single non-union rate prevailed (as defined in 29 CFR 1.2) for this classification in the survey or that the rate was derived by computing a weighted average rate based on all the rates reported in the survey for that classification. As a weighted average rate includes all rates reported in the survey, it may include both union and non-union rates. Example: SUFL2022-007 6/27/2024. SU indicates the rate is a single non-union prevailing rate or a weighted average of survey data for that classification. FL indicates the State of Florida. 2022 is the year of the survey on which these classifications and rates are based. The next number, 007 in the example, is an internal number used in producing the wage determination. The date, 6/27/2024 in the example, indicates the survey completion date for the classifications and rates under that identifier.

?SU? wage rates typically remain in effect until a new survey is conducted. However, the Wage and Hour Division (WHD) has the discretion to update such rates under 29 CFR 1.6(c)(1).

State Adopted Rate Identifiers

The ""SA"" identifier indicates that the classifications and prevailing wage rates set by a state (or local) government were adopted under 29 C.F.R 1.3(g)-(h). Example: SAME2023-007 01/03/2024. SA reflects that the rates are state adopted. ME refers to the State of Maine. 2023 is the year during which the state completed the survey on which the listed classifications and rates are based. The next number, 007 in the example, is an internal number used in producing the wage determination. The date, 01/03/2024 in the example, reflects the date on which the classifications and rates under the ?SA? identifier took effect under state law in the state from which the rates were adopted.

WAGE DETERMINATION APPEALS PROCESS

1) Has there been an initial decision in the matter? This can be:

- a) a survey underlying a wage determination
- b) an existing published wage determination
- c) an initial WHD letter setting forth a position on a wage determination matter
- d) an initial conformance (additional classification and rate) determination

On survey related matters, initial contact, including requests for summaries of surveys, should be directed to the WHD Branch of Wage Surveys. Requests can be submitted via email to davisbaconinfo@dol.gov or by mail to:

Branch of Wage Surveys
Wage and Hour Division
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210

Regarding any other wage determination matter such as conformance decisions, requests for initial decisions should be directed to the WHD Branch of Construction Wage Determinations. Requests can be submitted via email to BCWD-Office@dol.gov or by mail to:

Branch of Construction Wage Determinations
Wage and Hour Division
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210

2) If an initial decision has been issued, then any interested party (those affected by the action) that disagrees with the decision can request review and reconsideration from the Wage

and Hour Administrator (See 29 CFR Part 1.8 and 29 CFR Part 7).
Requests for review and reconsideration can be submitted via
email to dba.reconsideration@dol.gov or by mail to:

Wage and Hour Administrator
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210

The request should be accompanied by a full statement of the
interested party's position and any information (wage payment
data, project description, area practice material, etc.) that
the requestor considers relevant to the issue.

3) If the decision of the Administrator is not favorable, an
interested party may appeal directly to the Administrative
Review Board (formerly the Wage Appeals Board). Write to:

Administrative Review Board
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210.

END OF GENERAL DECISION"

(End of Summary of Changes)

HVAC AND CONTROLS REPLACEMENT F/20824
 Wright-Patterson AFB, OH P2#513948

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25 05 11.02 05/21, CHG 1: 08/23 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS (METERING)
25 05 11.03 05/21, CHG 1: 08/23 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS (LIGHTING)

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26 20 00	08/23	INTERIOR DISTRIBUTION SYSTEM
26 51 00	05/20, CHG 2: 11/21	INTERIOR LIGHTING

DIVISION 28 - ELECTRONIC SAFETY AND SECURITY

28 31 70	08/20	INTERIOR FIRE ALARM SYSTEM, ADDRESSABLE
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-- End of Project Table of Contents --

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Wright-Patterson AFB, OH P2#513948

SECTION 00 80 00.00 06

SPECIAL PROVISIONS
04/21, CHG 1: 06/23

PART 1 GENERAL

Attachments to this specification are as follows:

Construction Project Sign Details
Project Submittal Register

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY (DA)

AR 530-1 Operation Security

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety and Occupational Health
(SOH) Requirements

ERDC/ITL TR-12-1 (2015) A/E/C Graphics Standard, Release 2.0

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1926.59 Hazard Communication

1.2 SUBMITTALS

Government approval/acceptance is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. The following shall be submitted in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

SF1413 Statement and Acknowledgement

SD-02 Shop Drawings

Mechanical/Electrical Room Layout; G

SD-04 Samples

Equipment Warranty Identification Tags; G

SD-05 Design Data

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Progress Photographs

SD-07 Certificates

Warranty of Construction

NO ASBESTOS - CONTAINING MATERIAL (ACM) CERTIFICATION

Construction Phase:

Certification for each individual product installed and identified to contain mineral fibers that no asbestos-containing materials were installed

Documentation to show that the products containing mineral fiber materials have been microscopically examined by an AIHA- or NVLAP-certified laboratory and the lab has determined that the material does not contain asbestos

Insurance;

Sales and Use Tax;

SD-11 Closeout Submittals

Preliminary (Working) As-Built Drawings; G

Final As-Built Drawings; G

CAD Working As-Built Drawings; G

Equipment-in-Place List

Maintenance and Parts Data

Warranty Management Plan; G

Completion of BUILDER Update Form

1.3 COMMENCEMENT, PROSECUTION AND COMPLETION OF WORK

- a. Refer to clause 52.211-12 "LIQUIDATED DAMAGES" in Section 00 70 00 for the amount of Liquidated Damages for the project.
- b. Refer to clause 52.211-10 "Commencement, Prosecution, and Completion of Work" in Section 00 70 00 for a notification of significant contract dates.
- c. After approval of the required bonds, the Government will issue an Administrative Notice-to-Proceed (NTP) to allow for preconstruction activities to be initiated in accordance with the contract requirements. The contract period of performance specified in Contract Clause 52.211-10 will begin upon issuance of Administrative NTP. No physical onsite work will be permitted during the Administrative NTP period. All required preconstruction activities and submittals must be submitted and accepted by the Government, where applicable, within the Administrative NTP period of 45 days. The Government will issue a Construction NTP to allow commencement of all other contract work after the Administrative NTP period, or after the contractor's completion of all work specified in the Administrative NTP. Failure to complete all

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required preconstruction activities may result in the contract being terminated for default in accordance with Contract Clause 52.249-10 Default (Fixed-Price Construction).

1.3.1 Additional Requirements/Clarifications of Work Included Within the Contract

- a. The time stated in clause 52.211-10 "Commencement, Prosecution, and Completion of Work" in Section 00 70 00 for completion shall include installation of Government-furnished furniture, if required, as well as as-built drawings, O&M manuals, operational tests/reports/training/instructions, equipment lists.
- b. Government equipment and furniture will be moved by the Government before sequences are commenced. Government will require 45 days notice prior to the commencement of work to begin process of removing equipment and furniture in designated areas, including Rooms 111, 114, and 121. Any equipment to remain in place during construction is noted on the contract documents and shall be protected by the Contractor, where indicated. Sequence 3 (Bid Option 1) will require specific attention to items to remain that will require protection during construction. Attachment 1 is provided to better describe the level of effort that may be required. Those items that will require protection are identified as "KTR" responsibility, or "Gov't" responsibility. If indicated as "KTR", the Contractor is responsible for provided protection to the noted equipment or furniture. The means and method of said protection is at the discretion of the Contractor, but will be approved by the Government prior to the commencement of work.

1.4 NOT USED

1.5 NOT USED

1.6 CONTRACT DRAWINGS AND SPECIFICATIONS

In addition to DFARS 252.236-7001 "Contract Drawings and Specifications" in Section 00 70 00 the following will apply:

- a. After Award or no later than Administrative Notice to Proceed (ADMIN NTP), the Government will furnish the Contractor electronic file(s) containing all technical contract documents in electronic media only. This electronic file(s) will include a complete set of drawing files and technical specification files which have all amendments included. The electronic files(s) will contain drawing files in PDF format along with technical specifications in PDF format. These PDF files are the contract documents that represent the construction requirements of the contract, and are being provided for the Contractor's use in printing paper copies of contract documents.
- b. In addition, native CAD files(this includes, but not limited to, all source files, models, custom fonts and linetypes, plot files, and images used to create the contract drawings) are provided in accordance with the "AS-BUILT DOCUMENTS" paragraph for the Contractor's use in maintaining and preparing as-built plans. If another CAD Program is used other than the Using Agency's System, all native CAD files that were generated with that software and all support files will also be included. Only native files are to be used for As-Built preparation and information.
- c. Native files are to be used for As-Built preparation only. The PDF

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files are the contract documents that represent the construction requirements of the contract.

1.7 AS-BUILT DOCUMENTS FOR DESIGN BID BUILD PROJECTS

1.7.1 General

This section covers the completion of final as-built drawings, as a requirement of the contract. The Contractor is responsible for maintaining paper copy working as-built drawings during the construction phase. These paper copy drawings will be used by the Contractor to prepare, maintain and submit the final as-built drawings

1.7.1.1 As-Built Drawings

An as-built drawing is a contract construction drawing revised to reflect the final as-built conditions of the project because of modifications, changes, corrections to the project design required during construction, submittals and extensions of design. The terms "drawings," "contract drawings," "drawing files," "working as-built drawings" and "final as-built drawings" refer to contract drawings that are revised to be used for the "RECORD DRAWING AS-BUILTS".

1.7.1.2 Government-Furnished Files

- a. The Contractor will be provided electronic files at the beginning of construction for use during the construction phase which are to be maintained during construction for the preparation of as-builts. The Contractor shall be responsible to print two full size paper copies. The Contractor shall enter changes and corrections on two sets of paper full size construction plans on a weekly basis in accordance with Paragraph "Maintenance of Working As-Built Drawings" in this section.
- b. The Contractor is required to prepare final as-built drawings utilizing the native files provided by the Government. If translation is required, the original design models (BIM or CAD) shall be updated to As-Built conditions and then appropriately translated. Updating translated drawings will not be accepted. The contractor shall update the CAD working as-built drawings, in accordance with paragraph "Maintenance of Working As-Built Drawings", on a quarterly basis and submit them to the COR for independent Government review. Both paper and electronic documents shall be available at all times and shall be provided promptly to the Contracting Officer's Representatives when requested. The Contractor shall be responsible for backup of electronic files during construction and for controlling release of information.

1.7.2 Withholding

Maintenance of working as-builts is considered part of the value of the facilities being constructed and will not be paid for as a separate line item. All costs in conjunction with periodic as-built maintenance and final preparation shall be considered a subsidiary obligation of the Contractor.

1.7.2.1 Failure to Maintain

If the Contractor fails to maintain the working as-built drawings as specified herein, the Contracting Officer will deduct from the monthly progress payment an amount up to 10% or which, in the Contracting Officer's

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judgment, represents the estimated cost of bringing the as-built documents up to date. This monthly deduction will continue until an agreement can be reached between the Contracting Officer and the Contractor regarding the accuracy and completeness of working as-built documents. This includes conversion of submittals and other miscellaneous documents.

1.7.2.2 As-Builts Prepared by Contractor

The Contractor is required to prepare and provide final as-built drawings. The Contractor shall include an activity in the cost-loaded schedule for the final As-Built drawing submittal in the amount defined in the following paragraph. See Section 01 32 01.00 06, PROJECT SCHEDULE, para "Basis for Payment and Cost Loading". This amount shall be withheld and not paid until the final As-Built drawing submittal has been accepted by the Government.

Withholding for the final as-built drawing submittal shall be in the amount of: 1% for contract awards less than \$5,000,000; \$50,000 for contracts awarded from \$5,000,000 to \$10,000,000; or \$100,000 for contracts awarded greater than \$10,000,000. Withholding shall be withheld until the final as-built drawing submittal has been approved and accepted by the Government.

1.7.3 Maintenance of Working As-Built Drawings

The Contractor shall revise two (2) sets of paper drawings by red-line process to show the as-built conditions during the prosecution of the project. These as-built marked drawings shall be kept current on a weekly basis and available on the jobsite at all times. Changes in the work from the contract or additional information which is uncovered in the course of construction shall be accurately and neatly recorded as they occur by means of details and notes. Changes must be reflected on all sheets that the change affects. The working as-built marked drawings will be jointly reviewed for accuracy and completeness by the Contracting Officer and the Contractor before submission of each monthly pay estimate. The working as-built drawings shall show the following information if applicable to the project, but not be limited thereto:

- a. The actual location, kinds and sizes of all sub-surface utility lines. In order that the location of these lines and appurtenances may be determined in the event the surface openings or indicators become covered over or obscured, the as-built drawings shall show, by offset dimensions to two permanently fixed surface features, the end of each run including each change in direction. Valves, splice boxes and similar appurtenances shall be located by dimensioning along the utility run from a reference point. The average depth below the surface of each run shall also be recorded.
- b. The location and dimensions of any changes within the building structure.
- c. The correct alignments, grade elevations, typical cross section, earthwork, structures or utilities if any changes were made from contract plans.
- d. Additional as-built information that exceeds the detail shown on the Contract Drawings. These as-built conditions include those that reflect structural details, fabrication, erection, installation plans and placing details, pipe sizes, insulation material, dimensions of equipment foundations and layouts, equipment, sizes, mechanical and

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electrical room layouts and other extensions of design, that were not shown in the original contract documents because the exact details were not known until after the time of approved shop drawings. It is recognized that the shop drawing submittals (revised showing as-built conditions) will serve as the as-built record without actual incorporation into the contract drawings. The final as-built construction drawing shall reference the shop drawing file that includes the as-built information. In turn, the shop drawing shall reference the applicable construction as-built drawing. All such shop drawing submittals must include the paper copy and PDF of the drawings.

- e. The invert elevations and grades of any drainage structures or ditches installed or affected as part of the project construction.
- f. Changes or modifications which result from the final inspection.
- g. Where contract drawings present options, only the option selected for construction shall be shown on the final as-built drawings.
- h. Systems designed or enhanced by the Contractor, such as HVAC controls, fire alarms, fire sprinklers, fire protection, fire detection and irrigation systems and other related systems in this project, shall be incorporated into the as-built drawings to include detailed information for all aspects of the systems including wiring, piping, and equipment drawings.
- i. Room numbers shown on the contract drawings are selected for design convenience and may not represent the actual numbers intended for use by the end user. Final as-built drawings shall reflect actual room numbers adopted by the end user.
- j. Contract modification (change order price) shall include the Contractor's cost to change working and final as-built drawings to reflect modifications and compliance with the following procedures (See "Markings and Indicators"):
 - (1) Directions in the modification for posting descriptive changes shall be followed.
 - (2) A Revision Triangle shall be placed at the location of each deletion.
 - (3) For new details or sections which are added to a drawing, a Revision Triangle shall be placed by the detail or section title.
 - (4) For minor changes, a Revision Triangle shall be placed by the area changed on the drawing (each location).
 - (5) For major changes to a drawing, a Revision Triangle shall be placed by the title of the affected plan, section, or detail at each location.
 - (6) For changes to schedules or drawings, a Revision Triangle shall be placed either by the schedule heading or by the change in the schedule.

1.7.4 Preliminary (Working) As-Built Drawings

Six (6) weeks before Contract Completion Date, the Contractor shall submit

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one (1) set of the original paper working as-built drawings to the Contracting Officer for review and approval. These working as-built marked drawings shall be neat, legible and accurate. The review by Government personnel will be expedited to the maximum extent possible. If upon review, the working as-built drawings are found to contain errors and/or omissions, they will be returned to the Contractor for corrections. The Contractor shall complete the corrections and return the working as-built marked drawings to the Contracting Officer within fourteen (14) calendar days. Upon approval, the working as-built drawings will be returned to the Contractor for use in preparation of final as-built drawings.

1.7.5 Preparation of Final As-Built Drawings

The contract drawings shall be modified as may be necessary to correctly show the features of the project as it has been constructed by bringing the contract drawings into agreement with approved working as-built drawings, adding such additional drawings as may be necessary.

These final as-built drawings are part of the permanent records of the project and the Contractor shall be responsible for the protection and safety thereof until returned to the Contracting Officer. Any drawings damaged or lost by the Contractor shall be satisfactorily replaced by the Contractor at no expense to the Government.

When electronic CAD files are a part of the as-built process, a set of files shall be provided to the Government as a part of the Final As-Built submittal for a review to verify the correctness of the as-built markups and that all changes have been incorporated into the electronic files. Should errors be determined, the Contractor shall update the files and provide a corrected set of files within fourteen (14) calendar days of receipt of comments. An independent Government review, by the Louisville district As-Built Coordinator (CELRL-CDM-Q), will be made on the accepted files to determine compliance with the As-Built requirements of this section, National CAD Standards, and the AEC CAD Standards; and to verify graphic changes were done properly in preparing the electronic files. This review will require submission of electronic files, containing all the files needed to reproduce the contract drawings, a full size set of contract drawings in PDF format, all shop drawings in PDF format, a scanned set of the paper markups and the paper markups. Upon receipt of any comments from this independent review, the contractor shall update the electronic files and provide a corrected set of files within fourteen (14) calendar days of receipt of the comments.

In the event the Contractor accomplishes additional work which changes the as-built conditions of the facility, after submission and approval of the working as-built drawings, the Contractor shall be responsible for the addition of these changes to the working as-built drawings and also to the final as-built documents.

1.7.6 Markings and Indicators

Changes shall be annotated in accordance with ERDC/ITL TR-12-1 "A/E/C Graphics Standard_Release 2.0" at the following locations:

- a. Bottom of the revised detail.
- b. Right hand and bottom border aligned with the revised detail.
- c. The revision block of the title block.

Separate markings shall be made for each modification negotiated into the

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contract.

1.7.7 Construction Contract Specifications

Submit final PDF file record construction contract specifications, including revisions thereto, with submission of final as-built drawings.

1.7.8 Preparation of Other As-Built Documents

All other non-electronic documents which may include, for example, design analysis, catalog cuts, or certification documents that are not available in native electronic format shall be scanned and provided in an organized manner in Adobe PDF format.

1.7.9 Submittal of Final As-Built Documents

Within fourteen (14) calendar days of Final Acceptance meeting of the project, Final As-Built documents shall be provided to the Contracting Officer in the formats described in paragraph "Electronic File Use". The final as-built document submittal shall also include the approved preliminary paper working as-built drawings.

1.7.10 Partial Occupancy

For projects where portions of construction are to be occupied or activated before overall project completion, including portions of utility systems, as-built drawings for those portions of the facility being occupied or activated shall be supplied at the time the facility is occupied or activated. This same as-built information previously furnished must also be shown on the final set of as-built drawings at project completion.

1.7.11 Electronic File Use

Only personnel proficient in the preparation of CAD drawings shall be employed to modify the electronic contract drawings or prepare additional new electronic drawings. Additions and corrections to the contract drawings shall be equal in quality to that of the originals. Line work, line weights, lettering, layering conventions, and symbols shall be the same as the original line work, line weights, lettering, layering conventions, and symbols. If additional drawings are required, they shall be prepared using the specified electronic file format applying the same guidance specified for original drawings. Three dimensional (3D) elements shall be placed in files in their proper locations when using 3D files with spatially correct elements. If the Designer of Record used a different software than that requested by the Using Agency, the Designer of Record's files will be used for as-built purposes and then translated and/or exported, by the Contractor, to the Using Agency's system. The title block and drawing border to be used for any new final as-built drawings shall be identical to that used on the contract drawings. Additions and corrections to the contract drawings shall be accomplished using CAD media files supplied by the Government. All work by the Contractor shall be done on files in the format in which they are provided. Translation of files to a different format, for the purpose of As-Built production, and then retranslating back to the format originally provided, will not be acceptable. The original electronic files provided by the Government will be provided in the format compatible with the Using Agency. The Using Agency uses 2020 Autodesk Revit CAD software system. The Contractor shall be responsible for providing all program files and hardware necessary to prepare final as-built drawings. The Contracting Officer will review final as-built drawings for accuracy

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and the Contractor shall make all required corrections, changes, additions, and deletions.

- a. When final revisions have been completed, the cover sheet drawing shall show the wording "RECORD DRAWING AS-BUILT" followed by the name of the Contractor in letters at least 5 mm 3/16 inch high. All other contract drawings shall be marked in the bottom right-hand corner of each drawing either "AS-BUILT" drawing denoting no revisions on the sheet, or "REVISED AS-BUILT" denoting one or more revisions. As-Built drawings shall be dated with the Contract Completion Date in the revision block.
- b. After receipt by the Contractor of the approved working as-built drawings and the original contract drawings files the Contractor shall, within thirty (30) calendar days, make the final as-built submittal. This submittal shall consist of 2 sets of completed final as-built drawings on separate media consisting of both BIM model files, exported CAD models (both compatible with the Using Agency's system on electronic storage media identical to that supplied by the Government) and full size set in PDF format and the return of the approved marked up working as-built drawings. They shall be complete in all details and identical in form and function to the contract drawing files supplied by the Government. Any translations or adjustments necessary to accomplish this are the responsibility of the Contractor. The Government reserves the right to reject any files it deems incompatible with the required BIM or CAD software system. All paper drawings, drawing files and storage media submitted will become the property of the Government upon final approval. Failure to submit final as-built drawing files and marked drawings as specified shall be cause for withholding any payment due the Contractor under this contract. Approval and acceptance of final as-built drawings shall be accomplished before final payment is made to the Contractor.

1.8 NOT USED

1.8.1 Not Used

1.9 EQUIPMENT DATA, O&M, & REPAIR MANUALS WITH FIELD TRAINING REQUIREMENTS

1.9.1 Real Property Equipment

Equipment-in-Place Data

Contractor shall be required to make an Equipment-in-Place list of all installed equipment furnished under this contract. This list shall include all information usually listed on manufacturer's name plate. The Form is part of SPECIAL PROVISIONS and is included following the SPECIAL PROVISIONS, so to positively identify the piece of property. The list shall also include the cost of each piece of installed property F.O.B. construction site. For each of the items which are specified herein to be guaranteed for a specified period from the date of acceptance thereof, the following information shall be given: The name, serial and model number address of equipment supplier, or manufacturer originating the guaranteed item. The information shall also be provided in excel spreadsheet format with columns for above information in addition to floor, space id as listed in the drawings, system, and submittal register id number for the item. The Contractor's guarantee to the Government of these items will not be limited by the terms of any manufacturer's guarantee to the Contractor. Furnish the list as one (1) reproducible and three (3) copies, and in electronic

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format on CD to the Contracting Officer thirty (30) calendar days before completion of any segment of the contract work which has an incremental completion date.

Maintenance and Parts Data

The Contractor will be required to furnish a brochure, catalog cut, parts list, manufacturer's data sheet or other publication which will show detailed parts data on all other equipment subject to repair and maintenance procedures not otherwise required in Operations and Maintenance Manuals specified elsewhere in this contract. This information shall be provided electronically in PDF format with bookmarks for each piece of equipment with file name included in a separate column or linked worksheet in the equipment data excel spreadsheet as described in the paragraph above. Distribution of directives shall follow the same requirements as listed in paragraph above.

1.9.2 O&M and Repair Manuals

Withholding & Copies

The Contractor shall provide 6 electronic format copies on CD of the Equipment Operating, Maintenance, and Repair Manuals and two complete hardcopies; these requirements shall apply even through the Technical Specification section indicates otherwise. The manuals shall be prepared electronically in PDF format containing bookmarks for each table of contents item. The PDF file shall be referenced in a separate column or linked worksheet in the equipment data excel spreadsheet as described previously. Separate manuals shall be provided for each utility system as defined per the Technical Specification. Operations and Maintenance manuals shall be accepted/approved before field training or ninety (90) calendar days before substantial completion (whichever occurs earlier). An amount of \$20,000 shall be withheld until submittal and acceptance/approval of O&M manuals is complete. A draft outline and table of contents shall be submitted for acceptance/approval at 50% contract completion. See paragraph "EQUIPMENT DATA, O&M, & REPAIR MANUALS WITH FIELD TRAINING REQUIREMENTS" for detail O&M and Repair Manual format.

1.9.3 Field Training

1.9.3.1 Training Course

Contractor shall conduct a training course for the operating staff for each particular component and system. Where the training period is not identified by the technical specification, a minimum of 1 hour of training shall be provided for that component or system. Training shall only occur after the Manuals have been approved/ accepted by the Government and during normal working time, and shall start after the system or component is functionally completed. The field instructions shall cover all of the items contained in the Equipment Operating, Maintenance and Repair Manuals as identified per individual Technical Specifications. The training will include both classroom and "hands-on" training. The Contractor shall submit a lesson plan outlining the information to be discussed during training periods. This lesson plan will be submitted ninety (90) calendar days before contract completion and accepted/approved before the field training occurs. Training shall be documented by the Contractor and a list of attendees shall be furnished to the Government. Training audio/ video shall be digitally recorded on CDs or DVDs and shall be furnished to the Government within ten (10) calendar days following training.

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1.9.3.2 Training Recording

The Contractor shall provide all equipment, materials, and trained personnel required to visually and audibly record all site operations and maintenance (O&M) training sessions. The video technician/trainer shall be employed by a video production company that has been in business for a minimum of 2 years. The Contractor shall submit for acceptance by the Government, the resume of the technician/trainer and the video production company, and the proposed video format. The video format shall be one in wide use, and any software necessary to view the video shall be provided to the Government. Video shall be provided to the Government on DVD. Audio shall be adjusted, filtered or otherwise controlled to ensure the presenter can be understood at all times. Each system or piece of equipment shall be covered on a single DVD or set of DVDs, which shall be identified with a type written label showing the name of the project, equipment or system, and contract number. This same information shall be provided as an introduction on each DVD. When two or more DVDs are provided for a single system or piece of equipment, they shall be packaged as a set in an appropriate storage case. Provide three copies of each DVD(s) for each training session. Training DVDs shall be furnished to the Government within ten (10) working days following training.

1.9.4 Organization of O&M Manuals:

Assemble O&M data into a series of electronically indexed files using the current version of Adobe Acrobat or similar software capable of producing PDF file format. Organize the files in numerical sequence corresponding to the individual sections within the construction contract specifications.

- a. Provide one individual PDF file containing a table of contents for the entire O&M manual, corresponding to the table of contents for the contract specifications.
- b. Provide one individual PDF file for each two-digit numbered division in the contract specifications (e.g., one file for Division 03 Concrete, a separate file for Division 04 Masonry, etc.). At the front of each two-digit numbered division file, provide a table of contents listing each separate specification section within the Division, electronically bookmarked to the first page of O&M data for each section. Arrange PDF pages in sequence according to their specification section number. Include manufacturer's datasheets, operating instructions, and maintenance instructions. Include copies of final test reports, final commissioning reports, final certification reports, installer's instructions for the operator/user, or similar documents supplementing the manufacturer's information. Where manufacturer's datasheets describe multiple products or versions, mark the datasheets to clearly indicate the exact product or version provided. Indicate model name, number, configuration, color or style name, optional features selected, etc., as applicable.
- c. Use PDF files obtained directly from the manufacturer wherever applicable. Where PDF files must be locally produced, produce such files by direct electronic conversion from the electronic native files wherever applicable. PDF files may be produced by scanning hard copy documents only when electronic originals are not available; notify the Contracting Officer's representative of any such cases, and perform optical character recognition processing on the resulting scanned documents.
- d. Ensure all PDF files are searchable and legible when viewed on computer.

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monitor. Orient each PDF page such that the predominant information is displayed horizontally on a computer monitor, regardless of "portrait" vs "landscape" format of the page. Provide an electronic header on each page indicating the contract number, project; "W912" and "ZHTV number," and project name. Provide an electronic footer on each page indicating the two-digit numbered division and the sequential page number of each page.

1.10 AVAILABILITY OF UTILITIES

a. Availability and Use of Utility Services

The Government will furnish electric utilities free of charge for the Contractor's use during construction. The Contractor is responsible for maintaining all existing services and ensuring capacity of the service to safely provide the Contractor's additional loads.

b. Additional Requirements

- (1) Drinking Water may not be obtained from approved outlets by Contractor personnel.
- (2) Existing Washing Facilities in the building may not be used by Contractor employees.
- (3) Electrical service may be obtained through coordination with Ohio Edison and the Air Force.
- (4) If the Contractor requires a temporary trailer, the Contractor is responsible for all connection fees, but the government will pay for the utility usage.

1.10.1 Alterations to Utilities

Where changes and relocations of utility lines are noted to be performed by others, the Contractor shall give the Contracting Officer at least thirty (30) calendar days written notice in advance of the time that the change or relocation is required. In the event that, after the expiration of thirty (30) calendar days after the receipt of such notice by the Contracting Officer, such utility lines have not been changed or relocated and delay is occasioned to the completion of the work under contract, the Contractor may be entitled to a time extension equal to the period of time lost by the Contractor after the expiration of said thirty (30) calendar day period. Any modification to existing or relocated lines required as a result of the Contractor's method of operation shall be made wholly at the Contractor's expense and no additional time will be allowed for delays incurred by such modifications.

1.10.2 Interruptions of Utilities

- a. No utility services shall be interrupted by the Contractor to make connections, to relocate, or for any purpose without approval of the Contracting Officer.
- b. Request for Permission to shut down services shall be submitted in writing to the Contracting Officer not less than seventeen (17) working days before date of proposed interruption. The request shall give the following information:
 - (1) Nature of Utility. (Gas, L.P. or H.P., Water, etc.)

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- (2) Size of line and location of shutoff.
- (3) Buildings and services affected.
- (4) Hours and date of shutoff.
- (5) Estimated length of time services will be interrupted.

- c. Services shall not be shut off until receipt of approval of the proposed hours and date from the Contracting Officer.
- d. Shutoffs which will cause interruption of Government work operations as determined by the Contracting Officer shall be accomplished during regular non-work hours or on non-work days of the Using Agency without any additional cost to the Government.
- e. Operation of valves on water mains will be by persons of that utility. Where shutoff of water lines interrupts service to fire hydrants or fire sprinkler systems, the Contractor shall arrange his operations and have sufficient material and personnel available to complete the work without undue delay or to restore service without delay in event of emergency.
- f. Flow in gas mains which have been shut off shall not be restored until the Government inspector has determined that all items serviced by the gas line have been shut off.

1.11 NOT USED

1.12 PERFORMANCE OF WORK BY THE CONTRACTOR

- a. The requirements found in Section 00 70 00.
 - b. The "TOTAL AMOUNT OF WORK" is defined as total direct contract labor (including testing and layout personnel), exclusive of other general condition or field overhead personnel, material, or equipment.
 - c. Within seven calendar days after the award of any subcontract at any tier, either by himself or a subcontractor, the Contractor shall submit to the Contracting Officer a completed Standard Form SF1413 Statement and Acknowledgement (available at the GSA Forms Library, <https://www.gsa.gov/reference/forms#>). The form shall include the subcontractor's acknowledgement of the inclusion in his subcontract of the clauses in Section 00 70 00 of this contract entitled 52.222-4 "Contract Work Hours and Safety Standards - Overtime Compensation"; 52.222-8 "Payrolls and Basic Records"; 52.222-7 "Withholding of Funds"; 52.222-14 "Disputes Concerning Labor Standards"; 52.222-13 "Compliance with Construction Wage Rate Requirements and Related Regulations"; 52.222-6 "Construction Wage Rate Requirements" (formerly named "Davis-Bacon Act"); 52.222-9 "Apprentices and Trainees"; 52.222-10 "Compliance with Copeland Act Requirements"; 52.222-11 "Subcontracts (Labor Standards); 52.222-12 "Contract Termination - Debarment"; 52.222-15 "Certification of Eligibility". Nothing contained in this contract shall create any contractual relation between any subcontractor and the Government.
 - d. Veterans Employment Emphasis for U.S. Army Corps of Engineers Contracts
- In addition to complying with the requirements outlined in Subpart 22.13, provision 52.222-38, clause 52.222-35, clause 52.222-37, DFARS Subpart 22.13 and United States Department of Labor regulations, U.S. Army Corps of Engineers (USACE) contractors and subcontractors at all tiers are encouraged to promote the training and employment of U.S. veterans while

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performing under a USACE contract. While no set-aside, evaluation preference, or incentive applies to the solicitation or performance under the resultant contract, USACE contractors are encouraged to seek out highly qualified veterans to perform services under this contract. The following resources are available to assist USACE contractors in their outreach efforts:

(1) U.S. Department of Labor Veterans' Employment and Training Service (VETS): <https://www.dol.gov/vets/>

(2) Federal veteran employment information:
<https://www.fedshirevets.gov/>

(3) Veterans Opportunity to Work (VOW) Program:
<https://benefits.va.gov/vow/>

(4) U.S. Army Warrior Transition Command Employment Index:
<http://wct.army.mil/modules/employers/index.html>

(5) Hiring Our Heroes:
<https://www.uschamberfoundation.org/hiring-our-heroes>

1.13 SUPERINTENDENCE OF SUBCONTRACTORS

a. The Contractor shall be required to furnish the following, in addition to the superintendence required by clause 52.236-6 - "Superintendence By The Contractor" in Section 00 70 00.

(1) If more than 50 percent and less than 70 percent of the value of the contract work is subcontracted, one superintendent shall be provided at the site and on the Contractor's payroll to be responsible for coordinating, directing, inspecting and expediting the subcontract work.

(2) If 70 percent or more of the value of the work is subcontracted, the Contractor shall be required to furnish two such superintendents to be responsible for coordinating, directing, inspecting and expediting the subcontract work.

b. If the Contracting Officer, at any time after 50 percent of the subcontracted work has been completed, finds that satisfactory progress is being made, he may waive all or part of the above requirements for additional superintendence subject to the right of the Contracting Officer to reinstate such requirement if at any time during the progress of the remaining work he finds that satisfactory progress is not being made.

1.14 IDENTIFICATION OF EMPLOYEES.

a. The Contractor shall be responsible for furnishing an identification badge/card to each employee prior to the employees work on-site, and for requiring each employee engaged on the work to display identification as may be approved and directed by the Contracting Officer. All prescribed identification shall immediately be delivered to the Contracting Officer for cancellation upon release of the employee.

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1.15 NO ASBESTOS - CONTAINING MATERIAL (ACM) CERTIFICATION

1.15.1 Not Used

1.15.2 Construction Phase

Before final payment to the contractor, the contractor's project engineer/manager will sign and submit to the Government, on the contracting firm's letterhead, a dated copy of the following statement:

I hereby certify that to the best of my knowledge no asbestos-containing material (ACM) was used as a building material during this project. Furthermore, I understand that the Contracting Officer's Representative (COR) presumes that all materials marked "May Contain mineral fibers" are considered asbestos unless I either:

- (1) Submit a certification for each individual product installed and identified to contain mineral fibers that no asbestos-containing materials were installed.
- (2) Submit documentation to show that the products containing mineral fiber materials have been microscopically examined by an AIHA- or NVLAP-certified laboratory and the lab has determined that the material does not contain asbestos.

1.16 WARRANTY OF CONSTRUCTION

a. In addition to the requirements found in clause 52.246-21 Alt. I "WARRANTY OF CONSTRUCTION" in Section 00 70 00 the following shall be included:

- (1) This warranty shall continue for a period of 1 year from the date of final acceptance of the work. If the Government takes possession of any part of the work before final acceptance, this warranty shall continue for a period of 1 year from the date the Government takes possession.
- (2) Provide names, addresses, and telephone numbers of all subcontractors, equipment suppliers, or manufacturers with specific designation of their area of responsibilities if they are to be contacted directly on warranty corrections.

b. Warranty Management

(1) Warranty Management Plan

The Contractor shall develop a warranty management plan which shall contain information relevant to the clause "Warranty of Construction" in clause 52.246-21 Alt. I in Section 00 70 00. At least thirty (30) calendar days before the planned pre-warranty conference, the Contractor shall submit the warranty management plan for Government approval. The warranty management plan shall include all required actions and documents to assure that the Government receives all warranties to which it is entitled. The plan shall be in narrative form and contain sufficient detail to render it suitable for use by future maintenance and repair personnel, whether tradesmen, or of engineering background, not necessarily familiar with this contract. The term "status" as indicated below shall include due date and whether item has been submitted or was accomplished. Warranty information made

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available during the construction phase shall be submitted to the Contracting Officer for approval prior to each monthly pay estimate. Approved information shall be assembled in a binder and shall be turned over to the Government upon acceptance of the work. The construction warranty period shall begin on the date of project acceptance and shall continue for the full product warranty period. A joint 4 month and 9 month warranty inspection shall be conducted, measured from time of acceptance, by the Contractor, Contracting Officer and the Customer Representative. Information contained in the warranty management plan shall include, but shall not be limited to, the following:

(a) Roles and responsibilities of all personnel associated with the warranty process, including points of contact and telephone numbers within the organizations of the Contractors, subcontractors, manufacturers or suppliers involved.

(b) Listing and status of delivery of all Certificates of Warranty for extended warranty items, to include roofs, HVAC balancing, pumps, motors, transformers, and for all commissioned systems such as fire protection and alarm systems, sprinkler systems, lightning protection systems, etc.

(c) A list for each warranted equipment, item, and feature of construction or system indicating:

1. Name of item.
2. Model and serial numbers.
3. Location where installed.
4. Name and phone numbers of manufacturers or suppliers.
5. Names, addresses and telephone numbers of sources of spare parts.
6. Warranties and terms of warranty. This shall include one-year overall warranty of construction. Items which have extended warranties shall be indicated with separate warranty expiration dates.
7. Cross-reference to warranty certificates as applicable.
8. Starting point and duration of warranty period.
9. Summary of maintenance procedures required to continue the warranty in force.
10. Cross-reference to specific pertinent Operation and Maintenance manuals.
11. Organization, names and phone numbers of persons to call for warranty service.
12. Typical response time and repair time expected for various warranted equipment.

(d) The Contractor's plans for attendance at the 4 and 9 month post-construction warranty inspections conducted by the Government.

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(e) Procedure and status of tagging of all equipment covered by extended warranties.

(f) Copies of instructions to be posted near selected pieces of equipment where operation is critical for warranty and/or safety reasons.

c. Performance Bond

(1) The Contractor's Performance Bond will remain effective throughout the construction warranty period and warranty extensions.

(2) In the event the Contractor or his designated representative(s) fails to commence and diligently pursue any work required, and in a manner pursuant to the requirements thereof, the Contracting Officer shall have a right to demand that said work be performed under the Performance Bond by making written notice on the surety. If the surety fails or refuses to perform the obligation it assumed under the Performance Bond, the Contracting Officer shall have the work performed by others, and after completion of the work, may make demand for reimbursement of any or all expenses incurred by the Government while performing the work, including, but not limited to administrative expenses.

(3) In the event sufficient funds are not available to cover the construction warranty work performed by the Government at the Contractor's expense, the Contracting Officer will have the right to recoup expenses from the bonding company.

(4) Following oral or written notification of required warranty repair work, the Contractor will respond as dictated by para. 1.16e "Contractor's Response to Warranty Service Requirements". Written verification will follow oral instructions. Failure of the Contractor to respond will be cause for the Contracting Officer to proceed against the Contractor as outlined in the paragraph 1.16c(2) and/or (3) "Performance Bond" (2) and/or (3) above.

d. Pre-Warranty Conference

Prior to contract completion and at a time designated by the Contracting Officer, the Contractor shall meet with the Contracting Officer to develop a mutual understanding with respect to the requirements of this clause. Communication procedures for Contractor notification of warranty defects, priorities with respect to the type of defect, reasonable time required for Contractor response, and other details deemed necessary by the Contracting Officer for the execution of the construction warranty shall be established/reviewed at this meeting. In connection with these requirements and at the time of the Contractor's quality control completion inspection, the Contractor will furnish the name, telephone number and address of a licensed and bonded company which is authorized to initiate and pursue warranty work action on behalf of the Contractor. This point of contact will be located within the local service area of the warranted construction, will be continuously available, and will be responsive to Government inquiry on warranty work action and status. This requirement does not relieve the Contractor of any of his responsibilities in connection with other portions of this provision.

e. Contractor's Response to Warranty Service Requirements.

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Following oral or written notification by the Contracting Officer or an authorized representative of the installation designated in writing by the Contracting Officer, the Contractor shall respond to warranty service requirements in accordance with the "Warranty Service Priority List" and the three categories of priorities listed below. The Contractor shall submit a report on any warranty item that has been repaired during the warranty period. The report shall include the cause of the problem, date reported, corrective action taken, and when the repair was completed. If the Contractor does not perform the construction warranty within the timeframe specified, the Government will perform the work and backcharge the construction warranty payment item established.

- (1) First Priority Code 1 Perform on-site inspection to evaluate situation, determine course of action, initiate work within 24 hours and work continuously to completion or relief.
- (2) Second Priority Code 2 Perform on-site inspection to evaluate situation, determine course of action, initiate work within 48 hours and work continuously to completion or relief.
- (3) Third Priority Code 3 All other work to be initiated within (5) five work days and work continuously to completion or relief.
- (4) The "Warranty Service Priority List" is as follows:

- Code 1 Air Conditioning System
 - a. Buildings with computer equipment.
 - b. Barracks, mess halls, BOQ/BEQ (entire building down).
 - c. Troop medical and dental.
- Code 2 Air Conditioning Systems
 - a. Recreational support.
 - b. Air conditioning leak in part of building, if causing damage.
 - c. Air conditioning system not cooling properly
 - d. Admin buildings with ADP equipment not on priority list.
- Code 1 Doors
 - a. Overhead doors not operational.
- Code 1 Electrical
 - a. Power failure (entire area or any building operational after 1600 hours).
 - b. Traffic control devices.
 - c. Security lights
 - d. Smoke detectors and fire alarm systems
- Code 2 Electrical
 - a. Power failure (no power to a room or part of building).
 - b. Receptacle and lights.
- Code 3 Electrical
 - a. Street, parking area lights
- Code 1 Gas
 - a. Leaks and breaks.
 - b. No gas to family housing unit or cantonment area.

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- Code 1 Heat
 - a. Hospital/Medical facilities.
 - b. Commissary, Clubs and Main PX.
 - c. Army Reserve Projects, Training Bldg & OMS Administrative Areas of Bldg.
 - d. Area power failure affecting heat.
- Code 2 Heat
 - a. Medical storage.
 - b. Barracks.
 - c. Army Reserve Projects, Training Bldg & OMS Administrative Areas of Bldg.
- Code 3 Interior
 - a. Floor damage
 - b. Paint chipping or peeling
- Code 1 Intrusion Detection Systems
 - Finance, PX and Commissary, and high security areas.
- Code 2 Intrusion Detection Systems
 - Systems other than those listed under Code 1.
- Code 1 Kitchen Equipment
 - a. Dishwasher.
 - b. All other equipment hampering preparation of a meal.
- Code 2 Kitchen Equipment
 - All other equipment not listed under Code 1.
- Code 2 Plumbing
 - a. Flush valves not operating properly
 - b. Fixture drain, supply line commode, or water pipe leaking.
 - c. Commode leaking at base.
- Code 3 Plumbing
 - a. Leaking faucets
- Code 1 Refrigeration
 - a. Commissary.
 - b. Mess Hall, Army Reserve Projects.
 - c. Cold Storage.
 - d. Hospital.
 - e. Medical storage.
- Code 2 Refrigeration
 - Mess hall - other than walk-in refrigerators and freezers.
- Code 1 Roof Leaks
 - Temporary repairs will be made where major damage to property is occurring.
- Code 2 Roof Leaks
 - Where major damage to property is not occurring, check for location of leak during rain and complete repairs on a Code 2 basis.

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- Code 1 Sprinkler System
 - All sprinkler systems, valves, manholes, deluge systems, and air systems to sprinklers.
- Code 1 Swimming Pools
 - Chlorine leaks or broken pumps.
- Code 1 Tank Wash Racks (Bird Baths)
 - All systems which prevent tank wash.
- Code 1 Water (Exterior)
 - Normal operation of water pump station.
- Code 2 Water (Exterior)
 - No water to facility.
- Code 1 Water, Hot (and Steam)
 - a. Hospitals and Mess Halls.
 - b. Army Reserve Projects, Training Bldg & OMS Bldg.
 - c. BOQ, BEQ, barracks (entire building).
 - d. Medical and dental.
- Code 2 Water, Hot
 - No hot water in portion of building listed under Code 1 (items a through c)

(5) Should parts be required to complete the work and the parts are not immediately available, the Contractor shall have a maximum of 12 hours after arrival at the job site to provide the Contracting Officer or an authorized representative of the installation designated in writing by the Contracting Officer, with firm written proposals for emergency alternatives and temporary repairs for Government participation with the Contractor to provide emergency relief until the required parts are available on site for the Contractor to perform permanent warranty repair. The Contractors proposals shall include a firm date and time that the required parts shall be available on site to complete the permanent warranty repair. The Contracting Officer or an authorized representative of the installation designated in writing by the Contracting Officer, will evaluate the proposed alternatives and negotiate the alternative considered to be in the best interest of the Government to reduce the impact of the emergency condition. Alternatives considered by the Contracting Officer or an authorized representative of the installation designated in writing by the Contracting Officer will include the alternative for the Contractor to "Do Nothing" while waiting until the required parts are available to perform permanent warranty repair. Negotiating a proposal which will require Government participation and the expenditure of Government funds shall constitute a separate procurement action by the using service.

f. Equipment Warranty Identification Tags

(1) The Contractor at the time of installation shall provide warranty identification tags on all Contractor and Government furnished equipment which he has installed.

(a) The tags shall be suitable for interior and exterior locations, resistant to solvents, abrasion, and to fading caused by sunlight, precipitation, etc. These tags shall have a

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permanent pressure-sensitive adhesive back, and they shall be installed in a position that is easily (or most easily) noticeable. Contractor furnished equipment that has differing warranties on its components will have each component tagged.

(b) Sample tags shall be submitted for Government review and approval. These tags shall be filled out representative of how the Contractor will complete all other tags.

(c) Tags for Warrantied Equipment: The tag for this equipment shall be similar to the following. Exact format and size will be as approved.

EQUIPMENT WARRANTY CONTRACTOR FURNISHED EQUIPMENT	
MFG NAME	MODEL NO.
SERIAL NO.	
CONTRACT NO.	
CONTRACTOR NAME	
CONTRACTOR WARRANTY EXPIRES	
MFG WARRANTY (IES) EXPIRE	

EQUIPMENT WARRANTY GOVERNMENT FURNISHED EQUIPMENT	
MFG NAME	MODEL NO.
SERIAL NO.	
CONTRACT NO.	
CONTRACTOR NAME	
CONTRACTOR WARRANTY EXPIRES	
MFG WARRANTY (IES) EXPIRE	

(d) If the manufacturer's name (MFG), model number and serial number are on the manufacturer's equipment data plate and this data plate is easily found and fully legible, this information need not be duplicated on the equipment warranty tag. The Contractor warranty expires (warranty expiration date) and the final manufacturer's warranty expiration date will be determined as specified by clause 52.246-21 Alt. I "WARRANTY OF CONSTRUCTION" in Section 00 70 00.

(2) Execution. The Contractor will complete the required information on each tag and install these tags on the equipment by the time of and as a condition of final acceptance of the equipment.

(3) Payment. The work outlined above is a subsidiary portion of the contract work, and has a value to the Government approximating 5% of the value of the Contractor furnished equipment. The Contractor will assign a value of that amount in the breakdown for progress payments

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mentioned in the clause 52.232-5 "Payments Under Fixed-Price Construction Contracts" in Section 00 70 00.

(4) Equipment Warranty Tag Replacement. The Contractor's warranty with respect to work repaired or replaced shall run for one year from the date of repair or replacement. Such activity shall include an updated warranty identification tag on the repaired or replaced equipment. The tag shall be furnished and installed by the Contractor, and shall be identical to the original tag, except that the Contractor's warranty expiration date will be one year from the date of acceptance of the repair or replacement.

1.17 SALVAGE MATERIALS AND EQUIPMENT

The Contractor shall maintain adequate property control records for all materials or equipment specified in UFGS Section 02 41 00 DEMOLITION AND to be salvaged. These records may be in accordance with the Contractor's system of property control, if approved by the property administrator. The Contractor shall be responsible for the adequate storage and protection of all salvaged materials and equipment and shall replace, at no cost to the Government, all salvage materials and equipment which are broken or damaged during salvage operations as the result of his negligence, or while in his care.

1.18 NOT USED

1.19 PROJECT SIGN

a. General. The Contractor shall furnish and erect at the location directed one project sign. The sign shall be lettered on one side only and shall conform to the details shown as an attachment at the end of this specification section. Refer to the Construction Project Identification Sign details at the back of this section.

Project nomenclature shall be: WPAFB HVAC and Controls Replacement F/20824

Architect-Engineer name shall be: Tetra Tech, Inc.

Sponsor name shall be: GC Shall confirm with COR

b. Materials. Sign Panels can be constructed using either of the materials below:

(1) The sign panels shall be constructed of good sound materials suitable for the purpose. Lumber shall be salt treated softwood of No. 2 grade or better. Sizes shown are nominal. Plywood shall be 1/2-inch, B-B, marine grade and with dimensional lumber uprights and bracing. Screws shall be of commercial quality and of sizes shown.

(2) The sign Panels shall be fabricated using premium, furniture-grade exterior plywood laminated on both sides with factory-baked polyester painted aluminum surfaces and with dimensional lumber uprights and bracing. Screws shall be commercial quality and of the sizes shown.

c. Maintenance. The signs shall be maintained in good condition until completion of the contract, shall remain the property of the Contractor, and shall be removed from the site upon completion of work under the contract.

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d. Logos. The Corps of Engineers and the Army Star Logo, must be provided in sizes and styles as specified by the graphic formats shown in the attachment at the end of this section. The Contractor shall make no other changes..

e. Painting. The sign and posts shall be given one prime coat and two finish coats of gloss exterior-type enamel paint, either as specified in the Base Architectural Compatibility Guide or as approved by the COR.

Paint colors shall be as follow:

Black - Federal Standard 595a Color Number 27038
White - Federal Standard 595a Color Number 27875
Red - PANTONE 032

f. Payment. No separate payment will be made for furnishing and erecting the project sign(s) as specified and costs thereof shall be considered a subsidiary obligation of the Contractor.

1.20 TIME EXTENSIONS FOR UNUSUALLY SEVERE WEATHER

This provision specifies the procedure for the determination of time extensions for unusually severe weather in accordance with the clause 52.249-10 "Default (Fixed-Price Construction)" in Section 00 70 00. In order for the Contracting Officer to award a time extension under this clause, the following conditions must be satisfied:

The weather experienced at the project site during the contract period must be found to be unusually severe, that is, more severe than the adverse weather anticipated for the project location during any given month.

Indicate the location of the National Weather Service office closest to the site. See: <https://www.weather.gov/srh/nws/offices> The Contractor shall make his own investigations and determinations as to weather conditions at the site. Data may be obtained from various National Weather Service offices located generally at airports of principal cities nearest to this project being: Wilmington, OH.

The unusually severe weather must actually cause a delay to the completion of the project. The delay must be beyond the control and without the fault or negligence of the Contractor.

The following schedule of monthly anticipated adverse weather delays is based on National Oceanic and Atmospheric Administration (NOAA) or similar data for the project location and will constitute the base line for monthly weather time evaluations. The Contractor's progress schedule must reflect these anticipated adverse weather delays in weather dependent activities.

MONTHLY ANTICIPATED ADVERSE WEATHER DELAY WORK DAYS BASED ON FIVE (5) DAY WORK WEEK

JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
11	8	6	5	6	6	4	3	4	4	8	5

Upon acknowledgment of the Administrative Notice to Proceed (ADMIN NTP) and continuing throughout the contract, the Contractor will record on the daily CQC report, the occurrence of adverse weather and resultant impact to

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normally scheduled work. Actual adverse weather delay days must prevent work on critical activities for 50 percent or more of the Contractor's scheduled work day. The number of actual adverse weather delay days shall include days impacted by actual adverse weather (even if adverse weather occurred in previous month), be calculated chronologically from the first to the last day of each month, and be recorded as full days. If the number of actual adverse weather delay days exceeds the number of days anticipated listed above, the Contracting Officer will convert any qualifying delays to calendar days, giving full consideration for equivalent fair weather work days, and issue a modification in accordance with the clause 52.249-10 "Default (Fixed-Price Construction)" in Section 00 70 00.

1.21 INTERFERENCE WITH TRAFFIC AND PUBLIC AND PRIVATE PROPERTY

- a. The Contractor at all times shall dispose his plant and conduct the work in such manner as to cause as little interference as possible with private and public travel. Damage (other than that resulting from normal wear and tear) to roads, shall be repaired to as good a condition as they were prior to the beginning of work and to the satisfaction of the Contracting Officer.
- b. Under no circumstances will steel tracked equipment travel across any pavement, concrete or asphalt, on base. All steel tracked equipment will be transported on trailers to the jobsite.
- c. The Contractor shall schedule his work to cause the least amount of interference with base operations. Work schedules shall be subject to the approval of the Contracting Officer. Under no circumstances shall fire water system be out of service for more than 5 calendar days. Permission to interrupt any base roads and/or utility service shall be requested in writing a minimum of 10 calendar days prior to the desired date of interruption. Interruption shall not take place without specific approval from the Contracting Officer.

1.22 NOT USED.

1.23 NOT USED

1.24 AUTHORIZED WORK DAYS AND HOURS

- a. The standard workweek shall be Monday through Friday 0700-1600.
- b. Contractor may request days and hours outside of the Standard Work Week when necessary to perform particular task or to maintain or recover schedule. Request are to be made 5 days in advance to the COR and User and not less than that required for schedule Coordination.
- c. No work shall occur on federal holidays:

New Year's Day (January 1).
 Birthday of Martin Luther King, Jr. (Third Monday in January).
 Washington's Birthday (Third Monday in February).
 Memorial Day (Last Monday in May).
 Juneteenth National Independence Day (June 19).
 Independence Day (July 4).
 Labor Day (First Monday in September).
 Columbus Day (Second Monday in October).
 Veterans Day (November 11).

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Thanksgiving Day (Fourth Thursday in November).
Christmas Day (December 25).

- d. Authorized days and hours described above pertain to work inside the existing facilities and/or inside the secure construction laydown areas. Work outside secure areas may be scheduled as needed, subject to advance approval from the Contracting Officer's Representative and confirmation of availability of necessary services. Request are to be made 5 days in advance to the COR and User and not less than that required for schedule Coordination.
- e. Prior to the start of any construction operations, a schedule of work or operations in proper sequence shall be submitted by the contractor for approval by the Fire Protection Branch, Security Police, Safety Office, and the Contracting Officer so as to cause a minimum amount of disruption to the normal flow of traffic on streets, pedestrian travel, base security, and facility operations.
- f. Before starting any work where traffic will be impeded or hindered, especially fire trucks and security police vehicles, call and notify the Fire Protection Branch (extension 257-4075) and the Security Forces (257-6516 or 257-6517) 72 hours prior to placing barricades or starting any work. Under no circumstances place any barricades or start work until the Fire Protection Brance and Security Forces have approved the barricading of work areas.
- g. Furnish flagmen, barricades, and other warning devices as required to safely route traffic through the construction areas. Selected streets may be closed to traffic when necessary during the construction operations, upon prior approval of the Contracting Officer. When roads are closed, establish and mark safe detours for vehicles to follow. Detours, road closures and lane closures shall be marked and signed in accordance with "Work Zone Traffic Control-Standards and Guidelines" (US Department of Transportation; Federal Highway Administration) or with the Ohio Department of Transportation, Ohio Manual of Uniform Traffic Control Devices (MUTCD). Furnish the Contracting Officer with the names, addresses and telephone numbers of two employees who will be responsible for warning light and barricade maintenance.
- h. Gate Hours - (The information in this paragraph is for information only and is subject to change in particular before or after federal holidays and the latter part of December) In general only the 24 hour gates are open on federal holidays and weekends.

1. Normal gate operating hours are:

2. Gate 1A - 24 hours, Broad Street/Spruce Way

3. Gate 12A - 24 hours, Childlaw Road (Main Gate, Area A)

4. Gate 15A - SR-844/Skeel Avenue

5. Gate 26A(Commercial Vehicle Inspection Gate), Medway Road

6. Gate hours are subject to change and the Contractor is responsible for verifying them daily, sufficiently in advace of needing access, to properly and adequately plan and schedule delivery of materials, equipment, and personnel to mitigate impacts to contract performance. No equitable adjustment to the contract price ad performance period

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will be made for changes in the operational hours of access control points (e.g., entry gates, etc.). All costs and delays incurred because of a change in the hours of operation, or having to wait in line to gain entry, shall be absorbed by the Contractor.

1.25 NOT USED

1.26 COMPLIANCE WITH POST/BASE REGULATIONS

a. The site of the work is on a military reservation and all rules and regulations issued by the Commanding Officer covering general safety, security, sanitary requirements, pollution control and traffic regulations, shall be observed by the Contractor. Information regarding these requirements may be obtained by contacting the Contracting Officer, who will provide such information or assist in obtaining same from appropriate authorities.

b. Contractor personnel shall park only in areas authorized by the Contracting Officer.

1.27 NOT USED

1.28 ENGLISH-SPEAKING REPRESENTATIVE

At all times when any performance of the work at any site is being conducted by any employee of the Contractor or his subcontractors, the Contractor shall have a representative present at each site who has the capability of receiving instructions in the English language, fluently speaking the English language and explaining the work operations to persons performing the work, in the language that those performing the work are capable of understanding. The Contracting Officer shall have the right to determine whether the proposed representative has sufficient technical bilingual capabilities, and the Contractor shall immediately replace any individual not acceptable to the Contracting Officer.

1.29 SALES and USE TAX

Some states have tax exemptions for certain aspects of work when done for the federal government and the Contractor shall check with the state where the project is located for more information. If a sales tax exemption is applicable, the contractor is responsible for obtaining any required exemption certification.

1.30 CONTRACTOR SECURITY TRAINING/FACILITY ACCESS REQUIREMENTS

a. All contractor employees, to include subcontractor employees, shall be briefed on the local iWATCH program (training standards provided by the requiring activity ATO) by the contractor and all associated subcontractors. This local developed training will be used to inform employees of the types of behavior to watch for and instruct employees to report suspicious activity to the COR. The training shall be completed within seven (7) calendar days after commencing work on the site with the results reported to the COR NLT seven (7) calendar days after completion of training.

b. All contractor employees, to include subcontractor employees, shall comply with applicable installation, facility and area commander installation/facility access and local security policies and procedures (provided by government representative). The contractor shall also

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provide all information required for background checks to meet installation access requirements to be accomplished by installation Provost Marshal Office, Director of Emergency Services or Security Office. Contractor workforce must comply with all personal identity verification requirements as directed by DOD, HQDA and/or local policy. In addition to the changes otherwise authorized by clause 52.243-4 - "Changes" in Section 00 70 00 of this contract, should the Force Protection Condition (FPCON) at any individual facility or installation change, the Government may require changes in contractor security matters or processes.

- b. Per AR 530-1, Operations Security, all contractor employees and associated subcontractor employees must complete Level 1 OPSEC Training within thirty (30) calendar days of commencing work. Additionally, all contractor employees and associated subcontractor employees must complete annual OPSEC awareness training.
- c. Refer to clause 52.222-54: "Employment Eligibility Verification" in Section 00 70 00 for e-Verify requirements.

1.31 INSURANCE--WORK ON A GOVERNMENT INSTALLATION

In addition to the requirements of clause 52.228-5 "Insurance - Work on a Government Installation" found in Section 00 70 00 the following shall be provided:

- a. Coverage complying with State laws governing insurance requirements, such as those requirements pertaining to Workman's Compensation and Occupational Disease Insurance. Employer's Liability Insurance shall be furnished in limits of not less than \$100,000.00 except in states with exclusive or monopolistic funds.
- b. Comprehensive Automobile Liability Insurance for both bodily injury and property damage, shall be furnished in limits of not less than \$200,000.00 per person, \$500,000.00 per accident for bodily injury, and \$20,000.00 per accident for property damage. When the Financial Responsibility or Compulsory Insurance Law of the State, requires higher limits, the policy shall provide for coverage of at least those higher limits.
- c. Within seven calendar days after the award of contract, the Contractor shall submit to the Contracting Officer Evidence of Insurance for all insurance coverages. Dates of coverage shall be entered into RMS and maintained thru the contract duration. If and when new insurance is obtained because of expiration or renewal, new certificates shall be submitted to the Contracting Officer within seven days of obtainment and the new coverage information and dates entered into RMS.

1.32 AVAILABILITY OF SAFETY AND OCCUPATIONAL HEALTH (SOH) REQUIREMENTS MANUAL (EM 385-1-1)

As covered by clause 52.236-13 "Accident Prevention" in Section 00 70 00, compliance with EM 385-1-1 is a requirement for this contract. Copies may be downloaded from the following website:
<https://www.publications.usace.army.mil/USACE-Publications/Engineer-Manuals/>

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1.33 NOT USED

1.34 NOT USED

1.35 FIRE PROTECTION DURING CONSTRUCTION

The Contractor is alerted to the requirements of FAR 52.236-12 "Cleaning Up" and more specifically to the requirements for fire protection during construction spelled out in UFC 3-600-01, EM 385-1-1, and NFPA 241 Building Construction and Demolition Operations. This item must be covered in the submittal required under FAR 52.236-13 "Accident Prevention".

1.36 CONSTRUCTION HAZARD COMMUNICATION

The Contractor is required to comply with the requirements of the OSHA Hazard Communication Standard in alignment with the Globally Harmonized System (GHS) (29 CFR 1926.59). This standard is designed to inform workers of safe and appropriate methods of working with hazardous substances in the workplace. The standard has five requirements, and every hazardous or potentially hazardous substance used or stored in the work area is subject to all five. They are:

(1) Hazard Classification. Any company which produces or imports a chemical or compound must conduct a hazard classification of the substance to determine its potential health or physical hazard. The hazard evaluation consists of an investigation of all the available scientific evidence about the substance. The Contractor is required to assure that all producers (manufacturer/distributors) have performed these classifications and transmit the required information with any hazardous materials being used or stored on the project site. From the hazard classification, a substance may be classified as a health hazard or a physical hazard. These classifications are then further broken down into hazard categories according to the severity of the effect:

Health Hazards	Physical Hazards
Carcinogens	Combustible liquids
Irritants	Compressed gases
Sensitizers	Explosives
Corrosives	Flammables
Toxic substances	Organic peroxides
Highly toxic substances	Unstable substances
Substances harmful to specific organs or parts of the body	Water-reactive substances

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(2) Warning Labels. If a chemical is hazardous or potentially hazardous, the producer or importer must affix a label to every container of that chemical before it leaves his facility. The Contractor must assure these labels are attached and legible. The label must identify the hazard symbol/pictograms, signal words, hazard statements, product name or identifier (identify hazardous ingredients, where appropriate), precautionary statements and pictograms, supplier identification, and supplemental information. If the hazardous substance is transferred to another container, that container must then be labeled, tagged, or marked with the name of the chemical and the appropriate hazard warning. Warning labels shall be replaced immediately if they are defaced or removed.

(3) Safety Data Sheets. The producer or importer must also supply a safety data sheet (SDS) that follows the 16 heading format as defined by GHS. The Contractor must keep these available in the work area where the substance is used, so that the people using the substance can easily review important safety and health information, such as:

- (i) Emergency procedures for leaks, spills, fire and first aid.
- (ii) Precautions necessary for use, handling, and storage.
- (iii) Useful facts about the substance's physical or chemical properties.
- (iv) Regulatory information and any other pertinent information including information on preparation and revision of the SDS.

(4) Work Area Specific Training. Because of hazardous substance may react differently depending on how it is used or the environment of the work area, the Contractor must conduct work area specific training; special training which takes the Contractor's operations, environment, and work policies into consideration. Work area training presents:

The hazardous substances which are present in the work place and the hazards they pose.

Ways to protect against those hazards, such as protective equipment, emergency procedures, and safe handling.

Where the SDS's are kept, and an explanation of the labeling system.

Where the Contractor's written Hazard Communication Program is located.

(5) The Written Hazard Communication Program. In accordance with OSHA and the EM 385-1-1 requirements, the Contractor must prepare a written Hazard Communication Program. This document will be included in the Contractor's Accident Prevention Plan. This document states the hazardous or toxic agent inventory, how the Contractor plans to ensure that hazardous materials are appropriately labeled, how and where SDS's will be maintained, and how employees will be provided with specific information and training.

1.37 MECHANICAL/ELECTRICAL ROOM LAYOUT

Detailed mechanical/electrical room layout drawings shall be submitted for

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approval in accordance with Section 01 33 00 SUBMITTAL PROCEDURES. Layout drawings shall show location and maintenance clearances for all mechanical/electrical room equipment, and all utility runs/chases for mechanical, electrical, telephone and other similar systems. Drawings shall be submitted at the same time as the submittals for the equipment to be located within the mechanical/electrical room.

1.38 RED ZONE MEETING

Approximately sixty (60) calendar days prior to Contract Required Completion Date, the Contractor and the Government's project delivery team will conduct what is known as the Red Zone Meeting to discuss the close-out process, to schedule the events and review responsibilities for actions necessary to produce a timely physical, as well as fiscal, project close-out. The Red Zone meeting derives its name from the football term used to describe the team effort to move the ball the last 20 yards into the end zone. The close-out of a construction project sometimes can be equally as hard and most definitely requires the whole team's efforts.

1.39 PARTNERING

In order to most effectively accomplish this contract, the Government proposes to form a partnership with the Contractor to develop a cohesive building team. It is anticipated that this partnership would involve Project Delivery Team members from the Corps of Engineers, Program Sponsor, facility user representatives, the Contractor, primary subcontractors, and the designers. The partnership will draw upon the strength of each organization in an effort to achieve a project that is without any safety mishaps, conforms to the Contract, stays within budget and on schedule.

The Government encourages partnering to be initiated near the beginning of the Contract and endure through the life of the Contract. This partnership would be bilateral in membership and participation will be mandatory.

1.40 PROGRESS PHOTOGRAPHS

Monthly digital photography shall be performed between the first and fifth of each month, delivered no later than the 10th of each month taken as described herein. A minimum of six views from different positions shall be taken as directed to show, inasmuch as possible, work accomplished during the previous month, and a minimum of six views shall be taken of the completed work. Additional views and positions may be required by the Contracting Officer to depict the work done.

Digital photographs are required to be furnished to the COR. The Contractor shall furnish digital photographs (in pdf) depicting the progress of the work during construction and after final inspection by the Contracting Officer of the conditions at the completion of the contract.

No separate payment will be made for these services and all costs in connection thereto shall be considered a subsidiary obligation of the Contractor.

1.41 DAMAGE TO WORK

Version 1

The responsibility for damage to any part of the permanent work shall be as set forth in clause 52.236-7 "Permits And Responsibilities" in Section 00

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70 00. Except as herein provided, damage to all work (including temporary construction), utilities, materials, equipment and plant shall be repaired to the satisfaction of the Contracting Officer at the Contractor's expense, regardless of the cause of such damage.

1.42 NOT USED

1.43 NOT USED

1.44 NOT USED

1.45 NOT USED

1.46 NOT USED

1.47 NOT USED

1.48 FINAL CLEANING

Clean the premises in accordance with clause 52.236-12 "Cleaning Up" in Section 00 70 00 and additional requirements state here. Remove stains, foreign substances, and temporary labels from surfaces. Vacuum carpet and soft surfaces. Clean equipment and fixtures to a sanitary condition. Clean or replace filters of operating equipment if cleaning is not possible or practicable. Remove debris from roofs, drainage systems, gutters, and downspouts. Sweep paved areas and rake clean landscaped areas. Remove waste, surplus materials, and rubbish from the site. Remove all temporary structures, barricades, project signs, fences and construction facilities. A list of completed clean-up items shall be submitted on the day of final inspection.

1.49 BASIS FOR SETTLEMENT OF PROPOSALS

Actual costs will be used to determine equipment costs for a settlement proposal submitted on the total cost basis under FAR 49.206-2(b) "Basis for Settlement of Proposals". In evaluating a termination settlement proposal using the total cost basis, the following principles will be applied to determine allowable equipment costs:

a. Actual costs for each piece of equipment, or groups of similar serial or series equipment, need not be available in the contractor's accounting records to determine local actual equipment costs.

b. If equipment costs have been allocated to a contract using predetermined rates, those charges will be adjusted to actual costs.

c. Recorded job costs adjusted for unallowable expenses will be used to determine equipment operating expenses.

d. Ownership costs (depreciation) will be determined using the contractor's depreciation schedule (subject to the provisions of FAR 31.205-11 "Depreciation").

f. License, taxes, storage and insurance costs are normally recovered as an indirect expense and unless the contractor charges these costs directly to contracts, they will be recovered through the indirect expense rate.

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1.50 BUILDER UPDATING

a. The BUILDER Sustainment Management System is an application used by DoD to manage facility data to support operations, maintenance, and future capital investment decisions. The Contractor shall complete a BUILDER Update Form to document the changes to facility data resulting from work under this contract, to the extent applicable to the following thirteen Unifomat II building systems: A10 Foundations, B10 Superstructure, B20 Exterior Enclosure, B30 Roofing, C10 Interior Construction, C20 Stairs, C30 Interior Finishes, D10 Conveying, D20 Plumbing, D30 HVAC, D40 Fire, D50 Electrical, and G20 Site Improvements. A partially completed form is attached as "Appendix A - BUILDER Update Form" to this section. Following contract award, the Government will provide the BUILDER Update Form in Excel spreadsheet format.

1.50.1 Data Completion

The BUILDER Update Form provided by the Government is pre-populated with the types, quantities, and locations of equipment expected to be removed, installed, or replaced under this contract. The Contractor shall update the pre-populated data where necessary to reflect as-built conditions, and for each record shall add the installation date and the manufacturer's name, model number, and serial number as applicable for the installed item. See further instructions provided with the BUILDER Update Form.

1.50.2 Submittals

Quarterly, update and submit a copy of the Working BUILDER Update Form in Excel spreadsheet format, reflecting as-built information for work completed to date. Submit a Final BUILDER Update Form in Excel spreadsheet format upon completion of work. With each submittal, include a signed memorandum from the Quality Control Manager certifying the form accurately reflects as-built information for work completed to date.

1.51 Final Acceptance Testing

1.51.1 General

Where work involves fire suppression and/or fire alarm systems, comply with testing requirements in this section and in the applicable technical specification section(s). In the event of conflict between this section and the applicable technical specification section(s), the more stringent requirement shall govern.

1.51.2 Request for Final Acceptance Testing

At least 14 days prior to the proposed start of the Final Acceptance Test for fire suppression and/or fire alarm systems, submit a Request for Final Acceptance Testing. Include a signed memorandum from the Quality Control Manager, applicable system completion certificates, and a testing plan.

a. The Quality Control Manager shall certify the system(s) have passed the Contractor's quality control following the same testing protocols described in the final testing plan, and shall certify all requirements of this section have been met. Include confirmation that each addressable device has been labeled with its assigned address, as-built drawings are complete, and all quality control measures and preliminary tests have been successfully completed with any deficiencies resolved and re-tested.

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b. Completion certificates shall include NFPA 13 Material and Test Certificate, NFPA 72 Fire Alarm System Record of Completion, and/or other similar documentation as applicable.

c. The testing plan shall identify the devices to be tested, the sequence in which they will be tested, the procedures and equipment to be used, applicable NFPA or other standards, the names and qualifications of the Contractor's installing technician(s) and/or engineer(s) who shall conduct the test, and the communication devices to be used during the test.

1.51.3 Test Administration

On the date of Final Acceptance Testing, provide one hard-copy set of as-built drawings at the project site for Government reference during the test, and provide two hard-copy sets of the applicable inspection and testing forms for use by the Fire Department and the Contracting Officer's representative. Conduct the test as detailed in the testing plan

1.52 WORK CLEARANCES AND COORDINATION

1.52.1 Digging Clearances

All excavation, trenching, soil boring, directional boring, or similar work requires advance Government approval. At least 17 days prior to the planned work, mark the site with stakes or paint, provide a detailed description and location map to the Government project inspector, and request Government preparation of an Air Force Form 103, Base Civil Engineering Work Clearance Request. The Government shall process the Form 103 to obtain coordination from utility system owners, and shall notify the Contractor of the outcome and, if approved as planned, the date upon which work is approved to proceed. Do not proceed with work prior to receipt of the completed and signed AF Form 103. When contacting OUPS they will only mark certain Utilities, The Contractor may need to coordinate GPR or other means to locate Utilities at contractors cost.

1.52.2 Building Systems Service Outages

Where work requires shutdown or modification of fire sprinklers, HVAC systems, electrical distribution, and/or other building systems or utilities, follow procedures as described above for digging clearances. The applicable system owner shall operate all controls or valves. Do not proceed with work prior to receipt of the completed and signed AF Form 103

1.52.3 Safe Clearance from Underground Utilities

Where work is required near underground utilities, the Contractor shall physically verify underground utility locations, including utility depth, by hand digging using wood or fiberglass handled tools when any adjacent construction work is expected to come within 3 feet of the underground system.

1.52.4 Written Notices

Where work is required near underground utilities, the Contractor shall physically verify underground utility locations, including utility depth, by hand digging using wood or fiberglass handled tools when any adjacent construction work is expected to come within 3 feet of the underground

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system.

1.52.5 Form

It is the requesters responsibility to be on site with the 103 form in hand at the time of the marking of utilities to get the 103 signed. Work shall be started within 30 days. If not; requester/designee is responsible for removal of flags. Flags must be removed upon completion of project/dig by requester/designee. If work is not complete within 90 days a new 103 request is required. A signed copy of the 103 needs to be given to 88 CES/CSU and available on site or electronically.

1.53 HOT WORK

Hot Work is defined by NFPA 51B as work involving burning, welding, or a similar operation that is capable of initiating fires or explosions. Obtain and maintain on site a Hot Work Permit prior to performing any Hot Work. Hot Work Permits are valid for a maximum of 24 hours; longer operations require a new permit each day. Hot Work Permits shall be obtained via either of the following methods:

a. Government-issued permits: Call the WPAFB Fire and Emergency Services Flight (WPFES) Hot Work line, 937-904-3157, 24 hours (one work day) in advance and follow the instructions provided by WPFES.

b. Contractor-issued permits: Provide an individual or individuals to serve as Permit Authorizing Individual(s) (PAI), obtain training from WPFES, and have the PAI issue permits for Hot Work under this contract. Call the Fire Prevention office at 937-257-4075 to request PAI training, and provide the name and contact information of the PAI to WPFES and to the Contracting Officer's Representative. WPFES will train and provide supporting materials to the Contractor's PAI, review the area in which Hot Work will be required, and issue a Permit Authorized Designated Area letter. Once training is complete and the Permit Authorized Designated Area letter has been issued, the PAI may issue Hot Work Permits for work under this contract within that designated area.

The Contractor may, at the Contractor's option, use either of the above methods. For either method, maintain a file of all Hot Work Permits issued for work under this contract. Perform all Hot Work in accordance with NFPA 51B and the WPAFB Contractors Guide for Safeguarding Building Construction Operations from Fire Hazards. Determine site-specific hazards and appropriate fire prevention and fire protection measures. Ensure appropriate fire extinguishing equipment is readily available on site and personnel are trained in its operation. Quantity and type of extinguishing equipment shall be as determined by the Contractor but not less than one type 2A:10B:C dry chemical extinguisher nor less than as may be required by other sections of this contract. Maintain a fire watch on site for a minimum of one hour after completion of Hot Work.

1.54 CRANES AND OTHER TALL EQUIPMENT

a. For any crane or other item 30 feet tall or above, submit a Crane Usage Request at least 60 days prior to planned use. Identify the item's maximum height above ground level, the location(s) in which the item will be positioned, the expected dates of its presence, and (where applicable) any obstruction marking and lighting. Provide a crane range diagram or similar drawing depicting how the item will be configured and operated to stay at or below the identified maximum

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height above ground level. If the crane will be lowered when not in active use, identify the expected daily hours of operation.

b. For any crane or other item tall enough to encroach upon airfield "imaginary surfaces" as described by 14 CFR 77.21, provide obstruction marking and lighting in accordance with FAA AC 70/7460-1. Items lower than 140 feet above ground level and at least 2000 feet from the nearest runway centerline are unlikely to encroach upon airfield imaginary surfaces, but the Government shall make a determination on a case-by-case basis.

c. When the Government determines a crane or other item will encroach upon airfield imaginary surfaces, the Government shall obtain appropriate waivers to airfield criteria and shall make appropriate notifications to the Federal Aviation Administration. The Contractor shall not raise the item into position prior to receipt of Government confirmation that all applicable waivers and notifications have been completed.

1.55 LABELS

Labels shall be provided for electrical outlets, light switches, and any electrical components and junction boxes. Labels shall be provided for all Mechanical Piping, HVAC Equipment, Access Doors, and Miscellaneous HVAC Equipment. Label shall be coordinated with the COR. Samples shall be provided for the COR to review and accept. Samples shall be provided to the COR for review.

1.56 ENERGY/UTILITY METERING

This contract involves provision of metering for energy or utility systems (electric, gas, water, steam, high temperature hot water, or centrally-produced heating hot water or chilled water).

When the contract requires provision of energy/utility metering, all such meters shall communicate only with the Advanced Meter Reading System (AMRS).

Other specification sections or drawings within this contract may refer to the Energy Management Control System (EMCS), Industrial Control System (ICS), Building Automation System (BAS), or similar terms. In the context of monitoring or control of HVAC, lighting, or other building systems, all such references shall be interpreted to refer to the YARS EMCS. In the context of measuring energy usage, all such references shall be interpreted to refer to the YARS AMRS.

The Contractor shall program energy/utility meter systems to communicate with AMRS via the Civil Engineer Community of Interest Network Enclave (CE COINE), using the Government's existing Schneider Electric Power Monitoring Expert (PME) user interface. At least 60 days prior to planned connection to AMRS, request Government identification of an IP address for AMRS and Government assistance in obtaining access to the PME user interface.

1.57 CONTRACTOR SUPPLY AND USE OF ELECTRONIC SOFTWARE FOR PROCESSING WAGE REQUIREMENT STATUTE CERTIFIED LABOR PAYROLLS

a. The contractor is encouraged to use a commercially-available electronic system to process and submit certified payrolls electronically to the

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Government. The requirements for preparing, processing and providing certified labor payrolls are established by the Wage Rate Requirements statute.

- b. If the contractor elects to use an electronic payroll processing system, then the contractor shall be responsible for obtaining and providing for all access, licenses, and other services required to provide for receipt, processing, certifying, electronically transmitting to the Government, and storing weekly payrolls and other data required for the contractor to comply with the Wage Rate Requirements statute. When the contractor uses an electronic payroll system, the electronic payroll service shall be used by the contractor to prepare, process, and maintain the relevant payrolls and basic records during all work under this construction contract and the electronic payroll service shall be capable of preserving these payrolls and related basic records for the required 3 years after contract completion. If the contractor chooses to use an electronic payroll system, then the contractor shall obtain and provide electronic system access to the Government, as required to comply with the Wage Rate Requirements over the duration of this construction contract. The access shall include electronic review access by the Government contract administration office to the electronic payroll processing system used by the contractor.
- c. The contractor's provision and use of an electronic payroll processing system shall meet the following basic functional criteria:
 - (1) commercially available;
 - (2) compliant with appropriate Wage Rate Requirements statute payroll provisions in the Federal Acquisition Regulation (FAR);
 - (3) able to accommodate the required numbers of employees and subcontractors planned to be employed under the contract
 - (4) capable of producing an Excel spreadsheet-compatible electronic output of weekly payroll records for export in an Excel spreadsheet to be imported into the contractor's Quality Control System (QCS) version of Resident Management System (RMS), that in turn shall export payroll data to the Government's RMS;
 - (5) demonstrated security of data and data entry rights;
 - (6) ability to produce contractor-certified electronic versions of weekly payroll data;
 - (7) ability to identify erroneous entries and track the date/time of all versions of the certified Wage Rate Requirements statute payrolls submitted to the government over the life of the contract;
 - (8) capable of generating a durable record copy, that is, a CD or DVD and PDF file record of data from the system database at end of the contract closeout. This durable record copy of data from the electronic payroll processing system shall be provided to the Government during contract closeout.
 - (9) Contractors will upload weekly pay roll records into RMS-Contractor Pay Roll section in the Quality Control System (QCS) version of Resident Management System (RMS), that in turn shall export payroll data to the Government's RM.
- d. All contractor-incurred costs related to the contractor's provision and use of an electronic payroll processing service shall be included in the contractor's price for the overall work under the contract. The costs for compliance with the Wage Rate Requirements statute by using electronic payroll processing services shall not be a separately bid or

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reimbursed item under this contract.

PART 2 PRODUCTS NOT USED

PART 3 EXECUTION NOT USED

-- End of Section --

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CONTRACT NO.

TITLE AND LOCATION
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CONTRACTOR

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SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

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CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION		DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		00 80 00.00 06	Documentation to show that the products containing mineral fiber materials have been microscopically examined by an AIHA- or NVLAP-certified laboratory and the lab has determined that the material does not contain asbestos	1.15.2													
			Insurance	1.31													
			Sales and Use Tax	1.29													
			SD-11 Closeout Submittals														
			Preliminary (Working) As-Built Drawings	1.7.4	G												
			Final As-Built Drawings	1.7.1	G												
			CAD Working As-Built Drawings	1.7.1.2	G												
			Equipment-in-Place List	1.9.1													
			Maintenance and Parts Data	1.9.1													
			Warranty Management Plan	1.16	G												
			Completion of BUILDER Update Form														
		01 11 00	SD-01 Preconstruction Submittals Work Plan		G												
		01 32 01.00 06	SD-01 Preconstruction Submittals														
			Project Scheduler Qualifications	1.3	G												
			Preliminary Project Schedule	3.4.1	G												
			Initial Project Schedule	3.4.2	G												

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						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION		DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 32 01.00 06	Periodic Schedule Update	3.6.2	G												
		01 33 00	SD-01 Preconstruction Submittals														
			Submittal Register	1.8	G												
		01 33 29.00 06	SD-01 Preconstruction Submittals														
			Sustainability Action Plan	1.5.1.1													
			GBI GPCP	1.4													
			Design Phase High Performance and Sustainable Building Checklist														
			Sustainability Progress Report	1.5.1.4													
			SD-11 Closeout Submittals														
			Final Sustainability eNotebook	1.5.1.2	G												
			Amended Final Sustainability eNotebook	1.5.1.2	G												
			Final High Performance and Sustainable Building Checklist	1.5.1.2	G												
		01 35 26.00 06	SD-01 Preconstruction Submittals														
			Accident Prevention Plan (APP)	1.7	G												
			Fatigue Management Plan														
			Bloodborne Pathogen Plan														
			Exposure Control Plan														
			Hearing Conservation Program														
			Respiratory Protection Plan														
			Health Hazard Control Program														
			Hazard Communication Program														
			Heat Stress Monitoring Plan														
			Cold Stress Monitoring Plan														

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(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 35 26.00 06	Indoor Air Quality Management Plan														
			Crystalline Silica Assessment														
			Fire Prevention Plan														
			Hazardous Energy Control Plan														
			Standard Pre-Lift Plan (LHE)														
			Critical Lift Plan - LHE														
			Fall Protection and Prevention Plan														
			SD-06 Test Reports														
			Reports	1.12													
			Accident/Mishap Investigation														
			Reports														
			Monthly Exposure Reports	1.12.3													
			Regulatory Citations and Violations	1.12.4													
			SD-07 Certificates														
			Hot work permit														
		01 35 43	SD-01 Preconstruction Submittals														
			Environmental Coordinator	1.4.3													
			Designation and Application for RCRA Training														
			Application For Authorization - Environmental Restoration Site	3.3.3	G												
			Disturbance														
			Erosion Control Plan	3.2.2	G												

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(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 35 43	Storm Water Pollution Prevention Plan (SWPPP)	3.2.3	G												
			Notice of Intent (NOI) for Coverage Under NPDES Permit	3.2.3													
			Sanitary or Wastewater Permit to Install (PTI)	3.2.4	G												
			Drinking Water Plans Approval	3.2.5	G												
			Ohio EPA Notification of Demolition and Renovation/Abatement	3.1.1													
			Asbestos Work Plan	3.1.2	G												
			Fuel-Fired Equipment Startup Information	3.1.10													
			HAZMAT Inventory	3.6.2													
			Site Specific Spill Plan	3.6.2	G												
			WPAFB Worksheet 1433 - Hazardous Waste Storage Permit	3.7.2													
			Permit Applications for UST Work	3.8	G												
			SD-06 Test Reports														
			Asbestos Abatement Clearance Air Sampling Report	3.1.4													
			HVAC System Water Analysis	3.2.6	G												
			Quarterly Recycling and Solid Waste Summary Report	3.5.3													
			Refrigerant Record	3.6.5													
			SD-07 Certificates														

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						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION		DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 35 43	Certification of Recovered and Biobased Products	1.5.2													
			Section 608 Technician Certification	3.6.5													
			Refrigerant Recovery Equipment Certification	3.6.5													
			Certificate of Conformity	3.1.9													
			STI SP001 Inspector's Training Certification														
			SD-11 Closeout Submittals														
			Green Products Determination Form (GPDF)	1.5.3													
			Notice of Termination (NOT)	3.2.3													
			STI SP001 Aboveground Storage Tank Inspection Report														
			STI SP131 Underground Storage Tank Inspection Report														
		01 45 04.10 06	SD-01 Preconstruction Submittals														
			Construction Quality Control Plan	3.3	G RO												
		01 45 35	SD-01 Preconstruction Submittals														
			SIOR Letter of Acceptance		G												
			Project Manual		G												
			Written NDT Practices														
			SD-06 Test Reports														
			Special Inspection Daily Reports	3.1.1													
			Special Inspection Daily Reports	3.1.2													

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		01 45 35	Special Inspection Biweekly Reports	3.1.1													
			SD-07 Certificates														
			AISC Certified Steel Fabricator	2.1													
			Steel Truss Plant Quality Assurance Program														
			Wood Truss Plant Quality Assurance Program														
			AC472 Accreditation														
			Certified Plant														
			Certificate of Compliance	2.1													
			Special Inspector of Record														
			Special Inspector	1.5													
			Qualification Records for Nondestructive Testing Technicians														
			SD-11 Closeout Submittals														
			Interim Report		G												
			Comprehensive Final Report	3.1.2	G												
			Final Report		G												
		01 46 00.00 06	SD-01 Preconstruction Submittals														
			Project Schedule	1.7	G DO												
			SD-06 Test Reports														
			PVT Procedures		G DO												
			PVT Report		G DO												
			Trend Log Report														

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		01 46 00.00 06	SD-07 Certificates														
			Certificate of Readiness	1.6	G DO												
			SD-10 Operation and Maintenance														
			Data														
			Systems Training	3.2	G DO												
			Training Plan	3.3	G RO												
			Training Attendance Rosters	3.3	G RO												
			Systems Manual		G DO												
			Maintenance and Service Life	3.4	G DO												
			Plans														
		01 50 00	SD-01 Preconstruction Submittals														
			Construction Site Plan	1.3	G												
			Traffic Control Plan	3.3.1	G												
			Haul Road Plan		G												
			Contractor Computer	1.6.1.4													
			Cybersecurity Compliance														
			Statements														
			Contractor Temporary Network	1.6.6													
			Cybersecurity Compliance														
			Statements														
			SD-06 Test Reports														
			Backflow Preventer Tests	3.4													
			SD-07 Certificates														
			Backflow Tester	1.4.1													
			Backflow Preventers	1.4													

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		01 78 23	SD-10 Operation and Maintenance														
			Data														
			Training Plan	3.1.1	G												
			Training Outline	3.1.3	G												
			Training Content	3.1.2	G												
			Operation And Maintenance	3.2.1	G												
			Manual, Progress Submittal														
			Operation And Maintenance	3.2.2	G												
			Manual, Prefinal Submittal														
			Operation And Maintenance	3.2.3	G												
			Manual, Final Submittal														
			SD-11 Closeout Submittals														
			Training Video Recording	3.1.4	G												
			Validation of Training Completion	3.1.6	G												
			Training Plan	3.1.1	G												
			Record Drawings And Utility	1.6.6.7	G												
			Systems														
		02 41 00	SD-01 Preconstruction Submittals														
			Demolition Plan	1.2.2	G												
			Existing Conditions	1.11													
			SD-07 Certificates														
			Notification	1.7													
			Notification of Demolition and														
			Renovation Form														
			SD-11 Closeout Submittals														
			Receipts	3.3.4													

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		03 30 00	SD-01 Preconstruction Submittals														
			Concrete Curing Plan	1.6.3.1													
			Quality Control Program	1.6.5	G												
			Quality Control Personnel	1.6.6													
			Certifications														
			Quality Control Organizational	1.6.6													
			Chart														
			Laboratory Accreditation	1.6.8													
			Maturity Method Data	3.3.8													
			SD-02 Shop Drawings														
			Reinforcing Steel	1.6.2.1	G												
			SD-03 Product Data														
			Joint Sealants	2.4.5	S												
			Joint Filler	2.4.4	S												
			Formwork Materials	2.1													
			Cementitious Materials	2.3.1	S												
			Vapor Retarder	2.4.6													
			Concrete Curing Materials	2.4.1													
			Reinforcement	2.6	S												
			Admixtures	2.3.4													
			Reinforcing Fibers	2.6.6													
			Mechanical Reinforcing Bar	2.6.2													
			Connectors														
			Waterstops	2.2.2													
			Local/Regional Materials	1.8.1	S												

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		03 30 00	Biodegradable Form Release Agent	2.2.3													
			Pumping Concrete	1.6.3.2													
			Finishing Plan	1.6.3.4													
			Nonshrink Grout	2.4.2													
			SD-05 Design Data														
			Concrete Mix Design	1.6.1.1	G												
			SD-06 Test Reports														
			Concrete Mix Design	1.6.1.1	G												
			Fly Ash	1.6.4.1													
			Pozzolan	1.6.4.1													
			Slag Cement	1.6.4.2													
			Aggregates	1.6.4.3													
			Fiber-Reinforced Concrete	1.6.4.4													
			Compressive Strength Tests	3.13.3.3													
			Air Content	3.13.3.4													
			Slump Tests	3.13.3.1													
			Water	2.3.2													
			SD-07 Certificates														
			Reinforcing Bars	2.6.1													
			Welder Qualifications	1.9													
			Silica Fume Manufacturer's Representative	1.6.3.3													
			Safety Data Sheets	1.6.3.5													
			Forest Stewardship Council (FSC) Certification	1.8.2													

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		03 30 00	Field Testing Technician and Testing Agency	1.6.6.2													
			Environmental Product Declaration	1.8.3	S												
			Energy Star Energy Performance Score	1.8.4	S												
			SD-08 Manufacturer's Instructions														
			Joint Sealants	2.4.5	S												
			Curing Compound	2.4.1													
		04 20 00	SD-02 Shop Drawings														
			Detail Drawings	3.4.1.1	G												
			SD-03 Product Data														
			Hot Weather Procedures	1.5.1													
			Cold Weather Procedures	1.5.2													
			Clay or Shale Brick	2.2.2													
			Cementitious Materials	2.4.1.1													
			SD-04 Samples														
			Clay or Shale Brick	2.2.2	G												
			Admixtures for Masonry Mortar	2.4.1.3	G												
			Anchors, Ties, and Bar Positioners	2.6.2	G												
			Joint Reinforcement	2.6.3	G												
			Clay Masonry Expansion-Joint Materials	2.6.6	G												
			SD-05 Design Data														
			Masonry Compressive Strength	2.1.2	G												

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		04 20 00	Bracing Calculations	3.2.5	G												
			Structural Lintel Calculations	3.2.6	G												
			SD-06 Test Reports														
			Efflorescence Test	2.2.2.1.3													
			Prism Tests	3.6.1.1													
			Single-Wythe Masonry Wall	3.6.1.2													
			Water Penetration Test														
			SD-07 Certificates														
			Special Masonry Inspector	1.3.1													
			Qualifications														
			Clay or Shale Brick	2.2.2													
			Cementitious Materials	2.4.1.1													
			Admixtures for Masonry Mortar	2.4.1.3													
			Admixtures for Grout	2.4.2.2													
			Anchors, Ties, and Bar	2.6.2													
			Positioners														
			Joint Reinforcement	2.6.3													
			SD-08 Manufacturer's Instructions														
			Admixtures for Masonry Mortar	2.4.1.3													
			Admixtures for Grout	2.4.2.2													
		05 05 23.16	SD-01 Preconstruction Submittals														
			Welding Quality Assurance Plan	3.2													
			SD-03 Product Data														
			Welding Procedure Qualifications	1.3													
			Welder, Welding Operator, and	1.3.3													
			Tacker Qualification														

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		05 05 23.16	Previous Qualifications	1.3.2													
			Pre-Qualified Procedures														
			Welding Electrodes and Rods	2.2													
			SD-06 Test Reports														
			Nondestructive Testing	3.3													
			Weld Inspection Log	3.2													
			SD-07 Certificates														
			Certified Welding Procedure	1.3.1													
			Specifications (WPS)														
			Certified Brazing Procedure	1.3.1													
			Specifications (BPS)														
			Certified Procedure Qualification	1.3.1													
			Records (PQR)														
			Certified Welder Performance	1.3.1													
			Qualifications (WPQ)														
			Certified Brazer Performance	1.3.1													
			Qualifications (BPQ)														
			Certified Welding Inspector	1.3.4													
			Nondestructive Testing	1.3.4													
			Personnel														
		05 12 00	SD-01 Preconstruction Submittals														
			Erection and Erection Bracing	1.3.1.1	G												
			Drawings														
			SD-02 Shop Drawings														
			Fabrication Drawings	1.3.2	G												
			SD-03 Product Data														

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		05 12 00	Shop Primer	2.6.2													
			Welding Electrodes and Rods	2.4.1													
			Direct Tension Indicator Washers	2.3.2.3													
			Non-Shrink Grout	2.4.2													
			Tension Control Bolts														
			Recycled Content for Structural Steel	2.2.1	S												
			Recycled Content for Structural Steel Tubing		S												
			Recycled Content for Steel Pipe		S												
			SD-05 Design Data														
			Design Calculations for Steel Connections	1.3.3	G												
			SD-06 Test Reports														
			Class B Coating	2.6.2													
			Bolts, Nuts, and Washers	2.3													
			Weld Inspection Reports	3.7.1.2													
			Direct Tension Indicator Washer Inspection Reports	3.7.2.1													
			Bolt Testing Reports	3.7.3.1													
			SD-07 Certificates														
			Steel	2.2													
			Bolts, Nuts, and Washers	2.3													
			AISC Structural Steel Fabricator Quality Certification														

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		05 12 00	AISC Structural Steel Erector Quality Certification														
			Welding Procedures and Qualifications	1.3.4.1													
			Welding Electrodes and Rods	2.4.1													
			Certified Welding Inspector	3.7.1.1													
			NDT Technician	3.7.1.2													
			Welding Procedure Specifications (WPS)	3.4													
		05 50 13	SD-02 Shop Drawings														
			Bollards/Pipe Guards	2.7	G												
			Angles and Plates	2.8	G												
			Roof Hatches	2.9	G												
			SD-03 Product Data														
			Corner Guards	2.4													
			Floor Gratings	2.6	G												
			Roof Hatches	2.9	G												
		05 52 00	SD-02 Shop Drawings														
			Fabrication Drawings	1.2.1	G												
			Fabrication Drawings	1.5.1	G												
			SD-03 Product Data														
			Concrete Inserts		G												
			Masonry Anchorage Devices		G												
			Protective Coating	2.1.2	G												
			Steel Railings and Handrails	2.2.4	G												

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		05 52 00	Anchorage and Fastening Systems	1.2.1	G												
			SD-05 Design Data														
			Delegated Design Submittal	1.5.2	G												
			SD-08 Manufacturer's Instructions														
			Installation Instructions	3.2													
		07 60 00	SD-02 Shop Drawings														
			Exposed Sheet Metal	2.1.1	G												
			Expansion Joints	3.1.15	G												
			Flashing for Roof Drains	3.1.13	G												
			Base Flashing	3.1.8	G												
			Counterflashing	3.1.9	G												
			Flashing at Roof Penetrations and Equipment Supports	3.1.16	G												
			Reglets	2.1.10	G												
			Copings	3.1.18	G												
			SD-04 Samples														
			Finish Samples	1.4.2	G												
			SD-07 Certificates														
			Warranty on Finishes	2.1.7.1	G												
			SD-08 Manufacturer's Instructions														
			Instructions for Installation	1.4.3													
			SD-10 Operation and Maintenance Data														
			Cleaning and Maintenance	1.4.3	G												
		07 84 00	SD-02 Shop Drawings														

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		07 84 00	Firestopping System	2.1													
			SD-03 Product Data														
			Firestopping Materials	2.2													
			SD-06 Test Reports														
			Inspection	3.3													
			SD-07 Certificates														
			Inspector Qualifications	1.5.2													
			Firestopping Materials	2.2													
			Installer Qualifications	1.5.1													
		07 92 00	SD-03 Product Data														
			Sealants	2.1													
			Primers	2.2													
			Bond Breakers	2.3													
			Backstops	2.4													
			Caulking	2.5													
			Cleaning Solvents	2.6													
			Field Adhesion	3.1													
			SD-07 Certificates														
			Indoor Air Quality For Interior	2.1.1													
			Sealants														
			Indoor Air Quality For Interior	2.1.3													
			Floor Joint Sealants														
			Indoor Air Quality For Interior	2.1.4													
			Acoustical Sealants														
			Indoor Air Quality For Interior	2.5													
			Caulking														

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		08 91 00	SD-02 Shop Drawings														
			Wall Louvers	1.4													
			SD-03 Product Data														
			Metal Wall Louvers	2.2													
			SD-04 Samples														
			Wall Louver Samples	1.5	G												
		09 29 00	SD-03 Product Data														
			Cementitious Backer Units	1.4.2													
			Glass Mat Water-Resistant	2.1.4													
			Gypsum Tile Backing Board														
			Water-Resistant Gypsum	2.1.3													
			Backing Board														
			Accessories	2.1.8													
			Gypsum Board	2.1.1													
			Recycled Content for Gypsum	2.1.1													
			Board														
			Recycled Content for Paper	2.1.1													
			Facing and Gypsum Cores														
			VOC Content of Joint Compound	2.1.6													
			SD-06 Test Reports														
			ASTM E90 Factory Test Report	3.6													
			ASTM E90 Factory Test Report	3.9													
			ASTM E336 Field Test Report	3.9													
			SD-07 Certificates														
			Asbestos Free Materials	2.1													

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		09 29 00	Indoor Air Quality for Gypsum Board	2.1.1													
			SD-08 Manufacturer's Instructions														
			Safety Data Sheets	2.1													
			SD-10 Operation and Maintenance														
			Data														
			Manufacturer Maintenance	2.1													
			Instructions														
		09 51 00	SD-02 Shop Drawings														
			Drawings	2.1													
			SD-03 Product Data														
			Acoustical Ceiling Systems	3.1													
			Acoustical Performance	2.1.1													
			SD-04 Samples														
			Acoustical Units	2.2	G												
			Acoustical Ceiling Tiles	2.1	G												
		09 90 00	SD-02 Shop Drawings														
			Piping Identification	3.10													
			SD-03 Product Data														
			Coating	2.1													
			Product Data Sheets	2.1													
			SD-04 Samples														
			Color	2.2	G												
			SD-07 Certificates														
			Qualification Testing	1.6.6.2													

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		09 90 00	Indoor Air Quality for Paints and Primers	1.6.5													
			Indoor Air Quality for Consolidated Latex Paints	1.6.5													
			SD-08 Manufacturer's Instructions														
			Application	1.6.2													
			Mixing	2.1													
			Manufacturer's Safety Data	1.8.1													
			Sheets														
			SD-10 Operation and Maintenance														
			Data														
			Coatings	2.1	G												
		10 44 16	SD-02 Shop Drawings														
			Fire Extinguishers	2.1.1													
			Accessories	Part 2													
			Cabinets	Part 2													
			Wall Brackets	2.2.2													
			Schedule	1.5													
			SD-03 Product Data														
			Fire Extinguishers	2.1.1													
			Accessories	Part 2													
			Cabinets	Part 2													
			Wall Brackets	2.2.2													
			Replacement Parts List	3.2.1													
			SD-07 Certificates														
			Fire Extinguishers Certifications	2.1.1													

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		10 44 16	Manufacturer's Warranty with	1.4													
			Inspection Tag														
		21 13 13	SD-01 Preconstruction Submittals														
			Qualified Fire Protection	1.2.3	G												
			Engineer (QFPE)														
			Sprinkler System Designer	1.4.2.1	G												
			Sprinkler System Installer	1.4.2.2	G												
			SD-02 Shop Drawings														
			Shop Drawing	1.4.1.1	G AE												
			SD-03 Product Data														
			Pipe	2.2.1.1	G												
			Fittings	2.2.1.2	G												
			Sprinklers	2.3	G												
			Pipe Hangers and Supports	2.2.3	G												
			Nameplates	2.1.2	G												
			Accessories	2.4	G												
			SD-05 Design Data														
			Hydraulic Calculations	1.2.1.2	G AE												
			SD-06 Test Reports														
			Test Procedures	3.6.1	G												
			SD-07 Certificates														
			Verification of Compliant	3.6.2.1	G												
			Installation														
			Request for Government Final	3.6.2.2	G												
			Test														

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		21 13 13	SD-10 Operation and Maintenance Data														
			Operating and Maintenance (O&M) Instructions	1.4.1.4	G												
			Spare Parts	1.6	G												
			SD-11 Closeout Submittals														
			As-built drawings	3.8													
		22 00 00	SD-02 Shop Drawings														
			Plumbing System	3.5.1	G												
			SD-03 Product Data														
			Backflow Prevention Assemblies	3.5.1.1													
			Press Fittings	2.1.1	G												
			Welding	1.6.1													
			Plumbing System	3.5.1													
			SD-06 Test Reports														
			Tests, Flushing and Disinfection	3.5													
			Test of Backflow Prevention Assemblies	3.5.1.1													
			SD-07 Certificates														
			Materials and Equipment	1.4													
			SD-10 Operation and Maintenance Data														
			Plumbing System	3.5.1	G												
		23 01 30.41	SD-01 Preconstruction Submittals														
			Record of Existing Conditions	3.1.1.1													
			Coordination Plan	3.1.1.2													

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		23 01 30.41	NADCA Firm	1.4.1													
			NADCA Team Assistants	1.4.1													
			NADCA Air System Cleaning Specialist (ASCS)	1.4.1													
			NADCA Supervisor Qualifications	1.4.1													
			Records of Experience in the Field of HVAC System Cleaning	1.4.2													
			NADCA Work Execution Schedule	1.5.1													
			SD-03 Product Data														
			Safety Data Sheets (SDS)	1.4.3													
			SD-06 Test Reports														
			Testing Procedures Summary	3.3.2													
			Post-Project Report	3.3.2	G												
		23 03 00.00 20	SD-03 Product Data														
			Equipment Labels	2.1.1													
			Pipe Labels	2.1.2													
			Duct Labels	2.1.3													
			SD-11 Closeout Submittals														
			Valve Schedules	2.1.4													
		23 05 15	SD-01 Preconstruction Submittals														
			Material, Equipment, and Fixture Lists	1.2	G												
			SD-02 Shop Drawings														
			Record Drawings	1.2	G												
			Connection Diagrams	1.2	G												

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		23 05 15	Coordination Drawings	1.2	G												
			Fabrication Drawings	1.2	G												
			Installation Drawings	3.1	G												
			SD-03 Product Data														
			Pipe and Fittings	2.1	G												
			Piping Specialties	2.2	G												
			Valves	2.3	G												
			Miscellaneous Materials	2.4	G												
			Supporting Elements	2.5	G												
			Equipment Foundation Data	1.2	G												
			SD-04 Samples														
			Manufacturer's Standard Color Charts	1.2	G												
			SD-05 Design Data														
			Pipe and Fittings	2.1	G												
			Piping Specialties	2.2	G												
			Valves	2.3	G												
			SD-06 Test Reports														
			Hydrostatic Tests	3.1													
			Air Tests	3.1													
			Valve-Operating Tests	3.1													
			Drainage Tests	3.1													
			Pneumatic Tests	3.1													
			Non-Destructive Electric Tests	3.1													
			System Operation Tests	3.1													
			SD-07 Certificates														

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		23 05 15	Record of Satisfactory Field Operation	1.4.2													
			List of Qualified Permanent Service Organizations	1.4.3													
			Listing of Product Installations	1.2													
			Records of Existing Conditions	1.2													
			Surface Resistance	3.1													
			Shear and Tensile Strengths	3.1													
			Temperature Ratings	3.1													
			Bending Tests	3.1													
			Flattening Tests	3.1													
			Transverse Guided Weld Bend Tests	3.1													
			SD-10 Operation and Maintenance Data														
			Operation and Maintenance Manuals	3.9	G												
		23 05 48.19	SD-02 Shop Drawings														
			Coupling and Bracing	3.1													
			Flexible Couplings or Joints	3.2													
			Equipment Restraint	2.1													
			SD-03 Product Data														
			Coupling and Bracing	3.1													
			Flexible Couplings Or Joints	3.2													
			Equipment Restraint	2.1													
			Anchor Bolts	3.6													

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		23 05 48.19	Vibration Isolators	2.1.2													
			SD-06 Test Reports														
			Anchor Bolts	3.6													
		23 05 93	SD-02 Shop Drawings														
			TAB Schematic Drawings and	1.3.3	G												
			Report Forms														
			SD-03 Product Data														
			Equipment and Performance	1.3	G												
			Data														
			SD-06 Test Reports														
			Completed Pre-Final DALT	3.3.5	G												
			Report														
			Certified Final DALT Report	3.3.8	G												
			SD-07 Certificates														
			Independent TAB Agency and	1.5.1	G												
			Personnel Qualifications														
		23 07 00	SD-02 Shop Drawings														
			MICA Plates	3.2.2.4													
			Pipe Insulation Systems	2.3													
			Pipe Insulation Systems	3.2													
			Duct Insulation Systems	3.3													
			Recycled content for insulation	2.3.1	S												
			materials														
			SD-03 Product Data														
			Pipe Insulation Systems	2.3													
			Pipe Insulation Systems	3.2													

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		23 07 00	Duct Insulation Systems	3.3													
			SD-07 Certificates														
			Indoor air quality for adhesives	2.2.1	S												
			SD-08 Manufacturer's Instructions														
			Pipe Insulation Systems	2.3													
			Pipe Insulation Systems	3.2													
			Duct Insulation Systems	3.3													
		23 08 00	SD-03 Product Data														
			Test Equipment	2.1	G												
			SD-06 Test Reports														
			Pipe Flushing, Testing, And	1.5	G												
			Water Treatment Reports														
			Completed Pre-Functional	3.3	G												
			Checklists														
			Seasonal Test Report	3.12	G												
			SD-07 Certificates														
			Certificate Of Readiness	1.6	G												
		23 09 00	SD-02 Shop Drawings														
			DDC Contractor Design Drawings	3.2	G												
			Draft As-Built Drawings	3.2	G												
			Final As-Built Drawings	3.2	G												
			SD-03 Product Data														
			Programming Software	1.8.3	G												
			Controller Application Programs	1.8.4	G												
			Configuration Software	1.8.1	G												
			Controller Configuration Settings	1.8.2	G												

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		23 09 00	Manufacturer's Product Data	2.2	G												
			Niagara Framework Supervisory Gateway Backups	1.8.5	G												
			Niagara Framework Engineering Tool	1.8.6	G												
			SD-06 Test Reports														
			Pre-Construction Quality Control (QC) Checklist	1.9.1	G												
			Post-Construction Quality Control (QC) Checklist	1.9.2	G												
			Start-Up Testing Report	3.4.2	G												
			PVT Procedures	3.5.1	G												
			PVT Report	3.5.3	G												
			SD-10 Operation and Maintenance Data														
			Operation and Maintenance (O&M) Instructions	3.6	G												
			Training Documentation	3.8.1	G												
			SD-11 Closeout Submittals														
			Enclosure Keys	2.5	G												
			Password Summary Report	3.1.6.1	G												
			Closeout Quality Control (QC) Checklist	1.9.3	G												
		23 21 13.00 20	SD-02 Shop Drawings														
			Hot water heating system	1.3.1													
			SD-03 Product Data														

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		23 21 13.00 20	Pumps	2.3.1													
			Expansion tanks	2.3.2													
			Backflow preventers														
			External air separation tanks	2.3.3													
			Hot water heating pipe	2.1.1													
			SD-06 Test Reports														
			Hydrostatic test of piping system	3.3.1													
			Auxiliary equipment and accessory tests	3.3.2													
			SD-07 Certificates														
			Backflow preventer certification	1.5.4													
			Report of prior installations	1.5.2.1													
			Welding procedures	1.5.2.2													
			Welder's qualifications	1.5.2.3													
		23 23 00	SD-02 Shop Drawings														
			Refrigerant Piping System	2.2	G												
			SD-03 Product Data														
			Refrigerant Piping System	2.2													
			Refrigerant Piping Tests	3.5													
			Verification of Dimensions	3.1													
			SD-06 Test Reports														
			Refrigerant Piping Tests	3.5													
			SD-07 Certificates														
			Service Organization	2.1													
			SD-10 Operation and Maintenance														
			Data														

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		23 23 00	Maintenance	1.5	G												
			Operation and Maintenance	3.4	G												
			Manuals														
			Demonstrations	3.4	G												
		23 25 00	SD-03 Product Data														
			Water Analysis	2.1.3	G												
			Spare Parts	1.3													
			Field Instructions	3.4.1													
			Tests	3.3.1	G												
			Training Course	3.4.1	G												
		23 30 00	SD-02 Shop Drawings														
			Detail Drawings	1.4.4	G												
			SD-03 Product Data														
			Metallic Flexible Duct	2.8.1.1													
			Insulated Nonmetallic Flexible	2.8.1.2													
			Duct Runouts														
			Duct Connectors	2.8.1.2													
			Duct Access Doors	2.8.2													
			Fire Dampers	2.8.3													
			Manual Balancing Dampers	2.8.4													
			Sound Attenuation Equipment	2.8.8													
			Diffusers	2.8.9.1													
			Registers and Grilles	2.8.9.2													
			Louvers	2.8.10													
			Centrifugal Fans	2.9.1.1													
			In-Line Centrifugal Fans	2.9.1.2													

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						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION		DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		23 30 00	Room Fan-Coil Units	2.10.1	G												
			Variable Volume, Single Duct	2.10.2.1	G												
			Terminal Units														
			Test Procedures	1.4.5													
			Diagrams	1.2.1.2	G												
			Indoor Air Quality for Duct	2.8.1	S												
			Sealants														
			SD-06 Test Reports														
			Performance Tests	3.10													
			Damper Acceptance Test	3.8													
			SD-07 Certificates														
			Ozone Depleting Substances	1.4.3													
			Technician Certification														
			SD-08 Manufacturer's Instructions														
			Manufacturer's Installation	3.2													
			Instructions														
			Operation and Maintenance	3.12.2													
			Training														
			SD-10 Operation and Maintenance														
			Data														
			Operation and Maintenance	3.12.1	G												
			Manuals														
			Fire Dampers	2.8.3	G												
			Manual Balancing Dampers	2.8.4	G												
			Centrifugal Fans	2.9.1.1	G												
			In-Line Centrifugal Fans	2.9.1.2	G												

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(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		23 30 00	Room Fan-Coil Units	2.10.1	G												
			Variable Volume, Single Duct	2.10.2.1	G												
			Terminal Units														
			SD-11 Closeout Submittals														
			Indoor Air Quality During	3.11	S												
			Construction														
		23 52 43.00 20	SD-02 Shop Drawings														
			Wiring Diagram	1.3.2.2													
			SD-03 Product Data														
			Boilers	2.1	G												
			Boiler Trim and Control	2.2	G												
			Equipment														
			SD-06 Test Reports														
			Operational Tests	3.4.1													
			Water Analysis	1.3.3													
			SD-07 Certificates														
			Boilers	2.1													
			Boiler Trim and Control	2.2													
			Equipment														
			SD-08 Manufacturer's Instructions														
			Boilers	2.1													
			SD-10 Operation and Maintenance														
			Data														
			Boilers	2.1	G												
			SD-11 Closeout Submittals														
			Boilers	2.1	G												

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						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	MAILED TO CONTR/ DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		23 64 10	SD-03 Product Data														
			Water Chiller	2.1	G												
			Verification of Dimensions	1.6.1													
			System Performance Tests	3.5													
			Demonstrations	3.6													
			Refrigerant	2.5.1													
			Water Chiller - Field Acceptance	3.4.1													
			Test Plan														
			SD-06 Test Reports														
			Field Acceptance Testing	3.4													
			Water Chiller - Field Acceptance	3.4.2													
			Test Report														
			System Performance Tests	3.5													
			SD-07 Certificates														
			Refrigeration System	3.1.8													
			Ozone Depleting Substances	1.3.1													
			Technician Certification														
			SD-08 Manufacturer's Instructions														
			Water Chiller - Installation	3.1													
			Instructions														
			SD-10 Operation and Maintenance														
			Data														
			Operation and Maintenance	3.6	G												
			Manuals														
			SD-11 Closeout Submittals														

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		23 64 10	Indoor Air Quality During Construction	3.3	S												
		23 64 26	SD-03 Product Data														
			Calibrated Balancing Valves	2.4.5	G												
			Expansion Joints	2.5.9	G												
			Pumps	2.6	G												
			Combination Strainer and Pump Suction Diffuser	2.5.3													
			Expansion Tanks	2.7													
			Air Separator Tanks	2.8													
			Water Treatment Systems	2.9	G												
			SD-06 Test Reports														
			Piping Welds NDE Report	3.1.1.3													
			Pressure Tests Reports	3.4.2													
			SD-07 Certificates														
			Employer's Record Documents (For Welding)	3.1.1.1													
			Welding Procedures and Qualifications	3.1.1.2													
			SD-08 Manufacturer's Instructions														
			Lesson plan for the Instruction Course	3.5	G												
			SD-10 Operation and Maintenance Data														
			Water Treatment Systems	2.9	G												
			Calibrated Balancing Valves	2.4.5	G												

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		23 64 26	Expansion Joints	2.5.9	G												
			Pumps	2.6	G												
			Combination Strainer and Pump Suction Diffuser	2.5.3	G												
			Expansion Tanks	2.7	G												
			Air Separator Tanks	2.8	G												
		23 74 33	SD-02 Shop Drawings														
			Coils	2.4	G												
			Controls	2.7													
			Unit Cabinet	2.2.1	G												
			Casing	2.2.2	G												
			Installation Drawings	3.1	G												
			SD-03 Product Data														
			Equipment and Performance Data	Part 1	G												
			Air-Conditioning Systems	3.2.1	G												
			Coils	2.4	G												
			Fans	2.3	G												
			Controls	2.7	G												
			Unit Cabinet	2.2.1	G												
			Casing	2.2.2	G												
			Air Filters	2.5	G												
			Energy Recovery Devices	2.6	G												
			SD-06 Test Reports														
			System Performance Tests	3.2.2													
			SD-07 Certificates														

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		23 74 33	List of Product Installations	1.4													
			Manufacturer's Warranty	1.9													
			SD-08 Manufacturer's Instructions														
			Manufacturer's Installation	3.1													
			Instructions														
			Operation and Maintenance	3.3.2													
			Training														
			SD-10 Operation and Maintenance														
			Data														
			Operation and Maintenance	3.3.1													
			Manuals, Data Package 2														
		23 81 00	SD-03 Product Data														
			Spare Parts	3.7.1													
			Posted Instructions	3.4													
			System Performance Tests	3.6													
			Training	3.4	G												
			Inventory	1.5													
			Manufacturer's Standard Catalog	2.1													
			Data														
			SD-06 Test Reports														
			Refrigerant Tests, Charging, and	3.5													
			Start-Up														
			System Performance Tests	3.6													
			SD-07 Certificates														
			Service Organizations	3.7.2													

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		23 81 00	SD-10 Operation and Maintenance Data														
			Operation and Maintenance Manuals	3.4	G												
			SD-11 Closeout Submittals														
			Ozone Depleting Substances	2.1.2.3	S												
		23 82 00.00 20	SD-03 Product Data														
			Unit Heaters	2.1	G												
			SD-10 Operation and Maintenance Data														
			Unit Heaters	2.1													
		25 05 11.01	SD-01 Preconstruction Submittals														
			Wireless and Wired Broadcast Communication Request	3.2.1.2	G												
			Device Account Lock Exception Request	3.3.2	G												
			Multiple Ethernet Connection Device Request	3.2.3.2	G												
			Contractor Computer Cybersecurity Compliance Statements	1.9.1.6	G												
			Contractor Temporary Network Cybersecurity Compliance Statements	1.9.6	G												
			Cybersecurity Interconnection Schedule	1.7.2	G												

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		25 05 11.01	Proposed STIG and SRG Applicability Report	1.7.1	G												
			SD-02 Shop Drawings														
			Network Communication Report	1.7.3	G												
			Cybersecurity Riser Diagram	1.7.6	G												
			SD-03 Product Data														
			Control System Cybersecurity Documentation	1.7.8	G												
			SD-06 Test Reports														
			Control System Cybersecurity Testing Procedures	3.12.1	G												
			Control System Cybersecurity Testing Report	3.12.3	G												
			SD-07 Certificates														
			Software Licenses	1.8	G												
			SD-11 Closeout Submittals														
			Confidential Password Report	3.4.4.2	G												
			Enclosure Keys	3.3.5	G												
			Software and Configuration Backups	1.7.5	G												
			System Maintenance Tool Software	3.9	G												
			Control System Scanning Tools	3.10.1	G												
			STIG, SRG and Vendor Guide	1.7.7	G												
			Compliance Result Report														
			Control System Inventory Report	1.7.4	G												

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		25 05 11.02	SD-01 Preconstruction Submittals														
			Wireless and Wired Broadcast	3.2.1.3	G												
			Communication Request														
			Device Account Lock Exception	3.3.2	G												
			Request														
			Multiple Ethernet Connection	3.2.3.2	G												
			Device Request														
			Contractor Computer	1.9.1.6	G												
			Cybersecurity Compliance														
			Statements														
			Contractor Temporary Network	1.9.6	G												
			Cybersecurity Compliance														
			Statements														
			Cybersecurity Interconnection	1.7.1	G												
			Schedule														
			SD-02 Shop Drawings														
			Network Communication Report	1.7.2	G												
			Cybersecurity Riser Diagram	1.7.5	G												
			SD-03 Product Data														
			Control System Cybersecurity	1.7.6	G												
			Documentation														
			SD-06 Test Reports														
			Control System Cybersecurity	3.11.1	G												
			Testing Procedures														
			Control System Cybersecurity	3.11.3	G												
			Testing Report														

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(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		25 05 11.02	SD-07 Certificates														
			Software Licenses	1.8	G												
			SD-11 Closeout Submittals														
			Confidential Password Report	3.4.3.2	G												
			Enclosure Keys	3.3.5													
			Software and Configuration	1.7.4	G												
			Backups														
			System Maintenance Tool	3.8	G												
			Software														
			Control System Scanning Tools	3.9.1	G												
			Control System Inventory Report	1.7.3	G												
		25 05 11.03	SD-01 Preconstruction Submittals														
			Wireless and Wired Broadcast	3.2.1.3	G												
			Communication Request														
			Device Account Lock Exception	3.3.2	G												
			Request														
			Multiple Ethernet Connection	3.2.3.2	G												
			Device Request														
			Contractor Computer	1.9.1.6	G												
			Cybersecurity Compliance														
			Statements														
			Contractor Temporary Network	1.9.6	G												
			Cybersecurity Compliance														
			Statements														
			Cybersecurity Interconnection	1.7.1	G												
			Schedule														

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		25 05 11.03	SD-02 Shop Drawings														
			Network Communication Report	1.7.2	G												
			Cybersecurity Riser Diagram	1.7.5	G												
			SD-03 Product Data														
			Control System Cybersecurity	1.7.6	G												
			Documentation														
			SD-06 Test Reports														
			Control System Cybersecurity	3.11.1	G												
			Testing Procedures														
			Control System Cybersecurity	3.11.3	G												
			Testing Report														
			SD-07 Certificates														
			Software Licenses	1.8	G												
			SD-11 Closeout Submittals														
			Confidential Password Report	3.4.3.2	G												
			Enclosure Keys	3.3.5	G												
			Software and Configuration	1.7.4	G												
			Backups														
			System Maintenance Tool	3.8	G												
			Software														
			Control System Scanning Tools	3.9.1	G												
			Control System Inventory Report	1.7.3	G												
		26 20 00	SD-02 Shop Drawings														
			Panelboards	2.8	G												
			Wireways	2.17	G												
			Marking Strips	3.1.10.1	G												

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		26 20 00	SD-03 Product Data														
			Circuit Breakers	2.8.3	G												
			Switches	2.7	G												
			Enclosed Circuit Breakers	2.9	G												
			SD-06 Test Reports														
			600-volt Wiring Test	3.5.2													
			Grounding System Test	3.5.5													
			Ground-fault Receptacle Test	3.5.3													
			Arc-fault Receptacle Test	3.5.4													
			SD-10 Operation and Maintenance														
			Data														
			Electrical Systems	1.6.1	G												
		26 51 00	SD-02 Shop Drawings														
			Luminaire Drawings	1.5.1	G												
			Occupancy/Vacancy Sensor	1.5.9	G												
			Coverage Layout														
			Lighting Control System	1.7.2	G												
			One-Line Diagram														
			Sequence of Operation for	2.5.1	G												
			Lighting Control System														
			SD-03 Product Data														
			Luminaires	2.2	G												
			Light Sources	2.3	G												
			LED Drivers	2.4													
			Luminaire Warranty	1.6.1													
			Lighting Controls Warranty	1.6.2													

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		26 51 00	Local Area Controller	2.5.1.1.1													
			Switches	2.5.2.1													
			Occupancy/Vacancy Sensors	2.5.2.2													
			Exit Signs	2.6.1													
			Emergency Drivers	2.6.2	G												
			SD-05 Design Data														
			Luminaire Design Data	1.5.2	G												
			Photometric Plan	1.5.8	G												
			SD-06 Test Reports														
			ANSI/IES LM-79 Test Report	1.5.3													
			ANSI/IES LM-80 Test Report	1.5.4													
			ANSI/IES TM-21 Test Report	1.5.5													
			ANSI/IES TM-30 Test Report	1.5.6													
			Occupancy/Vacancy Sensor	3.2.1.1													
			Verification Test														
			SD-07 Certificates														
			LED Driver and Dimming Switch	1.5.7													
			Compatibility Certificate														
			SD-10 Operation and Maintenance														
			Data														
			Lighting System	1.7.1	G												
			Lighting Control System	1.7.2	G												
			Maintenance Staff Training Plan	3.3.2.1	G												
			End-User Training Plan	3.3.2.2	G												
		28 31 70	SD-01 Preconstruction Submittals														

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CONTRACT NO.

TITLE AND LOCATION

USACE WPAFB, HVAC & Controls Replacement F/20824

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION OR REVIEW	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION		DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		28 31 70	Qualified Fire Protection Engineer (QFPE)	1.3.2	G												
			Fire alarm system designer	1.9.2.1	G												
			Supervisor	1.9.2.2	G												
			Technician	1.9.2.3	G												
			Installer	1.9.2.4	G												
			Test Technician	1.9.2.5	G												
			Fire Alarm System Site-Specific Software Acknowledgement	1.7	G												
			SD-02 Shop Drawings														
			Nameplates	1.9.1.3	G AE												
			Wiring Diagrams	1.9.1.4	G AE												
			System Layout	1.9.1.5	G AE												
			Notification Appliance Data	1.9.1.6	G AE												
			Initiating devices	1.9.1.7	G AE												
			Amplifier Data	1.9.1.8	G AE												
			Battery Power	1.9.1.9	G AE												
			Voltage Drop Calculations	1.9.1.10	G AE												
			SD-03 Product Data														
			Amplifiers	2.3	G												
			Tone Generators	2.3	G												
			Manual Stations	2.4	G												
			Smoke Detectors	2.5	G												
			Addressable Interface Devices	2.6	G												
			Addressable Control Modules	2.7	G												
			Notification Appliances	2.8	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

USACE WPAFB, HVAC & Controls Replacement F/20824

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION REVIEW	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION		DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		28 31 70	Batteries	2.10.1	G												
			Battery Chargers	2.10.2	G												
			Supplemental Notification	2.10.1.1	G												
			Appliance Circuit Panels														
			Auxiliary Power Supply Panels	2.10.1.1	G												
			Surge Protective Devices	2.11	G												
			Alarm Wiring	2.12.1	G												
			Back Boxes and Conduit	3.3.4	G												
			Ceiling Bridges	3.2.5	G												
			Terminal Cabinets	3.3.2	G												
			Environmental Enclosures or	2.13	G												
			Guards														
			Document Storage Cabinet	3.12.3	G												
			SD-06 Test Reports														
			Test Procedures	3.8.1	G												
			SD-07 Certificates														
			Verification of Compliant	3.8.2.1	G												
			Installation														
			Request for Government Final	3.8.2.2	G												
			Test														
			SD-10 Operation and Maintenance														
			Data														
			Operation and Maintenance	1.9.1.12	G												
			(O&M) Instructions														
			Instruction of Government	3.11	G												
			Employees														

The use of signs to identify Corps managed or supervised design, construction, and rehabilitation projects - both for military and civil works - is an important part of efforts to keep the public informed of Corps work. For this purpose, a construction project sign package has been adopted. This package consists of two signs: one for project identification and the other to show on-the-job safety performance of the contractor.

These two signs are to be displayed side by side and mounted for reading by passing viewers. Exact placement location will be designated by the contracting officer's representative.

The panel sizes and graphic formats have been standardized for visual consistency throughout all Corps operations.

Panels are fabricated using HDO plywood or aluminum with dimensional lumber uprights and bracing. The sign faces are nonreflective vinyl.

All legends are to be die-cut or computer-cut in the sizes and typefaces specified and applied to the white panel background following the graphic formats shown on pages 16-2 thru 16-4. The Communication Red panel on the left side of the construction project sign with Corps Signature (reverse version) is screen-printed onto the white background.

Displays of the signs are shown on the following pages. Mounting and fabrication details are provided on page 16-5.

Special applications or situations not covered in these guidelines should be referred to the district Sign Program Manager.

Following are two samples of the Construction Project Identification sign showing how this panel is adaptable for use to identify either Civil or Military works projects. The graphic format for this 4'x 6' sign panel follows the legend guidelines and layout as specified below. The large 4'x 4' section of the panel on the right is to be white with black legend. The 2'x 4' section of the sign on the left

with the full Corps Signature (reverse version) is to be screen-printed Communication Red on the white background. The designation of a sponsor in the area indicated is optional with Military or Civil Works construction signs. Signs may list one sponsoring entity. If agreement on a sponsor designation cannot be achieved, the area should be left blank.

This sign is to be placed with the Safety Performance sign shown on the following page. Mounting and fabrication details are provided on page 16-4.

Special applications or situations not covered in these guidelines should be referred to the district Sign Program Manager.

ATTACHMENT A: Example Graphic of Signage with Dimensions for MILCON project



ATTACHMENT B: Example Graphic of Signage with Dimensions for Civil Works Project



Each contractor's safety record is to be posted on Corps managed or supervised construction projects and mounted with the Construction Project Identification sign specified on page 16-2.

The graphic format, color, size and typeface used on the sign are to be reproduced exactly as specified below. The

title with First Aid logo in the top section of the sign, and the performance record captions are standard for all signs of this type. Legend groups 2 and 3 below identify the project and the contractor and are to be placed on the sign as shown.

Safety record numbers are mounted on individual metal plates and are screw-

mounted to the background to allow for daily revisions to posted safety performance record.

Special applications or situations not covered in these guidelines should be referred to the district Sign Program Manager.

Legend Group 1: Standard two-line title "Safety is a Job Requirement" with 8" (outside diameter) Safety Green first aid logo.
Color: To match Pantone system 347
Typeface: 3" Helvetica Bold
Color: Black

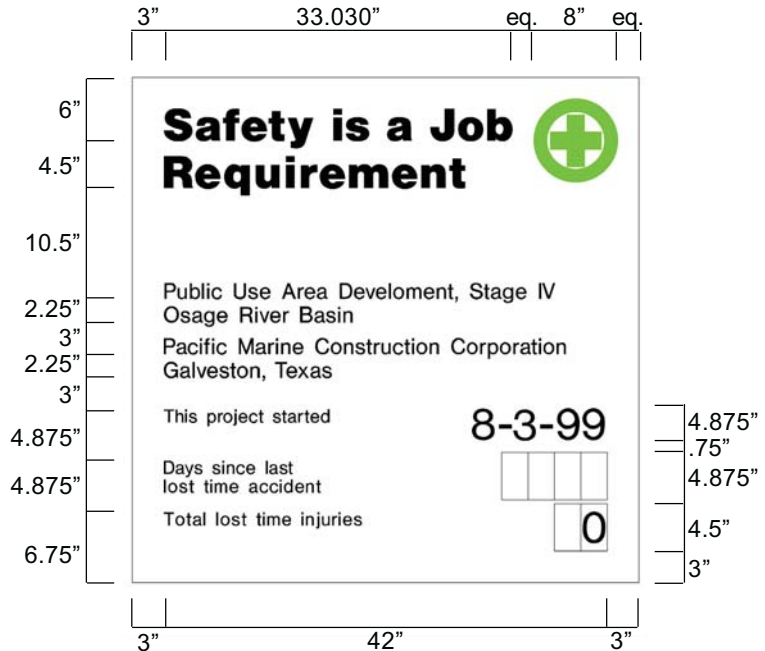
Legend Group 2: One- to two-line project title legend describes the work being done under this contract and name of host project.
Color: Black
Typeface: 1.5" Helvetica Regular
Maximum line length: 42"

Legend Group 3: One- to two-line identification: name of prime contractor and city, state address. Color: Black
Typeface: 1.5" Helvetica Regular
Maximum line length: 42"

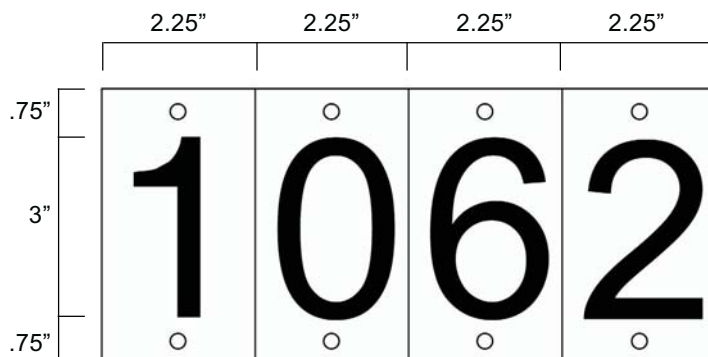
Legend Group 4: Standard safety record captions as shown.
Color: Black
Typeface: 1.25" Helvetica Regular

Replaceable numbers are to be mounted on white .060 aluminum plates and screw-mounted to background.
Color: Black
Typeface: 3" Helvetica Regular
Plate size: 2.5" x 4.5"

All typography is flush left and rag right, upper and lower case with initial capitals only as shown. Letter- and word-spacing to follow Corps standards as specified in Appendix D.



Sign Type	Legend Size (A)	Panel Size	Post Size	Specification Code	Mounting Height	Color Bkg/Lgd
CID-02	various	4'x4'	4"x4"	HDO-3	48"	WH/BK-SG



All Construction Project Identification signs and Safety Performance signs are to be fabricated and installed as described below. The signs are to be erected at a location designated by the contracting officer representative and shall conform to the size, format, and typographic standards shown on pages 16-2 thru 16-4. Detailed specifications

for HDO plywood panel preparation are provided in Appendix B.

For additional information on the proper method to prepare sign panel graphics, contact the district Sign Program Manager.

Shown below the mounting diagram is a panel layout grid with spaces provided for project information. Photocopy this page and use as a worksheet when preparing sign legend orders.

The sign panels are to be fabricated from .75" High Density Overlay Plywood. Panel preparation to follow HDO specifications provided in Appendix B.

Sign graphics to be prepared on a white nonreflective vinyl film with positionable adhesive backing.

All graphics except for the Communication Red background with Corps Signature on the project sign are to be die-cut or computer-cut nonreflective vinyl, prespaced legends prepared in the sizes and typefaces specified and applied to the background panel following the graphic formats shown on pages 16-2 and 16-3.

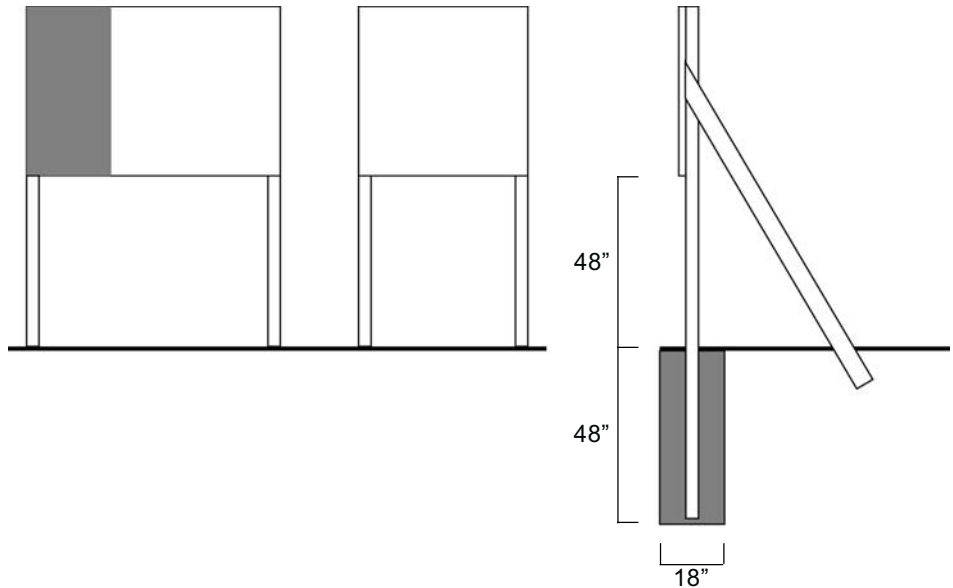
The 2'x 4' Communication Red panel (to match Pantone system 032) with full Corps Signature (reverse version) is to be screen-printed on the white background. Identification of the district or division may be applied under the signature with white cut vinyl letters prepared to Corps standards.

Drill and insert six (6) .375" T-nuts from the front face of the HDO sign panel. Position holes as shown. Flange of T-nut to be flush with sign face.

Apply graphic panel to prepared HDO plywood panel following manufacturers' instructions.

Sign uprights to be structural grade 4" x 4" treated Douglas Fir or Southern Yellow Pine, No.1 or better. Post to be 12' long. Drill six (6) .375" mounting holes in uprights to align with T-nuts in sign panel. Countersink (.5") back of hole to accept socket head cap screw (4" x .375").

Assemble sign panel and uprights. Imbed assembled sign panel and uprights in 4' hole. Local soil conditions and/or wind loading may require bolting additional 2" x 4" struts on inside face of uprights to reinforce installation as shown.



Construction Project Identification Sign Legend Group 1: Corps Relationship

1. _____
2. _____

Legend Group 2: Division/District Name

1. _____
2. _____

Legend Group 2a: Military/Civil Works Sponsor

1. _____
2. _____

Legend Group 3: Project Title

1. _____
2. _____
3. _____

Legend Group 4: Facility Name

1. _____
2. _____

Legend Group 5: Contractor/A&E

1. _____
2. _____
3. _____
4. _____
5. _____

Legend Group 5b: Contractor/A&E

1. _____
2. _____
3. _____
4. _____
5. _____

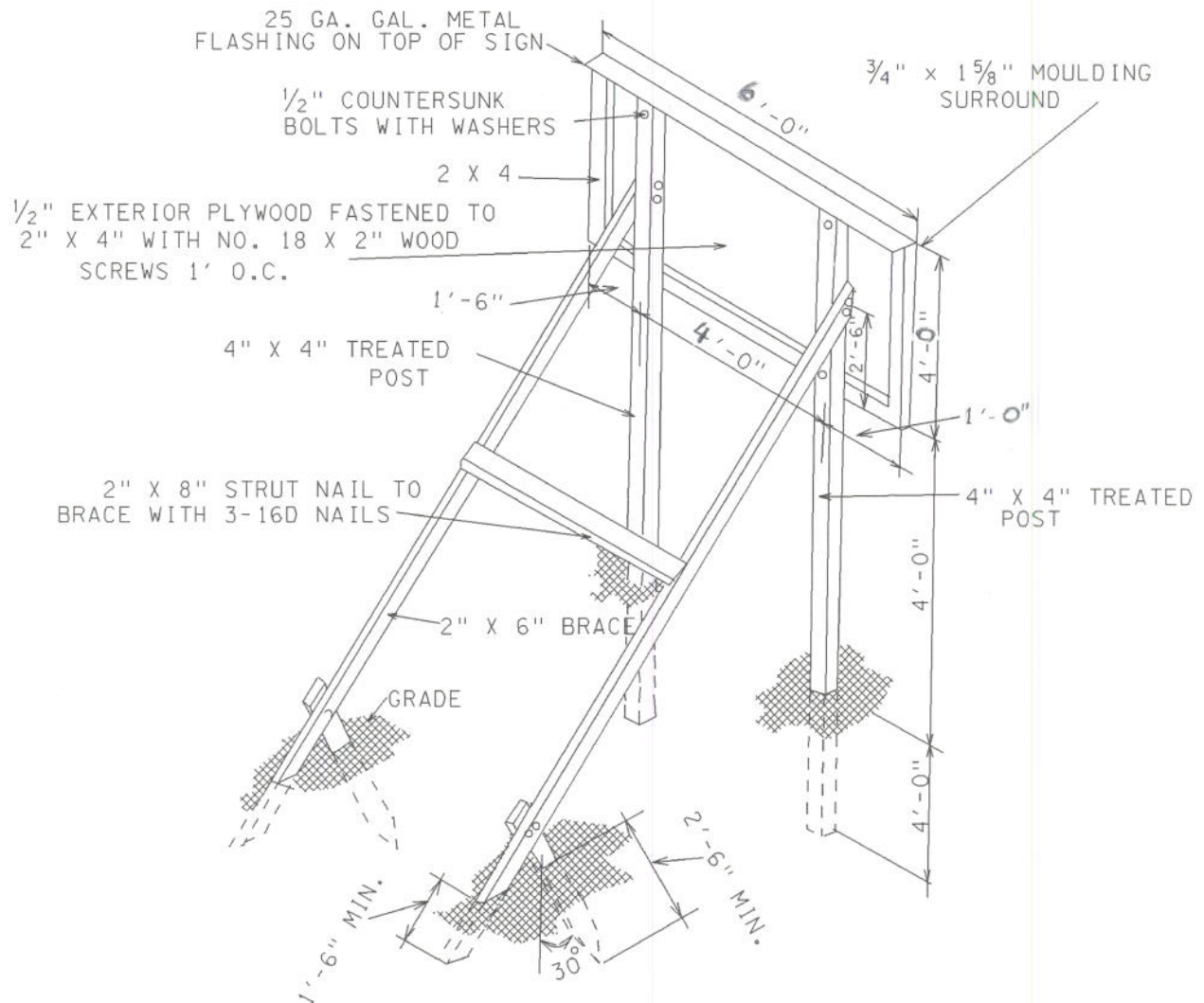
Safety Performance Sign

Legend Group 2: Project Title

1. _____
2. _____

Legend Group 3: Contractor/A&E

1. _____
2. _____



CONSTRUCTION SIGN ISOMETRIC
ERECTION DETAILS

HVAC AND CONTROLS REPLACEMENT F/20824
Wright-Patterson AFB, OH P2#513948

SECTION 01 11 00

SUMMARY OF WORK
08/15, CHG 2: 08/21

PART 1 GENERAL

1.1 SUBMITTALS

SD-01 Preconstruction Submittals

Work Plan; G
Shall be submitted for equipment disconnection, reconnection, and protection.

1.2 WORK COVERED BY CONTRACT DOCUMENTS

1.2.1 Project Description

The work includes replacement of HVAC distribution systems, and as required, the removal of existing ceilings, lighting, and incidental related work.

1.2.2 Location

The work is located at in Building 20824 on Ascani Street on Wright Patterson Air Force Base, Area B, approximately as indicated. The exact location is indicated on the Drawings.

1.2.3 Sequencing

The government will have occupancy and full functional use of either the Horizontal Test Track area or the Vertical Test Track area at all times. The work shall be sequenced so that only one area is down at a time.

See Attachment 1 at the end of this section that provides necessary information about the work in Sequence 3.

1.3 OCCUPANCY OF PREMISES

Building(s) will be occupied during performance of work under this Contract.

Before work is started, arrange with the Contracting Officer a sequence of procedure, means of access, space for storage of materials and equipment, and use of approaches, corridors, and stairways.

1.4 EXISTING WORK

In addition to FAR 52.236-9 Protection of Existing Vegetation, Structures, Equipment, Utilities, and Improvements:

- a. Remove or alter existing work in such a manner as to prevent injury or damage to any portions of the existing work which remain.
- b. Repair or replace portions of existing work which have been altered during construction operations to match existing or adjoining work, as approved by the Contracting Officer. At the completion of operations,

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existing work must be in a condition equal to or better than that which
existed before new work started.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

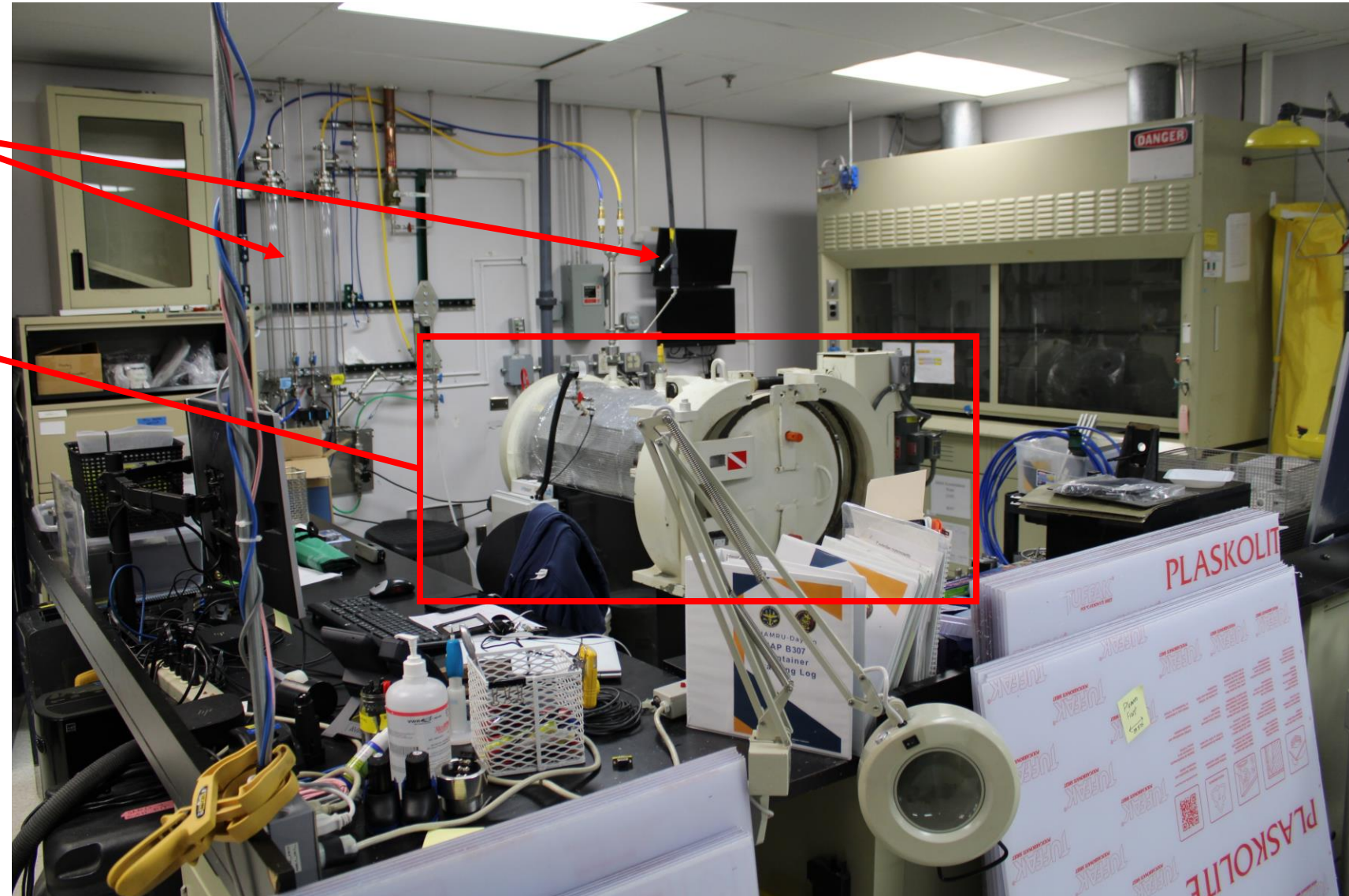
Room 111

Equipment on wall will need protected. -KTR

Hyperbaric chamber cannot be moved, needs protected. -KTR

Countertops need covered. -KTR

Most items on the countertops can be moved if necessary. -Gov't

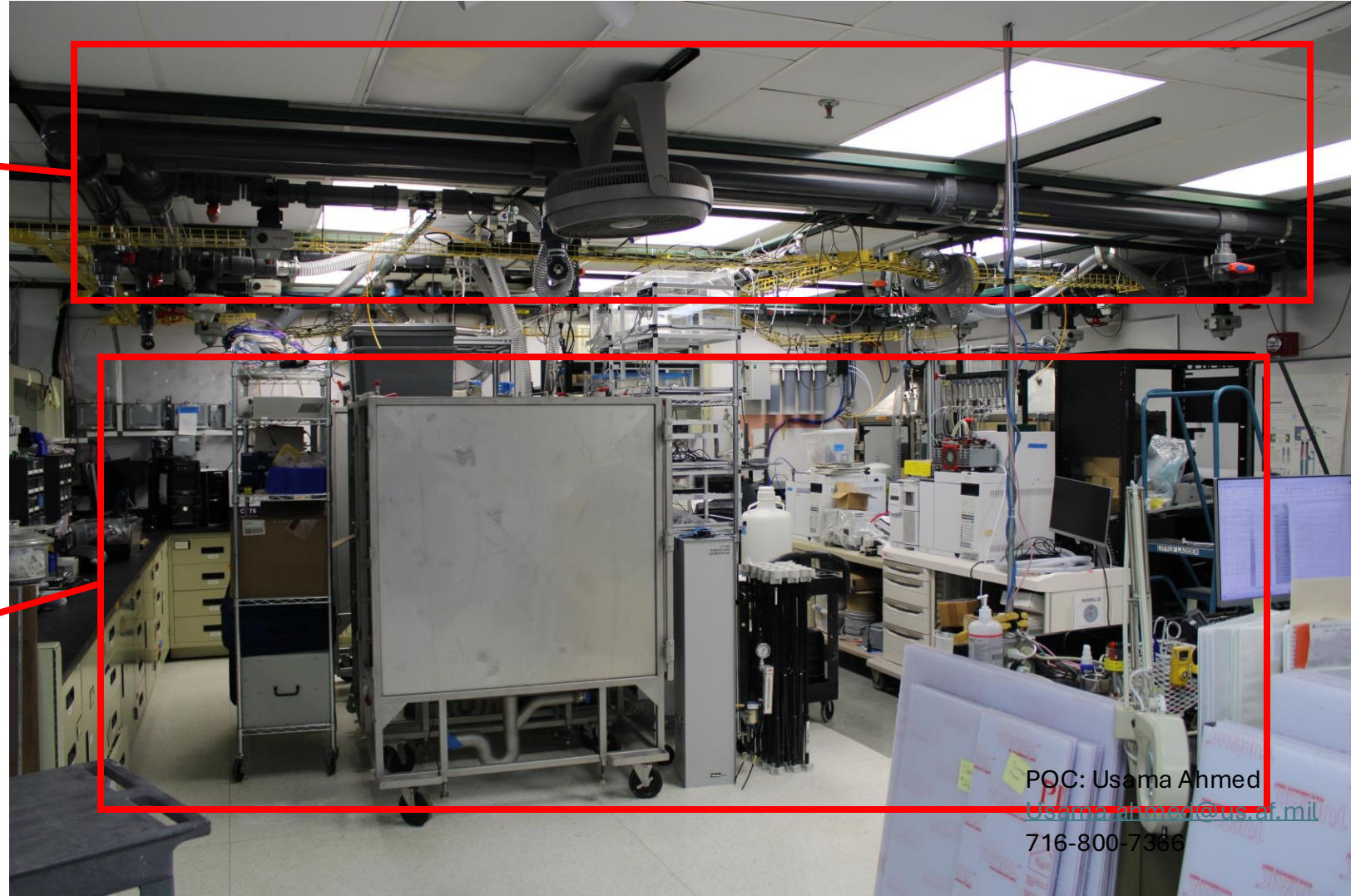


Room 111 continued

Pipe and wire tray can be removed from the Unistrut grid. The Unistrut grid is attached to the truss structure of the building and cannot be disassembled by us.

All items in this space are mobile and can be moved to a different area while HVAC work is ongoing. These items will likely need to be shuffled around as work proceeds in this room as we lack space to remove them completely. -Gov't

Countertop areas around the open area will need covered. -KTR



POC: Usama Ahmed
usama.ahmed@us.af.mil
716-800-7366

Room 111 continued

Countertops need covered.-KTR

Most items on the countertops
can be moved if necessary.
Carts and chairs will be
removed. -Gov't



POC: Usama Ar
Usama.armed@us.af.mil
716-800-7366

Room 111 continued

Left and right countertops need covered. -KTR

Most items on the countertops can be moved if necessary. -Gov't

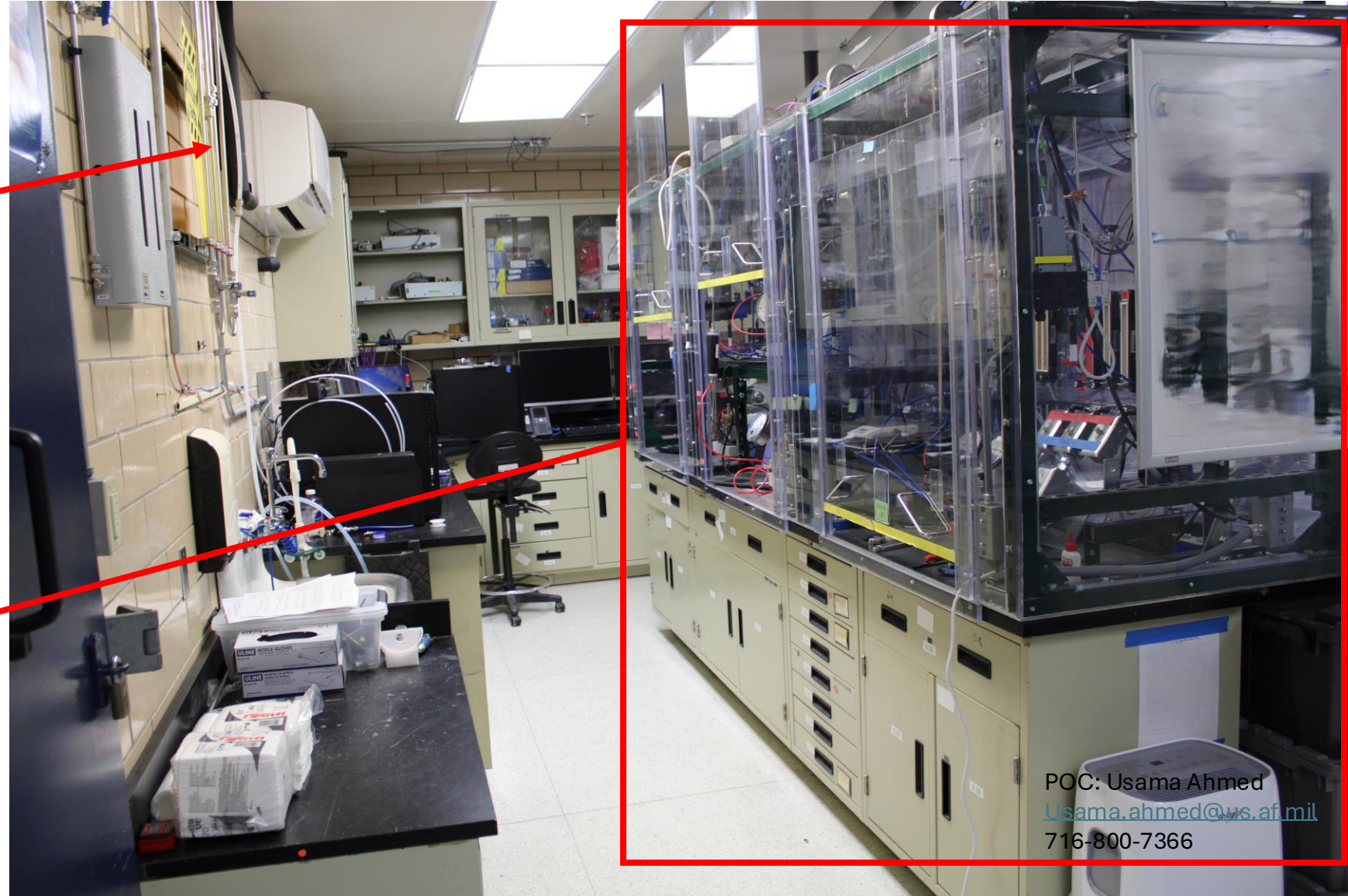


POC: Usama Ahmed
Usama.ahmed@us.af.mil
716-800-7366

Room 114 continued

Minisplit will need protected. -KTR

This 3-pair set of hoods are not able to be moved or deconstructed. They will need protected in place. -KTR



Room 114 continued

Ball valves will be disconnected. -Gov't

This is the other side view of 3-pair set of hoods.

Hood with cylinder cabinet next to it.



POC: Usama Ahmed
Usama.ahmed@us.af.mil
716-800-7366

Room 114 continued

This hood and the cylinder cabinet cannot be moved, will need protected. -KTR

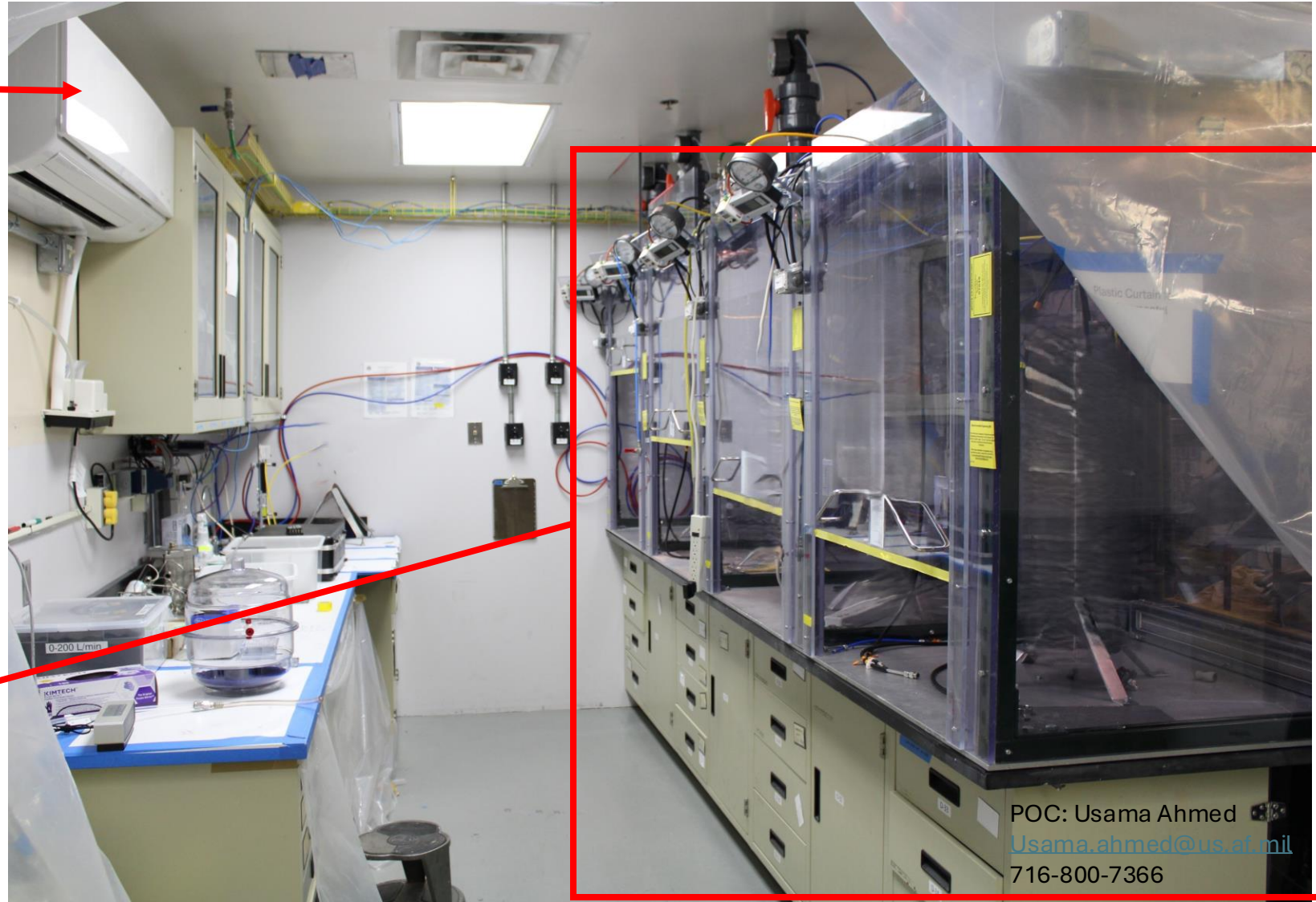
3-pair set of hoods.



Room 114 continued

Minisplit will need protected. -KTR

This 4-pair set of hoods are not able to be moved or deconstructed. They will need protected in place. -KTR



Room 121

**Minisplit and
screen will need
protected -KTR**

This room has approximately 5ft of space between the ceiling and the Unistrut grid. The Unistrut structure itself cannot be disassembled. The items being stored on top of the grid can be moved to allow workers access. Things attached to the underside of the grid can be removed if necessary. -Gov't



POC: Usama Ahmed
Usama.ahmed@us.af.mil
716-800-7366

Room 121 continued

These chambers and tanks below the Unistrut grid are not moveable, will need protected. -KTR



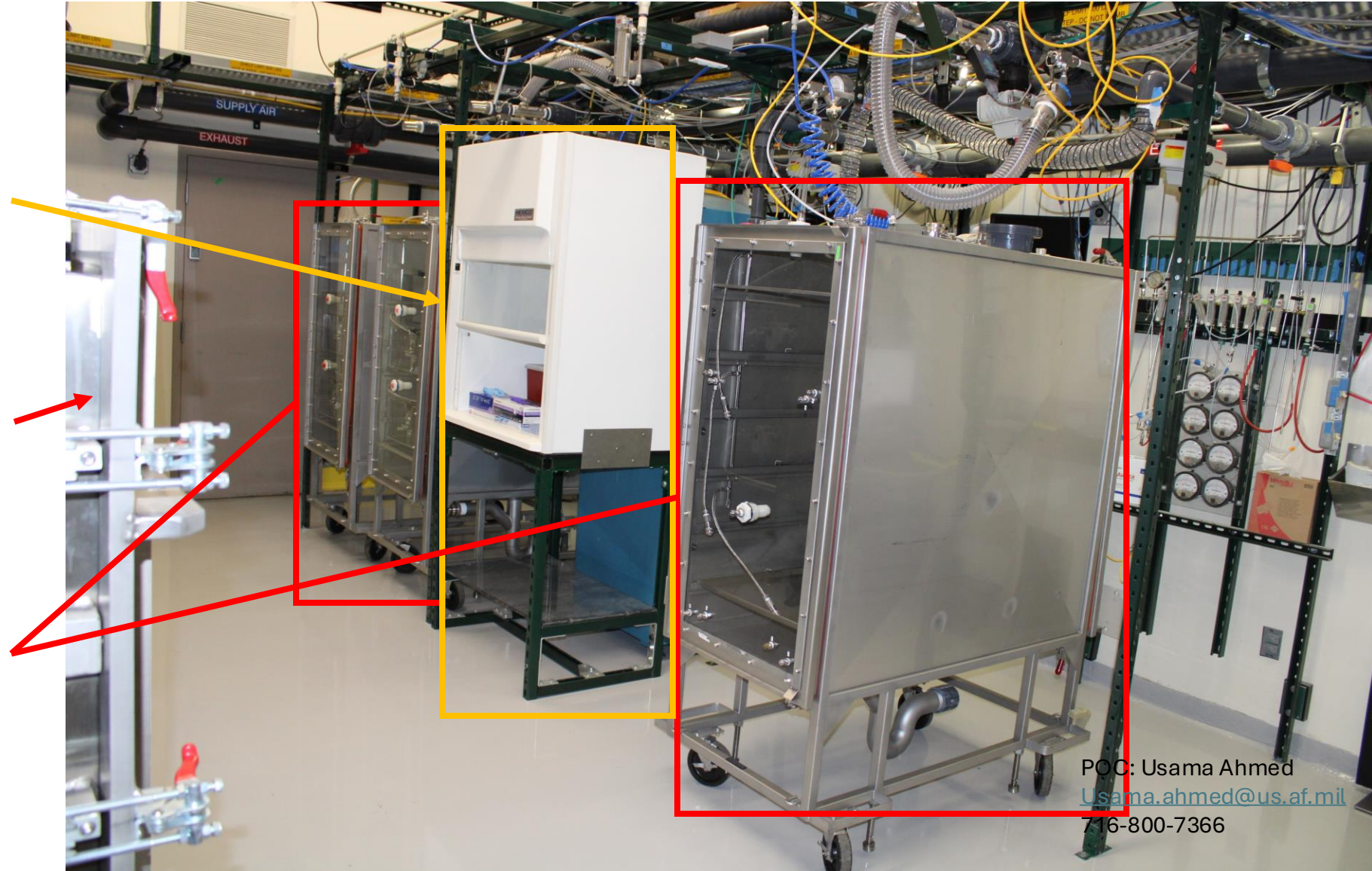
POC: Usama Ahmed
Usama.ahmed@us.af.mil
716-800-7366

Room 121 continued

This hood and the cylinder cabinet behind it cannot be moved, will need protected.-KTR

Stainless chamber from previous slide.

These chambers can be moved. -Gov't



POC: Usama Ahmed
Usama.ahmed@us.af.mil
716-800-7366

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SECTION 01 32 01.00 06

PROJECT SCHEDULE
10/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AACE INTERNATIONAL (AACE)

AACE 29R-03	(2011) Forensic Schedule Analysis
AACE 52R-06	(2006) Time Impact Analysis - As Applied in Construction
AACE 84R-13	(2015) Planning and Accounting for Adverse Weather

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 67-17	(2017) Schedule Delay Analysis
------------	--------------------------------

U.S. ARMY CORPS OF ENGINEERS (USACE)

ER 1-1-11	(2017) Administration -- Project Schedules
-----------	--

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Project Scheduler Qualifications; G

Preliminary Project Schedule; G

Initial Project Schedule; G

Periodic Schedule Update; G

1.3 PROJECT SCHEDULER QUALIFICATIONS

Designate an authorized representative to be responsible for the preparation of the schedule and all required updating and production of reports. The Designated Project Scheduler must have prepared and maintained at least three previous construction schedules for projects of similar size and complexity to this contract, using Primavera P6.. Representative must have a comprehensive knowledge of CPM scheduling principles and application this person shall not be the Quality Control

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Manger for the contract.

PART 2 PRODUCTS

2.1 SOFTWARE

The scheduling software utilized to produce and update the schedules required herein must be capable of meeting all requirements of this specification.

2.1.1 Government Default Software

The Government uses Primavera P6. Ensure exported schedule files are compatible with the CURRENT version of P6 used by the Government.

2.1.2 Contractor Software

Scheduling software used by the contractor must be commercially available from the software vendor for purchase with vendor software support agreements available. The software routine used to create the required sdef file must be created and supported by the software manufacturer.

2.1.2.1 Primavera

If Primavera P6 is selected for use, provide the "xer" export file in a version of P6 importable by the Government system. Verify at the SEKO meeting which version of P6 is in use by the Government. Export the schedule in a version of P6 no newer than that used by the Government.

2.1.2.2 Other Than Primavera

If the Contractor chooses software other than Primavera P6, that is compliant with this specification, provide for the Government's use two licenses, two computers, and training for two Government employees in the use of the software. These computers will be stand-alone and not connected to Government network. Computers and licenses will be returned at project completion.

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

Prepare for approval a Project Schedule, as specified herein, pursuant to FAR Clause 52.236-15 Schedules for Construction Contracts. Show in the schedule the proposed sequence to perform the work and dates contemplated for starting and completing all schedule activities. The scheduling of the entire project is required. The scheduling of design and construction is the responsibility of the Contractor. Contractor management personnel must actively participate in its development. Designers, Subcontractors, and suppliers working on the project must also contribute to developing and maintaining an accurate Project Schedule. Provide a schedule that is a forward planning as well as a project monitoring tool. Use the Critical Path Method (CPM) of network calculation to generate all Project Schedules. Prepare each Project Schedule using the Precedence Diagram Method (PDM).

3.1.1 Scheduling Expectations Kickoff Meeting(SEKO)

In conjunction with the post award conference or pre construction

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conference the Government and the Contractor will hold a separate and distinct Scheduling Expectations Kickoff Meeting(SEKO). The intent of this meeting is for the Government to review with the contractor the requirements of the scheduling specification, communicate the Government's expectations regarding schedule development and management, and answer any questions the Contractor may have regarding the schedule requirements.

3.2 BASIS FOR PAYMENT AND COST LOADING

The schedule is the basis for determining contract earnings during each update period and therefore the amount of each progress payment. The aggregate value of all activities coded to a contract CLIN must equal the value of the CLIN. Match the progress payment "Pay Period Thru" date in RMS to the schedule data date.

3.2.1 Activity Cost Loading

- a. Activity cost loading must be reasonable and without front-end loading. Activities with a negative cost loading is not allowed. Provide additional documentation to demonstrate reasonableness if requested by the Contracting Officer.
- b. Coordinate with the Contracting Officer to request approval for, and assign costs, to the respective procurement activity for which payment will be requested for materials/equipment in advance of installation. Any such payment request for approved material/equipment advanced payment shall be supported by a paid invoice or as determined by the Contracting Officer.

3.2.2 Withholdings / Payment Rejection

Failure to meet the requirements of this specification may result in the disapproval of the preliminary, initial, or periodic schedule updates and subsequent rejection of payment requests until compliance is met.

In the event that the Contracting Officer directs schedule revisions and those revisions have not been included in subsequent Project Schedule revisions or updates, the Contracting Officer may withhold 10 percent of pay request amount from each payment period until such revisions to the project schedule have been made.

3.3 PROJECT SCHEDULE DETAILED REQUIREMENTS

3.3.1 Level of Detail Required

Develop the Project Schedule to the appropriate level of detail to address major milestones and to allow for satisfactory project planning and execution. Failure to develop the Project Schedule to an appropriate level of detail will result in its disapproval. The Contracting Officer will consider, but is not limited to, the following characteristics and requirements to determine appropriate level of detail:

3.3.2 Activity Durations

Reasonable activity durations are those that allow the progress of ongoing activities to be accurately determined between update periods. Non-procurement activities Original Durations (OD) are not to exceed 20 workdays or 30 calendar days.

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3.3.3 Design and Permit Activities

Include design and permit activities with the necessary conferences and follow-up actions and design package submission dates. Include the design schedule in the project schedule, showing the sequence of events involved in carrying out the project design tasks within the specific contract period. Provide a detailed level of scheduling sufficient to identify all major design tasks, including those that control the flow of work. Also include review and correction periods associated with each item.

3.3.4 Procurement Activities

Include in the schedule activities associated with the critical submittals and their approvals, procurement, fabrication, and delivery of long lead materials, equipment, fabricated assemblies, and supplies. Long lead procurement activities are those with an anticipated procurement sequence of over 90 calendar days.

3.3.5 Mandatory Activities

Include the following activities in the initial project schedule and all updates.

- a. Submission, review and acceptance of SD-01 Preconstruction Submittals (individual activity for each).
- b. Submission, review and acceptance of design packages.
- c. Submission of mechanical/electrical/information systems layout drawings.
- d. Submission and approval of O & M manuals.
- e. Submission and approval of as-built drawings.
- f. Submission and approval of DD1354 data and installed equipment lists.
- g. Submission and approval of testing and air balance (TAB).
- h. Submission of TAB specialist design review report.
- i. Submission and approval of fire protection specialist.
- j. Submission and approval of Building Commissioning Plan, Controls testing plan, test data, and reports: Develop the schedule logic associated with testing and commissioning of mechanical systems to a level of detail consistent with the contract commissioning requirements. All tasks associated with building testing and commissioning will be completed prior to submission of building commissioning report and subsequent contract completion.

These activities may include but are not limited to: air and water balancing, building commissioning - functional performance testing, controls testing, and performance verification testing. Include other systems testing, if required. Include additional commissioning requirements noted elsewhere in the Contract as required in the schedule and discuss during the SEKO meeting, noted in paragraph SCHEDULING EXPECTATIONS KICKOFF MEETING (SEKO).

- k. Contractor's pre-final inspection.

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- l. Correction of punch list from Contractor's pre-final inspection.
- m. Government's pre-final inspection.
- n. Correction of punch list from Government's pre-final inspection.
- o. Final inspection.

3.3.6 Government Activities

Show in schedule, Government and other agency activities that could impact progress. These activities include, but are not limited to approvals, acceptance, design reviews, environmental permit approvals by State regulators, inspections, utility tie-in, Government Furnished Equipment (GFE) and Notice to Proceed (NTP) for phasing requirements.

3.3.7 Standard Activity Coding Dictionary

Use the activity coding structure defined in the Standard Data Exchange Format (SDEF) in ER 1-1-11. This exact structure is mandatory. Develop and assign all Activity Codes to activities as detailed herein.

The SDEF format is as follows:

Field	Activity Code	Length	Description
1	WRKP	3	Workers per day
2	RESP	4	Responsible party
3	AREA	4	Area of work
4	MODF	6	Modification Number
5	BIDI	6	Bid Item (CLIN)
6	PHAS	2	Phase of work
7	CATW	1	Category of work
8	FOW	20	Feature of work*
*Some systems require that FEATURE OF WORK values be placed in several activity code fields. The notation shown is for Primavera P6. Refer to the specific software guidelines with respect to the FEATURE OF WORK field requirements.			

3.3.7.1 Workers Per Day (WRKP)

Assign Workers per Day for all field construction or direct work activities, if directed by the Contracting Officer. Workers per day is based on the average number of workers expected each day to perform a task for the duration of that activity.

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3.3.7.2 Responsible Party Coding (RESP)

Assign responsibility code for all activities to the Prime Contractor, Subcontractor(s) or Government agency(ies) responsible for performing the activity.

- a. Activities coded with a Government Responsibility code include, but are not limited to: Government approvals, Government design reviews, environmental permit approvals by State regulators, Government Furnished Property/Equipment (GFP) and Notice to Proceed (NTP) for phasing requirements.
- b. Activities cannot have more than one Responsibility Code. Examples of acceptable activity code values are: DOR (for the designer of record); ELEC (for the electrical subcontractor); MECH (for the mechanical subcontractor); and GOVT (for USACE).

3.3.7.3 Area of Work Coding (AREA)

Assign Work Area code to activities based upon the work area in which the activity occurs. Define work areas based on resource constraints or space constraints that would preclude a resource, such as a particular trade or craft work crew from working in more than one work area at a time due to restraints on resources or space. Examples of Work Area Coding include different areas within a floor of a building, different floors within a building, and different buildings within a complex of buildings. Activities cannot have more than one Work Area Code.

3.3.7.4 Modification Number (MODF)

Assign a Modification Number Code to any activity or sequence of activities added to the schedule as a result of a Contract Modification, when approved by Contracting Officer. Key all Code values to the Government's modification numbering system. An activity can have only one Modification Number Code.

3.3.7.5 Bid Item Coding (BIDI)

Assign a Bid Item Code to all activities using the Contract Line Item Number (CLIN) to which the activity belongs, even when an activity is not cost loaded. An activity can have only one BIDI Code.

3.3.7.6 Phase of Work Coding (PHAS)

Assign Phase of Work Code to all activities. Examples of phase of work are design phase, procurement phase and construction phase. Each activity can have only one Phase of Work code.

- a. Code proposed fast track design and construction phases proposed to allow filtering and organizing the schedule by fast track design and construction packages.
- b. If the contract specifies phasing with separately defined performance periods, identify a Phase Code to allow filtering and organizing the schedule accordingly.

3.3.7.7 Category of Work Coding (CATW)

Assign a Category of Work Code to all activities. Category of Work Codes

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include, but are not limited to design, design submittal, design reviews, review conferences, permits, construction submittal, procurement, fabrication, weather sensitive installation, non-weather sensitive installation, start-up, and testing activities. Each activity can have no more than one Category of Work Code.

3.3.7.8 Feature of Work Coding (FOW)

Assign a Feature of Work Code to appropriate activities based on the Definable Feature of Work to which the activity belongs based on the approved QC plan.

Definable Feature of Work is defined in Section 01 45 04.10 06 CONTRACTOR QUALITY CONTROL. An activity can have only one Feature of Work Code.

3.3.8 Contract Milestones and Constraints

Milestone activities are to be used for significant project events including, but not limited to, project phasing, project start and end activities, or interim completion dates. The use of artificial float constraints such as "zero free float" or "zero total float" are prohibited.

Mandatory constraints that ignore or effect network logic are prohibited. No constrained dates are allowed in the schedule other than those specified herein. Submit additional constraints to the Contracting Officer for approval on a case by case basis.

3.3.8.1 Project Start Date Milestone and Constraint

The first activity in the project schedule must be a start milestone titled "NTP Acknowledged," which must have a "Start On" constraint date equal to the date that the NTP is acknowledged.

3.3.8.2 End Project Finish Milestone and Constraint

The last activity in the schedule must be a finish milestone titled "End Project."

Constrain the project schedule to the Contract Completion Date in such a way that if the schedule calculates an early finish, then the float calculation for "End Project" milestone reflects positive float on the longest path. If the project schedule calculates a late finish, then the "End Project" milestone float calculation reflects negative float on the longest path.

3.3.8.3 Interim Completion Dates and Constraints

Constrain contractually specified interim completion dates to show negative float when the calculated late finish date of the last activity in that phase is later than the specified interim completion date.

3.3.8.3.1 Start Phase

Use a start milestone as the first activity for a project phase. Call the start milestone "Start Phase X" where "X" refers to the phase of work.

3.3.8.3.2 End Phase

Use a finish milestone as the last activity for a project phase. Call the

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finish milestone "End Phase X" where "X" refers to the phase of work.

3.3.9 Adverse Weather

Ensure anticipated adverse weather is accounted for in the Project Schedule. AACE 84R-13 provides methodologies for techniques in planning for adverse weather. The REQUIRED methodology is the use of a weather calendar.

3.3.10 Calendars

Schedule activities on a Calendar to which the activity logically belongs. Develop calendars to accommodate any contract defined work period such as a 7-day calendar for Government Acceptance activities, concrete cure times, etc. Develop the default Calendar to match the physical work plan with non-work periods identified including weekends and holidays. Develop Seasonal Calendar(s) and assign to seasonally affected activities as applicable.

3.3.11 Open Ended Logic

Only two open ended activities are allowed: the first activity "NTP Acknowledged" is to have no predecessor logic, and the last activity -"End Project" is to have no successor logic. Except as noted above all activities are to have at least a start-to-start or finish-to-start relationship with its predecessor(s) and a finish-to-finish or finish-to-start relationship with its successor(s).

Predecessor open ended logic may be allowed in a time impact analyses upon the Contracting Officer's approval.

3.3.12 Default Progress Data Disallowed

Actual Start and Finish dates must not automatically update with default mechanisms included in the scheduling software. Updating of the percent complete and the remaining duration of any activity must be independent functions. Disable program features that calculate one of these parameters from the other. Activity Actual Start (AS) and Actual Finish (AF) dates assigned during the updating process must match those dates provided in the Contractor Quality Control Reports. Failure to document the AS and AF dates in the Daily Quality Control report will result in disapproval of the Contractor's schedule.

3.3.13 Out-of-Sequence Progress

Activities that have progressed before all preceding logic has been satisfied (Out-of-Sequence Progress) will be allowed only on a case-by-case basis subject to approval by the Contracting Officer. Propose logic corrections to eliminate out of sequence progress or justify not changing the sequencing for approval prior to submitting an updated project schedule. Address out of sequence progress or logic changes in the Narrative Report and in the periodic schedule update meetings.

3.3.14 Added and Deleted Activities

Do not delete activities from the project schedule or add new activities to the schedule without approval from the Contracting Officer. Activity ID and description changes are considered new activities and cannot be changed without Contracting Officer approval.

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3.3.15 Original Durations

Activity Original Durations (OD) must be reasonable to perform the work item. OD changes are prohibited unless justification is provided and approved by the Contracting Officer.

3.3.16 Leads, Lags, and Start to Finish Relationships

Lags must be reasonable as determined by the Government and not used in place of realistic original durations, must not be in place to artificially absorb or create float, or to replace proper schedule logic.

- a. Leads (negative lags) are prohibited.
- b. Start to Finish (SF) relationships are prohibited.

3.3.17 Retained Logic

Schedule calculations must retain the logic between predecessors and successors ("retained logic" mode) even when the successor activity(s) starts and the predecessor activity(s) has not finished (out-of-sequence progress). Software features that in effect sever the tie between predecessor and successor activities when the successor has started and the predecessor logic is not satisfied ("progress override") are not be allowed.

3.3.18 Percent Complete

Update the percent complete for each activity started, based on the realistic assessment of earned value. Activities which are complete but for remaining minor punch list work and which do not restrain the initiation of successor activities may be declared 100 percent complete to allow for proper schedule management. Budgeted cost of activity is to be reduced by the amount needed to correct deficiency. The activity referenced in paragraph COST LOADING of CLOSEOUT ACTIVITIES must have its budgeted cost increased by this amount.

3.3.19 Remaining Duration

Update the remaining duration for each activity based on the number of estimated workdays it will take to complete the activity. Remaining duration may not mathematically correlate with percentage found under paragraph entitled Percent Complete. The remaining duration for unstarted activities are not to be less than its original duration.

3.3.20 Cost Loading of Closeout Activities

Cost load the "Correction of punch list from Government pre-final inspection" activity(ies) not less than 1 percent of the present contract value. Activity(ies) may be declared 100 percent complete upon the Government's verification of completion and correction of all punch list work identified during Government pre-final inspection(s).

3.3.20.1 As-Built Drawings

Refer to LRL Section 00 80 00.00 06 regarding AS-BUILT DOCUMENTS for D/B or D/B/B contracts for a determination of withholding for final as-built drawing submittals.

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3.3.20.2 O & M Manuals

Refer to LRL Section 00 80 00.00 06 regarding EQUIPMENT DATA, O&M & REPAIR MANUALS WITH FIELD TRAINING REQUIREMENTS for a determination of withholding for final acceptance/approval of O&M Manuals.

3.3.21 Early Completion Schedule and the Right to Finish Early

An Early Completion Schedule is an Initial Project Schedule (IPS) that indicates all scope of the required contract work will be completed before the contractually required completion date.

- a. No IPS indicating an Early Completion will be accepted without being fully resource-loaded (including crew sizes and manhours) and the Government agreeing that the schedule is reasonable and achievable.
- b. The Government is under no obligation to accelerate work items it is responsible for to ensure that the early completion is met nor is it responsible to modify incremental funding (if applicable) for the project to meet the contractor's accelerated work.

3.4 PROJECT SCHEDULE SUBMISSIONS

- a. Provide the submissions as described below. The data reports and network diagrams required for each submission are contained in paragraph SUBMISSION REQUIREMENTS. If the Contractor fails or refuses to furnish the information and schedule updates as set forth herein, the Contractor will be deemed not to have provided an estimate upon which a progress payment can be made.
- b. Review comments made by the Government on the schedule(s) do not relieve the Contractor from compliance with requirements of the Contract Documents.

3.4.1 Preliminary Project Schedule Submission

Within 15 calendar days after the NTP is acknowledged, submit the Preliminary Project Schedule defining the planned operations detailed for the first 90 calendar days for approval. The approved Preliminary Project Schedule will be used for payment purposes not to exceed 90 calendar days after NTP. Completely cost load the Preliminary Project Schedule to balance the contract award CLINS shown on the Price Schedule. The Preliminary Project Schedule may be summary in nature for the remaining performance period. It must be early start and late finish constrained and logically tied as specified. The Preliminary Project Schedule forms the basis for the Initial Project Schedule specified herein and must include all of the required plan and program preparations, submissions and approvals identified in the contract (for example, Quality Control Plan, Safety Plan, and Environmental Protection Plan) as well as design activities, planned submissions of all early design packages, permitting activities, design review conference activities, and other non-construction activities intended to occur within the first 90 calendar days. Government acceptance of the associated design package(s) and all other specified Program and Plan approvals must occur prior to any planned construction activities. Activity code any activities that are summary in nature after the first 90 calendar days with Bid Item (CLIN) code (BIDI), Responsibility Code (RESP) and Feature of Work code (FOW).

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3.4.2 Initial Project Schedule Submission

Submit the Initial Project Schedule for approval within 42 calendar days after notice to proceed is issued. The schedule must demonstrate a reasonable and realistic sequence of activities which represent all work through the entire contract performance period. Include in the design-build schedule detailed design and permitting activities, including but not limited to identification of individual design packages, design submission, reviews and conferences; permit submissions and any required Government actions; and long lead item acquisition prior to design completion. Also cover in the initial design-build schedule the entire construction effort with as much detail as is known at the time but, as a minimum, include all construction start and completion milestones, and detailed construction activities through the dry-in, including all activity coding and cost loading. Include the remaining construction, including cost loading, but it may be scheduled summary in nature. As the design proceeds and design packages are developed, fully detail the remaining construction activities concurrent with the monthly schedule updating process. Constrain construction activities by Government acceptance of associated designs. When the design is complete, incorporate into the then approved schedule update all remaining detailed construction activities that are planned to occur after the dry-in milestone. No payment will be made for work items not fully detailed in the Project Schedule.

3.4.2.1 Design Package Schedule Submission

With each design package submitted to the Government, submit a fragment schedule extracted from the then current Preliminary, Initial or Updated schedule which covers the activities associated with that Design Package including construction, procurement and permitting activities.

3.4.3 Periodic Schedule Updates

Update the Project Schedule on a regular basis, monthly at a minimum. Provide a draft Periodic Schedule Update for review at the schedule update meetings as prescribed in the paragraph PERIODIC SCHEDULE UPDATE MEETINGS. These updates will enable the Government to assess Contractor's progress. Update the schedule to include detailed construction activities as the design progresses, but not later than the submission of the final un-reviewed design submission for each separate design package. The Contracting Officer may require submission of detailed schedule activities for any distinct construction that is started prior to submission of a final design submission if such activity is authorized.

- a. Update information including Actual Start Dates (AS), Actual Finish Dates (AF), Remaining Durations (RD), and Percent Complete. Updated information is subject to the approval of the Government at the meeting.
- b. AS and AF dates must match the date(s) reported on the Contractor's Quality Control Report for an activity start or finish.

3.5 SUBMISSION REQUIREMENTS

Submit the following items for the Preliminary Schedule, Initial Schedule, and every Periodic Schedule Update throughout the life of the project:

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3.5.1 Electronic Scheduling Data

Provide electronic scheduling data containing the current project schedule and all previously submitted schedules in the format of the scheduling software (e.g. .xer). Also include the Narrative Report and all required Schedule Reports. The electronic file is to identify the type of schedule (Preliminary, Initial, Update), full contract number, Data Date and schedule file name. Each schedule must have a unique file name and use project specific settings. The file naming convention is to be determined at SEKO meeting.

3.5.2 Narrative Report

Provide a Narrative Report with each schedule submission. The Schedule Narrative Report is a "stand alone" report in PDF or Word format. Title the report "Schedule Narrative Report". Show the Report date, schedule name, and contract number on the first page. Indicate who is responsible for the schedule update and narrative report. Provide a brief description of the Project Scope. The Narrative Report is expected to communicate to the Government the thorough analysis of the schedule output and the plans to compensate for problems, either current or potential, which are revealed through that analysis. Include the following information at a minimum in the Narrative Report:

- a. Identify and discuss the work scheduled to start in the next update period.
- b. Identify and describe Progress and Non-Progress updates made to the schedule.
 - o Progress Only updates are considered to be: Actual Start (AS), Actual Finish (AF), Reduced Remaining Duration (RD) changes, and Increased Percent Complete.
 - o Non-Progress updates are considered to be: Addition or Deletion of Activities; Split or combined activities using new activity IDs; Addition or deletion of logic links; Changes to lag values of logic links; Addition, Deletion, or Changes to constraints; Changes to OD; Increases in RD such that RD becomes greater than OD; Changes to RD not accompanied by changes to Percent Complete; Increase in RD of activities that have not started; Changes to calendar assignments; Changes to holiday assignments within a pre-existing calendar.
- c. Describe activities along the critical path in addition to activities on the next float path after the critical path.
- d. Describe current and anticipated problem areas, delaying factors and their impact, and an explanation of corrective actions taken or required to be taken. Identify the party responsible for delay, if applicable.
- e. Identify and explain why activities, based on their calculated late dates, should have either started or finished during the update period but did not.
- f. Identify and discuss all schedule changes by activity ID and activity name including what specifically was changed and why the change was needed. Include at a minimum new and deleted activities, logic

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changes, duration changes, calendar changes, lag changes, resource changes, and actual start and finish date changes.

- g. Identify and explain why out-of-sequence work was performed and how the schedule was corrected to show as-built condition of out-of-sequence work.
- h. If the longest path changes from a monthly update to the next, provide an explanation of the factors driving the change.
- i. Respond to USACE Schedule Review Comments from the previous update(s).

3.5.3 Schedule Reports

The format, filtering, organizing, and sorting for each schedule report will be as directed by the Contracting Officer. Typically, reports contain Activity Numbers, Activity Description, Original Duration, Remaining Duration, Early Start Date, Early Finish Date, Late Start Date, Late Finish Date, Total Float, Actual Start Date, Actual Finish Date, and Percent Complete. Provide the reports electronically in .pdf format. The following lists typical reports that will be requested:

3.5.3.1 Activity Report

List of all activities sorted according to activity number.

3.5.3.2 Logic Report

List of detailed predecessor and successor activities for every activity in ascending order by activity number.

3.5.3.3 Total Float Report

A list of all incomplete activities sorted in ascending order of total float. List activities which have the same amount of total float in ascending order of Early Start Dates. Do not show completed activities on this report.

3.5.3.4 Earnings Report by CLIN

A compilation of the Total Earnings on the project from the NTP to the data date, which reflects the earnings of activities based on the agreements made in the schedule update meeting defined herein. Provided a complete schedule update has been furnished, this report serves as the basis of determining progress payments. Group activities by CLIN number and sort by activity number. Provide a total CLIN percent earned value, CLIN percent complete, and project percent complete. The printed report must contain the following for each activity: the Activity Number, Activity Description, Original Budgeted Amount, Earnings to Date, Earnings this period, Total Quantity, Quantity to Date, and Percent Complete (based on cost).

3.5.3.5 Scheduling and Leveling Report

Provide a Scheduling/Leveling Report generated from the current project schedule being submitted.

3.5.3.6 Three Week Look Ahead & Detail Schedule

- a. Prepare and issue a 3-week Look Ahead schedule for each weekly meeting

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using a 3-week look-ahead filter to show all activities including critical activities, planned start, finish, and total float.

- b. In addition, provide a more detailed day-to-day plan of upcoming work derived, but maintained separately, from the project schedule. The detailed plan is typically created and maintained by the Project Superintendent. The detailed plan information shall correspond with the Activity Numbers and Activity Description of the monthly schedule updates. It shall be attached to the corresponding QC Daily Report each week. The detailed plan activities must not exceed 5 working days in duration.
- c. The activities shall include a sufficient level of detail to show upcoming outages, closures, testing, preparatory & initial meetings, prime and subcontractor crew assignments (including the name of the subcontractors and/or suppliers as necessary), anticipated crew size, material & equipment as necessary, and specific areas where work is to take place (e.g - POV parking; Ductbank between X & Y; Mech Room ###; 1st floor lobby, Area A Roof, Critical Lifts, etc.)

3.5.4 Network Diagram

The Network Diagram is required for the Preliminary, Initial and Periodic Updates. Depict and display the order and interdependence of activities and the sequence in which the work is to be accomplished. The Contracting Officer will use, but is not limited to, the following conditions to review compliance with this paragraph:

3.5.4.1 Continuous Flow

Show a continuous flow from left to right with no arrows from right to left. Show the activity number, description, duration, and estimated earned value on the diagram.

3.5.4.2 Project Milestone Dates

Show dates on the diagram for start of project, any contract required interim completion dates, and contract completion dates.

3.5.4.3 Critical Path

Show all activities on the critical path. The critical path is defined as the longest path.

3.5.4.4 Grouping and Sorting

Group and sort activities as directed to assist in the understanding of the activity sequence. Typically, this flow will group activities by major elements of work, category of work, work area and/or responsibility.

3.5.4.5 Cash Flow / Schedule Variance Control (SVC) Diagram

With each schedule submission, provide a SVC diagram showing 1) Cash Flow S-Curves indicating planned project cost based on projected early and late activity finish dates, and 2) Earned Value to-date.

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3.6 PERIODIC SCHEDULE UPDATE

3.6.1 Periodic Schedule Update Meetings

Conduct periodic schedule update meetings for the purpose of reviewing the proposed Periodic Schedule Update, Narrative Report, Schedule Reports, and progress payment. Conduct meetings at least monthly within five days of the proposed schedule data date. Provide a computer with the scheduling software loaded and a projector which allows all meeting participants to view the proposed schedule during the meeting. The Contractor's authorized scheduler must organize, group, sort, filter, perform schedule revisions as needed and review functions as requested by the Contractor and/or Government. The Contractor scheduler must attend in person unless virtual attendance is approved by Contracting Officer. The meeting is a working interactive exchange which allows the Government and Contractor the opportunity to review the updated schedule on a real time and interactive basis. The meeting will last no longer than 8 hours. Submit schedule file and narrative to the Government a minimum of two workdays in advance of the meeting. The Contractor's Project Manager must attend the meeting with the authorized representative of the Contracting Officer. Superintendents, foremen and major subcontractors must attend the meeting as required to discuss the project schedule and work. Following the schedule update meeting, make corrections to the schedule and resubmit, following the required schedule naming convention. Include only those changes approved by the Government in the schedule review meeting. Only approved schedules may be used to generate invoices for payment.

3.6.2 Update Submission Following Progress Meeting

Submit the complete Periodic Schedule Update of the Project Schedule containing all approved progress, revisions, and adjustments, pursuant to paragraph SUBMISSION REQUIREMENTS not later than 4 workdays after the periodic schedule update meeting.

3.7 WEEKLY PROGRESS MEETINGS

- a. The Government and the Contractor shall meet weekly (or as otherwise mutually agreed to) between the meetings described in paragraph PERIODIC SCHEDULE UPDATE MEETINGS for the purpose of jointly reviewing the actual progress of the project as compared to the as planned progress and to review planned activities for the upcoming three weeks. The then current and approved schedule update shall be used for the purposes of this meeting and for the production and review of reports. The Contractor's Project Manager and the Authorized Representative of the Contracting Officer shall attend. The Contractor shall provide the planned activities for the next three weeks based on a Primavera filter of the current approved schedule. Any activity that has not started or finished as planned shall be listed as a QC deficiency in RMS.

The Contractor's Project Manager and the Authorized Representative of the Contracting Officer shall attend. The weekly progress meeting will address the status of RFI's, RFP's and Submittals. At the weekly progress meeting, address the status of approved schedule progress, RFIs, RFPs and Submittals. Prior to beginning work on specific work elements of a Project, the Contractor shall confer with the COR and agree on a sequence of procedures and means of access to premises and buildings; space for storage of materials and equipment; delivery of materials; and use of approaches, use of corridors, stairways, and

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similar means of passage. Contractor shall provide minutes for the Weekly Progress Meeting, minutes to be attached with the QC Daily Report.

- b. Provide a bar chart produced by the scheduling software, organized by Total Float and Sorted by Early Start Date, and a three week "look-ahead" schedule by filtering all schedule activities to show only current ongoing activities and activities scheduled to start during the upcoming two weeks, organized by Work Area Code (AREA) and sorted by Early Start Date.

3.8 REQUESTS FOR TIME EXTENSIONS

ASCE 67-17 provides delay analysis guidelines. If there is a conflict between the contract and guidance provided in ASCE 67-17, the contract will govern. Provide justification of delay to the Contracting Officer in accordance with the contract provisions and clauses for approval, within 10 days of a delay occurring. Also prepare a time impact analysis for each Government request for proposal (RFP).

3.8.1 Justification of Delay

Provide a description of the event(s) that caused the delay and/or impact to the work. As part of the description, identify all schedule activities impacted. Show that the event that caused the delay/impact was the responsibility of the Government. Provide a time impact analysis that demonstrates the effects of the delay or impact on the project completion date, or interim completion date(s). Evaluate multiple impacts chronologically; each with its own justification of delay. With multiple impacts consider any concurrency of delay. A time extension and the schedule fragnet becomes part of the project schedule and all future schedule updates upon approval by the Contracting Officer.

3.8.2 Changes That Do Not Cause Delay

If it is determined that a change does not constitute a schedule time extension, a fragnet as described in paragraph FRAGMENTARY NETWORK, is to be created and inserted into the Project schedule.

3.8.3 Time Impact Analysis (Prospective Analysis)

Prepare a time impact analysis for approval by the Contracting Officer based on industry standard AACE 52R-06. Utilize a copy of the last approved schedule prior to the first day of the impact or delay for the time impact analysis. If Contracting Officer determines the time frame between the last approved schedule and the first day of impact is too great, prepare an interim updated schedule to perform the time impact analysis. Unless approved by the Contracting Officer, no other changes may be incorporated into the schedule being used to justify the time impact.

- a. Each TIA must be in both narrative and schedule form. The narrative must define the scope and conditions of the change; provide start and finish dates of impact; predecessor and successor activity to the impact period; responsible party; describe how it originated and how it impacts the schedule. The schedule submission must consist of three native files:

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- (1) Fragnet used to define the scope of the changed condition,
- (2) Most recent accepted schedule update as of the time of the proposal that has been updated to show all activity progress as of the time of the impact start date.
- (3) The impacted schedule that has the fragnet inserted into the updated schedule calculated to show the new projected completion date.

- b. All TIAs shall include a description of mitigation considerations describing efforts undertaken to minimize or eliminate impact and/or delay. The TIAs shall determine the apportionment of the overall delay assignable to each individual delay. The associated narrative must clearly describe the findings in a chronological listing beginning with the earliest delay event.

3.8.4 Forensic Schedule Analysis (Retrospective Analysis)

Prepare an analysis for approval by the Contracting Officer based on methodologies identified in industry standard AACE 29R-03. Identify which methodology was chosen and explain why it is more appropriate than other methodologies available.

3.8.5 Fragmentary Network (Fragnet)

Prepare a proposed fragnet for time impact analysis consisting of a sequence of new activities that are proposed to be added to the project schedule to demonstrate the influence of the delay or impact to the project's contractual dates. Clearly show how the proposed fragnet is to be tied into the project schedule including all predecessors and successors to the fragnet activities. The proposed fragnet must be approved by the Contracting Officer prior to incorporation into the project schedule.

3.8.6 Time Extension

The Contracting Officer must approve the Justification of Delay including the time impact analysis before a time extension will be granted. No time extension will be granted unless the delay consumes all available Project Float and extends the projected finish date ("End Project" milestone) beyond the Contract Completion Date. The time extension will be in calendar days.

Actual delays that are found to be caused by the Contractor's own actions, which result in a calculated schedule delay, are not a cause for an extension to the performance period, completion date, or any interim milestone date.

3.8.7 Impact to Early Completion Schedule

No extended overhead will be paid for delay prior to the original Contract Completion Date for an Early Completion IPS unless the Contractor actually performed work in accordance with that Early Completion Schedule. The Contractor must show that an early completion was achievable had it not been for the impact.

3.9 FAILURE TO ACHIEVE PROGRESS

Should the progress fall behind the approved project schedule for reasons

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other than those that are excusable within the terms of the contract, the Contracting Officer may require provision of a written recovery plan for approval. The plan must detail how progress will be made-up to include which activities will be accelerated by adding additional crews, longer work hours, extra workdays, etc.

3.9.1 Artificially Improving Progress

Artificially improving progress by means such as, but not limited to, revising the schedule logic, modifying, or adding constraints, shortening activity durations, or changing calendars in the project schedule is prohibited. Indicate assumptions made and the basis for any logic, constraint, duration, and calendar changes used in the creation of the recovery plan. Any additional resources, manpower, or daily and weekly work hour changes proposed in the recovery plan must be evident at the work site and documented in the daily report along with the Schedule Narrative Report.

3.9.2 Failure to Perform

Failure to perform work and maintain progress in accordance with the supplemental recovery plan may result in an interim and final unsatisfactory performance rating and may result in corrective action directed by the Contracting Officer pursuant to FAR 52.236-15 Schedules for Construction Contracts, FAR 52.249-10 Default (Fixed-Price Construction), and other contract provisions.

3.9.3 Recovery Schedule

Should the Contracting Officer find it necessary, submit a recovery schedule pursuant to FAR 52.236-15 Schedules for Construction Contracts.

3.10 OWNERSHIP OF FLOAT

- a. Except for the provision given in the paragraph IMPACT TO EARLY COMPLETION SCHEDULE, float available in the schedule, at any time, belongs to the Project and is available for Contractor and Government use. This includes activity and project float. Activity float is the number of workdays that an activity can be delayed without causing a delay to the "End Project" finish milestone. Project float (if applicable) is the number of work days between the projected early finish and the "End Project" finish.
- b. Use of float suppression techniques, such as: preferential sequencing (arranging critical path through activities more susceptible to Government caused delay); lag logic restraints; zero total or free float constraints, extended activity times, or imposing constraint dates other than as required by the contract, shall be cause for rejection of the project schedule or its updates. The use of Resource Leveling (or similar software features) used for the purpose of artificially adjusting activity durations to consume float and influence the critical path is expressly prohibited.

3.11 TRANSFER OF SCHEDULE DATA INTO RESIDENT MANAGEMENT SYSTEM

Ensure schedule data is uploaded to RMS. This data is additional supporting data in a form and detail required by the Contracting Officer pursuant to FAR 52.232-5 Payments under Fixed-Price Construction Contracts. The receipt of a proper payment request pursuant to FAR

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52.232-27 Prompt Payment for Construction Contracts is contingent upon the Government receiving both acceptable and approvable hard copies and matching electronic versions of the application for progress payment.

3.12 PRIMAVERA P6 MANDATORY REQUIREMENTS

Ensure Primavera P6 settings provide a schedule capable of fulfilling the requirements of the contract. The following settings are mandatory and required in all schedule submissions to the Government:

- a. Activity Codes must be Project Level, not Global or EPS level.
- b. Calendars must be Project Level, not Global or Resource level.
- c. Activity Duration Types must be set to "Fixed Duration & Units".
- d. Percent Complete Types must be set to "Physical".
- e. Time Period Admin Preferences must remain the default "8.0 hr/day, 40 hr/week, 172 hr/month, 2000 hr/year". Set Calendar Work Hours/Day to 8.0 Hour days.
- f. Set Schedule Option for defining Critical Activities to "Longest Path".
- g. Set Schedule Option for defining progressed activities to "Retained Logic".
- h. Set up cost loading using a single lump sum non-labor resource. The Price/Unit must be \$1/hr, Default Units/Time must be "8h/d", and settings "Auto Compute Actuals" and "Calculate costs from units" un-selected.
- i. Activity ID's must not exceed 10 characters.
- j. Activity Names must have a verb-noun structure and contain the most defining and detailed description within the first 30 characters.
- k. The daily ending hour for all work calendars must be the same.

-- End of Section --

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SECTION 01 33 00

SUBMITTAL PROCEDURES

08/18, CHG 4: 02/21

PART 1 GENERAL

1.1 SUMMARY

1.1.1 Submittal Information

The Contracting Officer may request submittals in addition to those specified when deemed necessary to adequately describe the work covered in the respective sections. Each submittal is to be complete and in sufficient detail to allow ready determination of compliance with contract requirements.

Units of weights and measures used on all submittals are to be the same as those used in the contract drawings.

1.1.2 Project Type

The Contractor's Quality Control (CQC) System Manager are to check and approve all items before submittal and stamp, sign, and date indicating action taken. Proposed deviations from the contract requirements are to be clearly identified. Include within submittals items such as: Contractor's, manufacturer's, or fabricator's drawings; descriptive literature including (but not limited to) catalog cuts, diagrams, operating charts or curves; test reports; test cylinders; samples; O&M manuals (including parts list); certifications; warranties; and other such required submittals.

1.1.3 Submission of Submittals

Schedule and provide submittals requiring Government approval before acquiring the material or equipment covered thereby. Pick up and dispose of samples not incorporated into the work in accordance with manufacturer's Safety Data Sheets (SDS) and in compliance with existing laws and regulations.

1.2 DEFINITIONS

1.2.1 Submittal Descriptions (SD)

Submittal requirements are specified in the technical sections. Examples and descriptions of submittals identified by the Submittal Description (SD) numbers and titles follow:

SD-01 Preconstruction Submittals

Submittals that are required prior to or at the start of construction (work) or the next major phase of the construction on a multiphase contract.

Preconstruction Submittals include schedules and a tabular list of locations, features, and other pertinent information regarding products, materials, equipment, or components to be used in the work.

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Certificates Of Insurance

Surety Bonds

List Of Proposed Subcontractors

List Of Proposed Products

Baseline Network Analysis Schedule (NAS)

Submittal Register

Schedule Of Prices Or Earned Value Report

Accident Prevention Plan

Work Plan

Quality Control (QC) plan

Environmental Protection Plan

SD-02 Shop Drawings

Drawings, diagrams and schedules specifically prepared to illustrate some portion of the work.

Diagrams and instructions from a manufacturer or fabricator for use in producing the product and as aids to the Contractor for integrating the product or system into the project.

Drawings prepared by or for the Contractor to show how multiple systems and interdisciplinary work will be coordinated.

SD-03 Product Data

Catalog cuts, illustrations, schedules, diagrams, performance charts, instructions and brochures illustrating size, physical appearance and other characteristics of materials, systems or equipment for some portion of the work.

Samples of warranty language when the contract requires extended product warranties.

SD-04 Samples

Fabricated or unfabricated physical examples of materials, equipment or workmanship that illustrate functional and aesthetic characteristics of a material or product and establish standards by which the work can be judged.

Color samples from the manufacturer's standard line (or custom color samples if specified) to be used in selecting or approving colors for the project.

Field samples and mock-ups constructed on the project site establish standards ensuring work can be judged. Includes assemblies or portions of assemblies that are to be incorporated into the project and those that will be removed at conclusion of the work.

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SD-05 Design Data

Design calculations, mix designs, analyses or other data pertaining to a part of work.

SD-06 Test Reports

Report signed by authorized official of testing laboratory that a material, product or system identical to the material, product or system to be provided has been tested in accord with specified requirements. Unless specified in another section, testing must have been within three years of date of contract award for the project.

Report that includes findings of a test required to be performed on an actual portion of the work or prototype prepared for the project before shipment to job site.

Report that includes finding of a test made at the job site or on sample taken from the job site, on portion of work during or after installation.

Investigation reports

Daily logs and checklists

Final acceptance test and operational test procedure

SD-07 Certificates

Statements printed on the manufacturer's letterhead and signed by responsible officials of manufacturer of product, system or material attesting that the product, system, or material meets specification requirements. Must be dated after award of project contract and clearly name the project.

Document required of Contractor, or of a manufacturer, supplier, installer or Subcontractor through Contractor. The document purpose is to further promote the orderly progression of a portion of the work by documenting procedures, acceptability of methods, or personnel qualifications.

Confined space entry permits

Text of posted operating instructions

SD-08 Manufacturer's Instructions

Preprinted material describing installation of a product, system or material, including special notices and (SDS) concerning impedances, hazards and safety precautions.

SD-10 Operation and Maintenance Data

Data provided by the manufacturer, or the system provider, including manufacturer's help and product line documentation, necessary to maintain and install equipment, for operating and maintenance use by facility personnel.

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Data required by operating and maintenance personnel for the safe and efficient operation, maintenance and repair of the item.

Data incorporated in an operations and maintenance manual or control system.

SD-11 Closeout Submittals

Documentation to record compliance with technical or administrative requirements or to establish an administrative mechanism.

Submittals required for Guiding Principle Validation (GPV).

Special requirements necessary to properly close out a construction contract. For example, Record Drawings and as-built drawings. Also, submittal requirements necessary to properly close out a major phase of construction on a multi-phase contract.

1.2.2 Approving Authority

Office or designated person authorized to approve the submittal.

1.2.3 Work

As used in this section, on-site and off-site construction required by contract documents, including labor necessary to produce submittals, construction, materials, products, equipment, and systems incorporated or to be incorporated in such construction. In exception, excludes work to produce SD-01 submittals.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Submittal Register; G

1.4 SUBMITTAL CLASSIFICATION

1.4.1 Government Approved (G)

Government approval is required for extensions of design, critical materials, variations, equipment whose compatibility with the entire system must be checked, and other items as designated by the Government.

Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, submittals are considered to be "shop drawings."

1.4.2 For Information Only

Submittals not requiring Government approval will be for information only. Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, they are not considered to be "shop drawings."

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1.4.3 Sustainability Reporting Submittals (S)

Submittals for Guiding Principle Validation (GPV). These submittals are for information only and for use as specified in Section 01 33 29.00 06 SUSTAINABILITY REPORTING.

Schedule submittals for these items throughout the course of construction as provided; do not wait until closeout.

1.5 PREPARATION

1.5.1 Transmittal Form

Use the ENG Form 4025-R transmittal form for submitting both Government-approved and information-only submittals. Submit in accordance with the instructions on the reverse side of the form. These forms or similar forms are included in the RMS CM software that the Contractor is required to use for this contract. Properly complete this form by filling out all the heading blank spaces and identifying each item submitted. Exercise special care to ensure proper listing of the specification paragraph and sheet number of the contract drawings pertinent to the data submitted for each item.

1.5.2 Submittal Format

1.5.2.1 Format of SD-01 Preconstruction Submittals

When the submittal includes a document that is to be used in the project, or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

1.5.2.2 Format for SD-02 Shop Drawings

Provide shop drawings not less than 8 1/2 by 11 inches nor more than 30 by 42 inches, except for full-size patterns or templates. Prepare drawings to accurate size, with scale indicated, unless another form is required. Ensure drawings are suitable for reproduction and of a quality to produce clear, distinct lines and letters, with dark lines on a white background.

- a. Include the nameplate data, size, and capacity on drawings. Also include applicable federal, military, industry, and technical society publication references.
- b. Dimension drawings, except diagrams and schematic drawings. Prepare drawings demonstrating interface with other trades to scale. Use the same unit of measure for shop drawings as indicated on the contract drawings. Identify materials and products for work shown.

Submit an electronic copy of drawings in PDF format.

1.5.2.2.1 Drawing Identification

Include on each drawing the drawing title, number, date, and revision numbers and dates, in addition to information required in paragraph IDENTIFYING SUBMITTALS.

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Number drawings in a logical sequence. Each drawing is to bear the number of the submittal in a uniform location next to the title block. Place the Government contract number in the margin, immediately below the title block, for each drawing.

Reserve a blank space, no smaller than 4 inches on the right-hand side of each sheet for the Government disposition stamp.

1.5.2.3 Format of SD-03 Product Data

Present product data submittals for each section. Include a table of contents, listing the page and catalog item numbers for product data.

Indicate, by prominent notation, each product that is being submitted; indicate the specification section number and paragraph number to which it pertains.

1.5.2.3.1 Product Information

Supplement product data with material prepared for the project to satisfy the submittal requirements where product data does not exist. Identify this material as developed specifically for the project, with information and format as required for submission of SD-07 Certificates.

Provide product data in units used in the Contract documents. Where product data are included in preprinted catalogs with another unit, submit the dimensions in contract document units, on a separate sheet.

1.5.2.3.2 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.5.2.3.3 Data Submission

Collect required data submittals for each specific material, product, unit of work, or system into a single submittal that is marked for choices, options, and portions applicable to the submittal. Mark each copy of the product data identically. Partial submittals will not be accepted for expedition of the construction effort.

Submit the manufacturer's instructions before installation.

1.5.2.4 Format of SD-04 Samples

1.5.2.4.1 Sample Characteristics

Furnish samples in the following sizes, unless otherwise specified or unless the manufacturer has prepackaged samples of approximately the same

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size as specified:

- a. Sample of Equipment or Device: Full size.
- b. Sample of Materials Less Than 2 by 3 inches: Built up to 8 1/2 by 11 inches.
- c. Sample of Materials Exceeding 8 1/2 by 11 inches: Cut down to 8 1/2 by 11 inches and adequate to indicate color, texture, and material variations.
- d. Sample of Linear Devices or Materials: 10 inch length or length to be supplied, if less than 10 inches. Examples of linear devices or materials are conduit and handrails.
- e. Sample Volume of Nonsolid Materials: Pint. Examples of nonsolid materials are sand and paint.
- f. Color Selection Samples: 2 by 4 inches. Where samples are specified for selection of color, finish, pattern, or texture, submit the full set of available choices for the material or product specified. Sizes and quantities of samples are to represent their respective standard unit.
- g. Sample Panel: 4 by 4 feet.
- h. Sample Installation: 100 square feet.

1.5.2.4.2 Sample Incorporation

Reusable Samples: Incorporate returned samples into work only if so specified or indicated. Incorporated samples are to be in undamaged condition at the time of use.

Recording of Sample Installation: Note and preserve the notation of any area constituting a sample installation, but remove the notation at the final clean-up of the project.

1.5.2.4.3 Comparison Sample

Samples Showing Range of Variation: Where variations in color, finish, pattern, or texture are unavoidable due to nature of the materials, submit sets of samples of not less than three units showing extremes and middle of range. Mark each unit to describe its relation to the range of the variation.

When color, texture, or pattern is specified by naming a particular manufacturer and style, include one sample of that manufacturer and style, for comparison.

1.5.2.5 Format of SD-05 Design Data

Provide design data and certificates on 8 1/2 by 11 inch paper.

1.5.2.6 Format of SD-06 Test Reports

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

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1.5.2.7 Format of SD-07 Certificates

Provide design data and certificates on 8 1/2 by 11 inch paper.

1.5.2.8 Format of SD-08 Manufacturer's Instructions

Present manufacturer's instructions submittals for each section. Include the manufacturer's name, trade name, place of manufacture, and catalog model or number on product data. Also include applicable federal, military, industry, and technical-society publication references. If supplemental information is needed to clarify the manufacturer's data, submit it as specified for SD-07 Certificates.

Submit the manufacturer's instructions before installation.

1.5.2.8.1 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.5.2.9 Format of SD-09 Manufacturer's Field Reports

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.5.2.10 Format of SD-10 Operation and Maintenance Data (O&M)

Comply with the requirements specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA for O&M Data format.

1.5.2.11 Format of SD-11 Closeout Submittals

When the submittal includes a document that is to be used in the project or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

1.5.3 Source Drawings for Shop Drawings

1.5.3.1 Source Drawings

The entire set of source drawing files (DWG) will not be provided to the Contractor. Request the specific Drawing Number for the preparation of shop drawings. Only those drawings requested to prepare shop drawings will be provided. These drawings are provided only after award.

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1.5.3.2 Terms and Conditions

Data contained on these electronic files must not be used for any purpose other than as a convenience in the preparation of construction data for the referenced project. Any other use or reuse is at the sole risk of the Contractor and without liability or legal exposure to the Government. The Contractor must make no claim, and waives to the fullest extent permitted by law any claim or cause of action of any nature against the Government, its agents, or its subconsultants that may arise out of or in connection with the use of these electronic files. The Contractor must, to the fullest extent permitted by law, indemnify and hold the Government harmless against all damages, liabilities, or costs, including reasonable attorney's fees and defense costs, arising out of or resulting from the use of these electronic files.

These electronic source drawing files are not construction documents. Differences may exist between the source drawing files and the corresponding construction documents. The Government makes no representation regarding the accuracy or completeness of the electronic source drawing files, nor does it make representation to the compatibility of these files with the Contractor hardware or software. The Contractor is responsible for determining if any conflict exists. In the event that a conflict arises between the signed and sealed construction documents prepared by the Government and the furnished source drawing files, the signed and sealed construction documents govern. Use of these source drawing files does not relieve the Contractor of the duty to fully comply with the contract documents, including and without limitation the need to check, confirm and coordinate the work of all contractors for the project. If the Contractor uses, duplicates or modifies these electronic source drawing files for use in producing construction data related to this contract, remove all previous indication of ownership (seals, logos, signatures, initials and dates).

1.5.4 Electronic File Format

Provide submittals in electronic format, with the exception of material samples required for SD-04 Samples items. In addition to the electronic submittal, provide three hard copies of the submittals. Compile the submittal file as a single, complete document, to include the Transmittal Form described within, and also separately attach the native files which were used to create PDF. The attached files should include the original digital files used to create the submittal. Name the electronic submittal file specifically according to its contents, and coordinate the file naming convention with the Contracting Officer. Electronic files must be of sufficient quality that all information is legible. Use PDF as the electronic format, unless otherwise specified or directed by the Contracting Officer. Generate PDF files from original documents with bookmarks so that the text included in the PDF file is searchable and can be copied. If documents are scanned, optical character resolution (OCR) routines are required. Index and bookmark files exceeding 30 pages to allow efficient navigation of the file. When required, the electronic file must include a valid electronic signature or a scan of a signature.

E-mail electronic submittal documents smaller than 10MB to an e-mail address as directed by the Contracting Officer. Provide electronic documents over 10 MB on an optical disc or through an electronic file sharing system such as the DOD SAFE Web Application located at the following website: <https://safe.apps.mil/>.

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1.6 QUANTITY OF SUBMITTALS

1.6.1 Number of SD-01 Preconstruction Submittal Copies

Unless otherwise specified, submit three sets of administrative submittals.

1.6.2 Number of SD-04 Samples

- a. Submit two samples, or two sets of samples showing the range of variation, of each required item. One approved sample or set of samples will be retained by the approving authority and one will be returned to the Contractor.
- b. Submit one sample panel or provide one sample installation where directed. Include components listed in the technical section or as directed.
- c. Submit one sample installation, where directed.
- d. Submit one sample of nonsolid materials.

1.7 INFORMATION ONLY SUBMITTALS

Submittals without a "G" designation must be certified by the QC manager and submitted to the Contracting Officer for information-only. Provide information-only submittals to the Contracting Officer a minimum of 30 calendar days prior to the Preparatory Meeting for the associated Definable Feature of Work (DFOW). Approval of the Contracting Officer is not required on information only submittals. The Contracting Officer will mark "receipt acknowledged" on submittals for information and will return only the transmittal cover sheet to the Contractor. Normally, submittals for information only will not be returned. However, the Government reserves the right to return unsatisfactory submittals and require the Contractor to resubmit any item found not to comply with the contract. This does not relieve the Contractor from the obligation to furnish material conforming to the plans and specifications; will not prevent the Contracting Officer from requiring removal and replacement of nonconforming material incorporated in the work; and does not relieve the Contractor of the requirement to furnish samples for testing by the Government laboratory or for check testing by the Government in those instances where the technical specifications so prescribe.

1.8 PROJECT SUBMITTAL REGISTER

A sample Project Submittal Register showing items of equipment and materials for when submittals are required by the specifications is provided as "Appendix A - Submittal Register."

1.8.1 Submittal Management

Prepare and maintain a submittal register, as the work progresses. Do not change data that is output in columns (c), (d), (e), and (f) as delivered by Government; retain data that is output in columns (a), (g), (h), and (i) as approved. As an attachment, provide a submittal register showing items of equipment and materials for which submittals are required by the specifications. This list may not be all-inclusive and additional submittals may be required. Maintain a submittal register for the project in accordance with Section 01 45 00.15 10 RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE(RMS CM). The Government will provide the initial submittal

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register in electronic format with the following fields completed, to the extent that will be required by the Government during subsequent usage.

Column (c): Lists specification section in which submittal is required.

Column (d): Lists each submittal description (SD Number. and type, e.g., SD-02 Shop Drawings) required in each specification section.

Column (e): Lists one principal paragraph in each specification section where a material or product is specified. This listing is only to facilitate locating submitted requirements. Do not consider entries in column (e) as limiting the project requirements.

Thereafter, the Contractor is to track all submittals by maintaining a complete list, including completion of all data columns and all dates on which submittals are received by and returned by the Government.

1.8.2 Preconstruction Use of Submittal Register

Submit the submittal register. Include the QC plan and the project schedule. Verify that all submittals required for the project are listed and add missing submittals. Coordinate and complete the following fields on the register submitted with the QC plan and the project schedule:

Column (a) Activity Number: Activity number from the project schedule.

Column (g) Contractor Submit Date: Scheduled date for the approving authority to receive submittals.

Column (h) Contractor Approval Date: Date that Contractor needs approval of submittal.

Column (i) Contractor Material: Date that Contractor needs material delivered to Contractor control.

1.8.3 Contractor Use of Submittal Register

Update the following fields in the Government-furnished submittal register program or equivalent fields in the program used by the Contractor with each submittal throughout the contract.

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (j) Action Code (k): Date of action used to record Contractor's review when forwarding submittals to QC.

Column (l) Date submittal transmitted.

Column (q) Date approval was received.

1.8.4 Approving Authority Use of Submittal Register

Update the following fields:

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Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (l) Date submittal was received.

Column (m) through (p) Dates of review actions.

Column (q) Date of return to Contractor.

1.8.5 Action Codes

1.8.5.1 Contractor Action Codes

DESIGN BID BUILD SUBMITTALS			
Submittal Classifications shown in UFGS Sections	Submittal Classification	Corresponding SpecsIntact Submittal Register Code which is populated in the SI Submittal Register. Software Limitations: (The software shows one character delineation in the SpecsIntact Submittal Register)	RMS - The following Submittal Classifications are populated in RMS when the SpecsIntact Submittal Data File is pulled into RMS)
G	Submittal requires Government	G	GA
BLANK	Submittal is For Information Only (FIO)	BLANK	FIO
S	Submittal is for documentation of Sustainable requirements	S	S/FIO

1.8.6 Delivery of Copies

Submit an updated electronic copy of the submittal register to the Contracting Officer with each invoice request. Provide an updated Submittal Register monthly regardless of whether an invoice is submitted.

1.9 VARIATIONS

Variations from contract requirements require Contracting Officer approval pursuant to contract Clause FAR 52.236-21 Alt. I Specifications and Drawings for Construction, and will be considered where advantageous to the Government.

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1.9.1 Considering Variations

Discussion of variations with the Contracting Officer before submission of a variation submittal will help ensure that functional and quality requirements are met and minimize rejections and resubmittals. For variations that include design changes or some material or product substitutions, the Government may require an evaluation and analysis by a licensed professional engineer hired by the contractor.

Specifically point out variations from contract requirements in a transmittal letter. Failure to point out variations may cause the Government to require rejection and removal of such work at no additional cost to the Government.

1.9.2 Proposing Variations

When proposing variation, deliver a submittal, clearly marked as a "VARIATION" to the Contracting Officer, with documentation illustrating the nature and features of the variation including any necessary technical submittals and why the variation is desirable and beneficial to Government. If lower cost is a benefit, also include an estimate of the cost savings. In addition to documentation required for variation, include the submittals required for the item. Clearly mark the proposed variation in all documentation.

Specifically point out variations from contract requirements in a transmittal letter. Failure to point out variations may cause the Government to require rejection and removal of such work at no additional cost to the Government.

Check the column "variation" of ENG Form 4025 for submittals that include variations proposed by the Contractor. Set forth in writing the reason for any variations and note such variations on the submittal. The Government reserves the right to rescind inadvertent approval of submittals containing unnoted variations.

1.9.3 Warranting that Variations are Compatible

When delivering a variation for approval, the Contractor warrants that this contract has been reviewed to establish that the variation, if incorporated, will be compatible with other elements of work.

1.9.4 Review Schedule Extension

In addition to the normal submittal review period Add after submittal 30 day review period, a period of 30 calendar days will be allowed for the Government to consider submittals with variations.

1.10 SCHEDULING

Schedule and submit concurrently product data and shop drawings covering component items forming a system or items that are interrelated. Submit pertinent certifications at the same time. No delay damages or time extensions will be allowed for time lost in late submittals. Allow an additional 14 calendar days for review and approval of submittals for refrigeration and HVAC control systems.

- a. Coordinate scheduling, sequencing, preparing, and processing of submittals with performance of work so that work will not be delayed by

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submittal processing. The Contractor is responsible for additional time required for Government reviews resulting from required resubmittals. The review period for each resubmittal is the same as for the initial submittal.

- b. Submittals required by the contract documents are listed on the submittal register. If a submittal is listed in the submittal register but does not pertain to the contract work, the Contractor is to include the submittal in the register and annotate it "N/A" with a brief explanation. Approval by the Contracting Officer does not relieve the Contractor of supplying submittals required by the contract documents but that have been omitted from the register or marked "N/A."
- c. Resubmit the submittal register and annotate it monthly with actual submission and approval dates. When all items on the register have been fully approved, no further resubmittal is required.

Contracting Officer review will be completed within 30 calendar days after the date of submission.

1.11 GOVERNMENT APPROVING AUTHORITY

When the approving authority is the Contracting Officer, the Government will:

- a. Note the date on which the submittal was received.
- b. Review submittals for approval within the scheduling period specified and only for conformance with project design concepts and compliance with contract documents.
- c. Identify returned submittals with one of the actions defined in paragraph REVIEW NOTATIONS and with comments and markings appropriate for the action indicated.

Upon completion of review of submittals requiring Government approval, stamp and date submittals. 1 copies of the submittal will be retained by the Contracting Officer and 1 copyies of the submittal will be returned to the Contractor.

1.11.1 Review Notations

Submittals will be returned to the Contractor with the following notations:

- a. Submittals marked "approved" or "accepted" authorize proceeding with the work covered.
- b. Submittals marked "approved as noted" or "approved, except as noted, resubmittal not required," authorize proceeding with the work covered provided that the Contractor takes no exception to the corrections.
- c. Submittals marked "not approved," "disapproved," or "revise and resubmit" indicate incomplete submittal or noncompliance with the contract requirements or design concept. Resubmit with appropriate changes. Do not proceed with work for this item until the resubmittal is approved.
- d. Submittals marked "not reviewed" indicate that the submittal has been previously reviewed and approved, is not required, does not have evidence of being reviewed and approved by Contractor, or is not

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complete. A submittal marked "not reviewed" will be returned with an explanation of the reason it is not reviewed. Resubmit submittals returned for lack of review by Contractor or for being incomplete, with appropriate action, coordination, or change.

- e. Submittals marked "receipt acknowledged" indicate that submittals have been received by the Government. This applies only to "information-only submittals" as previously defined.

1.12 DISAPPROVED SUBMITTALS

Make corrections required by the Contracting Officer. If the Contractor considers any correction or notation on the returned submittals to constitute a change to the contract drawings or specifications, give notice to the Contracting Officer as required under the FAR clause titled CHANGES. The Contractor is responsible for the dimensions and design of connection details and the construction of work. Failure to point out variations may cause the Government to require rejection and removal of such work at the Contractor's expense.

If changes are necessary to submittals, make such revisions and resubmit in accordance with the procedures above. No item of work requiring a submittal change is to be accomplished until the changed submittals are approved.

1.13 APPROVED SUBMITTALS

The Contracting Officer's approval of submittals is not to be construed as a complete check, and indicates only that

Approval or acceptance by the Government for a submittal does not relieve the Contractor of the responsibility for meeting the contract requirements or for any error that may exist, because under the Quality Control (QC) requirements of this contract, the Contractor is responsible for ensuring information contained within each submittal accurately conforms with the requirements of the contract documents.

After submittals have been approved or accepted by the Contracting Officer, no resubmittal for the purpose of substituting materials or equipment will be considered unless accompanied by an explanation of why a substitution is necessary.

1.14 APPROVED SAMPLES

Approval of a sample is only for the characteristics or use named in such approval and is not to be construed to change or modify any contract requirements. Before submitting samples, provide assurance that the materials or equipment will be available in quantities required in the project. No change or substitution will be permitted after a sample has been approved.

Match the approved samples for materials and equipment incorporated in the work. If requested, approved samples, including those that may be damaged in testing, will be returned to the Contractor, at its expense, upon completion of the contract. Unapproved samples will also be returned to the Contractor at its expense, if so requested.

Failure of any materials to pass the specified tests will be sufficient cause for refusal to consider, under this contract, any further samples of

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the same brand or make as that material. The Government reserves the right to disapprove any material or equipment that has previously proved unsatisfactory in service.

Samples of various materials or equipment delivered on the site or in place may be taken by the Contracting Officer for testing. Samples failing to meet contract requirements will automatically void previous approvals. Replace such materials or equipment to meet contract requirements.

1.15 WITHHOLDING OF PAYMENT

Payment for materials incorporated in the work will not be made if required approvals have not been obtained.

1.16 CERTIFICATION OF SUBMITTAL DATA

Certify the submittal data as follows on Form ENG 4025: "I certify that the above submitted items had been reviewed in detail and are correct and in strict conformance with the contract drawings and specifications except as otherwise stated.

____ NAME OF CONTRACTOR _____ SIGNATURE OF CONTRACTOR

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

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SECTION 01 33 29.00 06

SUSTAINABILITY REPORTING

11/17

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING
ENGINEERS (ASHRAE)

ASHRAE 189.1 (2014; ERTA 1 2017) Standard for the
Design of High-Performance Green Buildings
Except Low-Rise Residential Buildings

U.S. DEPARTMENT OF AGRICULTURE (USDA)

FSRIA 9002 Farm Security and Rural Investment Act
Section 9002 (USDA BiopREFERRED Program)

U.S. DEPARTMENT OF ENERGY (DOE)

Energy Star (1992; R 2006) Energy Star Energy
Efficiency Labeling System (FEMP)

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

SNAP (2016) EPA's Significant New Alternatives
Policy Program

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 247 Comprehensive Procurement Guideline for
Products Containing Recovered Materials

1.2 SUMMARY

This specification includes general requirements and procedures for this project to be constructed and documented per the federally mandated High Performance and Sustainable Building Guiding Principles (GP), GBI Guiding Principles Compliance Program for New Construction (DoD version), and other requirements identified in this specification.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following that "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with LRL Section 01 33 00 Submittal Procedures:

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SD-01 Preconstruction Submittals

Sustainability Action Plan

GBI GPCP

Design Phase High Performance and Sustainable Building Checklist

Sustainability Progress Report

SD-05 Design Data

SD-11 Closeout Submittals

Final Sustainability eNotebook; G,

Amended Final Sustainability eNotebook; G,

Final High Performance and Sustainable Building Checklist; G,

1.4 GBI Guiding Principles Compliance Professional (GBI GPCP)

A GBI Green Building Compliance Professional (GBI GPCP), certified by the Green Building Institute, on the Contractor's Quality Control Staff must assemble documentation and ensure that Guiding Principles Compliance documentation is prepared in accordance with this specification section. Identify the GBI GPCP on the Contractor's Quality Control Staff in accordance with specification LRL Section 01 45 04.10 06 CONTRACTOR QUALITY CONTROL. Provide a statement of qualifications of the GBI GPCP submitted to the Government with the QC Plan. Includes name and date of expiration of the current credential.

The GPCP must review all requests for information, Contractor proposals, contract modifications, deviations from the design documents, or any other items that could potentially impact the compliance of the project. Bring any actions that could negatively impact the ability to comply with Guiding Principles to the attention of the Contracting Officer.

1.5 Guiding Principles Compliance

Provide construction related sustainability documentation to verify compliance with Federal High Performance and Sustainable Building requirements. Refer to paragraph Guiding Principle Requirements for requirements and associated documentation to include in the Sustainability eNotebook.

1.5.1 Construction Phase

1.5.1.1 Sustainability Action Plan

Provide a Sustainability Action Plan to demonstrate planned methods to comply with the Guiding Principles requirements identified herein. Provide 3 copies submitted to the Government on compact disk with the QC Plan. The Government will consider an interim Sustainability Action Plan for the first 60 calendar days of the Contractor's operation. Construction will be permitted to begin only after acceptance of the plans or acceptance of an interim plan applicable to the work to be started related to the Guiding Principles requirements identified in the interim plan. Include a detailed description of all activities that relate to accomplishing each project

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Guiding Principle requirement, including construction practices and procurement practices. Include any plans required by the Guiding Principle Requirements including the Indoor Air Quality Plan and Waste Management Plan. Include a schedule for completion and documentation of each requirement. Include the template to be used for Sustainability Progress Reports, a plan for indoctrinating employees to sustainable goals and responsibilities, a team structure identifying responsibility for documentation, and a process for reviewing sustainability related submittals and documentation.

1.5.1.2 Sustainability eNotebook

Provide and maintain a comprehensive Sustainability eNotebook to document compliance with the Guiding Principles requirements identified herein. All materials requirements must be updated each month. The Sustainability eNotebook must contain all required data to support compliance with the Guiding Principles requirements. The Sustainability eNotebook must be in the form of an Adobe PDF file; bookmarked at each requirement, and sub-bookmarked at each document. Include the Final High performance and Sustainable Building Checklist. Include all "S" submittals through the specification sections of this contract.

Provide 3 copies of the Final Sustainability eNotebook and Final High Performance and Sustainable Building Checklist to the Government on DVD 30 calendar days prior to the Contract Required Completion Date. Final progress payment retainage may be held by the Contracting Officer until the documentation is complete. Provide an Amended Final Sustainability eNotebook including any updates to documentation based on post-occupancy corrections.

1.5.1.3 High Performance and Sustainable Building Checklist

Complete the AF Sustainability Requirements Scoresheet shown in Attachment A as required in paragraph Sustainability eNotebook. Obtain the electronic copy of the Army Energy & Sustainability Record Card from the Contracting Officer's Representative. If any deviations from contract result in a change in status to compliance with a listed requirement in the checklist, update the status of that item in the checklist.

1.5.1.4 Progress Report

Submit a monthly Sustainability Progress Report, on the same day each month, that identifies the Guiding Principles requirements completed and documented to date. The Sustainability Progress Report must also include documentation completed to date. The Government will compare the Sustainability Progress Report and the available data in the current Sustainability eNotebook to the schedule in the Sustainability Action Plan. Failure to adhere to the schedule for completion of requirements, including the associated documentation, in the Sustainability Action Plan will be cause for withholding the monthly progress payment until the information is updated and in accordance with the Sustainability Action Plan.

1.6 Guiding Principles Requirements

Incorporate each of the following Guiding Principles requirements into the project construction, and provide documentation that proves compliance with each listed requirement. For each of the following paragraphs that require the use of products listed on Government-required websites, provide documentation of the process used to select products, or process

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used to determine why listed products do not meet project performance requirements.

1.6.1 Commissioning

Comply with the requirements of LRL section 01 46 00.00 06 TOTAL BUILDING COMMISSIONING.

1.6.2 Energy Efficient Products

Provide only energy-using products that are Energy Star rated, or have the Federal Energy Management Program (FEMP) recommended efficiency for products with established Energy Star or FEMP requirements. Provide only energy using products that meet FEMP requirements for low standby power consumption. Energy efficient products can be found at: <https://energy.gov/eere/femp/federal-energy-management-program> and <https://www.energystar.gov/>. Provide the following documentation:

Proof that products are labeled energy efficient and comply with the cited requirements.

1.6.3 Indoor Water Use

Provide only water-consuming products that are EPA WaterSense labeled, or water fixtures available that meet the requirements of ASHRAE 189.1 Section 6.3.2, whichever is most water-efficient. Provide the following documentation:

For products available with EPA WaterSense labeling, proof that fixtures are labeled EPA WaterSense or Energy Star; for all other fixtures, proof they comply with the cited efficiency requirements.

1.6.4 Reduce Volatile Organic Compound (VOC) (Low Emitting Materials)

Meet the requirements of Table 3-1 at the end of this specification. Provide the following documentation:

Provide certifications or labels that demonstrate compliance with cited requirements.

1.6.5 Indoor Air Quality During Construction

Prior to construction, create indoor air quality plan. Implement IAQ plan during construction and flush building air before occupancy.

For new construction and for renovation of unoccupied existing buildings, indoor air quality plan must meet the requirements of ASHRAE 189.1 Section 10.3.1.4. (Indoor Air Quality (IAQ) Construction Management), with maximum outdoor air consistent with achieving relative humidity no greater than 60 percent.

Provide documentation showing that after construction ends and prior to occupancy, HVAC filters were replaced and building air was flushed out in accordance with the cited standard.

1.6.6 Recycled Content

Comply with 40 CFR 247. Refer to <https://www.epa.gov/smm/comprehensive-procurement-guidelines-construction-products>

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for assistance identifying products cited in 40 CFR 247. Selected products must comply with non-proprietary requirements of the Federal Acquisition Regulation, and must meet performance requirements. Provide the following documentation:

- a. Manufacturers' documents stating the recycled content by material, or written justification for claiming one of the exceptions allowed on the cited website.
- b. Substitutions: Submit for Government approval, proposed alternative products or systems that provide equivalent performance and appearance and have greater contribution to project recycled content requirements. For all such proposed substitutions, submit with the Sustainability Action Plan accompanied by product data demonstrating equivalence.

1.6.7 Bio-Based Products

Provide products and material composed of the highest percentage of biobased materials (including rapidly renewable resources and certified sustainably harvested products), consistent with FSRIA 9002 USDA Biopreferred Program, to the maximum extent possible without jeopardizing the intended end use or detracting from the overall quality delivered to the end user. Use only supplies and materials of a type and quality that conform to applicable specifications and standards.

Comply with FSRIA 9002 USDA BioPreferred Program. Refer to <https://www.biopreferred.gov/BioPreferred/> for the product categories and BioPreferred Catalog. Selected products must comply with non-proprietary requirements of the Federal Acquisition Regulation, and must meet performance requirements. Provide the following documentation:

USDA Biopreferred label for each product; for bio-based products used on project but not listed with Biopreferred program, provide bio-based content and percentage.

1.6.8 Ozone Depleting Substances

Meet the requirements of ASHRAE 189.1 Section 9.3.3 Refrigerants for no CFC-based refrigerants in heating ventilation, air conditioning and refrigeration systems. Where feasible, use products from U.S. EPA Significant New Alternatives Policy (SNAP) (<https://www.epa.gov/snap>) or meet the criteria of SNAP. Provide the following documentation:

- a. MSDS sheets for all refrigerants.
- b. Provide label for each product meeting the cited standards.

1.6.9 Waste Material Management (Recycling - Construction)

Divert construction debris from landfill disposal where markets or on-site recycling exists, and provide documentation.

A minimum of 60% of nonhazardous construction and demolition waste material generated prior to the issuance of the final certificate of occupancy shall be diverted from disposal in landfills and incinerators by recycling or reuse. Reuse includes donation of materials to charitable organization, salvage of existing materials onsite, and packaging materials returned to the manufacturer, shipper, or other source that will reuse the packaging in

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future shipments. Excavated soil and land clearing debris shall not be included in the calculation. Calculations are allowed to be done by either weight or volume, but shall be consistent throughout. Specific areas on the construction site shall be designated for collection of recyclable and reusable materials. Off-site storage and sorting of materials shall be allowed. Diversion efforts shall be tracked by the contractor throughout the construction process and documentation provided to the Government prior to contract closeout.

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION - NOT USED

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3.1 SUSTAINABILITY IMPLEMENTATION MEETING

The GBI GPCP, QC System Manager, and Contracting Officer's Representative must meet within 60 calendar days after notice to proceed to review project sustainability goals with respect to the Sustainability Action Plan, identify potential difficulties related to meeting the Guiding Principle Requirements, and discuss mitigation strategies.

3.2 Sustainability Pre-Closeout Meeting

The GBI GPCP, QC System Manager, and Contracting Officer's Representative must meet 60 calendars days prior to Contract Required Completion Date to review completion status of sustainability requirements including level of completion of Guiding Principles Requirements documentation and the High Performance and Sustainable Building Checklist. Discuss any potential difficulties in completing outstanding requirements and documentation. Provide a copy of the Sustainability eNotebook with up-to-date documentation.

3.3 TABLE 3-1 VOLATILE ORGANIC COMPOUNDS (VOC) (LOW EMITTING MATERIALS) REQUIREMENTS

Refer to following table, based on ASHRAE 189.1 section 8.4.2 (Materials), for compliance criteria.

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TABLE 3-1 Volatile Organic Compounds (VOC) (Low Emitting Materials) Requirements				
MATERIAL CATEGORY	EMISSIONS REQUIREMENT		MATERIALS WITH ADDED VOC REQUIREMENT	MATERIAL CATEGORY
Adhesives and Sealants	CDPH/EHLB/Standard method V1.1 (California Section 01350) (Use "office" or "classroom" space limits for all applications)	or	Adhesives (carpet, resilient, wood flooring; panel; primers) Sealants (acoustical; firestop; HVAC Air duct; primers) Caulks	SCAQMD Rule 1168 (Use "other" category for HVAC duct sealant) (for firestop adhesive, UFC 3-600-01 overrides conflicting requirements)
			Aerosol adhesives	Section 3 of Green Seal Standard GS-36 (except: cleaners, solvent cements, and primers used with plastic piping and conduit in plumbing, fire suppression, and electrical systems; HVAC air duct sealants when the application space air temp is less than 40 F (4.5 C) .
Paints and Coatings	CDPH/EHLB/Standard method V1.1 (California Section 01350) (Use "office" or "classroom" space limits for all applications)	or	Flat and nonflat topcoats, primers, undercoaters, and anti-corrosive coatings	Green Seal Standard GS-11

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TABLE 3-1 Volatile Organic Compounds (VOC) (Low Emitting Materials) Requirements				
MATERIAL CATEGORY	EMISSIONS REQUIREMENT		MATERIALS WITH ADDED VOC REQUIREMENT	MATERIAL CATEGORY
Paints and Coatings	CDPH/EHLB/Standard method V1.1 (California Section 01350) (Use "office" or "classroom" space limits for all applications)	or	Concrete/masonry sealers (waterproofing concrete/masonry sealers), concrete curing compounds, dry fog coatings, faux finishing coatings, fire resistive coatings, floor coatings, graphic arts (sign) coatings, industrial maintenance coatings, mastic texture coatings, metallic pigmented coatings, multicolor coatings, pretreatment wash primers, reactive penetrating sealers, recycled coatings, shellacs (clear and opaque), specialty primers, stains, wood coatings (clear wood finishes), wood preservatives, and zinc primers	California Air Resources Board (CARB) Suggested Control Measure for Architectural Coatings or SCAQMD Rule 1113

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TABLE 3-1 Volatile Organic Compounds (VOC) (Low Emitting Materials) Requirements				
MATERIAL CATEGORY	EMISSIONS REQUIREMENT		MATERIALS WITH ADDED VOC REQUIREMENT	MATERIAL CATEGORY
Paints and Coatings	CDPH/EHLB/Standard method V1.1 (California Section 01350) (Use "office" or "classroom" space limits for all applications)	or	Basement specialty coatings, high-temperature coatings, low solids coatings, stone consolidants, swimming-pool coatings, tub-and tile-refining coatings, and waterproofing membranes	California Air Resources Board (CARB) Suggested Control Measure for Architectural Coatings
Floor Covering Materials	For carpet, all locations: CDPH/EHLB/Standard Method V1.1 (California Section 01350) or label for Section 9 of CDPH/EHLB/Standard Method V1.1 (California Section 01350)		none	none

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TABLE 3-1 Volatile Organic Compounds (VOC) (Low Emitting Materials) Requirements

MATERIAL CATEGORY	EMISSIONS REQUIREMENT		MATERIALS WITH ADDED VOC REQUIREMENT	MATERIAL CATEGORY
Composite Wood, Wood Structural Panel, and Agrifiber Products particleboard medium density fiberboard (MDF) wheatboard strawboard panel substrates door cores no added urea-formaldehyde resins including laminating adhesives for composite wood and agrifiber assemblies	Third-party certification (approved by CARB) of California Air Resource Board's (CARB) regulation Airborne Toxic Control Measure to Reduce Formaldehyde Emissions from Composite Wood Products CDPH/EHLB/Standard method V1.1 (California Section 01350) (Use "office" or "classroom" space limits for all applications) (except: Structural panel components such as plywood, particle board, wafer board, and oriented strand board identified as "EXPOSURE 1," "EXTERIOR," or "HUD-APPROVED" are considered acceptable for interior use.)		none	none

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TABLE 3-1 Volatile Organic Compounds (VOC) (Low Emitting Materials) Requirements

MATERIAL CATEGORY	EMISSIONS REQUIREMENT		MATERIALS WITH ADDED VOC REQUIREMENT	MATERIAL CATEGORY
Office Furniture Systems and Seating installed prior to occupancy	ANSI/BIFMA X7.1 ANSI/BIFMA X7.1: (95 percent of installed office furniture system workstations and seating units) Section 7.6.2 of ANSI/BIFMA e3 (50 percent of office furniture system workstations and		none	none
Ceiling and Wall Systems ceiling and wall insulation acoustical ceiling panels tackable wall panels gypsum wall board and panels wall coverings	CDPH/EHLB/Standard method V1.1 (California Section 01350) (Use "office" or "classroom" space limits for all applications)		none	none

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Attachment A
HIGH PERFORMANCE AND SUSTAINABLE BUILDING CHECKLIST

-- End of Section --

Air Force Sustainability Requirements Scoresheet

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HPSB COMPLIANCE (Updated Jan 2017)

* required entry

General Information



Federal
Requirements
Complete

ZHTV231013	Project ID (e.g. ABCD12345)
000000000000000000	Real Property Unique ID (RPUID)
20824	Facility Number
711th HPW Laboratory	Building Name
Wright-Patterson AFB	Installation
Dayton	City
OH	State
Yes	CONUS
AFMC	MAJCOM
USACE	Construction Agent
Hendricks, Randall	AFCEC DM/CM (Last Name, First Name)
\$7,500,000.00	PA
48,000	Building Size (SF)
2026	Program Year (FY####)
Design Complete	Project Phase
09/19/23	Design Started (MM/DD/YY)
12/01/27	BOD (MM/DD/YY)
GBI GP	Guiding Principles Compliance Certification Method
	Date Project Registered (MM/DD/YY)
	Date Project Certified (MM/DD/YY)
100%	HPSB Compliant
0%	Energy Efficiency Achieved (% below ANSI/ASHRAE/IESNA Standard 90.1-2013)
2017V1	Scoresheet version

Color Coding: See Instructions Tab for more detail

Drop-Down Box	Yes or N/A
No Entry Required	No
Custom Entry	Recommended not Required

90.1-2013

HPSB I: Employ Integrated Design Principles (UFC 1-200-02 para 2-2)

Total Points	2	Possible Points	2
Yes	HPSB I.1	Integrated Design	1
Yes	HPSB I.2	Commissioning	1

HPSB II: Optimize Energy Performance (UFC 1-200-02 para 2-3)

Total Points	5	Possible Points	5
Yes	HPSB II.1	Energy Efficiency	1

Yes	Reduce energy use 30% below ANSI/ASHRAE/IESNA Standard 90.1-2013 or IECC, or if not - achieve maximum energy efficiency that is lifecycle cost effective
0.0%	Insert percentage below ANSI/ASHRAE/IESNA Standard 90.1-2013 or IECC, in terms of energy use (e.g. 32)
218.6	Insert building energy intensity (kBtu/yr-sqft) calculated IAW 10 CFR 433
Yes	Roof Attributes (Recommended)
0	Select roof types (Check below)

- ☐ Cool roof
 ☐ Solar electric
 ☐ Solar Passive
 ☐ Green roof
 ☐ Solar thermal

Energy Efficient Products

Yes

Yes HPSB II.2

On-site Renewable Energy

Yes	Installed renewable energy elements or projects were not lifecycle cost effective
1	Renewable energy types (check below)

- ☐ Solar PV
 ☐ Geothermal
 ☐ Hydro
 ☐ Waste to Energy
 ☐ Solar CP
 ☐ GSHP
 ☐ Wind
 ☒ Renewables were not lifecycle cost effective

0.0	Insert generation capacity (kW)
0.0%	Insert percentage of total building

N/A HPSB II.3

On-site Renewable Energy - Solar Hot Water Heater System

N/A	Installed solar hot water heater system or found installation not lifecycle cost effective
0.0	Insert generation capacity (MMBtu/yr)
0.0%	Insert percentage of demand

Yes HPSB II.4

Metering

Yes	Electric Metering: Select N/A if no service
Yes	Natural Gas Metering: Select N/A if no service
Yes	Steam Metering: Select N/A if no service

HPSB III: Protect and Conserve Water (UFC 1-200-02 para 2-4)

Total Points	6	Possible Points	6
N/A	HPSB III.1	Indoor Water	1
N/A		Indoor Water Metering	1

Air Force Sustainability Requirements Scoresheet

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HPSB COMPLIANCE (Updated Jan 2017)

* required entry

N/A	HPSB III.2	Outdoor Water	1
N/A		Outdoor Water Metering	1
N/A	HPSB III.3	Alternative Water	1
N/A	HPSB III.4	Stormwater Management (LID Documentation per UFC 3-210-10)	1
		0.0	Change in Impervious Area (SF)
		\$0.00	Pre-Award Cost Estimate (\$)
		N/A	Project addressed EISA 438
			EISA Technical Constraints
		☐ Retaining stormwater impact receiving water flow ☐ Shallow bedrock, contaminated soil, high ground water table, underground utilities ☐ Soil infiltration capacity limited Structural, plumbing, and other mods not feasible	
		☐ Site too small to infiltrate significant volume ☐ Non-potable water demand to small ☐ Other	
		☐ State or local restrict water ☐ State or local restrict use of green	
		0.0%	Percent Increase in Stormwater Runoff for 95 Percentile Storm (%) - or- Percent Increase in Stormwater Runoff from continuous simulation model, published data, studies, or other established tools (Reference UFC 3-210-10 Figure 2-1 Implementation of EISA Section 438)
		On-Site	LID Features Locations
		0	Integrated Management Practices Employed
		☐ Bio-Retention ☐ Dry Wells ☐ Filter Strips ☐ Grassed Swells	
		☐ Infiltration Trench ☐ Inlet Pollution Removal Device ☐ Permeable Pavement/Pavers ☐ Rain Barrels/Cistern	
		☐ Soil Amendments ☐ Tree Box Filters ☐ Vegetated Buffers ☐ Vegetat	
		☐ Other	
		Final LID Construction Cost (\$)	
		Post Construction Analysis (Name of DOR)	
HPSB IV: Enhance Indoor Environmental Quality (UFC 1-200-02 para 2-5)			
Total Points		8	Possible Points 8
Yes	HPSB IV.1	Thermal Comfort	1
Yes	HPSB IV.2	Ventilation	1
N/A	HPSB IV.3	Daylighting	1
Yes	HPSB IV.4	Moisture Control	1
Yes	HPSB IV.5	Low Emitting Materials	1
Yes	HPSB IV.6	Protect Indoor Air Quality during Construction	1
Yes	HPSB IV.7	Environmental Tobacco Smoke Control	1
N/A	HPSB IV.8	Occupant Health and Wellness	1
HPSB V: Reduce Environmental Impact of Materials (UFC 1-200-02 para 2-6)			
Total Points		5	Possible Points 5
Yes	HPSB V.1	Recycled Content	1
N/A	HPSB V.2	Biologically-based Products	1
Yes	HPSB V.3	Ozone Depleting Substances	1
Yes	HPSB V.4	Waste and Materials Management - Recycling	1
Yes	HPSB V.5	Waste and Materials Management - Divert 60% from Disposal	1
		Yes	60% or greater diverted
		60.0%	Insert percentage diverted from landfill
HPSB VI: Address Climate Change Risk (UFC 1-200-02 para 2-7)			
Total Points		1	Possible Points 1
Yes	HPSB VI.1	Address Climate Change Risk	1
		Possible Points 27	
27	Federal Requirements - Yes or N/A		
0	Federal Requirements - No		
100%	Percentage of Federal Requirements Met		

Federal Requirements for High Performance and Sustainable Buildings (HPSB) & UFC 1-200-02

Instructions: Provide a common or project specific justification for an element to be non-applicable, when completed, the Scoresheet tab will allow an N/A response.

Justification for Non-Applicable Answers		Common Justification	Project Specific Justification	Complete?
HPSB I: Employ Integrated Design Principles (UFC 1-200-02, 2-2)				
HPSB I.1	Integrated Design			Applicable
HPSB I.2	Commissioning			Applicable
HPSB II: Optimize Energy Performance (UFC 1-200-02, 2-3)				
HPSB II.1	Energy Efficiency			Applicable
HPSB II.1	Energy Efficient Products			Applicable
HPSB II.2	On-site Renewable Energy			Applicable
HPSB II.3	On-site Renewable Energy - Solar Hot Water Heater System	Domestic water heating not in scope		Yes
HPSB II.4	Metering			
HPSB II.4	Electric Metering: Select N/A if no service			Applicable
HPSB II.4	Natural Gas Metering: Select N/A if no service			Applicable
HPSB II.4	Steam Metering: Select N/A if no service			Applicable
HPSB III: Protect and Conserve Water (UFC 1-200-02, 2-4)				
HPSB III.1	Indoor Water	Water systems are not a part of the scope.		Yes
HPSB III.1	Indoor Water Metering: Select N/A if no service	Water systems are not a part of the scope.		Yes
HPSB III.2	Outdoor Water	Water systems are not a part of the scope.		Yes
HPSB III.2	Outdoor Water Metering: Select N/A if no service	Water systems are not a part of the scope.		Yes
HPSB III.3	Alternative Water	Water systems are not a part of the scope.		Yes
HPSB III.4	Stormwater Management			
HPSB III.4	Project addressed EISA 438	Not applicable - no site work		Yes
HPSB IV: Enhance Indoor Environmental Quality (UFC 1-200-02, 2-5)				
HPSB IV.1	Thermal Comfort			Applicable
HPSB IV.2	Ventilation			Applicable
HPSB IV.3	Daylighting	Adjustment of the fenestration is out of scope		Yes
HPSB IV.4	Moisture Control			Applicable
HPSB IV.5	Low Emitting Materials			Applicable
HPSB IV.6	Protect Indoor Air Quality during Construction			Applicable
HPSB IV.8	Occupant Health and Wellness	No applicable to the scope		Yes
HPSB V: Reduce Environmental Impact of Materials (UFC 1-200-02, 2-6)				
HPSB V.1	Recycled Content			Applicable
HPSB V.2	Biologically based products	The project scope does not offer the use of biologically based products		Yes
HPSB V.3	Ozone Depleting Substances			Applicable
HPSB V.4	Waste and Materials Management - Recycling			Applicable
HPSB V.5	Waste and Materials Management - Divert 60% from Disposal			Applicable

HPSB VI: Address Climate Change Riskd (UFC 1-200-02, 2-7)			
HPSB VI.1	Address Climate Change Risk		Applicable

High Performance Sustainable Building Requirements

References

Executive Order (EO) 13693, Planning for Federal Sustainability in the Next Decade

HPSB I: Employ Integrated Design Principles	
Integrated Design HPSB I.1 UFC 1-200-02 para 2-2.1	Incorporate the following planning and evaluation into the integrated design, as described in ASHRAE 189.1 Informative Appendix F (Integrated Design). Follow the steps of design optimization, as applicable, in ASHRAE 189.1 Section F1.1.1 (Charrette Process). Use a collaborative, integrated planning and design team, composed of user, government support staff, and appropriate professionals, to identify requirements and to establish performance goals for siting, energy, water, materials, indoor environmental quality, and other comprehensive design goals. Ensure incorporation of these goals throughout the design and lifecycle of the building, including deconstruction. Evaluate the site and building components to determine whether passive and natural design strategies and features are cost effectively incorporated before the active and mechanical systems are designed. Incorporate these features where applicable. Take into account site attributes, including climate and local and regional context, which impact the design of the building. During the site selection process, meet the requirements of UFC 2-100-01. See Appendix B “Best Practices” for desirable site characteristics. During the planning and design process meet the requirements of applicable UFCs, and use the following site development considerations and passive strategies: • Site design elements that ensure safe and convenient pedestrian access. • Meet the requirements of UFC 3-201-02. • Incorporate results of site analysis, in order to design the building, focusing on orientation, configuration and massing. • Orient building to maximize energy efficiency, passive solar and daylighting potential. • Select, design and integrate into the overall building, high performance and sustainable systems (e.g., HVAC, plumbing, water heating systems, lighting systems, control systems, elevators, building envelope and fire protection systems). • Promote opportunities for occupants to voluntarily increase physical movement such as making stairwells a desirable option for circulation and active workstations.
Commissioning HPSB I.2 UFC 1-200-02 para 2-2.2	In order to verify design and performance, and ensure that the Government requirements are met, employ commissioning practices appropriate to the size and complexity of the building and its system components. This must include an experienced commissioning provider, who should be independent of the project design and construction team, and the operations team. The choice of either contracted services or Government personnel to serve as the commissioning provider will be determined at project level. Meet the requirements of ASHRAE 189.1 Section 10.3.1.2 (Building Project Commissioning), with the following modifications: • For buildings and systems that are less complex, commissioning may be tailored as determined by the DOD Component AHJ. • “Schematic design” is the design charrette or similar conceptual design activity. • Documentation as described in ASHRAE 55 Section 6.2 is not required. For Air Force projects, the Project Delivery Team must determine the level of commissioning activities required.
Requirement Source Document(s)	Executive Order 13693 UFGS 01 91 00.15, Total Building Commissioning
HPSB II: Optimize Energy Performance	

High Performance Sustainable Building Requirements**Energy Efficiency****HPSB II.1****UFC 1-200-02 para 2-3.1****Energy Efficiency -**

Base energy efficiency design decisions on life-cycle cost as indicated in Chapter 1 of UFC 1-200-02.

Commercial and Multi-Family High-Rise Residential Buildings:

- Meet the requirements of ASHRAE 90.1. Use ASHRAE 90.1(2010) for all projects with design starts before November 6, 2016. Use ASHRAE 90.1(2013) thereafter.
- Design the building to achieve at least 30% energy consumption reduction from ASHRAE 90.1 baseline.
- If a 30% reduction is not LCCE, modify the design of the proposed building to achieve an energy consumption level at the highest level of energy efficiency that is LCCE.
- Determine energy consumption levels for both the ASHRAE Baseline Building and proposed building by using the Performance Rating Method found in appendix G of ASHRAE 90.1, except the formula for calculating the Performance Rating.

Replace the formula in G1.2 with the following:

Percentage improvement = $100 \times ((\text{Baseline building consumption} - \text{Receptacle and process loads}) - (\text{Proposed building consumption} - \text{Receptacle and process loads})) / (\text{Baseline building consumption} - \text{Receptacle and process loads})$

Low-Rise Residential Buildings:

- Meet the requirements of International Energy Conservation Code (IECC).
- Design the building to achieve at least 30% energy consumption reduction from the IECC baseline using the Simulated Performance Alternative found in Section 405 of the IECC.
- If a 30% reduction is not LCCE, modify the design of the proposed building to achieve an energy consumption level at the highest level of energy efficiency that is LCCE.

Renovations:

- Renovation projects that replace everything above the foundation must either apply 2-3.1.1 or 2-3.1.2 as applicable.
- All other renovations choose one of the following options:
 1. Reduce measured building energy use by at least 30%, below FY 2003 energy use baseline.
 2. Reduce measured building energy use by at least 20% below FY 2015 energy use baseline.
 3. Reduce modeled energy use (from all sources including renewable energy) by 20% compared to the ASRHAE 90.1 baseline building design.

• If none of the reduction choices is life-cycle cost-effective, modify the design of the proposed building system(s) to achieve an energy consumption level at the highest level of energy efficiency that is life-cycle cost-effective.

Energy Efficient Products**HPSB II.1****UFC 1-200-02 para 2-3.1.4****Energy Efficient Products –**

Per EISA 2007 Section 525, acquire products that are ENERGY STAR®-qualified or meet FEMP-designated efficiency requirements in all covered product categories. Select products based on life cycle cost, not initial cost. Link to EPA sites: <http://www.energystar.gov/> or <http://www1.eere.energy.gov/femp/>

Per EISA 2007 Section 524, provide commercially available, off-the-shelf products that use no more than 1 watt in their standby mode.

Requirement Source Document(s) [10 CFR 433, EPAAct 05](#)

Links [Energy Star®](#)
[Federal Energy Management Program](#)

On-site Renewable Energy**HPSB II.2****UFC 3-440-01 para 2-3.2**

Provide on-site renewable energy systems in accordance with ASHRAE 189.1 Section 7.4.1.1 (On-Site Renewable Energy Systems) and UFC 3-440-01 where LCCE, considering climate, infrastructure condition, mission compatibility, and effects on base wide electrical system (grid) power quality. Exception: Do not use purchase of renewable energy certificates (RECs) as a substitute for the Section 7.4.1.1 new building requirement.

Air Force will utilize an installation-level solution to renewable energy systems and will not require ASHRAE 189.1 Section 7.3.2 if a building-level solution is not LCCE.

High Performance Sustainable Building Requirements	
Requirement Source Document(s):	Executive Order 13693
On-Site Renewable Energy – Solar Hot Water Heater System HPSB II.3 UFC 1-200-02 para 2-3.2.1	Per EISA 2007 Section 523, meet at least 30% of the annual domestic hot water requirement through the installation of solar water heating unless SDHW is not LCCE. If 30% is not LCCE, modify the design of the proposed system to achieve the highest level of solar water heating that is LCCE.
Requirement Source Document(s):	
Measurement and Verification HPSB II.4 UFC 1-200-02 para 2-3.4	A utility meter must be installed at each building, for each utility serving the building (e.g., district steam, district hot and chilled water, electricity, natural gas, fuel oil, etc.) in the standard units of the measure. Meters must be connected to a base wide energy and utility monitoring and control system using the installation's advanced metering protocols. The installation of meters is required per DODI 4170.11, and as amended by DOD Utilities Meter Policy, 16 April 2013.
Requirement Source Document(s):	Federal Leadership in High Performance and Sustainable Buildings MOU
HPSB III: Protect and Conserve Water	
Indoor Water HPSB III.1 UFC 1-200-02 para 2-4.1	Indoor Water – Base water efficiency design decisions on life-cycle cost as indicated in Chapter 1 of UFC 1-200-02. • Meet the requirements of ASHRAE 189.1 Section 6.3.2 (Building Water Use Reduction), which incorporates USEPA WaterSense-labeled products. Water closet replacements in renovations may have a flush value of up to 1.6 GPF (6.1 LPF) to accommodate existing plumbing capacity. • Meet the requirements of ASHRAE 189.1 Section 6.4.2 (Building Water Use Reduction). • Meet the requirements of ASHRAE 189.1 Section 6.4.3 (Special Water Features).
Indoor Water Metering HPSB III.1 UFC 1-200-02 para 2-4.1.1	Indoor Water Metering -- Install advanced water meters to monitor building indoor potable water consumption, as required by DOD Utilities Meter Policy, 16 April 2013.
Requirement Source Document(s):	EPA Act 1992
Outdoor Water HPSB III.2 UFC 1-200-02 para 2-4.2	Outdoor Water -- • Limit potable irrigation water use by 50% compared to the conventional methods, and when LCCE, by using the methodologies in ASHRAE 189.1: Section 6.3.1 (Mandatory Provisions: Site Water Use Reduction) and either Section 6.4.1 (Prescriptive Option: Site Water Use Reduction) or Section 6.5.1 (Performance Option: Site Water Use Reduction). • For existing systems, if a building has a single water meter, reduce indoor and outdoor potable water use combined by at least 20% compared to building water use in 2007. Compare results to a baseline building, using the EPA WaterSense landscape water budget tool version 1.01 or later, or a Component approved tool. • Show preference for irrigation contractors who are certified through a WaterSense labeled program, or other industry-recognized credentialing programs. • Refer to UFC 3-201-02 for additional requirements.
Outdoor Water Metering HPSB III.2 UFC 1-200-02 para 2-4.2.1	Water metering for landscaping – When potable water is used, provide water meters for irrigation systems serving more than 25,000 square feet of landscape, when life-cycle cost-effective. Install advanced water meters to monitor outdoor potable water consumption, as required by DOD Utilities Meter Policy, 16 April 2013. For all other irrigation systems, separate water meters for locations with outdoor water use are encouraged.
Requirement Source Document(s):	Executive Order 13693

High Performance Sustainable Building Requirements	
Alternative Water HPSB III.3 UFC 1-200-02 para 2-4.3	Where life-cycle cost-effective and permitted by local laws and regulations, use alternative water sources, such as harvested rainwater, treated wastewater, air handler condensate capture, grey water, and reclaimed water.
Requirement Source Document(s):	
Stormwater Management HPSB III.4 UFC 3-210-10 para 2-4.3.1	The criteria and design standards in UFC 3-210-10 are required for the planning, design and construction of all Department of Defense (DoD) projects in the United States, United States Territories and Possessions of the United States that meet both of the following conditions: 1) The project includes construction or expansion of one or more buildings as part of its primary scope (i.e., primary facilities vice supporting facilities). 2) The "footprint" is greater than 5,000 gross square feet (464.5 square meters). "Footprint" consists of all new impervious surfaces associated with the building(s), including both building area and pavement area of associated supporting facilities (such as parking and sidewalks). "Footprint" does not include existing building area to be renovated, existing pavement area to be resurfaced, or new pavement area other than supporting facilities associated with the building(s). For projects in the United States, United States Territories, and Possessions of the United States that do not meet the applicability requirements above, LID techniques apply to the extent practical.
Requirement Source Document(s):	EISA 2007 Sec 438
HPSB IV: Enhance Indoor Environmental Quality	
Thermal Comfort HPSB IV.1 UFC 1-200-02 para 2-5.1	Meet the requirements of ASHRAE 55. Compliance with standards by providing passive (non-mechanical) thermal comfort methods are allowed and encouraged as described in paragraph entitled, "Integrated Design" in UFC 1-200-02. Exception: For Medical Treatment Facilities, refer to UFC 4-510-01 Medical Military Facilities for thermal comfort criteria.
Requirement Source Document(s):	Executive Order 13693
Ventilation HPSB IV.2 UFC 1-200-02 para 2-5.1	Meet the requirements of ASHRAE 62.1 (62.2 for low-rise residential). Exception: For Medical Treatment Facilities, refer to UFC 4-510-01 Medical Military Facilities for ventilation criteria.
Requirement Source Document(s):	Executive Order 13693
Daylighting HPSB IV.3 UFC 1-200-02 para 2-5.2	All regularly occupied spaces located on the exterior wall must have vision fenestration. Where mission precludes vision fenestration for regularly occupied spaces located on the exterior wall, daylighting is still required. All classrooms and offices must have daylighting. Meet the requirements of ASHRAE 189.1 Section 8.4.1.2 (Minimum Sidelighting effective Aperture for Office Spaces and Classrooms) or Section 8.5.1.2 (Usable Daylight Illuminance in Office Spaces and Classrooms). Provide automated lighting controls in accordance with UFC 3-530-01. Exception: For Medical Treatment Facilities, refer to UFC 4-510-01 Medical Military Facilities for additional daylighting criteria.
Requirement Source Document(s):	Executive Order 13693
Moisture Control HPSB IV.4 UFC 1-200-02 para 2-5.3.1	Establish and implement a moisture control strategy for controlling moisture flows and condensation to prevent building damage, minimize mold contamination, and reduce health risks related to moisture. Meet the requirements of ASHRAE 189.1 Section 10.3.1.5 (Moisture Control), UFC 3-410-01, Chapter 3, Sections 3-2 and 3-3 (Ventilation Air), and UFC 3-101-01 Chapter 3 (Building Envelope Requirements). Refer to Appendix B "Protect Indoor Air Quality" for best practices.

High Performance Sustainable Building Requirements	
Requirement Source Document(s):	Executive Order 13693
Low Emitting Materials HPSB IV.5 UFC 1-200-02 para 2-5.3.2	Specify materials and products with low or no pollutant emissions, including composite wood products, adhesives, sealants, interior paints and finishes, carpet systems, and furnishings. Meet the requirements of ASHRAE 189.1 Section 8.4.2 (Prescriptive Option: Materials). Exception: Exclude compliance with 8.4.2, first sentence.
Requirement Source Document(s):	Executive Order 13693
Protect Indoor Air Quality during Construction HPSB IV.6 UFC 1-200-02 para 2-5.3.3	For new construction and for renovation of unoccupied existing buildings, comply with ASHRAE 189.1 Section 10.3.1.4 (Indoor Air Quality (IAQ) Construction Management), with maximum outdoor air consistent with achieving relative humidity no greater than 60%. For renovation of occupied existing buildings, comply with ANSI/SMACNA 008-2008, 2nd Edition, SMACNA IAQ Guidelines for Occupied Buildings Under Construction.
Requirement Source Document(s):	Executive Order 13693
Environmental Tobacco Smoke Control HPSB IV.7 UFC 1-200-02 para 2-5.3.4	Prohibit smoking within the building and within a minimum of 50 feet (15.24 meters) of all building entrances, operable windows, and building ventilation intakes. Verify if more stringent facility criteria or Installation policy applies.
Requirement Source Document(s):	Federal Register: December 22, 2008 (Volume 73, Number 246) - EO 13058
Occupant Health and Wellness HPSB IV.8 UFC 1-200-02 para 2-5.4	Promote opportunities for occupants to voluntarily increase physical movement such as making stairwells a desirable option for circulation, active workstations, fitness centers, and bicycle commuter facilities. Support occupant health by considering options such as providing convenient access to healthy dining options, potable water, daylight, plants, and exterior views. Indicate in the Integrated Design Process how these efforts were evaluated.
Requirement Source Document(s):	
HPSB V: Reduce Environmental Impact of Materials	
Recycled Content HPSB V.1 UFC 1-200-02 para 2-6.1.1	Use RCRA Section 6002 compliant products that meet or exceed EPA's recycled content recommendations, available on EPA's Comprehensive Procurement Guideline web site at http://www.epa.gov/
Requirement Source Document(s):	RCRA 2002, Sec 6002
Biologically-based Products HPSB V.2 UFC 1-200-02 para 2-6.1.2	Per Section 9002 of the Farm Security and Rural Investment Act, specify products composed of the highest percentage of biobased content consistent with the USDA BioPreferred Program, if products meet performance requirements and are available at a reasonable cost. Exceptions taken to biobased product procurement must be documented. A preference for purchasing products with the highest biobased content per USDA recommendations for designated product categories must be included in all applicable solicitations. USDA's biobased product designations and biobased content (which includes certified sustainably-harvested and rapidly renewable resources) recommendations are available on USDA's BioPreferred web site at http://www.biopreferred.gov/
Requirement Source Document(s):	FSRIA 2002 Section 9002

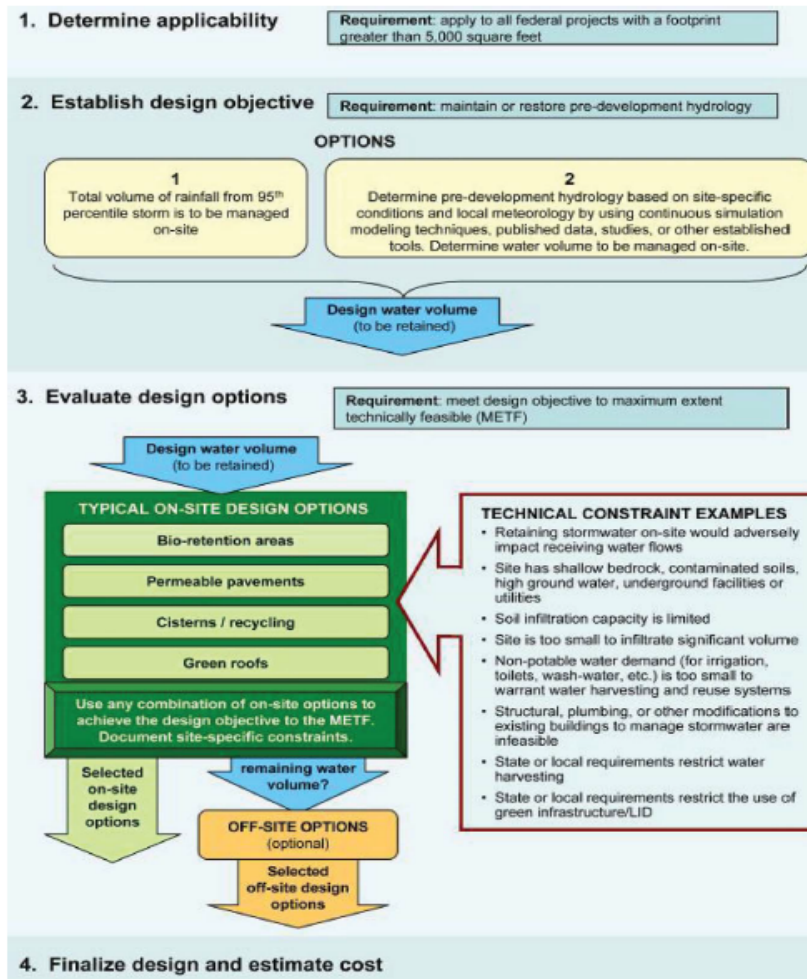
High Performance Sustainable Building Requirements	
Environmentally Preferable Products HPSB V.3 UFC 1-200-02 para 2-6.1.4	Meet the requirements of ASHRAE 189.1 Section 9.3.3 (Refrigerants), if equipment or systems using ozone depleting substances are included in the project. Do not use ozone depleting substances (ODS) or high Global Warming Potential (GWP) chemicals where EPAs Significant New Alternative Policy (SNAP) has identified acceptable substitutes or where other environmentally preferable products are available for use in construction, repair or end-of-life replacements: www.epa.gov/snap Exceptions: Refer to UFC 3-600-01 for fire protection system requirements.
Requirement Source Document(s):	Executive Order 13693
Waste and Materials Management - Recycling HPSB V.4 UFC 1-200-02 para 2-6.2.1	Meet the requirements of ASHRAE 189.1 Section 9.3.4.1 (Storage and Collection of Recyclables – Recyclables), where markets or onsite recycling exist.
Requirement Source Document(s):	Executive Order 13693
Waste and Materials Management, Divert 60% from Disposal HPSB V.5 UFC 1-200-02 para 2-6.2.2	Divert minimum 60% of nonhazardous construction and demolition waste material from landfills.
Requirement Source Document(s):	Executive Order 13693
	UFGS 01 74 19, Construction and Demolition Waste Management
	HPSB VI: Address Climate Change Risk
Address Climate Change Risk HPSB VI.1 UFC 1-200-02 para 2-7	Provide building design solutions responsive to any Government-provided projections of climate change projection and determination of acceptable risk. For a building located in a floodplain of concern, provide design solutions which mitigate both impact on the floodplain, and impact of the design flood event on building function and occupants, consistent with mission criticality.

UFC 3-210-10

1 June 2015

Change 1, 1 February 2016

Figure 2-1 Implementation of EISA Section 438



AF Sustainable Building Requirements Legend

Gray Text	Recommended (Federal Requirement not fully defined at this time)
Light Gray Cell	Cell is populated by other cells. User is not required to enter values.
Light Green Cell (When used, conditional formatting changes cell to Green or Red depending on entry)	Cell provides a drop-down box for the user to select which LEED Credits and HPSB Requirements that the project is attempting. When a "Yes" or "No" is selected the cell follows the stoplight convention to visually represent progress. The stoplight convention is also used for "Yes" or "No" totals. Some light green boxes also have an option for "N/A" when used for subquestions
	Cell uses a drop down box and user is designating "Yes" or "N/A" when applicable
	Cell uses a drop down box and user is designating "No"
Light Yellow Cell	Cell is for custom entry on how the project is pursuing HPSB requirements. These are subquestions and also general information questions. Boxes that appear when cell is selected give further instruction on entering data. Some cells restrict values that can be entered - which is explained if incorrectly entered

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SECTION 01 35 26.00 06

GOVERNMENT SAFETY REQUIREMENTS
05/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

- ASSP A10.32 (2013) Fall Protection
- ASSP A10.34 (2021) Protection of the Public on or
Adjacent to Construction Sites
- ASSP Z359.1 (2020) The Fall Protection Code

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

- NFPA 241 (2022) Standard for Safeguarding
Construction, Alteration, and Demolition
Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

- EM 385-1-1 (2024) Safety and Health Requirements
Manual

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

- 29 CFR 1926 Safety and Health Regulations for
Construction
- 29 CFR 1926.500 Fall Protection

1.2 SUBMITTALS

Government approval/acceptance is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. The following must be submitted in accordance with LRL Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Accident Prevention Plan (APP); G

Fatigue Management Plan

Bloodborne Pathogen Plan

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Exposure Control Plan
Hearing Conservation Program
Respiratory Protection Plan
Health Hazard Control Program
Hazard Communication Program
Heat Stress Monitoring Plan
Cold Stress Monitoring Plan
Indoor Air Quality Management Plan
Crystalline Silica Assessment
Fire Prevention Plan
Hazardous Energy Control Plan
Standard Pre-Lift Plan (LHE)
Critical Lift Plan - LHE
Fall Protection and Prevention Plan

SD-06 Test Reports

Reports
Accident/Mishap Investigation Reports
Monthly Exposure Reports
Regulatory Citations and Violations

SD-07 Certificates

Hot work permit
Submit one copy of each permit/certificate attached to each daily Quality Control Report.

1.3 DEFINITIONS

- a. Site Safety and Health Officer (SSHO). The qualified or competent person who is responsible for the on-site safety and health management required for the contract project work.
- b. Competent Person, Fall Protection: A person designated in writing by the employer to be responsible for immediate supervising, implementing and monitoring of the fall protection program, who through training, knowledge and experience in fall protection and rescue systems and equipment, is capable of identifying, evaluating and addressing existing and potential fall hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

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c. High Visibility Accident. Any mishap which may generate publicity and/or high visibility.

d. Near Miss. A potentially serious act of commission or omission that leads to an uncontrolled or unanticipated incident. The incident could have resulted in injury, illness, death, and/or property damage (including environmental damage) but was avoided by chance, and/or immediate, real-time mitigation.

e. Medical Treatment. Treatment administered by a physician or by registered professional personnel under the standing orders of a physician. Medical treatment does not include first aid treatment even though provided by a physician or registered personnel.

f. Qualified Person, Fall Protection: A person with a recognized degree or professional certificate and with extensive knowledge, training and experience in the fall protection and rescue field who is capable of designing, analyzing, evaluating and specifying fall protection and rescue systems; shall have an advanced understanding of the regulatory requirements, physical sciences and engineering principles that affect equipment and systems for FP and rescue; be able to calculate forces generated by an arrested fall, the total loading and the deflection of the fall arrest anchorage, the impact on the structural members to which the fall arrest system is attached and shall be able to determine safe locations of anchorages; shall supervise the design, selection, installation and inspection of certified anchorages and horizontal lifelines.

g. Recordable Injury, Illness, or Property Damage. Any work-related accident that results in:

- (1) Death, regardless of the time between the injury and death, or the length of the illness;
- (2) Days away from work (any time lost after day of injury/illness onset);
- (3) Fatal injury / illness;
- (4) Permanent totally disabling injury/illness;
- (5) Permanent partial disabling injury/illness;
- (6) One(1) or more persons hospitalized as inpatients as a result of a single occurrence;
- (7) \$5,000 or greater accidental property damage;
- (8) Three(3) or more individuals become ill or have a medical condition which is suspected to be related to a site condition, or a hazardous or toxic agent on the site;
- (9) Restricted work;
- (10) Transfer to another job;
- (11) Medical treatment beyond first aid;
- (12) Loss of consciousness; or

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(13) A significant injury or illness diagnosed by a physician or other licensed health care professional, even if it did not result in (1) through (13) above.

h. Reportable Accident. An occurrences that causes an injury, occupational illness, or property damage of any kind.

i. "USACE" property and equipment specified in USACE EM 385-1-1 should be interpreted as Government property and equipment.

j. Weight Handling Equipment (WHE) Accident. A WHE accident occurs when any one or more of the six elements in the operating envelope fails to perform correctly during operation, including operation during maintenance or testing resulting in personnel injury or death; material or equipment damage; dropped load; derailment; two-blocking; overload; and/or collision, including unplanned contact between the load, crane, and/or other objects. A dropped load, derailment, two-blocking, overload and collision are considered accidents even though no material damage or injury occurs. A component failure (e.g., motor burnout, gear tooth failure, bearing failure) is not considered an accident solely due to material or equipment damage unless the component failure results in damage to other components (e.g., dropped boom, dropped load, roll over, etc.).

k. Low-slope roof. A roof having a slope less than or equal to 4 in 12 (vertical to horizontal).

l. Steep roof. A roof having a slope greater than 4 in 12 (vertical to horizontal).

k. Certified Safety Professional/Certified Industrial Hygienist Qualifications

(1) Certified Construction Health & Safety Technician (CHST). An individual who is currently certified by the Board of Certified Safety Professionals.

(2) Certified Industrial Hygienist (CIH). An individual who is currently certified by the American Board of Industrial Hygiene.

(3) Certified Safety Professional (CSP). An individual who is currently certified by the Board of Certified Safety Professionals.

(4) Certified Safety Trained Supervisor (STS). An individual who is currently certified by the Board of Certified Safety Professionals.

(5) Associate Safety Professional (ASP). An individual who is currently certified by the Board of Certified Safety Professionals.

1.4 REGULATORY REQUIREMENTS

In addition to the detailed requirements included in the provisions of this contract, work performed shall comply with USACE EM 385-1-1, federal, state, local, host nation laws, ordinances, criteria, rules and regulations as listed above. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting work. Where the requirements of this specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements shall apply.

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1.5 SITE QUALIFICATIONS, DUTIES AND MEETINGS

1.5.1 Personnel Qualifications

1.5.1.1 Site Safety and Health Officer Qualifications (SSHO)

a. A Site Safety and Health Officer (SSHO) and alternate(s) must be provided at the work site at all times and will be a member of the onsite work organization and be responsible for overall management of the safety and occupational health program. The SSHO will have the authority to act in all safety matters for the Contractor at all times to perform safety and occupational health management, surveillance, inspections, and safety enforcement for the Contractor. The SSHO and alternate(s) will be employed by the Prime Contractor and must report to a corporate safety official or other corporate official not engaged in quality control or supervision.

(1) Completed, as a minimum, the 30-hour OSHA Construction Industry safety class with current First Aid and CPR Training / AED.

(2) Either a person with 10 years of demonstratable SSHO experience on similar projects or a College graduate with Five (5) years of Construction Industry safety experience on similar projects in supervising or managing general or industry construction (managing safety programs or processes or conducting hazard analyses and developing controls).

b. The SSHO must be:

Assigned as SSHO, but may have duties as the CQC System Manager will not be the Superintendent.

24 hours of Competent Person Fall Protection Training.

8 hours of Competent Person in Scaffolding.

c. To insure that safety and health conditions are maintained/enforced at all times, and a SSHO is present at all times, the Contractor will designate one or more alternates to perform the safety and health requirements stated herein to cover any period when the SSHO can not be present, such as during absences for vacations/extended sickness, or when there are multiple shifts that requires additional coverage. The alternate(s) will meet the same qualifications, training, education requirements as the SSHO.

d. The Contractor must identify the SSHO and alternate(s) for this project and will submit qualifications to the Government in resume form for acceptance. A copy of the letter to the SSHO and alternate(s) signed by an authorized official of the firm describing responsibilities and delegating authority to stop work when safety or occupational health of workers is compromised must be provided to the Government.

e. Acceptance of the Contractor's SSHO and alternate(s) is required prior to the start of construction. Acceptance is conditional and will be predicated on satisfactory performance during construction. The Government reserves the right to require the Contractor to make changes to operations including removal of personnel, as necessary, to obtain a safe work site. At no time will the job be permitted to operate without a SSHO on duty at the work site.

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f. Duties of the SSHO will include, at a minimum, the following in addition to the duties now listed per other paragraphs of this Section:

- (1) Prepare the Contractor's Accident Prevention Plan and associated work Safety Plan(s), and Activity Hazard Analysis for each definable feature of work;
- (2) Provide safety indoctrination to all construction site visitors;
- (3) Ensure the Contractor's accepted Accident Prevention Plan is carried out;
- (4) Ensure that Contractor/subcontractor employees have been provided with all relevant training as required per the EM 385-1-1 and as outlined in the accepted Accident Prevention plan.
- (5) Conducts daily walk through of the site ensuring work is being accomplished safely, personnel protective equipment meets applicable OSHA/EPA requirements, and occupational health is not compromised;
- (6) Attend and participate in all preparatory and initial quality control phase meetings;
- (7) Ensure weekly worker safety meetings are conducted and relevant to current and upcoming work
- (8) Ensure monthly supervisory safety meetings are conducted and relevant to current and future work;
- (9) Ensure accident investigations are conducted and the required reports are completed and provided. Maintain the OSHA Form 300 for prime contractor
- (10) Produce a DailySafety Report of activities performed in accordance with other paragraphs of this section. The Daily Safety Report will include observations made, accident and/or near-misses experienced, and ensure this report is attached to the daily Quality Control Report.
- (11) Provide minutes for weekly and monthly safety meetings, minutes to be attached with the Daily Safety Report.

1.5.2 Personnel Duties

1.5.2.1 Site Safety and Health Officer (SSHO)

a. Conduct daily safety and health inspections and maintain a written log which includes area/operation inspected, date of inspection, identified hazards, recommended corrective actions, estimated and actual dates of corrections. Safety inspection logs shall be attached to the Contractors' daily quality control report.

c. Maintain applicable safety reference material on the job site.

d. Attend the pre-construction conference, pre-work safety conference, pre-work meetings including preparatory inspection meeting, and periodic in-progress meetings.

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- e. Implement and enforce accepted APPS and AHAs.
- f. Maintain a safety and health deficiency tracking system that monitors outstanding deficiencies until resolution. A list of unresolved safety and health deficiencies shall be posted on the safety bulletin board.
- g. Ensure sub-contractor compliance with safety and health requirements.
- h. Other duties as identified per LRL Section 01 45 04.10 06 Contractor Quality Control. Failure to perform the above duties shall result in dismissal of the SSHO, and/or CQC System Manager, and/or superintendent and a project work stoppage. The project work stoppage will remain in effect pending approval of a suitable replacement.

1.5.3 Meetings

1.5.3.1 Pework Safety Conference

- a. The purpose of the prework safety conference is for the Contractor and the Contracting Officer's representatives to become acquainted and explain the functions and operating procedures of their respective organizations and to reach mutual understanding relative to the administration of the overall project's APP before the initiation of work. The prework safety conference will not be scheduled until a preliminary APP is submitted and accepted.
- b. Contractor representatives who have a responsibility or significant role in accident prevention on the project must attend the prework safety conference. At a minimum, the attendees will include the project superintendent, Site Safety and Health Officer, Quality Control System Manager, and any other assigned safety and health professionals who participated in the development of the APP (including the Activity Hazard Analyses (AHAs) and special plans, program and procedures associated with it).
- c. The Contractor will discuss the details of the submitted APP to include incorporated plans, programs, procedures and a listing of anticipated AHAs that will be developed and implemented during the performance of the contract. The list of Definable Features of Work will be reviewed at the conference and an agreement will be reached between the Contractor and the Contracting Officer's representative to determine applicable AHAs to be prepared. . In addition, a schedule for the preparation, submittal, review, and acceptance of AHAs will be established to preclude project delays.
- d. The prework safety conference will not be scheduled until a preliminary APP is submitted and accepted. Work will not begin until after a Pework Safety Conference has been conducted.
- e. The functions of a prework safety conference, may take place at the Post-Award Kickoff meeting for Design Build Contracts.

1.5.3.2 Weekly Safety Meetings

Must be conducted and documented as required by EM 385-1-1, Chapter 2-8(4) (a-c). Documentation as Required in EM 385-1-1, Chapter 2-8(4) (b) will be attached to the Contractors' daily quality control report.

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1.5.3.3 Work Phase Meetings

The appropriate AHA will be reviewed and attendance documented by the Contractor at the preparatory, initial, and follow-up control phases of quality control inspection in accordance with LRL Section 01 45 04.10 06 CONTRACTOR QUALITY CONTROL. The analysis will be used during daily inspections to ensure the implementation and effectiveness of safety and health controls; and the results reported on the Contractors' daily quality control Report.

1.6 TRAINING

1.6.1 New Employee Indoctrination

New employees (prime and sub-contractor) will be provided indoctrination training as outlined in EM 385-1-1, Chapter 2-3(6) (a-f). Documentation of this orientation will be kept on file at the project site and provided to the Contracting Officer upon request.

1.6.2 Periodic Training

Provide Safety and Health Training in accordance with USACE EM 385-1-1 and the accepted APP. Documentation of training will be kept on file at the project site and will be provided to the Contracting Officer upon request.

1.6.3 Training on Activity Hazard Analysis (AHA)

Prior to beginning a new control phase, training will be provided to all affected employees to include a review of the AHA to be implemented.

1.7 ACCIDENT PREVENTION PLAN (APP)

a. The Contractor must use a qualified person to prepare the written site-specific APP. Prepare the APP in accordance with the format and requirements of USACE EM 385-1-1 and as supplemented herein. Cover all paragraph and subparagraph elements in USACE EM 385-1-1, Appendix A, "Minimum Basic Outline for Accident Prevention Plan".

b. Submit the APP to the Contracting Officer thirty (30) calendar days prior to the date of the prework safety conference for acceptance. Work cannot proceed without an accepted APP. The Contracting Officer reviews and comments on the Contractor's submitted APP and accepts it when it meets the requirements of the contract provisions and EM 385-1-1.

c. Acceptance is conditional and will be predicated on satisfactory performance during the construction. Once accepted by the Contracting Officer, the APP and attachments will be enforced as part of the contract. Disregarding the provisions of this contract or the accepted APP will be cause for stopping of work, at the discretion of the Contracting Officer, until the matter has been rectified. Work cannot proceed without an accepted APP.

d. Once work begins, the APP must be evaluated for updates throughout the life of the project to ensure it is site specific. Changes, revisions, or updates to the accepted APP must be reviewed and approved by the contractor and submitted to the KO or COR for acceptance. Should hazards outside the accepted APP become evident, all necessary action(s) must be taken to restore and maintain safe working conditions in order to safeguard onsite personnel, visitors, the public (as defined by ASSP A10.34,) and the environment.

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e. Copies of the accepted plan will be maintained at the contractor's job site office.

f. The APP must be continuously reviewed and amended, as necessary, throughout the life of the contract. Activities not identified in the currently accepted APP will be incorporated into the plan as they are discovered and resubmitted for acceptance. Work related to that activity will not begin until the APP is updated and accepted.

1.8 ACTIVITY HAZARD ANALYSIS (AHA)

a. The Activity Hazard Analysis (AHA) must be completed in accordance with USACE EM 385-1-1 requirements, using ENG Form 6206, current edition. Submit the AHA for review at least fifteen (15) calendar days prior to the start of each definable feature of work.

b. An AHA will be developed by the Contractor for every operation involving a type of work presenting hazards not experienced in previous project operations or where a new work crew or subcontractor is to perform work. The AHA must be completed in accordance with EM 385-1-1, Chapter 2-6.

c. The analysis should be used as part of the daily inspections to ensure the implementation and effectiveness of the activity's safety and health controls.

d. The AHA list will be reviewed periodically (at least monthly) at the Contractor supervisory safety meeting and updated as necessary when procedures, scheduling, or hazards change.

e. Activity hazard analyses will be evaluated for updates throughout the life of the project to ensure they are site specific. AHAs will be updated as necessary to provide an effective response to changing work conditions and activities. Changes, revisions, or updates to the accepted AHAs must be reviewed and approved by the contractor, to include the on-site superintendent, site safety and health officer and competent persons, and submitted to the KO or COR for acceptance.

f. The activity hazard analyses will be developed using the project schedule as the basis for the activities performed. Any activities listed on the project schedule will require an AHA. The AHAs will be developed by the contractor, supplier or subcontractor and provided to the prime contractor for submittal to the Contracting Officer.

1.9 DISPLAY OF SAFETY INFORMATION

Within seven (7) calendar day after commencement of work, erect a safety bulletin board at the job site. The safety bulletin board will include information and be maintained as required by EM 385-1-1, Chapter 2.

1.10 SITE SAFETY REFERENCE MATERIALS

Maintain safety-related references applicable to the project, including those listed in the article "References", all applicable equipment manufacturer's manuals, and those required per the EM 385-1-1.

1.11 EMERGENCY MEDICAL TREATMENT

Contractors will arrange for their own emergency medical treatment.

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Government has no responsibility to provide emergency medical treatment.

1.12 REPORTS

Submit reports in accordance with the requirements of this paragraph. All reports must be maintained on the project site and submitted as part of the Contractors' daily quality control report.

1.12.1 Accident Notification

Notify the Contracting Officer as soon as practical, but not later than eight hours, after any accident involving a fatality, in-patient hospitalizations, amputation, eye loss, or property damage over \$600,000. For all other accidents and near-misses, notify the contracting officer no later than 24 hours. Accident and near-miss notification will be made using the ENG Form 3394 (2021). Unless instructed otherwise, preserve the conditions and evidence on the accident site until the Government investigation team arrives on-site and Government investigation is conducted.

1.12.2 Accident Reports

The Prime Contractor will conduct an accident investigation and determine causal factors for all accidents and near-misses. Completed accident and near-miss investigations will be recorded using the ENG Form 3394 (2021) and provide the report to the Contracting Officer within seven (7) calendar day(s) of the accident. All Accident and Near-Miss Investigations will be conducted in accordance with EM 385-1-1, Chapter 2. The Contracting Officer will provide copies of any required or special forms.

1.12.3 Monthly Exposure Reports

Monthly exposure reporting to the Contracting Officer must be provided monthly using ENG Form 4025. This report is a compilation of employee-hours worked each month for all site workers, both prime and subcontractor.

1.12.4 Regulatory Citations and Violations

Notify the Contracting Officer immediately of any OSHA or other regulatory agency inspection or visit, and provide the Contracting Officer with a copy of each citation, report, and contractor response. Correct violations and citations promptly and provide written corrective actions to the Contracting Officer.

PART 2 PRODUCTS NOT USED

PART 3 EXECUTION

3.1 CONSTRUCTION AND/OR OTHER WORK

The Contractor must comply with USACE EM 385-1-1, NFPA 241, the APP, the AHA, Federal and/or State OSHA regulations, and other related submittals and installation/activity fire and safety regulations. The most stringent standard will prevail.

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3.1.1 Hazardous Material Use and Exclusions

Each hazardous material must receive approval by the Authority Having Jurisdiction (AHJ) prior to being brought onto the job site or prior to any other use in connection with this contract. Any exclusions must be coordinated through the AHJ. The Contracting Officer must be notified of any exclusions provided by the AHJ. See 01 57 19.00 06 Temporary Environmental Controls and Permits

3.1.2 Unforeseen Hazardous Material

The design should have identified materials such as PCB, lead paint, and friable and non-friable asbestos. If additional material, not indicated, that may be hazardous to human health upon disturbance during construction operations is encountered, stop that portion of work and notify the Contracting Officer immediately. Within fourteen (14) calendar days the Government will determine if the material is hazardous. If material is not hazardous or poses no danger, the Government will direct the Contractor to proceed without change. If material is hazardous and handling of the material is necessary to accomplish the work, the Government will issue a modification pursuant to FAR 52.243-4 - Changes and FAR 52.236-2 - Differing Site Conditions.

3.1.3 Unanticipated Discovery of Ordnance and Explosives

If during the course of construction operations, any unanticipated or unplanned discovery of Munitions and Explosives of Concern (MEC), Explosive Media, Chemical Warfare Media (CWM), or chemical agent contaminated media (CACM) occurs, all work must cease, personnel must withdraw from the affected area and the Contracting Officer's Representative (COR) must be contacted for further information and direction. Refer ER 385-1-95 and EM 385-1-97 for additional requirements.

3.2 PRE-OUTAGE COORDINATION MEETING

Contractors are required to apply for utility outages through the Installation Public Utilities Company..Special requirements for electrical outage requests are contained 00 08 00.00 06, 01 11 00. Work must be performed in accordance with EM 385-1-1, Chapter 12, and NFPA 70 E, and any applicable regulations. Prior to beginning work on the utility system requiring shut down, the Contractor must attend a pre-outage coordination meeting with the Contracting Officer and the Installation representative and Public Utilities representative to review the scope of work and the lock-out/tag-out procedures for worker protection. No work will be performed on energized electrical circuits unless proof is provided that no other means exist. A pre outage coordination meeting will be conducted per the three phase inspection process, to include the AHA.

3.3 FALL HAZARD PROTECTION AND PREVENTION PROGRAM

The Contractor will establish a fall protection and prevention program, for the protection of all employees exposed to fall hazards. The program must include company policy, identify responsibilities, education and training requirements, fall hazard identification, prevention and control measures, inspection, storage, care and maintenance of fall protection equipment and rescue and evacuation procedures.

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3.3.1 Training

As part of the Fall Hazard Protection and Prevention Program, the Contractor will provide training to individuals who might be exposed to fall hazards on the site. Training will be in accordance with USACE EM 385-1-1, Chapter 21

3.3.2 Fall Protection Equipment and Systems

The Contractor must enforce use of the fall protection equipment and systems designated for each specific work activity in the Fall Protection and Prevention Plan and/or AHA at all times when an employee is exposed to a fall hazard or on a surface 6 feet or more above lower levels. Fall protection systems such as guardrails/toeboards, personnel fall arrest system, safety nets, etc., are required when working within 6 feet of any leading edge and employees will be protected from fall hazards as specified in EM 385-1-1, Section 21. In addition to the required fall protection systems, safety skiff, personal floatation devices, life rings etc., are required when working above or next to water in accordance with USACE EM 385-1-1, Section 21. Personal fall arrest systems are required when working from an articulating or extendible boom, swing stages, or suspended platform. In addition, personal fall arrest systems are required when operating other equipment such as scissor lifts if the work platform is capable of being positioned outside the wheelbase. The need for tying-off in such equipment is to prevent ejection of the employee from the equipment during raising, lowering, or travel. Fall protection must comply with 29 CFR 1926.500, Subpart M, USACE EM 385-1-1 and ASSP A10.32.

3.3.2.1 Personal Fall Arrest Equipment

Personal fall arrest equipment, systems, subsystems, and components must meet ASSP Z359.1. Only a full-body harness with a shock-absorbing lanyard or self-retracting lanyard is an acceptable personal fall arrest body support device. Body belts may only be used as a positioning device system (for uses such as steel reinforcing assembly and in addition to an approved fall arrest system). Harnesses must have a fall arrest attachment affixed to the body support (usually a Dorsal D-ring) and specifically designated for attachment to the rest of the system. Only locking snap hooks and carabiners will be used. Webbing, straps, and ropes shall be made of synthetic fiber. The maximum free fall distance when using fall arrest equipment shall not exceed (6 feet). The total fall distance and any swinging of the worker (pendulum-like motion) that can occur during a fall shall always be taken into consideration when attaching a person to a fall arrest system.

3.3.3 Fall Protection for Roofing Work

Fall protection controls must be implemented based on the type of roof being constructed and work being performed. The roof area to be accessed will be evaluated for its structural integrity including weight-bearing capabilities for the projected loading.

a. Low Sloped Roofs:

- (1) For work within (6 feet) of an edge, on low-slope roofs, personnel shall be protected from falling by use of personal fall arrest systems, guardrails, or safety nets.
- (2) For work greater than (6 feet) from an edge, warning lines,

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if utilized, must be erected and installed in accordance with
29 CFR 1926.500 and USACE EM 385-1-1, Chapter 21-8.i.

b. Steep-Sloped Roofs: Work on steep-sloped roofs requires a personal fall arrest system, guardrails with toe-boards, or safety nets. This requirement also includes residential or housing type construction.

3.3.4 Existing Anchorage

Existing anchorages, to be used for attachment of personal fall arrest equipment, must be certified (or re-certified) by a qualified person for fall protection in accordance with ASSP Z359.1. Existing horizontal lifeline anchorages must be certified (or re-certified) by a registered professional engineer with experience in designing horizontal lifeline systems.

3.3.5 Horizontal Lifelines

Horizontal lifelines will be designed, installed, certified and used under the supervision of a qualified person for fall protection as part of a complete fall arrest system which maintains a safety factor of 2 (29 CFR 1926.500).

3.3.6 Guardrails and Safety Nets

Guardrails and safety nets will be designed, installed and used in accordance with EM 385-1-1 and 29 CFR 1926 Subpart M.

-- End of Section --

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SECTION 01 35 43

GENERAL ENVIRONMENTAL PROTECTION REQUIREMENTS (WPAFB)
12/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

CAA Section 608	Section 608 of the Clean Air Act, Stationary Refrigeration and Air Conditioning
42 USC Chapter 85	Air Pollution Prevention and Control
29 CFR 1910	Occupational Safety and Health Standards
29 CFR 1926	Safety and Health Regulations for Construction
40 CFR 60	Standards of Performance for New Stationary Sources
40 CFR 61 Subpart M	National Emission Standard for Asbestos
40 CFR 68	Chemical Accident Prevention Provisions
40 CFR 82	Protection of Stratospheric Ozone
40 CFR 89	Control of Emissions from Nonroad Compression-Ignition Engines
40 CFR 240.101	Solid Wastes: Definitions
40 CFR 247	Comprehensive Procurement Guideline for Products Containing Recovered Materials
40 CFR 261	Identification and Listing of Hazardous Waste
40 CFR 262	Standards Applicable to Generators of Hazardous Waste
40 CFR 273	Standards for Universal Waste Management
40 CFR 745	Lead-Based Paint Poisoning Prevention in Certain Residential Structures
40 CFR 761	Polychlorinated Biphenyls (PCBs) Manufacturing, Processing, Distribution in Commerce, and Use Prohibitions

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40 CFR 763	Asbestos
48 CFR 52.223-1	Biobased Product Certification
48 CFR 52.223-2	Affirmative Procurement of Biobased Products Under Service and Construction Contracts
48 CFR 52.223-4	Recovered Material Certification

U.S. ENVIRONMENTAL PROTECTION AGENCY

Title III List of Lists	Consolidated List of Chemicals Subject to the Emergency Planning and Community Right-to-Know Act (EPCRA), Comprehensive Environmental Response, Compensation and Liability Act (CERCLA), and Section 112(r) of the Clean Air Act
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OHIO ADMINISTRATIVE CODE

OAC 1301:7-9	Underground Storage Tanks
OAC 3745-17	Particulate Matter Standards
OAC 3745-20	Asbestos Emission Control
OAC 3745-22	Asbestos Hazard Abatement
OAC 3745-42	Permits to Install and Plan Approvals for Water Pollution Control
OAC 3745-91	Public Water System: Plans Approval
OAC 3745-513	Authorization To Disturb Land Where a Waste Facility Was Operated

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

AHRI Guideline K	(2009) Guideline for Containers for Recovered Non-Flammable Fluorocarbon Refrigerants
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1.2 DEFINITIONS

1.2.1 Green Procurement.

The purchase of environmentally preferable products and services in accordance with federal requirements for recycled and biobased content.

1.2.2 Hazardous Materials (HAZMAT)

For the purposes of this document, the term HAZMAT includes all items covered under the EPCRA reporting requirement (including the Title III List of Lists), the OSHA Hazard Communication (HAZCOM) Standard, and all Class I and Class II ODSs.

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1.2.3 Hazardous Wastes (HW)

Wastes which are listed by chemical name in 40 CFR 261 Subpart D, or exhibit one or more characteristics described in 40 CFR 261 Subpart C.

1.2.4 Initial Accumulation Point (IAP)

A permitted HW collection point at or near an area where HW is generated and under the control of the operator of the process that generates the waste.

1.2.5 HAZMART

The organization, function, or location which performs the HAZMAT Tracking Activity. The HAZMART is the only entity on WPAFB authorized to issue government-owned HAZMAT from any source.

1.2.6 Manifest or Uniform HW Manifest

EPA form 8700-22 and, if necessary, EPA form 8700-22A, originated and signed by the generator in accordance with the instructions included in the appendix to 40 CFR 262.

1.2.7 Safety Data Sheet (SDS)

A summary of safety, health, and emergency response information about a manufactured product obtained from the product manufacturer or distributor, and prepared in accordance with 29 CFR 1910.1200, the OSHA HAZCOM Standard. The SDS includes information about the chemical constituents, their hazards, and personal protective equipment to be used.

1.2.8 Special Waste

As defined by 40 CFR 240.101, a non-hazardous solid waste requiring handling other than that normally used for municipal solid waste.

1.2.9 Toxics and Toxic Waste

For the purposes of this section, Toxics refers to Lead Based Paint (LBP) as defined in 40 CFR 745 and Polychlorinated Biphenyls (PCBs) as defined in 40 CFR 761. Toxic Waste refers to these materials when processed for disposal.

1.2.10 Universal Waste

A subset of Hazardous Wastes, described by and managed in accordance with 40 CFR 273, including batteries, pesticides, mercury-containing thermostats or other mercury-containing devices, and fluorescent, HID, mercury vapor, low pressure sodium, and other lamps.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

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Environmental Coordinator Designation and Application for RCRA
Training

Application For Authorization - Environmental Restoration Site
Disturbance; G

Erosion Control Plan; G

Storm Water Pollution Prevention Plan (SWPPP); G

Notice of Intent (NOI) for Coverage Under NPDES Permit

Sanitary or Wastewater Permit to Install (PTI); G

Drinking Water Plans Approval; G

Ohio EPA Notification of Demolition and Renovation/Abatement

Asbestos Work Plan; G

Fuel-Fired Equipment Startup Information

HAZMAT Inventory

Site Specific Spill Plan; G

WPAFB Worksheet 1433 - Hazardous Waste Storage Permit

Permit Applications for UST Work; G

SD-06 Test Reports

Asbestos Abatement Clearance Air Sampling Report

HVAC System Water Analysis; G

Quarterly Recycling and Solid Waste Summary Report

Refrigerant Record

SD-07 Certificates

Certification of Recovered and Biobased Products

Section 608 Technician Certification

Refrigerant Recovery Equipment Certification

Certificate of Conformity

STI SP001 Inspector's Training Certification

SD-11 Closeout Submittals

Green Products Determination Form (GPDF)

Notice of Termination (NOT)

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STI SP001 Aboveground Storage Tank Inspection Report

STI SP131 Underground Storage Tank Inspection Report

1.4 GENERAL PROVISIONS

1.4.1 Base Environmental Authority

The 88th Civil Engineer Group Environmental Branch, 88 CEG/CEIE, establishes environmental compliance policy, administers environmental programs, and coordinates environment protection requirements for WPAFB. 88 CEG/CEIE consists of the Compliance Section, 88 CEG/CEIEC, and the Environmental Assets Section, 88 CEG/CEIEA. 88 CEG/CEIE is the designated environmental spokesperson for WPAFB and is the only office authorized to interface with environmental regulatory agencies.

1.4.2 Contractor's Environmental Coordinator

The Contractor shall designate a primary and alternate environmental coordinator. The environmental coordinator shall serve as the Contractor's single, integrated point of contact for all environmental questions and requirements. This individual shall be responsible for providing the Contractor's response to all environmental requirements and shall be authorized to direct the Contractor's organization to respond to environment requirements.

1.4.3 Resource Conservation and Recovery Act (RCRA) Training

Within 15 days after receipt of Notice to Proceed (NTP) submit an Environmental Coordinator Designation and Application for RCRA Training, listing the names and telephone numbers of the designated environmental coordinator and alternate. The phone numbers shall include both the usual business telephone number and a 24-hour emergency telephone number where the individual can be reached at any time for an environmental emergency. If the designated individuals have completed the WPAFB Installation RCRA Training course within the previous 12 months, indicate the date of prior completion. The RCRA Training is a one-hour course that explains local procedures and requirements for the management of HAZMATs and HW. Upon receipt of the application, 88 CEG/CEIE will contact the individuals to schedule the course.

1.4.4 Environmental Management System

The Contractor shall review the **Fundamentals of an Environmental Management System** document provided during the Preconstruction Conference following contract award, and shall sign and return the accompanying attendance form. For additional information contact 88 CEG/CEIE at 937-257-7152.

1.4.5 Permits

Unless otherwise stated, the Contractor shall obtain all necessary environmental permits and licenses and shall be responsible for payment of applicable fees. The Contractor shall comply with all terms and conditions of permits.

1.4.6 Competent Person

The Contractor shall provide a competent person on-site who is capable of identifying existing and predictable hazards in the workplace, and who has

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authority to take prompt corrective measures. For asbestos, a "competent person" is one who has been trained pursuant to OAC 3745-22 and 40 CFR 763, subpart E, Appendix C.

1.4.7 Maintenance of Environmental Controls

The Contractor shall identify existing and predictable sources of pollution related to work under this contract, and shall establish and maintain in good working order equipment, facilities, structures, or procedures as necessary to prevent or control any such pollution.

1.4.8 Prohibited Materials

The Contractor shall not provide nor install any asbestos containing material (ACM), nor any equipment of any kind which contains a Class I or Class II Ozone Depleting Substance (ODS). In the event that a technical specification section requires provision or installation of ACM or Class I or Class II ODS, notify the Contracting Officer's representative.

1.4.9 Recyclable Materials

Other sections within this contract may address title to or disposition of recyclable materials generated from demolition or other work under this contract. In the event of conflict between this section and others, this section shall take precedence. Refer to SOLID WASTE AND RECYCLING, below.

1.4.10 No Release of Other Requirements

Nothing in this section shall relieve the Contractor of any requirement specified in federal, state, or local laws or regulations.

1.5 GREEN PROCUREMENT

1.5.1 General

Provide materials that comply with federal requirements for recycled content and biobased content, in accordance with 40 CFR 247; the USDA BioPreferred Program; 48 CFR 52.223-2; and the technical specification sections in this contract. Refer to <https://www.epa.gov/smm/comprehensive-procurement-guideline-cpg-program> for assistance identifying products cited in 40 CFR 247. Refer to <https://www.biopreferred.gov/BioPreferred/> for assistance identifying products in the USDA BioPreferred Program.

1.5.2 Certification of Recovered and Biobased Products

Prior to beginning work, submit in electronic PDF format Certification of Recovered and Biobased Products, as required by 48 CFR 52.223-4 and 48 CFR 52.223-1. Include the following information:

- a. Project name, "ZHTV" project number, contract number, Contractor name, license number, Contractor address, Contractor signature, and date.
- b. Certification as follows:

"I hereby certify the percentage of recovered materials content for EPA-designated items, and the percentage of biobased materials content for USDA-designated items, to be used in the performance of the contract will be at least the amount required by applicable

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specifications or other contractual requirements."

1.5.3 Green Products Determination Form (GPDF)

At least four weeks prior to the planned date of project completion, submit in electronic PDF format a Green Products Determination Form (GPDF). Include the following information:

- a. Project name, "ZHTV" project number, contract number, Contractor name, license number, Contractor address, Contractor signature, and date.
- b. Certification as follows:

"I hereby certify the requisition/procurement of all materials listed on this form comply with current EPA standards for recycled/recovered materials content, and current USDA standards for biobased materials content. The following exemptions may apply:

- (1) The product does not meet appropriate performance standards;
- (2) The product is not available within a reasonable time frame;
- (3) The product is not available competitively (from two or more sources);
- (4) The product is only available at an unreasonable price (compared with a comparable non-recycled/non-biobased content product)."

- c. For each product used in the project that has a requirement or option of containing recycled content, or a requirement or option of containing biobased content, list the technical specification number, product name, total dollar value of the product, and the percentage of post-industrial recycled content, post-consumer recycled content, or percentage of biobased content. Where recycled or biobased content was required or available but not used, indicate the applicable exemption (1, 2, 3, or 4 as indicated above) and comments.

1.6 EXISTING CONDITIONS

1.6.1 Assumed Presence of Lead and Asbestos

All existing paint shall be assumed to contain lead unless proven otherwise by laboratory testing. All existing door and window caulking and glazing compounds shall be assumed to contain asbestos unless proven otherwise by laboratory testing.

1.6.2 Environmental Survey

The project area has been inspected in accordance with OAC 3745-20-02. The inspection report is attached as "Appendix A - Environmental Survey" to this section.

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PART 2 PRODUCTS: NOT USED

PART 3 EXECUTION

3.1 PREVENTION OF AIR POLLUTION

3.1.1 Ohio EPA Notification of Demolition and Renovation/Abatement

This contract does not involve the demolition of load-supporting structural member(s) of a facility; the disturbance of at least 260 linear feet, 160 square feet, or 35 cubic feet of regulated asbestos-containing materials (ACM); or the abatement of at least 50 linear feet or 50 square feet of friable ACM.

When the contract requires demolition of load-supporting structural member(s) of a facility; the disturbance of at least 260 linear feet, 160 square feet, or 35 cubic feet of regulated ACM; or the abatement of at least 50 linear feet or 50 square feet of friable ACM, submit to the Ohio Environmental Protection Agency at least 10 days prior to planned start of work a Notification of Demolition and Renovation/Abatement with applicable fee payment. Notification forms and instructions are available at epa.ohio.gov/asbestos. Submit a copy of the completed notification form to the Contracting Officer, for forwarding to the 88 CEG/CEIE Air Quality Program Manager.

3.1.2 Asbestos Containing Material (ACM)

Existing asbestos-containing materials have not been identified within the project boundaries.

When asbestos-containing materials have been identified within the project boundaries, submit an Asbestos Work Plan. Include the following information:

- a. Acknowledgment of the Environmental Survey appendix to this section. If the Contractor has observed discrepancies or potential discrepancies between the Environmental Survey and the conditions on the project site, include a description as applicable.
- b. Site plan or facility plan drawings showing locations of regulated areas, critical barriers, containments, or decon facilities as applicable. Indicate the locations, types, quantities, and disposition (to be removed, to be encapsulated in place, to remain undisturbed, etc.) of ACM.
- c. Written narrative of work procedures for abatement, containerization, manifesting, transport, disposal, and clearance air sampling and testing. Include full contact information and applicable qualifications, licensing, certification, landfill NESHAP air permit information, etc., for each involved firm or organization.
- d. Names, roles/responsibilities, and training and Ohio licensing or certification documentation for all asbestos abatement personnel.
- e. Expected date of notification of Demolition and Renovation/Abatement to Ohio EPA, as described above. If notification has already been completed, provide a copy in the Asbestos Work Plan and disregard requirement to submit it separately.

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3.1.3 Discovery of ACM

If the Contractor encounters suspected ACM not identified in the contract documents, the Contractor shall immediately notify the Contracting Officer. Any material that is suspected of containing asbestos shall not be disturbed until the Contracting Officer has determined the content of the material and the proper handling procedures, if required. All asbestos operations shall be accomplished in accordance with this section.

3.1.4 Abatement of ACM

- a. All work shall comply with 29 CFR 1926.1101, 40 CFR 61 Subpart M, OAC 3745-20, and OAC 3745-22, and shall be performed only by a contractor licensed by the State of Ohio as an Asbestos Hazard Abatement Contractor.
- b. Provide clearance air sampling in accordance with OAC 3745-22 and submit an Asbestos Abatement Clearance Air Sampling Report prior to removal of containment. Clearance air sampling shall be performed only by an Ohio-certified asbestos hazard abatement air-monitoring technician, Ohio-certified asbestos hazard evaluation specialist, or industrial hygienist certified by the American Board of Industrial Hygiene. The Report shall include written documentation of successful clearance, including the names and certifications of involved personnel and full details of the sampling and testing methodologies, the results, and applicable standards to which the results were compared.

3.1.5 Disposal of ACM

- a. All disposal of ACM shall conform to OAC 3745-20-05.
- b. All Regulated ACM (friable ACM or non-friable ACM that has or is likely to become friable; see 40 CFR 61.141 for full definition) shall be disposed of in an Ohio waste disposal site meeting the requirements of OAC 3745-20-06 or an out of state waste disposal site meeting the requirements of 40 CFR 61.154.
- c. Coordinate waste inspection and preparation of waste shipment with the 88 CEG/CEIE Hazardous Waste Program Manager at 257-0023 or 257-5531 at least two full working days prior to scheduled transport. Provide details of the type and amount of ACM, the container type, and the contact information of the transporter and the receiving landfill. The Hazardous Waste Program Manager or designated representative will prepare the shipping manifest, inspect the load, and provide the manifest to the transporter.
- d. After disposal of Regulated ACM, return the manifest signed by the landfill to the Contracting Officer for forwarding to the Hazardous Waste Program Manager within 35 calendar days of the date the waste was accepted by the transporter.
- e. Non-Regulated ACM (Category I non-friable material that has not and is not expected to become friable during removal, transport, or disposal) may be disposed of in a C&D or municipal landfill. Provide a copy of all shipping papers for non-regulated waste to the Contracting Officer for forwarding to the Hazardous Waste Program Manager, within 45 days of completion of all disposal activities.

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3.1.6 Particulate Matter Standards

The Contractor shall not cause or permit any fugitive dust that violates the opacity limits of OAC 3745-17 or 20% opacity as a three-minute average. Reasonably available control measures shall be used to prevent fugitive dust from becoming airborne. Such measures shall include, but not be limited to, one or more of the following:

- a. Use of water or other suitable dust suppression chemicals for demolition and construction operations and the clearing of land. If water is used, it shall be reapplied at frequent intervals to keep all parts of the disturbed area at least damp at all times.
- b. Use of adequate containment methods during sandblasting or other similar operations.
- c. The periodic application of water or other suitable dust suppression chemicals or the use of canvas or other covering for all materials stockpiles, except temporary stockpiles on-site for 30 days or less.
- d. Covering at all times of open-bodied vehicles when transporting materials likely to become airborne.
- e. Prompt removal, in such a manner as to minimize re-suspension, of earth or other material from paved streets onto which earth or other materials has been deposited by trucking or earth moving equipment or erosion by water or other means.

3.1.7 Open Burning

No open burning of any construction waste or unsalvageable materials shall be permitted.

3.1.8 Contractor-Operated Fuel Tanks

If the Contractor brings a fuel tank on base for refueling of company vehicles, the Contractor shall keep a log of the fuel throughput for this tank, whether it is gasoline or diesel. The Contractor shall also notify the 88 CEG/CEIE Air Quality Program Manager (AQPM) if they are bringing such a tank on base. If more than 6,000 gallons of gasoline will be dispensed in any calendar year, the Contractor shall coordinate with the AQPM and Ohio EPA to obtain a Permit To Install for this operation. At the conclusion of the contract, the Contractor shall provide a copy of their throughput logs to 88 CEG/CEIE AQPM.

3.1.9 Stationary Combustion Engines

- a. This contract does not involve provision of a stationary combustion engine or device (generator, air compressor, pump, etc) driven by a combustion engine.
- b. If work involves provision of a combustion engine or engine-driven device, within 14 days of Notice To Proceed contact the 88 CEG/CEIE Air Quality PM to confirm required permits have been issued. Installation shall not occur without confirmation from 88 CEG/CEIE of applicable permit coverage.
- c. The Contractor shall only install combustion engines that meet the application emissions standards and fuel requirements as outlined in

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40 CFR 60, Subpart IIII (compression ignition) or Subpart JJJJ (spark ignition). With each stationary engine the Contractor shall provide the manufacturer's Certificate of Conformity (40 CFR 89) for that engine to the 88 CEG/CEIE Air Quality PM.

3.1.10 Fuel-Fired Equipment

- a. This contract does not involve provision of a fuel-fired boiler, water heater, humidifier, or similar equipment.
- b. If the design input rating is 10 million BTU/hr or greater, within 14 days of Notice To Proceed contact the 88 CEG/CEIE Air Quality PM to confirm required permits have been issued. Installation shall not occur without confirmation from 88 CEG/CEIE of applicable permit coverage.
- c. At least 60 days prior to installation, submit the following Fuel-Fired Equipment Startup Information. The 88 CEG/CEIE Air Quality PM will use the information to make required notifications to OEPA and USEPA.

(1) Description (steam boiler, hot water boiler, hot water heater, humidifier, etc)

(2) Heat input rating (BTU/hr)

(3) Tank capacity in gallons

(4) Fuel type

(5) Manufacturer and model number, and serial number if available

(6) Estimated date of installation, and estimated date of startup (date the burner will be lit for the first time)

(7) Building number and room number where unit will be installed

(8) Construction contractor point of contact and phone number

3.2 PROTECTION OF WATER RESOURCES

3.2.1 General

- a. The Contractor shall not pollute any river, stream, storm drainage structure, or body of water with chemicals, petroleum products, eroded soil, construction wastes, sewage, or other harmful materials. See also HAZARDOUS MATERIALS MANAGEMENT requirements in this specification section. All construction sites are subject to inspection by WPAFB personnel to ensure all pollution control measures are adequately installed and maintained.
- b. Source Water Protection: The Contractor shall utilize best management practices to ensure proper storage, handling, use or production of regulated substances to prevent their introduction into the ground water aquifer.

3.2.2 Disturbing Less Than 1 Acre of Soil

Before any soil disturbing activity of less than 1 acre, submit an Erosion Control Plan through the contract administrator to 88 CEG/CEIE, Water

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Quality Program Manager (WQPM). The Erosion Control Plan shall show the Contractor's scheme for controlling erosion at the job site, including a site plan showing the area(s) to be disturbed and the placement of erosion control structures. The plan shall state the total acreage of soils to be exposed, including soil stockpiles, and shall include adequate measures to:

- a. Reduce by the greatest extent practicable the area and duration of exposure of bare soils.
- b. Protect readily erodible soils by use of temporary vegetation, or seeding and mulch, or by accelerating the establishment of permanent vegetation.
- c. Retard the rate of runoff from construction site.
- d. Trap sediment resulting from construction using silt fences, straw bale structures, other inlet protection, or sediment ponds. This includes pump discharges resulting from dewatering operations. Construction/installation details shall be included for sediment control structures.
- e. Provide a schedule for inspection and maintenance of all erosion control measures including any time periods where construction operations are suspended for any reason.

3.2.3 Disturbing 1 Acre or More

- a. Prepare and submit a Storm Water Pollution Prevention Plan (SWPPP) meeting the requirements of the current Ohio EPA General Permit for Storm Water Discharges from Small and Large Construction Activities. Include drawings and narratives as required to describe the construction site, potential sources of pollution in storm water discharges from the site, best management practices to control pollution, and other information as may be required by the general permit. Sign, certify, and maintain on-site the SWPPP as required by the general permit.
- b. Prepare a Notice of Intent (NOI) for Coverage Under NPDES Permit using on-line tools available at the Ohio EPA eBusiness Center website, at <https://ebiz.epa.ohio.gov/>. The Contractor shall be named as the applicant and shall fulfill all responsibilities of the permittee as outlined in the Ohio EPA general storm water permit for construction activities. Submit a copy of the NOI for 88 CEG/CEIE approval prior to formal submission to Ohio EPA. Following receipt of 88 CEG/CEIE approval, submit the NOI with applicable fees to Ohio EPA.
- c. Within 45 days of completion of soil disturbing activity, prepare a Notice of Termination (NOT), submit to Ohio EPA, and submit a copy to 88 CEG/CEIE. This action terminates Ohio EPA permit coverage, but does not signify Government concurrence or acceptance of all contractual obligations.

3.2.4 Sanitary or Wastewater Permit to Install (PTI)

- a. For any construction or modification of a sanitary sewer line (excluding the construction or modification of a lateral from an existing main to an individual facility), prepare and submit for 88 CEG/CEIE processing a PTI in accordance with OAC 3745-42. Use forms as available from the Ohio EPA website at

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<http://www.epa.ohio.gov/dsw/pti/index.aspx>. Leave "Applicant" information blank, to be completed by 88 CEG/CEIE. Provide a check in the amount of applicable fees made out to Ohio EPA, and documentation as follows:

- (1) Three copies of PTI Form A and corresponding Form B.
- (2) Five copies of the Project Detailed Plan.
- (3) Three copies of the Project Specifications.
- (4) Three copies of the Anti-degradation Permit application (if required).

- b. 88 CEG/CEIE shall complete the PTI, forward to Ohio EPA, and upon Ohio EPA approval provide a record copy to the Contractor through the Contracting Officer's representative. Allow a minimum of two months for WPAFB and Ohio EPA review and approval. Delays can be minimized through preplanning and careful attention to detail in preparation of the package.

3.2.5 Drinking Water Plans Approval

- a. For any construction or modification of a drinking water system, including tapping into or rerouting an existing water main, prepare and submit for 88 CEG/CEIE processing a Plans Approval package in accordance with OAC 3745-91. Use forms as available from the Ohio EPA website at <http://epa.ohio.gov/ddagw/engineering.aspx>. Leave "Applicant" information blank, to be completed by 88 CEG/CEIE. Provide a check in the amount of applicable fees made out to Ohio EPA, and documentation as follows:
 - (1) Four sets of construction drawings (certified by a P.E.)
 - (2) Two sets of specifications
 - (3) Two copies of the Ohio EPA Water Supply Data Sheet
 - (4) Two copies of the Project Summary Sheet
 - (5) Two copies of the Project Summary Letter
- b. 88 CEG/CEIE shall complete the Drinking Water Plans Approval package, forward to Ohio EPA, and upon Ohio EPA approval provide a record copy to the Contractor through the Contracting Officer's representative. Allow a minimum of two months for WPAFB and Ohio EPA review and approval. Delays can be minimized through preplanning and careful attention to detail in preparation of the package.

3.2.6 HVAC System Water Analysis

This contract does not involve draining existing HVAC piping.

Where work involves draining existing HVAC piping, 30 days prior to start of work take a representative sample of the HVAC system water and analyze it for the total level of each of the following: Arsenic, cadmium, chromium, copper, cyanide, lead, mercury, molybdenum, nickel, phosphorus, selenium, silver, sulfide, zinc, oil and grease, chemical oxygen demand (COD), and biochemical oxygen demand (BOD). Ensure the lab performing the

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analyses is certified by Ohio EPA to analyze for the listed parameters. Submit the lab's HVAC System Water Analysis report. The Contracting Officer's representative, in coordination with 88 CEG/CEIE, shall review the report to determine whether the HVAC System Water can be drained directly into the sanitary sewer or must be disposed of as Hazardous Waste in accordance with this section.

When the determination allows HVAC System Water to be drained to the sanitary sewer, such work shall be considered part of this contract. When disposing of HVAC System Water as Hazardous Waste is required, the disposal shall be addressed by the Government as a new requirement beyond the terms of the original contract.

3.3 PROTECTION OF LAND RESOURCES

3.3.1 Preserve Existing Resources

The Contractor shall confine all activities to areas designated by the contract drawings and specifications. Prior to beginning any exterior construction work the Contractor shall identify the land resources to be preserved within the work area. Except in areas indicated on the drawings or specified to be cleared, the Contractor shall not remove, drill, cut, deface, injure, or destroy land resources including trees, shrubs, vines, grasses, topsoil, and land forms.

3.3.2 Restore Damaged Resources

All disturbed soil areas that previously supported vegetation, regardless of the type of vegetation, shall be restored. If the area is to be seeded, it shall consist of perennial species (not annual) and the work shall be started and completed either in March through mid-May or in September through mid-October. If seeding cannot be accomplished during these times, a mulch or temporary cover crop such as rye shall be placed on the disturbed area to prevent erosion in the interim.

3.3.3 Landfill, Earthfill, or Waste Burial Sites

This contract involves site work within 300 feet of a boundary of a Landfill, Earthfill, or Waste Burial site. Where the contract documents indicate the boundary of a Landfill, Earthfill, or Waste Burial Site, and where grading, filling, excavation, drilling, or similar earthwork is required within 300 feet of such a site, submit an Application for Authorization - Environmental Restoration Site Disturbance in accordance with OAC 3745-513-400 at least 60 days prior to the proposed start of work. Contact the AFCEC/CZOM Installation Restoration Program office at 937-257-2201 for details.

Any excavation or ground disturbance (penetrating, digging, scraping, movement of heavy equipment, etc.) will require a OEPA Application for Authorization - Environmental Restoration Site Disturbance IAW OAC 3745-513 and soil management plan. IAW AFI 32-7020, "6.4. Construction At or Near Contaminated Sites. Costs associated with the actions to address contamination generated during a construction project (military construction or non-military construction), routine operation, management, or maintenance are not environmental restoration account eligible. Costs associated with a need to change ERP-generated timing actions to address such contamination are also not ERA eligible. These costs are funded as part of the construction project. This includes the handling, mitigation, and disposal or other disposition of contamination discovered before or

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during the construction activity."

3.4 CONTROL OF PCB COMPOUNDS

- a. All work involving the handling of PCBs shall be performed in strict conformance to the applicable requirements of 40 CFR 761.
- b. Any PCB waste generated as part of a renovation or demolition shall be managed in accordance with 40 CFR 761 Sub-Part D. Process all PCB shipment manifests through 88 CEG/CEIEC.
- c. Fluorescent Light Ballasts (FLB). FLBs shall be visually inspected IAW 40 CFR 761 to determine if they are PCB-containing FLBs prior to removal for disposal. Work involving the removal and disposal of PCB containing FLBs shall be conducted in accordance with 40 CFR 761.60 (b)(2)(i) and (iv). Note: Disposal in accordance with 40 CFR 761.60 (ii) is not authorized for PCB FLBs from WPAFB.

3.5 SOLID WASTE AND RECYCLING

3.5.1 Waste Collection and Disposal

- a. Litter shall be controlled and containerized at all times. Care shall be exercised to ensure no litter is lost from any vehicle while in transit to or from the construction site.
- b. Manage all wastes in strict conformance with applicable laws and regulations. Dispose of all waste materials generated by this work outside the limits of WPAFB, except as noted below.

3.5.2 Recyclable Materials

The Contractor shall participate in the Qualified Recycling Program (QRP) for recyclable wastes generated on the installation. Collect recyclable materials from the work site and deliver them to the Wright-Patterson AFB Recycling Center. Minimize the amount of non-recyclables mixed into any delivery to the Recycling Center.

- a. Recycle metals including steel, cast iron, piping, fittings, conduit, communication wire, electrical wire, brass, copper, aluminum, lead plates or shielding, and non-PCB fluorescent light ballasts. Cut metal into pieces which fit in a 30-yard container. Metals of different types may be comingled in a single container.
- b. Recycle clean cardboard and paper products. Deliver cardboard and paper products separately, each in their own container.
- c. Other materials may be recyclable if they can be cost-effectively segregated from demolition debris. Confirm the recyclability of other materials with 88 CEG/CEIE through the Contracting Officer's representative.
- d. Provide the general contractor name, "ZHTV" project number, contract number, and building number to the 88 CEG/CEIE Recycling Program Manager at the time of or before the first recycling drop-off. All trucks carrying recyclables shall be weighed in at Building 741 in Area B on the recycling center scales full and then empty. If recycling center scales are inoperable, the Contractor shall use Defense Logistics Agency (DLA) scales located at Building 741 in Area B. The

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Contractor will get a weight ticket so that the weights of recyclables can be included by the Contractor on their quarterly diversion report. The recycling center is open M-F 0700 to 1530 and closed on holidays and some other predetermined worker training days. Call 937-257-4889 to ensure the recycling center is open for business.

3.5.3 Waste Diversion

- a. Minimize solid waste generation, maximize pollution prevention processes, and maximize landfill diversion efforts through source reduction, reuse of materials, and recycling. Divert at least 60% of construction and demolition waste coming from the construction site. In cases where 60% is infeasible, provide written explanation as to why this amount could not be diverted.
- b. Track solid waste generation and disposal, and submit a Quarterly Recycling and Solid Waste Summary Report (WPAFB Diversion Report) no later than the 10th day of each January, April, July, and October, covering the preceding three months. The Government will provide a copy of the report form during the Preconstruction Conference following contract award. Include the following:
 - (1) Tonnage of construction and demolition materials recycled, and associated cost to recycle
 - (2) Tonnage of construction and demolition materials reused, and associated cost to reuse
 - (3) Tonnage of construction and demolition materials mulched, and associated cost to mulch
 - (4) Tonnage of construction and demolition materials converted from waste to energy, and associated cost to convert from waste to energy
 - (5) Tonnage of construction and demolition materials disposed of at a landfill, and associated cost to landfill
 - (6) Tonnage of construction and demolition materials incinerated, and associated cost to incinerate
 - (7) Narrative explanation if overall diversion rate is less than 60%

3.6 HAZARDOUS MATERIALS MANAGEMENT

3.6.1 General

The Contractor shall minimize the risk of spills of any hazardous material, and in the event of a spill shall contain, report, and clean up the spill and restore any damaged area. Use, handle, and store all HAZMATs in accordance with applicable state and federal laws. Follow manufacturer's recommendations for control of temperature, humidity, cleanliness, and material handling relative to storage and use of all HAZMATs. Furnish chemical security cabinets, gas cylinder storage cages, locks, secondary containment, spill response equipment, fire extinguishers, personal protective equipment, or other items as appropriate to safely and securely manage HAZMATs at the project site. Ensure site personnel are familiar with safe handling and emergency response procedures for hazardous materials. Upon completion of work, remove all HAZMAT.

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3.6.2 HAZMAT Inventory and Site Specific Spill Plan

- a. Prior to bringing solvents, adhesives, paints, fuels, oils, chemicals, or other HAZMAT onto WPAFB, submit a HAZMAT Inventory listing each proposed type of hazardous material, along with the container size (volume) and the expected quantity of each size of container applicable to each material. Submit a Site Specific Spill Plan (SSSP) detailing routine management and emergency response procedures for each material. The Government will provide a copy of the SSSP format during the Preconstruction Conference following contract award; for additional information contact 88 CEG/CEIE at 937-257-5899.
- b. If proposed volumes exceed threshold quantities defined in 40 CFR 68, notify 88 CEG/CEIE at 937-257-5899 at least 30 working days prior to bringing the materials on-site. Provide other information as may be requested by 88 CEG/CEIE to support installation risk management planning in accordance with the Clean Air Act Section 112(r).
- c. Update the HAZMAT Inventory and SSSP as necessary if additional materials are added over the course of work, and post the updated inventory and SSSP at the material storage location(s). An SDS for each HAZMAT brought onto WPAFB shall be readily available at the project site or in the possession of the user prior to the HAZMAT being used.

3.6.3 Labeling

Individually label each HAZMAT container with a durable adhesive label or tag, printed electronically or filled out legibly with a pen capable of withstanding diverse climate/weather conditions. If paper labels are used, provide lamination or similar protection to prevent deterioration. Promptly replace labels which become lost, worn, faded, or defaced in any manner. If a HAZMAT container is too small for placement of the label, the item can be stored in a plastic bag with the label attached or affixed to the container by other means as long as the label stays with the product. Labels shall provide the following information:

- a. Contractor's company name and address
- b. Contractor's point of contact name and phone number
- c. "ZHTV" project number and contract number

3.6.4 Spill Response

- a. If a spill occurs beyond the Contractor's ability to immediately and safely contain, notify the Fire Department by calling 937-257-9111 or by calling 911 and asking the dispatcher to transfer you to the WPAFB Emergency Communications Center. Also notify the Contracting Officer's representative.
- b. In addition to emergency notifications as described above, notify the Fire Department and the Contracting Officer's representative of any spill of more than one gallon of fuel or oil; any quantity of hazardous chemicals; or any quantity of any hazardous material in a sewer, drain, storm drainage structure, or other area with the potential to pollute the environment.
- c. Following any spill resulting from the Contractor's actions or

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inactions, the Contractor shall clean up spilled material and restore the area as determined by the Contracting Officer in coordination with 88 CEG/CEIE.

3.6.5 Ozone Depleting Substances (ODS)

- a. This contract does not involve removal of or work on equipment containing ODS.
- b. Recover all refrigerants prior to removing air conditioning, refrigeration, and other equipment containing refrigerants. Recovery of ODS shall comply with CAA Section 608; 42 USC Chapter 85, Air Pollution Prevention and Control; and 40 CFR 82, Protection of Stratospheric Ozone.
- c. Technicians shall be certified in accordance with CAA Section 608 and 40 CFR 82.161. Submit Section 608 Technician Certification prior to commencing work.
- d. Recovery equipment shall be certified by an EPA-approved testing organization in accordance with 40 CFR 82.158. Submit Refrigerant Recovery Equipment Certification prior to commencing work.
- e. Place recovered ODS in cylinders meeting AHRI Guideline K suitable for the type of ODS, filled to no more than 80 percent capacity, and provide appropriate labeling.
- f. All recovered Class I ODS and all recovered R-22/HCFC-22 refrigerant shall remain the property of the Government. Turn properly filled and labeled cylinders over to the Government as directed by the Contracting Officer in coordination with 88 CEG/CEIE.
- g. Recovered Class II ODS in a quantity of 20 pounds or more shall remain the property of the Government. Turn properly filled and labeled cylinders over to the Government as directed by the Contracting Officer in coordination with 88 CEG/CEIE.
- h. Recovered Class II ODS in a quantity less than 20 pounds, and other refrigerants not described above, shall be removed.
- i. Submit a Refrigerant Record within one working day of completion of work involving refrigerants. List the facility number, the type of refrigeration equipment, the type of refrigerant, the process affecting refrigerant (installation, recharging, removal, etc), the quantity or quantities added, recovered, or lost (as applicable), and the disposition (put back into the equipment, turned over to the Government, removed by the Contractor, or lost).

3.7 HAZARDOUS WASTE MANAGEMENT

3.7.1 Hazardous, Toxic, or Special Waste Disposal

The Contractor shall manage in accordance with applicable laws and this section any wastes which cannot be disposed of in a "RCRA Subtitle D" Municipal Solid Waste or Construction & Demolition Debris Landfill.

- a. Asbestos waste shall be handled in accordance with "Disposal of Asbestos Containing Materials (ACM)" in this section.

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- b. PCB waste shall be disposed of in an EPA-approved waste disposal site meeting the requirements of 40 CFR 761.60.
- c. All other hazardous wastes including Lead Based Paint debris shall be managed as specified below.
- d. For non-hazardous Special Wastes, or for any waste for which the Contractor is uncertain of proper disposal requirements, notify the Contracting Officer and 88 CEG/CEIE.

3.7.2 Initial Accumulation Point (IAP)

The Contractor shall establish an IAP if any hazardous or other regulated wastes are to be stored on WPAFB for greater than 24 hours. The Contractor shall schedule an IAP site visit with an 88 CEG/CEIE representative prior to waste storage by calling 937-257-0023 or 257-5531, and requesting inspection of the proposed IAP storage area. For each IAP, the Contractor shall apply in writing through the contract administrator to 88 CEG/CEIE for a WPAFB Worksheet 1433 - Hazardous Waste Storage Permit. The permit shall include the proposed IAP location, description of the regulated waste, estimated quantity of waste to be generated, and the type(s) of waste containers to be stored. The Contractor shall manage the IAP in accordance with the WPAFB HW Management Plan and shall furnish all containers, labels, locks, security, chemical storage cabinets, secondary containment, spill response equipment, fire extinguishers, personnel protective equipment, and other materials as may be necessary. A maximum of 55 gallons of HW or one quart of acutely HW may be stored in an IAP at any one time. The Contractor's environmental coordinator and alternate shall serve as the primary and alternate managers for each IAP. The IAP manager shall properly complete and maintain all required documentation associated with the IAP including, but not limited to, container labeling, SDSs, waste profile sheets, container tracking logs, weekly inspection logs, SSSPs, and RCRA training certificates (see "Resource Conservation and Recovery Act (RCRA) Training" in this section).

3.7.3 HW Container Storage and Labeling

Store and label HW containers in accordance with applicable laws and as specified below. These requirements address the labeling and security of all items containing HW including drums, bottles, boxes, cans, barrels, bags, and miscellaneous containers. The Contractor shall maintain a running inventory of material, waste, and empty containers.

3.7.3.1 Identification Labeling

Individually label each container with a durable adhesive label or tag, printed electronically or filled out legibly with a pen capable of withstanding diverse climate/weather conditions. If paper labels are used, provide lamination or similar protection to prevent deterioration. Promptly replace labels which become lost, worn, faded, or defaced in any manner. If a hazardous waste container is too small for placement of the label, the item can be stored in a plastic bag with the label attached or affixed to the container by other means as long as the label stays with the product. Labels shall provide the following information:

- a. Contractor's company name and address
- b. Contractor's point of contact name and phone number

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- c. "ZHTV" project number and contract number

3.7.3.2 Content Labeling

Individually label each container in accordance with 40 CFR 262.32. All containers shall have one of the following acceptable labels properly filled out and in good condition:

- a. NON-REGULATED WASTE (Blue label - NON-RCRA but DOT regulated, i.e., sodium hydroxide solid)
- b. NON HAZARDOUS WASTE (Green Label - NON-RCRA / NON-DOT)
- c. HAZARDOUS WASTE
- d. EMPTY
- e. UNIVERSAL WASTE

3.7.3.3 Storage

All containers shall be secured so that they are under control of the Contractor and to prevent access to unauthorized personnel. Containers shall be stored indoors or in an area not accessible by the general base population. If more than 12 empty containers are stored in a single area, they shall be clearly sectioned off and identified as empty.

3.7.3.4 Discovery of unknown materials

If containers of unknown constituents are found, notify 88 CEG/CEIE immediately and mark the unknown container(s) with the following information:

- a. Name and phone number of container owner, if known
- b. Date 88 CEG/CEIE was notified

3.7.4 HW Disposal and Turn-In

3.7.4.1 Hazardous Waste

The Government will dispose of properly containerized and labeled hazardous waste generated by the Contractor in the course of work under this contract. Contact 88 CEG/CEIE at 937-257-0023 or 257-5531 prior to accumulation of more than 55 gallons of HW or more than 1 quart of acutely HW, and again as necessary prior to completion of work, to arrange pickup. Send the following information via email to the CEIE HW Program Manager, with a copy to the Contracting Officer's representative:

- a. Primary or alternate IAP manager's name and phone number
- b. Name of Contractor
- c. Facility number (5-digit number) where waste was generated
- d. Common or generic name of waste
- e. Constituent chemicals and, if known, their percentages in the waste

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- f. EPA waste codes IAW 40 CFR 261
- g. Description of process generating waste
- h. Container type (i.e. metal drum, cardboard box, glass bottle, etc)
- i. Container size (volume of container, not volume of the waste)
- j. Number of containers requiring pick-up
- k. Physical state (solid / liquid / gas / sludge)
- l. pH of material
- m. Flashpoint of material
- n. Copy of SDS or sample results

3.7.4.2 Waste Petroleum Products

If an existing infrastructure component contains a petroleum product, and removal or alteration of the component results in the petroleum product becoming a waste to be disposed of, coordinate applicable disposal requirements with the 88 CEG/CEIE HW Program Manager and the Contracting Officer's representative.

3.7.5 HW Recycling

Any HW that is being offered for recycling (e.g., batteries, solvent waste being sent for recycling, wipes being sent for laundering) may be handled outside of the normal HW disposal channels. The Contractor shall obtain approval in writing from the Contracting Officer's representative in coordination with 88 CEG/CEIE prior to commencing any such operations. All recycled HW shall be accompanied with a manifest signed by 88 CEG/CEIE prior to removal from WPAFB.

3.8 ABOVEGROUND AND UNDERGROUND STORAGE TANKS

- a. This contract does not involve provision or alteration of aboveground storage tanks (AST) or underground storage tanks (UST).
- b. All tanks shall be constructed of material that is compatible with the product to be stored.
- c. All secondary containment areas for tank and piping systems shall be constructed of material compatible with the product to be stored. Containment areas shall also be impervious and capable of holding product and preventing contamination of soil and water in the event of a spill or release.
- d. All tanks shall be labeled as to product stored and associated hazards. All labeling shall meet applicable regulatory requirements.
- e. For ASTs, containment shall be provided by either external secondary containment of sufficient capacity to hold at least 110% of the total volume of the tank system or, alternatively, the tank shall have a double-wall with interstitial monitoring capability. The exception to this requirement is for ASTs holding extremely hazardous substances, such as sulfuric acid. Tanks intended to store HAZMATs, such as

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sulfuric acid, shall have a double-wall and external secondary containment. Containment for piping shall be either double-wall or installation of appropriate containment or diversionary structures or equipment to prevent a release from reaching surface water before cleanup can occur.

- f. Temporary storage tanks brought onto base shall also have acceptable, impervious secondary containment or a double wall for the tank system, including piping. Temporary tanks shall be properly labeled as to product stored and hazards and inspected regularly and the inspection documented. Accumulated storm water in secondary containment areas shall be inspected for the presence of product stored prior to discharge and the inspection documented. Any amount of product present in secondary containment areas shall be removed promptly and prior to discharge of any accumulated water. Oil-only absorbent pads can be used to remove small amounts (e.g., sheen) of petroleum product from accumulated storm water. Leaking tanks shall be repaired without delay. Users of temporary tanks shall receive required oil-handlers training prior to bringing the tank onto WPAFB. Oil handler training is provided through the 88 CEG/CEIE STPM.
- g. Dispensing of petroleum products shall be accomplished over an area impervious to the product and capable of preventing direct contact of the product with soil in the event of a spill or release.
- h. All tanks shall have spill protection around the fill pipe and overflow protection in the form of a high level alarm with both an audible and visual signal and an automatic shutoff to stop flow at 90% capacity of the tank unless otherwise authorized through the 88 CEG/CEIE STPM.
- i. All ASTs shall have an automatic tank gauging system capable of monitoring product and water levels inside the tank and the interstitial space for double-wall tanks.
- j. All USTs storing petroleum products shall meet requirements outlined in OAC 1301:7-9. All USTs shall have double wall construction and shall be equipped with an automatic tank gauging system for interstitial, product level, and water level monitoring. The UST system shall have a method, or combination of methods, for release detection that can detect a release from any portion of the tank and the connecting underground piping that routinely contains product. The UST system shall be equipped with both a visual and audible high level alarm to prevent overfills and a high liquid level pump cutoff device to stop flow at 90% full.
- k. A State of Ohio Bureau of Underground Storage Tank Regulations (BUSTR) permit is required prior to engaging in any of the following with respect to USTs: An installation, an upgrade or modification, a permanent abandonment, a permanent removal, a replacement, a repair, a change in service or a temporary closure. Contact 88 CEG/CEIE STPM (937-257-5899) for additional information regarding permits.
- l. The 88 CEG/CEIE STPM provides UST permitting support and submits tank permit and registration forms to the appropriate regulatory agencies. Contractors shall not submit any forms to the regulatory agencies unless first authorized to do so by the 88 CEG/CEIE STPM. Permit Applications for UST Work shall be submitted to 88 CEG/CEIE STPM at least 60 days prior to tank installation and contain design drawings of the UST system per Ohio BUSTR requirements. An UST registration form

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shall be submitted to the 88 CEG/CEIE STPM within 5 days after the tank is put into service.

3.9 CULTURAL RESOURCES

3.9.1 Inadvertent Discovery of Artifacts

If cultural or archaeological objects are encountered during a project, the Contractor shall immediately cease work in the discovery area; secure but not move the objects; and notify the Contracting Officer's representative and the 88 CEG/CEIE Cultural Resources Manager (CRM). The Government shall provide an in situ evaluation of the resources by a qualified archaeologist. Based on recommendations from the archaeologist and the State Historic Preservation Office, the CRM will advise the Contracting Officer's representative on further actions regarding the treatment of the objects.

3.9.2 Inadvertent Discovery of Human Remains

If human remains or objects are encountered during a project, the Contractor shall immediately cease work in the discovery area; secure but not move the objects; and notify the Contracting Officer's representative and the 88 CEG/CEIE CRM. The CRM shall in turn notify the installation commander, as the primary management authority, and the appropriate Native American organizations of the discovery. All activity in the area of the discovery shall be suspended for up to 30 days after certification that the appropriate Native American organizations have been notified. Disposition and control of the remains or objects shall follow the requirements of Section 3, Subsections (a) and (b), of the Native American Graves Protection and Repatriation Act and the procedures set forth for this in the current version of the WPAFB Integrated Cultural Resources Management Plan (ICRMP). Contact the 88 CEG/CEIE CRM at 937-257-1374 for more information about the ICRMP.

3.10 NATURAL RESOURCES

3.10.1 Removing Trees

Tree removal shall take place only within the period of 1 October through 31 March. If tree removal must be accomplished at other times, the Contractor shall comply with direction provided by the Contracting Officer in coordination with the 88 CEG/CEIE Natural Resources Manager. Procedural requirements pertaining to tree removal outside the approved period may be extensive and time-consuming.

3.10.2 Projects located within the Huffman Retarding Basin

- a. This contract does not involve site work within the boundary of the Huffman Retarding Basin as administered by the Miami Conservancy District (MCD).

-- End of Section --



Hazardous Materials Assessment Report

Facility 20824

Wright-Patterson AFB, Ohio



Project No. 103IG9132
September 2024

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1.0 INTRODUCTION

Tetra Tech was contracted by the Department of Defense (DoD) to perform a pre-renovation Hazardous Materials Assessment of Facility 20824 (Subject Property) at Wright Patterson Air Force Base (WPAFB) in Dayton, Montgomery County, Ohio (See Figures 1 and 2).

Tetra Tech prepared this Hazardous Materials Assessment Report for use by the DoD to identify actual and potential environmental hazards and/or conditions that would require proper removal and/or management prior to conducting demolition activities at the Subject Property.

1.1 Purpose

The objective of these survey activities was to identify potential environmental hazards that may require proper management prior to conducting demolition activities. Per the DoD request for proposal, the assessment focused primarily on evaluating the presence of lead-based paint (LBP), asbestos-containing materials (ACM), and other potentially hazardous materials associated with the Subject Property building. Information about the building is provided below:

Building	# of Floors	Total Area in Square Feet
Facility 20824	2	48,900

Tetra Tech performed a site reconnaissance and sampling effort to evaluate potential LBP, ACM and other hazardous materials associated with the Subject Property on November 1, 2023 and May 16, 2024. Site access was provided by Scott Hansen, Facility Operations Specialist.

1.2 Limitations and Exceptions

Tetra Tech did not employ destructive sampling technique (i.e., breaching walls, floors, and hard ceilings) in order to obtain samples. Therefore, we cannot state that all potential hazardous materials were identified. Thus, during any future construction or demolition activities, if suspect hazardous materials are discovered that were not identified in this report, work on the Subject Property should cease and the suspect materials should be evaluated further to determine appropriate methods of handling them.

2.0 SITE DESCRIPTION

Wright-Patterson Air Force Base is located in Dayton, Montgomery County, Ohio (Figures 1 and 2). The address is 2241 Avionics Circle, Wright-Patterson AFB, Ohio 45433.

Constructed in 1955, Facility 20824, located in Area B is a two story is a concrete framed research facility. Facility 20824 contains one full first floor, a partial second floor, and a mezzanine. The facility totals 48,846 gross square feet (GSF) and houses research activities exclusively. It has a Building Condition Index of 70. The facility has undergone minimal renovations since it was constructed. The facilities low slope, built-up roof was replaced in 1995. The next roof replacement is programmed for 2027.

The facility supports the 711th Human Performance Wing (711 HPW) providing space for Department of Defense (DoD) Research & Development (R&D). The 711th Human Performance Wing (711 HPW) is comprised of the Airman Systems Directorate (AFRL/RH) and the United States Air Force School of Aerospace Medicine (USAFSAM). The 711 HPW functions as a joint DoD Center of Excellence for human performance, sustainment and readiness, optimization, and enhancement through partnerships with the Naval Medical Research Unit-Dayton and nearby universities, industry, and medical institutions. The 711 HPW leads the development, integration, and delivery of Airman-centric research, education, and consultation enabling the Air Force to achieve responsive and effective vigilance, reach, and power on a global scale. Facility 20824 provides both medical laboratory and administrative space for 711 HPW/RH, as well as NAMRU-D research missions.

3.0 FIELD INVESTIGATION ACTIVITIES & RESULTS

Tetra Tech's field investigation activities consisted of three parts: ACM survey, LBP survey, and an environmental hazards evaluation. The assessment was conducted by Tetra Tech's Ohio accredited inspectors Vicky Farmer and Wes Williams (see Appendix A for Inspector Credentials).

3.1 Asbestos Containing Materials Survey

Asbestos was one of the first hazardous air pollutants regulated under the air toxics program. On March 31, 1971, United States Environmental Protection Agency (U.S. EPA) identified asbestos as a hazardous pollutant, and on April 6, 1973, U.S. EPA promulgated in the United States Code of Federal Regulations (CFR) Title 40 Chapter I, Subchapter C, Part 61, Subpart M, the Asbestos National Emissions Standards for Hazardous Air Pollutants (NESHAP). NESHAP defines ACM as a material containing more than 1% asbestos. NESHAP specifies work practices for asbestos to be followed during demolitions and renovations of all facilities, including, but not limited to, structures, installations, and buildings (excluding residential buildings that have four or fewer dwelling units). The regulations require a thorough inspection where the demolition or renovation operation will occur.

However, material identified to be less than 1% asbestos may be regulated by the Occupational Safety and Health Administration (OSHA). OSHA regulates renovation and demolition activities involving building materials which contain **any** amount of asbestos under the Construction Industry standard found in 29 Code of Federal Regulations 1926.1101.

As is common practice with asbestos inspections the number of potential ACM samples collected, and quantities were determined by Tetra Tech using Asbestos Hazard Emergency Response Act (AHERA) requirements. Even though the AHERA regulations were promulgated specifically to address asbestos in schools, it provides more specific guidelines for the collection of samples, which is utilized by most asbestos inspectors to ensure ACM are properly identified and quantified. According to AHERA there are three main kinds of materials, which include Surfacing Material, Thermal system Insulation (TSI), and Miscellaneous material. TSI may include, but is not limited to, insulation materials found on piping, fittings, boilers, tanks, ducts, or various components of heating and cooling systems. The classification bears implications for number of samples to be taken. For Surfacing material, there's the 3-5-7 rule, meaning 3 samples from less than 1,000 square feet area, 5 samples from 1,000 to 5,000 square feet area, and 7 samples from greater than 5,000 square feet area. For Thermal system material, with some exceptions, 3 samples should be taken for each homogeneous area. For Miscellaneous material, at least one sample should be taken from each homogeneous material.

Tetra Tech's field investigation activities consisted of identifying homogeneous areas of suspect ACM, collecting bulk samples, and quantifying identified ACM for each homogenous area. Tetra Tech identified homogeneous areas of suspect ACM by similar color and texture and collected samples from unique homogeneous areas of suspect ACM identified in the buildings. Tetra Tech estimated the amount of identified ACM in square footage by measuring ACM dimensions, where possible. Quantities of ACM used in this report should be considered as reasonable estimates. Tetra Tech initially classified each suspect ACM as friable or non-friable. A friable material is defined as any material that, when dry, can be pulverized by hand pressure. Ceiling tile and pipe insulation are examples of friable materials; undamaged floor tile and cement board are examples of non-friable materials. Pipe insulation can be treated as non-friable if the

outer covering is intact and undamaged and the insulation is not exposed. Tetra Tech identified the following suspect ACM homogeneous areas within the buildings:

Thermal System Insulation		
TSI Pipe Insulation		
Miscellaneous Materials		
Expansion Joint	Drywall	Foundation Caulk
Cove Base Molding Mastic	Floor Tile & Mastic	Window Caulk
Carpet Mastic	Joint Compound	Window Glazing
Ceiling Tile	Door Caulk	Fire Doors
Roofing Materials	Tower Siding	

Tetra Tech understands that fire doors are present on site. Fire doors were not sampled but should be considered Presumed ACM (PACM). As this was a pre-renovation survey, destructive sampling techniques were not utilized, and roofing materials and siding on the tower were not sampled. Therefore, the roofing and siding material should be considered PACM until it can be sampled. Potential transite paneling was observed on the exterior of the building's tower. This could not be accessed and is therefore PACM.

The TSI observed within the building was marked as asbestos containing and therefore not re-sampled. Tetra Tech gathered quantities only for the TSI.

During the field assessment, Tetra Tech inspected the building; visual observations including condition of potential ACM and potential for disturbance were recorded, and digital photographs of sampled or presumed ACM were taken. The photographic log containing select site photographs is included as Appendix B.

Tetra Tech collected bulk samples consistent with guidelines and requirements of AHERA (40 CFR 763.86) from each unique homogeneous area of suspect ACM in accordance with the U.S. EPA simplified sampling scheme. These sampling procedures provide basic guidelines for ensuring that suspect ACM was randomly sampled, and that a statistically significant number of samples was obtained for analysis. Between three (3) to seven (7) samples were taken from each homogenous area.

Samples of suspect ACM were sent to EMSL Analytical, Inc. (EMSL) located in Indianapolis, Indiana for bulk analysis via U.S. EPA Method 600/R-93-116 using Polarized Light Microscopy (PLM). EMSL is currently accredited by the U.S. Department of Commerce, National Voluntary Laboratory Accreditation Program (NVLAP) for Asbestos Fiber Analysis (see Appendix A for Inspector and Laboratory Credentials). Samples consisting of multiple suspect ACMs, such as floor tile and mastic, were submitted to the laboratory as one sample; however, EMSL separated and analyzed each distinct matrix for ACM content and identified each layer as "001a", "001b", etc. with their laboratory sample identification. The EMSL Laboratory Analytical Report is included in Appendix C. A summary of sample locations and descriptions are provided in Table 1. Approximate sample locations are depicted on the Figures 3a and 3b and are depicted by either red, blue, or green dots.

Samples identified as containing asbestos material or "positive" via the PLM analysis are listed in Table 2. Samples listed in Table 1 and NOT in Table 2 should be considered non-ACM based on the definition of an ACM as any material possessing asbestos with a greater than 1% composition. Copies of the EMSL laboratory analytical results for the ACM assessment are

included in Appendix C. The areas where the positive ACM was identified within the building are also shown as red dots on Figures 3a and 3b.

3.1.1 Asbestos Inspection Laboratory Results

Analytical results indicate that asbestos is present throughout the building in the following materials specifically described in Table 2:

- TSI (previously sampled);
- Grey Floor Tile (FT);
- Floor Tile Mastic (M) (was identified as <1%); and
- Carpet Mastic (CM) (was identified as <1%).

Sampled materials such as floor tile and mastic were often found layered underneath other materials, such as carpet mastic, making identification of like materials difficult, if not impossible.

Table 2 provides the percent and type of asbestos, friability, General Condition, and quantity of the material identified (including PACM) in each bulk sample collected during the asbestos inspection.

Note that several mastics identified were found to be less than 1% asbestos, which by definition (U.S. EPA), is not considered to be ACM. However, these materials do fall under the jurisdiction of the Occupational Health and Safety Administration (OSHA) for potential worker exposure and OSHA considers *any* amount to be asbestos. Table 2 also includes these less than 1% materials.

3.2 Lead-Based Paint Surveys

The Department of Health and Human Services (DHHS) has determined that lead and lead compounds are reasonably anticipated to be human carcinogens and the U.S. EPA has determined that lead is a probable human carcinogen. Therefore, lead containing materials must be properly managed in placed and/or properly abated and disposed during renovation or demolition activities. The purpose of these surveys was twofold: first, to determine if lead is present in any of the Subject Property buildings; and second, if detected, at what concentrations.

The building was evaluated to determine the number of homogeneous areas of painted surfaces. First, a Tetra Tech Lead Risk Assessor conducted an initial survey of each building to identify the different painted surfaces by color and listed them in a sample log. Tetra Tech utilized an appropriately calibrated Thermo Niton XL5 X-Ray Fluorescence (XRF) Analyzer to determine the concentration of lead in the paint on painted surfaces on the interior and exterior of the facilities.

The XRF was calibrated using a National Institute of Standards and Technology (NIST) calibration standard located within the base of the instrument's charging station. The XRF was calibrated before use. The calibration check result was internally determined by the instrument to be within +/- 20% of the stated value of the standard in order to pass calibration. The

instrument was recalibrated approximately sporadically throughout the assessment, or within four hours of initial calibration.

The U.S. EPA defines lead-based paint as equal to or greater than 0.5% by weight or 1.0 milligram per square centimeter (mg/cm²).

3.2.1 Lead-Based Paint Results

The survey was conducted by Mr. Wes Williams (see Appendix A for Inspector Credentials). Prior to the survey, the XRF was calibrated by placing it on the base and using the “auto-cal” function built in to the XRF. The Thermo Niton XL5 XRF Analyzer passed calibration and was utilized in the survey. The building was surveyed, and observable surfaces were analyzed; a total of 41 XRF screening samples were collected.

Results are presented in Table 3 along with sample names, descriptions, and locations. Note that Table 3 and Figures show the total samples collected, however those readings associated with calibration and/or testing of the equipment are not included. Figures 4a and 4b show the locations of the XRF screening samples shown with either blue or pink dots.

Note that some rooms were not assessable during the XRF assessment phase of this evaluation.

Of the 41 data points collected throughout the building (Table 3), **none** were found to contain LBP above 1 mg/cm².

3.3 Evaluation of Other Hazardous Materials

During the site inspection, Tetra Tech visually evaluated the interior of the building for other potentially hazardous materials that would require appropriate management prior to any renovation activities. Potential hazardous materials that were evaluated during the inspections included, but were not limited to, Polychlorinated Biphenyl (PCB) containing equipment, mold, fuel storage tanks, mercury containing devices, exit signs, smoke detectors, Chlorofluorocarbon (CFC) containing equipment, fluorescent lighting ballasts/fixtures, cleaning materials, paints, oils, and solvents. A summary of these findings are included in Table 4.

3.3.1 PCB Equipment

Tetra Tech did not observe any obvious PCB containing equipment (i.e. transformers, hydraulic oils, etc.) during the site survey. However, see Section 3.3.5 below for potential PCB containing light ballasts. For the purposes of this evaluation, equipment should be assumed to contain PCBs.

3.3.2 Mold

Tetra Tech did not observe obvious evidence of mold growing in the areas surveyed.

3.3.3 Storage Tanks

Tetra Tech visually inspected the interior and the exterior of the building for any evidence of

aboveground storage tanks (ASTs) or underground storage tanks (USTs). Tetra Tech did not observe any obvious evidence indicating the current or past presence of ASTs or USTs.

3.3.4 Mercury Containing Devices

Tetra Tech visually inspected the interior of the building for the presence of thermometers and thermostats. Tetra Tech observed 13 potential mercury-containing devices, including thermostats throughout the building and thermometers associated with the building's air handlers. Additionally, see section 3.3.5 below for potential mercury-containing fluorescent lights.

3.3.5 Fluorescent Lighting

Fluorescent bulbs often contain mercury and ballasts may contain PCBs. Fluorescent light ballasts and bulbs were observed throughout the building. There was no obvious information available as to whether all the fluorescent lights contained mercury or if the ballasts contained PCBs (i.e., non-PCB labeling). Therefore, for the purposes of this evaluation, all fluorescent lights should be assumed to be mercury-containing and all light ballasts assumed to contain PCBs.

3.3.6 Potentially Hazardous Containers/Materials

Two containers of hydraulic fluid were observed within the garage area, along with hazardous materials cabinets by the Manikin Testing Lab. The cabinets contained various small (5-gallon or less) containers. Compressed nitrogen was also observed to be stored within the building.

3.3.7 Exit Signs

Certain "Exit" signs that glow in the dark often contain a radioactive gas called tritium. These signs do not require electricity or batteries and can be used where it is hard to install electric signs, such as above doors. They serve an important safety function by marking exits to be used during power outages and emergencies. The signs pose little or no threat to public health and safety or security. Facilities that use tritium "Exit" signs are considered "general licensees," meaning they do not need a specific license to use the signs. The signs are considered "generally licensed devices," and are designed to be inherently safe so they can be used without the need for radiation training.

However, the signs are subject to certain regulatory requirements. Most important is the requirement for proper disposal, because a damaged sign could contaminate the immediate area and require expensive cleanup.

Tetra Tech observed exit signs in the building. The signs appeared to be in good condition, however there was no obvious indication to determine if they contained tritium.

3.3.8 Smoke Detectors

Smoke detectors typically fall within two categories: photoelectric and ionization. When it comes to disposal, old photoelectric detectors can be safely put in the trash, so long as you remove the battery first.

Ionization-Based Smoke Detectors contain a small amount of Americium 241, a radioactive isotope. When in use, these detectors are perfectly safe, as the radioactive material is shielded by a metal chamber inside the device. If dismantled, however, the radioactive material can become exposed.

Tetra Tech observed smoke detectors in the building. The smoke detectors appeared to be in good condition, however there was no obvious indication to determine if they were ionization type.

3.3.9 CFCs

Tetra Tech observed evidence of an HVAC system on all floors throughout the building during the site visit. The units were not able to be inspected, and therefore the presence/absence of CFCs was not able to be determined.

3.3.10 Other Potentially Hazardous Materials

Tetra Tech observed “white goods” such as microwaves, stoves, furnaces, and water heaters throughout the building. Out of use appliances can pose a variety of risks and some require proper removal and disposal.

In addition, Tetra Tech also observed electronics such as computer monitors and printers, and televisions that were in use in the building.

Electronic products can contain forms of heavy metals (i.e., mercury, copper, barium, nickel and arsenic), semi-metals, and various chemical compounds that may be considered hazardous once products are no longer in use and require disposal. If disposal is required, products should be properly managed and disposed in accordance with local, State and Federal regulations.

4.0 RECOMMENDATIONS

A pre-renovation Hazardous Materials Assessment was completed at Facility 20824 to identify potential environmental hazards that may require proper management in place and/or during demolition activities. Based on the survey, Tetra Tech recommends the following:

4.1 ACM

In total, seven (7) types of ACM and PACM were identified throughout the building:

1. Floor tile;
2. TSI;
3. Floor tile mastic (< 1%);
4. Carpet mastic (< 1%);
5. Fire Rated Doors (PACM);
6. Exterior Siding/Paneling (PACM); and
7. Roofing Materials (PACM).

See Table 2 and the previously identified figures.

ACM Recommendations

The TSI is considered to be "Friable asbestos material" per the Ohio Administrative Code (OAC) 3745-20-01. According to OAC 3745-20-01, friable asbestos material means any material containing more than one per cent asbestos by area, as determined using the method specified in 40 CFR Part 763, Subpart E, Appendix E, Section 1 PLM, that, when dry can be crumbled, pulverized, or reduced to powder by hand pressure. Friable asbestos materials are considered to be regulated asbestos containing material (RACM), which means it must be properly removed by a certified abatement professional licensed asbestos abatement contractor or appropriately managed in place per local, State, and Federal guidelines.

Floor Tile, mastics, and roofing materials (PACM) are considered Category I non-friable asbestos-containing materials, per the Ohio Administrative Code (OAC) 3745-20-01. According to OAC 3745-20-01 "Category I nonfriable asbestos-containing material" means asbestos-containing packing, gaskets, resilient floor covering, and asphalt roofing products containing more than 1% asbestos as determined using the method specified in 40 CFR Part 763, Subpart E, Appendix E, Section 1, PLM.

Fire rated doors (PACM)s and exterior siding/paneling (PACM) would be considered Category II non-friable asbestos containing materials, per OAC 3745-20-01. According to OAC 3745-20-01 "Category II nonfriable asbestos-containing material" means any material, excluding Category I nonfriable asbestos-containing material, containing more than 1% asbestos as determined using the method specified in 40 CFR Part 763, Subpart E, Appendix E, Section 1, PLM, that, when dry, cannot be crumbled, pulverized, or reduced to powder by hand pressure.

Category I or II ACM that becomes damaged during removal or abatement resulting in small fragments four square inches or less is considered friable or a RACM.

NESHAP defines asbestos-containing material as a material containing more than 1% asbestos. However, materials identified to be less than 1% asbestos (window caulk and carpet mastic) may

be regulated by the Occupational Safety and Health Administration (OSHA). OSHA regulates renovation and demolition activities involving building materials which contain **any** amount of asbestos under the Construction Industry standard found in 29 CFR 1926.1101.

In general, ACM was observed to be in good condition. Therefore, the ACM can be left in place and managed through an asbestos management plan until demolition occurs. The disturbance or abatement of the ACM should be conducted by a licensed asbestos abatement contractor and in accordance with local, State, and Federal regulations.

Asbestos-containing materials can be managed with an Operations and Maintenance (O&M) Plan. Although not specifically required by regulation, an O&M Plan can incorporate elements that are required to comply with 29 CFR 1926.1101 (OSHA Construction Asbestos Standard) and 29 CFR 1910.1001 (OSHA Asbestos General Industry Standard). Elements of this Plan could include notification and labeling procedures, documentation of the identified asbestos, training, recordkeeping procedures, surveillance program, requirements for maintenance, renovation or demolition activities. Tetra Tech recommends that an O&M Plan be developed to properly manage the identified asbestos at the facility.

4.2 Lead-Based Paint

Throughout the building, no detections of lead were found above the U.S. EPA definition of lead-based paint. No additional work is required.

4.3 Fluorescent Lights

Fluorescent bulbs often contain mercury and ballasts may contain PCBs. Fluorescent light ballasts and bulbs were observed throughout the building. Several bulbs and ballasts were checked when possible. No identifying information was found to indicate whether the ballasts contained PCBs or the bulbs were mercury-containing. Therefore, for the purposes of this evaluation, all fluorescent lights should be assumed to be mercury-containing and all light ballasts assumed to contain PCBs. Bulbs and ballasts must be properly managed and disposed in accordance with local, State, and Federal regulations during maintenance, renovation, or demolition activities.

4.4 Mercury Containing Devices

Tetra Tech observed potential mercury-containing devices, including thermostats throughout the building and thermometers associated with the building's air handlers. Additionally, see section 3.3.5 for potential mercury-containing fluorescent lights. Mercury containing devices must be properly managed and disposed in accordance with local, State, and Federal regulations during maintenance, renovation, or demolition activities.

4.5 Potential Hazardous Containers/Materials

Tetra Tech observed minimal evidence of hazardous materials. However, hazardous materials must be properly labeled, stored, and managed (i.e., secondary containment, Safety Data Sheets, inventory, sections, etc.). Tetra Tech recommends managing these materials in accordance with applicable site, local, state and federal regulations and guidelines.

4.6 Exit Signs

Tetra Tech observed exit signs throughout the building. The signs appeared to be new and in good condition. There was no obvious indication that the signs contained tritium (stickers, labeling, etc.); therefore, the signs should be assumed to contain tritium and should be handled and disposed of accordingly.

4.7 Smoke Detectors

Tetra Tech observed smoke detectors in the building. The smoke detectors appeared new and in good condition. There was no obvious indication that the detectors were ionization (labeling, stickers, etc.). Therefore, these detectors should be assumed to be ionization-based smoke detectors until their disposal, at which time they should either be tested or disposed of accordingly.

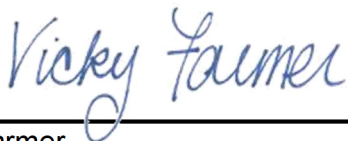
4.8 CFCs

Tetra Tech observed evidence of an HVAC system; the presence/absence of CFCs was not able to be confirmed. Tetra Tech recommends managing these materials appropriately at the end of their useful life.

4.9 Other Potentially Hazardous Materials

Tetra Tech observed electronics such as computer monitors and printers, and televisions that were in use in the building.

Electronic products can contain forms of heavy metals (i.e., mercury, copper, barium, nickel and arsenic), semi-metals, and various chemical compounds that may be considered hazardous once products are no longer in use and require disposal. Tetra Tech recommends managing these materials appropriately at the end of their useful life. If disposal is required, products should be properly managed and disposed in accordance with local, State and Federal regulations.

5.0 SIGNATURES OF ENVIRONMENTAL PROFESSIONALS

September 11, 2024

Vicky Farmer
Ohio Asbestos Hazard Evaluation Specialist

Date

Qualifications

Ms. Farmer is an Environmental Scientist for the Tetra Tech Cincinnati, Ohio office and is an Ohio Accredited Asbestos Inspector and Management Planner. She graduated in 2001 with a Bachelor of Science from Morehead State University. Her experience includes a variety of Phase I Environmental Site Assessments, and numerous asbestos and lead based paint surveys for commercial and industrial buildings within the state of Ohio and throughout the Midwest.



September 11, 2024

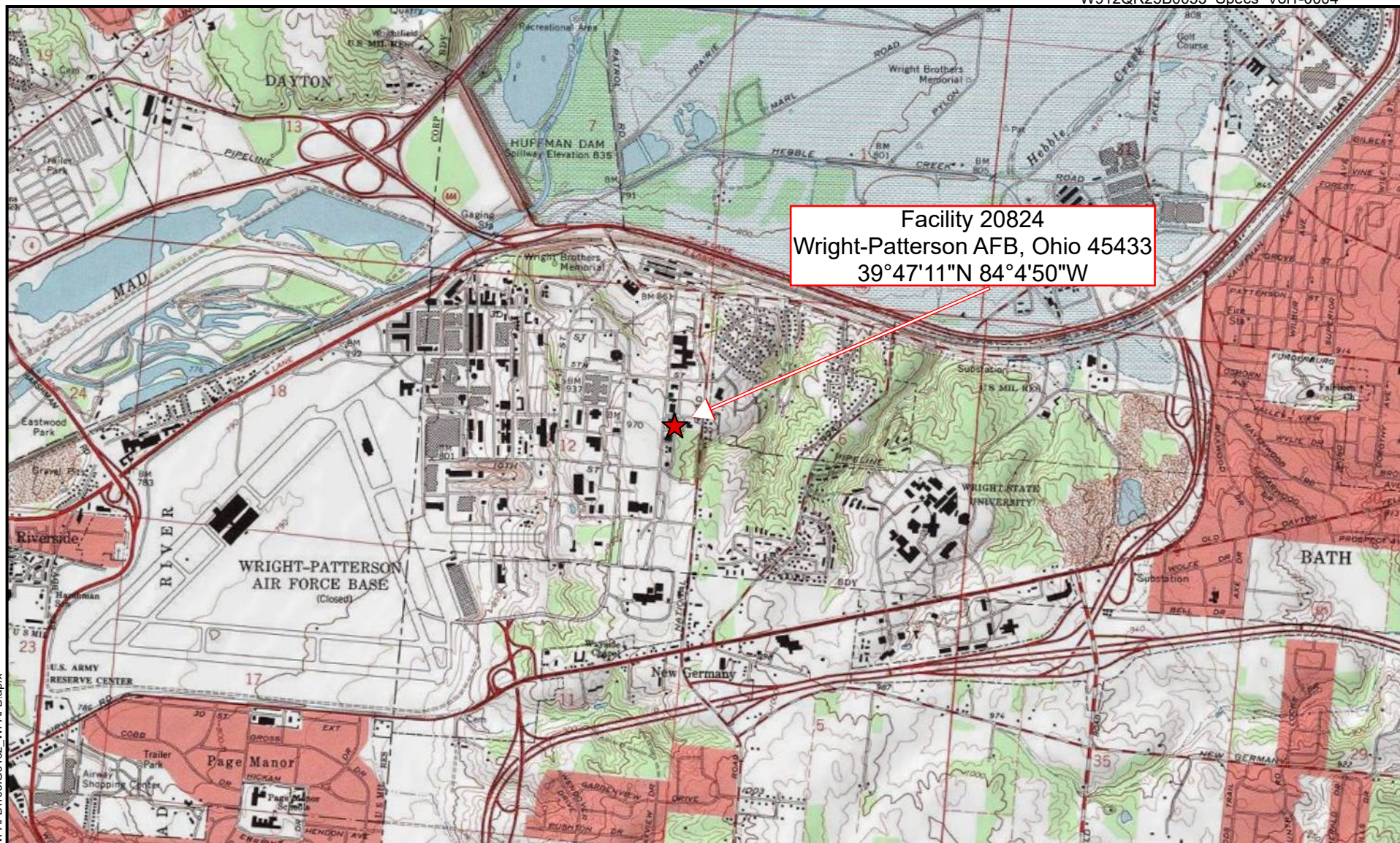
Douglas J. McIlvain, M. En., CHMM
Senior Environmental Scientist

Date

Qualifications

Mr. Douglas J. McIlvain is a Senior Environmental Scientist and Program Manager for Tetra Tech in Cincinnati, Ohio. Mr. McIlvain has over 29 years of experience in the environmental field and is a Certified Hazardous Materials Manager. He holds a Bachelor's degree in Zoology and a Master's degree in Environmental Science from Miami University in Oxford, Ohio. Mr. McIlvain's experience includes performing and/or managing hundreds of Phase I and Phase II Environmental Site Assessments, asbestos surveys, lead surveys, mold assessments, and brownfields redevelopment for both public and private sector clients at a wide variety of residential, commercial, and industrial sites throughout the United States. Mr. McIlvain is currently or has previously served as the Program Manager for a variety of environmental, planning, training, and energy support services provided to the USACE, DOD, and Ohio Army National Guard.

Figures



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Legend

★ Site Property Location

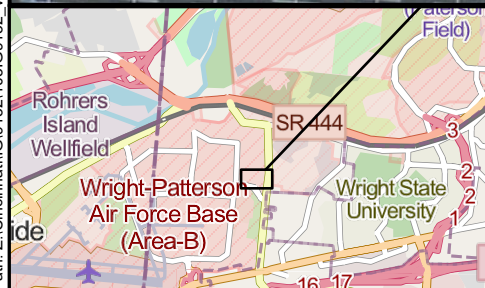
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FeetFacility 20824
2241 Avionics Circle
Wright-Patterson AFB, Ohio 45433Figure 1
Site Location Map

Prepared for: USACE

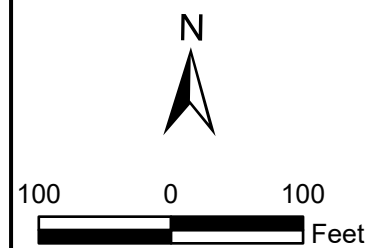
Prepared by: Bridget Taylor

Date Saved: 6/12/2024

Coordinate System: NAD 1983 2011 StatePlane Ohio South FIPS 3402 Ft US

**Legend**

★ Site Property Location



Facility 20824
2241 Avionics Circle
Wright-Patterson AFB, Ohio 45433

Figure 2
Site Layout Map



Prepared for: USACE

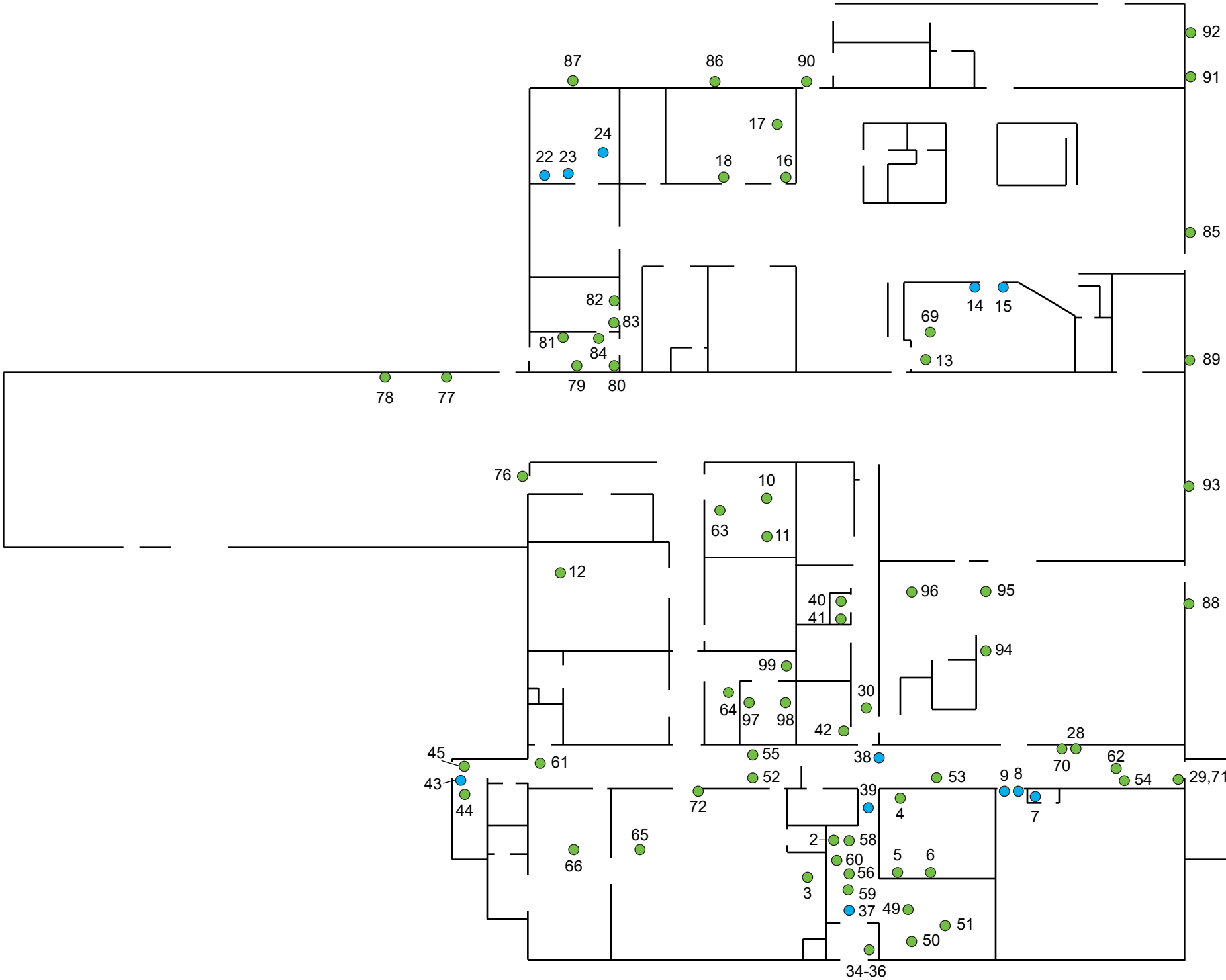
Prepared by: Bridget Taylor

Coordinate System: NAD 1983 2011 StatePlane Ohio South FIPS 3402 Ft US

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Number	Sample ID	Description
2	824-CM1-2	Carpet mastic
3	824-CM1-3	
4	824-CM2-1	
5	824-CM2-2	Carpet mastic
6	824-CM2-3	
7	824-CM3-1	Carpet mastic
8	824-CM3-2	
9	824-CM3-3	
10	824-CM4-1	Carpet mastic
11	824-CM4-2	
12	824-CM4-3	
13	824-CM5-1	Carpet mastic
14	824-CM5-2	
15	824-CM5-3	
16	824-CM6-1	Carpet mastic
17	824-CM6-2	
18	824-CM6-3	
22	824-CM8-1	Carpet mastic
23	824-CM8-2	
24	824-CM8-3	
28	824-DWJC1-1	Drywall / Joint Compound
29	824-DWJC1-2	
30	824-DWJC1-3	Floor tile
34	824-FT1-1	
35	824-FT1-2	
36	824-FT1-3	Floor tile
37	824-FT2-1	
38	824-FT2-2	
39	824-FT2-3	Floor tile
40	824-FT3-1	
41	824-FT3-2	
42	824-FT3-3	Floor tile
43	824-FT4-1	
44	824-FT4-2	
45	824-FT4-3	Ceiling tile
49	824-CT1-1	
50	824-CT1-2	
51	824-CT1-3	Ceiling tile
52	824-CT2-1	
53	824-CT2-2	
54	824-CT2-3	

Number	Sample ID	Description
55	824-CT3-1	Ceiling tile
56	824-CT3-2	
58	824-CT4-1	Ceiling tile
59	824-CT4-2	
60	824-CT4-3	Ceiling tile
61	824-CT5-1	
62	824-CT5-2	
63	824-CT5-3	Ceiling tile
64	824-CT6-1	
65	824-CT6-2	Ceiling tile
66	824-CT6-3	
69	824-CT7-3	Ceiling tile
70	824-CBM1-1	Cove base mastic
71	824-CBM1-2	
72	824-CBM1-3	Cove base mastic
76	824-CBM3-1	
77	824-CBM3-2	
78	824-CBM3-3	Cove base mastic
79	824-CBM4-1	
80	824-CBM4-2	Cove base mastic
81	824-CBM4-3	
82	824-CBM5-1	Cove base mastic
83	824-CBM5-2	
84	824-CBM5-3	Window caulk
85	824-WC-1	
86	824-WC-2	
87	824-WC-3	Door caulk
88	824-DC-1	
89	824-DC-2	Door caulk
90	824-DC-3	
91	824-FC-1	Foundation caulk
92	824-FC-2	
93	824-FC-3	
94	824-EJ-1	Expansion joint
95	824-EJ-2	
96	824-EJ-3	Ceiling tile
97	824-CT8-1	
98	824-CT8-2	
99	824-CT8-3	



Legend

- Negative ACM Sampling Location
- Less than 1% ACM Sampling Location
- Positive ACM Sampling Location

Note

- ACM - Asbestos Containing Materials
- PACM - Presumed Asbestos Containing Materials



Not to scale

Facility 20824 - First Floor
2241 Avionics Circle
Wright-Patterson AFB, Ohio 45433

Figure 3a
Asbestos Sample Location Map



Prepared For: USACE

Prepared by: Bridget Taylor

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Number	Sample ID	Description
1	824-CM1-1	Carpet mastic
19	824-CM7-1	Carpet mastic
20	824-CM7-2	
21	824-CM7-3	
25	824-CM9-1	Carpet mastic
26	824-CM9-2	
27	824-CM9-3	
31	824-DWJC2-1	Drywall / Joint compound
32	824-DWJC2-2	
33	824-DWJC2-3	
46	824-FT5-1	Floor tile
47	824-FT5-2	
48	824-FT5-3	
57	824-CT3-3	Ceiling tile
67	824-CT7-1	Ceiling tile
68	824-CT7-2	
73	824-CBM2-1	Cove base mastic
74	824-CBM2-2	
75	824-CBM2-3	



Legend

- Negative ACM Sampling Location
- Less than 1% ACM Sampling Location
- Positive ACM Sampling Location

Note

- ACM - Asbestos Containing Materials
- PACM - Presumed Asbestos Containing Materials



Not to scale

Facility 20824 - Second Floor
2241 Avionics Circle
Wright-Patterson AFB, Ohio 45433

Figure 3b
Asbestos Sample Location Map



Prepared For: USACE

Prepared by: Bridget Taylor

Number	Location	Component
1	Main entrance	Wall
2	Conference room	Wall
3	Conference room door	Door
4	Elevator mechanical room	Door
5	Elevator exterior - 1st floor	Door
6	Door marked 110	Door
7	Door marked 104	Door
8	Room 104	Wall
9	Door marked 107	Door
10	Room 107	Wall
11	Room 107	Wall
12	Door marked 132	Door
13	Door to southwest stairwell	Door
14	Southwest stairwell	Wall
15	Door/exit from SW stairwell	Door
16	Southwest stairwell handrail	Handrail
17	Door marked 126	Door
18	Door to shower/mother's room	Door
19	Door marked 127	Door
20	Storage garage	Wall
21	Storage garage	Floor
22	Storage garage	Door
23	Storage garage	Handrail
24	Storage garage	Wall
25	Hallway to lab	Wall
26	Room 124	Wall
27	Door marked 123	Door
28	Calibration and bench area	Wall
29	Calibration and bench area	Guardrail
30	Calibration and bench area	Floor
31	Calibration and bench area	Floor
32	Room 123	Wall
33	Calibration and bench area	Wall
34	Room 134	Wall
35	Room 133	Door jam



Legend

- Negative LXP XRF Screening Location
- Positive LBP XRF Screening Location

Note

- LBP - Lead-Based Paint
- XRF - X-Ray Fluorescence



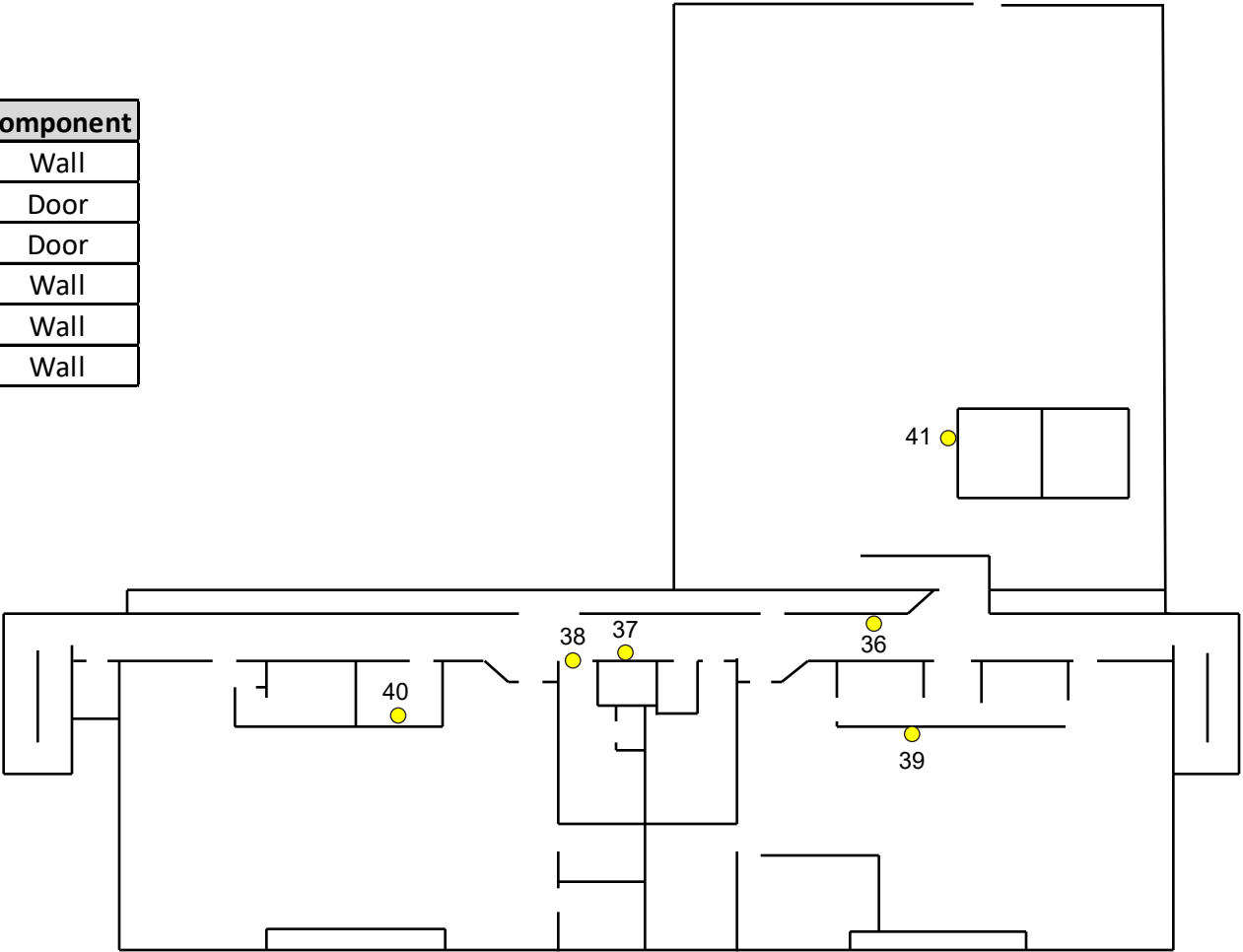
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Facility 20824 - First Floor
2241 Avionics Circle
Wright-Patterson AFB, Ohio 45433

Figure 4a
LBP XRF Screening Locations



Number	Location	Component
36	Hallway of second floor	Wall
37	Elevator exterior - 2nd floor	Door
38	Restroom door	Door
39	Room 206	Wall
40	Room 203	Wall
41	Mezzanine	Wall



Legend

- Negative LXP XRF Screening Location
- Positive LBP XRF Screening Location

Note

- LBP - Lead-Based Paint
- XRF - X-Ray Fluorescence



Not to scale

Facility 20824 - Second Floor
2241 Avionics Circle
Wright-Patterson AFB, Ohio 45433

Figure 4b
LBP XRF Screening Locations



Prepared For: USACE

Prepared by: Bridget Taylor

Tables

Table 1 - Asbestos Sampling Locations and Descriptions

Sample ID	Laboratory ID	Figure ID	Material	Color/Description	Location	Friable/Non-Friable
824-CM1-1-Mastic	162410144-001	1	Multi-layered - Carpet mastic, floor tile, floor tile mastic	Clear carpet mastic over tan and gray floor tile with clear/black mastic	Entryway and rooms 224 and 225	Non-Friable
824-CM1-1-Floor Tile	162410144-001A	1				
824-CM1-2-Mastic	162410144-002	2				
824-CM1-2-Floor Tile	162410144-002A	2				
824-CM1-2-Mastic	162410144-002B	2				
824-CM1-3-Mastic	162410144-003	3				
824-CM1-3-Floor Tile	162410144-003A	3				
824-CM1-3-Mastic	162410144-003B	3				
824-CM2-1	162410144-0004	4	Carpet mastic	Gray/tan mastic	Room 104	Non-Friable
824-CM2-2	162410144-0005	5				
824-CM2-3	162410144-0006	6				
824-CM3-1	162410144-0007	7	Carpet mastic	Black/yellow mastic	Room 107	Non-Friable
824-CM3-2	162410144-0008	8				
824-CM3-3	162410144-0009	9				
824-CM4-1	162410144-0010	11	Carpet mastic	Tan/black mastic	Rooms 122 and 124	Non-Friable
824-CM4-2	162410144-0011	12				
824-CM4-3	162410144-0012	13				
824-CM5-1	162410144-0013	13	Carpet mastic	Gray/black/tan mastic	Room 136	Non-Friable
824-CM5-2	162410144-0014	14				
824-CM5-3	162410144-0015	15				
824-CM6-1	162410144-0016	16	Carpet mastic	Tan mastic	Rooms 138 and 142	Non-Friable
824-CM6-2	162410144-0017	17				
824-CM6-3	162410144-0018	18				
824-CM7-1	162410144-0019	19	Carpet mastic	Tan mastic	Room 141, upper offices	Non-Friable
824-CM7-2	162410144-0020	20				
824-CM7-3	162410144-0021	21				
824-CM8-1	162410144-0022	22	Carpet mastic	Black/tan mastic	Room 140	Non-Friable
824-CM8-2	162410144-0023	23				
824-CM8-3	162410144-0024	24				
824-CM9-1	162410144-0025	25	Carpet mastic	Tan/green mastic	Second floor hallway	Non-Friable
824-CM9-2	162410144-0026	26				
824-CM9-3	162410144-0027	27				

Table 1 - Asbestos Sampling Locations and Descriptions

Sample ID	Laboratory ID	Figure ID	Material	Color/Description	Location	Friable/Non-Friable
824-DWJC1-1-Drywall	162410144-0028	28	Multi-layered drywall with joint compound	Brown/white sheetrock with white joint compound	Throughout first floor	Non-Friable
824-DWJC1-1-Joint Compound	162410144-0028A	28				
824-DWJC1-2-Drywall	162410144-0029	29				
824-DWJC1-2-Joint Compound	162410144-0029A	29				
824-DWJC1-2-Mastic	162410144-0029B	29				
824-DWJC1-3-Joint Compound	162410144-0030A	30				
824-DWJC-2-1-Joint Compound	162410144-0031A	31	Multi-layered drywall with joint compound	Brown/white sheetrock with white joint compound	Throughout second floor	Non-Friable
824-DWJC-2-1-Mastic	162410144-0031B	31				
824-DWJC-2-2-Joint Compound	162410144-0032A	32				
824-DWJC-2-3-Drywall	162410144-0033	33				
824-DWJC-2-3-Foam	162410144-0033B	33				
824-FT1-1-Floor Tile	162410144-0034	34	Multi-layered - Floor tile and mastic	Gray floor tile with gray/tan/black mastic	Lobby	Non-Friable
824-FT1-1-Mastic	162410144-0034A	34				
824-FT1-2-Floor Tile	162410144-0035	35				
824-FT1-2-Mastic	162410144-0035A	35				
824-FT1-3-Floor Tile	162410144-0036	36				
824-FT1-3-Mastic	162410144-0036A	36				
824-FT2-1-Floor Tile	162410144-0037	37	Multi-layered - Floor tile and mastic	12x12 Tan floor tile, black mastic	First floor hallway, vestibule	Non-Friable
824-FT2-1-Mastic	162410144-0037A	37				
824-FT2-2-Floor Tile	162410144-0038	38				
824-FT2-2-Mastic	162410144-0038A	38				
824-FT2-3-Floor Tile	162410144-0039	39				
824-FT2-3-Mastic	162410144-0039A	39				
824-FT3-1-Floor Tile	162410144-0040	40	Multi-layered - Floor tile and mastic	12x12 Beige floor tile, tan mastic	Outside first floor bathrooms	Non-Friable
824-FT3-1-Mastic	162410144-0040A	40				
824-FT3-2-Floor Tile	162410144-0041	41				
824-FT3-2-Mastic	162410144-0041A	41				
824-FT3-3-Floor Tile	162410144-0042A	42				
824-FT4-1-Floor Tile	162410144-0043	43	Multi-layered - Floor tile and mastic	12x12 Gray floor tile, gray/tan/black mastic	Room 139, hallway by south stairwell, north stairwell	Non-Friable
824-FT4-1-Mastic	162410144-0043A	43				
824-FT4-2-Floor Tile	162410144-0044	44				
824-FT4-2-Mastic	162410144-0044A	44				
824-FT4-3-Floor Tile	162410144-0045	45				
824-FT4-3-Mastic	162410144-0045A	45				

Table 1 - Asbestos Sampling Locations and Descriptions

Sample ID	Laboratory ID	Figure ID	Material	Color/Description	Location	Friable/Non-Friable
824-FT5-1-Floor Tile	162410144-0046	46	Multi-layered - Floor tile and mastic	12x12 Gray floor tile, clear mastic	Second floor hallway	Non-Friable
824-FT5-2-Floor Tile	162410144-0047	47				
824-FT5-3-Floor Tile	162410144-0048	48				
824-FT5-3-Mastic	162410144-0048A	48				
824-CT1-1	162410144-0049	49	Ceiling tile	2x2 deep wormhole ceiling tile	Conference room	Friable
824-CT1-2	162410144-0050	50				
824-CT1-3	162410144-0051	51				
824-CT2-1	162410144-0052	52	Ceiling tile	2x2 wormhole/pin hole ceiling tile	Throughout first floor	Friable
824-CT2-2	162410144-0053	53				
824-CT2-3	162410144-0054	54				
824-CT3-1	162410144-0055	55	Ceiling tile	2x4 work track ceiling tile	Entryway and upper offices	Friable
824-CT3-2	162410144-0056	56				
824-CT3-3	162410144-0057	57				
824-CT4-1	162410144-0058	58	Ceiling tile	2x4 wormhole/pin hole smooth ceiling tile	Entryway	Friable
824-CT4-2	162410144-0059	59				
824-CT4-3	162410144-0060	60				
824-CT5-1	162410144-0061	61	Ceiling tile	2x4 cratered/pin hole ceiling tile	Hallway, Room 124	Friable
824-CT5-2	162410144-0062	62				
824-CT5-3	162410144-0063	63				
824-CT6-1	162410144-0064	64	Ceiling tile	Drywall ceiling tile	Rooms 111 and 130, labs	Friable
824-CT6-2	162410144-0065	65				
824-CT6-3	162410144-0066	66				
824-CT7-1-Ceiling Tile	162410144-0067	67	Ceiling tile	1x1 pin hole ceiling tile	Rooms 136, 224, abd 225	Friable
824-CT7-1-Mastic	162410144-0067A	67				
824-CT7-2-Ceiling Tile	162410144-0068	68				
824-CT7-2-Mastic	162410144-0068A	68				
824-CT7-3-Ceiling Tile	162410144-0069	69				
824-CT7-3-Mastic	162410144-0069A	69				
824-CBM1-1	162410144-0070	70	Cove base mastic	Tan/brown mastic	First floor, throughout	Non-Friable
824-CBM1-2-Mastic	162410144-0071	71				
824-CBM1-2-Mastic	162410144-0071A	71				
824-CBM1-3-Mastic	162410144-0072	72				
824-CBM1-3-Mastic	162410144-0072A	72				

Table 1 - Asbestos Sampling Locations and Descriptions

Sample ID	Laboratory ID	Figure ID	Material	Color/Description	Location	Friable/Non-Friable
824-CBM2-1-Mastic	162410144-0073	73	Cove base mastic	Tan/brown mastic	Second floor, throughout	Non-Friable
824-CBM2-1-Mastic	162410144-0073A	73				
824-CBM2-2-Mastic	162410144-0074	74				
824-CBM2-2-Mastic	162410144-0074A	74				
824-CBM2-3-Mastic	162410144-0075	75				
824-CBM2-3-Mastic	162410144-0075A	75				
824-CBM3-1	162410144-0076	76	Cove base mastic	Brown mastic	Room 133	Non-Friable
824-CBM3-2	162410144-0077	77				
824-CBM3-3	162410144-0078	78				
824-CBM4-1	162410144-0079	79	Cove base mastic	Tan mastic	Rooms 137 and 141, bathroom by 141	Non-Friable
824-CBM4-2	162410144-0080	80				
824-CBM4-3	162410144-0081	81				
824-CBM5-1	162410144-0082	82	Cove base mastic	Tan mastic	Room 138	Non-Friable
824-CBM5-2	162410144-0083	83				
824-CBM5-3	162410144-0084	84				
824-WC-1	162410144-0085	85	Window caulk	Brown caulk	Exterior windows	Non-Friable
824-WC-2	162410144-0086	86				
824-WC-3	162410144-0087	87				
824-DC-1	162410144-0088	88	Door caulk	Gray/white caulk	Exterior doors	Non-Friable
824-DC-2	162410144-0089	89				
824-DC-3	162410144-0090	90				
824-FC-1	162410144-0091	91	Foundation caulk	Gray caulk	Exterior foundation	Non-Friable
824-FC-2	162410144-0092	92				
824-FC-3	162410144-0093	93				
824-EJ-1	162410144-0094	94	Expansion joint	Black/gray caulk	Storage garage	Non-Friable
824-EJ-2	162410144-0095	95				
824-EJ-3	162410144-0096	96				
824-CT8-1-Ceiling Tile	162410144-0097	97	Ceiling tile	1x1 pin hole ceiling tile	Room 130, drill press room	Friable
824-CT8-1-Mastic	162410144-0097A	97				
824-CT8-2-Ceiling Tile	162410144-0098	98				
824-CT8-2-Mastic	162410144-0098A	98				
824-CT8-3-Ceiling Tile	162410144-0099	99				
824-CT8-3-Mastic	162410144-0099A	99				
TSI Pipe Runs	Not sampled	NA	Pipe runs	Pipe wrap and elbows	Visible in garage and mezzanine	Friable
Fire Rated Doors	Not sampled	NA	Fire Doors	Various	Throughout	Non-Friable

Table 1 - Asbestos Sampling Locations and Descriptions

Sample ID	Laboratory ID	Figure ID	Material	Color/Description	Location	Friable/Non-Friable
Roofing	Not sampled	NA	Roofing	Flat black built-up roofing	Exterior	Non-Friable
Exterior siding/paneling	Not sampled	NA	Paneling	Gray panels	Exterior on the tower	Non-Friable

Notes

CBM = Cove Base Molding

CM = Carpet Mastic

CT = Ceiling Tile

DC = Door Caulk

DWJC = Drywall Joint Compound

EJ = Expansion Joint

FC = Foundation Caulk

FT = Floor Tile

TSI = Thermal System Insulation

WC = Window Caulk

Sources:

Tetra Tech Field Notes

EMSL Laboratory Analytical Reports

Note that Figure ID corresponds to the sample location on the figures.

Table 2 - Positive Asbestos Sampling Results Locations

Sample ID	Laboratory ID	Figure ID	Material	Color/Description	Location	Friable/Non Friable	Condition	Approximate LF or SF	% and Type of Asbestos
824-CM1-1-Floor Tile	162410144-001A	1	Floor tile	Gray floor tile	Rooms 224 and 225	Non-Friable	Good	820 SF	2.8% Chrysotile
TSI	Not sampled	NA	Pipe runs	Pipe wrap and elbows	Visible in garage and mezzanine	Friable	Good	350 lf of pipe run, 953 elbows	Not sampled by Tt; previously sampled and labeled as ACM
824-CM3-1	162410144-0007	7	Carpet mastic	Black/yellow mastic	Room 107	Non-Friable	Good	1,515 SF	<1% Chrysotile
824-CM3-2	162410144-0008	8							<1% Chrysotile
824-CM3-3	162410144-0009	9							<1% Chrysotile
824-CM4-1	162410144-0010	10	Carpet mastic	Tan/black mastic	Rooms 122 and 124	Non-Friable	Good	1,353 SF	<1% Chrysotile
824-CM5-2	162410144-0014	14	Carpet mastic	Tan/black mastic	Room 136	Non-Friable	Good	737 SF	<1% Chrysotile
824-CM5-3	162410144-0015	15							<1% Chrysotile
824-CM8-1	162410144-0022	22	Carpet mastic	Black/tan mastic	Room 140	Non-Friable	Good	421 SF	<1% Chrysotile
824-CM8-2	162410144-0023	23							<1% Chrysotile
824-CM8-3	162410144-0024	24							<1% Chrysotile
824-FT2-1-Mastic	162410144-0037A	37	Mastic	Black mastic	First floor hallway, vestibule	Non-Friable	Good	600 SF	0.3% Chrysotile
824-FT2-2-Mastic	162410144-0038A	38							<0.25% Chrysotile
824-FT2-3-Mastic	162410144-0039A	39							0.7% Chrysotile
824-FT4-1-Mastic	162410144-0043A	43	Mastic	Tan/black mastic	Room 139, hallway by south stairwell, north stairwell	Non-Friable	Good	1,600 SF	<1% Chrysotile
Fire Rated Doors	Not sampled	NA	Fire Doors	Various	Throughout	Non-Friable	Good	Unknown	Not sampled, PACM
Roofing	Not sampled	NA	Roofing	Flat black built-up roofing	Exterior	Non-Friable	Good	Unknown	Not sampled, PACM
Exterior siding/paneling	Not sampled	NA	Paneling	Gray panels	Exterior on the tower	Non-Friable	Good	Unknown	Not sampled, PACM

Notes

CM = Carpet Mastic

FT = Floor Tile

NA = Not analyzed

LF = Linear Feet

SF= Square Feet

AHERA = Asbestos Hazard Emergency Response Act

PACM = Presumed asbestos containing material

Note that Figure ID corresponds to the sample location on the figures.

Approximate quantity of ACM materials was determined in the field by Tetra Tech professionals.

NESHAP defines asbestos-containing material as a material containing more than 1% asbestos

Note that materials identified to be less than 1% asbestos may be regulated by OSHA

Sources:

Tetra Tech Field Notes

EMSL Laboratory Analytical Reports

National Emission Standards for Hazardous Air Pollutants (NESHAP)

Asbestos Hazard Emergency Act (AHERA)

Occupational Safety and Health Administration (OSHA)

Condition Evaluation:

Good = Little to no damage observed

Fair = 10% of overall material damaged, or 25% localized

Poor = Over 10%/25% damaged

Table 3 - XRF Screening Results

Date	Reading No.	Location	Component	Substrate	Color	Condition	Pb (mg/cm ²)	Result (+/-)
11/1/2023	1	Main entrance	Wall	Drywall	Tan	Good	ND	-
11/1/2023	2	Conference room	Wall	Drywall	Tan	Good	ND	-
11/1/2023	3	Conference room door	Door	Metal	Gray	Good	ND	-
11/1/2023	4	Elevator mechanical room	Door	Metal	Gray	Good	ND	-
11/1/2023	5	Elevator exterior - 1st floor	Door	Metal	Gray	Good	ND	-
11/1/2023	6	Door marked 110	Door	Metal	Gray	Good	ND	-
11/1/2023	7	Door marked 104	Door	Metal	Gray	Good	ND	-
11/1/2023	8	Room 104	Wall	Drywall	Tan	Good	ND	-
11/1/2023	9	Door marked 107	Door	Metal	Gray	Good	ND	-
11/1/2023	10	Room 107	Wall	Drywall	Blue-gray	Good	ND	-
11/1/2023	11	Room 107	Wall	Drywall	Dark gray	Good	ND	-
11/1/2023	12	Door marked 132	Door	Metal	Gray	Good	ND	-
11/1/2023	13	Door to southwest stairwell	Door	Metal	Tan	Good	ND	-
11/1/2023	14	Southwest stairwell	Wall	Concrete block	Tan	Good	ND	-
11/1/2023	15	Door/exit from SW stairwell	Door	Metal	Gray	Good	ND	-
11/1/2023	16	Southwest stairwell handrail	Handrail	Metal	Brown	Good	ND	-
11/1/2023	17	Door marked 126	Door	Metal	Brown	Good	ND	-
11/1/2023	18	Door to shower/mother's room	Door	Metal	Dark gray	Good	ND	-
11/1/2023	19	Door marked 127	Door	Metal	Light gray	Good	ND	-
11/1/2023	20	Storage garage	Wall	Concrete	White	Good	ND	-
11/1/2023	21	Storage garage	Floor	Concrete	Yellow	Fair	ND	-
11/1/2023	22	Storage garage	Door	Metal	Gray	Good	ND	-
11/1/2023	23	Storage garage	Handrail	Metal	Gray	Good	ND	-
11/1/2023	24	Storage garage	Wall	Wood	Gray	Good	ND	-
11/1/2023	25	Hallway to lab	Wall	Drywall	White	Good	ND	-
11/1/2023	26	Room 124	Wall	Drywall	Cream	Good	ND	-
11/1/2023	27	Door marked 123	Door	Metal	Dark gray	Good	ND	-
11/1/2023	28	Calibration and bench area	Wall	Concrete	Cream	Good	ND	-
11/1/2023	29	Calibration and bench area	Guardrail	Metal	Yellow	Good	ND	-
11/1/2023	30	Calibration and bench area	Floor	Concrete	Blue-gray	Good	ND	-
11/1/2023	31	Calibration and bench area	Floor	Metal	Red	Good	ND	-
11/1/2023	32	Room 123	Wall	Concrete	Gray	Good	ND	-
11/1/2023	33	Calibration and bench area	Wall	Tile	Cream	Good	ND	-
11/1/2023	34	Room 134	Wall	Concrete	Brown	Good	ND	-
11/1/2023	35	Room 133	Door jam	Metal	Light gray	Good	ND	-
11/1/2023	36	Hallway of second floor	Wall	Drywall	Gray-white	Good	ND	-
11/1/2023	37	Elevator exterior - 2nd floor	Door	Metal	Dark gray	Good	ND	-
11/1/2023	38	Restroom door	Door	Metal	Dark gray	Good	ND	-
11/1/2023	39	Room 206	Wall	Drywall	Gray	Good	ND	-
11/1/2023	40	Room 203	Wall	Drywall	Cream	Good	ND	-
11/1/2023	41	Mezzanine	Wall	Tile	Tan	Good	ND	-

Notes:

No. = Number

mg/cm² = milligrams per square centimeter

sec. = seconds

Pb = lead

XRF = X-Ray Fluorescence Analyzer

+/- = positive/negative

ND = Non detect

Sources:

Tetra Tech Field Notes

XRF Readings

Note the Reading No. also is the sample location shown on the lead figures.

The U.S. EPA defines that lead-based paint as equal to or greater than 0.5% by weight or 1 milligram per square centimeter (mg/cm²) requires appropriate removal and disposal.

Table 4 - Hazardous Materials Assessment Summary

Potential Hazardous Material	Location & Number Observed		
	All Rooms	Garage	By testing lab
Mercury Equipment	13	--	--
PCBs Equipment	--	--	--
PCBs (Lights)	82	--	--
Fluorescent Bulbs	167	--	--
Mercury (Lights)	--	--	--
Other Hazardous Materials**	--	2	3
Exit Signs	23	--	--
Smoke Detectors	6	--	--
Fire Extinguishers	--	--	--
Computer Monitors	145	--	--
Printers / Scanners	11	--	--
Appliances	7	--	--
Storage Tanks	--	--	--

Notes:

Mercury = Mercury containing equipment/devices observed

Mercury (Lights) = potential lighting equipment that contains mercury

-- = Not Observed

CFCs = Chlorofluorocarbons

PCB = Polychlorinated Biphenyls

* Tetra Tech surveyed with the assumption that all lightbulbs were fluorescent and all ballasts were potentially PCB-containing.

Appendix A

Inspector Credentials



Mike DeWine, Governor
Jon Husted, Lt. Governor
Anne M. Vogel, Director

2/26/2024

Victoria Farmer
2847 Urwiler Ave
Cincinnati, OH 45211

RE: Evaluation Specialist
Certification Number: ES35298
Expiration Date: 4/11/2025

Dear Victoria Farmer:

This letter and enclosed certification card approves your request to be certified as an asbestos Evaluation Specialist. You must present your card upon request at any project site while performing duties. Copies of cards are not acceptable as proof of certification.

This certification may be revoked by the Director of the Ohio Environmental Protection Agency (EPA) for violation of any of the requirements of 3745-22 or 3745-20 of the Ohio Administrative Code.

If you have any questions, please contact the Asbestos Program at 614-644-0226 or by email at asbestoslicensing@epa.ohio.gov.

Sincerely,

Brandon M. Schwendeman

Brandon Schwendeman
Manager, Business Operations Support Section
Ohio EPA - Division of Air Pollution Control



Suite 700 • P.O. Box 1049 • Columbus, OH 43216-1049
• (614) 644-3020 • (614) 644-3184 (fax)



This certification is issued pursuant to Revised Code Chapter 3710 and Ohio Administrative Code Chapter 3745-22.

Your certification card is valid for a period of one (1) year, as indicated by the expiration date on the card.

Your card must be present on any project site where you are conducting asbestos-related work.

All questions regarding your certification should be directed to 614-644-0226 or asbestoslicensing@epa.ohio.gov

If found please return card to:

Ohio EPA, DAPC
P.O. Box 1049
Columbus, OH 43216





Appendix B

Photographic Log

ACM Sample Photos

					
Date: 5/16/24	Sample ID: 824-CM1-1	Photographer: V. Farmer	Date: 5/16/24	Sample ID: 824-CM3-1	Photographer: V. Farmer
Positive asbestos sample – floor tile underneath carpet in Room 224.			Asbestos sample found to be less than 1%.		
					
Date: 5/16/24	Sample ID: 824-CM5-2	Photographer: V. Farmer	Date: 5/16/24	Sample ID: 824-CM8-1	Photographer: V. Farmer
Asbestos sample found to be less than 1%.			Asbestos sample found to be less than 1%.		

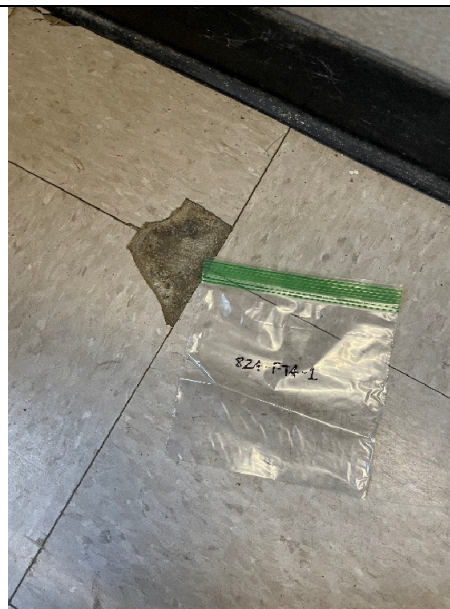


Date:
5/16/24

Sample ID:
824-FT2

Photographer:
W. Williams

Asbestos sample found to be less than 1% (mastic associated with the floor tile).



Date:
5/16/24

Sample ID:
824-FT4-1

Photographer:
V. Farmer

Asbestos sample found to be less than 1% (mastic associated with the floor tile).



Date:
5/16/24

Sample ID:
PACM Siding

Photographer:
V. Farmer

View of the potential asbestos-containing siding on the building's tower.



Date:
11/1/23

Sample ID:
TSI

Photographer:
W. Williams

A view of a TSI pipe run labeled as asbestos containing.



Date:
11/1/23

Sample ID:
TSI

Photographer:
W. Williams

A view of a TSI pipe run labeled as asbestos containing.



Date:
11/1/23

Sample ID:
TSI

Photographer:
W. Williams

Additional pipe runs within the building.

Appendix C

Laboratory Analytical Reports



EMSL Analytical, Inc.

6340 CastlePlace Dr. Indianapolis, IN 46250

Tel/Fax: (317) 803-2997 / (317) 803-3047

<http://www.EMSL.com> / indianapolislab@emsl.com

EMSL Order: 162410144
W912QR25B0053 Specs_Vol1-0004

Customer ID: PRCE50

Customer PO:

Project ID:

Attention: Vicky Farmer
Tetra Tech EM Inc (PRC)
250 West Court Street
Suite 200-W
Cincinnati, OH 45202

Project: WPAFB Bldg 824

Phone: (513) 333-3666

Fax: (513) 241-0354

Received Date: 05/20/2024 9:38 AM

Analysis Date: 05/25/2024 - 05/28/2024

Collected Date:

Test Report: Asbestos Analysis of Bulk Materials via AHERA Method 40CFR 763 Subpart E Appendix E supplemented with EPA 600/R-93/116 using Polarized Light Microscopy

Sample	Description	Appearance	<u>Non-Asbestos</u>		<u>Asbestos</u>
			% Fibrous	% Non-Fibrous	% Type
824-CM1-1-Mastic 162410144-0001	Carpet Mastic	Clear Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CM1-1-Floor Tile 162410144-0001A	Carpet Mastic	Gray Non-Fibrous Homogeneous		97% Non-fibrous (Other)	3% Chrysotile
824-CM1-2-Mastic 162410144-0002	Carpet Mastic	Clear Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CM1-2-Floor Tile 162410144-0002A	Carpet Mastic	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CM1-2-Mastic 162410144-0002B	Carpet Mastic	Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CM1-3-Mastic 162410144-0003	Carpet Mastic	Clear Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CM1-3-Floor Tile 162410144-0003A	Carpet Mastic	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CM1-3-Mastic 162410144-0003B	Carpet Mastic	Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CM2-1 162410144-0004	Carpet Mastic	Gray/Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CM2-2 162410144-0005	Carpet Mastic	Gray/Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CM2-3 162410144-0006	Carpet Mastic	Gray/Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CM3-1 162410144-0007	Carpet Mastic	Black/Yellow Non-Fibrous Homogeneous		100% Non-fibrous (Other)	<1% Chrysotile
824-CM3-2 162410144-0008	Carpet Mastic	Black/Yellow Non-Fibrous Homogeneous		100% Non-fibrous (Other)	<1% Chrysotile
824-CM3-3 162410144-0009	Carpet Mastic	Black/Yellow Non-Fibrous Homogeneous		100% Non-fibrous (Other)	<1% Chrysotile
824-CM4-1 162410144-0010	Carpet Mastic	Tan/Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	<1% Chrysotile
824-CM4-2 162410144-0011	Carpet Mastic	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected

Initial report from: 05/28/2024 11:27:26



EMSL Analytical, Inc.

6340 CastlePlace Dr. Indianapolis, IN 46250

Tel/Fax: (317) 803-2997 / (317) 803-3047

<http://www.EMSL.com / indianapolislabs@emsl.com>

EMSL Order: 162410144
W912QR25B0053 Specs_Vol1-0004

Customer ID: PRCE50

Customer PO:

Project ID:

Test Report: Asbestos Analysis of Bulk Materials via AHERA Method 40CFR 763 Subpart E Appendix E supplemented with EPA 600/R-93/116 using Polarized Light Microscopy

Sample	Description	Appearance	Non-Asbestos		Asbestos
			% Fibrous	% Non-Fibrous	% Type
824-CM4-3 162410144-0012	Carpet Mastic	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CM5-1 162410144-0013	Carpet Mastic	Gray/Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CM5-2 162410144-0014	Carpet Mastic	Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	<1% Chrysotile
824-CM5-3 162410144-0015	Carpet Mastic	Tan/Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	<1% Chrysotile
824-CM6-1 162410144-0016	Carpet Mastic	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CM6-2 162410144-0017	Carpet Mastic	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CM6-3 162410144-0018	Carpet Mastic	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CM7-1 162410144-0019	Carpet Mastic	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CM7-2 162410144-0020	Carpet Mastic	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CM7-3 162410144-0021	Carpet Mastic	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CM8-1 162410144-0022	Carpet Mastic	Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	<1% Chrysotile
824-CM8-2 162410144-0023	Carpet Mastic	Tan/Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	<1% Chrysotile
824-CM8-3 162410144-0024	Carpet Mastic	Tan/Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	<1% Chrysotile
824-CM9-1 162410144-0025	Carpet Mastic	Tan/Green Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CM9-2 162410144-0026	Carpet Mastic	Tan/Green Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CM9-3 162410144-0027	Carpet Mastic	Tan/Green Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-DWJC1-1-Drywall 162410144-0028	Drywall / Joint Compound	Brown/White Fibrous Heterogeneous	5% Cellulose	90% Gypsum 5% Non-fibrous (Other)	None Detected
824-DWJC1-1-Joint Compound 162410144-0028A	Drywall / Joint Compound	White Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-DWJC1-2-Drywall 162410144-0029	Drywall / Joint Compound	Brown/White Fibrous Heterogeneous	15% Cellulose	80% Gypsum 5% Non-fibrous (Other)	None Detected

Initial report from: 05/28/2024 11:27:26



EMSL Analytical, Inc.

6340 CastlePlace Dr. Indianapolis, IN 46250

Tel/Fax: (317) 803-2997 / (317) 803-3047

<http://www.EMSL.com / indianapolislaboratory@emsl.com>

EMSL Order: 162410144
W912QR25B0053 Specs_Vol1-0004

Customer ID: PRCE50

Customer PO:

Project ID:

Test Report: Asbestos Analysis of Bulk Materials via AHERA Method 40CFR 763 Subpart E Appendix E supplemented with EPA 600/R-93/116 using Polarized Light Microscopy

Sample	Description	Appearance	Non-Asbestos		Asbestos
			% Fibrous	% Non-Fibrous	% Type
824-DWJC1-2-Joint Compound	Drywall / Joint Compound	White Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
162410144-0029A					
824-DWJC1-2-Mastic	Drywall / Joint Compound	Brown/Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
162410144-0029B					
824-DWJC1-3-Drywall	Drywall / Joint Compound				Layer Not Present
162410144-0030					
824-DWJC1-3-Joint Compound	Drywall / Joint Compound	White Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
162410144-0030A					
824-DWJC-2-1-Drywall	Drywall / Joint Compound				Layer Not Present
162410144-0031					
824-DWJC-2-1-Joint Compound	Drywall / Joint Compound	White Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
162410144-0031A					
824-DWJC-2-1-Mastic	Drywall / Joint Compound	Brown/Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
162410144-0031B					
824-DWJC-2-2-Drywall	Drywall / Joint Compound				Layer Not Present
162410144-0032					
824-DWJC-2-2-Joint Compound	Drywall / Joint Compound	White Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
162410144-0032A					
824-DWJC-2-3-Drywall	Drywall / Joint Compound	Brown/White Fibrous Heterogeneous	55% Cellulose <1% Glass	40% Gypsum 5% Non-fibrous (Other)	None Detected
162410144-0033					
824-DWJC-2-3-Joint Compound	Drywall / Joint Compound				Layer Not Present
162410144-0033A					
824-DWJC-2-3-Foam	Drywall / Joint Compound	Orange Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
162410144-0033B					
824-FT1-1-Floor Tile	Floor Tile and Mastic	Gray Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
162410144-0034					
824-FT1-1-Mastic	Floor Tile and Mastic	Gray/Tan/Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
162410144-0034A					
824-FT1-2-Floor Tile	Floor Tile and Mastic	Gray Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
162410144-0035					
824-FT1-2-Mastic	Floor Tile and Mastic	Gray/Tan/Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
162410144-0035A					
824-FT1-3-Floor Tile	Floor Tile and Mastic	Gray Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
162410144-0036					

Initial report from: 05/28/2024 11:27:26



EMSL Analytical, Inc.

6340 CastlePlace Dr. Indianapolis, IN 46250

Tel/Fax: (317) 803-2997 / (317) 803-3047

<http://www.EMSL.com / indianapolislabs@emsl.com>

EMSL Order: 162410144
W912QR25B0053 Specs_Vol1-0004

Customer ID: PRCE50

Customer PO:

Project ID:

Test Report: Asbestos Analysis of Bulk Materials via AHERA Method 40CFR 763 Subpart E Appendix E supplemented with EPA 600/R-93/116 using Polarized Light Microscopy

Sample	Description	Appearance	Non-Asbestos		Asbestos
			% Fibrous	% Non-Fibrous	% Type
824-FT1-3-Mastic 162410144-0036A	Floor Tile and Mastic	Tan/Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-FT2-1-Floor Tile 162410144-0037	Floor Tile and Mastic	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-FT2-1-Mastic 162410144-0037A	Floor Tile and Mastic	Black Non-Fibrous Homogeneous		98% Non-fibrous (Other)	2% Chrysotile
824-FT2-2-Floor Tile 162410144-0038	Floor Tile and Mastic	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-FT2-2-Mastic 162410144-0038A	Floor Tile and Mastic	Black Non-Fibrous Homogeneous		98% Non-fibrous (Other)	2% Chrysotile
824-FT2-3-Floor Tile 162410144-0039	Floor Tile and Mastic	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-FT2-3-Mastic 162410144-0039A	Floor Tile and Mastic	Black Non-Fibrous Homogeneous		98% Non-fibrous (Other)	2% Chrysotile
824-FT3-1-Floor Tile 162410144-0040	Floor Tile and Mastic	Beige Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-FT3-1-Mastic 162410144-0040A	Floor Tile and Mastic	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-FT3-2-Floor Tile 162410144-0041	Floor Tile and Mastic	Beige Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-FT3-2-Mastic 162410144-0041A	Floor Tile and Mastic	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-FT3-3-Floor Tile 162410144-0042	Floor Tile and Mastic	Beige Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-FT3-3-Mastic 162410144-0042A	Floor Tile and Mastic				Insufficient Material
824-FT4-1-Floor Tile 162410144-0043	Floor Tile and Mastic	Gray Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-FT4-1-Mastic 162410144-0043A	Floor Tile and Mastic	Tan/Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	<1% Chrysotile
824-FT4-2-Floor Tile 162410144-0044	Floor Tile and Mastic	Gray Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-FT4-2-Mastic 162410144-0044A	Floor Tile and Mastic	Gray/Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-FT4-3-Floor Tile 162410144-0045	Floor Tile and Mastic	Gray Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-FT4-3-Mastic 162410144-0045A	Floor Tile and Mastic	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected

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EMSL Order: 162410144
W912QR25B0053 Specs Vol1-0004

Customer ID: PRCE50

Customer PO:

Project ID:

Test Report: Asbestos Analysis of Bulk Materials via AHERA Method 40CFR 763 Subpart E Appendix E supplemented with EPA 600/R-93/116 using Polarized Light Microscopy

Sample	Description	Appearance	Non-Asbestos		Asbestos
			% Fibrous	% Non-Fibrous	% Type
824-FT5-1-Floor Tile 162410144-0046	Floor Tile and Mastic	Gray Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-FT5-1-Mastic 162410144-0046A	Floor Tile and Mastic				Layer Not Present
824-FT5-2-Floor Tile 162410144-0047	Floor Tile and Mastic	Gray Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-FT5-2-Mastic 162410144-0047A	Floor Tile and Mastic				Layer Not Present
824-FT5-3-Floor Tile 162410144-0048	Floor Tile and Mastic	Gray Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-FT5-3-Mastic 162410144-0048A	Floor Tile and Mastic	Clear Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CT1-1 162410144-0049	Ceiling Tile	Gray/White Fibrous Homogeneous	60% Cellulose 20% Min. Wool	15% Perlite 5% Non-fibrous (Other)	None Detected
824-CT1-2 162410144-0050	Ceiling Tile	Gray/White Fibrous Homogeneous	60% Cellulose 20% Min. Wool	15% Perlite 5% Non-fibrous (Other)	None Detected
824-CT1-3 162410144-0051	Ceiling Tile	Gray/White Fibrous Homogeneous	60% Cellulose 20% Min. Wool	15% Perlite 5% Non-fibrous (Other)	None Detected
824-CT2-1 162410144-0052	Ceiling Tile	Gray/White Fibrous Homogeneous	60% Cellulose 20% Min. Wool	15% Perlite 5% Non-fibrous (Other)	None Detected
824-CT2-2 162410144-0053	Ceiling Tile	Gray/White Fibrous Homogeneous	60% Cellulose 20% Min. Wool	15% Perlite 5% Non-fibrous (Other)	None Detected
824-CT2-3 162410144-0054	Ceiling Tile	Gray/White Fibrous Homogeneous	60% Cellulose 20% Min. Wool	15% Perlite 5% Non-fibrous (Other)	None Detected
824-CT3-1 162410144-0055	Ceiling Tile	Gray/White Fibrous Homogeneous	70% Cellulose 10% Min. Wool	15% Perlite 5% Non-fibrous (Other)	None Detected
824-CT3-2 162410144-0056	Ceiling Tile	Gray/White Fibrous Homogeneous	70% Cellulose 10% Min. Wool	15% Perlite 5% Non-fibrous (Other)	None Detected
824-CT3-3 162410144-0057	Ceiling Tile	Gray/White Fibrous Homogeneous	70% Cellulose 10% Min. Wool	15% Perlite 5% Non-fibrous (Other)	None Detected
824-CT4-1 162410144-0058	Ceiling Tile	Gray/White Fibrous Homogeneous	60% Cellulose 20% Min. Wool	15% Perlite 5% Non-fibrous (Other)	None Detected
824-CT4-2 162410144-0059	Ceiling Tile	Gray/White Fibrous Homogeneous	60% Cellulose 20% Min. Wool	15% Perlite 5% Non-fibrous (Other)	None Detected
824-CT4-3 162410144-0060	Ceiling Tile	Gray/White Fibrous Homogeneous	60% Cellulose 20% Min. Wool	15% Perlite 5% Non-fibrous (Other)	None Detected
824-CT5-1 162410144-0061	Ceiling Tile	Gray/White Fibrous Homogeneous	40% Cellulose 40% Min. Wool	15% Perlite 5% Non-fibrous (Other)	None Detected

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EMSL Order: 162410144
W912QR25B0053 Specs_Vol1-0004

Customer ID: PRCE50

Customer PO:

Project ID:

Test Report: Asbestos Analysis of Bulk Materials via AHERA Method 40CFR 763 Subpart E Appendix E supplemented with EPA 600/R-93/116 using Polarized Light Microscopy

Sample	Description	Appearance	Non-Asbestos		Asbestos
			% Fibrous	% Non-Fibrous	% Type
824-CT5-2 162410144-0062	Ceiling Tile	Gray/White Fibrous Homogeneous	40% Cellulose 40% Min. Wool	15% Perlite 5% Non-fibrous (Other)	None Detected
824-CT5-3 162410144-0063	Ceiling Tile	Gray/White Fibrous Homogeneous	40% Cellulose 40% Min. Wool	15% Perlite 5% Non-fibrous (Other)	None Detected
824-CT6-1 162410144-0064	Ceiling Tile	Brown/White Fibrous Heterogeneous	70% Cellulose <1% Glass	20% Gypsum <1% Mica 10% Non-fibrous (Other)	None Detected
824-CT6-2 162410144-0065	Ceiling Tile	Brown/White Fibrous Heterogeneous	20% Cellulose 2% Glass	70% Gypsum 2% Mica 6% Non-fibrous (Other)	None Detected
824-CT6-3 162410144-0066	Ceiling Tile	Brown/White Fibrous Heterogeneous	40% Cellulose <1% Glass	50% Gypsum <1% Mica 10% Non-fibrous (Other)	None Detected
824-CT7-1-Ceiling Tile 162410144-0067	Ceiling Tile	White Fibrous Homogeneous	95% Min. Wool	5% Non-fibrous (Other)	None Detected
824-CT7-1-Mastic 162410144-0067A	Ceiling Tile	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CT7-2-Ceiling Tile 162410144-0068	Ceiling Tile	White Fibrous Homogeneous	95% Min. Wool	5% Non-fibrous (Other)	None Detected
824-CT7-2-Mastic 162410144-0068A	Ceiling Tile	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CT7-3-Ceiling Tile 162410144-0069	Ceiling Tile	White Fibrous Homogeneous	95% Min. Wool	5% Non-fibrous (Other)	None Detected
824-CT7-3-Mastic 162410144-0069A	Ceiling Tile	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CBM1-1 162410144-0070	Cove Base (Mastic Only)	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CBM1-2-Mastic 162410144-0071	Cove Base (Mastic Only)	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CBM1-2-Mastic 162410144-0071A	Cove Base (Mastic Only)	Brown Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CBM1-3-Mastic 162410144-0072	Cove Base (Mastic Only)	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CBM1-3-Mastic 162410144-0072A	Cove Base (Mastic Only)	Brown Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CBM2-1-Mastic 162410144-0073	Cove Base (Mastic Only)	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CBM2-1-Mastic 162410144-0073A	Cove Base (Mastic Only)	Brown Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CBM2-2-Mastic 162410144-0074	Cove Base (Mastic Only)	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected

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EMSL Order: 162410144
W912QR25B0053 Specs_Vol1-0004

Customer ID: PRCE50

Customer PO:

Project ID:

Test Report: Asbestos Analysis of Bulk Materials via AHERA Method 40CFR 763 Subpart E Appendix E supplemented with EPA 600/R-93/116 using Polarized Light Microscopy

Sample	Description	Appearance	Non-Asbestos		Asbestos
			% Fibrous	% Non-Fibrous	% Type
824-CBM2-2-Mastic 162410144-0074A	Cove Base (Mastic Only)	Brown Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CBM2-3-Mastic 162410144-0075	Cove Base (Mastic Only)	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CBM2-3-Mastic 162410144-0075A	Cove Base (Mastic Only)	Brown Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CBM3-1 162410144-0076	Cove Base (Mastic Only)	Brown Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CBM3-2 162410144-0077	Cove Base (Mastic Only)	Brown Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CBM3-3 162410144-0078	Cove Base (Mastic Only)	Brown Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CBM4-1 162410144-0079	Cove Base (Mastic Only)	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CBM4-2 162410144-0080	Cove Base (Mastic Only)	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CBM4-3 162410144-0081	Cove Base (Mastic Only)	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CBM5-1 162410144-0082	Cove Base (Mastic Only)	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CBM5-2 162410144-0083	Cove Base (Mastic Only)	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CBM5-3 162410144-0084	Cove Base (Mastic Only)	Tan Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-WC-1 162410144-0085	Window Caulk	Brown Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-WC-2 162410144-0086	Window Caulk	Brown Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-WC-3 162410144-0087	Window Caulk	Brown Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-DC-1 162410144-0088	Door Caulk	Gray/White Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-DC-2 162410144-0089	Door Caulk	Gray/White Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-DC-3 162410144-0090	Door Caulk	Gray/White Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-FC-1 162410144-0091	Foundation Caulk	Gray Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected

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 EMSL Order: 162410144
 W912QR25B0053 Specs_Vol1-0004

Customer ID: PRCE50

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**Test Report: Asbestos Analysis of Bulk Materials via AHERA Method 40CFR 763 Subpart E
 Appendix E supplemented with EPA 600/R-93/116 using Polarized Light Microscopy**

Sample	Description	Appearance	<u>Non-Asbestos</u>		<u>Asbestos</u>
			% Fibrous	% Non-Fibrous	% Type
824-FC-2 162410144-0092	Foundation Caulk	Gray Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-FC-3 162410144-0093	Foundation Caulk	Gray Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-EJ-1 162410144-0094	Expansion Joint	Black Fibrous Homogeneous	60% Cellulose	40% Non-fibrous (Other)	None Detected
824-EJ-2 162410144-0095	Expansion Joint	Gray/Black Fibrous Homogeneous	60% Cellulose	40% Non-fibrous (Other)	None Detected
824-EJ-3 162410144-0096	Expansion Joint	Gray/Black Fibrous Homogeneous	80% Cellulose	20% Non-fibrous (Other)	None Detected
824-CT8-1-Ceiling Tile 162410144-0097	Ceiling Tile	Gray/White Fibrous Homogeneous	25% Cellulose 70% Min. Wool	5% Non-fibrous (Other)	None Detected
824-CT8-1-Mastic 162410144-0097A	Ceiling Tile	Brown Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CT8-2-Ceiling Tile 162410144-0098	Ceiling Tile	Gray/White Fibrous Homogeneous	25% Cellulose 70% Min. Wool	5% Non-fibrous (Other)	None Detected
824-CT8-2-Mastic 162410144-0098A	Ceiling Tile	Brown Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
824-CT8-3-Ceiling Tile 162410144-0099	Ceiling Tile	Gray/White Fibrous Homogeneous	25% Cellulose 70% Min. Wool	5% Non-fibrous (Other)	None Detected
824-CT8-3-Mastic 162410144-0099A	Ceiling Tile	Brown Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected

Analyst(s)

Maggie Hayden (9)

Paul Rihm (105)

Ross Matlock (18)

 Asbestos Laboratory Manager
 or Other Approved Signatory

EMSL maintains liability limited to cost of analysis. Interpretation and use of test results are the responsibility of the client. This report relates only to the samples reported above, and may not be reproduced, except in full, without written approval by EMSL. EMSL bears no responsibility for sample collection activities or analytical method limitations. The report reflects the samples as received. Results are generated from the field sampling data (sampling volumes and areas, locations, etc.) provided by the client on the Chain of Custody. Samples are within quality control criteria and met method specifications unless otherwise noted. The above analyses were performed in general compliance with Appendix E to Subpart E of 40 CFR (previously EPA 600/M4-82-020 "Interim Method") but augmented with procedures outlined in the 1993 ("final") version of the method. This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST or any agency of the federal government. Non-friable organically bound materials present a problem matrix and therefore EMSL recommends gravimetric reduction prior to analysis. Unless requested by the client, building materials manufactured with multiple layers (i.e. linoleum, wallboard, etc.) are reported as a single sample. Estimation of uncertainty is available on request.

Samples analyzed by EMSL Analytical, Inc. Indianapolis, IN NVLAP Lab Code 200188-0, AZ0939, CA 2575, CO AL-15132, TX 300262, A2LA Accredited - Certificate #2845.25

Initial report from: 05/28/2024 11:27:26


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W912QR25B0093-Specs_Vol 1920094

EMSL Order: 162410144

CustomerID: PRCE50

CustomerPO:

ProjectID:

Attn: **Vicky Farmer**
Tetra Tech EM Inc (PRC)
250 West Court Street
Suite 200-W
Cincinnati, OH 45202

Phone: (513) 241-0149
 Fax: (513) 241-0354
 Received: 5/20/2024 09:38 AM
 Analysis Date: 5/29/2024
 Collected:

 Project: **WPAFB Bldg 824**

Test Report: Asbestos Analysis of Bulk Materials via AHERA Method 40CFR 763
Subpart E Appendix E supplemented with EPA 600/R-93/116 using Polarized Light
Microscopy with Gravimetric Reduction. Quantitation using 400 Point Count Procedure.

SAMPLE ID	DESCRIPTION	APPEARANCE	(%) Matrix Organic Acid		NON- ASBESTOS % Fibrous	NON- ASBESTOS % NON-FIBROUS	ASBESTOS % TYPES
824-CM1-1- Floor Tile 162410144-0001A	Carpet Mastic	Gray Non-Fibrous Homogeneous	23.7	23.2		50.3 Non-fibrous (other)	2.8 Chrysotile
824-FT2-1- Mastic 162410144-0037A	Floor Tile and Mastic	Black Non-Fibrous Homogeneous	88.9	4.4		6.3 Non-fibrous (other)	0.3 Chrysotile
824-FT2-2- Mastic 162410144-0038A	Floor Tile and Mastic	Black Non-Fibrous Homogeneous	75.6	20.7		3.6 Non-fibrous (other)	<0.25 Chrysotile
824-FT2-3- Mastic 162410144-0039A	Floor Tile and Mastic	Black Non-Fibrous Homogeneous	42.5	24.4		32.4 Non-fibrous (other)	0.7 Chrysotile

Analyst(s)

Maggie Hayden (4)

 Asbestos Laboratory Manager
 or other approved signatory

EMSL maintains liability limited to cost of analysis. Interpretation and use of test results are the responsibility of the client. This report relates only to the samples reported above, and may not be reproduced, except in full, without written approval by EMSL. EMSL bears no responsibility for sample collection activities or analytical method limitations. The report reflects the samples as received. Results are generated from the field sampling data (sampling volumes and areas, locations, etc.) provided by the client on the Chain of Custody. Samples are within quality control criteria and met method specifications unless otherwise noted. The above analyses were performed in general compliance with Appendix E to Subpart E of 40 CFR (previously EPA 600/M4-82-020 "Interim Method") but augmented with procedures outlined in the 1993 ("final") version of the method. This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST or any agency of the federal government. Unless requested by the client, building materials manufactured with multiple layers (i.e. linoleum, wallboard, etc.) are reported as a single sample. Some samples may contain asbestos fibers present in dimensions below PLM resolution limits. EMSL suggests that samples reported as <0.25% or none detected undergo additional analysis via TEM. Estimation of uncertainty is available on request.

Samples analyzed by EMSL Analytical, Inc. Indianapolis, IN NVLAP Lab Code 200188-0, A2LA Accredited - Certificate #2845.25

Initial report from 05/29/2024 11:36:19

HVAC AND CONTROLS REPLACEMENT F/20824
Wright-Patterson AFB, OH P2#513948

SECTION 01 42 00

SOURCES FOR REFERENCE PUBLICATIONS

02/19, CHG 1: 08/23

PART 1 GENERAL

1.1 REFERENCES

Various publications are referenced in other sections of the specifications to establish requirements for the work. These references are identified in each section by document number, date and title. The document number used in the citation is the number assigned by the standards producing organization (e.g., ASTM B564 Standard Specification for Nickel Alloy Forgings). However, when the standards producing organization has not assigned a number to a document, an identifying number has been assigned for reference purposes.

1.2 ORDERING INFORMATION

The addresses of the standards publishing organizations whose documents are referenced in other sections of these specifications are listed below, and if the source of the publications is different from the address of the sponsoring organization, that information is also provided.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
Two Park Avenue
New York, NY 10016-5990
Ph: 800-843-2763
Fax: 973-882-1717
E-mail: customercare@asme.org
Internet: <https://www.asme.org/>

AMERICAN WATER WORKS ASSOCIATION (AWWA)
6666 W. Quincy Avenue
Denver, CO 80235 USA
Ph: 303-794-7711 or 800-926-7337
Fax: 303-347-0804
Internet: <https://www.awwa.org/>

ASTM INTERNATIONAL (ASTM)
100 Barr Harbor Drive, P.O. Box C700
West Conshohocken, PA 19428-2959
Ph: 610-832-9500
Fax: 610-832-9555
E-mail: service@astm.org
Internet: <https://www.astm.org/>

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)
1 Batterymarch Park
Quincy, MA 02169-7471
Ph: 800-344-3555
Fax: 800-593-6372
Internet: <https://www.nfpa.org>

U.S. ARMY (DA)
Army Publishing Directorate
9301 Chapek Rd., Bldg 1458

HVAC AND CONTROLS REPLACEMENT F/20824
Wright-Patterson AFB, OH P2#513948

Fort Belvoir, VA 22060-5447
Ph: 703-614-3727
E-mail: usarmy.pentagon.hqda-apd.mbx.customer-service@mail.mil
Internet: <https://armypubs.army.mil/>

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)
1200 Pennsylvania Avenue, N.W.
Washington, DC 20004
Ph: 202-564-4700
Internet: <https://www.epa.gov>
--- Some EPA documents are available only from:
National Technical Information Service (NTIS)
5301 Shawnee Road
Alexandria, VA 22312
Ph: 703-605-6060 or 1-800-363-2068
Fax: 703-605-6880
TDD: 703-487-4639
E-mail: info@ntis.gov
Internet: <https://www.ntis.gov/>

U.S. FEDERAL HIGHWAY ADMINISTRATION (FHWA)
1200 New Jersey Ave., SE
Washington, DC 20590
Ph: 202-366-4000
E-mail: ExecSecretariat.FHWA@dot.gov
Internet: <https://www.fhwa.dot.gov/>
Order from:
Superintendent of Documents
U.S. Government Publishing Office (GPO)
732 N. Capitol Street, NW
Washington, DC 20401
Ph: 202-512-1800 or 866-512-1800
Bookstore: 202-512-0132
Internet: <https://www.gpo.gov/>

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)
8601 Adelphi Road
College Park, MD 20740-6001
Ph: 866-272-6272
Internet: <https://www.archives.gov/>
Order documents from:
Superintendent of Documents
U.S. Government Publishing Office (GPO)
732 N. Capitol Street, NW
Washington, DC 20401
Ph: 202-512-1800 or 866-512-1800
Bookstore: 202-512-0132
Internet: <https://www.gpo.gov/>

PART 2 PRODUCTS

Not used

PART 3 EXECUTION

Not used

-- End of Section --

HVAC AND CONTROLS REPLACEMENT F/20824
Wright-Patterson AFB, OH P2#513948

SECTION 01 45 00.15 10

RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM)

11/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety and Occupational Health
(SOH) Requirements

1.2 MEASUREMENT AND PAYMENT

The work of this section is not measured for payment. The Contractor is responsible for the work of this section, without any direct compensation other than the payment received for contract items.

1.3 CONTRACT ADMINISTRATION

The Government will use the Resident Management System (RMS) to assist in its monitoring and administration of this contract. The Government accesses the system using the Government Mode of RMS (RMS GM) and the Contractor accesses the system using the Contractor Mode (RMS CM). The term RMS will be used in the remainder of this section for both RMS GM and RMS CM. The joint Government-Contractor use of RMS facilitates electronic exchange of information and overall management of the contract. The Contractor accesses RMS to record, maintain, input, track, and electronically share information with the Government throughout the contract period in the following areas:

- Administration
- Finances
- Quality Control
- Submittal Monitoring
- Scheduling
- Closeout
- Import/Export of Data

1.3.1 Correspondence and Electronic Communications

For ease and speed of communications, exchange correspondence and other documents in electronic format to the maximum extent feasible. Contracting Officer will determine which documents require submission of paper copies.

1.3.2 Other Factors

Other portions of this document have a direct relationship to the reporting accomplished through RMS. Particular attention is directed to FAR 52.236-15 Schedules for Construction Contracts; FAR 52.232-27 Prompt Payment for Construction Contracts; FAR 52.232-5 Payments Under Fixed-Priced Construction Contracts; Section 01 32 01.00 06 PROJECT

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SCHEDULE; Section 01 33 00 SUBMITTAL PROCEDURES; Section 01 35 26.00 06 GOVERNMENT SAFETY REQUIREMENTS; and Section 01 45 04.10 06 CONTRACTOR QUALITY CONTROL.

1.4 RMS SOFTWARE

RMS is a web based application. Download, install and be able to utilize the latest version of RMS within 7 calendar days of receipt of the Notice to Proceed. RMS software, user manuals, access and installation instructions, program updates and training information are available from the RMS website (<https://rms.usace.army.mil>). The Government and the Contractor will have different access authorities to the same contract database through RMS. The common database will be updated automatically each time a user finalizes an entry or change.

1.5 CONTRACT DATABASE - GOVERNMENT

The Government will enter the basic contract award data in RMS prior to granting the Contractor access. The Government entries into RMS will generally be related to submittal reviews, correspondence status, and Quality Assurance(QA)comments, as well as other miscellaneous administrative information.

1.6 CONTRACT DATABASE - CONTRACTOR

Contractor entries into RMS establish, maintain, and update data throughout the duration of the contract. Contractor entries generally include prime and subcontractor information, daily reports, submittals, RFI's, schedule updates and payment requests. RMS includes the ability to import attachments and export reports in many of the modules, including submittals. The Contractor responsibilities for entries in RMS typically include the following items:

1.6.1 Administration

1.6.1.1 Contractor Information

Enter all current Contractor administrative data and information into RMS within 7 calendar days of receiving access to the contract in RMS. This includes, but is not limited to, Contractor's name, address, telephone numbers, management staff, and other required items.

1.6.1.2 Subcontractor Information

Enter all missing subcontractor administrative data and information into RMS CM within seven calendar days of receiving access to the Contract in RMS or within seven calendar days of the signing of the subcontractor agreement for agreements signed at a later date. This includes name, trade, address, phone numbers, and other required information for all subcontractors. A subcontractor is listed separately for each trade to be performed.

1.6.1.3 Correspondence

Identify all Contractor correspondence to the Government with a serial number. Prefix correspondence initiated by the Contractor's site office with "S". Prefix letters initiated by the Contractor's home (main) office with "H". Letters are numbered starting from 0001. (e.g., H-0001 or S-0001). The Government's letters to the Contractor will be prefixed with

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"C" or "RFP".

1.6.1.4 Equipment

Enter and maintain a current list of equipment planned for use or being used on the jobsite, including the most recent and planned equipment inspection dates in accordance with Section 01 45 04.10 06 CONTRACTOR QUALITY CONTROL.

1.6.1.5 Reports

Track the status of the project using the reports available in RMS. The value of these reports reflects the quality of the data input. These reports include but are not limited to the Progress Payment Request worksheet, Quality Control (QC) comments, Deficiency Tracking Log, Submittal Register Status, and Three-Phase Control worksheets in accordance with Sections Section 01 32 01.00 06 PROJECT SCHEDULE, 01 33 00 SUBMITTAL PROCEDURES, and 01 45 04.10 06 CONTRACTOR QUALITY CONTROL.

1.6.1.6 Request For Information (RFI)

Create and track all Requests For Information (RFI) in the RMS Administration Module for Government review and response. Provide status at weekly progress meeting in accordance with Section 01 32 01.00 06 PROJECT SCHEDULE.

1.6.2 Finances

1.6.2.1 Pay Activity Data

Develop and enter a list of pay activities in conjunction with the project schedule in accordance with Section 01 32 01.00 06 PROJECT SCHEDULE. The sum of pay activities equals the total Contract amount, including modifications. Each pay activity must be assigned to a Contract Line Item Number (CLIN). The sum of the activities assigned to a CLIN equals the amount of each CLIN.

1.6.2.2 Payment Requests

Prepare all progress payment requests using RMS. Update the work completed under the Contract at least monthly, measured as percent or as specific quantities. After the update, generate a payment request and prompt payment certification using RMS. Submit the signed prompt payment certification and payment request as well as supporting data, either electronically or by hard copy.

1.6.3 Quality Control (QC)

Enter and track implementation of the 3-phase QC Control System, QC testing, transferred and installed property and warranties in RMS. Prepare daily reports, identify and track deficiencies, document progress of work, and support other Contractor QC requirements in RMS. Maintain all data on a daily basis. Ensure that RMS reflects all quality control methods, tests and actions contained within the Contractor Quality Control (CQC) Plan and Government review comments. Provide within seven calendar days of Government acceptance of the CQC Plan.

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1.6.3.1 Quality Control (QC) Reports

The Contractor's Quality Control (QC) Daily Report in RMS is the official report. The Contractor can use other supplemental formats to record QC data, but information from any supplemental formats are to be consolidated and entered into the RMS QC Daily Report. Any supplemental information may be entered into RMS as an attachment to the report. QC Daily Reports must be finalized and signed in RMS within 24 hours after the date covered by the report.

1.6.3.2 Deficiency Tracking.

Use the QC Daily Report Module to enter and track deficiencies. Deficiencies identified and entered into RMS by the Contractor or the Government will be sequentially numbered with a QC or QA prefix for tracking purposes. Enter each deficiency into RMS the same day that the deficiency is identified. Monitor, track and resolve all QC and QA entered deficiencies. A deficiency is not considered to be corrected until the Government indicates concurrence in RMS.

1.6.3.3 Three-Phase Control Meetings

Maintain scheduled and actual dates and times of preparatory and initial control meetings in RMS, and upload meeting minutes of the meetings. Worksheets for the three-phase control meetings are generated within RMS.

1.6.3.4 Labor and Equipment Hours

Enter labor and equipment exposure hours on a daily basis. Roll up the labor and equipment exposure data into a monthly exposure report.

1.6.3.5 Accident/Safety Reporting

Both the Contractor and the Government enter safety related comments in RMS as a deficiency. The Contractor must monitor, track and show resolution for safety issues in the QC Daily Report area of the RMS QC Module. In addition, follow all reporting requirements for accidents and incidents as required in EM 385-1-1, Section 01 35 26.00 06 GOVERNMENT SAFETY REQUIREMENTS and as required by any other applicable Federal, State or local agencies.

1.6.3.6 Definable Features of Work

Enter each feature of work, as defined in the approved CQC Plan, into the RMS QC Module. A feature of work may be associated with a single or multiple pay activities, however a pay activity is only to be linked to a single feature of work.

1.6.3.7 Activity Hazard Analysis

Import activity hazard analysis electronic document files into the RMS QC Module utilizing the document package manager. Prepare activity hazard analysis in accordance with Section 01 35 26.00 06 GOVERNMENT SAFETY REQUIREMENTS.

1.6.4 Submittal Management

Enter all current submittal register data and information into RMS within seven calendar days of receiving access to the Contract in RMS. Contractor

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is responsible for entering Activity Numbers for each submittal item corresponding to their Schedule. The Contractor will import the information shown in the submittal register following the specification Section 01 33 00 SUBMITTAL PROCEDURES unless otherwise directed by the Contracting Officer. Group electronic submittal documents into transmittal packages to send to the Government, except very large electronic files, samples, spare parts, mock ups, color boards, or where hard copies are specifically required. Track transmittals and update the submittal register in RMS on a daily basis throughout the duration of the Contract.

Contractor may request submittals to be added to RMS throughout the lifecycle of the project. Approval/disapproval of Contractor requests is at the discretion of the Contracting Officer. Submittals entry into RMS must be updated daily or as practical for performance of the Contract, as approved by the Contracting Officer.

1.6.5 Schedule

Enter and update the contract project schedule in RMS by either manually entering all schedule data or by importing the Standard Data Exchange Format (SDEF) file, based on the requirements in Section 01 32 01.00 06 PROJECT SCHEDULE.

1.6.6 Closeout

Closeout documents, processes and forms are managed and tracked in RMS by both the Contractor and the Government. Ensure that all closeout documents are entered, completed and documented within RMS.

1.7 IMPLEMENTATION

Use of RMS as described in the preceding paragraphs is mandatory. Ensure that sufficient resources are available to maintain contract data within the RMS system. RMS is an integral part of the Contractor's required management of quality control.

1.8 NOTIFICATION OF NONCOMPLIANCE

Take corrective action within seven calendar days after receipt of notice of RMS non-compliance by the Contracting Officer.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

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SECTION 01 45 04.10 06

CONTRACTOR QUALITY CONTROL
04/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D3740	(2019) Minimum Requirements for Agencies Engaged in the Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction
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ASTM E329	(2018) Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection
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1.2 PAYMENT

Separate payment will not be made for providing and maintaining an effective Quality Control program, and all costs associated therewith shall be included in the applicable unit prices or lump-sum prices contained in the Bidding Schedule.

1.3 SUBMITTALS

Government approval/acceptance is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval, or for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. The following shall be submitted in accordance with LRL Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Construction Quality Control Plan; G, RO

PART 2 PRODUCTS (Not Applicable)

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

The Contractor is responsible for quality control and shall establish and maintain an effective quality control system in compliance with FAR 52.246-12 - Inspection of Construction. The quality control system shall consist of plans, procedures, and organization necessary to produce an end product which complies with the contract requirements. The system shall cover all design and construction operations, both onsite and offsite, and shall be keyed to the proposed construction sequence. The site project

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superintendent will be held responsible for the quality of work on the job and is subject to removal by the Contracting Officer for non-compliance with the quality requirements specified in the contract. The site project superintendent in this context shall be the highest level manager responsible for the overall construction activities at the site, including quality and production. The site project superintendent shall maintain a physical presence at the site at all times, except as otherwise acceptable to the Contracting Officer, and shall be responsible for all construction and construction related activities at the site.

3.2 NOT USED

3.3 CONSTRUCTION QUALITY CONTROL PLAN (CQCP)

The Contractor shall furnish for review by the Government, not later than thirty (30) days after receipt of notice to proceed, the Contractor Construction Quality Control (CQC) Plan proposed to implement the requirements of FAR 52.246-12 - Inspection of Construction. The plan shall identify personnel, procedures, control, instructions, tests, records, and forms to be used. The Government will consider an interim plan for the first thirty (30) days of operation. Construction will be permitted to begin only after acceptance of the CQC Plan or acceptance of an interim plan applicable to the particular feature of work to be started. Work outside of the features of work included in an accepted interim plan will not be permitted to begin until acceptance of a CQC Plan or another interim plan containing the additional features of work to be started.

3.3.1 Content of the CQC Plan

The CQC Plan shall include, as a minimum, the following to cover all construction operations, both onsite and offsite, including work by subcontractors, fabricators, suppliers, and purchasing agents:

- a. A description of the quality control organization, including a chart showing lines of authority and acknowledgment that the CQC staff shall implement the three phase control system for all aspects of the work specified. The staff shall include a CQC System Manager who shall report to someone higher in the Contractor's organization than the project superintendent, shall not be the superintendent.
- b. The name, qualifications (in resume format), duties, responsibilities, and authorities of each person assigned a CQC function.
- c. A copy of the letter to the CQC System Manager signed by an authorized official of the firm which describes the responsibilities and delegates sufficient authorities to adequately perform the functions of the CQC System Manager, including authority to stop work which is not in compliance with the contract. The CQC System Manager shall issue letters of direction to all other various quality control representatives outlining duties, authorities, and responsibilities. Copies of these letters shall also be furnished to the Government.
- d. Procedures for scheduling, reviewing, certifying, and managing submittals, including those of subcontractors, offsite fabricators, suppliers, and purchasing agents. These procedures shall be in accordance with LRL Section 01 33 00 SUBMITTAL

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PROCEDURES.

- e. Control, verification, and acceptance testing procedures for each specific test to include the test name, specification paragraph requiring test, feature of work to be tested, test frequency, and person responsible for each test. (Laboratory facilities will be approved by the Contracting Officer.)
- f. Procedures for tracking preparatory, initial, and follow-up control phases and control, verification, and acceptance tests including documentation.
- g. Procedures for tracking construction deficiencies from identification through acceptable corrective action. These procedures shall establish verification that identified deficiencies have been corrected.
- h. Reporting procedures, including proposed reporting formats.
- i. A list of the definable features of work. A definable feature of work is a task which is separate and distinct from other tasks, has separate control requirements, and may be identified by different trades or disciplines, or it may be work by the same trade in a different environment. Although each section of the specifications may generally be considered as a definable feature of work, there are frequently more than one definable features under a particular section. This list will be agreed upon during the coordination meeting.

3.3.2 Acceptance of Plan

Acceptance of the Contractor's plan is required prior to the start of construction. Acceptance is conditional and will be predicated on satisfactory performance during the construction. The Government reserves the right to require the Contractor to make changes in his CQC Plan and operations including removal of personnel, as necessary, to obtain the quality specified.

3.3.3 Notification of Changes

After acceptance of the CQC Plan, the Contractor shall notify the Contracting Officer in writing of any proposed change. Proposed changes are subject to acceptance by the Contracting Officer.

3.4 COORDINATION MEETING

After the Preconstruction Conference, before start of construction, and prior to acceptance by the Government of the CQC Plan, the Contractor shall meet with the Contracting Officer or Authorized Representative and discuss the Contractor's quality control system. The CQC Plan shall be submitted for review a minimum of thirty (30) calendar days prior to the Coordination Meeting. During the meeting, a mutual understanding of the system details shall be developed, including the forms for recording the CQC operations, control activities, testing, administration of the system for both onsite and offsite work, and the interrelationship of Contractor's Management and control with the Government's Quality Assurance. Minutes of the meeting shall be prepared by the Government and signed by both the Contractor and the Contracting Officer. The minutes shall become a part of the contract file. There may be occasions when subsequent conferences will be called by

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either party to reconfirm mutual understandings and/or address deficiencies in the CQC system or procedures which may require corrective action by the Contractor.

3.4.1 Subcontractor CQC Orientation

Before a Subcontractor begins work on the jobsite, the CQC System Manager will train the Subcontractor and answer any questions pertaining to quality control operations. This requirement is waived only if a Subcontractor attended the initial coordination meeting described above. A record of the orientation shall be documented in the QC Report.

3.5 CONSTRUCTION QUALITY CONTROL ORGANIZATION

3.5.1 Personnel Requirements

a. The requirements for the CQC organization are a CQC System Manager and sufficient number of additional qualified personnel to ensure safety and contract compliance. A Site Safety Health Officer (SSHO) will be required for this contract. See LRL Section 01 35 26.00 06 GOVERNMENT SAFETY REQUIREMENTS for the SSHO qualifications and duties.

b. Personnel identified in the technical provisions as requiring specialized skills to assure the required work is being performed properly will also be included as part of the CQC organization. The Contractor's CQC staff shall maintain a presence at the site at all times during progress of the work and have complete authority and responsibility to take any action necessary to ensure contract compliance. The CQC staff shall be subject to acceptance by the Contracting Officer. The Contractor shall provide adequate office space, filing systems and other resources as necessary to maintain an effective and fully functional CQC, and safety/health organization. Complete records of all letters, material submittals, shop drawing submittals, schedules and all other project documentation shall be promptly furnished to the CQC organization by the Contractor. The CQC organization shall be responsible to maintain these documents and records at the site at all times and made available to the SSHO, except as otherwise acceptable to the Contracting Officer.

3.5.2 CQC System Manager Qualifications and Duties

a. The Contractor shall identify as CQC System Manager an individual within the onsite work organization who shall be responsible for overall management of CQC and have the authority to act in all CQC matters for the Contractor. The CQC System Manager shall be a graduate engineergraduate architect, or a graduate of construction management, with a minimum of 5 years construction experience on construction similar to this contract, or a construction person with a minimum of 10 years in related quality management work.

b. This CQC System Manager shall be employed by the Prime Contractor and be on the site at all times during construction. Alternate(s) for the CQC System Manager shall be identified in the CQC Plan to serve in the event of the CQC System Manager's absence. The requirements for the alternates shall be the same as for the designated CQC System Manager.

c. The CQC System Manager may have duties as the SSHO. The CQC System Manager shall not be the superintendent.

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3.5.3 CQC Personnel

a. In addition to CQC personnel specified elsewhere in the contract, the Contractor shall provide as part of the CQC organization specialized personnel to assist the CQC System Manager for the following areas identified per Experience Matrix Table. These individuals may be employees of the prime or subcontractor. These individuals identified per the Experience Matrix Table, shall be responsible to the CQC System Manager; be physically present at the construction site during work on their areas of responsibility; have the necessary education and/or experience in accordance with the experience matrix listed herein. These individuals in the Experience Matrix Table may perform other duties but must be allowed sufficient time to perform their assigned quality control duties as described in the Quality Control Plan.

b. The word "graduate" below indicates an individual possessing a four-year college degree accredited in the respective field listed-with experience obtained following graduation in the type of work being performed on the project.

Experience Matrix Table

<u>AREA</u>		<u>QUALIFICATIONS</u>
a.	Mechanical	Graduate Mechanical Engineer with 5 yrs related experience or person with 10 yrs related experience. This person shall be employed by the Prime Contractor or by the Sub-Contractor and shall not be the CQCM.
b.	Electrical	Graduate Electrical Engineer with 5 yrs related experience or person with 10 yrs related experience. This person shall be employed by the Prime Contractor or the Sub-Contractor and Shall not be the CQCM.
c.	IT/Communications	BICSI Certified RCDD Registered Communication Distribution Designer)with 10 yrs related experience. This person shall be employed by Sub Contractor and Shall not be the COCM.

3.5.4 Additional Requirement

In addition to the above experience and/or education requirements the CQC System Manager and Alternate(s) shall have completed and passed the course entitled "Construction Quality Management For Contractors" within the last 5 years. A copy of the certification shall be provided with the CQCP. This course is periodically offered by the Associated Builders and Constructors, Inc., or Associated General Contractor, Inc., and the U.S. Army Corps of Engineers.

3.5.5 Organizational Changes

The Contractor shall maintain the CQC staff at full strength at all times.

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When it is necessary to make changes to the CQC staff, the Contractor shall revise the CQC Plan to reflect the changes and submit the changes to the Contracting Officer for acceptance.

3.6 SUBMITTALS AND DELIVERABLES

Submittals, if needed, shall be made as specified in LRL Section 01 33 00 SUBMITTAL PROCEDURES. The CQC organization shall be responsible for certifying that all submittals and deliverables are in compliance with the contract requirements. When LRL Section 01 46 00.00 06 TOTAL BUILDING COMMISSIONING is included in the contract, the submittals required by those sections shall be coordinated with LRL Section 01 33 00 SUBMITTAL PROCEDURES to ensure adequate time is allowed for each type of submittal required.

3.7 CONTROL

Contractor Quality Control is the means by which the Contractor ensures that the construction, to include that of subcontractors and suppliers, complies with the requirements of the contract. At least three phases of control shall be conducted by the CQC System Manager for each definable feature of work as follows:

3.7.1 Preparatory Phase

This phase shall be performed prior to beginning work on each definable feature of work, after all required plans/documents/materials are approved/accepted, and after copies are at the work site. This phase shall include:

- a. A review of each paragraph of applicable specifications, reference codes, and standards. A copy of those sections of referenced codes and standards applicable to that portion of the work to be accomplished in the field shall be made available by the Contractor at the preparatory inspection. These copies shall be maintained in the field and available for use by Government personnel until final acceptance of the work.
- b. A review of the contract drawings.
- c. A check to assure that all materials and/or equipment have been tested, submitted, and approved.
- d. Examination of the work area to assure that all required preliminary work has been completed and is in compliance with the contract.
- e. A physical examination of required materials, equipment, and sample work to assure that they are on hand, conform to approved shop drawings or submitted data, and are properly stored.
- f. A review of the appropriate activity hazard analysis to assure safety requirements are met.
- g. Discussion of procedures for controlling quality of the work including repetitive deficiencies. Document construction tolerances and workmanship standards for that feature of work.
- h. A check to ensure that the portion of the plan for the work to be

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performed has been accepted by the Contracting Officer.

- i. Resolve all differences.
- j. Discussion of the initial control phase.
- k. Review of provisions that have been made to provide required control inspection and testing.
- l. Review of the CQC plan, specifically its organization chart and delegation letters. Insure all required members of the CQC organization for this feature of work are qualified, have been appointed, accepted and have requisite authority delegated.
- m. The Government shall be notified at least 72 hours in advance of beginning the preparatory control phase. This phase shall include a meeting conducted by the CQC System Manager and attended by the superintendent, other CQC personnel (as applicable), and the foreman responsible for the definable feature. The results of the preparatory phase actions shall be documented by separate minutes prepared by the CQC System Manager and attached to the daily CQC report. The Contractor shall instruct applicable workers as to the acceptable level of workmanship required in order to meet contract specifications.
- m. Review of special inspections, the statement of special inspections, and the schedule of special inspections.

3.7.2 Initial Phase

This phase shall be accomplished at the beginning of a definable feature of work. The following shall be accomplished:

- a. A check of work to ensure that it is in full compliance with contract requirements. Review minutes of the preparatory meeting.
- b. Verify adequacy of controls to ensure full contract compliance. Verify required control inspection and testing.
- c. Establish level of workmanship and verify that it meets minimum acceptable workmanship standards. Compare with required sample panels as appropriate.
- d. Resolve all differences.
- e. Check safety to include compliance with and upgrading of the safety plan and activity hazard analysis. Review the activity analysis with each worker.
- f. The Government shall be notified at least 72 hours in advance of beginning the initial phase. Separate minutes of this phase shall be prepared by the CQC System Manager and attached to the daily CQC report. Exact location of initial phase shall be indicated for future reference and comparison with follow-up phases.
- g. The initial phase should be repeated for each new crew to work onsite, or any time acceptable specified quality standards are not being met.

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- h. Coordinate scheduled work with special inspections, SPECIAL INSPECTIONS, the Statement of Special Inspections, and the Schedule of Special Inspections.

3.7.3 Follow-up Phase

Daily checks shall be performed to assure control activities, including control testing, are providing continued compliance with contract requirements, until completion of the particular feature of work. The checks shall be made a matter of record in the CQC documentation. Final follow-up checks shall be conducted and all deficiencies corrected prior to the start of additional features of work which may be affected by the deficient work. The Contractor shall not build upon nor conceal non-conforming work.

3.7.4 Additional Preparatory and Initial Phases

Additional preparatory and initial phases shall be conducted on the same definable features of work if: the quality of on-going work is unacceptable; if there are changes in the applicable CQC staff, onsite production supervision or work crew; if work on a definable feature is resumed after a substantial period of inactivity; or if other problems develop.

3.8 TESTS

3.8.1 Testing Procedure

The Contractor shall perform specified or required tests to verify that control measures are adequate to provide a product which conforms to contract requirements. Upon request, the Contractor shall furnish to the Government duplicate samples of test specimens for possible testing by the Government. Testing includes operation and/or acceptance tests when specified. For QC testing of construction materials including soil, rock, aggregate, asphalt, concrete, and steel, the Contractor shall procure the services of a Corps of Engineers (COE) validated testing laboratory or establish a COE validated testing laboratory at the project site. Technical specifications included in the contract that require materials testing by an approved commercial testing laboratory shall be intended to mean by a COE validated laboratory. The Contractor shall perform the following activities and record and provide the following data:

- a. Verify that testing procedures comply with contract requirements.
- b. Verify that facilities and testing equipment are available and comply with testing standards.
- c. Check test instrument calibration data against certified standards.
- d. Verify that recording forms and test identification control number system, including all of the test documentation requirements, have been prepared.
- e. Results of all tests taken, both passing and failing tests, shall be recorded on the CQC report for the date taken. Specification paragraph reference, location where tests were taken, and the sequential control number identifying the test shall be given. If approved by the Contracting Officer, actual test reports may be submitted later with a reference to the test number and date

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taken. An information copy of tests performed by an offsite or commercial test facility shall be provided directly to the Contracting Officer. Failure to submit timely test reports as stated may result in nonpayment for related work performed and disapproval of the test facility for this contract.

3.8.2 Testing Laboratories

3.8.2.1 Capability Check

The Government reserves the right to check laboratory equipment in the proposed laboratory for compliance with the standards set forth in the contract specifications and to check the laboratory technician's testing procedures and techniques. Laboratories utilized for testing soils, concrete, asphalt, and steel shall meet criteria detailed in ASTM D3740 and ASTM E329.

3.8.2.2 Capability Recheck

If the selected laboratory fails the capability check, the Contractor will be assessed a charge of \$1,375.00 to reimburse the Government for each succeeding recheck of the laboratory or the checking of a subsequently selected laboratory. Such costs will be deducted from the contract amount due the Contractor.

3.8.3 Onsite Laboratory

The Government reserves the right to utilize the Contractor's control testing laboratory and equipment to make assurance tests, and to check the Contractor's testing procedures, techniques, and test results at no additional cost to the Government.

3.8.4 Furnishing or Transportation of Samples for Testing

Costs incidental to the transportation of samples or materials shall be borne by the Contractor. Samples of materials for test verification and acceptance testing by the Government shall be delivered to the Corps of Engineers Division Laboratory, f.o.b., at the following address:

For delivery by mail:

Geotechnical & Structures Laboratory
Material Testing Center (GS-E)
U.S. Army Engineer Research and Development Center
3909 Halls Ferry Road
Vicksburg, MS 39180-6199

Coordination for each specific test, exact delivery location, and dates will be made through the Area Office.

3.9 COMPLETION INSPECTION

3.9.1 Punch-Out Inspection

Near the end of the work, or any increment of the work established by a time stated in the FAR 52.211-10 - Commencement, Prosecution, and Completion of Work, or by the specifications, the CQC System Manager shall conduct an inspection of the work. A punch list of items which do not conform to the approved drawings and specifications shall be prepared and

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included in the CQC documentation, as required by paragraph DOCUMENTATION. The list of deficiencies shall include the estimated date by which the deficiencies will be corrected. The CQC System Manager or staff shall make a second inspection to ascertain that all deficiencies have been corrected. Once this is accomplished, the Contractor shall notify the Government that the facility is ready for the Government Pre-Final inspection.

3.9.2 Pre-Final Inspection

The Government will perform the pre-final inspection to verify that the facility is complete and ready to be occupied. A Government Pre-Final Punch List may be developed as a result of this inspection. The Contractor's CQC System Manager shall ensure that all items on this list have been corrected before notifying the Government, so that a Final inspection with the customer can be scheduled. Any items noted on the Pre-Final inspection shall be corrected in a timely manner. These inspections and any deficiency corrections required by this paragraph shall be accomplished within the time slated for completion of the entire work or any particular increment of the work if the project is divided into increments by separate completion dates.

3.9.3 Final Acceptance Inspection

The Contractor's Quality Control Inspection personnel, plus the superintendent or other primary management person, and the Contracting Officer's Representative shall be in attendance at the final acceptance inspection. Additional Government personnel including, but not limited to, those from Base/Post Civil Facility Engineer user groups, and major commands may also be in attendance. The final acceptance inspection will be formally scheduled by the Contracting Officer based upon results of the Pre-Final inspection. Notice shall be given to the Contracting Officer at least fourteen (14) days prior to the final acceptance inspection and shall include the Contractor's assurance that all specific items previously identified to the Contractor as being unacceptable, along with all remaining work performed under the contract, will be complete and acceptable by the date scheduled for the final acceptance inspection. Failure of the Contractor to have all contract work acceptably complete for this inspection will be cause for the Contracting Officer to bill the Contractor for the Government's additional inspection cost in accordance with FAR 52.246-12 - Inspection of Construction.

3.10 DOCUMENTATION

The Contractor shall maintain current records providing factual evidence that required quality control activities and/or tests have been performed. These records shall include the work of subcontractors and suppliers and shall be on an acceptable form that includes, as a minimum, the following information:

- a. Contractor/subcontractor and their area of responsibility.
- b. Operating plant/equipment with hours worked, idle, or down for repair.
- c. Work performed each day, giving location, description, and by whom. When Network Analysis (NAS) is used, identify each phase of work performed each day by NAS activity number.

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- d. Test and/or control activities performed with results and references to specifications/drawings requirements. The control phase shall be identified (Preparatory, Initial, Follow-up). List of deficiencies noted, along with corrective action.
- e. Quantity of materials received at the site with statement as to acceptability, storage, and reference to specifications/drawings requirements.
- f. Submittals and deliverables reviewed, with contract reference, by whom, and action taken.
- g. Offsite surveillance activities, including actions taken.
- h. Job safety evaluations stating what was checked, results, and instructions or corrective actions.
- i. Instructions given/received and conflicts in plans and/or specifications.
- j. Contractor's verification statement.
- k. These records shall indicate a description of trades working on the project; the number of personnel working; weather conditions encountered; and any delays encountered. These records shall cover both conforming and deficient features and shall include a statement that equipment and materials incorporated in the work and workmanship comply with the contract. The original and one copy of these records in report form shall be furnished to the Government daily within 24 hours after the date covered by the report. All calendar days shall be accounted for throughout the life of the contract. Reports shall be signed and dated by the CQC System Manager. The report from the CQC System Manager shall include copies of test reports and copies of reports prepared by all subordinate quality control personnel.
- l. Deficiency Tracking System. The Contractor shall maintain a cumulative list of deficiencies identified for the duration of the project. Deficiencies to be listed include those failures, Government oral observations and Notifications of Noncompliance. The list shall be maintained at the project site. Copies of updated listings shall be submitted to the Government at least every thirty (30) days.

3.11 NOTIFICATION OF NONCOMPLIANCE

The Contracting Officer will notify the Contractor of any detected noncompliance with the foregoing requirements. The Contractor shall take immediate corrective action after receipt of such notice. Such notice, when delivered to the Contractor at the work site, shall be deemed sufficient for the purpose of notification. If the Contractor fails or refuses to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders shall be made the subject of claim for extension of time or for excess costs or damages by the Contractor.

-- End of Section --

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SECTION 01 45 35

SPECIAL INSPECTIONS

05/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC (2024) International Building Code

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-301-01 (2023; with Change 1, 2023) Structural Engineering

1.2 GENERAL REQUIREMENTS

Perform Special Inspections in accordance with the Statement of Special Inspections, Schedule of Special Inspections, Chapter 17 of ICC IBC, and UFC 3-301-01. The Statement of Special Inspections and Schedule of Special Inspections are included as an attachment to this specification. Special Inspections are to be performed by an independent third party and are intended to ensure that the work of the Prime Contractor is in accordance with the Contract Documents and applicable building codes. Special inspections do not take the place of the three phases of control inspections performed by the Contractor's QC Manager or any testing and inspections required by other sections of the specifications.

1.3 DEFINITIONS

1.3.1 Continuous Special Inspections

Continuous Special Inspections is the constant monitoring of specific tasks by a special inspector. These inspections must be carried out continuously over the duration of the particular tasks.

1.3.2 Perform

Perform these tasks for each weld, fastener or bolted connection, and noted required verification.

1.3.3 Observe

Observe these items randomly during the course of each work day to insure that applicable requirements are being met. Observe these Special Inspections items on a periodic daily basis. Operations need not be delayed pending these inspections.

1.3.4 Special Inspector (SI)

A qualified person retained by the Contractor and approved by the

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Contracting Officer as having the competence necessary to inspect a particular type of construction requiring Special Inspections. The SI must be an independent third party hired directly by the Prime Contractor.

1.3.5 Associate Special Inspector (ASI)

A qualified person who assists the SI in performing Special Inspections but who must perform inspection under the direct supervision of the SI and who cannot perform inspections without the SI on site.

1.3.6 Third Party

A Special inspector must not be an employee of the Contractor or of any Sub-Contractor performing the work to be inspected.

1.3.7 Contracting Officer

The Government official having overall authority for administrative contracting actions. Certain contracting actions may be delegated to the Contracting Officer's Representative (COR).

1.3.8 Quality Control (QC) Manager

An individual retained by the Prime Contractor and qualified in accordance with the Section 01 45 04.10 06 CONTRACTOR QUALITY CONTROL having the overall responsibility for the Contractor's QC organization.

1.3.9 Structural Engineer of Record (SER)

A registered design professional contracted by the Government as an A/E responsible for the overall design and review of submittal documents prepared by others. The SER is registered or licensed to practice their respective design profession as defined by the statutory requirements of the professional registration laws in the state in which the design professional works. The SER is also referred to as the Engineer of Record (EOR) in design documents.

1.3.10 Statement of Special Inspections (SSI)

A document developed by the SER identifying the materials, systems, components and work required to have Special Inspections. This statement is included at the end of this specification.

1.3.11 Schedule of Special Inspections (SSI)

A schedule which lists each of the required Special Inspections, the extent to which each Special Inspection is to be performed, and the required frequency for each in accordance with ICC IBC Chapter 17. This schedule is included at the end of this specification.

1.3.12 Definable Feature of Work (DFOW)

An inspection group that is separate and distinct from other inspection groups, having inspection requirements or inspectors that are unique.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for

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Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

SIOR Letter of Acceptance; G

Special Inspections Project Manual; G

Special Inspections Agency's Written NDT Practices with method and evidence of regular equipment calibration where applicable

SD-06 Test Reports

Special Inspection Daily Reports

Special Inspection Biweekly Reports

SD-07 Certificates

AISC Certified Steel Fabricator

Steel Truss Plant Quality Assurance Program

Wood Truss Plant Quality Assurance Program

AC472 Accreditation

Precast Concrete Institute (PCI) Certified Plant

Certificate of Compliance

Special Inspector of Record Qualifications

Special Inspector Qualifications

Qualification Records for Nondestructive Testing Technicians

SD-11 Closeout Submittals

Interim Report of Special Inspections from each DFO; G

Comprehensive Final Report of Special Inspections; G

Final Report of Structural Observations where required; G

1.5 SPECIAL INSPECTOR QUALIFICATIONS

Submit qualifications for each special inspector according to the following certification requirements. The Government will consider alternate equivalent certification upon request. Without written acceptance deviations to the following qualifications are not approved.

1.5.1 Steel Construction and High Strength Bolting

1.5.1.1 Special Inspector

- a. ICC Structural Steel and Bolting Special Inspector certificate with one year of related experience, or

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- b. Registered Professional Engineer with 3 years of related experience

1.5.1.2 Associate Special Inspector

Engineer-In-Training with one year of related experience.

1.5.2 Welding Structural Steel

1.5.2.1 Special Inspector

- a. ICC Structural Welding Special Inspector certificate with one year of related experience, or

- b. AWS Certified Welding Inspector

1.5.2.2 Associate Special Inspector

AWS Certified Associate Welding Inspector

1.5.3 Nondestructive Testing of Welds

1.5.3.1 Special Inspector

NDT Level III Certificate

1.5.3.2 Associate Special Inspector

NDT Level II Certificate plus one year of related experience

1.5.4 Cold Formed Steel Framing

1.5.4.1 Special Inspector

- a. ICC Structural Steel and Bolting Special Inspector certificate with one year of related experience, or

- b. ICC Commercial Building Inspector with one year of experience, or

- c. Registered Professional Engineer with 3 years related experience

1.5.4.2 Associate Special Inspector

Engineer-In-Training with one year of related experience.

1.5.5 Concrete Construction

1.5.5.1 Special Inspector

- a. ICC Reinforced Concrete Special Inspector Certificate with one year of related experience, or

- b. ACI Concrete Construction Special Inspector, or

- c. Registered Professional Engineer with 3 years of related experience

1.5.5.2 Associate Special Inspector

- a. ACI Concrete Construction Special Inspector-in-Training, or

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- b. Engineer-In-Training with one year of related experience

1.5.6 Prestressed Concrete Construction

1.5.6.1 Special Inspector

- a. ICC Prestressed Concrete Special Inspector Certificate with one year of related experience, or
- b. PCI Quality Control Technician/Inspector Level II Certificate with one year of related experience, or
- c. Registered Professional Engineer with 3 years of related experience

1.5.6.2 Associate Special Inspector

- a. PCI Quality Control Technician/Inspector Level I Certificate with one year of related experience, or
- b. Engineer-In-Training with one year of related experience

1.5.7 Post-tensioned Concrete Construction

1.5.7.1 Special Inspector

- a. PTI Level 2 Unbonded PT Inspector Certificate, or
- b. Registered Professional Engineer with 3 years of related experience

1.5.7.2 Associate Special Inspector

- a. PTI Level 1 Unbonded PT Inspector Certificate with one year of related experience, or
- b. Engineer-In-Training with one year of related experience

1.5.8 Masonry Construction

1.5.8.1 Special Inspector

- a. ICC Structural Masonry Special Inspector Certificate with one year of related experience, or
- b. Registered Professional Engineer with 3 years of related experience

1.5.8.2 Associate Special Inspector

Engineer-In-Training with one year of related experience.

1.5.9 Wood

1.5.9.1 Special Inspector

- a. ICC Commercial Building Inspector Certificate with one year of related experience, or
- b. ICC Residential Building Inspector with one year of experience, or

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- c. Registered Professional Engineer with 3 years of related experience

1.5.9.2 Associate Special Inspector

Engineer-In-Training with one year of related experience.

1.5.10 Verification of Site Soil Condition, Fill Placement and Load-Bearing Requirements

1.5.10.1 Special Inspector

- a. ICC Soils Special Inspector Certificate with one year of related experience, or
- b. National Institute for Certification in Engineering Technologies (NICET) Soils Technician Level II Certificate in Construction Material Testing, or
- c. Geologist-In-Training with 3 years of related experience, or
- d. Registered Professional Engineer with 3 years of related experience

1.5.10.2 Associate Special Inspector

- a. NICET Soils Technician Level I Certificate in Construction Material Testing with one year of related experience, or
- b. Engineer-In-Training with one year of related experience

1.5.11 Deep Foundations

1.5.11.1 Special Inspector

- a. NICET Soils Technician Level II Certificate in Construction Material Testing, or
- b. Geologist-In-Training with 3 years of related experience, or
- c. Registered Professional Engineer with 3 years of related experience

1.5.11.2 Associate Special Inspector

- a. NICET Soils Technician Level I Certificate in Construction Material Testing with one year of related experience, or
- b. NICET Geotechnical Engineering Technician Level I Construction or Generalist Certificate with one year of related experience, or
- c. Engineer-In-Training with one year of related experience

1.5.12 Sprayed Fire-Resistive Material

1.5.12.1 Special Inspector

- a. ICC Spray-applied Fireproofing Special Inspector Certificate, or
- b. ICC Fire Inspector I Certificate with one year of related experience, or
- c. Registered Professional Engineer or Architect with related experience

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1.5.12.2 Associate Special Inspector

Engineer-In-Training with one year of related experience

1.5.13 Intumescent Fire-Resistive Coatings

1.5.13.1 Special Inspector

- a. ICC Spray-applied Fireproofing Special Inspector Certificate, or
- b. ICC Fire Inspector I Certificate with one year of related experience, or
- c. Registered Professional Engineer or Architect with related experience

1.5.13.2 Associate Special Inspector

Engineer-In-Training with one year of related experience.

1.5.14 Exterior Insulation and Finish System (EIFS)

1.5.14.1 Special Inspector

- a. Association of the Wall and Ceiling Industry (AWCI) EIFS Inspector Certificate, or
- b. Exterior Design Institute Certificate, or
- c. Registered Professional Engineer or Architect with related experience

1.5.14.2 Associate Special Inspector

Engineer-In-Training with one year of related experience.

1.5.15 Fire-Resistant Penetrations and Joints

1.5.15.1 Special Inspector

- a. Passed the UL Firestop Exam with one year of related experience, or
- b. Passed the FM Firestop Exam with one year of related experience, or
- c. Passed the International Firestop Council Exam with one year of related experience, or
- d. Registered Professional Engineer with related experience

1.5.15.2 Associate Special Inspector

Engineer-In-Training with one year of related experience.

1.5.16 Smoke Control

1.5.16.1 Special Inspector

- a. Associated Air Balance Council (AABC) Technician Certification with one year of related experience, or
- b. Registered Professional Engineer with related experience

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1.5.16.2 Associate Special Inspector

Engineer-In-Training with one year of related experience.

PART 2 PRODUCTS

2.1 SPECIAL INSPECTION OF FABRICATED ITEMS

Special Inspection of fabricator's work performed in the fabricator's shop is required to be inspected in accordance with the Statement of Special Inspections and the Schedule of Special Inspections unless the fabricator is certified by the approved agency to perform such work without Special Inspections. Submit the following certification to the Contracting Officer for information to allow work performed in the fabricator's shop to not be subjected to Special Inspections.

AISC Certified Steel Fabricator.

At the completion of fabrication, submit a certificate of compliance, to be included with the comprehensive final report of Special Inspections, stating that the materials supplied and work performed by the fabricator are in accordance with the construction documents.

PART 3 EXECUTION

3.1 RESPONSIBILITIES

3.1.1 Quality Control Manager

- a. Supervise all Special Inspectors required by the Contract Documents and the IBC.
- b. Verify the qualifications of all the Special Inspectors.
- c. Verify the qualifications of all fabricators.
- d. Maintain a 3-ring binder for the Special Inspector's special inspection daily reports and special inspection biweekly reports. This binder must be located in a conspicuous place in the project trailer/office to allow review by the Contracting Officer and the SER.
- e. Maintain a rework items list that includes discrepancies noted on the Special Inspector's daily report.

3.1.2 Special Inspectors

- a. Inspect all elements of the project which the special inspector is qualified to inspect and which are identified in the Schedule of Special Inspections.
- b. Attend preparatory phase meetings related to the Definable Feature of Work (DFOW) which the Special Inspector is qualified to inspect.
- f. Submit a copy of the special inspection daily reports to the QC Manager.
- g. Report discrepancies that are observed during Special Inspections to the QC Manager for correction. If discrepancies are not corrected

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before the special inspector leaves the site the observed discrepancies must be documented in the daily report.

- h. Submit a biweekly Special Inspection Report until all inspections are complete. A report is required for each biweekly period in which Special Inspections activity occurs, and must include the following:
 - (1) A brief summary of the work performed during the reporting time frame.
 - (2) Changes from and discrepancies with the drawings, specifications that were observed during the reporting period.
 - (3) Discrepancies which were resolved or corrected.
 - (4) A list of nonconforming items requiring resolution.
 - (5) All applicable test result, including nondestructive testing reports.
- j. At the completion of the project, submit a comprehensive final report of Special Inspections that documents the Special Inspections completed for the project and corrections of all discrepancies noted in the daily reports. The comprehensive final report of Special Inspections must be signed and dated and it must indicate the certification of the Special Inspector qualifying them to conduct the inspection.

3.2 DEFECTIVE WORK

Check work as it progresses, but failure to detect any defective work or materials must in no way prevent later rejection if defective work or materials are discovered, nor obligate the Contracting Officer to accept the defective work or materials.

-- End of Section --

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SECTION 01 46 00.00 06

TOTAL BUILDING COMMISSIONING

04/20

PART 1 GENERAL

Commission the building systems listed herein.

1.1 SYSTEMS TO BE COMMISSIONED

Commission the following systems:

Heating, Ventilating, Air Conditioning, and Refrigeration Systems (HVAC)

Building Automation System

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING
ENGINEERS (ASHRAE)

ASHRAE 180 (2018) Standard Practice for Inspection
and Maintenance of Commercial Building
HVAC Systems

ASHRAE 202 (2018) Commissioning Process for Buildings
and Systems

ASSOCIATED AIR BALANCE COUNCIL (AABC)

ACG Commissioning Guideline (2005) Commissioning Guideline

NATIONAL ENVIRONMENTAL BALANCING BUREAU (NEBB)

NEBB Commissioning Standard (2018) Whole Building Technical
Commissioning of New Construction

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION
(SMACNA)

ANSI/SMACNA 014-2013 (2013) HVAC Systems Commissioning Manual,
2nd Edition

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29.00 06 SUSTAINABILITY REPORTING. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

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SD-01 Preconstruction Submittals

Project Schedule; G, DO

Project construction schedule which includes commissioning milestone activities. Submit within 14 calendar days following the Construction Commissioning Coordination Meeting. Submit one hard copy and an electronic copy.

SD-06 Test Reports

PVT Procedures; G, DO

Submit no later than 14 calendar days prior to Performance Verification Tests. Submit one hard copy and an electronic copy.

PVT Report; G, DO

Submit no later than 30 calendar days prior to Functional Performance Tests. Submit three hard copies and an electronic copy.

Trend Log Report

Submit one hard copy and an electronic copy no later than 14 calendar days prior to Functional Performance Tests.

SD-07 Certificates

Certificate of Readiness; G, DO

Submit no later than 14 calendar days prior to Functional Performance Tests. Submit one hard copy and an electronic copy.

SD-10 Operation and Maintenance Data

Systems Training; G, DO

Submit two copies of the Systems Training recording no later than 14 calendar days following completing of the Systems Training.

Training Plan; G, RO

Submit one hard copy and an electronic copy no later than 30 calendar days prior to the associated training.

Training Attendance Rosters; G, RO

Submit one hard copy and an electronic copy no later than 7 calendar days following the completion of the training for each system to be commissioned.

Systems Manual; G, DO

Submit Systems Manual no later than 30 calendar days following completion of Functional Performance Tests. Submit three hard copies and an electronic copy.

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Maintenance and Service Life Plans; G, DO

Submit the Maintenance Plan and Service Life Plan no later than 30 calendar days following the completion of Functional Performance tests. Submit three hard copies and an electronic copy.

1.4 COMMISSIONING SPECIALISTS

1.4.1 Lead Commissioning Specialist (CxC)

The Contracting Officer's designated representative shall be the Lead Commissioning Specialist.

1.5 COMMISSIONING STANDARD

Comply with the requirements of one of the acceptable standards unless otherwise stated herein. The acceptable standards are ACG Commissioning Guideline, NEBB Commissioning Standard, ANSI/SMACNA 014-2013, or ASHRAE 202.

- a. Implement all recommendations and suggested practices contained in the Commissioning Standard and electrical test standards.
- b. Use the Commissioning Standard for all aspects of Commissioning, including calibration of instruments.
- c. Where the instrument manufacturer calibration recommendations are more stringent than those listed in the Commissioning Standard, adhere to the manufacturer calibration recommendations.
- d. Use any new requirements, recommendations, and procedures published or adopted prior to contract solicitation by the body responsible for the Commissioning Standard.

1.6 CERTIFICATE OF READINESS

Prior to scheduling Functional Performance Tests for each system, the Contractor shall issue a Certificate of Readiness certifying that the building system is ready for Functional Performance Testing. The Certificate of Readiness must include, for each system to be commissioned, all equipment and system start-up reports; Performance Verification Test Reports. The Contractor; the Lead Commissioning Specialist; the Contractor's Quality Control Representative; and the Mechanical, Electrical, Controls, and TAB subcontractor representatives must sign and date the Certificate of Readiness. Do not schedule and perform Functional Performance Tests prior to Government approval of the Certificate of Readiness.

1.7 PROJECT SCHEDULE

Include the following tasks in the Project Schedule provided in accordance with section 01 32 01.00 06 PROJECT SCHEDULE. Ensure sufficient time is scheduled to accommodate the requirements of this specification section. The order of items listed are not intended to imply a specified sequence:

- a. Pre-Construction Conference.
- b. Electrical system energization.

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- c. Manufacturer Equipment Start-Up for each system to be commissioned.
- d. Testing, Adjusting, and Balancing (TAB)
- e. TAB VerificationField Acceptance Testing
- f. Performance Verification Tests
- g. HVAC Trend Log Report
- h. Pre-Functional Checklist Submittal
- i. Functional Performance Testing
- j. Training for each system to be commissioned
- k. Systems Manual, Maintenance Plan, and Service Life Plan Submission
- l. Seasonal Tests

1.8 FUNCTIONAL PERFORMANCE TEST PREREQUISITES

Complete the following prior to starting Functional Performance Tests of the mechanical systems:

- a. All equipment and systems have been completed, cleaned, flushed, disinfected, calibrated, tested, and operate in accordance with the contract documents and construction plans and specifications.
- b. Testing, Adjusting, and Balancing has been completed and the Testing, Adjusting, and Balancing Report has been submitted and approved in accordance with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.
- c. Performance Verification Tests of the controls systems have been completed and the Performance Verification Tests Report has been submitted and approved in accordance with Specification Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.
- d. The manufacturer's startup Checklists have been submitted and approved.
- e. The Certificate of Readiness for the mechanical systems has been submitted and approved.

PART 2 PRODUCTS (NOT APPLICABLE)

PART 3 EXECUTION

3.1 CONSTRUCTION PHASE

3.1.1 Construction Commissioning Coordination Meeting

The Lead Commissioning Specialist must lead a Construction Commissioning Coordination Meeting to discuss the commissioning process including contract requirements, lines of communication, roles and responsibilities, schedules, documentation requirements, inspection and test procedures, and logistics as specified in this specification section. The Contractor's Superintendent or Project Manager, the Contractor's Quality Control

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Representative, and the Government must attend this meeting. Invite the User and Base Civil Engineer Representative to attend this meeting.

3.1.2 Construction Submittals

Provide all submittals associated with the systems to be commissioned, including shop drawings; equipment submittals; test plans, procedures, and reports; and resubmittals to the Commissioning Specialists. The Commissioning Specialists must review the submittals to the extent necessary verify that the equipment and system installation will comply with the contract requirements and the requirements of the Basis of Design and the Owner's Project Requirements Document.

3.1.3 Inspection and Testing

Demonstrate that all system components have been installed, that each control device and item of equipment operates, and that the systems operate and perform in accordance with contract documents and the Owner's Project Requirements. Requirements in related Sections are independent from the requirements of this Section and do not satisfy any of the requirements specified in this specification section. Provide all materials, services, and labor required to perform HVAC system trend logs, and Functional Performance Tests.

3.1.3.1 Commissioning Team

3.1.3.1.1 Commissioning Coordinator

Provide a commissioning representative for each sub-contractor associated with the systems to be commissioned. Each commissioning representative is responsible for coordination of their respective sub-contractor's execution of the commissioning activities and participation in the inspection and testing required by this specification section.

Designate team members to participate in the building envelope inspections, Pre-Functional checks, and the Functional Performance Testing specified herein.

3.1.3.1.2 Mechanical Systems Team

The following team members must participate in Pre-Functional Checks and Functional Performance Testing of mechanical systems. The CxC, QAR, CQC, MC, EC, and CC must participate in Demand Response Tests:

Designation	Function
QAR/CxC	Contracting Officer's Quality Assurance Representative/Commsioning Authority
CQC	Contractor's Quality Control Personnel
MC	Contractor's Mechanical Commissioning Representative
EC	Contractor's Electrical Commissioning Representative
CC	Contractor's Controls Commissioning Representative

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Designation	Function
CCC	Contractor's Commissioning Coordinator

3.1.3.1.3 Other Pre-Functional and Functional Performance Participants

The following may participate as team members during Pre-Functional Checks and Functional Performance Testing:

Designation	Function
User	Using Agent's Representative

3.1.3.2 Testing, Adjusting, and Balancing (TAB) Verification

The Lead Commissioning Specialist must witness the TAB Field Acceptance Testing performed in accordance with UFGS section 23 05 93 TESTING, ADJUSTING, AND BALANCING OF HVAC. Identify any deficiencies in the Issues Log

The Lead Commissioning Specialist must witness the TAB Field Acceptance Testing performed in accordance with UFGS section 23 05 93 TESTING, ADJUSTING, AND BALANCING OF HVAC.

3.1.3.3 Building Control System Performance Verification Tests

The controls contractor must perform a Performance Verification Test (PVT) of the building control systems demonstrating that the control systems are in compliance with the requirements of the construction contract and accepted design. Show, step-by-step, the actions and results demonstrating that the systems perform in accordance with the sequences of operation. A one-point accuracy check will be performed for each sensor.

Prepare and submit PVT Procedures that list the step-by-step procedures to be performed during the tests and the expected results from each step that demonstrate contract and accepted design compliance. Start the Performance Verification Tests only after approval of the PVT Procedures.

Provide a PVT Report documenting all tests performed during the PVT and the results. Document all failures and associated repairs in the PVT Report.

3.1.3.4 Tests

3.1.3.4.1 Government's Functional Performance Tests

Schedule personnel to attend the Functional Performance Tests for each system only after the Certificate of Readiness has been approved by the Government for the system and all deficiencies identified through any prior review, inspection, or test activity have been corrected. Functional Performance Tests must be performed with the Contracting Officer's Quality Assurance Representative present. The Lead Commissioning Specialist must lead and document all Functional Performance Tests for the systems to be commissioned with the Contractor and appropriate sub-contractors performing the Functional Performance Tests. The representatives listed in the paragraph "Commissioning Team" must attend the Functional Performance Tests. Abort the Functional Performance Tests when any required commissioning team member is not present for the test.

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3.1.3.4.1.1 Acceptance

The Contractor's Quality Control Representative and the Lead Commissioning Specialist must indicate acceptance after the equipment and systems are free of deficiencies. See Section 01 32 01.00 06, PROJECT SCHEDULE, PARA "Basis for Payment and Cost Loading." This amount shall be withheld and not paid until the final Commissioning report submittal has been accepted by the Government.

Withholding for the final commissioning report submittal shall be in the amount of: 1% for contract awards less than \$5,000,000; \$50,000 for contracts awarded from \$5,000,000 to \$10,000,000; or \$100,000 for contracts awarded greater than \$10,000,000. Withholding shall be withheld until the final commissioning report submittal has been approved and accepted by the Government.

3.1.3.4.2 Seasonal Tests

Regardless of the season, perform initial Functional Performance Tests of equipment and systems at the time of system completion. Develop and implement means of artificial loading to demonstrate, to a reasonable level of confidence, the ability of the HVAC systems to handle peak seasonal loads.

In addition to the initial Functional Performance Tests, perform Functional Performance Tests of the HVAC systems during peak heating and cooling seasons during outdoor air condition design extremes. Schedule the seasonal Functional Performance Tests in coordination with the Government.

Systems may be partially accepted by the Government prior to seasonal testing if they comply with all construction contract requirements that can be tested during the initial Functional Performance Tests. All Functional Performance Test procedures must be completed prior to full systems acceptance.

3.1.3.4.3 Aborted Tests and Re-Testing

Abort Functional Performance Tests or Seasonal Tests if any deficiency prevents successful completion of the test or if any required commissioning team member is not present for the test. Reimburse the Government for all costs associated with effort lost caused by re-testing due to test failures and aborted tests. These costs must include salary, travel costs, and per diem for Government commissioning team members. Re-test only after all deficiencies identified during the original tests have been corrected.

3.1.3.4.3.1 100% Sample

Systems or items of equipment for which 100% sample are tested fail if one or more of the test procedures results in a discovery of a deficiency and the deficiency can not be resolved within 5 minutes during the test. Upon test failure, abort the test of the system or item of equipment. Schedule a re-test, in coordination with the Government, only after all deficiencies for all failed equipment and systems have been corrected.

Re-test to the extent necessary to confirm that the deficiencies have been corrected. At the sole discretion of the Government, all test procedures for the failed item of equipment or system must be repeated to confirm that no deficiencies remain within that equipment or system.

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3.2 SYSTEMS TRAINING

The training specified by the specification sections associated with commissioned systems must be provided by factory certified technicians or trainers. Include both demonstration of proper equipment and system operation both at the equipment and classroom training. For the classroom training, include proper operating and maintenance procedures, preventative maintenance requirements and procedures, trouble-shooting procedures, and calibration frequency and procedures. Include identification of the equipment and system warranties and procedures for correction under the warranties. Include a review of the draft systems manual, maintenance plan, and service life plans.

Visibly and audibly record the systems training. All instruction on the recording must be clear and intelligible.

3.3 TRAINING PLAN

Develop a training plan which identifies all training required by specification sections associated with commissioned systems. Include a matrix listing each training requirement, content of the training, the trainer name, trainer contact information, and schedule and location of training.

Document training attendance using Training Attendance Rosters and provide completed attendance rosters to the Lead Commissioning Specialist and the Government.

3.4 MAINTENANCE AND SERVICE LIFE PLANS

Prepare and submit a Maintenance Plan for the project mechanical, electrical, plumbing, and fire protection systems. Prepare the Maintenance Plan in accordance with ASHRAE 180 for heating, ventilation, air conditioning, and refrigeration systems. Develop required inspection and maintenance tasks similar to Section 5 of ASHRAE 180 for the other commissioned systems and fire protection systems.

Prepare and submit a Service Life Plan for the building envelope, structural systems, and site hardscape that includes the following for each assembly or component:

- a. A description of each including the materials or products.
- b. The estimated service life, in years.
- c. The estimated maintenance frequency and description of maintenance tasks.
- d. The point of maintenance access for the components with estimated service life less than service life of the building.

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APPENDIX A - OWNER'S PROJECT REQUIREMENTS DOCUMENT

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OWNER'S PROJECT REQUIREMENTS DOCUMENT

Project: Project, Location, PN #####

Approved: _____
Name Design Agent's Representative Date

Name Owner's Representative Date

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OWNER'S PROJECT REQUIREMENTS DOCUMENT

Contents

1. Owner and User Requirements
 - a. Primary Purpose, Program, and Use
 - b. Project History
 - c. Broad Goals
 - i. Future Expansion
 - ii. Flexibility
 - iii. Quality of Materials
 - iv. Construction Costs
 - v. Operational Costs
2. Environmental and Sustainability Goals
 - a. LEED or Green Globes Goal
 - b. Other
3. Energy Efficiency Goals
 - a. Goals/Policy
 - b. Systems and Feature Energy Impact
4. Indoor Environmental Quality Requirements
 - a. Space Type 1
 - i. Intended Use
 - ii. Occupancy Schedule
 - iii. Environmental Requirements
 - iv. Occupant System Control Ability
 - v. Type of Lighting
 - vi. After-hour Use Accommodation
 - b. Space Type 2
 - i. Intended Use
 - ii. Occupancy Schedule
 - iii. Environmental Requirements
 - iv. Occupant System Control Ability
 - v. Type of Lighting
 - vi. After-hour Use Accommodation
5. Equipment and System Expectations
 - a. HVAC Systems
 - i. Quality and Reliability
 - ii. Type
 - iii. Automation
 - iv. Flexibility
 - v. Maintenance Requirements
 - b. Lighting Systems
 - i. Quality and Reliability
 - ii. Type
 - iii. Automation
 - iv. Flexibility
 - v. Maintenance Requirements
 - c. Domestic Hot Water Systems
 - i. Quality and Reliability
 - ii. Type
 - iii. Automation
 - iv. Flexibility
 - v. Maintenance Requirements

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Contents (continued)

- d. On-site Power Systems
 - i. Quality and Reliability
 - ii. Type
 - iii. Automation
 - iv. Flexibility
 - v. Maintenance Requirements
- e. Other Systems
 - i. Quality and Reliability
 - ii. Type
 - iii. Automation
 - iv. Flexibility
 - v. Maintenance Requirements
- 6. Building Occupant and O&M Personnel Requirements
 - a. Facility Operation
 - b. UMCS (EMCS or FMCS)
 - c. Occupant Training and Orientation
 - d. O&M Staff Training and Orientation

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1. Owner and User Requirements

a. Primary Purpose, Program, and Use

Explain the purpose, program, and use of the facility. (i.e. Army Reserve Center used for training reserve units. Training includes spaces such as weapons, medical, vehicle repair, cooking, etc.)

b. Project History

Explain the history of the project related to design/construction (i.e. D/B/B, D/B, IDIQ, JOC, COE in-house, A/E, etc.). Explain any additional project background that would impact energy/sustainability goals.

c. Broad Goals

i. Future Expansion: Explain goals related to potential future expansion.

ii. Flexibility: Explain goals related to flexibility for layout and use of the building. (i.e. high rate of office churn, expected frequency of renovation, etc.)

iii. Quality of Materials: Explain goals related to quality of materials. (i.e. highest quality materials, 50 yr life, 25 yr life, highest quality within budget, etc.)

iv. Construction Costs: Explain goals related to construction costs. (i.e. how low can you go, set project amount, select simplest systems for low cost, etc.)

v. Operational Costs: Explain goals related to operational costs. (i.e. low utilities based on water and energy conservation, trade-off allowable on maintenance costs to reduce utility cost, utility cost unimportant compared to construction cost, etc.)

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2. Environmental and Sustainability Goals

a. LEED/Green Globes Goal

Set LEED/Green Globes goal and explain sustainable features permissible or preferred to be incorporated. Explain relative importance of LEED/Green Globes goal within project scope. Indicate requirement from service or agency specific criteria and policy.

b. Other

Explain any special sustainability or environmental goals associated with the project. Identify specific sustainability features that may be required or desired. (i.e. hydro-power, solar power, on-site water treatment, on-site water infiltration, impervious cover reduction, parking capacity, etc.)

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3. Energy Efficiency Goals

a. Goals/Policy

Explain the specific project goals and requirements regarding energy efficiency. Incorporate the requirements of UFC 1-200-02 High Performance and Sustainable Building Requirements and/or other relevant agency policies.

b. Systems and Feature Energy Impacts

Identify and explain envelope, system, or site and building features that will be incorporated to maximize energy efficiency. Identify features that must be incorporated that will reduce or limit energy efficiency.

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4. Indoor Environmental Quality Requirements

a. Space Type 1

i. Intended Use: Explain how the space will be used (i.e. classroom occasionally used as conference room).

ii. Occupancy Schedule: Describe the occupancy including number of people at various times (i.e. drill weekend-maximum capacity, weekdays-20 percent; or 0700-0900 - none, 0900-1400 - 30 people, 1400-1600 - none).

iii. Environmental Requirements: Describe the environmental requirements of the space. Include description of temperatures, humidity levels, ventilation rates, air quality, lighting levels, or any other specific parameters desired (i.e. 75 deg F, 50 percent rh, 30 fc, etc.).

iv. Occupant System Control Ability: Describe the desired level of control the occupants will have over the thermal comfort and lighting systems. (i.e. adjustable thermostat for every person, adjustable thermostat in all private offices, no adjustable thermostats, adjustable thermostat in senior rank also controlling other offices, occupancy sensors for lighting, adjustable dimming, etc.)

v. Type of Lighting: Describe the type of lighting desired (i.e. task lighting with minimal overhead, maximize daylight with dimming on overhead, accent lighting, particular fixtures, etc.).

vi. After-hour Use Accommodations: Describe whether and how often the space may be used after hours. Describe the systems that activate when an occupant uses the building after-hours. Describe the level of control of after-hour use HVAC.

(Example: Space is rarely used after-hours by few occupants. HVAC and lighting system should activate when occupants enter after-hours. The HVAC operation will be limited to that required to provide heating, A/C, and ventilation to the occupied space alone.) (Example: Space is rarely used after-hours by few occupants. Lighting and heating systems should activate. Ventilation and cooling should remain in normal after-hour operation.)

b. Space Type 2

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5. Equipment and System Expectations

a. HVAC Systems

i. Quality and Reliability: Explain the level of quality and reliability required of the HVAC systems.

(Example: Equipment efficiency should meet ASHRAE _____ and FEMP/Energy Star requirements. Due to critical nature of facility, additional redundancy in the cooling and heating systems is required, i.e. multiple chillers, boilers, and pumps.) (Example: No specific quality or reliability requirements specified. Equipment should remain serviceable over life of building or to the extent typical of the type of equipment.)

ii. Type: Explain the type of equipment desired.

(Example: Boilers should be condensing type. Use hydronic heating and cooling. Use self-contained A/C units in computer rooms.)

iii. Automation: Explain the level of automation in the HVAC System desired.

(Example: Single loop HVAC systems permissible. Use packaged controls only.) (Example: Control HVAC systems from DDC system connected to the base UMCS.) (Example: Boilers should have packaged controls connected to the DDC system.)

iv. Flexibility: Describe the desired level of flexibility of the HVAC system.

(Example: System should accommodate frequent office layout changes including private office wall movement.) (Example: Layout will remain mostly unchanged; no flexibility required.) (Example: Accommodate potential for conference and classrooms to change to offices.)

v. Maintenance Requirements: Describe the level of maintenance available or the requirements of the equipment regarding maintainability.

(Example: Equipment should be located to allow easy maintenance access. Equipment vendors or repair service should be able to respond within 24 hrs.)

b. Lighting Systems

i. Quality and Reliability: Explain the level of quality and reliability required of the lighting system controls.

(Example: The building lighting system should meet ASHRAE 90.1 - IP requirements.)

ii. Type: Explain the type of lighting or control equipment desired.

(Example: High-efficiency fluorescent lamps with high-efficiency ballasts will be specified. Indirect lighting will be used in all office and classroom spaces. Lighting foot-candle levels may be reduced to 45 foot-candles in lieu of the typical 50 foot-candles when indirect lighting is used.)

iii. Automation: Explain the level of automation in the lighting control

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system desired.

(Example: Provide occupancy sensors in restrooms, corridors, and storage areas.)

iv. Flexibility: Describe the desired level of flexibility of the lighting system and control systems.

(Example: Provide dual level switching in classrooms and conference rooms.)

v. Maintenance Requirements: Describe the level of maintenance available or the requirements of the equipment regarding maintainability.

(Example:)

c. Domestic Hot Water Systems

i. Quality and Reliability: Explain the level of quality and reliability required of the domestic hot water systems.

(Example: Equipment efficiency should meet ASHRAE and FEMP/Energy Star requirements. Due to critical nature of facility, additional redundancy in the water heating systems is required, i.e. multiple hot water heaters and circulation pumps.) (Example: No specific quality or reliability requirements specified. Equipment should remain serviceable over life of building or to the extent typical of the type of equipment.)

ii. Type: Explain the type of equipment desired.

(Example: Gas-fired storage tank water heater with mixing valve for temperature control.) (Example: Instantaneous electric water heater at lavatories.) (Example: Instantaneous electric water heater with integral control system for eyewash/showers.)

iii. Automation: Explain the level of automation in the domestic hot water control system desired.

(Example: Occupancy schedule control for recirculation loop and gas burner. Connect package controls to DDC system.)

iv. Flexibility: Describe the desired level of flexibility of the domestic hot water systems.

(Example: No anticipated changes to restroom layout; no additional flexibility required.)

v. Maintenance Requirements: Describe the level of maintenance available or the requirements of the equipment regarding maintainability.

(Example: Equipment should be located to allow easy maintenance access. Equipment vendors or repair service should be able to respond within 24 hrs.)

d. On-site Power Systems

i. Quality and Reliability: Explain the level of quality and reliability required of the on-site power system.

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- ii. Type: Explain the type of on-site power system desired.
 - iii. Automation: Explain the level of automation in the on-site power system desired.
 - iv. Flexibility: Describe the desired level of flexibility of the on-site power system.
 - v. Maintenance Requirements: Describe the level of maintenance available or the requirements of the on-site power system regarding maintainability.
- e. Other Systems
- i. Quality and Reliability: Explain the level of quality and reliability required of the system.
 - ii. Type: Explain the type of system desired.
 - iii. Automation: Explain the level of automation in the system desired.
 - iv. Flexibility: Describe the desired level of flexibility of the system.
 - v. Maintenance Requirements: Describe the level of maintenance available or the requirements of the system regarding maintainability.

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6. Building Occupant and O&M Personnel Requirements

a. Facility Operation

Describe how the facility will be operated. Who operates the facility? Who maintains the facility? Who pays the utility bills?

b. UMCS (EMCS or FMCS)

Will the building be tied to an UMCS/EMCS/FMCS? What system will be connected to? Provide information regarding connection requirements, protocols, and control, scheduling and monitoring points.

c. Occupant Training and Orientation

How much training and orientation is desired for building occupants? Will training need to be provided for all systems? To what extent do the occupants need to understand and use the systems?

d. O&M Staff Training and Orientation

How much training and orientation is desired for building occupants? Will training need to be provided for all systems? To what extent do the occupants need to understand and use the systems?

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APPENDIX B - BASIS OF DESIGN

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APPENDIX C - DESIGN PHASE COMMISSIONING PLAN

-- End of Section --

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SECTION 01 50 00

TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS

11/20, CHG 2: 08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C511 (2017; R 2021) Reduced-Pressure Principle
Backflow Prevention Assembly

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023; ERTA 4 2023) National Electrical
Code

NFPA 241 (2022) Standard for Safeguarding
Construction, Alteration, and Demolition
Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety and Occupational Health
(SOH) Requirements

U.S. FEDERAL HIGHWAY ADMINISTRATION (FHWA)

MUTCD (2009; Rev 2012) Manual on Uniform Traffic
Control Devices

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Construction Site Plan; G

Traffic Control Plan; G

Haul Road Plan; G

Contractor Computer Cybersecurity Compliance Statements

Contractor Temporary Network Cybersecurity Compliance Statements

SD-06 Test Reports

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Backflow Preventer Tests

SD-07 Certificates

Backflow Tester Certification

Backflow Preventers Certificate of Full Approval

1.3 CONSTRUCTION SITE PLAN

Prior to the start of work, submit for Government approval a site plan showing the locations and dimensions of temporary facilities (including layouts and details, equipment and material storage area (onsite and offsite), and access and haul routes, avenues of ingress/egress to the fenced area and details of the fence installation. Identify any areas which may have to be graveled to prevent the tracking of mud. Indicate if the use of a supplemental or other staging area is desired. Show locations of safety and construction fences, site trailers, construction entrances, trash dumpsters, temporary sanitary facilities, and worker parking areas.

1.4 BACKFLOW PREVENTERS CERTIFICATE

1.4.1 Backflow Tester Certificate

Prior to testing, submit to the Contracting Officer certification issued by the State or local regulatory agency attesting that the backflow tester has successfully completed a certification course sponsored by the regulatory agency. Tester must not be affiliated with a company participating in other phases of this Contract.

1.4.2 Backflow Prevention Training Certificate

Submit a certificate recognized by the State or local authority that states the Contractor has completed at least 10 hours of training in backflow preventer installations. The certificate must be current.

1.5 DOD CONDITION OF READINESS (COR)

DOD will set the Condition of Readiness (COR) based on the weather forecast for sustained winds 50 knots (58 mph) or greater. Contact the Contracting Officer for the current COR setting.

Monitor weather conditions a minimum of twice a day and take appropriate actions according to the approved Emergency Plan in the accepted Accident Prevention Plan, EM 385-1-1 Chapter 01 Emergency Planning and the instructions below.

Unless otherwise directed by the Contracting Officer, comply with:

- a. Condition FOUR (Sustained winds of 58 mph or greater expected within 72 hours): Normal daily jobsite cleanup and good housekeeping practices. Collect and store in piles or containers scrap lumber, waste material, and rubbish for removal and disposal at the close of each work day. Maintain the construction site including storage areas, free of accumulation of debris. Stack form lumber in neat piles less than 3.3 feet high. Remove all debris, trash, or objects that could become missile hazards. Review requirements pertaining to "Condition THREE" and continue action as necessary to attain "Condition FOUR" readiness.

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Contact Contracting Officer for weather and COR updates and completion of required actions.

- b. Condition THREE (Sustained winds of 58 mph or greater expected within 48 hours): Maintain "Condition FOUR" requirements and commence securing operations necessary for "Condition ONE" which cannot be completed within 18 hours. Cease all routine activities which might interfere with securing operations. Commence securing and stow all gear and portable equipment. Make preparations for securing buildings. Reinforce or remove formwork and scaffolding. Secure machinery, tools, equipment, materials, or remove from the jobsite. Expend every effort to clear all missile hazards and loose equipment from general base areas. Contact Contracting Officer for weather and COR updates and completion of required actions. Review requirements pertaining to "Condition TWO" and continue action as necessary to attain "Condition THREE" readiness.
- c. Condition TWO (Sustained winds of 58 mph or greater expected within 24 hours): Secure the jobsite, and leave Government premises.
- d. Condition ONE. (Sustained winds of 58 mph or greater expected within 12 hours): Contractor access to the jobsite and Government premises is prohibited.

1.6 CYBERSECURITY DURING CONSTRUCTION

{For Reference Only: This subpart (and its subparts) relates to AC-18, SA-3, CCI-00258.} Meet the following requirements throughout the construction process.

1.6.1 Contractor Computer Equipment

Contractor owned computers may be used for construction. When used, contractor computers must meet the following requirements:

1.6.1.1 Operating System

The operating system must be an operating system currently supported by the manufacturer of the operating system. The operating system must be current on security patches and operating system manufacturer required updates.

1.6.1.2 Anti-Malware Software

The computer must run anti-malware software from a reputable software manufacturer. Anti-malware software must be a version currently supported by the software manufacturer, must be current on all patches and updates, and must use the latest definitions file. All computers used on this project must be scanned using the installed software at least once per day.

1.6.1.3 Passwords and Passphrases

The passwords and passphrases for all computers must be changed from their default values. Passwords must be a minimum of eight characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.6.1.4 Contractor Computer Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Computer

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Cybersecurity Compliance Statements for each company using contractor owned computers. Contractor Computer Cybersecurity Compliance Statements must use the template published at <http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>. Each Statement must be signed by a cybersecurity representative for the relevant company.

1.6.2 Temporary IP Networks

Temporary contractor-installed IP networks may be used during construction. When used, temporary contractor-installed IP networks must meet the following requirements:

1.6.2.1 Network Boundaries and Connections

The network must not extend outside the project site and must not connect to any IP network other than IP networks provided under this project or Government furnished IP networks provided for this purpose. Any and all network access from outside the project site is prohibited.

1.6.3 Government Access to Network

Government personnel, as defined, prescribed, and identified by the Contracting Officer, must be allowed to have complete and immediate access to the network at any time in order to verify compliance with this specification. Or if there is a Government agency that's responsible, identify that agency.

1.6.4 Temporary Wireless IP Networks

In addition to the other requirements on temporary IP networks, temporary wireless IP (WiFi) networks must not interfere with existing wireless network and must use WPA2 security. Network names (SSID) for wireless networks must be changed from their default values.

1.6.5 Passwords and Passphrases

The passwords and passphrases for all network devices and network access must be changed from their default values. Passwords must be a minimum 8 characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.6.6 Contractor Temporary Network Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Temporary Network Cybersecurity Compliance Statements for each company implementing a temporary IP network. Contractor Temporary Network Cybersecurity Compliance Statements must use the template published at <http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>. Each Statement must be signed by a cybersecurity representative for the relevant company. If no temporary IP networks will be used, provide a single copy of the Statement indicating this.

PART 2 PRODUCTS

2.1 TEMPORARY SIGNAGE

2.1.1 Bulletin Board

Prior to the commencement of work activities, provide a clear weatherproof

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covered bulletin board not less than 36 by 48 inches in size for displaying the Equal Employment Opportunity poster, a copy of the wage decision contained in the Contract, Wage Rate Information poster, Safety and Health Information as required by EM 385-1-1 Chapter 02, and other information approved by the Contracting Officer. Coordinate requirements herein with 01 35 26.00 06 GOVERNMENTAL SAFETY REQUIREMENTS. Locate the bulletin board at the project site in a conspicuous place easily accessible to all employees, and in location as approved by the Contracting Officer.

2.1.2 Project Identification Signs

The requirements for the signs, their content, and location are as indicated. Erect signs within 15 days after receipt of the notice to proceed. Correct the data required by the safety sign daily, with light colored metallic or non-metallic numerals.

2.1.3 Warning Signs

Post temporary signs, tags, and labels to give workers and the public adequate warning and caution of construction hazards according to the EM 385-1-1 Chapter 04. Attach signs to the perimeter fencing every 150 feet warning the public of the presence of construction hazards. Signs must require unauthorized persons to keep out of the construction site. Correct the data required by safety signs daily. Post signs at all points of entry designating the construction site as a hard hat area.

2.2 TEMPORARY TRAFFIC CONTROL

2.2.1 Barricades

Erect and maintain temporary barricades to limit public access to hazardous areas. Barricades are required whenever safe public access to paved areas such as roads, parking areas or sidewalks is prevented by construction activities or as otherwise necessary to ensure the safety of both pedestrian and vehicular traffic. Securely place barricades clearly visible with adequate illumination to provide sufficient visual warning of the hazard during both day and night.

2.3 FENCING

Provide fencing along the construction site and at all open excavations and tunnels to control access by unauthorized personnel. Safety fencing must be highly visible to be seen by pedestrians and vehicular traffic. All fencing must meet the requirements of EM 385-1-1. Remove the fence upon completion and acceptance of the work.

2.3.1 Polyethylene Mesh Safety Fencing

Temporary safety fencing must be a high visibility orange colored, high density polyethylene grid, a minimum of 48 inches high and maximum mesh size of 2 inches. Fencing must extend from the grade to a minimum of 48 inches above the grade and be tightly secured to T-posts spaced as necessary to maintain a rigid and taut fence. Fencing must remain rigid and taut with a minimum of 200 pounds of force exerted on it from any direction with less than 4 inches of deflection.

2.3.2 Chain Link Panel Fencing

Temporary panel fencing must be galvanized steel chain link panels 6 feet

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high. Multiple fencing panels may be linked together at the bases to form long spans as needed. Each panel base must be weighted down using sand bags or other suitable materials in order for the fencing to withstand anticipated winds while remaining upright. Fencing must remain rigid and taut with a minimum of 200 pounds of force exerted on it from any direction with less than 4 inches of deflection.

2.3.3 Post-Driven Chain Link Fencing

Temporary post-driven fencing must be galvanized chain link fencing 6 feet high supported by an tightly secured to galvanized steel posts driven below grade. Fence posts must be located on minimum 10 foot centers. Posts may be set in various surfaces such as sand, soil, asphalt or concrete as necessary. Chain link fencing must remain rigid and taut with a minimum of 200 pounds of force exerted on it from any direction with less than 4 inches of deflection. Completely remove fencing and posts at the completion of construction and restore surfaces disturbed or damaged to its original condition. Locate and identify underground utilities prior to setting fence posts. Equip fence with a lockable gate. Gate must remain locked when construction personnel are not present.

2.4 TEMPORARY WIRING

Provide temporary wiring in accordance with EM 385-1-1 Chapter 11, NFPA 241 and NFPA 70. Include monthly inspection and testing of all equipment and apparatus.

2.5 BACKFLOW PREVENTERS

Certificate of Full Approval from FCCCHR List, University of Southern California, attesting that the design, size and make of each backflow preventer has satisfactorily passed the complete sequence of performance testing and evaluation for the respective level of approval. Certificate of Provisional Approval is not acceptable.

Reduced pressure principle type conforming to the applicable requirements AWWA C511. Provide backflow preventers complete with 150 pound flanged , ductile iron, brass mounted gate valve and strainer, 304 stainless steel or bronze, internal parts.

PART 3 EXECUTION

3.1 EMPLOYEE PARKING

Construction Contract employees must park privately owned vehicles in an area designated by the Contracting Officer. Employee parking must not interfere with existing and established parking requirements of the Government installation.

3.2 AVAILABILITY AND USE OF UTILITY SERVICES

3.2.1 Temporary Utilities

Provide temporary utilities required for construction. Materials may be new or used, must be adequate for the required usage, not create unsafe conditions, and not violate applicable codes and standards.

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3.2.2 Sanitation

Provide and maintain within the construction area minimum field-type sanitary facilities in accordance with EM 385-1-1 Chapter 35, Section 02. Locate the facilities behind the construction fence or out of the public view. Clean units and empty wastes at least once a week or more frequently into a municipal, district, or station sanitary sewage system, or remove waste to a commercial facility. Obtain approval from the system owner prior to discharge into a municipal, district, or commercial sanitary sewer system. Penalties or fines associated with improper discharge will be the responsibility of the Contractor. Coordinate with the Contracting Officer and follow station regulations and procedures when discharging into the station sanitary sewer system. Maintain these conveniences at all times. Include provisions for pest control and elimination of odors. Government toilet facilities will not be available to Contractor's personnel.

3.2.3 Fire Protection

Provide temporary fire protection equipment for the protection of personnel and property during construction. Remove debris and flammable materials weekly to minimize potential hazards.

3.3 TRAFFIC PROVISIONS

3.3.1 Maintenance of Traffic

- a. Conduct operations in a manner that will not close a thoroughfare or interfere with traffic on railways or highways except with written permission of the Contracting Officer at least 15 calendar days prior to the proposed modification date, and provide a Traffic Control Plan for Government approval detailing the proposed controls to traffic movement for approval. The plan must be in accordance with State and local regulations and the MUTCD, Part VI. Make all notifications and obtain all permits required for modification to traffic movements outside Station's jurisdiction.. Contractor may move oversized and slow-moving vehicles to the worksite provided requirements of the highway authority have been met.
- b. Conduct work so as to minimize obstruction of traffic, and maintain traffic on at least half of the roadway width at all times. Obtain approval from the Contracting Officer prior to starting any activity that will obstruct traffic.
- c. Provide, erect, and maintain, at Contractor's expense, lights, barriers, signals, passageways, detours, and other items, that may be required by the Life Safety Signage, overhead protection authority having jurisdiction.
- d. Provide cones, signs, barricades, lights, or other traffic control devices and personnel required to control traffic. Do not use foil-backed material for temporary pavement marking because of its potential to conduct electricity during accidents involving downed power lines.

3.3.2 Protection of Traffic

Maintain and protect traffic on all affected roads during the construction period except as otherwise specifically directed by the Contracting Officer. Measures for the protection and diversion of traffic, including

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the provision of watchmen and flagmen, erection of barricades, placing of lights around and in front of equipment the work, and the erection and maintenance of adequate warning, danger, and direction signs, will be as required by the State and local authorities having jurisdiction. Provide self-illuminated (lighted) barricades during hours of darkness. Brightly-colored (orange) vests are required for all personnel working in roadways. Protect the traveling public from damage to person and property. Minimize the interference with public traffic on roads selected for hauling material to and from the site. Investigate the adequacy of existing roads and their allowable load limit. Contractor is responsible for the repair of damage to roads caused by construction operations.

3.3.3 Rush Hour Restrictions

Do not interfere with the peak traffic flows preceding and during normal operations without notification to and approval by the Contracting Officer.

3.3.4 Dust Control

Dust control methods and procedures must be approved by the Contracting Officer..

3.4 REDUCED PRESSURE BACKFLOW PREVENTERS

Provide an approved reduced pressure backflow prevention assembly at each location where the Contractor taps into the Government potable water supply.

Perform backflow preventer tests using test equipment, procedures, and certification forms conforming to those outlined in the latest edition of the Manual of Cross-Connection Control published by the FCCCHR Manual. Test and tag each reduced pressure backflow preventer upon initial installation (prior to continued water use) and monthly thereafter. Tag must contain the following information: make, model, serial number, dates of tests, results, maintenance performed, and signature of tester. Record test results on certification forms conforming to requirements cited earlier in this paragraph.

3.5 CONTRACTOR'S TEMPORARY FACILITIES

Temporary facilities must meet requirements as identified in EM 385-1-1 Chapter 04.

Contractor is responsible for security of their property. Provide adequate outside security lighting at the temporary facilities. Trailers must be anchored to resist high winds and meet applicable state or local standards for anchoring mobile trailers. Coordinate anchoring with EM 385-1-1 Chapter 04. The Contract Clause entitled "FAR 52.236-10, Operations and Storage Areas" and the following apply:

3.5.1 Quality Control Manager Records and Field Office

Provide on the jobsite an office with approximately 200 square feet of useful floor area for the exclusive use of the QC Manager. Provide a weathertight structure with adequate heating and cooling, toilet facilities, lighting, ventilation, a 4 by 8 foot plan table, a standard size office desk and chair, computer station, and working communications facilities. Provide either a 1,500 watt radiant heater and a window-mounted air conditioner rated at 9,000 Btus minimum or a

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window-mounted heat pump of the same minimum heating and cooling ratings. Provide a door with a cylinder lock and windows with locking hardware. Make utility connections. Locate as directed. File quality control records in the office and make available at all times to the Government. After completion of the work, remove the entire structure from the site.

3.5.2 Storage Area

Construct a temporary 6 foot high chain link fence around trailers and materials. Include plastic strip inserts, colored brown, so that visibility through the fence is obstructed. Fence posts may be driven, in lieu of concrete bases, where soil conditions permit. Do not place or store trailers, materials, or equipment outside the fenced area unless such trailers, materials, or equipment are assigned a separate and distinct storage area by the Contracting Officer away from the vicinity of the construction site but within the installation boundaries. Trailers, equipment, or materials must not be open to public view with the exception of those items which are in support of ongoing work on the current day. Do not stockpile materials outside the fence in preparation for the next day's work. Park mobile equipment, such as tractors, wheeled lifting equipment, cranes, trucks, and like equipment within the fenced area at the end of each work day.

Keep fencing in a state of good repair and proper alignment. If the Contractor elects to traverse grassed or unpaved areas which are not established roadways with construction equipment or other vehicles, cover the grassed or unpaved areas with a layer of gravel as necessary to prevent rutting and the tracking of mud onto paved or established roadways; gravel gradation must be at the Contractor's discretion.. Mow and maintain grass located within the boundaries of the construction site for the duration of the project. Grass and vegetation along fences, structures, under trailers, and in areas not accessible to mowers must be edged or trimmed neatly.

3.5.3 Supplemental Storage Area

Upon request, and pending availability, the Contracting Officer will designate another or supplemental area for the use and storage of trailers, equipment, and materials. This area may not be in close proximity of the construction site but will be within the installation boundaries. Maintain the area in a clean and orderly fashion and secured if needed to protect supplies and equipment. Utilities will not be provided to this area by the Government.

3.5.4 Appearance of Trailers

- a. Trailers must be roadworthy and comply with all appropriate state and local vehicle requirements. Trailers which are rusted, have peeling paint or are otherwise in need of repair will not be allowed on Installation property. Trailers must present a clean and neat exterior appearance and be in a state of good repair.
- b. Maintain the temporary facilities. Failure to do so will be sufficient reason to require their removal at the Contractor's expense.

3.5.5 Safety Systems

Protect the integrity of all installed safety systems or personnel safety devices. Obtain prior approval from the Contracting Officer if entrance

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into systems serving safety devices is required. If it is temporarily necessary to remove or disable personnel safety devices in order to accomplish Contract requirements, provide alternative means of protection prior to removing or disabling any permanently installed safety devices or equipment and obtain approval from the Contracting Officer.

3.5.6 Weather Protection of Temporary Facilities and Stored Materials

Take necessary precautions to ensure that roof openings and other critical openings in the building are monitored carefully. Take immediate actions required to seal off such openings when rain or other detrimental weather is imminent, and at the end of each workday. Ensure that the openings are completely sealed off to protect materials and equipment in the building from damage.

3.5.6.1 Building and Site Storm Protection

When a warning of gale force winds is issued, take precautions to minimize danger to persons, and protect the work and nearby Government property. Precautions must include, but are not limited to, closing openings; removing loose materials, tools and equipment from exposed locations; and removing or securing scaffolding and other temporary work. Close openings in the work when storms of lesser intensity pose a threat to the work or any nearby Government property.

3.6 TEMPORARY PROJECT SAFETY FENCING

As soon as practicable, but not later than 15 days after the date established for commencement of work, furnish and erect temporary project safety fencing at the work site. Maintain the safety fencing during the life of the Contract and, upon completion and acceptance of the work, remove from the work site.

3.7 CLEANUP

Remove construction debris, waste materials, packaging material and the like from the work site daily. Any dirt or mud which is tracked onto paved or surfaced roadways must be cleaned away. Store all salvageable materials resulting from demolition activities within the fenced area described above or at the supplemental storage area. Neatly stack stored materials not in trailers, whether new or salvaged.

3.8 RESTORATION OF STORAGE AREA

Upon completion of the project remove the bulletin board, signs, barricades, haul roads, and all other temporary products from the site. After removal of trailers, materials, and equipment from within the fenced area, remove the fence. Restore areas used during the performance of the Contract to the original or better condition. Remove gravel used to traverse grassed areas and restore the area to its original condition, including top soil and seeding as necessary.

-- End of Section --

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SECTION 01 78 23

OPERATION AND MAINTENANCE DATA

05/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING
ENGINEERS (ASHRAE)

ASHRAE GUIDELINE 1.4 (2019) Preparing Systems Manuals for
Facilities

ASTM INTERNATIONAL (ASTM)

ASTM E1971 (2005; R 2011) Standard Guide for
Stewardship for the Cleaning of Commercial
and Institutional Buildings

ASTM E2166 (2016; R 2023) Standard Practice for
Organizing and Managing Building Data

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-10 Operation and Maintenance Data

Training Plan; G

Training Outline; G

Training Content; G

Operation And Maintenance Manual, Progress Submittal; G

Operation And Maintenance Manual, Prefinal Submittal; G

Operation And Maintenance Manual, Final Submittal; G

SD-11 Closeout Submittals

Training Video Recording; G

Validation of Training Completion; G

Training Plan; G

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Record Drawings And Utility Systems; G

1.3 MEETINGS

To assure that Operation and Maintenance Manual and Facility Data Workbook requirements are being met through the duration of the project, organize the following meetings and discuss the subsequent topics:

1.3.1 Pre-Construction Meeting

At a minimum, discuss the following:

- a. The requirement for Operation and Maintenance Manuals and Facility Data deliverables under this contract including coordination meetings
- b. Processes and method of gathering Facility Data information during construction
- c. Primary roles and responsibilities associated with the development and delivery of the Operation and Maintenance Manuals and Facility Data deliverables, and
- d. Identify and agree upon a date and attendance list for the meetings described below:

1.3.2 Operation and Maintenance Manual and Facility Data Workbook Coordination Meeting

Facilitate a meeting after the Pre-Construction Meeting prior to the submission of the Operation and Maintenance Manual Progress Submittal. Meeting attendance must include the Contractor's Operation and Maintenance Manual and Facility Data Workbook Preparer, Quality Control Manager, the Commissioning Authority (CxA), the Government's Design Manager (DM), Contracting Officer's Representative, and Government's facility data reviewer. Include any Mechanical, Electrical, and Fire Protection Sub-Contractors.

The purpose of this meeting is to reach a mutual understanding of the scope of work concerning the contract requirements for Operation and Maintenance Manual and coordinate the efforts necessary by both the Government and Contractor to ensure an accurate collection, preparation and timely Government review of Operation and Maintenance Manual.

1.3.3 Submittal Coordination Meeting

Facilitate a meeting following submission and Government review of each design or progress submittal of the Operation and Maintenance Manual and Facility Data Workbook.

- a. Include personnel from the Coordination meeting and any additional personnel identified.
- b. The purpose of this meeting is to demonstrate ongoing compliance with the requirements identified in this specification. Discuss Government review comments and unresolved items preventing completion and Government approval of the Operation and Maintenance Manuals and Facility Data Workbook (FDW).

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- c. The applicable deliverables, along with Government remarks associated with review of these submittals serve as the primary guide and agenda for this meeting.

1.3.4 Facility Turnover Meeting

Include Operation and Maintenance Manual in NAVFAC Red Zone (NRZ) facility turnover meetings.

1.4 FACILITY DATA WORKBOOK

Develop an editable, electronic spreadsheet based on the equipment in the Operation and Maintenance Manuals that contains the information required to start a preventive maintenance program. As a minimum, provide Facility Data Workbook as a list of system equipment, location installed, warranty expiration date, manufacturer, model, and serial number.

1.5 OPERATION AND MAINTENANCE MANUAL MEDIA

Assemble Operation and Maintenance Manual into an electronically bookmarked file using the most current version of Adobe Acrobat or similar software capable of producing PDF file format. Provide compact disks (CD) or data digital versatile disk (DVD) as appropriate, so that each one contains operation, maintenance and record files, project record documents, and training videos. Include a complete bookmarked operation and maintenance directory.

1.5.1 CD or DVD Label and Disk Holder or Case

Provide the following information on the disk label and disk holder or case:

- a. Building Number
- b. Project Title
- c. Activity and Location
- d. Construction Contract Number
- e. Prepared For: (Contracting Agency)
- f. Prepared By: (Name, title, phone number and email address)
- g. Include the disk content on the disk label
- h. Date
- i. Virus scanning program used

1.5.2 O&M Manual Tabbed Hard Copy

Provide a hard copy of the O&M manual upon completion of the project. Provide tabs for each section and subsection for ease of navigation by the user.

1.6 O&M MANUAL CONTENT

Organize the bookmarked Operation and Maintenance Manual into the following Parts in accordance with ASHRAE GUIDELINE 1.4, and as modified and detailed

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below. Word template for O&M Manual is available at:
<https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-01-78-23>.

1.6.1 Part 1: Executive Summary

Provide a summary of the information found in the O&M manual including the purpose of the manual and a description of the manual's organization.

1.6.2 Part 2: Facility Design and Construction

1.6.2.1 General Facility and Systems Description

Provide an overview of the intent for design and use of the facility. Provide a PDF of the Record Drawings and bookmarked using the sheet title and sheet number. Include uncluttered 11 by 17 inches floor plans with room numbers, type or function of space, and overall facility dimensions on the floor plans. Do not include items such as construction instructions, references, or frame numbers.

Detail the overall dimensions of the facility, number of floors, foundation type, expected number of occupants, and facility Category Code list and generally describe all the facility systems and any special building features (for example, HVAC Controls, Sprinkler Systems, Cranes, Elevators, and Generators). Include photographs marked up and labeled to show key operating components and the overall facility appearance.

1.6.2.2 Basis of Design

Provide a copy of the contract Basis of Design.

1.6.2.3 Contract Documents, RFP, Amendments, and Modifications

Provide the contract construction documents complete, to include specifications, drawings, Request for Proposal, amendments, and modifications.

1.6.2.4 Room Inventory of Real Property and Finishes

Provide a list of installed equipment furnished under this contract. Include all information usually listed on manufacturer's name plate. Include, as applicable, the following information for each piece of equipment installed: description of item, all dimensions, location by room number, model number, serial number, capacity, name and address of manufacturer, name and address of equipment supplier, condition, spare parts list, manufacturer's catalog, and warranty. Real property includes, but is not limited to, floor coverings, wall surfaces, ceiling surfaces, windows, roofing, HVAC filters, plumbing fixtures, and lighting fixtures. Submit the final list 30 days after transfer of the completed facility.

Include spatial data defining actual net square footage and data of each room. Also include the room finish schedule including room names and numbers. Include schedules in the construction drawings in the room inventory. Add a column to each schedule to record what was provided by the contractor during construction. Provide a PDF of room inventory. Key the designations to the related area depicted on the contract drawings. List the following data:

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RECORD OF DESIGNATED EQUIPMENT AND MATERIALS DATA				
Description	Specification Section	Manufacturer and Catalog, Model, and Serial Number	Composition and Size	Where Used

1.6.3 Part 3: Facilities, Systems, and Assemblies Information

1.6.3.1 Organization

Bookmark information in this section using the current version of ASTM E2166 Uniformat II, UFGS numbers, and document type as outlined in the example below. Bookmark/tab each item to the third level for easy navigation of the manual.

Example as shown in Table below:

PARTS AND SUBPART NUMBERING	
3.1	B20 EXTERIOR CLOSURE (System)
3.1.1	B2030 EXTERIOR DOORS (Subsystem)
3.1.1.1	B2030110 GLAZED DOORS (Component)
3.1.1.1.1	Applicable specifications List in UFGS Format
3.1.1.1.2	Manufacturer's Operations and Maintenance Data
3.1.1.1.3	Approved Submittal
3.1.1.1.4	Coordination/Shop Drawings
3.1.1.1.5	Sequence of Operation for Operating Equipment
3.1.1.1.6	Testing Equipment Information and Performance Data
3.1.1.1.7	Routine Maintenance Requirements
3.1.1.1.8	Repair Procedures
3.1.1.1.9	Emergency Procedures & Locations of Applicable
3.1.1.1.10	Warranties
3.1.1.1.11	Record Drawings and Utility Systems
3.1.1.1.12	Contractor / Supplies Listing and Contact Information

1.6.3.2 Related Specifications

Reference each specification related to the subsystem in this section, and locate the actual specification section in Part 2 of the O&M Manual. List

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specifications in table format as shown in the below example.

UFGS Number	Specification Title	Page Spec Begins in Part 2

1.6.3.3 Manufacturer's Operations and Maintenance Data

Provide a copy of all manufacturer specifications and cutsheets. Provide text-searchable, high-quality document files from the manufacturer's online or electronic documentation. Color documents are preferred. Provide documents specific to the product(s) installed under this Contract. Provide identification and coverage for the parts of each component, assembly, subassembly, and accessory of the end items subject to replacement. Provide Uniformat II Level 3 identification for D20, D30, D40 installed equipment. When possible, do not submit document files containing multiple product catalogs from the same manufacturer, or product data from multiple manufacturers in the same files. Provide documents directly from the manufacturer whenever possible. Do not provide scanned copies of hardcopy documents. Provide identification and coverage for the parts of each component, assembly, subassembly, and accessory of the end items subject to replacement. Include special hardware requirements, such as requirement to use high-strength bolts and nuts. Identify parts by make, model, serial number, and source of supply to allow reordering without further identification. Provide clear and legible illustrations, drawings, and exploded views to enable easy identification of the items. When illustrations omit the part numbers and description, both the illustrations and separate listing must show the index, reference, or key number that will cross-reference the illustrated part to the listed part. Group the parts shown in the listings by components, assemblies, and subassemblies in accordance with the manufacturer's standard practice. Parts data may cover more than one model or series of equipment, components, assemblies, subassemblies, attachments, or accessories, such as typically shown in a master part catalog.

1.6.3.4 Approved Submittals and Certificates

Provide a copy of all submittals documented with the required approval as applicable for each UFGS specification listed in the table outlined in applicable specifications. Include copies of SD-07 Certificates submittals documented with the required approval, SD-08 Manufacturer's Instructions submittals documented with the required approval, and SD-10 Operation and Maintenance Data submittals documents with the required approval.

1.6.3.5 Approved Coordination/Shop Drawings

Drawings, diagrams and schedules specifically prepared to illustrate some portion of the work. Diagrams and instructions from a manufacturer or fabricator for use in producing the product and as aids to the Contractor for integrating the product or system into the project. Drawings prepared by or for the Contractor to show how multiple systems and interdisciplinary work will be coordinated.

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1.6.4 Sequence of Operation for Operating Equipment

Provide record one-line diagrams for each floor, delineating mechanical equipment location within the building. Provide specific instructions, procedures, and illustrations for the following phases of operation for the installed model and features of each system:

1.6.4.1 Safety Precautions and Hazards

List personnel hazards and equipment or product safety precautions for operating conditions. List all residual hazards identified in the Activity Hazard Analysis provided under Section 01 35 26.00 06 GOVERNMENTAL SAFETY REQUIREMENTS. Provide recommended safeguards for each identified hazard. Specify if any certifications or licenses are required to operate the equipment.

1.6.4.2 Operator Prestart

Provide procedures required to install, set up, and prepare each system for use.

1.6.4.3 Startup, Shutdown, and Post-Shutdown Procedures

Provide narrative description for Startup, Shutdown and Post-shutdown operating procedures including the control sequence for each procedure.

1.6.4.4 Normal Operations

Provide Control Diagrams with data to explain operation and control of systems and specific equipment. Provide narrative description of Normal Operating Procedures.

1.6.4.5 Emergency Operations

Provide Emergency Procedures for equipment malfunctions to permit a short period of continued operation or to shut down the equipment to prevent further damage to systems and equipment. Provide Emergency Shutdown Instructions for fire, explosion, spills, or other foreseeable contingencies. Provide guidance and procedures for emergency operation of utility systems including required valve positions, valve locations and zones or portions of systems controlled.

1.6.4.6 Operator Service Requirements

Provide instructions for services to be performed by the operator such as lubrication, adjustment, inspection, and recording gauge readings.

1.6.4.7 Environmental Conditions

Provide a list of Environmental Conditions (temperature, humidity, and other relevant data) that are best suited for the operation of each product, component or system. Describe conditions under which the item equipment should not be allowed to run.

1.6.4.8 Operating Log

Provide forms, sample logs, and instructions for maintaining necessary operating records.

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1.6.4.9 Additional Requirements for Equipment Control Systems

Provide Data Package 5 and the following for all control systems:

- a. Provide a narrative description on how to perform and apply functions, features, modes, and other operations, including unoccupied operation, seasonal changeover, manual operation, and alarms. Include detailed technical manual for programming and customizing control loops and algorithms.
- b. Submit complete controls equipment schedules, full as-built sequence of operations, wiring and logic diagrams, Input/Output Tables, equipment schedules, copies of checkout tests and calibrations performed by the Contractor (not Cx tests), and all associates information.
- c. Full points list. Provide a listing of rooms with the following information for each room:
 - (1) Floor
 - (2) Room number
 - (3) Room name
 - (4) Air handler unit ID
 - (5) Reference drawing number
 - (6) Air terminal unit tag ID
 - (7) Heating or cooling valve tag ID
 - (8) Minimum cfm
 - (9) Maximum cfm
- d. Full print out of all schedules and set points after testing and acceptance of the system.
- e. Full as-built print out of software program.
- f. Marking of system sensors and thermostats on the as-built floor plan and mechanical drawings with their control system designations.

1.6.4.10 Testing Equipment Information and Performance Data

Include information on test equipment required to perform specified tests and on special tools needed for the operation, maintenance, and repair of components. Provide final set points.

Include completed prefunctional checklists, functional performance test forms, and monitoring reports. Include recommended schedule for retesting and blank test forms. Provide final set points.

1.6.5 Routine Maintenance Requirements

1.6.5.1 Preventive Maintenance Plan, Schedule, and Procedures

Provide manufacturer's schedule for routine preventive maintenance,

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inspections, condition monitoring (predictive tests) and adjustments required to ensure proper and economical operation and to minimize repairs. Provide instructions stating when the systems should be retested. Provide manufacturer's projection of preventive maintenance work-hours on a daily, weekly, monthly, and annual basis including requirements by type of activity. For periodic calibrations, provide manufacturer's specified frequency and procedures for each separate operation.

- a. Define the anticipated time required to perform each of each test (work-hours), test apparatus, number of personnel identified by responsibility, and a testing validation procedure permitting the record operation capability requirements within the schedule. Provide a remarks column for the testing validation procedure referencing operating limits of time, pressure, temperature, volume, voltage, current, acceleration, velocity, alignment, calibration, adjustments, cleaning, or special system notes. Delineate procedures for preventive maintenance, inspection, adjustment, lubrication and cleaning necessary to minimize repairs.
- b. Repair requirements must inform operators how to check out, troubleshoot, repair, and replace components of the system. Include electrical and mechanical schematics and diagrams and diagnostic techniques necessary to enable operation and troubleshooting of the system after acceptance.

1.6.5.2 Lubrication Data

Include the following preventive maintenance lubrication data, in addition to instructions for lubrication required under paragraph OPERATOR SERVICE REQUIREMENTS:

- a. A table showing recommended lubricants for specific temperature ranges and applications.
- b. Charts with a schematic diagram of the equipment showing lubrication points, recommended types and grades of lubricants, and capacities. Provide procedural instructions for Oil Sampling for all equipment.
- c. A Lubrication Schedule showing service interval frequency.

1.6.6 Repair Procedures

Provide instructions and a list of tools required to repair or restore the product or equipment to proper condition or operating standards. Provide manufacturer's recommended procedures and instructions for correcting problems and making repairs for the installed model and features of each system. Include potential environmental and indoor air quality impacts of recommended maintenance procedures and materials. Specify if any certifications or licenses are required to repair the equipment.

1.6.6.1 Troubleshooting Guides and Diagnostic Techniques

Provide step-by-step procedures to promptly isolate the cause of typical malfunctions. Describe clearly why the checkout is performed and what conditions are to be sought. Identify tests or inspections and test equipment required to determine whether parts and equipment may be reused or require replacement.

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1.6.6.2 Wiring Diagrams and Control Diagrams

Provide point-to-point drawings of wiring and control circuits including factory-field interfaces. Provide a complete and accurate depiction of the actual job specific wiring and control work. On diagrams, number electrical and electronic wiring and pneumatic control tubing and the terminals for each type, identically to actual installation configuration and numbering.

1.6.6.3 Removal and Replacement Instructions

Provide step-by-step procedures and a list of required specialty tools and supplies for removal, replacement, disassembly, and assembly of components, assemblies, subassemblies, accessories, and attachments. Provide tolerances, dimensions, settings and adjustments required. Use a combination of text and illustrations.

1.6.6.4 Repair Work-Hours

Provide manufacturer's projection of repair work-hours including requirements by type of craft. Identify, and tabulate separately, repair that requires the equipment manufacturer to complete or to participate.

1.6.6.5 Warranty Information

List and explain the various warranties and clearly identify the servicing and technical precautions prescribed by the manufacturers or contract documents in order to keep warranties in force. Identify if replacement of a subassembly, attachment or accessory requires the entire assembly to be replaced. Include warranty information for primary components of the system. Provide copies of warranties.

1.6.6.6 Extended Warranty Information

List all warranties for products, equipment, components, and sub-components whose duration exceeds one year. For each warranty listed, indicate the applicable specification section, duration, start date, end date, and the point of contact for warranty fulfillment. Also, list or reference the specific operation and maintenance procedures that must be performed to keep the warranty valid. Provide copies of warranties.

1.6.6.7 Record Drawings and Utility Systems

The record drawings are the final compilation of actual conditions reflected in the as-built drawings. Provide record drawings.

Using Record Source Drawings, show and document details of the actual installation of the utility systems, annotate and highlight the Operation and Maintenance information. Provide the following drawings at a large enough scale to differentiate designated isolation units from surrounding valves and switches.

1.6.6.8 Personnel Training Requirements

Provide information available from the manufacturers that is needed for use in training designated personnel to properly operate and maintain the equipment and systems.

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1.6.6.9 Contractor / Supplier Listing and Contact Information

Provide a list that includes the name, address, telephone number, email and website of the General Contractor and each Subcontractor who installed the product or equipment, or system. For each item, also provide the name address and telephone number of the manufacturer's representative and service organization that can provide replacements most convenient to the project site. Provide the name, address, and telephone number of the product, equipment, and system manufacturers.

1.6.7 Part 4: Facility Operations

1.6.7.1 Completed Facility Operating Plan

Provide a plan that documents that procedures for the operation of systems and assemblies in the facility. The systems that should be included in the Operating Plan include, but are not limited to:

- a. Electrical systems and equipment
- b. Mechanical systems and equipment
- c. Fire Protection systems and equipment
- d. Control Systems and equipment
- e. Architectural and Structural systems, fixtures, structures, and equipment
- f. Vertical transportation such as elevators and escalators

1.6.7.2 Testing Equipment and Special Tool Information

Include information on test equipment required to perform specified tests and on special tools needed for the operation, maintenance, and repair of components. Provide final set points.

1.6.7.3 Testing and Performance Data

Include completed prefunctional checklists, functional performance test forms, and monitoring reports. Include recommended schedule for retesting and blank test forms. Provide final set points.

1.6.7.4 Approved Field Test Reports and Manufacturer's Field Reports

Compile and provide approved Field Test Reports (SD-06) and Manufacturer's Field Reports (SD-09) submittals.

1.6.7.5 Maintenance Plans, Procedures, Checklists, Records, and Spare Parts Inventory

1.6.7.5.1 Maintenance Schedules

Include recommended maintenance schedules for systems and equipment.

1.6.7.5.2 Ongoing Commissioning Operational and Maintenance Record Keeping

Include ongoing commissioning and optimization procedures and documentation to monitor and improve the performance of facility systems.

1.6.7.5.3 Janitorial and Cleaning Plans and Procedures

Include a copy of facility cleaning and janitorial plan with procedures and intended chemicals and equipment.

Provide environmentally friendly cleaning recommendations in accordance with ASTM E1971.

1.6.7.6 Utility Record Drawings

1.6.7.6.1 Utility Schematic Diagrams

Provide a one-line schematic diagram for each utility system such as power, water, wastewater, and gas/fuel. Schematic diagram must show from the point where the utility line is connected to the mainline up to the 5 foot connection point to the facility. Indicate location or area designation for route of transmission or distribution lines; locations of duct banks, manholes/handholes or poles; isolation units such as valves and switches; and utility facilities such as pump stations, lift stations, and substations.

1.6.7.6.2 Enlarged Connection and Cutoff Plans

Provide enlarged floor plans and provide information between the 5 foot utilities connection point and where utilities connect to facility distribution. Enlarge floor plans/elevations of the rooms where the utility enters the building and indicate on these plans the locations of the main interiors and exterior connection and cutoff points for the utilities. Also enlarge floor plans/elevations of the rooms where equipment is located. Include enough information to enable someone unfamiliar with the facility to locate the connection and cutoff points. Indicate designations such as room number, panel number, circuit breaker, or valve number of each utility and equipment connection and cutoff point, and what that connection and cutoff point controls.

1.6.7.6.2.1 Description of Utility Metering and Monitoring Systems

Provide in narrative format a description of the utility metering and monitoring systems. Include locations, function, and related systems.

1.6.7.6.2.2 Procedures for Tracking Utility Use and Reporting

Procedures for usage reporting and tracking in support of establishing and monitoring utility budgets and costs, and in developing annual energy reports.

1.6.7.6.2.3 One-Line Diagrams and Meter Location of System

Provide one-line diagrams and design drawings that highlight meter locations on the site.

1.6.7.6.3 Spare Parts and Supply Lists

Provide lists of spare parts and supplies required for repair to ensure continued service or operation without unreasonable delays. Special consideration is required for facilities at remote locations. List spare parts and supplies that have a long lead-time to obtain.

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1.6.8 Part 5: Training

Provide a copy of training plans used for each type of equipment along with training materials used, arranged in specification sequence. Provide a copy of training records, sign-in sheets, and agendas. Include training and documentation on the updating and continued use of the O&M Manual.

1.6.9 Part 6: Cx Project Report and TAB Report

Provide the final Cx Plan and complete Cx reports with evaluation and testing forms and records for each building system. Include relevant commissioned system assemblies test reports including installers checklists of assemblies. Provide all Cx Progress Reports, issues and resolutions logs with resolution or status of each item, and a list of any open items and seasonal or additional testing required.

1.6.10 Part 7: Regulatory Requirements

Provide information describing regulatory and policies compliance requirements or provide a reference to where it is stored.

1.6.11 Part 8: Permits

Provide information requiring frequently asked questions and associated answers or provide a reference to where it is stored.

1.6.12 Part 9: Operations and Maintenance Manual Approval

Provide a signed document stating that the project O&M Manual has been reviewed and confirming agreement with the approach it presents. Include contact information for the signer for coordination of any future changes.

1.7 SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES

Provide the O&M data packages specified in individual technical sections. O&M Data Packages are one of the components of the O&M Manual. The information required in each type of data package follows:

1.7.1 Package Quality

Documents must be fully legible. Operation and Maintenance data must be consistent with the manufacturer's standard brochures, schematics, printed instructions, general operating procedures, and safety precautions.

1.7.2 Data Package 1

- a. Safety precautions and hazards
- b. Cleaning recommendations
- c. Maintenance and repair procedures
- d. Warranty information
- e. Extended warranty information
- f. Contractor information
- g. Spare parts and supply list

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1.7.3 Data Package 2

- a. Safety precautions and hazards
- b. Normal operations
- c. Environmental conditions
- d. Lubrication data
- e. Preventive maintenance plan, schedule, and procedures
- f. Cleaning recommendations
- g. Maintenance and repair procedures
- h. Removal and replacement instructions
- i. Spare parts and supply list
- j. Parts identification
- k. Warranty information
- l. Extended warranty information
- m. Contractor information

1.7.4 Data Package 3

- a. Safety precautions and hazards
- b. Operator prestart
- c. Startup, shutdown, and post-shutdown procedures
- d. Normal operations
- e. Emergency operations
- f. Environmental conditions
- g. Operating log
- h. Lubrication data
- i. Preventive maintenance plan, schedule, and procedures
- j. Cleaning recommendations
- k. Troubleshooting guides and diagnostic techniques
- l. Wiring diagrams and control diagrams
- m. Maintenance and repair procedures
- n. Removal and replacement instructions

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- o. Spare parts and supply list
- p. Product submittal data
- q. O&M submittal data
- r. Parts identification
- s. Warranty information
- t. Extended warranty information
- u. Testing equipment and special tool information
- v. Testing and performance data
- w. Contractor information
- x. Field test reports

1.7.5 Data Package 4

- a. Safety precautions and hazards
- b. Operator prestart
- c. Startup, shutdown, and post-shutdown procedures
- d. Normal operations
- e. Emergency operations
- f. Operator service requirements
- g. Environmental conditions
- h. Operating log
- i. Lubrication data
- j. Preventive maintenance plan, schedule, and procedures
- k. Cleaning recommendations
- l. Troubleshooting guides and diagnostic techniques
- m. Wiring diagrams and control diagrams
- n. Repair procedures
- o. Removal and replacement instructions
- p. Spare parts and supply list
- q. Repair work-hours
- r. Product submittal data
- s. O&M submittal data

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- t. Parts identification
- u. Warranty information
- v. Extended warranty information
- w. Personnel training requirements
- x. Testing equipment and special tool information
- y. Testing and performance data
- z. Contractor information
- aa. Field test reports

1.7.6 Data Package 5

- a. Safety precautions and hazards
- b. Operator prestart
- c. Start-up, shutdown, and post-shutdown procedures
- d. Normal operations
- e. Environmental conditions
- f. Preventive maintenance plan, schedule, and procedures
- g. Troubleshooting guides and diagnostic techniques
- h. Wiring and control diagrams
- i. Maintenance and repair procedures
- j. Removal and replacement instructions
- k. Spare parts and supply list
- l. Product submittal data
- m. Manufacturer's instructions
- n. O&M submittal data
- o. Parts identification
- p. Testing equipment and special tool information
- q. Warranty information
- r. Extended warranty information
- s. Testing and performance data
- t. Contractor information

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u. Field test reports

1.7.7 Changes to Submittals

Provide manufacturer-originated changes or revisions to submitted data if a component of an item is so affected subsequent to acceptance of the O&M Data. Submit changes, additions, or revisions required by the Contracting Officer for final acceptance of submitted data within 30 calendar days of the notification of this change requirement.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 TRAINING

Prior to acceptance of the facility by the Contracting Officer for Beneficial Occupancy, provide comprehensive training for the systems and equipment specified in the technical specifications. The training must be targeted for the building maintenance personnel, and applicable building occupants. Instructors must be well-versed in the particular systems that they are presenting. Address aspects of the Operation and Maintenance Manual submitted. Training must include classroom or field lectures based on the system operating requirements. The location of classroom training requires approval by the Contracting Officer.

3.1.1 Training Plan

Submit a written training plan to the Contracting Officer for approval at least 60 calendar days prior to the scheduled training. Training plan must be approved by the Quality Control Manager (QC) prior to forwarding to the Contracting Officer. Also, coordinate the training schedule with the Contracting Officer and . Include within the plan the following elements:

- a. Equipment included in training
- b. Intended audience
- c. Location of training
- d. Dates of training
- e. Objectives
- f. Outline of the information to be presented and subjects covered including description
- g. Start and finish times and duration of training on each subject
- h. Methods (e.g. classroom lecture, video, site walk-through, actual operational demonstrations, written handouts)
- i. Instructor names and instructor qualifications for each subject
- j. List of texts and other materials to be furnished by the Contractor that are required to support training

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- k. Description of proposed software to be used for video recording of training sessions.

3.1.1.2 Training Content

The core of this training must be based on manufacturer's recommendations and the operation and maintenance information. The QCCxA is responsible for overseeing and approving the content and adequacy of the training. Spend 95 percent of the instruction time during the presentation on the OPERATION AND MAINTENANCE DATA. Include the following for each system training presentation:

- a. Start-up, normal operation, shutdown, unoccupied operation, seasonal changeover, manual operation, controls set-up and programming, troubleshooting, and alarms.
- b. Relevant health and safety issues.
- c. Discussion of how the feature or system is environmentally responsive. Advise adjustments and optimizing methods for energy conservation.
- d. Design intent.
- e. Use of O&M Manual Files.
- f. Review of control drawings and schematics.
- g. Interactions with other systems.
- h. Special maintenance and replacement sources.
- i. Tenant interaction issues.

3.1.1.3 Training Outline

Provide the Operation and Maintenance Manual Files (Bookmarked PDF) and a written course outline listing the major and minor topics to be discussed by the instructor on each day of the course to each trainee in the course. Provide the course outline 14 calendar days prior to the training.

3.1.1.4 Training Video Recording

Record classroom training session(s) on video. Provide to the Contracting Officer two copies of the training session(s) in DVD video recording format. Capture within the recording, in video and audio, the instructors' training presentations including question and answer periods with the attendees. The recording camera(s) must be attended by a person during the recording sessions to assure proper size of exhibits and projections during the recording are visible and readable when viewed as training.

3.1.1.5 Unresolved Questions from Attendees

If, at the end of the training course, there are questions from attendees that remain unresolved, the instructor must send the answers, in writing, to the Contracting Officer for transmittal to the attendees, and the training video must be modified to include the appropriate clarifications.

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3.1.6 Validation of Training Completion

Ensure that each attendee at each training session signs a class roster daily to confirm Government participation in the training. At the completion of training, submit a signed validation letter that includes a sample record of training for reporting what systems were included in the training, who provided the training, when and where the training was performed, and copies of the signed class rosters. Provide two copies of the validation to the Contracting Officer, and one copy to the Operation and Maintenance Manual Preparer for inclusion into the Manual's documentation.

3.1.7 Quality Control Coordination

Coordinate this training with the QC in accordance with Section 01 45 04.10 06 QUALITY CONTROL.

3.2 SUBMITTAL SCHEDULING

3.2.1 Operation and Maintenance Manual, Progress Submittal

Submit the Progress submittal when construction is approximately 50 percent complete, to the Contracting Officer for approval. Provide Operation and Maintenance Manual Files (Bookmarked PDF). Include the elements and portions of system construction completed up to this point. The purpose of this submittal is to verify progress is in accordance with contract requirements as discussed during the Operation and Maintenance Manual Coordination Meeting.

3.2.2 Operation and Maintenance Manual, Prefinal Submittal

Submit the 100 percent submittal of the Operation and Maintenance Prefinal Submittal to the Contracting Officer for approval within 60 calendar days of the Beneficial Occupancy Date (BOD). This submittal must provide a complete, working document that can be used to operate and maintain the facility. Any portion of the submittal that is incomplete or inaccurate requires the entire submittal to be returned for correction. Any discrepancies discovered during the Government's review of Operation and Maintenance Progress submittal must be corrected prior to the Prefinal submission. The Prefinal Submittal must include Operation and Maintenance Manual Files (Bookmarked PDF).

3.2.3 Operation and Maintenance Manual, Final Submittal

Submit completed Operation and Maintenance Manual Files (Bookmarked PDF) and one hard copy of the Operation and Maintenance Manuals. The Final submittal is due at BOD. Any discrepancies discovered during the Government's review of the Prefinal submittal, including the Field Verification, must be corrected prior to the Final submission.

-- End of Section --

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SECTION 02 41 00

DEMOLITION

08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

AHRI Guideline K (2009) Guideline for Containers for
Recovered Non-Flammable Fluorocarbon
Refrigerants

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO M 145 (1991; R 2012) Standard Specification for
Classification of Soils and Soil-Aggregate
Mixtures for Highway Construction Purposes

AASHTO T 180 (2017) Standard Method of Test for
Moisture-Density Relations of Soils Using
a 4.54-kg (10-lb) Rammer and a 457-mm
(18-in.) Drop

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.6 (2006) Safety & Health Program
Requirements for Demolition Operations -
American National Standard for
Construction and Demolition Operations

CARPET AND RUG INSTITUTE (CRI)

CRI 104 (2015) Carpet Installation Standard for
Commercial Carpet

CRI 105 (2015) Carpet Installation Standard for
Residential Carpet

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety and Occupational Health
(SOH) Requirements

U.S. DEFENSE LOGISTICS AGENCY (DLA)

DLA 4145.25 (Jun 2000; Reaffirmed Oct 2010) Storage
and Handling of Liquefied and Gaseous
Compressed Gases and Their Full and Empty
Cylinders;
<https://www.dla.mil/Portals/104/Documents/DispositionSer>

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/ddsr/docs/cylinderjointpub.pdf

U.S. DEPARTMENT OF DEFENSE (DOD)

DOD 4000.25-1-M	(2006) MILSTRIP - Military Standard Requisitioning and Issue Procedures
MIL-STD-129	(2014; Rev R; Change 1 2018; Change 2 2019; Change 3 2023) Military Marking for Shipment and Storage

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 61	National Emission Standards for Hazardous Air Pollutants
40 CFR 82	Protection of Stratospheric Ozone
49 CFR 173.301	Shipment of Compressed Gases in Cylinders and Spherical Pressure Vessels

1.2 PROJECT DESCRIPTION

1.2.1 Definitions

1.2.1.1 Demolition

Demolition is the process of tearing apart and removing any feature of a facility together with any related handling and disposal operations.

1.2.1.2 Deconstruction

Deconstruction is the process of taking apart a facility with the primary goal of preserving the value of all useful building materials.

1.2.1.3 Demolition Plan

Demolition Plan is the planned steps and processes for managing demolition activities and identifying the required sequencing activities and disposal mechanisms.

1.2.1.4 Deconstruction Plan

Deconstruction Plan is the planned steps and processes for dismantling all or portions of a structure or assembly, to include managing sequencing activities, storage, re-installation activities, salvage and disposal mechanisms.

1.2.2 Demolition Plan

Prepare a Demolition Plan and submit proposed demolition, and removal procedures for approval before work is started. Include in the plan procedures for careful removal and disposition of materials specified to be salvaged, coordination with other work in progress. Identify components and materials to be salvaged for reuse or recycling with reference to paragraph Existing Facilities to be Removed. Append tracking forms for all removed materials indicating type, quantities, condition, destination, and end use.. Include statements affirming Contractor inspection of the existing roof deck and its suitability to perform as a safe working

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platform or if inspection reveals a safety hazard to workers, state provisions for securing the safety of the workers throughout the performance of the work. Provide procedures for safe conduct of the work in accordance with EM 385-1-1. Plan must be approved by prior to work beginning.

1.2.3 General Requirements

Do not begin demolition or deconstruction until authorization is received from the Contracting Officer. The work of this section is to be performed in a manner that maximizes the value derived from the salvage and recycling of materials. Remove rubbish and debris from the project site; do not allow accumulations inside or outside the building. The work includes demolition, , salvage of identified items and materials, and removal of resulting rubbish and debris. Remove rubbish and debris from Government property daily, unless otherwise directed. Store materials that cannot be removed daily in areas specified by the Contracting Officer. In the interest of occupational safety and health, perform the work in accordance with EM 385-1-1, Section 23, Demolition, and other applicable Sections.

1.3 ITEMS TO REMAIN IN PLACE

Comply with FAR 52.236-9 to protect existing vegetation, structures, equipment, utilities, and improvements. Coordinate the work of this section with all other work indicated. Construct and maintain shoring, bracing, and supports as required. Ensure that structural elements are not overloaded. Increase structural supports or add new supports as may be required as a result of any cutting, removal, or demolition work performed under this contract. Do not overload structural elements. Provide new supports and reinforcement for existing construction weakened by demolition, or removal work. Repairs, reinforcement, or structural replacement require approval by the Contracting Officer prior to performing such work.

1.3.1 Existing Construction Limits and Protection

Do not disturb existing construction beyond the extent indicated or necessary for installation of new construction. Provide temporary shoring and bracing for support of building components to prevent settlement or other movement. Provide protective measures to control accumulation and migration of dust and dirt in all work areas. Remove dust, dirt, and debris from work areas daily.

1.3.2 Weather Protection

For portions of the building to remain, protect building interior and materials and equipment from the weather at all times. Where removal of existing roofing is necessary to accomplish work, have materials and workmen ready to provide adequate and temporary covering of exposed areas.

1.3.3 Trees

Protect trees within the project site which might be damaged during demolition, and which are indicated to be left in place, by a 6 foot high fence. Erect and secure fence a minimum of 5 feet from the trunk of individual trees or follow the outer perimeter of branches or clumps of trees. Replace any tree designated to remain that is damaged during the work under this contract with like-kind or as approved by the Contracting Officer.

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1.3.4 Utility Service

Maintain existing utilities indicated to stay in service and protect against damage during demolition operations. Prior to start of work, utilities serving each area of alteration or removal will be shut off by the Government and disconnected and sealed by the Contractor .

1.3.5 Facilities

Protect electrical and mechanical services and utilities. Where removal of existing utilities and pavement is specified or indicated, provide approved barricades, temporary covering of exposed areas, and temporary services or connections for electrical and mechanical utilities. Floors, roofs, walls, columns, pilasters, and other structural components that are designed and constructed to stand without lateral support or shoring, and are determined to be in stable condition, must remain standing without additional bracing, shoring, or lateral support until demolished, unless directed otherwise by the Contracting Officer. Ensure that no elements determined to be unstable are left unsupported and place and secure bracing, shoring, or lateral supports as may be required as a result of any cutting, removal, or demolition work performed under this contract.

1.4 BURNING

The use of burning at the project site for the disposal of refuse and debris will not be permitted . Where burning is permitted, adhere to federal, state, and local regulations.

1.5 AVAILABILITY OF WORK AREAS

Areas in which the work is to be accomplished will be available at contract award.

1.6 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Demolition Plan; G

Existing Conditions

SD-07 Certificates

Notification

Notification of Demolition and Renovation Form

SD-11 Closeout Submittals

Receipts

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1.7 QUALITY ASSURANCE

Submit timely notification of demolition projects to Federal, State, regional, and local authorities in accordance with 40 CFR 61, Subpart M. Notify the Regional Office of the United States Environmental Protection Agency (USEPA) and the Contracting Officer in writing 10 working days prior to the commencement of work in accordance with 40 CFR 61, Subpart M. Comply with federal, state, and local hauling and disposal regulations. In addition to the requirements of the "Contract Clauses," conform to the safety requirements contained in ASSP A10.6. Comply with the Environmental Protection Agency requirements specified. Use of explosives will not be permitted.

1.7.1 Dust and Debris Control

Prevent the spread of dust and debris to occupied portions of the building and avoid the creation of a nuisance or hazard in the surrounding area. Do not use water if it results in hazardous or objectionable conditions such as, but not limited to, ice, flooding, or pollution. Vacuum and dust the work area daily.

1.8 PROTECTION

1.8.1 Traffic Control Signs

a. Where pedestrian and driver safety is endangered in the area of removal work, use traffic barricades with flashing lights. Notify the Contracting Officer prior to beginning such work.

1.8.2 Protection of Personnel

Before, during and after the demolition work continuously evaluate the condition of the structure being demolished and take immediate action to protect all personnel working in and around the project site. No area, section, or component of floors, roofs, walls, columns, pilasters, or other structural element will be allowed to be left standing without sufficient bracing, shoring, or lateral support to prevent collapse or failure while workmen remove debris or perform other work in the immediate area.

1.9 FOREIGN OBJECT DAMAGE (FOD)

Aircraft and aircraft engines are subject to FOD from debris and waste material lying on airfield pavements. Remove all such materials that may appear on operational aircraft pavements due to the Contractor's operations. If necessary, the Contracting Officer may require the Contractor to install a temporary barricade at the Contractor's expense to control the spread of FOD potential debris. Provide a barricade consisting of a fence covered with a fabric designed to stop the spread of debris. Anchor the fence and fabric to prevent displacement by winds or jet/prop blasts. Remove barricade when no longer required.

1.10 RELOCATIONS

Perform the removal and reinstallation of relocated items as indicated with workmen skilled in the trades involved. Repair or replace items to be relocated which are damaged by the Contractor with new undamaged items as approved by the Contracting Officer.

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1.11 EXISTING CONDITIONS

Before beginning any demolition or deconstruction work, survey the site and examine the drawings and specifications to determine the extent of the work. Record existing conditions in the presence of the Contracting Officer or the Contracting Officer's Representative showing the condition of structures and other facilities adjacent to areas of alteration or removal. Photographs or electronic images with a minimum resolution of 3072 x 2304 pixels, capable of a print resolution of 300 dpi, will be acceptable as a record of existing conditions. Include in the record the elevation of the top of foundation walls, finish floor elevations, possible conflicting electrical conduits, plumbing lines, alarms systems, the location and extent of existing cracks and other damage and description of surface conditions that exist prior to starting work. It is the Contractor's responsibility to verify and document all required outages which will be required during the course of work, and to note these outages on the record document. Submit survey results to the Contracting Officer or the Contracting Officer's Representative.

PART 2 PRODUCTS

2.1 FILL MATERIAL

- a. Comply with excavating, backfilling, and compacting procedures for soils used as backfill material to fill basements, voids, depressions or excavations resulting from demolition or deconstruction of structures. Provide fill material consisting of waste products from demolition or deconstruction until all waste appropriate for this purpose is consumed.
- b. Provide fill material conforming to the definition of satisfactory soil material as defined in AASHTO M 145, Soil Classification Groups A-1, A-2-4, A-2-5 and A-3. In addition, fill material must be free from roots and other organic matter, trash, debris, frozen materials, and stones larger than 2 inches in any dimension.
- c. Proposed fill material must be sampled and tested by an approved soil testing laboratory, as follows:

Soil classification	AASHTO M 145
Moisture-density relations	AASHTO T 180, Method B or D

PART 3 EXECUTION

3.1 EXISTING FACILITIES TO BE REMOVED

Inspect and evaluate existing structures onsite for reuse. Disassemble existing construction scheduled to be removed for reuse. Dismantled and removed materials are to be separated, set aside, and prepared as specified, and stored or delivered to a collection point for reuse, remanufacture, recycling, or other disposal, as specified. Designate materials for reuse onsite whenever possible.

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3.1.1 Utilities and Related Equipment

3.1.1.1 General Requirements

Do not interrupt existing utilities serving occupied or used facilities, except when authorized in writing by the Contracting Officer. Do not interrupt existing utilities serving facilities occupied and used by the Government except when approved in writing and then only after temporary utility services have been approved and provided. Do not begin demolition or deconstruction work until all utility disconnections have been made. Shut off and cap utilities for future use, as indicated.

3.1.1.2 Disconnecting Existing Utilities

Remove existing utilities , as indicated and terminate in a manner conforming to the nationally recognized code covering the specific utility and approved by the Contracting Officer. When utility lines are encountered but are not indicated on the drawings, notify the Contracting Officer prior to further work in that area. Remove meters and related equipment and deliver to a location in accordance with instructions of the Contracting Officer.

3.1.2 Miscellaneous Metal

Salvage shop-fabricated items such as access doors and frames, steel gratings, metal ladders, wire mesh partitions, metal railings, metal windows and similar items as whole units. Salvage light-gage and cold-formed metal framing, such as steel studs, steel trusses, metal gutters, roofing and siding, metal toilet partitions, toilet accessories and similar items. Scrap metal is the Contractor's property. Recycle scrap metal as part of demolition and deconstruction operations. Provide separate containers to collect scrap metal and transport to a scrap metal collection or recycling facility, in accordance with the Waste Management Plan.

3.1.3 Carpentry

Salvage for recycle lumber, millwork items, and finished boards, and sort by type and size. Remove windows, doors, frames, and cabinets, and similar items as whole units, complete with trim and accessories. Brace the open end of door frames to prevent damage.

3.1.4 Carpet

Remove existing carpet for reclamation in accordance with manufacturer recommendations and as follows. Remove used carpet in large pieces, roll tightly, and pack neatly in a container. Remove adhesive according to recommendations of the Carpet and Rug Institute (CRI). Provide adhesive removal solvents in compliance with CRI 104/CRI 105. Recycle removed carpet cushion.

3.1.5 Acoustic Ceiling Tile

Remove, neatly stack, and recycle acoustic ceiling tiles. Recycling may be available with manufacturer. Otherwise, give priority to a local recycling organization. Recycling is not required if the tiles contain or may have been exposed to asbestos material.

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3.1.6 Air Conditioning Equipment

Remove air conditioning, refrigeration, and other equipment containing refrigerants without releasing chlorofluorocarbon refrigerants to the atmosphere in accordance with the Clean Air Act Amendment of 1990.

3.1.7 Cylinders and Canisters

Remove all fire suppression system cylinders and canisters and dispose of in accordance with the paragraph entitled "Disposal of Ozone Depleting Substance (ODS)."

3.1.8 Locksets on Swinging Doors

Remove all locksets from all swinging doors indicated to be removed and disposed of. Deliver the locksets and related items to a designated location for receipt by the Contracting Officer after removal.

3.1.9 Mechanical Equipment and Fixtures

Disconnect mechanical hardware at the nearest connection to existing services to remain, unless otherwise noted. Disconnect mechanical equipment and fixtures at fittings. Remove service valves attached to the unit. Salvage each item of equipment and fixtures as a whole unit; listed, indexed, tagged, and stored. Salvage each unit with its normal operating auxiliary equipment. Transport salvaged equipment and fixtures, including motors and machines, to a designated storage area as directed by the Contracting Officer. Do not remove equipment until approved. Do not offer low-efficiency equipment for reuse; provide to recycling service for disassembly and recycling of parts.

3.1.9.1 Preparation for Storage

Remove water, dirt, dust, and foreign matter from units; drain tanks, piping and fixtures; if previously used to store flammable, explosive, or other dangerous liquids, steam clean interiors. Seal openings with caps, plates, or plugs. Secure motors attached by flexible connections to the unit. Change lubricating systems with the proper oil or grease.

3.1.9.2 Piping

Disconnect piping at unions, flanges and valves, and fittings as required to reduce the pipe into straight lengths for practical storage. Store salvaged piping according to size and type. If the piping that remains can become pressurized due to upstream valve failure, attach end caps, blind flanges, or other types of plugs or fittings with a pressure gage and bleed valve to the open end of the pipe to ensure positive leak control. Carefully dismantle piping that previously contained gas, gasoline, oil, or other dangerous fluids, with precautions taken to prevent injury to persons and property. Store piping outdoors until all fumes and residues are removed. Box prefabricated supports, hangers, plates, valves, and specialty items according to size and type. Wrap sprinkler heads individually in plastic bags before boxing. Classify piping not designated for salvage, or not reusable, as scrap metal.

3.1.9.3 Ducts

Classify removed duct work as scrap metal.

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3.1.9.4 Fixtures, Motors and Machines

Remove and salvage fixtures, motors and machines associated with plumbing, heating, air conditioning, refrigeration, and other mechanical system installations. Salvage, box and store auxiliary units and accessories with the main motor and machines. Tag salvaged items for identification, storage, and protection from damage. Classify broken, damaged, or otherwise unserviceable units and not caused to be broken, damaged, or otherwise unserviceable as debris to be disposed of by the Contractor. Salvage and crush porcelain plumbing fixtures unsuitable for reuse.

3.1.10 Electrical Equipment and Fixtures

Salvage motors, motor controllers, and operating and control equipment that are attached to the driven equipment. Salvage wiring systems and components. Box loose items and tag for identification. Disconnect primary, secondary, control, communication, and signal circuits at the point of attachment to their distribution system.

3.1.10.1 Fixtures

Remove and salvage electrical fixtures. Salvage unprotected glassware from the fixture and salvage separately. Salvage incandescent, mercury-vapor, and fluorescent lamps and fluorescent ballasts manufactured prior to 1978, boxed and tagged for identification, and protected from breakage.

3.1.10.2 Electrical Devices

Remove and salvage switches, switchgear, transformers, conductors including wire and nonmetallic sheathed and flexible armored cable, regulators, meters, instruments, plates, circuit breakers, panelboards, outlet boxes, and similar items. Box and tag these items for identification according to type and size.

3.1.10.3 Wiring Ducts or Troughs

Remove and salvage wiring ducts or troughs. Dismantle plug-in ducts and wiring troughs into unit lengths. Remove plug-in or disconnecting devices from the busway and store separately.

3.1.10.4 Conduit and Miscellaneous Items

Salvage conduit except where embedded in concrete or masonry. Consider corroded, bent, or damaged conduit as scrap metal. Sort straight and undamaged lengths of conduit according to size and type. Classify supports, knobs, tubes, cleats, and straps as debris to be removed and disposed.

3.2 CONCURRENT EARTH-MOVING OPERATIONS

Do not begin excavation, filling, and other earth-moving operations that are sequential to demolition or deconstruction work in areas occupied by structures to be demolished or deconstructed until all demolition and deconstruction in the area has been completed and debris removed. Fill holes, open basements and other hazardous openings.

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3.3 DISPOSITION OF MATERIAL

3.3.1 Title to Materials

Except for salvaged items specified in related Sections, and for materials or equipment scheduled for salvage, all materials and equipment removed and not reused or salvaged, become the property of the Contractor and must be removed from Government property. Materials approved for storage by the Contracting Officer must be removed before completion of the contract. Title to materials resulting from demolition and deconstruction, and materials and equipment to be removed, is vested in the Contractor upon approval by the Contracting Officer. The Government will not be responsible for the condition or loss of, or damage to, such property after contract award. Showing for sale or selling materials and equipment on site is prohibited.

3.3.2 Reuse of Materials and Equipment

Remove and store materials and equipment indicated to be reused or relocated to prevent damage, and reinstall as the work progresses. Coordinate the re-use of materials and equipment with the re-use requirements. Capture re-use of materials in the diversion calculations for the project.

3.3.3 Salvaged Materials and Equipment

Remove materials and equipment that are indicated to be removed by the Contractor and that are to remain the property of the Government, and deliver to a storage site, as directed by the Contracting Officer within ten miles of the work site.

- a. Salvage items and material to the maximum extent possible.
- b. Store all materials salvaged for the Contractor as approved by the Contracting Officer and remove from Government property before completion of the contract. Coordinate the salvaged materials with tracking requirements. Capture salvaged materials in the diversion calculations for the project.
- c. Remove salvaged items to remain the property of the Government in a manner to prevent damage, and packed or crated to protect the items from damage while in storage or during shipment. Items damaged during removal or storage must be repaired or replaced to match existing items. Properly identify the contents of containers. Deliver the following items reserved as property of the Government to the areas designated.

3.3.4 Disposal of Ozone Depleting Substance (ODS)

Class I and Class II ODS are defined in Section, 602(a) and (b), of The Clean Air Act. Prevent discharge of Class I and Class II ODS to the atmosphere. Place recovered ODS in cylinders meeting AHRI Guideline K suitable for the type ODS (filled to no more than 80 percent capacity) and provide appropriate labeling. Recovered ODS must be removed from Government property and disposed of in accordance with 40 CFR 82. Dispose products, equipment and appliances containing ODS in a sealed, self-contained system (e.g. residential refrigerators and window air conditioners) in accordance with 40 CFR 82. Submit Receipts or bills of lading, as specified. Submit a shipping receipt or bill of lading for all

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containers of ozone depleting substance (ODS) shipped to the Defense Depot, Richmond, Virginia.

3.3.4.1 Special Instructions

No more than one type of ODS is permitted in each container. Apply a warning/hazardous label to the containers in accordance with Department of Transportation regulations. Provide a tag with the following information on all cylinders including but not limited to fire extinguishers, spheres, or canisters containing an ODS:

- a. Activity name and unit identification code
- b. Activity point of contact and phone number
- c. Type of ODS and pounds of ODS contained
- d. Date of shipment
- e. National stock number (for information, call (804) 279-4525).

3.3.4.2 Fire Suppression Containers

Deactivate fire suppression system cylinders and canisters with electrical charges or initiators prior to shipment. Also, safety caps must be used to cover exposed actuation mechanisms and discharge ports on these special cylinders.

3.3.5 Transportation Guidance

Ship all ODS containers in accordance with MIL-STD-129, DLA 4145.25 (also referenced one of the following: Army Regulation 700-68, Naval Supply Instruction 4440.128C, Marine Corps Order 10330.2C, and Air Force Regulation 67-12), 49 CFR 173.301, and DOD 4000.25-1-M.

3.3.6 Unsalvageable and Non-Recyclable Material

Dispose of unsalvageable and non-recyclable noncombustible material as directed by the Contracting Officer and specified. Dispose of unsalvageable and non-recyclable combustible material off the site.

3.4 CLEANUP

Remove debris and rubbish from project site and similar excavations. Remove and transport the debris in a manner that prevents spillage on streets or adjacent areas. Apply local regulations regarding hauling and disposal.

3.5 DISPOSAL OF REMOVED MATERIALS

3.5.1 Burning on Government Property

Burning of materials removed from demolished and deconstructed structures will not be permitted on Government property.

3.5.2 Removal to Spoil Areas on Government Property

Transport noncombustible materials removed from demolition structures to designated spoil areas on Government property.

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3.5.3 Removal from Government Property

Transport waste materials removed from demolished structures, except waste soil, from Government property for legal disposal. Dispose of waste soil as directed.

3.6 REUSE OF SALVAGED ITEMS

Recondition salvaged materials and equipment designated for reuse before installation. Replace items damaged during removal and salvage operations or restore them as necessary to usable condition.

-- End of Section --

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SECTION 03 30 00

CAST-IN-PLACE CONCRETE

02/19, CHG 7: 02/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 117	(2010; Errata 2011) Specifications for Tolerances for Concrete Construction and Materials and Commentary
ACI 121R	(2008) Guide for Concrete Construction Quality Systems in Conformance with ISO 9001
ACI 301	(2016) Specifications for Structural Concrete
ACI 302.1R	(2015) Guide for Concrete Floor and Slab Construction
ACI 304.2R	(2017) Guide to Placing Concrete by Pumping Methods
ACI 304R	(2000; R 2009) Guide for Measuring, Mixing, Transporting, and Placing Concrete
ACI 305.1	(2014) Specification for Hot Weather Concreting
ACI 305R	(2020) Guide to Hot Weather Concreting
ACI 306.1	(1990; R 2002) Standard Specification for Cold Weather Concreting
ACI 306R	(2016) Guide to Cold Weather Concreting
ACI 308.1	(2011) Specification for Curing Concrete
ACI SP-2	(2007; Abstract: 10th Edition) ACI Manual of Concrete Inspection
ACI SP-15	(2011) Field Reference Manual: Standard Specifications for Structural Concrete ACI 301-05 with Selected ACI References

AMERICAN HARDBOARD ASSOCIATION (AHA)

AHA A135.4	(1995; R 2004) Basic Hardboard
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AMERICAN WELDING SOCIETY (AWS)

AWS D1.4/D1.4M (2011) Structural Welding Code -
Reinforcing Steel

ASTM INTERNATIONAL (ASTM)

ASTM A53/A53M (2022) Standard Specification for Pipe,
Steel, Black and Hot-Dipped, Zinc-Coated,
Welded and Seamless

ASTM A184/A184M (2019) Standard Specification for Welded
Deformed Steel Bar Mats for Concrete
Reinforcement

ASTM A615/A615M (2022) Standard Specification for Deformed
and Plain Carbon-Steel Bars for Concrete
Reinforcement

ASTM A706/A706M (2022a) Standard Specification for
Low-Alloy Steel Deformed and Plain Bars
for Concrete Reinforcement

ASTM A934/A934M (2022) Standard Specification for
Epoxy-Coated Prefabricated Steel
Reinforcing Bars

ASTM A996/A996M (2016) Standard Specification for
Rail-Steel and Axle-Steel Deformed Bars
for Concrete Reinforcement

ASTM A1044/A1044M (2016a) Standard Specification for Steel
Stud Assemblies for Shear Reinforcement of
Concrete

ASTM A1064/A1064M (2022) Standard Specification for
Carbon-Steel Wire and Welded Wire
Reinforcement, Plain and Deformed, for
Concrete

ASTM C31/C31M (2023) Standard Practice for Making and
Curing Concrete Test Specimens in the Field

ASTM C33/C33M (2023) Standard Specification for Concrete
Aggregates

ASTM C39/C39M (2023) Standard Test Method for
Compressive Strength of Cylindrical
Concrete Specimens

ASTM C42/C42M (2020) Standard Test Method for Obtaining
and Testing Drilled Cores and Sawed Beams
of Concrete

ASTM C78/C78M (2022) Standard Test Method for Flexural
Strength of Concrete (Using Simple Beam
with Third-Point Loading)

ASTM C94/C94M (2023) Standard Specification for

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Ready-Mixed Concrete

ASTM C136/C136M	(2019) Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates
ASTM C143/C143M	(2020) Standard Test Method for Slump of Hydraulic-Cement Concrete
ASTM C150/C150M	(2022) Standard Specification for Portland Cement
ASTM C172/C172M	(2017) Standard Practice for Sampling Freshly Mixed Concrete
ASTM C173/C173M	(2023) Standard Test Method for Air Content of Freshly Mixed Concrete by the Volumetric Method
ASTM C231/C231M	(2022) Standard Test Method for Air Content of Freshly Mixed Concrete by the Pressure Method
ASTM C260/C260M	(2010a; R 2016) Standard Specification for Air-Entraining Admixtures for Concrete
ASTM C311/C311M	(2022) Standard Test Methods for Sampling and Testing Fly Ash or Natural Pozzolans for Use in Portland-Cement Concrete
ASTM C494/C494M	(2019; E 2022) Standard Specification for Chemical Admixtures for Concrete
ASTM C595/C595M	(2023) Standard Specification for Blended Hydraulic Cements
ASTM C618	(2023; E 2023) Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete
ASTM C803/C803M	(2023) Standard Test Method for Penetration Resistance of Hardened Concrete
ASTM C845/C845M	(2018) Standard Specification for Expansive Hydraulic Cement
ASTM C873/C873M	(2015) Standard Test Method for Compressive Strength of Concrete Cylinders Cast in Place in Cylindrical Molds
ASTM C900	(2015) Standard Test Method for Pullout Strength of Hardened Concrete
ASTM C920	(2018) Standard Specification for Elastomeric Joint Sealants
ASTM C989/C989M	(2022) Standard Specification for Slag Cement for Use in Concrete and Mortars
ASTM C1012/C1012M	(2018b) Standard Test Method for Length

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	Change of Hydraulic-Cement Mortars Exposed to a Sulfate Solution
ASTM C1017/C1017M	(2013; E 2015) Standard Specification for Chemical Admixtures for Use in Producing Flowing Concrete
ASTM C1074	(2019) Standard Practice for Estimating Concrete Strength by the Maturity Method
ASTM C1077	(2017) Standard Practice for Agencies Testing Concrete and Concrete Aggregates for Use in Construction and Criteria for Testing Agency Evaluation
ASTM C1107/C1107M	(2020) Standard Specification for Packaged Dry, Hydraulic-Cement Grout (Nonshrink)
ASTM C1116/C1116M	(2023) Standard Specification for Fiber-Reinforced Concrete
ASTM C1157/C1157M	(2023) Standard Performance Specification for Hydraulic Cement
ASTM C1218/C1218M	(2020c) Standard Test Method for Water-Soluble Chloride in Mortar and Concrete
ASTM C1240	(2020) Standard Specification for Silica Fume Used in Cementitious Mixtures
ASTM C1260	(2023) Standard Test Method for Potential Alkali Reactivity of Aggregates (Mortar-Bar Method)
ASTM C1293	(2008; R 2015) Standard Test Method for Determination of Length Change of Concrete Due to Alkali-Silica Reaction
ASTM C1567	(2023) Standard Test Method for Potential Alkali-Silica Reactivity of Combinations of Cementitious Materials and Aggregate (Accelerated Mortar-Bar Method)
ASTM C1602/C1602M	(2022) Standard Specification for Mixing Water Used in Production of Hydraulic Cement Concrete
ASTM C1778	(2016) Standard Guide for Reducing the Risk of Deleterious Alkali-Aggregate Reaction in Concrete
ASTM D1751	(2018) Standard Specification for Preformed Expansion Joint Filler for Concrete Paving and Structural Construction (Nonextruding and Resilient Bituminous Types)
ASTM D1752	(2018) Standard Specification for

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Preformed Sponge Rubber, Cork and Recycled
 PVC Expansion Joint Fillers for Concrete
 Paving and Structural Construction

ASTM D2628 (1991; R 2016) Standard Specification for
 Preformed Polychloroprene Elastomeric
 Joint Seals for Concrete Pavements

ASTM D2835 (1989; R 2017) Standard Specification for
 Lubricant for Installation of Preformed
 Compression Seals in Concrete Pavements

ASTM D5759 (2012; R 2020) Characterization of Coal
 Fly Ash and Clean Coal Combustion Fly Ash
 for Potential Uses

ASTM D6690 (2015) Standard Specification for Joint
 and Crack Sealants, Hot Applied, for
 Concrete and Asphalt Pavements

ASTM E96/E96M (2022a; E 2023) Standard Test Methods for
 Gravimetric Determination of Water Vapor
 Transmission Rate of Materials

ASTM E329 (2023) Standard Specification for Agencies
 Engaged in Construction Inspection,
 Testing, or Special Inspection

ASTM E1745 (2017; R 2023) Standard Specification for
 Water Vapor Retarders Used in Contact with
 Soil or Granular Fill under Concrete Slabs

CONCRETE REINFORCING STEEL INSTITUTE (CRSI)

CRSI 10MSP (2018) Manual of Standard Practice

CRSI RB4.1 (2016) Supports for Reinforcement Used in
 Concrete

FOREST STEWARDSHIP COUNCIL (FSC)

FSC STD 01 001 (2015) Principles and Criteria for Forest
 Stewardship

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 14025 (2006) Environmental Labels and
 Declarations

ISO 21930 (2017) Sustainability in Buildings and
 Civil Engineering Works

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST PS 1 (2009) DOC Voluntary Product Standard PS
 1-07, Structural Plywood

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NSF INTERNATIONAL (NSF)

NSF PCR (2021) Product Category Rule for Portland,
Blended, Masonry, Mortar, and Plastic
(Stucco) Cements

U.S. ARMY CORPS OF ENGINEERS (USACE)

COE CRD-C 104 (1980) Method of Calculation of the
Fineness Modulus of Aggregate

COE CRD-C 572 (1974) Corps of Engineers Specifications
for Polyvinylchloride Waterstops

1.2 DEFINITIONS

- a. "Cementitious material" as used herein must include all portland cement, pozzolan, fly ash, slag cement, and silica fume. Added lime, including the extra lime in Type IL blended cement, isn't considered a "cementitious material".
- b. "Exposed to public view" means situated so that it can be seen from eye level from a public location after completion of the building. A public location is accessible to persons not responsible for operation or maintenance of the building.
- c. "Chemical admixtures" are materials in the form of powder or fluids that are added to the concrete to give it certain characteristics not obtainable with plain concrete mixes.
- d. "Supplementary cementing materials" (SCM) include coal fly ash, silica fume, slag cement, natural or calcined pozzolans, and ultra-fine coal ash when used in such proportions to replace the portland cement that result in improvement to sustainability and durability and reduced cost.
- e. "Design strength" (f'c) is the specified compressive strength of concrete at time(s) specified in this section to meet structural design criteria.
- f. "Mass Concrete" is any concrete system that approaches a maximum temperature of 158 degrees F within the first 72 hours of placement. In addition, it includes all concrete elements with a section thickness of 3 feet or more regardless of temperature.
- g. "Mixture proportioning" is the process of designing concrete mixture proportions to enable it to meet the strength, service life and constructability requirements of the project while minimizing the initial and life-cycle cost.
- h. "Mixture proportions" are the masses or volumes of individual ingredients used to make a unit measure (cubic meter or cubic yard) of concrete.
- i. "Pozzolan" is a siliceous or siliceous and aluminous material, which in itself possesses little or no cementitious value but will, in finely divided form and in the presence of moisture, chemically react with calcium hydroxide at ordinary temperatures to form compounds possessing cementitious properties.

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- j. "Workability (or consistence)" is the ability of a fresh (plastic) concrete mix to fill the form/mould properly with the desired work (vibration) and without reducing the concrete's quality. Workability depends on water content, chemical admixtures, aggregate (shape and size distribution), cementitious content and age (level of hydration).

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 00 80 00.00 06 SPECIAL PROVISIONS:

SD-01 Preconstruction Submittals

Concrete Curing Plan

Quality Control Program; G

Quality Control Personnel Certifications

Quality Control Organizational Chart

Laboratory Accreditation

Maturity Method Data

SD-02 Shop Drawings

Reinforcing Steel; G

SD-03 Product Data

Joint Sealants; S

Joint Filler; S

Formwork Materials

Cementitious Materials; S

Vapor Retarder

Concrete Curing Materials

Reinforcement; S

Admixtures

Reinforcing Fibers

Mechanical Reinforcing Bar Connectors

Waterstops

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Local/Regional Materials; S

Biodegradable Form Release Agent

Pumping Concrete

Finishing Plan

Nonshrink Grout

SD-05 Design Data

Concrete Mix Design; G

SD-06 Test Reports

Concrete Mix Design; G

Fly Ash

Pozzolan

Slag Cement

Aggregates

Fiber-Reinforced Concrete

Compressive Strength Tests

Air Content

Slump Tests

Water

SD-07 Certificates

Reinforcing Bars

Welder Qualifications

Silica Fume Manufacturer's Representative

Safety Data Sheets

Forest Stewardship Council (FSC) Certification

Field Testing Technician and Testing Agency

Environmental Product Declaration; S

Energy Star Energy Performance Score; S

SD-08 Manufacturer's Instructions

Joint Sealants; S

Curing Compound

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1.4 MODIFICATION OF REFERENCES

Accomplish work in accordance with ACI publications except as modified herein. Consider the advisory or recommended provisions to be mandatory. Interpret reference to the "Building Official," the "Structural Engineer," and the "Architect/Engineer" to mean the Contracting Officer.

1.5 DELIVERY, STORAGE, AND HANDLING

Follow ACI 301, ACI 304R and ASTM A934/A934M requirements and recommendations. Do not deliver concrete until vapor retarder, forms, reinforcement, embedded items, and chamfer strips are in place and ready for concrete placement. Do not store concrete curing compounds or sealers with materials that have a high capacity to adsorb volatile organic compound (VOC) emissions. Do not store concrete curing compounds or sealers in occupied spaces.

1.5.1 Reinforcement

Store reinforcement of different sizes and shapes in separate piles or racks raised above the ground to avoid excessive rusting. Protect from contaminants such as grease, oil, and dirt. Ensure bar sizes can be accurately identified after bundles are broken and tags removed.

1.6 QUALITY ASSURANCE

1.6.1 Design Data

1.6.1.1 Concrete Mix Design

Sixty days minimum prior to concrete placement, submit a mix design for each strength and type of concrete. Submit all documentation required in ACI 301 Section 4 and as specified in this section. Submit a complete list of materials including type; brand; source and amount of cement, supplementary cementitious materials, fibers, and admixtures; and applicable reference specifications. Submit mill test and all other test for cement, supplementary cementitious materials, aggregates, and admixtures. Provide documentation of maximum nominal aggregate size, gradation analysis, percentage retained and passing sieve, and a graph of percentage retained verses sieve size. Provide mix proportion data using at least three different water-cementitious material ratios for each type of mixture, which produce a range of strength encompassing those required for each type of concrete required. If source material changes, resubmit mix proportion data using revised source material. Provide only materials that have been proven by trial mix studies to meet the requirements of this specification, unless otherwise approved in writing by the Contracting Officer. Indicate clearly in the submittal where each mix design is used when more than one mix design is submitted. Resubmit data on concrete components if the qualities or source of components changes. Required average strength can be documented by field experience if field strength test data are available and represent a single group of at least 10 consecutive strength tests for one mixture, using materials and conditions similar to those expected for work, and encompassing a period of not less than 45 days. The average of field strength tests shall equal or exceed fcr'. Changes in materials, conditions, and proportions within the test record shall not have been more closely restricted than those for the proposed work. Test records shall not be more than 24 months old. Obtain mix design approval from the contracting officer prior to concrete

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placement.

1.6.2 Shop Drawings

1.6.2.1 Reinforcing Steel

Indicate bending diagrams, assembly diagrams, splicing and laps of bars, shapes, dimensions, and details of bar reinforcing, accessories, and concrete cover. Do not scale dimensions from structural drawings to determine lengths of reinforcing bars. Reproductions of contract drawings are unacceptable.

1.6.3 Control Submittals

1.6.3.1 Concrete Curing Plan

Submit proposed materials, methods and duration for curing concrete elements in accordance with ACI 308.1.

1.6.3.2 Pumping Concrete

Submit proposed materials and methods for pumping concrete. Submittal must include mix designs, pumping equipment including type of pump and size and material for pipe, and maximum length and height concrete is to be pumped.

1.6.3.3 Silica Fume Manufacturer's Representative

The manufacturer's representative must be present at mix plant to ensure proper mix, including high range water reducer, and batching methods during the first 3 days of concrete mix preparation and placement. After which the manufacturer's representative must designate a representative at the concrete producer's plant to ensure the concrete mix procedures meet the silica fume manufacturer's recommendations. Representative to attend and advise at finishing of sample slab.

1.6.3.4 Finishing Plan

Submit proposed material and procedures to be used in obtaining the finish for the floors. Include qualification of person to be used for obtaining floor tolerance measurement, description of measuring equipment to be used, and a sketch showing lines and locations the measuring equipment will follow.

1.6.3.5 Safety Data Sheets

Submit Safety Data Sheets (SDS) for all materials that are regulated for hazardous health effects. SDS must be readily accessible during each work shift to employees when they are at the construction site.

1.6.4 Test Reports

1.6.4.1 Fly Ash and Pozzolan

Submit test results in accordance with ASTM C618 for fly ash and pozzolan. Submit test results performed within 6 months of submittal date.

1.6.4.2 Slag Cement

Submit test results in accordance with ASTM C989/C989M for slag cement.

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Submit test results performed within 6 months of submittal date.

1.6.4.3 Aggregates

Submit test results in accordance with ASTM C33/C33M, and ASTM C1293 or ASTM C1567 as required in the paragraph titled ALKALI-AGGREGATE REACTION.

1.6.4.4 Fiber-Reinforced Concrete

Test to determine flexural toughness index I5 in accordance with ASTM C1116/C1116M.

1.6.5 Quality Control Program

Develop and submit for approval a concrete quality control program in accordance with the guidelines of ACI 121R and as specified herein. The plan must include approved laboratories. Provide direct oversight for the concrete qualification program inclusive of associated sampling and testing. All quality control reports must be provided to the Contracting Officer, Quality Manager and Concrete Supplier. Maintain a copy of ACI SP-15 and CRSI 10MSP at project site.

1.6.6 Quality Control Personnel Certifications

The Contractor must submit for approval the responsibilities of the various quality control personnel, including the names and qualifications of the individuals in those positions and a quality control organizational chart defining the quality control hierarchy and the responsibility of the various positions. Quality control personnel must be employed by the Contractor.

Submit American Concrete Institute certification for the following:

- a. CQC personnel responsible for inspection of concrete operations.
- b. Lead Foreman or Journeyman of the Concrete Placing, Finishing, and Curing Crews.
- c. Field Testing Technicians: ACI Concrete Field Testing Technician, Grade I.

1.6.6.1 Quality Manager Qualifications

The quality manager must hold a current license as a professional engineer in a U.S. state or territory with experience on at least five similar projects. Evidence of extraordinary proven experience may be considered by the Contracting Officer as sufficient to act as the Quality Manager.

1.6.6.2 Field Testing Technician and Testing Agency

Submit data on qualifications of proposed testing agency and technicians for approval by the Contracting Officer prior to performing testing on concrete.

- a. Work on concrete under this contract must be performed by an ACI Concrete Field Testing Technician Grade 1 qualified in accordance with ACI SP-2 or equivalent. Equivalent certification programs must include requirements for written and performance examinations as stipulated in ACI SP-2.

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- b. Testing agencies that perform testing services on reinforcing steel must meet the requirements of ASTM E329.
- c. Testing agencies that perform testing services on concrete materials must meet the requirements of ASTM C1077.

1.6.7 Laboratory Qualifications for Concrete Qualification Testing

The concrete testing laboratory must have the necessary equipment and experience to accomplish required testing. The laboratory must meet the requirements of ASTM C1077 and be Cement and Concrete Reference Laboratory (CCRL) inspected.

1.6.8 Laboratory Accreditation

Laboratory and testing facilities must be provided by and at the expense of the Contractor. The laboratories performing the tests must be accredited in accordance with ASTM C1077, including ASTM C78/C78M and ASTM C1260. The accreditation must be current and must include the required test methods, as specified. Furthermore, the testing must comply with the following requirements:

- a. Aggregate Testing and Mix Proportioning: Aggregate testing and mixture proportioning studies must be performed by an accredited laboratory and under the direction of a registered professional engineer in a U.S. state or territory competent in concrete materials who is competent in concrete materials and must sign all reports and designs.
- b. Acceptance Testing: Furnish all materials, labor, and facilities required for molding, curing, testing, and protecting test specimens at the site and in the laboratory. Furnish and maintain boxes or other facilities suitable for storing and curing the specimens at the site while in the mold within the temperature range stipulated by ASTM C31/C31M.
- c. Contractor Quality Control: All sampling and testing must be performed by an approved, onsite, independent, accredited laboratory.

1.7 ENVIRONMENTAL REQUIREMENTS

Provide space ventilation according to material manufacturer recommendations, at a minimum, during and following installation of concrete curing compound and sealer. Maintain one of the following ventilation conditions during the curing period or for 72 hours after installation:

- a. Supply 100 percent outside air 24 hours a day.
- b. Supply airflow at a rate of 6 air changes per hour, when outside temperatures are between 55 degrees F and 84 degrees F and humidity is between 30 percent and 60 percent.
- c. Supply airflow at a rate of 1.5 air changes per hour, when outside air conditions are not within the range stipulated above.

1.7.1 Submittals for Environmental Performance

- a. Provide data indication the percentage of post-industrial pozzolan (fly

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ash, slag cement) cement substitution as a percentage of the full product composite by weight.

- b. Provide data indicating the percentage of post-industrial and post-consumer recycled content aggregate.
- c. Provide product data indicating the percentage of post-consumer recycled steel content in each type of steel reinforcement as a percentage of the full product composite by weight.
- d. Provide product data stating the location where all products were manufactured
- e. For projects using FSC certified formwork, provide chain-of-custody documentation for all certified wood products.
- f. For projects using reusable formwork, provide data showing how formwork is reused.
- g. Provide SDS product information data showing that form release agents meet any environmental performance goals such as using vegetable and soy based products.
- h. Provide SDS product information data showing that concrete adhesives meet any environmental performance goals including low emitting, low volatile organic compound products.

1.8 SUSTAINABLE DESIGN REQUIREMENTS

1.8.1 Local/Regional Materials

See Section 01 33 29.00 06 SUSTAINABILITY REPORTING for cumulative total local material requirements. Concrete materials may be locally available. Submit documentation indicating distance between manufacturing facility and the project site. Indicate distance of raw material origin from the project site. Indicate relative dollar value of local/regional materials to total dollar value of products included in project.

1.8.2 Forest Stewardship Council (FSC) Certification

Use FSC-certified wood where specified. Provide letter of certification signed by lumber supplier. Indicate compliance with FSC STD 01 001 and identify certifying organization. Submit FSC certification numbers; identify each certified product on a line-item basis. Submit copies of invoices bearing the FSC certification numbers.

1.8.3 Environmental Product Declarations

Submit a product-specific Type III (third-party verified) Environmental Product Declaration EPD) that is based on the NSF International's Product Category Rule (PCR) for Concrete NSF 1112-19; NSF PCR and conforms with ISO 14025 and ISO 21930, for each concrete mix used on this project. Submit a product-specific Type III EPD that is based on the NSF International's PCR for Portland, Blended, Masonry, Mortar, and Plastic (Stucco) Cements and conforms with ISO 14025 and ISO 21930, for each cement product used on this project.

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1.8.4 Energy Star Energy Performance Scores

Submit Energy Star Energy Performance Score for each supplying cement plant producing cement for this project.

1.9 QUALIFICATIONS FOR WELDING WORK

Welding procedures must be in accordance with AWS D1.4/D1.4M.

Verify that Welder qualifications are in accordance with AWS D1.4/D1.4M for welding of reinforcement or under an equivalent qualification test approved in advance. Welders are permitted to do only the type of welding for which each is specifically qualified.

PART 2 PRODUCTS

2.1 FORMWORK MATERIALS

- a. Form-facing material in contact with concrete must be lumber, plywood, tempered concrete-form-grade hardboard, metal, plastic, or treated paper that creates specified appearance and texture of concrete surface. Submit product information on proposed form-facing materials if different from that specified herein.
- b. Design formwork, shores, reshores, and backshores to support loads transmitted to them and to comply with applicable building code requirements.
- c. Design formwork to withstand pressure resulting from placement and vibration of concrete and to maintain specified tolerances.
- d. Design formwork to accommodate waterstop materials in joints at locations indicated in Contract Documents.
- e. Provide temporary openings in formwork if needed to facilitate cleaning and inspection.
- f. Design formwork joints to inhibit leakage of mortar.
- g. Limit deflection of facing materials for concrete surfaces exposed to view to 1/400 of center-to-center spacing of facing supports.
- h. Submit product information on proposed form-facing materials if different from that specified herein.
- i. Submit procedure for reshoring and backshoring, including drawings signed and sealed by a licensed design engineer. Include on shop drawings the formwork removal procedure and magnitude of construction loads used for design of reshoring or backshoring system. Indicate in procedure the magnitude of live and dead loads assumed for required capacity of the structure at time of reshoring or backshoring.
- j. Submit manufacturer's product data on form liner proposed for use with each formed surface.

2.1.1 Wood Forms

Use lumber as follows. Provide lumber that is square edged or tongue-and-groove boards, free of raised grain, knotholes, or other surface

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defects. Provide plywood that complies with NIST PS 1, B-B concrete form panels or better or AHA A135.4, hardboard for smooth form lining.

2.1.1.1 Concrete Form Plywood (Standard Rough)

Provide plywood that conforms to NIST PS 1, B-B, concrete form, not less than 5/8-inch thick.

2.1.1.2 Overlaid Concrete Form Plywood (Standard Smooth)

Provide plywood that conforms to NIST PS 1, B-B, high density form overlay, not less than 5/8-inch thick.

2.1.2 Carton Forms

Moisture resistant treated paper faces, biodegradable, structurally sufficient to support weight of wet concrete until initial set. Provide carton forms that contain a minimum of 5 percent post-consumer recycled content, or a minimum of 20 percent post-industrial recycled content.

2.1.3 Steel Forms

Provide steel form surfaces that do not contain irregularities, dents, or sags.

2.2 FORMWORK ACCESSORIES

- a. Use commercially manufactured formwork accessories, including ties and hangers.
- b. Form ties and accessories must not reduce the effective cover of the reinforcement.

2.2.1 Form Ties

- a. Use form ties with ends or end fasteners that can be removed without damage to concrete.
- b. Where indicated in Contract Documents, use form ties with integral water barrier plates or other acceptable positive water barriers in walls.
- c. The breakback distance for ferrous ties must be at least 2 in. for Surface Finish-2.0 or Surface Finish-3.0, as defined in ACI 301.
- d. Submit manufacturer's data sheet on form ties.

2.2.2 Waterstops

Submit manufacturer's data sheet on waterstop materials and splices.

2.2.2.1 PVC Waterstop

Polyvinylchloride waterstops must conform to COE CRD-C 572.

2.2.3 Biodegradable Form Release Agent

- a. Provide form release agent that is colorless, biodegradable, and with a low (maximum of 55 grams/liter (g/l)) VOC content.

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- b. Provide product that does not bond with, stain, or adversely affect concrete surfaces and does not impair subsequent treatments of concrete surfaces.
- c. Provide form release agent that reduces formwork moisture absorption, and does not contain diesel fuel, petroleum-based lubricating oils, waxes, or kerosene. Submit documentation indicating type of biobased material in product and biobased content. Indicate relative dollar value of biobased content products to total dollar value of products included in project.
- d. Submit manufacturer's product data on formwork release agent for use on each form-facing material.

2.2.4 Chamfer Materials

Use lumber materials with dimensions of 3/4 x 3/4 in.

2.2.5 Construction and movement joints

- a. Submit details and locations of construction joints in accordance with the requirements herein.
- b. Locate construction joints within middle one-third of spans of slabs, beams, and girders. If a beam intersects a girder within the middle one-third of girder span, the distance between the construction joint in the girder and the edge of the beam must be at least twice the width of the larger member.
- c. Locate construction joints in walls and columns at underside of slabs, beams, or girders and at tops of footings or slabs.
- d. Make construction joints perpendicular to main reinforcement.
- e. Provide movement joints where indicated in Contract Documents or in accepted alternate locations.
- f. Submit location and detail of movement joints if different from those indicated in Contract Documents.
- g. Submit manufacturer's data sheet on expansion joint materials.
- h. Provide keyways where indicated in Contract Documents. Longitudinal keyways indicated in Contract Documents must be at least 1-1/2 in. deep, measured perpendicular to the plane of the joint.

2.2.6 Other Embedded items

Use sleeves, inserts, anchors, and other embedded items of material and design indicated in Contract Documents.

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2.3 CONCRETE MATERIALS

2.3.1 Cementitious Materials

2.3.1.1 Portland Cement

- a. Unless otherwise specified, provide cement that conforms to ASTM C150/C150M Type I, II or I/II.
- b. Use one brand and type of cement for formed concrete having exposed-to-view finished surfaces.
- c. Submit information along with evidence demonstrating compliance with referenced standards. Submittals must include types of cementitious materials, manufacturing locations, shipping locations, and certificates showing compliance.
- d. Cementitious materials must be stored and kept dry and free from contaminants.

2.3.1.2 Blended Cements

- a. Blended cements must conform to ASTM C595/C595M Type IP(MH), IS(MH) or IL(MH).
- b. Slag cement added to the Type IS blended cements must meet ASTM C989/C989M.
- c. The pozzolan added to the Type IP blended cements must meet ASTM C618 Class F, and must be interground with the cement clinker. The manufacturer must state in writing that the amount of pozzolan in the finished cement will not vary more than plus or minus 5 mass percent of the finished cement from lot-to-lot or within a lot. The percentage and type of pozzolan used in the blend must not change from that submitted for the aggregate evaluation and mixture proportioning.

2.3.1.3 Fly Ash

- a. ASTM C618, Class F, except that the maximum allowable loss on ignition must not exceed 6 percent.
- b. If fly ash is used it shall range from 15 to 25 percent by weight of cementitious material, provided the fly ash does not reduce the amount of cement in the concrete mix below the minimum requirements of local building codes. Where the use of fly ash cannot meet the minimum level, it shall not be used. Report the chemical analysis of the fly ash in accordance with ASTM C311/C311M. Evaluate and classify fly ash in accordance with ASTM D5759.

2.3.1.4 Slag Cement

ASTM C989/C989M, Grade 100 or 120.

2.3.1.5 Silica Fume

Silica fume must conform to ASTM C1240, including the optional limits on reactivity with cement alkalis. Silica fume may be furnished as a dry, densified material or as slurry. Proper mixing is essential to accomplish

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proper distribution of the silica fume and avoid agglomerated silica fume which can react with the alkali in the cement resulting in premature and extensive concrete damage. Supervision at the batch plant, finishing, and curing is essential. Provide at the Contractor's expense the services of a manufacturer's technical representative, experienced in mixing, proportioning, placement procedures, and curing of concrete containing silica fume. This representative must be present on the project prior to and during at least the first 4 days of concrete production and placement using silica fume. A High Range Water Reducing admixture (HRWRA) must be used when silica fume is the only pozzolan used.

2.3.1.6 Other Supplementary Cementitious Materials

Natural pozzolan must be raw or calcined and conform to ASTM C618, Class N, including the optional requirement for uniformity.

Ultra Fine Fly Ash (UFFA) and Ultra Fine Pozzolan (UFP) must conform to ASTM C618, Class F or N, and the following additional requirements:

- a. The strength activity index at 28 days of age must be at least 95 percent of the control specimens.
- b. The average particle size must not exceed 6 microns.
- c. The sum of SiO₂ plus Al₂O₃ plus Fe₂O₃ must be greater than 77 percent.

2.3.2 Water

- a. Water or ice must comply with the requirements of ASTM C1602/C1602M.
- b. Minimize the amount of water in the mix. Improve workability by adjusting the grading of the aggregate and using admixture rather than by adding water.
- c. Water must be potable; free from injurious amounts of oils, acids, alkalis, salts, organic materials, or other substances deleterious to concrete.
- d. Protect mixing water and ice from contamination during storage and delivery.
- e. Submit test report showing water complies with ASTM C1602/C1602M.

2.3.3 Aggregate

2.3.3.1 Normal-Weight Aggregate

- a. Aggregates must conform to ASTM C33/C33M.
- b. Aggregates used in concrete must be obtained from the same sources and have the same size range as aggregates used in concrete represented by submitted field test records or used in trial mixtures.
- c. Provide sand that is at least 50 percent natural sand.
- d. Store and handle aggregate in a manner that will avoid segregation and prevents contamination by other materials or other sizes of aggregates. Store aggregates in locations that will permit them to drain freely. Do not use aggregates that contain frozen lumps.

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- e. Submit types, pit or quarry locations, producers' names, aggregate supplier statement of compliance with ASTM C33/C33M, and ASTM C1293 expansion data not more than 18 months old.

2.3.4 Admixtures

- a. Chemical admixtures must conform to ASTM C494/C494M.
- b. Air-entraining admixtures must conform to ASTM C260/C260M.
- c. Chemical admixtures for use in producing flowing concrete must conform to ASTM C1017/C1017M.
- d. Do not use calcium chloride admixtures.
- e. Use a corrosion-inhibiting admixture for concrete classified under exposure category C2 concrete containing aggregate susceptible to ASR.
- f. Admixtures used in concrete must be the same as those used in the concrete represented by submitted field test records or used in trial mixtures.
- g. Protect stored admixtures against contamination, evaporation, or damage.
- h. To ensure uniform distribution of constituents, provide agitating equipment for admixtures used in the form of suspensions or unstable solutions. Protect liquid admixtures from freezing and from temperature changes that would adversely affect their characteristics.
- i. Submit types, brand names, producers' names, manufacturer's technical data sheets, and certificates showing compliance with standards required herein.

2.4 MISCELLANEOUS MATERIALS

2.4.1 Concrete Curing Materials

Provide concrete curing material in accordance with ACI 301 Section 5 and ACI 308.1 Section 2. Submit product data for concrete curing compounds. Submit manufactures instructions for placement of curing compound.

2.4.2 Nonshrink Grout

Nonshrink grout in accordance with ASTM C1107/C1107M.

2.4.3 Floor Finish Materials

2.4.3.1 Abrasive Aggregate for Nonslip Aggregate Finish

Aggregate must be packaged, factory-graded fused aluminum oxide grits, or it may be crushed emery containing not less than 40-percent aluminum oxide and not less than 25-percent ferric oxide. Aggregate must be rust proof and nonglazing and must be unaffected by freezing, moisture, and cleaning materials.

Aggregate must be well-graded in size from particles retained on No. 30 sieve 0.0236 inch to particles passing No. 8 sieve 0.0929 inch.

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2.4.3.2 Dry Materials for Colored Wear-Resistant Finish

Provide materials that are packaged, dry, and a combination of materials formulated for producing colored and wear-resistant monolithic surface treatments; they must include portland cement, graded-quartz aggregate, coloring pigments, and dispersing agents. Provide coloring pigments that are finely ground, nonfacing mineral oxides prepared especially for the purpose and interground with the cement.

2.4.4 Expansion/Contraction Joint Filler

ASTM D1751 or ASTM D1752 Type I or Type II. Material must be 1/2 inch thick, unless otherwise indicated.

2.4.5 Joint Sealants

Submit manufacturer's product data, indicating VOC content.

2.4.5.1 Horizontal Surfaces, 3 Percent Slope, Maximum

ASTM D6690 or ASTM C920, Type M, Class 25, Use T.

2.4.5.2 Vertical Surfaces Greater Than 3 Percent Slope

ASTM C920, Type M, Grade NS, Class 25, Use T.

2.4.5.3 Preformed Polychloroprene Elastomeric Type

ASTM D2628.

2.4.5.4 Lubricant for Preformed Compression Seals

ASTM D2835.

2.4.6 Vapor Retarder

ASTM E1745 Class C polyethylene sheeting, minimum 10 mil thickness or other equivalent material with a maximum permance rating of 0.04 perms per ASTM E96/E96M.

Consider plastic vapor retarders and adhesives with a high recycled content, low toxicity low VOC (Volatile Organic Compounds) levels.

2.4.7 Dovetail Anchor Slot

Preformed metal slot approximately 1 inch by 1 inch of not less than 22 gage galvanized steel cast in concrete. Coordinate actual size and throat opening with dovetail anchors and provide with removable filler material.

2.5 CONCRETE MIX DESIGN

2.5.1 Properties and Requirements

- a. Use materials and material combinations listed in this section and the contract documents.
- b. Cementitious material content must be adequate for concrete to satisfy the specified requirements for strength, w/cm, durability, and finishability described in this section and the contract documents.

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- c. Selected target slump must meet the requirements this section, the contract documents, and must not exceed 9 in. Concrete must not show visible signs of segregation.
- d. The target slump must be enforced for the duration of the project. Determine the slump by ASTM C143/C143M. Slump tolerances must meet the requirements of ACI 117.
- e. The nominal maximum size of coarse aggregate for a mixture must not exceed three-fourths of the minimum clear spacing between reinforcement, one-fifth of the narrowest dimension between sides of forms, or one-third of the thickness of slabs or toppings.
- f. Concrete must be air entrained for members assigned to Exposure Class F1, F2, or F3. The total air content must be in accordance with the requirements of the paragraph titled DURABILITY.
- g. Measure air content at the point of delivery in accordance with ASTM C173/C173M or ASTM C231/C231M.
- h. Concrete for slabs to receive a hard-troweled finish must not contain an air-entraining admixture or have a total air content greater than 3 percent.
- i. Concrete properties and requirements for each portion of the structure are specified in the table below. Refer to the paragraph titled DURABILITY for more details on exposure categories and their requirements.

	Minimum f'_c psi	Exposure Categories [^]	Miscellaneous Requirements
Interior above-grade beams, columns and elevated slabs	4000 at 28 days	S0; C0; W0; F0	Nominal maximum aggregate size must be 1 in.
Exterior slabs-on-ground	4500 at 28 days	S1; C1; W1; F2	Nominal maximum aggregate size must be 3/4 in.

2.5.2 Durability

2.5.2.1 Alkali-Aggregate Reaction

Do not use any aggregate susceptible to alkali-carbonate reaction (ACR). Use one of the three options below for qualifying concrete mixtures to reduce the potential of alkali-silica reaction (ASR):

- a. For each aggregate used in concrete, the expansion result determined in accordance with ASTM C1293 must not exceed 0.04 percent at one year.
- b. For each aggregate used in concrete, the expansion result of the aggregate and cementitious materials combination determined in

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accordance with ASTM C1567 must not exceed 0.10 percent at an age of 16 days.

- c. Alkali content in concrete (LBA) must not exceed 4 pounds per cubic yard for moderately reactive aggregate or 3 pounds per cubic yard for highly reactive aggregate. Reactivity must be determined by testing in accordance with ASTM C1293 and categorized in accordance with ASTM C1778. Alkali content is calculated as follows:

$$\text{LBA} = (\text{cement content, pounds per cubic yard}) \times (\text{equivalent alkali content of portland cement in percent}/100 \text{ percent})$$

2.5.2.2 Freezing and Thawing Resistance

- a. Provide concrete meeting the following requirements based on exposure class assigned to members for freezing-and-thawing exposure in Contract Documents:

Exposure class	Maximum w/cm*	Minimum $f'c$, psi	Air content	Additional Requirements
F0	N/A	2500		N/A
F1	0.55 (0.495 for mixes with Type IL cement)	3500	Depends on aggregate size	N/A
F2	0.45 (0.405 for mixes with Type IL cement)	4500	Depends on aggregate size	See limits on maximum cementitious material by mass
F3	0.40 (0.36 for mixes with Type IL cement)	5000	Depends on aggregate size	See limits on maximum cementitious material by mass
F3 plain concrete	0.45 (0.405 for mixes with Type IL cement)	4500	Depends on aggregate size	See limits on maximum cementitious material by mass

*The maximum w/cm limits are 90% normal limits for concrete mixes with blended Type IL cement, because Type IL blended cement can contain 10% more lime than Types I and II Portland Cement.

- b. Concrete must be air entrained for members assigned to Exposure Class F1, F2, or F3. The total air content must meet the requirements of the following table:

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Nominal maximum aggregate size, in.	Total air content, percent* [^]	
	Exposure Class F2 and F3	Exposure Class F1
3/8	7.5	6.0
1/2	7.0	5.5
3/4	6.0	5.0
1	6.0	4.5
1-1/2	5.5	4.5
2	5.0	4.0
3	5.5	3.5

*Tolerance on air content as delivered must be plus/minus 1.5 percent.

[^]For f'c greater than 5000 psi, reducing air content by 1.0 percentage point is acceptable.

- c. Submit documentation verifying compliance with specified requirements.
- d. For sections of the structure that are assigned Exposure Class F3, submit certification on cement composition verifying that concrete mixture meets the requirements of the following table:

Cementitious material	Maximum percent of total cementitious material by mass*
Fly ash or other pozzolans conforming to ASTM C618	25
Slag cement conforming to ASTM C989/C989M	50
Silica fume conforming to ASTM C1240	10
Total of fly ash or other pozzolans, slag cement, and silica fume	50 [^]
Total of fly ash or other pozzolans and silica fume	35 [^]

*Total cementitious material also includes ASTM C150/C150M, ASTM C595/C595M, ASTM C845/C845M, and ASTM C1157/C1157M cement. The maximum percentages above must include:

- i. Fly ash or other pozzolans present in ASTM C1157/C1157M or ASTM C595/C595M Type IP blended cement.
- ii. Slag cement present in ASTM C1157/C1157M or ASTM C595/C595M Type IS blended cement.

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iii. Silica fume conforming to ASTM C1240 present in
ASTM C1157/C1157M or ASTM C595/C595M Type IP blended cement.

^Fly ash or other pozzolans and silica fume must constitute no more than 25 percent and 10 percent, respectively, of the total mass of the cementitious materials.

2.5.2.3 Corrosion and Chloride Content

- a. Provide concrete meeting the requirements of the following table based on the exposure class assigned to members requiring protection against reinforcement corrosion in Contract Documents.
- b. Submit documentation verifying compliance with specified requirements.
- c. Water-soluble chloride ion content contributed from constituents including water, aggregates, cementitious materials, and admixtures must be determined for the concrete mixture by ASTM C1218/C1218M at age between 28 and 42 days.
- d. The maximum water-soluble chloride ion (Cl-) content in concrete, percent by mass of cement is as follows:

Exposure class	Maximum w/cm*	Minimum f'c, psi	Maximum water-soluble chloride ion (CL-) content in concrete, percent by mass of cement
Reinforced concrete			
C0	N/A	2500	1.00
C1	N/A	2500	0.30
C2	0.4	5000	0.15
Prestressed concrete			
C0	N/A	2500	0.06
C1	N/A	2500	0.06
C2	0.4	5000	0.06

2.5.2.4 Sulfate Resistance

- a. Provide concrete meeting the requirements of the following table based on the exposure class assigned to members for sulfate exposure.

Exposure class	Maximum w/cm	Minimum f'c, psi	Required cementitious materials-types			Calcium chloride admixture
			ASTM C150/C150M	ASTM C595/C595M	ASTM C1157/C1157M	
S0	N/A	2500	N/A	N/A	N/A	No restrictions

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Exposure class	Maximum w/cm	Minimum f'c, psi	Required cementitious materials-types			Calcium chloride admixture
			ASTM C150/C150M	ASTM C595/C595M	ASTM C1157/C1157M	
S1	0.50 (0.195 for mixes with Type IL cement)	4000	II*^	Types with (MS) designation	MS	No restrictions
S2	0.45 (0.405 for mixes with Type IL cement)	31 4500	V^	Types with (HS) designation	HS	Not permitted
S3	0.45 (0.405 for mixes with Type IL cement)	4500	V + pozzolan or slag cement**	Types with (HS) designation plus pozzolan or slag cement**	HS + pozzolan or slag cement**	Not permitted
S3	0.40 (0.36 for mixes with Type IL cement)	5000	V***	Types with (HS) designation	HS	Not permitted

* For seawater exposure, other types of portland cements with tricalcium aluminate (C3A) contents up to 10 percent are acceptable if the w/cm does not exceed 0.40.

** The amount of the specific source of the pozzolan or slag cement to be used shall be at least the amount determined by test or service record to improve sulfate resistance when used in concrete containing Type V cement. Alternatively, the amount of the specific source of the pozzolan or slag used shall not be less than the amount tested in accordance with ASTM C1012/C1012M and meeting the requirements maximum expansion requirements listed herein.

*** If Type V cement is used as the sole cementitious material, the optional sulfate requirement of 0.040 percent maximum expansion in ASTM C150/C150M shall be required.

^ Other available types of cement, such as Type III or Type I, are acceptable in exposure classes S1 or S2 if the C3A contents are less than 8 or 5 percent, respectively.

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- b. The maximum w/cm limits are 90% normal limits for concrete mixes with blended Type IL cement, because Type IL blended cement can contain 10% more lime than Types I and II Portland Cement.
- c. Alternative combinations of cementitious materials of those listed in this paragraph are acceptable if they meet the maximum expansion requirements listed in the following table:

Exposure class	Maximum expansion when tested using ASTM C1012/C1012M		
	At 6 months	At 12 months	At 18 months
S1	0.10 percent	N/A	N/A
S2	0.05 percent	0.10 percent [^]	N/A
S3	N/A	N/A	0.10 percent

[^]The 12-month expansion limit applies only when the measured expansion exceeds the 6-month maximum expansion limit.

2.5.2.5 Concrete Temperature

The temperature of concrete as delivered must not exceed 95°F.

2.5.2.6 Concrete permeability

- a. Provide concrete meeting the requirements of the following table based on exposure class assigned to members requiring low permeability in the Contract Documents.

Exposure class	Maximum w/cm*	Minimum f'c, psi	Additional minimum requirements
W0	N/A	2500	None
W1	0.5 (0.405 for mixes with Type IL cement)	4000	None

*The maximum w/cm limits are 90% normal limits for concrete mixes with blended Type IL cement, because Type IL blended cement can contain 10% more lime than Types I and II Portland Cement.

- b. Submit documentation verifying compliance with specified requirements.

2.5.3 Trial Mixtures

Trial mixtures must be in accordance to ACI 301.

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2.5.4 Ready-Mix Concrete

Provide concrete that meets the requirements of ASTM C94/C94M.

Ready-mixed concrete manufacturer must provide duplicate delivery tickets with each load of concrete delivered. Provide delivery tickets with the following information in addition to that required by ASTM C94/C94M:

- a. Type and brand cement
- b. Cement and supplementary cementitious materials content in 94-pound bags per cubic yard of concrete
- c. Maximum size of aggregate
- d. Amount and brand name of admixtures
- e. Total water content expressed by water: cementitious material ratio

2.6 REINFORCEMENT

- a. Bend reinforcement cold. Fabricate reinforcement in accordance with fabricating tolerances of ACI 117.
- b. Submit manufacturer's certified test report for reinforcement.
- c. Submit placing drawings showing fabrication dimensions and placement locations of reinforcement and reinforcement supports. Placing drawings must indicate locations of splices, lengths of lap splices, and details of mechanical and welded splices.
- d. Submit request with locations and details of splices not indicated in Contract Documents.
- e. Submit request to place column dowels without using templates.
- f. Submit request and procedure to field-bend or straighten reinforcing bars partially embedded in concrete at locations not indicated in Contract Documents. Field bending, if permitted, shall be performed using an approved and appropriate sized portable hydraulic device that makes ACI standard radius bends.
- g. Submit request for field cutting, including location and type of bar to be cut and reason field cutting is required.

2.6.1 Reinforcing Bars

- a. Reinforcing bars must be deformed, except spirals, load-transfer dowels, and welded wire reinforcement, which may be plain.
- b. ASTM A615/A615M with the bars marked S, Grade 60; or ASTM A996/A996M with the bars marked R, Grade 60, or marked A, Grade 60. Provide reinforcing bars that contain a minimum of 25 percent recycled content. See Section 01 33 29.00 06 SUSTAINABILITY REPORTING for cumulative total recycled content requirements.
- c. Reinforcing bars may contain post-consumer or post-industrial recycled content. Submit documentation indicating percentage of post-industrial and post-consumer recycled content per unit of product. Indicate

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relative dollar value of recycled content products to total dollar value of products included in project.

- d. Submit mill certificates for reinforcing bars.

2.6.1.1 Bar Mats

- a. Bar mats must conform to ASTM A184/A184M.

2.6.1.2 Headed Shear Stud Reinforcement

Headed studs and headed stud assemblies must conform to ASTM A1044/A1044M.

2.6.2 Mechanical Reinforcing Bar Connectors

- a. Provide 125 percent minimum yield strength of the reinforcement bar.
- b. Submit data on mechanical splices demonstrating compliance with this paragraph.

2.6.3 Wire

- a. Provide wire reinforcement that contains a minimum of 25 percent recycled content. See Section 01 33 29.00 06 SUSTAINABILITY REPORTING for cumulative total recycled content requirements. Wire reinforcement may contain post-consumer or post-industrial recycled content. Provide flat sheets of welded wire reinforcement for slabs and toppings.
- b. Plain or deformed steel wire must conform to ASTM A1064/A1064M.

2.6.4 Welded wire reinforcement

- a. Use welded wire reinforcement specified in Contract Documents and conforming to one or more of the specifications given herein.
- b. Plain welded wire reinforcement must conform to ASTM A1064/A1064M, with welded intersections spaced no greater than 12 in. apart in direction of principal reinforcement.
- c. Mechanical splices shall be the threaded type. Mechanical splices with set screws are not permitted.
- d. Deformed welded wire reinforcement must conform to ASTM A1064/A1064M, with welded intersections spaced no greater than 16 in. apart in direction of principal reinforcement.

2.6.5 Reinforcing Bar Supports

- a. Provide reinforcement support types within structure as required by Contract Documents. Reinforcement supports must conform to CRSI RB4.1. Submit description of reinforcement supports and materials for fastening coated reinforcement if not in conformance with CRSI RB4.1.
- b. Legs of supports in contact with formwork must be hot-dip galvanized, or plastic coated after fabrication, or stainless-steel bar supports.
- c. See Section 01 33 29.00 06 SUSTAINABILITY REPORTING for cumulative total recycled content requirements. Plastic and steel may contain post-consumer or post-industrial recycled content.

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2.6.6 Reinforcing Fibers

2.6.6.1 Synthetic Fibers

In addition to the requirements specified above, provide fiber reinforced concrete in accordance with ASTM C1116/C1116M Type III, synthetic fiber reinforced concrete, and as follows. Synthetic reinforcing fibers must be 100 percent virgin monofilament polypropylene fibers.

Provide fibers that have a specific gravity of 0.9, a minimum tensile strength of 70 ksi, graded per manufacturer, and specifically manufactured to an optimum gradation for use as concrete secondary reinforcement. Add fibers at the batch plant. Toughness indices must meet requirements for performance level I. Provide the services of a qualified technical representative to instruct the concrete supplier in proper batching and mixing of materials to be provided.

2.6.7 Dowels for Load Transfer in Floors

Provide greased dowels for load transfer in floors of the type, design, weight, and dimensions indicated. Provide dowel bars that are plain-billet steel conforming to ASTM A615/A615M, Grade 40. Provide dowel pipe that is steel conforming to ASTM A53/A53M.

2.6.8 Welding

- a. Provide weldable reinforcing bars that conform to ASTM A706/A706M and ASTM A615/A615M and Supplement S1, Grade 60, except that the maximum carbon content must be 0.55 percent.
- b. Comply with AWS D1.4/D1.4M unless otherwise specified. Do not tack weld reinforcing bars.
- c. Welded assemblies of steel reinforcement produced under factory conditions, such as welded wire reinforcement, bar mats, and deformed bar anchors, are allowed.

PART 3 EXECUTION

3.1 EXAMINATION

- a. Do not begin installation until substrates have been properly constructed; verify that substrates are level.
- b. If substrate preparation is the responsibility of another installer, notify Contracting Officer of unsatisfactory preparation before processing.
- c. Check field dimensions before beginning installation. If dimensions vary too much from design dimensions for proper installation, notify Contracting Officer and wait for instructions before beginning installation.

3.2 PREPARATION

Determine quantity of concrete needed and minimize the production of excess concrete. Designate locations or uses for potential excess concrete before the concrete is poured.

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3.2.1 General

- a. Surfaces against which concrete is to be placed must be free of debris, loose material, standing water, snow, ice, and other deleterious substances before start of concrete placing.
- b. Remove standing water without washing over freshly deposited concrete. Divert flow of water through side drains provided for such purpose.

3.2.2 Reinforcement and Other Embedded Items

- a. Secure reinforcement, joint materials, and other embedded materials in position, inspected, and approved before start of concrete placing.
- b. When concrete is placed, reinforcement must be free of materials deleterious to bond. Reinforcement with rust, mill scale, or a combination of both will be considered satisfactory, provided minimum nominal dimensions, nominal weight, and minimum average height of deformations of a hand-wire-brushed test specimen are not less than applicable ASTM specification requirements.

3.3 FORMS

- a. Provide forms, shoring, and scaffolding for concrete placement. Set forms mortar-tight and true to line and grade.
- b. Chamfer above grade exposed joints, edges, and external corners of concrete 0.75 inch. Place chamfer strips in corners of formwork to produce beveled edges on permanently exposed surfaces. Do not bevel reentrant corners or edges of formed joints of concrete at masonry ledges.
- c. Provide formwork with clean-out openings to permit inspection and removal of debris.
- d. Inspect formwork and remove foreign material before concrete is placed.
- e. At construction joints, lap form-facing materials over the concrete of previous placement. Ensure formwork is placed against hardened concrete so offsets at construction joints conform to specified tolerances.
- f. Provide positive means of adjustment (such as wedges or jacks) of shores and struts. Do not make adjustments in formwork after concrete has reached initial setting. Brace formwork to resist lateral deflection and lateral instability.
- g. Fasten form wedges in place after final adjustment of forms and before concrete placement.
- h. Provide anchoring and bracing to control upward and lateral movement of formwork system.
- i. Construct formwork for openings to facilitate removal and to produce opening dimensions as specified and within tolerances.
- j. Provide runways for moving equipment. Support runways directly on formwork or structural members. Do not support runways on

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reinforcement. Loading applied by runways must not exceed capacity of formwork or structural members.

- k. Position and support expansion joint materials, waterstops, and other embedded items to prevent displacement. Fill voids in sleeves, inserts, and anchor slots temporarily with removable material to prevent concrete entry into voids.
- l. Clean surfaces of formwork and embedded materials of mortar, grout, and foreign materials before concrete placement.

3.3.1 Coating

- a. Cover formwork surfaces with an acceptable material that inhibits bond with concrete.
- b. If formwork release agent is used, apply to formwork surfaces in accordance with manufacturer's recommendations before placing reinforcement. Remove excess release agent on formwork prior to concrete placement.
- c. Do not allow formwork release agent to contact reinforcement or hardened concrete against which fresh concrete is to be placed.

3.3.2 Reshoring

- a. Do not allow structural members to be loaded with combined dead and construction loads in excess of loads indicated in the accepted procedure.
- b. Install and remove reshores or backshores in accordance with accepted procedure.
- c. For floors supporting shores under newly placed concrete, either leave original supporting shores in place, or install reshores or backshores. Shoring system and supporting slabs must resist anticipated loads. Locate reshores and backshores directly under a shore position or as indicated on formwork shop drawings.
- d. In multistory buildings, place reshoring or backshoring over a sufficient number of stories to distribute weight of newly placed concrete, forms, and construction live loads.

3.3.3 Reuse

- a. Reuse forms providing the structural integrity of concrete and the aesthetics of exposed concrete are not compromised.
- b. Wood forms must not be clogged with paste and must be capable of absorbing high water-cementitious material ratio paste.
- c. Remove leaked mortar from formwork joints before reuse.

3.3.4 Forms for Standard Smooth Form Finish

Provide formwork in accordance with ACI 301 Section 5 with a surface finish, SF-3.0, for formed surfaces that are exposed to view. Do not provide mockup of concrete surface appearance and texture.

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3.3.5 Form Ties

- a. After ends or end fasteners of form ties have been removed, repair tie holes in accordance with ACI 301 Section 5 requirements.

3.3.6 Tolerances for Form Construction

- a. Construct formwork so concrete surfaces conform to tolerances in ACI 117.
- b. Position and secure sleeves, inserts, anchors, and other embedded items such that embedded items are positioned within ACI 117 tolerances.
- c. To maintain specified elevation and thickness within tolerances, install formwork to compensate for deflection and anticipated settlement in formwork during concrete placement. Set formwork and intermediate screed strips for slabs to produce designated elevation, camber, and contour of finished surface before formwork removal. If specified finish requires use of vibrating screeds or roller pipe screeds, ensure that edge forms and screed strips are strong enough to support such equipment.

3.3.7 Removal of Forms and Supports

- a. If vertical formed surfaces require finishing, remove forms as soon as removal operations will not damage concrete.
- b. Remove top forms on sloping surfaces of concrete as soon as removal will not allow concrete to sag. Perform repairs and finishing operations required. If forms are removed before end of specified curing period, provide curing and protection.
- c. Do not damage concrete during removal of vertical formwork for columns, walls, and sides of beams. Perform needed repair and finishing operations required on vertical surfaces. If forms are removed before end of specified curing period, provide curing and protection.
- d. Leave formwork and shoring in place to support construction loads and weight of concrete in beams, slabs, and other structural members until in-place required strength of concrete is reached.
- e. Form-facing material and horizontal facing support members may be removed before in-place concrete reaches specified compressive strength if shores and other supports are designed to allow facing removal without deflection of supported slab or member.

3.3.8 Strength of Concrete Required for Removal of Formwork

If removal of formwork, reshoring, or backshoring is based on concrete reaching a specified in-place strength, mold and field-cure cylinders in accordance with ASTM C31/C31M. Test cylinders in accordance with ASTM C39/C39M. Alternatively, use one or more of the methods listed herein to evaluate in-place concrete strength for formwork removal.

- a. Tests of cast-in-place cylinders in accordance with ASTM C873/C873M. This option is limited to slabs with concrete depths from 5 to 12 in.
- b. Penetration resistance in accordance with ASTM C803/C803M.
- c. Pullout strength in accordance with ASTM C900.

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- d. Maturity method in accordance with ASTM C1074. Submit maturity method data using project materials and concrete mix proportions used on the project to demonstrate the correlation between maturity and compressive strength of laboratory cured test specimens to the Contracting Officer.

3.4 WATERSTOP INSTALLATION AND SPLICES

- a. Provide waterstops in construction joints as indicated.
- b. Install formwork to accommodate waterstop materials. Locate waterstops in joints where indicated in Contract Documents. Minimize number of splices in waterstop. Splice waterstops in accordance with manufacturer's written instructions. Install factory-manufactured premolded mitered corners.
- c. Install waterstops to form a continuous diaphragm in each joint. Make adequate provisions to support and protect waterstops during progress of work. Protect waterstops protruding from joints from damage.

3.4.1 PVC Waterstop

Make splices by heat sealing the adjacent waterstop edges together using a thermoplastic splicing iron utilizing a non-stick surface specifically designed for waterstop welding. Reform waterstops at splices with a remolding iron with ribs or corrugations to match the pattern of the waterstop. The spliced area, when cooled, must show no signs of separation, holes, or other imperfections when bent by hand in as sharp an angle as possible.

3.5 PLACING REINFORCEMENT AND MISCELLANEOUS MATERIALS

- a. Unless otherwise specified, placing reinforcement and miscellaneous materials must be in accordance to ACI 301. Provide bars, welded wire reinforcement, wire ties, supports, and other devices necessary to install and secure reinforcement.
- b. Reinforcement must not have rust, scale, oil, grease, clay, or foreign substances that would reduce the bond. Rusting of reinforcement is a basis of rejection if the effective cross-sectional area or the nominal weight per unit length has been reduced. Remove loose rust prior to placing steel. Tack welding is prohibited.
- c. Nonprestressed cast-in-place concrete members must have concrete cover for reinforcement given in the following table:

Concrete Exposure	Member	Reinforcement	Specified cover, in.
Cast against and permanently in contact with ground	All	All	3

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Concrete Exposure	Member	Reinforcement	Specified cover, in.
Exposed to weather or in contact with ground	All	No. 6 through No. 18 bars	2
		No. 5 bar, W31 or D31 wire, and smaller	1-1/2
Not exposed to weather or in contact with ground	Slabs, joists, and walls	No. 14 and No. 18 bars	1-1/2
		No. 11 bar and smaller	3/4
	Beams, columns, pedestals, and tension ties	Primary reinforcement, stirrups, ties, spirals, and hoops	1-1/2

- d. Precast nonprestressed or prestressed concrete members manufactured under plant conditions must have concrete cover for reinforcement, ducts, and end fittings given in the following table:

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Concrete Exposure	Member	Reinforcement	Specified cover, in.
Exposed to weather or in contact with ground	Walls	No. 14 and No. 18 bars; tendons larger than 1-1/2 in. diameter	1-1/2
		No. 11 bars and smaller; W31 and D31 wire, and smaller; tendons and strands 1-1/2 in.	3/4
	All other	No. 14 and No. 18 bars; tendons larger than 1-1/2 in.	2
		No. 6 through No. 11 bars; tendons and strands larger than 5/8 in. diameter through 1-1/2 in.	1-1/2
		No. 5 bar, W31 or D31 wire, and smaller; tendons and strands 5/8 in. diameter and smaller	1-1/4

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Concrete Exposure	Member	Reinforcement	Specified cover, in.
Not exposed to weather n or in contact with ground	Slabs, joists, and walls	No. 14 and No. 18 bars; tendons larger than 1-1/2 in. diameter	1-1/4
		Tendons and strands 1-1/2 in. diameter and smaller	3/4
		No. 11 bar, W31 or D31	5/8
	Beams, columns, pedestals, and tension ties	Primary reinforcement	Greater of bar diameter and 5/8 and need not exceed 1-1/2
		Stirrups, ties, spirals, and hoops	3/8

3.5.1 General

Provide details of reinforcement that are in accordance with the Contract Documents.

3.5.2 Reinforcement Supports

Provide reinforcement support in accordance with CRSI RB4.1 and ACI 301 Section 3 requirements.

3.5.3 Splicing

As indicated in the Contract Documents. For splices not indicated follow ACI 301. Do not splice at points of maximum stress. Overlap welded wire reinforcement the spacing of the cross wires, plus 2 inches.

3.5.4 Future Bonding

Plug exposed, threaded, mechanical reinforcement bar connectors with a greased bolt. Provide bolt threads that match the connector. Countersink the connector in the concrete. Caulk the depression after the bolt is installed.

3.5.5 Setting Miscellaneous Material

Place and secure anchors and bolts, pipe sleeves, conduits, and other such items in position before concrete placement and support against

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displacement. Plumb anchor bolts and check location and elevation. Temporarily fill voids in sleeves with readily removable material to prevent the entry of concrete.

3.5.6 Fabrication

Shop fabricate reinforcing bars to conform to shapes and dimensions indicated for reinforcement, and as follows:

- a. Provide fabrication tolerances that are in accordance with ACI 117.
- b. Provide hooks and bends that are in accordance with the Contract Documents.

Reinforcement must be bent cold to shapes as indicated. Bending must be done in the shop. Rebending of a reinforcing bar that has been bent incorrectly is not be permitted. Bending must be in accordance with standard approved practice and by approved machine methods.

Deliver reinforcing bars bundled, tagged, and marked. Tags must be metal with bar size, length, mark, and other information pressed in by machine. Marks must correspond with those used on the placing drawings.

Do not use reinforcement that has any of the following defects:

- a. Bar lengths, depths, and bends beyond specified fabrication tolerances
- b. Bends or kinks not indicated on drawings or approved shop drawings
- c. Bars with reduced cross-section due to rusting or other cause

Replace defective reinforcement with new reinforcement having required shape, form, and cross-section area.

3.5.7 Placing Reinforcement

Place reinforcement in accordance with ACI 301.

For slabs on grade (over earth or over capillary water barrier) and for footing reinforcement, support bars or welded wire reinforcement on precast concrete blocks, spaced at intervals required by size of reinforcement, to keep reinforcement the minimum height specified above the underside of slab or footing.

For slabs other than on grade, supports for which any portion is less than 1 inch from concrete surfaces that are exposed to view or to be painted must be of precast concrete units, plastic-coated steel, or stainless steel protected bar supports. Precast concrete units must be wedge shaped, not larger than 3-1/2 by 3-1/2 inches, and of thickness equal to that indicated for concrete protection of reinforcement. Provide precast units that have cast-in galvanized tie wire hooked for anchorage and blend with concrete surfaces after finishing is completed.

Provide reinforcement that is supported and secured together to prevent displacement by construction loads or by placing of wet concrete, and as follows:

- a. Provide supports for reinforcing bars that are sufficient in number and have sufficient strength to carry the reinforcement they support, and

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in accordance with ACI 301 and CRSI 10MSP. Do not use supports to support runways for concrete conveying equipment and similar construction loads.

- b. Equip supports on ground and similar surfaces with sand-plates.
- c. Support welded wire reinforcement as required for reinforcing bars.
- d. Secure reinforcements to supports by means of tie wire. Wire must be black, soft iron wire, not less than 16 gage.
- e. Reinforcement must be accurately placed, securely tied at intersections, and held in position during placing of concrete by spacers, chairs, or other approved supports. Point wire-tie ends away from the form. Unless otherwise indicated, numbers, type, and spacing of supports must conform to the Contract Documents.
- f. Bending of reinforcing bars partially embedded in concrete is permitted only as specified in the Contract Documents.

3.5.8 Spacing of Reinforcing Bars

- a. Spacing must be as indicated in the Contract Documents.
- b. Reinforcing bars may be relocated to avoid interference with other reinforcement, or with conduit, pipe, or other embedded items. If any reinforcing bar is moved a distance exceeding one bar diameter or specified placing tolerance, resulting rearrangement of reinforcement is subject to preapproval by the Contracting Officer.

3.5.9 Concrete Protection for Reinforcement

Additional concrete protection must be in accordance with the Contract Documents.

3.5.10 Welding

Welding must be in accordance with AWS D1.4/D1.4M.

3.6 BATCHING, MEASURING, MIXING, AND TRANSPORTING CONCRETE

In accordance with ASTM C94/C94M, ACI 301, ACI 302.1R and ACI 304R, except as modified herein. Batching equipment must be such that the concrete ingredients are consistently measured within the following tolerances: 1 percent for cement and water, 2 percent for aggregate, and 3 percent for admixtures. Furnish mandatory batch ticket information for each load of ready mix concrete.

3.6.1 Measuring

Make measurements at intervals as specified in paragraphs SAMPLING and TESTING.

3.6.2 Mixing

- a. Mix concrete in accordance with ASTM C94/C94M, ACI 301 and ACI 304R.
- b. Machine mix concrete. Begin mixing within 30 minutes after the cement has been added to the aggregates. Place concrete within 90 minutes of

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either addition of mixing water to cement and aggregates or addition of cement to aggregates when the concrete temperature is 84 degrees F or less.

- c. Place concrete within 60 minutes when the concrete temperature is greater than 84 degrees except as follows: if set retarding admixture is used and slump requirements can be met, limit for placing concrete may remain at 90 minutes. Additional water may be added, provided that both the specified maximum slump and submitted water cementitious material ratio are not exceeded and the required concrete strength is still met. When additional water is added, as additional 30 revolutions of the mixer at mixing speed are required.
- d. Dissolve admixtures in the mixing water and mix in the drum to uniformly distribute the admixture throughout the batch. Do not reconstitute concrete that has begun to solidify.
- e. When fibers are used, add fibers together with the aggregates and never as the first component in the mixer. Fibers must be dispensed into the mixing system using appropriate dispensing equipment and procedure as recommended by the manufacturer.

3.6.3 Transporting

Transport concrete from the mixer to the forms as rapidly as practicable. Prevent segregation or loss of ingredients. Clean transporting equipment thoroughly before each batch. Do not use aluminum pipe or chutes. Remove concrete which has segregated in transporting and dispose of as directed.

3.7 PLACING CONCRETE

Place concrete in accordance with ACI 301 Section 5. Concrete shall be placed within 15 minutes of discharge into non-agitating equipment.

3.7.1 Footing Placement

Concrete for footings may be placed in excavations without forms upon inspection and approval by the Contracting Officer. Excavation width must be a minimum of 4 inches greater than indicated.

3.7.2 Pumping

ACI 304R and ACI 304.2R. Pumping must not result in separation or loss of materials nor cause interruptions sufficient to permit loss of plasticity between successive increments. Loss of slump in pumping equipment must not exceed 2 inches at discharge/placement. Do not convey concrete through pipe made of aluminum or aluminum alloy. Avoid rapid changes in pipe sizes. Limit maximum size of coarse aggregate to 33 percent of the diameter of the pipe. Limit maximum size of well-rounded aggregate to 40 percent of the pipe diameter. Take samples for testing at both the point of delivery to the pump and at the discharge end.

3.7.3 Cold Weather

Cold weather concrete must meet the requirements of ACI 306.1 unless otherwise specified. Do not allow concrete temperature to decrease below 50 degrees F. Obtain approval prior to placing concrete when the ambient temperature is below 40 degrees F or when concrete is likely to be subjected to freezing temperatures within 24 hours. Cover concrete and

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provide sufficient heat to maintain 50 degrees F minimum adjacent to both the formwork and the structure while curing. Limit the rate of cooling to 37 degrees F in any one hour and 50 degrees F per 24 hours after heat application.

3.7.4 Hot Weather

Hot weather concrete must meet the requirements of ACI 305.1 unless otherwise specified. Maintain required concrete temperature using Figure 4.2 in ACI 305R to prevent the evaporation rate from exceeding 0.2 pound of water per square foot of exposed concrete per hour. Cool ingredients before mixing or use other suitable means to control concrete temperature and prevent rapid drying of newly placed concrete. Shade the fresh concrete as soon as possible after placing. Start curing when the surface of the fresh concrete is sufficiently hard to permit curing without damage. Provide water hoses, pipes, spraying equipment, and water hauling equipment, where job site is remote to water source, to maintain a moist concrete surface throughout the curing period. Provide burlap cover or other suitable, permeable material with fog spray or continuous wetting of the concrete when weather conditions prevent the use of either liquid membrane curing compound or impervious sheets. For vertical surfaces, protect forms from direct sunlight and add water to top of structure once concrete is set.

3.7.5 Bonding

Surfaces of set concrete at joints, must be roughened and cleaned of laitance, coatings, loose particles, and foreign matter. Roughen surfaces in a manner that exposes the aggregate uniformly and does not leave laitance, loosened particles of aggregate, nor damaged concrete at the surface.

Obtain bonding of fresh concrete that has set as follows:

- a. At joints between footings and walls or columns, between walls or columns and the beams or slabs they support, and elsewhere unless otherwise specified; roughened and cleaned surface of set concrete must be dampened, but not saturated, immediately prior to placing of fresh concrete.
- b. At joints in exposed-to-view work; at vertical joints in walls; at joints near midpoint of span in girders, beams, supported slabs, other structural members; in work designed to contain liquids; the roughened and cleaned surface of set concrete must be dampened but not saturated and covered with a cement grout coating.
- c. Provide cement grout that consists of equal parts of portland cement and fine aggregate by weight with not more than 6 gallons of water per sack of cement. Apply cement grout with a stiff broom or brush to a minimum thickness of 1/16 inch. Deposit fresh concrete before cement grout has attained its initial set.

3.8 WASTE MANAGEMENT

Provide as specified in the Waste Management Plan and as follows.

3.8.1 Mixing Equipment

Before concrete pours, designate on-site area to be paved later in project

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for cleaning out concrete mixing trucks. Minimize water used to wash equipment.

3.8.2 Hardened, Cured Waste Concrete

Crush and reuse hardened, cured waste concrete as fill or as a base course for pavement.

3.8.3 Reinforcing Steel

Collect reinforcing steel and place in designated area for recycling.

3.8.4 Other Waste

Identify concrete manufacturer's or supplier's policy for collection or return of construction waste, unused material, deconstruction waste, and/or packaging material.

3.9 SURFACE FINISHES EXCEPT FLOOR, SLAB, AND PAVEMENT FINISHES

3.9.1 Defects

Repair surface defects in accordance with ACI 301 Section 5.

3.9.2 Not Against Forms (Top of Walls)

Surfaces not otherwise specified must be finished with wood floats to even surfaces. Finish must match adjacent finishes.

3.9.3 Formed Surfaces

3.9.3.1 Tolerances

Tolerances in accordance with ACI 117 and as indicated.

3.9.3.2 Standard Smooth Finish

Provide for surfaces exposed to public view a surface finish SF-3.0. Patch holes and defects in accordance with ACI 301.

3.9.4 Smooth-Rubbed and Grout-Cleaned Rubbed Finish

Provide a smooth-rubbed finish per ACI 301 Section 5 in the locations indicated. Provide a grout-cleaned rubbed finish per ACI 301 Section 5 in the locations indicated.

3.10 FLOOR, SLAB, AND PAVEMENT FINISHES AND MISCELLANEOUS CONSTRUCTION

In accordance with ACI 301 and ACI 302.1R, unless otherwise specified. Slope floors uniformly to drains where drains are provided. Depress the concrete base slab where quarry tile, ceramic tile, or embeded walk-off mats are indicated. Steel trowel and fine-broom finish concrete slabs that are to receive quarry tile, ceramic tile, or paver tile. Where straightedge measurements are specified, Contractor must provide straightedge.

3.10.1 Finish

Place, consolidate, and immediately strike off concrete to obtain proper contour, grade, and elevation before bleedwater appears. Permit concrete

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to attain a set sufficient for floating and supporting the weight of the finisher and equipment. If bleedwater is present prior to floating the surface, drag the excess water off or remove by absorption with porous materials. Do not use dry cement to absorb bleedwater. Grate tampers ("jitterbugs") shall not be used.

3.10.1.1 Broomed

Use on surfaces of exterior walks, platforms, patios, and ramps, unless otherwise indicated. Finish concrete in accordance with ACI 301 Section 5 for a broomed finish.

3.11 JOINTS

3.11.1 Construction Joints

Make and locate joints not indicated so as not to impair strength and appearance of the structure, as approved. Joints must be perpendicular to main reinforcement. Reinforcement must be continued and developed across construction joints. Locate construction joints as follows:

3.11.1.1 Construction Joints for Constructability Purposes

- a. In walls, at top of footing; at top of slabs on ground; at top and bottom of door and window openings or where required to conform to architectural details; and at underside of deepest beam or girder framing into wall.
- b. In columns or piers, at top of footing; at top of slabs on ground; and at underside of deepest beam or girder framing into column or pier.
- c. Near midpoint of spans for supported slabs, beams, and girders unless a beam intersects a girder at the center, in which case construction joints in girder must offset a distance equal to twice the width of the beam. Make transfer of shear through construction joint by use of inclined reinforcement.

Provide keyways at least 1-1/2-inches deep in construction joints in walls and slabs and between walls and footings; approved bulkheads may be used for slabs.

3.11.2 Isolation Joints in Slabs on Ground

- a. Provide joints at points of contact between slabs on ground and vertical surfaces, such as column pedestals, foundation walls, grade beams, and elsewhere as indicated.
- b. Fill joints with premolded joint filler strips 1/2 inch thick, extending full slab depth. Install filler strips at proper level below finish floor elevation with a slightly tapered, dress-and-oiled wood strip temporarily secured to top of filler strip to form a groove not less than 3/4 inch in depth where joint is sealed with sealing compound and not less than 1/4 inch in depth where joint sealing is not required. Remove wood strip after concrete has set. Contractor must clean groove of foreign matter and loose particles after surface has dried.

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3.11.3 Contraction Joints in Slabs on Ground

- a. Provide joints to form panels as indicated.
- b. Under and on exact line of each control joint, cut 50 percent of welded wire reinforcement before placing concrete.
- c. Sawcut contraction joints into slab on ground in accordance with ACI 301 Section 5.
- d. Sawcutting will be limited to within 12 hours after set and at 1/4 slab depth.

3.11.4 Sealing Joints in Slabs on Ground

- a. Contraction and control joints which are to receive finish flooring material must be sealed with joint sealing compound after concrete curing period. Slightly underfill groove with joint sealing compound to prevent extrusion of compound. Remove excess material as soon after sealing as possible.
- b. Sealed groove must be left ready to receive filling material that is provided as part of finish floor covering work.

3.12 CURING AND PROTECTION

Curing and protection in accordance with ACI 301 Section 5, unless otherwise specified. Begin curing immediately following form removal. Avoid damage to concrete from vibration created by blasting, pile driving, movement of equipment in the vicinity, disturbance of formwork or protruding reinforcement, and any other activity resulting in ground vibrations. Protect concrete from injurious action by sun, rain, flowing water, frost, mechanical injury, tire marks, and oil stains. Do not allow concrete to dry out from time of placement until the expiration of the specified curing period. Do not use membrane-forming compound on surfaces where appearance would be objectionable, on any surface to be painted, where coverings are to be bonded to the concrete, or on concrete to which other concrete is to be bonded. If forms are removed prior to the expiration of the curing period, provide another curing procedure specified herein for the remaining portion of the curing period. Provide moist curing for those areas receiving liquid chemical sealer, hardener, or epoxy coating. Allow curing compound/sealer installations to cure prior to the installation of materials that adsorb VOCs.

3.12.1 Requirements for Type III, High-Early-Strength Portland Cement

The curing periods are required to be not less than one-fourth of those specified for portland cement, but in no case less than 72 hours.

3.12.2 Curing Periods

ACI 301 Section 5, except 10 days for retaining walls, pavement or chimneys. Begin curing immediately after placement. Protect concrete from premature drying, excessively hot temperatures, and mechanical injury; and maintain minimal moisture loss at a relatively constant temperature for the period necessary for hydration of the cement and hardening of the concrete. The materials and methods of curing are subject to approval by the Contracting Officer.

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3.12.3 Curing Formed Surfaces

Accomplish curing of formed surfaces, including undersurfaces of girders, beams, supported slabs, and other similar surfaces by moist curing with forms in place for full curing period or until forms are removed. If forms are removed before end of curing period, accomplish final curing of formed surfaces by any of the curing methods specified above, as applicable.

3.12.4 Curing Unformed Surfaces

- a. Accomplish initial curing of unformed surfaces, such as monolithic slabs, floor topping, and other flat surfaces, by membrane curing.
- b. Accomplish final curing of unformed surfaces by any of curing methods specified, as applicable.
- c. Accomplish final curing of concrete surfaces to receive liquid floor hardener of finish flooring by moisture-retaining cover curing.

3.12.5 Temperature of Concrete During Curing

When temperature of atmosphere is 41 degrees F and below, maintain temperature of concrete at not less than 55 degrees F throughout concrete curing period or 45 degrees F when the curing period is measured by maturity. When necessary, make arrangements before start of concrete placing for heating, covering, insulation, or housing as required to maintain specified temperature and moisture conditions for concrete during curing period.

When the temperature of atmosphere is 80 degrees F and above or during other climatic conditions which cause too rapid drying of concrete, make arrangements before start of concrete placing for installation of wind breaks, of shading, and for fog spraying, wet sprinkling, or moisture-retaining covering of light color as required to protect concrete during curing period.

Changes in temperature of concrete must be uniform and not exceed 37 degrees F in any one hour nor 80 degrees F in any 24-hour period.

3.12.6 Protection from Mechanical Injury

During curing period, protect concrete from damaging mechanical disturbances, particularly load stresses, heavy shock, and excessive vibration and from damage caused by rain or running water.

3.12.7 Protection After Curing

Protect finished concrete surfaces from damage by construction operations.

3.13 FIELD QUALITY CONTROL

3.13.1 Aggregate Testing

3.13.1.1 Fine Aggregate

At least once during each shift when the concrete plant is operating, there shall be one sieve analysis and fineness modulus determination in accordance with ASTM C136/C136M and COE CRD-C 104 for the fine aggregate or

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for each fine aggregate if it is batched in more than one size or classification. The location at which samples are taken may be selected by the Contractor as the most advantageous for control. However, the Contractor is responsible for delivering fine aggregate to the mixer within specification limits. When the amount passing on any sieve is outside the specification limits, the fine aggregate shall be immediately resampled and retested. If there is another failure on any sieve, the fact shall be immediately reported to the Contracting Officer, concreting shall be stopped, and immediate steps taken to correct the grading.

3.13.1.2 Coarse Aggregate

At least once during each shift in which the concrete plant is operating, there shall be a sieve analysis in accordance with ASTM C136/C136M for each size of coarse aggregate. The location at which samples are taken may be selected by the Contractor as the most advantageous for production control. However, the Contractor shall be responsible for delivering the aggregate to the mixer within specification limits. A test record of samples of aggregate taken at the same locations shall show the results of the current test as well as the average results of the five most recent tests including the current test. The Contractor may adopt limits for control coarser than the specification limits for samples taken other than as delivered to the mixer to allow for degradation during handling. When the amount passing any sieve is outside the specification limits, the coarse aggregate shall be immediately resampled and retested. If the second sample fails on any sieve, that fact shall be reported to the Contracting Officer. Where two consecutive averages of 5 tests are outside specification limits, the operation shall be considered out of control and reported to the Contracting Officer. Concreting shall be stopped and immediate steps shall be taken to correct the grading.

3.13.2 Concrete Sampling

ASTM C172/C172M. Collect samples of fresh concrete to perform tests specified. ASTM C31/C31M for making test specimens.

3.13.3 Concrete Testing

3.13.3.1 Slump Tests

ASTM C143/C143M. Take concrete samples during concrete placement/discharge.

The maximum slump may be increased as specified with the addition of an approved admixture provided that the water-cementitious material ratio is not exceeded. Perform tests at commencement of concrete placement, when test cylinders are made, and for each 20 cubic yards (maximum) of concrete.

3.13.3.2 Temperature Tests

Test the concrete delivered and the concrete in the forms. Perform tests in hot or cold weather conditions (below 50 degrees F and above 80 degrees F) for each batch (minimum) or every 20 cubic yards (maximum) of concrete, until the specified temperature is obtained, and whenever test cylinders and slump tests are made.

3.13.3.3 Compressive Strength Tests

ASTM C39/C39M. Make six 6 inch by 12 inch test cylinders for each set of tests in accordance with ASTM C31/C31M, ASTM C172/C172M and applicable requirements of ACI 305R and ACI 306R. Take precautions to prevent

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evaporation and loss of water from the specimen. Test two 6 inch by 12 inch cylinders at 7 days, two 6 inch by 12 inch cylinders at 28 days, and hold two 6 inch by 12 inch cylinders in reserve. Take samples for strength tests for each concrete mixture placed each day not less than once a day, nor less than once for each 150 cubic yards of concrete, nor less than once for each 5000 square feet of surface area for slabs or walls. For the entire project, take no less than five sets of samples and perform strength tests for each mix design of concrete placed. Each strength test result must be the average of two 6 inch by 12 inch cylinders from the same concrete sample tested at 28 days. Concrete compressive tests must meet the requirements of this section, the Contract Document, and ACI 301. Retest locations represented by erratic core strengths. Where retest does not meet concrete compressive strength requirements submit a mitigation or remediation plan for review and approval by the Contracting Officer. Repair core holes with patching mortar. Match color of adjacent concrete.

3.13.3.4 Air Content

ASTM C173/C173M or ASTM C231/C231M for normal weight concrete. Test air-entrained concrete for air content at the same frequency as specified for slump tests.

3.13.3.5 Strength of Concrete Structure

The strength of the concrete structure will be considered to be deficient if any of the following conditions are identified:

- a. Failure to meet compressive strength tests as evaluated.
- b. Reinforcement not conforming to requirements specified.
- c. Concrete which differs from required dimensions or location in such a manner as to reduce strength.
- d. Concrete curing and protection of concrete against extremes of temperature during curing, not conforming to requirements specified.
- e. Concrete subjected to damaging mechanical disturbances, particularly load stresses, heavy shock, and excessive vibration.
- f. Poor workmanship likely to result in deficient strength.

Where the strength of the concrete structure is considered deficient submit a mitigation or remediation plan for review and approval by the contracting officer.

3.13.3.6 Non-Conforming Materials

Factors that indicate that there are non-conforming materials include (but not limited to) excessive compressive strength, inadequate compressive strength, excessive slump, excessive voids and honeycombing, concrete delivery records that indicate excessive time between mixing and placement, or excessive water was added to the mixture during delivery and placement. Any of these indicators alone are sufficient reason for the Contracting Officer to request additional sampling and testing.

Investigations into non-conforming materials must be conducted at the Contractor's expense. The Contractor must be responsible for the investigation and must make written recommendations to adequately mitigate

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or remediate the non-conforming material. The Contracting Officer may accept, accept with reduced payment, require mitigation, or require removal and replacement of non-conforming material at no additional cost to the Government.

3.13.3.7 Testing Concrete Structure for Strength

When there is evidence that strength of concrete structure in place does not meet specification requirements or there are non-conforming materials, make cores drilled from hardened concrete for compressive strength determination in accordance with ASTM C42/C42M, and as follows:

- a. Take at least three representative cores from each member or area of concrete-in-place that is considered potentially deficient. Location of cores will be determined by the Contracting Officer.
- b. Test cores after moisture conditioning in accordance with ASTM C42/C42M if concrete they represent is more than superficially wet under service.
- c. Air dry cores, (60 to 80 degrees F with relative humidity less than 60 percent) for 7 days before test and test dry if concrete they represent is dry under service conditions.
- d. Strength of cores from each member or area are considered satisfactory if their average compressive strength is equal to or greater than 85 percent of the design compressive strength, and if no single core strength is less than 75 percent of the design compressive strength. Additional testing of cores extracted from locations represented by erratic core strength results will be permitted only when approved by the Contracting Officer.

Fill core holes solid with patching mortar and finished to match adjacent concrete surfaces.

Correct concrete work that is found inadequate by core tests in a manner approved by the Contracting Officer.

3.14 REPAIR, REHABILITATION AND REMOVAL

Before the Contracting Officer accepts the structure the Contractor must inspect the structure for cracks, damage and substandard concrete placements that may adversely affect the service life of the structure. A report documenting these defects must be prepared which includes recommendations for repair, removal or remediation must be submitted to the Contracting Officer for approval before any corrective work is accomplished.

3.14.1 Repair of Weak Surfaces

Weak surfaces are defined as mortar-rich, rain-damaged, uncured, or containing exposed voids or deleterious materials. Concrete surfaces with weak surfaces less than 1/4 inch thick must be diamond ground to remove the weak surface. Surfaces containing weak surfaces greater than 1/4 inch thick must be removed and replaced or mitigated in a manner acceptable to the Contracting Officer.

3.14.2 Failure of Quality Assurance Test Results

Proposed mitigation efforts by the Contractor must be approved by the Contracting Officer prior to proceeding.

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-- End of Section --

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SECTION 04 20 00

UNIT MASONRY

11/15, CHG 2: 05/19

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

ACI SP-66 (2004) ACI Detailing Manual

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7 (2017) Minimum Design Loads for Buildings and Other Structures

ASTM INTERNATIONAL (ASTM)

ASTM A153/A153M (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

ASTM A167 (2011) Standard Specification for Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip

ASTM A270/A270M (2023) Standard Specification for Seamless and Welded Austenitic and Ferritic/Austenitic Stainless Steel Sanitary Tubing

ASTM A615/A615M (2022) Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement

ASTM A641/A641M (2019) Standard Specification for Zinc-Coated (Galvanized) Carbon Steel Wire

ASTM A653/A653M (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process

ASTM A951/A951M (2011) Standard Specification for Steel Wire for Masonry Joint Reinforcement

ASTM A996/A996M (2016) Standard Specification for Rail-Steel and Axle-Steel Deformed Bars for Concrete Reinforcement

ASTM A1008/A1008M (2021a) Standard Specification for Steel,

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	Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable
ASTM A1064/A1064M	(2022) Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete
ASTM B370	(2022) Standard Specification for Copper Sheet and Strip for Building Construction
ASTM C67/C67M	(2023) Standard Test Methods for Sampling and Testing Brick and Structural Clay Tile
ASTM C207	(2018) Standard Specification for Hydrated Lime for Masonry Purposes
ASTM C270	(2019a; E 2019) Standard Specification for Mortar for Unit Masonry
ASTM C476	(2023) Standard Specification for Grout for Masonry
ASTM C494/C494M	(2019; E 2022) Standard Specification for Chemical Admixtures for Concrete
ASTM C641	(2017) Standard Test Method for Iron Staining Materials in Lightweight Concrete Aggregates
ASTM C652	(2022) Standard Specification for Hollow Brick (Hollow Masonry Units Made from Clay or Shale)
ASTM C1019	(2019) Standard Test Method for Sampling and Testing Grout
ASTM C1314	(2014) Standard Test Method for Compressive Strength of Masonry Prisms
ASTM C1384	(2012a) Standard Specification for Admixtures for Masonry Mortars
ASTM D2000	(2018) Standard Classification System for Rubber Products in Automotive Applications
ASTM D2287	(2019) Nonrigid Vinyl Chloride Polymer and Copolymer Molding and Extrusion Compounds
ASTM E514/E514M	(2020) Standard Test Method for Water Penetration and Leakage Through Masonry

THE MASONRY SOCIETY (TMS)

TMS MSJC	(2016) Masonry Standard Joint Committee's (MSJC) Book - Building Code Requirements and Specification for Masonry Structures,
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Containing TMS 402/ACI 530/ASCE 5, TMS
602/ACI 530.1/ASCE 6, and Companion
Commentaries

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Reinforcement Detail Drawings; G

SD-03 Product Data

Hot Weather Procedures

Cold Weather Procedures

Clay or Shale Brick

Cementitious Materials

SD-04 Samples

Clay or Shale Brick; G

Admixtures for Masonry Mortar; G

Anchors, Ties, and Bar Positioners; G

Joint Reinforcement; G

Clay Masonry Expansion-Joint Materials; G

SD-05 Design Data

Masonry Compressive Strength; G

Bracing Calculations; G

Structural Lintel Calculations; G

SD-06 Test Reports

Efflorescence Test

Prism Tests

Single-Wythe Masonry Wall Water Penetration Test

SD-07 Certificates

Special Masonry Inspector Qualifications

Clay or Shale Brick

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Cementitious Materials

Admixtures for Masonry Mortar

Admixtures for Grout

Anchors, Ties, and Bar Positioners

Joint Reinforcement

SD-08 Manufacturer's Instructions

Admixtures for Masonry Mortar

Admixtures for Grout

1.3 QUALITY ASSURANCE

1.3.1 Special Masonry Inspector Qualifications

Refer to Section 01 45 35 SPECIAL INSPECTIONS for qualifications and responsibilities of the masonry special inspector.

1.4 DELIVERY, STORAGE, AND HANDLING

Deliver, store, handle, and protect material to avoid chipping, breakage, and contact with soil or contaminating material. Store and prepare materials in already disturbed areas to minimize project site disturbance and size of project site.

1.4.1 Masonry Units

Cover and protect masonry units from precipitation. Conform to handling and storage requirements of TMS MSJC.

- a. Do not stack pallets. Do not remove units from cartons until cartons are placed on scaffolds or in the location where units are to be laid.
- b. Mark prefabricated lintels on top sides to show either the lintel schedule number or the number and size of top and bottom bars.

1.4.2 Reinforcement, Anchors, and Ties

Store steel reinforcing bars, coated anchors, ties, and joint reinforcement above the ground. Maintain steel reinforcing bars and uncoated ties free of loose mill scale and loose rust.

1.4.3 Cementitious Materials, Sand and Aggregates

Deliver cementitious and other packaged materials in unopened containers, plainly marked and labeled with manufacturers' names and brands. Store cementitious material in dry, weathertight enclosures or completely cover. Handle cementitious materials in a manner that will prevent the inclusion of foreign materials and damage by water or dampness. Store sand and aggregates in a manner to prevent contamination and segregation.

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1.5 PROJECT/SITE CONDITIONS

Conform to TMS MSJC for hot and cold weather masonry erection.

1.5.1 Hot Weather Procedures

When ambient air temperature exceeds 100 degrees F, or exceeds 90 degrees F and the wind velocity is greater than 8 mph, comply with TMS MSJC Article 1.8 D for: preparation prior to conducting masonry work; construction while masonry work is in progress; and protection for newly completed masonry.

1.5.2 Cold Weather Procedures

When ambient temperature is below 40 degrees F, comply with TMS MSJC Article 1.8 C for: preparation prior to conducting masonry work; construction while masonry work is in progress; and protection for newly completed masonry.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

2.1.1 Design - Specified Compressive Strength of Masonry

The specified compressive strength of masonry, f'_m , is 2,000 psi net.

2.1.2 Performance - Verify Masonry Compressive Strength

Verify specified compressive strength of masonry using the "Unit Strength Method" of TMS MSJC. Submit calculations and certifications of unit and mortar strength.

Verify specified compressive strength of masonry using the "Prism Test Method" of TMS MSJC when the "Unit Strength Method" cannot be used. Submit test results.

2.2 MANUFACTURED UNITS

2.2.1 General Requirements

Do not change the source of materials, which will affect the appearance of the finished work, after the work has started except with Contracting Officer's approval. Submit test reports from an approved independent laboratory. Certify test reports on a previously tested material as the same materials as that proposed for use in this project. Submit certificates of compliance stating that the materials meet the specified requirements.

2.2.2 Clay or Shale Brick (WHERE OCCURS ON DRAWINGS)

2.2.2.1 General

2.2.2.1.1 Sample Submittal

Submit brick samples as specified, showing the color range and texture of clay or shale brick. Limit units used on the project to those that conform to the approved sample. Submit sample of colored mortar with applicable masonry unit and color samples of three stretcher units and one unit for

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each type of special shape.

2.2.2.1.2 Uniformity

Manufacture bricks at one time and from the same run. Deliver clay or shale brick units factory-blended to provide a uniform appearance and color range in the completed wall.

2.2.2.1.3 Efflorescence Test

Test clay brick that will be exposed to weathering for efflorescence in accordance with ASTM C67/C67M. Schedule tests far enough in advance of starting masonry work to permit retesting if necessary. Units meeting the definition of "effloresced" are subject to rejection.

2.2.2.2 Hollow Clay or Shale Brick

Provide hollow clay or shale brick that conforms to ASTM C652, Type HBS .

- a. Provide brick size of 4 inches thick, 2 2/3 inches high, and 8 inches long.
- b. Where vertical reinforcement is shown in hollow brick, provide hollow brick designed to provide precise vertical alignment of the cells, with minimum cell dimension of 2-1/2 inches.
- c. Provide hollow brick with minimum compressive strength of 1,500 psi.

2.2.3 Concrete Units

2.2.3.1 Aggregates

Test lightweight aggregates, and blends of lightweight and heavier aggregates in proportions used in producing the units, for stain-producing iron compounds in accordance with ASTM C641, visual classification method. Do not incorporate aggregates for which the iron stain deposited on the filter paper exceeds the "light stain" classification.

Use industrial waste by-products (air-cooled slag, cinders, or bottom ash), ground waste glass and concrete, granulated slag, and expanded slag in aggregates.

2.3 EQUIPMENT

2.3.1 Vibrators

Maintain at least one spare vibrator on site at all times.

2.3.2 Grout Pumps

Pumping through aluminum tubes is not permitted.

2.4 MATERIALS

2.4.1 Mortar Materials

2.4.1.1 Cementitious Materials

Provide cementitious materials that conform to those permitted by ASTM C270.

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2.4.1.2 Hydrated Lime and Alternates

Provide lime that conforms to one of the materials permitted by ASTM C207 for use in combination with portland cement, hydraulic cement, and blended hydraulic cement. Do not use lime in combination with masonry cement or mortar cement.

2.4.1.3 Admixtures for Masonry Mortar

In cold weather, use a non-chloride based accelerating admixture that conforms to ASTM C1384, unless Type III portland cement is used in the mortar.

In showers and kitchens, use mortar that contains a water-repellent admixture that conforms to ASTM C1384. Provide a water-repellent admixture, conforming to ASTM C1384 and of the same brand and manufacturer as the block's integral water-repellent, in the mortar used to place concrete masonry units that have an integral water-repellent admixture.

2.4.1.4 Aggregate and Water

Provide aggregate (sand) and water that conform to materials permitted by ASTM C270.

2.4.2 Grout and Ready-Mix Grout Materials

2.4.2.1 Cementitious Materials for Grout

Provide cementitious materials that conform to those permitted by ASTM C476.

2.4.2.2 Admixtures for Grout

Water-reducing admixtures that conform to ASTM C494/C494M Type F or G and viscosity-modifying admixtures that conform to ASTM C494/C494M Type S are permitted for use in grout. Other admixtures require approval by the Contracting Officer.

In cold weather, a non-chloride based accelerating admixture may be used subject to approval by the Contracting Officer; use accelerating admixture that is non-corrosive and conforms to ASTM C494/C494M, Type C.

2.4.2.3 Aggregate and Water

Provide aggregate (sand) and water that conform to materials permitted by ASTM A270/A270M.

2.5 MORTAR AND GROUT MIXES

2.5.1 Mortar Mix

- a. Provide mortar Type S unless specified otherwise herein.
- b. Use ASTM C270 Type S cement-lime mortar or mortar cement mortar for seismic-force-resisting elements indicated and in accordance with ASCE 7 (Chapter 13).
- c. Provide mortar that conforms to ASTM C270. Use Type S mortar for foundation walls and in piers.

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- d. Provide Type S mortar for non-load-bearing, non-shear-wall interior masonry.
- e. For preblended mortar, follow manufacturer's mixing instructions.

2.5.2 Grout and Ready Mix Grout Mix

Use grout that conforms to ASTM C476. Use conventional grout with a slump between 8 and 11 inches. Use self-consolidating grout with slump flow of 24 to 30 inches and a visual stability index (VSI) not greater than 1. Unless otherwise indicated provide minimum grout strength of 2000 psi in 28 days, as tested in accordance with ASTM C1019. Do not change proportions and do not use materials with different physical or chemical characteristics in grout for the work unless additional evidence is furnished that grout meets the specified requirements. Use ready-mixed grout that conforms to ASTM C476.

2.6 ACCESSORIES

2.6.1 Grout Barriers

Grout barriers for vertical cores that consist of fine mesh wire, fiberglass, or expanded metal.

2.6.2 Anchors, Ties, and Bar Positioners

2.6.2.1 General

- a. Fabricate anchors and ties without drips or crimps. Size anchors and ties to provide a minimum of 5/8 inch mortar cover from each face of masonry.
- b. Fabricate steel wire anchors and ties from wire conforming to ASTM A1064/A1064M and hot-dip galvanize in accordance with ASTM A153/A153M.
- c. Fabricate joint reinforcement in conformance with ASTM A951/A951M. Hot dip galvanize joint reinforcement in exterior walls and in interior walls exposed to moist environment in conformance with ASTM A153/A153M. Galvanize joint reinforcement in other interior walls in conformance with ASTM A641/A641M; coordinate with paragraph JOINT REINFORCEMENT below.
- d. Fabricate sheet metal anchors and ties in conformance with ASTM A1008/A1008M. Hot dip galvanize sheet metal anchors and ties in exterior walls and in interior walls exposed to moist environment in compliance with ASTM A153/A153M Class B. Galvanize sheet metal anchors and ties in other interior walls in compliance with ASTM A653/A653M, Coating Designation G60.
- e. Submit two anchors, ties and bar positioners of each type used, as samples.

2.6.2.2 Adjustable Anchors

2.6.2.2.1 Anchorage to Structural Steel

Provide hot-dip galvanized adjustable anchors for connecting masonry walls

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to the structural steel frame as detailed on the drawings .

2.6.2.2.2 Anchorage of Veneer to Light Gauge Steel or Concrete Backing

Use one of the following types of adjustable anchors to connect veneer to light gauge steel or concrete backing:

- a. sheet metal at least 7/8 inch wide, 0.06 inch thick, and with corrugations having a wavelength of 0.3 to 0.5 inch and an amplitude of 0.06 to 0.10 inch or bent, notched or punched to provide equivalent performance;
- b. wire anchors of minimum size W1.7 with ends bent to form a minimum 2 inches extension and without drips;
- c. or wire pintle anchors used in conjunction with joint reinforcement.

Do not exceed 1/16 inch clearance between connecting parts of the tie. Assemble adjustable anchors to prevent disengagement. Provide pintle anchors with one or more pintle legs of wire size W2.8 and an offset not exceeding 1-1/4 inch.

2.6.2.3 Veneer Anchor Screws

Provide screws for attachment of veneer anchors to cold-formed steel framing members of size as required by design to provide the needed pullout load capacity but not less than No. 12. Provide length of screws such that the screws penetrate the holding member by not less than 5/8 inch.

2.6.2.4 Bar Positioners

Factory-fabricate bar positioners, used to prevent displacement of reinforcing bars during the course of construction, from 9 gauge steel wire or equivalent, and hot-dip galvanized. Bar positioners must be suitable for intended use and be corrosion resistant steel. Bar positioners not fully contained within the wythe must be hot-dip galvanized.

2.6.3 Joint Reinforcement

Factory fabricate joint reinforcement in conformance with ASTM A951/A951M, welded construction. Provide ladder type joint reinforcement, having one longitudinal wire in the mortar bed of each face shell for hollow units and one wire for solid units and with all wires a minimum of 9 gauge. Size joint reinforcement to provide a minimum of 5/8 inch cover from each face. Space crosswires not more than 16 inches. Provide joint reinforcement for straight runs in flat sections not less than 10 feet long. Provide joint reinforcement with factory formed corners and intersections. If approved for use, joint reinforcement may be furnished with adjustable wall tie features. Submit one piece of each type used, including corner and wall intersection pieces, showing at least two cross wires.

2.6.4 Reinforcing Steel Bars

Provide reinforcing steel bars and rods conforming to ASTM A615/A615M or ASTM A996/A996M, Grade 60.

2.6.5 Concrete Masonry Control Joint Keys

Provide control joint keys of a factory fabricated solid section of natural

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or synthetic rubber (or combination thereof) conforming to ASTM D2000 M2AA-805 with a minimum durometer hardness of 80 or polyvinyl chloride conforming to ASTM D2287 Type PVC 654-4 with a minimum durometer hardness of 85. Form the control joint key with a solid shear section not less than 5/8 inch thick and 3/8 inch thick flanges, with a tolerance of plus or minus 1/16 inch, to fit neatly, but without forcing, in masonry unit jamb sash grooves.

2.6.6 Clay Masonry Expansion-Joint Materials

Provide backer rod and sealant, adequate to accommodate joint compression and extension equal to 50 percent of the width of the joint. Provide the backer rod of compressible rod stock of closed cell polyethylene foam, polyurethane foam, butyl rubber foam, or other flexible, nonabsorptive material as recommended by the sealant manufacturer. Provide sealant in conformance with Section 07 92 00 JOINT SEALANTS.

Submit one piece of each type of material used.

2.6.7 Through Wall Flashing and Weeps

2.6.7.1 General

Provide coated copper, copper or stainless steel sheet, self-adhesive rubberized sheet, or reinforced membrane sheet flashing

2.6.7.2 Coated-Copper Flashing

Provide 7 ounce, electrolytic copper sheet, uniformly coated on both sides with acidproof, alkaliproof, asphalt impregnated kraft paper or polyethylene sheets.

2.6.7.3 Copper or Stainless Steel Flashing

Provide copper sheet, complying with ASTM B370, minimum 16 ounce weight; or stainless steel, ASTM A167, Type 304 or 316, 0.015 inch thick, No. 2D finish.

2.6.7.4 Reinforced Membrane Flashing

Provide polyester film core with a reinforcing fiberglass scrim bonded to one side. Provide membrane that is impervious to moisture, flexible, is not affected by caustic alkalis, and after being exposed for not less than 1/2 hour to a temperature of 32 degrees F, shows no cracking when, at that temperature, it is bent 180 degrees over a 1/16 inch diameter mandrel and then bent at the same point over the same size mandrel in the opposite direction 360 degrees.

2.6.7.5 Rubberized Flashing

Provide self-adhesive rubberized asphalt sheet flashing consisting of 32-mil thick pliable and highly adhesive rubberized asphalt compound bonded completely and integrally to 8-mil thick, high density, cross-laminated polyethylene film to produce an overall thickness of 40 mils. Provide rubberized, asphalt-based mastic and surface conditioner that are each approved by flashing manufacturer for use with flashing material.

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2.6.7.6 Weep Ventilators

Provide weep ventilators that are prefabricated from stainless steel or plastic. Provide inserts with grill or louver-type openings designed to allow the passage of moisture from cavities and to prevent the entrance of insects, and with a rectangular closure strip to prevent mortar droppings from clogging the opening. Provide ventilators with compressible flanges to fit in a standard 3/8 inch wide mortar joint and with height equal to the nominal height of the unit.

2.6.7.7 Metal Drip Edge

Provide stainless steel drip edge, 15-mil thick, hemmed edges, with down-turned drip at the outside edge and upturned dam at the inside edge for use with membrane flashings.

PART 3 EXECUTION

3.1 EXAMINATION

Prior to start of work, verify the applicable conditions as set forth in TMS MSJC, inspection.

3.2 PREPARATION

3.2.1 Stains

Protect exposed surfaces from mortar and other stains. When mortar joints are tooled, remove mortar from exposed surfaces with fiber brushes and wooden paddles. Protect base of walls from splash stains by covering adjacent ground with sand, sawdust, or polyethylene.

3.2.2 Loads

Do not apply uniform loads for at least 12 hours or concentrated loads for at least 72 hours after masonry is constructed. Provide temporary bracing as required.

3.2.3 Concrete Surfaces

Where masonry is to be placed, clean concrete of laitance, dust, dirt, oil, organic matter, or other foreign materials and slightly roughen to provide a surface texture with a depth of at least 1/8 inch. Sandblast, if necessary, to remove laitance from pores and to expose the aggregate.

3.2.4 Shelf Angles

Adjust shelf angles as required to keep the masonry level and at the proper elevation.

3.2.5 Bracing

Provide bracing and scaffolding necessary for masonry work. Design bracing to resist wind pressure as required by OSHA and local codes and submit bracing calculations, sealed by a registered professional engineer. Do not remove bracing in less than 10 days.

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3.2.6 Structural Lintel Calculations

New openings in masonry walls shall receive structural lintels (pre-cast or steel) sized via delegated design submittal to fully support the remaining wall above them. Contractor to provide structural drawings or lintel schedule, signed, and sealed by a structural engineer with a USA state issued professional license.

3.3 ERECTION

3.3.1 General

- a. Coordinate masonry work with the work of other trades to accommodate built-in items and to avoid cutting and patching. Lay masonry units in running bond pattern. Lay facing courses level with back-up courses, unless the use of adjustable ties has been approved in which case the tolerances is plus or minus 1/2 inch. Adjust each unit to its final position while mortar is still soft and has plastic consistency.
- b. Remove and clean units that have been disturbed after the mortar has stiffened, and relay with fresh mortar. Keep air spaces, cavities, chases, expansion joints, and spaces to be grouted free from mortar and other debris. Select units to be used in exposed masonry surfaces from those having the least amount of chipped edges or other imperfections detracting from the appearance of the finished work.
- c. When necessary to temporarily discontinue the work, step (rack) back the masonry for joining when work resumes. Toothing may be used only when specifically approved by the Contracting Officer. Before resuming work, remove loose mortar and thoroughly clean the exposed joint. Cover the top of walls subjected to rain or snow with nonstaining waterproof covering or membrane when work is not in process. Extend the covering a minimum of 610 mm 2 feet down on each side of the wall and hold securely in place.
- d. Ensure that units being laid and surfaces to receive units are free of water film and frost. Lay solid units in a nonfurrowed full bed of mortar. Bevel mortar for veneer wythes and slope down toward the cavity side. Shove units into place so that the vertical joints are tight. Completely fill vertical joints between solid units with mortar, except where indicated at control, expansion, and isolation joints. Place hollow units so that mortar extends to the depth of the face shell at heads and beds, unless otherwise indicated. Mortar will be permitted to protrude up to 1/2 inch into the space or cells to be grouted. Provide means to prevent mortar from dropping into the space below or clean grout spaces prior to grouting.
- e. In multi-wythe construction with collar joints no more than 3/4 inch wide, bring up the inner wythe not more than 16 inches ahead of the outer wythe. Fill collar joints with mortar during the laying of the facing wythe, and do not lag the laying of the facing wythe by back-buttering each unit as it is laid.

3.3.1.1 Jointing

Tool mortar joints when the mortar is thumbprint hard. Tool horizontal joints after tooling vertical joints. Brush mortar joints to remove loose and excess mortar.

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3.3.1.1.1 Tooled Joints

Tool mortar joints in exposed exterior and interior masonry surfaces concave , using a jointer that is slightly larger than the joint width so that complete contact is made along the edges of the unit. Perform tooling so that the mortar is compressed and the joint surface is sealed. Use a jointer of sufficient length to obtain a straight and true mortar joint. No exterior joints are to be left un-tooled.

3.3.1.1.2 Flush Joints

Flush cut mortar joints in concealed masonry surfaces and joints at electrical outlet boxes in wet areas. Finish flush cut joints by cutting off the mortar flush with the face of the wall. Point joints in unparged masonry walls below grade tight. For architectural units, such as fluted units, completely fill both the head and bed joints and flush cut.

3.3.1.1.3 Door and Window Frame Joints

On the exposed interior side of exterior frames, rake joints between frames and abutting masonry walls to a depth of 3/8 inch. On the exterior side of exterior frames, rake joints between frames and abutting masonry walls to a depth of 3/8 inch.

3.3.1.1.4 Joint Widths

- a. Construct brick masonry with mortar joint widths equal to the difference between the specified and nominal dimensions of the unit, within tolerances permitted by TMS MSJC.
- b. Provide 3/8 inch wide mortar joints in concrete masonry, except for prefaced concrete masonry units.
- c. Provide 3/8 inch wide mortar joints on unfaced side of prefaced concrete masonry units and not less than 3/16 inch nor more than 1/4 inch wide on prefaced side.
- d. Maintain mortar joint widths within tolerances permitted by TMS MSJC

3.3.1.2 Cutting and Fitting

Use full units of the proper size wherever possible, in lieu of cut units. Locate cut units where they would have the least impact on the architectural aesthetic goals of the facility. Perform cutting and fitting, including that required to accommodate the work of others, by masonry mechanics using power masonry saws. Concrete masonry units may be wet or dry cut. Before being placed in the work, dry wet-cut units to the same surface-dry appearance as uncut units being laid in the wall. Provide cut edges that are clean, true and sharp.

- a. Carefully make openings in the masonry so that wall plates, cover plates or escutcheons required by the installation will completely conceal the openings and will have bottoms parallel with the masonry bed joints. Provide reinforced masonry lintels above openings over 12 inches wide for pipes, ducts, cable trays, and other wall penetrations, unless steel sleeves are used.
- b. Do not reduce masonry units in size by more than one-third in height and one-half in length. Do not locate cut products at ends of walls,

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corners, and other openings.

3.3.1.3 Unfinished Work

Rack back unfinished work for joining with new work. Toothing may be resorted to only when specifically approved by the Contracting Officer. Remove loose mortar and thoroughly clean the exposed joints before laying new work.

3.3.1.4 Clay Masonry Expansion Joints

Provide clay masonry expansion joints as indicated. Construct by filling with a compressible foam pad. Ensure that no mortar or other noncompressible materials are within the joint. Install backer rod and sealant in accordance with Section 07 92 00 JOINT SEALANTS.

3.3.1.5 Control Joints

Provide control joints in concrete masonry as indicated. Construct in accordance with the details shown on the Drawings. Form a continuous vertical joint at control joint locations, including through bond beams, by utilizing half blocks in alternating courses on each side of the joint. Interrupt the control joint key in courses containing continuous bond beam reinforcement.

Where mortar was placed in the joint, rake both faces of the control joints to a depth of 3/4 inch. Install backer rod and sealant on both faces in accordance with Section 07 92 00 JOINT SEALANTS.

3.3.1.6 Decorative Architectural Units

Place decorative masonry units with the patterned face shell properly aligned in the completed wall.

3.3.2 Clay or Shale Brick Masonry

3.3.2.1 Brick Placement

Blend all brick at the jobsite from several cubes to produce a uniform appearance when installed. An observable "banding" or "layering" of colors or textures caused by improperly mixed brick is unacceptable. Lay brick facing with the better face exposed. Lay brick in running bond with each course bonded at corners, unless otherwise indicated. Lay molded brick with the frog side down. Do not lay brick that is cored, recessed, or has other deformations in a manner that allows those deformations to be exposed to view; lay 100 percent solid units in these areas. Completely fill head and bed joints of solid units with mortar. Lay hollow units with mortar joints as specified for concrete masonry units.

Place exterior face of salvaged bricks towards the exterior.

3.3.2.2 Wetting of Units

Wet clay, shale brick, or hollow brick units having an initial rate of absorption of more than 1 gram per minute per square inch of bed surface in conformance with ASTM C67/C67M. Ensure that each unit is nearly saturated when wetted but surface dry when laid.

Test clay or shale brick daily on the job, prior to laying, as follows:

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Using a wax pencil, draw a circle the size of a quarter on five randomly selected bricks. Apply 20 drops of water with a medicine dropper to the surface within the circle on each brick. If the average time that the water is completely absorbed in the five bricks is less than 1-1/2 minutes, wet bricks represented by the five bricks tested.

3.3.2.3 Brick Sills

Lay brick on edge, slope not less than 3/4 inch downward to the outside, and project not less than 1/2 inch beyond the face of the wall to form a wash and drip. Fill all joints solidly with mortar and tool.

3.3.3 Anchored Veneer Construction

- a. Construct exterior masonry wythes to the thickness indicated on the drawings. Provide a minimum 1 inch air space behind the masonry veneer. Provide means to ensure that the cavity space and flashings are kept clean of mortar droppings and other loose debris. Maintain chases and raked-out joints free from mortar and debris.
- b. Place masonry in running bond pattern.
- c. For veneer over stud framing, do not install veneer until the exterior sheathing, moisture barrier, veneer anchors and flashing have been installed on the backing. Take extreme care to avoid damage to the moisture barrier and flashing during construction of the masonry veneer. Repair or replace portions of the moisture barrier and flashing that are damaged prior to completion of the veneer. Provide a continuous cavity as indicated.
- d. Provide anchors (ties) to connect the veneer to its backing in sufficient quantity to comply with the following requirements: maximum wall area per anchor {tie} of 2.67 square feet, and maximum vertical spacing of 25 inches, and maximum horizontal spacing of 32 inches. Provide additional anchors around openings larger than 16 inch in either direction. Space anchors around perimeter of opening at a maximum of 24 inches on center. Place anchors within 12 inches of openings. Anchors with drips are not permitted.
- e. With solid units, embed anchors in mortar joint and extend into the veneer a minimum of 1-1/2 inch, with at least 5/8 inch mortar cover to the outside face.
- f. With hollow units, embed anchors in mortar or grout and extend into the veneer a minimum of 1-1/2 inch, with at least 5/8 inch mortar or grout cover to outside face.

3.3.4 Reinforced, Single Wythe Concrete Masonry Units Walls

3.3.4.1 Concrete Masonry Unit Placement

- a. Fully bed units used to form piers, pilasters, columns, starting courses on footings, solid foundation walls, lintels, and beams, and where cells are to be filled with grout in mortar under both face shells and webs. Provide mortar beds under both face shells for other units. Mortar head joints for a distance in from the face of the unit not less than the thickness of the face shell.

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3.3.4.2 Preparation for Reinforcement

Lay units in such a manner as to preserve the unobstructed vertical continuity of cores to be grouted. Remove mortar protrusions extending 1/2 inch or more into cells before placing grout. Position reinforcing bars accurately as indicated before placing grout. Where vertical reinforcement occurs, fill cores solid with grout in accordance with paragraph PLACING GROUT in this Section.

3.3.5 ANCHORAGE

3.3.5.1 Anchorage to Concrete

Anchor masonry to the face of concrete columns, beams, or walls with dovetail anchors spaced not over 16 inches on centers vertically and 24 inches on center horizontally.

3.3.5.2 Anchorage to Structural Steel

Anchor masonry to vertical structural steel framing with adjustable steel wire anchors spaced not over 16 inches on centers vertically, and if applicable, not over 24 inches on centers horizontally.

3.3.5.3 Anchorage at Intersecting Walls

Provide wire mesh anchors at maximum 16 inches spacing at intersections of interior non-bearing masonry walls.

3.3.6 Lintels

3.3.6.1 Masonry Lintels

Construct masonry lintels with lintel units filled solid with grout in all courses and reinforced with a minimum of two No. 4 bars in the bottom course unless otherwise indicated. Extend lintel reinforcement beyond each side of masonry opening 40 bar diameters or 24 inches, whichever is greater. Support reinforcing bars in place prior to grouting and locate 1/2 inch above the bottom inside surface of the lintel unit.

3.3.7 Sills and Copings

Set sills and copings in a full bed of mortar with faces plumb and true. Slope sills and copings to drain water. Mechanically anchor copings and sills longer than 4 feet as indicated.

3.4 INSTALLATION

3.4.1 Bar Reinforcement Installation

3.4.1.1 Preparation

Submit detail drawings showing bar splice locations. Identify bent bars on a bending diagram and reference and locate such bars on the drawings. Show wall dimensions, bar clearances, and wall openings. Utilize bending details that conform to the requirements of ACI SP-66. No approval will be given to the shop drawings until the Contractor certifies that all openings, including those for mechanical and electrical service, are shown. If, during construction, additional masonry openings are required, resubmit the approved shop drawings with the additional openings shown.

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along with the proposed changes. Clearly highlight location of these additional openings. Provide wall elevation drawings with minimum scale of 1/4 inch per foot. Submit drawings including plans, elevations, and details of wall reinforcement; details of reinforcing bars at corners and wall intersections; offsets; tops, bottoms, and ends of walls; control and expansion joints; lintels; and wall openings.

Clean reinforcement of loose, flaky rust, scale, grease, mortar, grout, and other coatings that might destroy or reduce its bond prior to placing grout. Do not use bars with kinks or bends not shown on the approved shop drawings. Place reinforcement prior to grouting. Unless otherwise indicated, extend vertical wall reinforcement to within 2 inches of tops of walls.

3.4.1.2 Positioning Bars

- a. Accurately place vertical bars within the cells at the positions indicated on the drawings. Maintain a minimum clearance of 1/2 inch between the bars and masonry units. Provide minimum clearance between parallel bars of 1/2 inch between the bars and masonry units for coarse grout and a minimum clearance of 1/4 inch between the bars and masonry units for fine grout. Provide minimum clearance between parallel bars of 1 inch or one diameter of the reinforcement, whichever is greater. Vertical reinforcement may be held in place using bar positioners located near the ends of each bar and at intermediate intervals of not more than 192 diameters of the reinforcement or by other means to prevent displacement beyond permitted tolerances. As masonry work progresses, secure vertical reinforcement to prevent displacement beyond allowable tolerances.
- b. Wire column and pilaster lateral ties in position around the vertical reinforcing bars. Place lateral ties in contact with the vertical reinforcement and do not place in horizontal mortar bed joints.
- c. Position horizontal reinforcing bars as indicated. Stagger splices in adjacent horizontal bars, unless otherwise indicated.
- d. Form splices by lapping bars as indicated. Do not cut, bend or eliminate reinforcing bars. Foundation dowel bars may be field-bent when permitted by TMS MSJC.

3.4.1.3 Splices of Bar Reinforcement

Lap splice reinforcing bars as indicated. When used, provide welded or mechanical connections that develop at least 125 percent of the specified yield strength of the reinforcement.

3.4.2 Placing Grout

3.4.2.1 General

Fill cells containing reinforcing bars with grout. Solidly grout hollow masonry units in walls or partitions supporting plumbing, heating, or other mechanical fixtures, voids at door and window jambs, and other indicated spaces. Solidly grout cells under lintel bearings on each side of openings for full height of openings. Solidly grout walls below grade, lintels, and bond beams. Units other than open end units may require grouting each course to preclude voids in the units.

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Discard site-mixed grout that is not placed within 1-1/2 hours after water is first added to the batch or when the specified slump is not met without adding water after initial mixing. Discard ready-mixed grout that does not meet the specified slump without adding water other than water that was added at the time of initial discharge. Allow sufficient time between grout lifts to preclude displacement or cracking of face shells of masonry units. Provide a grout shear key between lifts when grouting is delayed and the lower lift loses plasticity. If blowouts, flowouts, misalignment, or cracking of face shells should occur during construction, tear down the wall and rebuild.

3.4.2.2 Vertical Grout Barriers for Multi-Wythe Composite Walls

In multi-wythe composite walls, provide grout barriers in the collar joint not more than 30 feet apart, or as required, to limit the horizontal flow of grout for each pour.

3.4.2.3 Horizontal Grout Barriers

Embed horizontal grout barriers in mortar below cells of hollow units receiving grout.

3.4.2.4 Grout Holes and Cleanouts

3.4.2.4.1 Grout Holes

Provide grouting holes in slabs, spandrel beams, and other in-place overhead construction. Locate holes over vertical reinforcing bars or as required to facilitate grout fill in bond beams. Provide additional openings spaced not more than 16 inches on centers where grouting of hollow unit masonry is indicated. Form such openings not less than 4 inches in diameter or 3 by 4 inches in horizontal dimensions. Upon completion of grouting operations, plug and finish grouting holes to match surrounding surfaces.

3.4.2.4.2 Cleanouts for Hollow Unit Masonry Construction

For hollow masonry units, provide cleanout holes at the bottom of every grout pour in cores containing vertical reinforcement when the height of the grout pour exceeds 5 feet 4 inches. Where all cells are to be grouted, construct cleanout courses using bond beam units in an inverted position to permit cleaning of all cells. Provide cleanout holes at a maximum spacing of 32 inches where all cells are to be filled with grout.

Establish a new series of cleanouts if grouting operations are stopped for more than 4 hours. Provide cleanouts not less than 3 by 3 inch by cutting openings in one face shell. Manufacturer's standard cutout units may be used at the Contractor's option. Do not cleanout holes until masonry work, reinforcement, and final cleaning of the grout spaces have been completed and inspected. For walls which will be exposed to view, close cleanout holes in an approved manner to match surrounding masonry.

3.4.2.5 Grout Placement

A grout pour is the total height of masonry to be grouted prior to erection of additional masonry. A grout lift is an increment of grout placement within a grout pour. A grout pour is filled by one or more lifts of grout.

- a. Lay masonry to the top of a pour permitted by TMS MSJC Table 7, based

on the size of the grout space and the type of grout. Prior to grouting, remove masonry protrusions that extend 1/2 inch or more into cells or spaces to be grouted. Provide grout holes and cleanouts in accordance with paragraph GROUT HOLES AND CLEANOUTS above when the grout pour height exceeds 5 feet 4 inches. Hold reinforcement, bolts, and embedded connections rigidly in position before grouting is started. Do not prewet concrete masonry units.

- b. Place grout using a hand bucket, concrete hopper, or grout pump to fill the grout space without segregation of aggregate. Operate grout pumps to produce a continuous stream of grout without air pockets, segregation, or contamination.
- c. If the masonry has cured at least 4 hours, grout slump is maintained between 10 to 11 inches, and no intermediate reinforced bond beams are placed between the top and bottom of the pour height, place conventional grout in lifts not exceeding 12 feet 8 inches. For the same curing and slump conditions but with intermediate bond beams, limit conventional grout lift to the bottom of the lowest bond beam that is more than 5 feet 4 inches above the bottom of the lift, but do not exceed 12 feet 8 inches. If masonry has not cured at least 4 hours or grout slump is not maintained between 10 to 11 inches, place conventional grout in lifts not exceeding 5 feet 4 inches.
- d. Consolidate conventional grout lift and reconsolidate after initial settlement before placing next lift. For grout pours that are 12 inches or less in height, consolidate and reconsolidate grout by mechanical vibration or puddling. For grout pours that are greater than 12 inches in height, consolidate and reconsolidate grout by mechanical vibration. Apply vibrators at uniformly spaced points not further apart than the visible effectiveness of the machine. Limit duration of vibration to time necessary to produce satisfactory consolidation without causing segregation. If previous lift is not permitted to set, dip vibrator into previous lift. Do not insert vibrators into lower lifts that are in a semi-solidified state. If lower lift sets prior to placement of subsequent lift, form a grout key by terminating grout a minimum of 1-1/2 inch below a mortar joint. Vibrate each vertical cell containing reinforcement in partially grouted masonry. Do not form grout keys within beams.
- e. If the masonry has cured 4 hours, place self-consolidating grout (SCG) in lifts not exceeding the pour height. If masonry has not cured for at least 4 hours, place SCG in lifts not exceeding 5 feet 4 inches. Do not mechanically consolidate self-consolidating grout. Place self-consolidating grout in accordance with manufacturer's recommendations.
- f. Upon completion of each day's grouting, remove waste materials and debris from the equipment, and dispose of outside the masonry.

3.4.3 Joint Reinforcement Installation

Install joint reinforcement at 16 inches on center unless otherwise indicated. Lap joint reinforcement not less than 6 inches. Install prefabricated sections at corners and wall intersections. Place the longitudinal wires of joint reinforcement in mortar beds to provide not less than 5/8 inch cover to either face of the unit.

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3.4.4 Flashing and Weeps

- a. Install through-wall flashing at obstructions in the cavity and where indicated on Drawings. Ensure continuity of the flashing at laps and inside and outside corners by splicing in a manner approved by the flashing manufacturer. Provide sealant below the drip edge of through-wall flashing.
- b. Wherever through-wall flashing occurs, provide weep holes to drain flashing to exterior at acceptable locations as indicated. Locate weeps not more than 24 inches on centers in mortar joints of the exterior wythe directly on the horizontal leg of through-wall flashing over foundations, bond beams, and any other horizontal interruptions of the cavity. Place weep holes perfectly horizontal or slightly canted downward to encourage water drainage outward and not inward. Other methods may be used for providing weeps when spacing is reduced to 16 inches on center and approved by the Contracting Officer. Maintain weeps free of mortar and other obstructions.

3.5 APPLICATION

3.5.1 Interface with Other Products

3.5.1.1 Built-In Items

Fill spaces around built-in items with mortar. Point openings around flush-mount electrical outlet boxes in wet locations with mortar. Embed anchors, ties, wall plugs, accessories, flashing, pipe sleeves and other items required to be built-in as the masonry work progresses. Fully embed anchors, ties and joint reinforcement in the mortar. Fill cells receiving anchor bolts and cells of the first course below bearing plates with grout, unless otherwise indicated.

3.5.1.2 Door and Window Frame Joints

On the exposed interior and exterior sides of exterior frames, rake joints between frames and abutting masonry walls to a depth of 3/8 inch.

3.5.1.3 Bearing Plates

Set bearing plates for beams, joists, joist girders and similar structural members to the proper line and elevation with damp-pack bedding mortar, except where non-shrink grout is indicated.

3.5.2 Tolerances

Lay masonry plumb, true to line, with courses level within the tolerances of TMS MSJC, Article 3.3 F.

3.6 FIELD QUALITY CONTROL

3.6.1 Tests

3.6.1.1 Prism Tests

Perform at least one prism test sample for each 5,000 square feet of wall but not less than three such tests for any building. Evaluate three prisms in each test. Fabricate, store, handle, and test prisms in accordance with ASTM C1314.

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Seven-day tests may be used provided the relationship between the 7- and 28-day strengths of the masonry is established by the tests of the materials used. If the compressive strength of any prism falls below the specified value by more than 500 psi, take steps to assure that the load-carrying capacity of the structure is not jeopardized. If the likelihood of low-strength masonry is confirmed and computations indicate that the load-carrying capacity may have been significantly reduced, tests of cores drilled, or prisms sawed, from the area in question may be required. In such case, take three specimens for each prism test more than 500 psi below the specified value. Masonry in the area in question will be considered structurally adequate if the average compressive strength of three specimens is equal to or exceeds the specified value. Additional testing of specimens extracted from locations represented by erratic core or prism strength test results will be permitted.

3.6.1.2 Single-Wythe Masonry Wall Water Penetration Test

Prepare a minimum of three specimens and cure for minimum 28 days prior to testing. Construct panels by the same methods, processes, and applications to be used on the project's construction site. Spray test for 6 hours on each specimen. If water is visible on back of test panels during the test and areas of dampness on the backside of the test panels do not exceed 25 percent of the wall area, the panels will be considered to have passed. Dampness is defined as any area of surface darkening or discoloration due to moisture penetration or accumulation below the observed surface.

Construct additional test panels for each failed test performed until three test panels pass the test. Factors that can affect test performance include materials, mixing, and quality of application and workmanship. Materials, mixing, and methods adjustments may be necessary in order to provide construction that passes the water penetration test. Document and record the test specimen construction materials and application and provide written test report in accordance with ASTM E514/E514M, supplemented by a detailed discussion of the specifics of test panel construction, application methods and processes used, quality of construction, and any variances or deviations that may have occurred between test panels during test panel construction. For failed test panels, identify in the supplemental report the variances, deficiencies or flaws that contributed to test panel failure and itemize the precautions to be taken in field construction of the masonry wall to prevent similar deficiencies and assure the wall construction replicates test panel conditions that pass the water penetration test. Submit the complete, certified test report, including supplemental report, to the Contracting Officer prior to start of single-wythe concrete masonry wall construction. Significant changes to materials, proportions, or construction techniques from those used in the passing water penetration test are grounds for performing new tests, at the discretion of the Contracting Officer.

3.6.2 Special Inspection

Perform special inspections and testing in accordance with Section 01 45 35 SPECIAL INSPECTIONS.

3.7 POINTING AND CLEANING

After mortar joints have attained their initial set, but prior to hardening, completely remove mortar and grout daubs and splashings from masonry-unit surfaces that will be exposed or painted. Before completion

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of the work, rake out defects in joints of masonry to be exposed or painted, fill with mortar, and tool to match existing joints. Immediately after grout work is completed, remove scum and stains that have percolated through the masonry work using a low pressure stream of water and a stiff bristled brush. Do not clean masonry surfaces, other than removing excess surface mortar, until mortar in joints has hardened. Leave masonry surfaces clean, free of mortar daubs, dirt, stain, and discoloration, including scum from cleaning operations, and with tight mortar joints throughout. Do not use metal tools and metal brushes for cleaning.

3.7.1 Dry-Brushing Concrete Masonry

Dry brush exposed concrete masonry surfaces at the end of each day's work and after any required pointing, using stiff-fiber bristled brushes.

3.7.2 Clay Brick Surfaces

Clean exposed clay brick masonry surfaces to obtain surfaces free of stain, dirt, mortar and grout daubs, efflorescence, and discoloration or scum from cleaning operations.

3.8 PROTECTION

Protect facing materials against staining. Cover top of walls with nonstaining waterproof covering or membrane to protect from moisture intrusion when work is not in progress. Continue covering the top of the unfinished walls until the wall is waterproofed with a complete roof or parapet system. Extend covering a minimum of 2 feet down on each side of the wall and hold securely in place. Before starting or resuming work, clean top surface of masonry in place of loose mortar and foreign material.

-- End of Section --

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SECTION 05 05 23.16

STRUCTURAL WELDING

08/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 360 (2016) Specification for Structural Steel Buildings

AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING (ASNT)

ANSI/ASNT CP-189 (2020) ASNT Standard for Qualification and Certification of Nondestructive Testing Personnel

AMERICAN WELDING SOCIETY (AWS)

AWS A2.4 (2012) Standard Symbols for Welding, Brazing and Nondestructive Examination

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding Code - Steel

AWS D1.3/D1.3M (2018) Structural Welding Code - Sheet Steel

AWS D1.4/D1.4M (2018) Structural Welding Code - Reinforcing Steel

AWS D14.4/D14.4M (2012) Specification for Welded Joints for Machinery and Equipment

AWS QC1 (2016) Specification for AWS Certification of Welding Inspectors

AWS Z49.1 (2021) Safety in Welding and Cutting and Allied Processes

ASTM INTERNATIONAL (ASTM)

ASTM E165/E165M (2023) Standard Practice for Liquid Penetrant Examination for General Industry

ASTM E709 (2021) Standard Guide for Magnetic Particle Testing

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S"

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classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Welding Quality Assurance Plan

SD-03 Product Data

Welding Procedure Qualifications

Welder, Welding Operator, and Tacker Qualification

Previous Qualifications

Pre-Qualified Procedures

Welding Electrodes and Rods

SD-06 Test Reports

Nondestructive Testing

Weld Inspection Log

SD-07 Certificates

Certified Welding Procedure Specifications (WPS)

Certified Brazing Procedure Specifications (BPS)

Certified Procedure Qualification Records (PQR)

Certified Welder Performance Qualifications (WPQ)

Certified Brazer Performance Qualifications (BPQ)

Certified Welding Inspector

Nondestructive Testing Personnel

1.3 QUALITY ASSURANCE

Except for pre-qualified (in accordance with AWS D1.1/D1.1M) and previously qualified procedures, each Contractor performing welding must record in detail and qualify the welding procedure specification for any welding procedure followed in the fabrication of weldments. Conform welding procedure qualifications to AWS D1.1/D1.1M and to the specifications in this section. Submit for approval copies of the welding procedure specification and the procedure qualification records for each type of welding being performed. Submission of the welder, welding operator, or tacker qualification test records is also required. Approval of any procedure, however, does not relieve the Contractor of the sole responsibility for producing a finished structure meeting all the specified requirements. Submit this information on the forms in Annex M of AWS D1.1/D1.1M. Individually identify and clearly reference on the detail drawings and erection drawings all welding procedure specifications, or suitably key them to the contract drawings. In case of conflict between

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this specification and AWS D1.1/D1.1M, this specification governs.

1.3.1 General Requirements

Fabricate work in an AISC Certified Fabrication Plant, Category BU. Erect work by an AISC Certified Erector, Category CSE.

a. For Structural Projects, provide documentation of the following:

- (1) Component Thickness 1/8 inch and greater: Qualification documents (WPS, PQR, and WPQ) in accordance with AWS D1.1/D1.1M.
- (2) Component Thickness Less than 1/8 inch: Qualification documents (WPS, PQR, and WPQ) in accordance with AWS D1.3/D1.3M.
- (3) Reinforcing Steel: Qualification documents (WPS, PQR, and WPQ) in accordance with AWS D1.4/D1.4M.

b. For other applications, provide documentation of the following:

- (1) Submit two copies of the Certified Welding Procedure Specifications (WPS), Certified Brazing Procedure Specifications (BPS) and Certified Procedure Qualification Records (PQR) to the Contracting Officer for approval .
- (2) Submit two copies of the Certified Welder Performance Qualifications (WPQ) and Certified Brazer Performance Qualifications (BPQ) to the Contracting Officer for approval within fifteen calendar days prior to any employee welding on the project material.
- (3) Machinery: Qualification documents (WPS, PQR, and WPQ) in accordance with AWS D14.4/D14.4M.

1.3.2 Previous Qualifications

Welding procedures previously qualified by test in accordance with AWS D1.1/D1.1M, may be accepted for this contract without re-qualification, upon receipt of the test results, if the following conditions are met:

- a. Testing was performed by an approved testing laboratory, technical consultant, or the Contractor's approved quality control organization.
- b. The qualified welding procedure conforms to the requirements of this specification and is applicable to welding conditions encountered under this contract.
- c. The welder, welding operator, and tacker qualification tests conform to the requirements of this specification and are applicable to welding conditions encountered under this contract.

1.3.3 Welder, Welding Operator, and Tacker Qualification

Each welder, welding operator, and tacker assigned to work on this contract must be qualified in accordance with the applicable requirements of AWS D1.1/D1.1M and as specified in this section. Welders, welding operators, and tackers who make acceptable procedure qualification test welds will be considered qualified for the welding procedure used within the applicable essential variables for welder qualification.

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1.3.3.1 Previous Personnel Qualifications

At the discretion of the Contracting Officer, welders, welding operators, and tackers qualified by test within the previous 6 months may be accepted for this contract without re-qualification if all the following conditions are met:

- a. Copies of the welding procedure specifications, the procedure qualification test records, and the welder, welding operator, and tacker qualification test records are submitted and approved in accordance with the specified requirements for detail drawings.
- b. Testing was performed by an approved testing laboratory, technical consultant, or the Contractor's approved quality control organization.
- c. The welder, welding operator, and tacker qualification tests conform to the requirements of this specification and are applicable to welding conditions encountered under this contract.

1.3.3.2 Certificates

Before assigning any welder, welding operator, or tacker to work under this contract, submit the names and certification that each individual is qualified as specified. State in the certification the type of welding and positions for which the welder, welding operator, or tacker is qualified, the code and procedure under which the individual is qualified, the date qualified, and the name of the firm and person certifying the qualification tests. Keep the certification current, on file, and furnish 3 copies.

1.3.3.3 Renewal of Qualification

Re-qualification of a welder or welding operator is required under any of the following conditions:

- a. It has been more than 6 months since the welder or welding operator has used the specific welding process for which he is qualified.
- b. There is specific reason to question the welder or welding operator's ability to make welds that meet the requirements of these specifications.
- c. The welder or welding operator was qualified by an employer other than those firms performing work under this contract, and a qualification test has not been taken within the past 12 months. Submit as evidence of conformance all records showing periods of employment, name of employer where welder, or welding operator, was last employed, and the process for which qualified.
- d. A tacker who passes the qualification test is considered eligible to perform tack welding indefinitely in the positions and with the processes for which he/she is qualified, unless there is some specific reason to question the tacker's ability or there has been a gap greater than 6 months since he/she last used the process. In such a case, the tacker is required to pass the prescribed tack welding test.

1.3.4 Inspector Qualification

Submit certificates indicating that certified welding inspectors meet the

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requirements of AWS QC1. Submit qualifications for nondestructive testing personnel in accordance with the requirements of ANSI/ASNT CP-189 for Levels I or II in the applicable nondestructive testing method. Level I inspectors must have direct supervision of a Level II inspector.

1.3.5 Symbols and Safety

Use symbols in accordance with AWS A2.4, unless otherwise indicated. Follow safe welding practices and safety precautions during welding in conformance with AWS Z49.1.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Conform the design of welded connections to AISC 360, unless otherwise indicated or specified. Material with welds will not be accepted unless the welding is specified or indicated on the drawings or otherwise approved. Perform welding as specified in this section, except where additional requirements are shown on the drawings or are specified in other sections. Do not commence welding until welding procedures, inspectors, nondestructive testing personnel, welders, welding operators, and tackers have been qualified and the submittals approved by the Contracting Officer. Perform all testing at or near the work site. Maintain records of the test results obtained in welding procedure, welder, welding operator, and tacker performance qualifications.

2.1.1 Pre-erection Conference

Hold a pre-erection conference prior to the start of the field welding, to bring all affected parties together and to gain a naturally clear understanding of the project and the Welding Procedure Specifications (WPS) (submitted for all welding, including welding done using pre-qualified procedures). Mandatory attendance is required by all Contractor's welding production and inspection personnel and appropriate Government personnel. Include as items for discussion: responsibilities of various parties; welding procedures and processes to be followed; welding sequence (both within a joint and joint sequence within the building); inspection requirements and procedures, both visual and nondestructive testing; welding schedule; and other items deemed necessary by the attendees.

2.2 WELDING EQUIPMENT AND MATERIALS

Provide all welding equipment, welding electrodes and rods, welding wire, and fluxes capable of producing satisfactory welds when used by a qualified welder or welding operator. Provide welding equipment and materials that comply with the applicable requirements of AWS D1.1/D1.1M. Submit product data on welding electrodes and rods.

PART 3 EXECUTION

3.1 WELDING OPERATIONS

3.1.1 Requirements

Conform workmanship and techniques for welded construction to the requirements of AWS D1.1/D1.1M and AISC 360. When AWS D1.1/D1.1M and the AISC 360 specification conflict, the requirements of AWS D1.1/D1.1M govern.

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3.1.2 Identification

Identify all welds in one of the following ways:

- a. Submit written records to indicate the location of welds made by each welder, welding operator, or tacker.
- b. Identify all work performed by each welder, welding operator, or tacker with an assigned number, letter, or symbol to identify welds made by that individual. The Contracting Officer may require welders, welding operators, and tackers to apply their symbol next to the weld by means of rubber stamp, felt-tipped marker with waterproof ink, or other methods that do not cause an indentation in the metal. Place the identification mark for seam welds adjacent to the weld at 3 foot intervals. Identification with die stamps or electric etchers is not allowed.

3.2 QUALITY CONTROL

Perform testing using an approved inspection or testing laboratory or technical consultant; or if approved, the Contractor's inspection and testing personnel may be used instead of the commercial inspection or testing laboratory or technical consultant. A Certified Welding Inspector must perform visual inspection on 100 percent of all welds. Document this inspection in the Visual Weld Inspection Log. Test 50% of CJP welds using ultrasonic testing per Table 6.2 of AWS D1.1/D1.1M. Randomly test 50% of all PJP and fillet welds or as indicated by magnetic particle or dye penetrant testing. Verify the welds conform to paragraph STANDARDS OF ACCEPTANCE. Conform procedures and techniques for inspection with applicable requirements of AWS D1.1/D1.1M, ASTM E165/E165M, and ASTM E709. Submit a Welding Quality Assurance Plan and records of tests and inspections.

3.3 STANDARDS OF ACCEPTANCE

Conform dimensional tolerances for welded construction, details of welds, and quality of welds with the applicable requirements of AWS D1.1/D1.1M and the contract drawings. Submit all records of nondestructive testing.

3.3.1 Nondestructive Testing

The welding is subject to inspection and tests in the mill, shop, and field. Inspection and tests in the mill or shop do not relieve the Contractor of the responsibility to furnish weldments of satisfactory quality. When materials or workmanship do not conform to the specification requirements, the Government reserves the right to reject material or workmanship or both at any time before final acceptance of the structure containing the weldment. Any indication of a defect is regarded as a defect, unless re-evaluation by nondestructive methods or by surface conditioning shows that no unacceptable defect is present. Submit all records of nondestructive testing in accordance with paragraph STANDARDS OF ACCEPTANCE.

3.3.2 Destructive Tests

Make all repairs when metallographic specimens are removed from any part of a structure. Employ only qualified welders or welding operators, and use the proper joints and welding procedures, including peening or heat treatment if required, to develop the full strength of the members and

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joints cut and to relieve residual stress.

3.4 GOVERNMENT INSPECTION AND TESTING

In addition to the inspection and tests performed by the Contractor for quality control, the Government will perform inspection and testing for acceptance to the extent determined by the Contracting Officer. The work may be performed by the Government's own forces or under a separate contract for inspection and testing. The Government reserves the right to perform supplemental nondestructive and destructive tests to determine compliance with paragraph STANDARDS OF ACCEPTANCE.

3.5 CORRECTIONS AND REPAIRS

If inspection or testing indicates defects in the weld joints, repair defective welds using a qualified welder or welding operator as applicable. Conduct corrections in accordance with the requirements of AWS D1.1/D1.1M and the specifications. Repair all defects in accordance with the approved procedures. Repair defects discovered between passes before additional weld material is deposited. Wherever a defect is removed and repair by welding is not required, blend the affected area into the surrounding surface to eliminate sharp notches, crevices, or corners. After a defect is thought to have been removed, and before re-welding, examine the area by suitable methods to ensure that the defect has been eliminated. Repaired welds must meet the inspection requirements for the original welds.

-- End of Section --

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SECTION 05 12 00

STRUCTURAL STEEL

08/18, CHG 2: 05/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 325	(2017) Steel Construction Manual
AISC 326	(2009) Detailing for Steel Construction
AISC 360	(2016) Specification for Structural Steel Buildings
AISC 420	(2010) Certification Standard for Shop Application of Complex Protective Coating Systems
AISC DESIGN GUIDE 10	(1997) Erection Bracing of Low-Rise Structural Steel Buildings
ANSI/AISC 303	(2022) Code of Standard Practice for Structural Steel Buildings and Bridges

AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING (ASNT)

ANSI/ASNT CP-189	(2020) ASNT Standard for Qualification and Certification of Nondestructive Testing Personnel
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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B46.1	(2020) Surface Texture, Surface Roughness, Waviness and Lay
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AMERICAN WELDING SOCIETY (AWS)

AWS A2.4	(2012) Standard Symbols for Welding, Brazing and Nondestructive Examination
AWS D1.1/D1.1M	(2020; Errata 1 2021) Structural Welding Code - Steel
AWS QC1	(2016) Specification for AWS Certification of Welding Inspectors

ASTM INTERNATIONAL (ASTM)

ASTM A6/A6M	(2024a) Standard Specification for General Requirements for Rolled Structural Steel
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Bars, Plates, Shapes, and Sheet Piling

ASTM A29/A29M	(2023) Standard Specification for General Requirements for Steel Bars, Carbon and Alloy, Hot-Wrought
ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A123/A123M	(2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A193/A193M	(2024) Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and Other Special Purpose Applications
ASTM A276/A276M	(2024) Standard Specification for Stainless Steel Bars and Shapes
ASTM A307	(2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
ASTM A563	(2021; E 2022a) Standard Specification for Carbon and Alloy Steel Nuts
ASTM A780/A780M	(2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings
ASTM A992/A992M	(2022) Standard Specification for Structural Steel Shapes
ASTM B695	(2021) Standard Specification for Coatings of Zinc Mechanically Deposited on Iron and Steel
ASTM C827/C827M	(2023) Standard Test Method for Change in Height at Early Ages of Cylindrical Specimens of Cementitious Mixtures
ASTM C1107/C1107M	(2020) Standard Specification for Packaged Dry, Hydraulic-Cement Grout (Nonshrink)
ASTM F436/F436M	(2019) Standard Specification for Hardened Steel Washers Inch and Metric Dimensions
ASTM F844	(2019; R 2024) Standard Specification for Washers, Steel, Plain (Flat), Unhardened for General Use
ASTM F959/F959M	(2017a; R 2023) Standard Specification for Compressible-Washer-Type Direct Tension Indicators for Use with Structural Fasteners, Inch and Metric Series
ASTM F1136/F1136M	(2011) Standard Specification for

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Zinc/Aluminum Corrosion Protective Coatings for Fasteners

ASTM F1554	(2020) Standard Specification for Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength
ASTM F2329/F2329M	(2015; R 2023) Standard Specification for Zinc Coating, Hot-Dip, Requirements for Application to Carbon and Alloy Steel Bolts, Screws, Washers, Nuts, and Special Threaded Fasteners
ASTM F2833	(2011; R 2017) Standard Specification for Corrosion Protective Fastener Coatings with Zinc Rich Base Coat and Aluminum Organic/Inorganic Type
ASTM F3125/F3125M	(2019) Standard Specification for High Strength Structural Bolts and Assemblies, Steel and Alloy Steel, Heat Treated, Inch Dimensions 120 ksi and 150 ksi Minimum Tensile Strength, and Metric Dimensions 830 MPa and 1040 MPa Minimum Tensile Strength

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC PA 1	(2024) Shop, Field, and Maintenance Coating of Metals
SSPC Paint 20	(2019) Zinc-Rich Primers (Type I, Inorganic, and Type II, Organic)
SSPC Paint 29	(2002; E 2004) Zinc Dust Sacrificial Primer, Performance-Based
SSPC SP 3	(2018) Power Tool Cleaning
SSPC SP 6/NACE No.3	(2007) Commercial Blast Cleaning

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-301-01	(2023; with Change 1, 2023) Structural Engineering
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U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR Part 1926, Subpart R	Steel Erection
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1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

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Erection and Erection Bracing Drawings; G

SD-02 Shop Drawings

Fabrication Drawings Including Details of Connections; G

SD-03 Product Data

Shop Primer

Welding Electrodes and Rods

Direct Tension Indicator Washers

Non-Shrink Grout

Tension Control Bolts

Recycled Content for Structural Steel; S

Recycled Content for Structural Steel Tubing; S

Recycled Content for Steel Pipe; S

SD-05 Design Data

Design Calculations for Steel Connections; G

SD-06 Test Reports

Class B Coating

Bolts, Nuts, and Washers

Weld Inspection Reports

Direct Tension Indicator Washer Inspection Reports

Bolt Testing Reports

SD-07 Certificates

Steel

Bolts, Nuts, and Washers

AISC Structural Steel Fabricator Quality Certification

AISC Structural Steel Erector Quality Certification

Welding Procedures and Qualifications

Welding Electrodes and Rods

Certified Welding Inspector

NDT Technician

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Welding Procedure Specifications (WPS)

1.3 QUALITY ASSURANCE

1.3.1 Preconstruction Submittals

1.3.1.1 Erection and Erection Bracing Drawings

Submit for record purposes. Indicate the sequence of erection, temporary shoring and bracing. The erection drawings must conform to ANSI/AISC 303. Erection drawings must be reviewed, stamped and sealed by a registered professional engineer.

1.3.2 Fabrication Drawing Requirements

Submit fabrication drawings for approval prior to fabrication. Prepare in accordance with ANSI/AISC 303, AISC 326 and AISC 325. Fabrication drawings must not be reproductions of contract drawings. Sign and seal fabrication drawings by a registered professional engineer. Include complete information for the fabrication and erection of the structure's components, including the location, type, and size of bolts, welds, member sizes and lengths, connection details, blocks, copes, and cuts. Use AWS A2.4 standard welding symbols. Clearly highlight any deviations from the details shown on the contract drawings highlighted on the fabrication drawings. Explain the reasons for any deviations from the contract drawings.

1.3.3 Delegated Connection Design

Design structural steel connection indicated in the contract documents per ANSI/AISC 303, Option 3, using the connection loads indicated. Submit design calculations for steel connections signed and sealed by a registered professional engineer.

1.3.4 Certifications

1.3.4.1 Welding Procedures and Qualifications

Prior to welding, submit certification for each welder stating the type of welding and positions qualified for, the code and procedure qualified under, date qualified, and the firm and individual certifying the qualification tests. If the qualification date of the welder or welding operator is more than 6 months old, the welding operator's qualification certificate must be accompanied by a current certificate by the welder attesting to the fact that he has been engaged in welding since the date of certification, with no break in welding service greater than 6 months.

Conform to all requirements specified in AWS D1.1/D1.1M.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Provide the structural steel system, including galvanizing, complete and ready for use. Provide structural steel systems including design, materials, installation, workmanship, fabrication, assembly, erection, inspection, quality control, and testing in accordance with ANSI/AISC 303, AISC 360, and UFC 3-301-01 except as modified in this contract.

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2.2 STEEL

2.2.1 Structural Steel

Wide flange and WT shapes, ASTM A992/A992M. Angles, Channels and Plates, ASTM A36/A36M. Provide structural steel containing a minimum of 80 percent recycled content. Submit data identifying percentage of recycled content for structural steel.

2.3 BOLTS, NUTS, AND WASHERS

Submit the certified manufacturer's mill reports which clearly show the applicable ASTM mechanical and chemical requirements together with the actual test results for the supplied fasteners.

2.3.1 Common Grade Bolts

2.3.1.1 Bolts

ASTM A307, Grade A, plain finish hot dipped zinc coating. The bolt heads and the nuts of the supplied fasteners must be marked with the manufacturer's identification mark, the strength grade and type specified by ASTM specifications.

2.3.1.2 Nuts

ASTM A563, Grade A, heavy hex style.

2.3.1.3 Washers

ASTM F844.

2.3.2 High-Strength Bolts

High strength bolts and nuts must be shipped together in the same shipping container. Fasteners indicated to be galvanized shall be tested by the supplier to show that the galvanized nut with the supplied lubricant provided may be rotated from the snug tight condition well in excess of the rotation required for pretensioned installation without stripping. The supplier shall supply nuts that have been lubricated and tested with the supplied bolts.

2.3.2.1 Bolts

ASTM F3125/F3125M, Grade A325M A325 , Type 3 Heavy Hex Head Style, plain finish hot dipped zinc coating in exterior locations and where indicated, mechanically deposited zinc coating in other locations. All bolts used in structural steel connections shall be A325 bolts, unless noted otherwise.

2.3.2.2 Nuts

ASTM A563, Grade and Style as specified in the applicable ASTM bolt standard.

2.3.2.3 Direct Tension Indicator Washers

ASTM F959/F959M. Provide ASTM B695, Class 55, Type 1 galvanizing. Submit product data for direct tension indicator washers.

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2.3.2.4 Washers

ASTM F436/F436M, plain carbon steel.

2.3.3 Foundation Anchorage

2.3.3.1 Anchor Rods

ASTM F1554 Gr 36 55, Class 1A. Stainless steel ASTM A193/A193M.

2.3.3.2 Anchor Nuts

ASTM A563, Grade A, hex style. Stainless steel ASTM A193/A193M.

2.3.3.3 Anchor Washers

ASTM F844. Stainless steel Type 304 conforming to ASTM A276/A276M.

2.3.3.4 Anchor Plate Washers

ASTM A36/A36M Stainless steel Type 304 conforming to ASTM A276/A276M.

2.4 STRUCTURAL STEEL ACCESSORIES

2.4.1 Welding Electrodes and Rods

AWS D1.1/D1.1M. Submit product data for welding electrodes and rods.

2.4.2 Non-Shrink Grout

ASTM C1107/C1107M, with no ASTM C827/C827M shrinkage. Grout must be nonmetallic. Submit product data for non-shrink grout.

2.4.3 Welded Shear Stud Connectors

ASTM A29/A29M, Grades 1010 through 1020. AWS D1.1/D1.1M, Table 7.1, Type B.

2.5 GALVANIZING

ASTM F2329/F2329M, ASTM F1136/F1136M, ASTM F2833 or ASTM B695 for threaded parts or ASTM A123/A123M for structural steel members, as applicable, unless specified otherwise galvanize after fabrication where practicable.

2.6 FABRICATION

Fabrication must be in accordance with the applicable provisions of AISC 325. Fabrication and assembly must be done in the shop to the greatest extent possible. Punch, subpunch and ream, or drill bolt holes perpendicular to the surface of the member.

Compression joints depending on contact bearing must have a surface roughness not in excess of 500 micro inch as determined by ASME B46.1, and ends must be square within the tolerances for milled ends specified in ASTM A6/A6M.

Shop splices of members between field splices will be permitted only where indicated on the Contract Drawings. Splices not indicated require the approval of the Contracting Officer.

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2.6.1 Markings

Prior to erection, identify members by a painted erection mark. Connecting parts assembled in the shop for reaming holes in field connections must be match marked with scratch and notch marks. Do not locate erection markings on areas to be welded. Do not locate match markings in areas that will decrease member strength or cause stress concentrations.

2.6.2 Shop Primer

SSPC Paint 20 or SSPC Paint 29, (zinc rich primer). Shop prime structural steel, except as modified herein, in accordance with SSPC PA 1. Do not prime steel surfaces embedded in concrete, galvanized surfaces, surfaces designed as part of a composite steel concrete section, or surfaces within 0.5 inch of the toe of the welds prior to welding (except surfaces on which metal decking and shear studs are to be welded). If flash rusting occurs, re-clean the surface prior to application of primer. Apply primer in accordance with endorsement "SPE-P1" of AISC 420 or approved equal NACE or SSPC certification to a minimum dry film thickness of 2.0 mil. Submit shop primer product data.

Prime slip critical surfaces with a Class B coating in accordance with AISC 325. Submit test report for Class B coating.

Prior to assembly, prime surfaces which will be concealed or inaccessible after assembly. Do not apply primer in foggy or rainy weather; when the ambient temperature is below 45 degrees F or over 95 degrees F; or when the primer may be exposed to temperatures below 40 degrees F within 48 hours after application, unless approved otherwise by the Contracting Officer. Repair damaged primed surfaces with an additional coat of primer.

2.6.2.1 Cleaning

SSPC SP 6/NACE No.3, except steel exposed in spaces above ceilings, attic spaces, furred spaces, and chases that will be hidden to view in finished construction may be cleaned to SSPC SP 3 when recommended by the shop primer manufacturer. Maintain steel surfaces free from rust, dirt, oil, grease, and other contaminants through final assembly.

2.7 DRAINAGE HOLES

Drill adequate drainage holes to eliminate water traps. Hole diameter must be 1/2 inch and location indicated on the detail drawings. Hole size and locations must not affect the structural integrity.

PART 3 EXECUTION

3.1 ERECTION

- a. Erection of structural steel, except as indicated in item b. below, must be in accordance with the applicable provisions of AISC 325, ANSI/AISC 303 and 29 CFR Part 1926, Subpart R.
- b. For low-rise structural steel buildings (60 feet tall or less and a maximum of 2 stories), erect the structure in accordance with AISC DESIGN GUIDE 10.

After final positioning of steel members, provide full bearing under base plates and bearing plates using nonshrink grout. Place nonshrink grout in

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accordance with the manufacturer's instructions.

3.1.1 STORAGE

Store the material out of contact with the ground in such manner and location as to minimize deterioration.

3.2 CONNECTIONS

Except as modified in this section, design connections indicated in accordance with AISC 360. Build connections into existing work. Do not tighten anchor bolts set in concrete with impact torque wrenches. Holes must not be cut or enlarged by burning. Bolts, nuts, and washers must be clean of dirt and rust, and lubricated immediately prior to installation.

3.2.1 Common Grade Bolts

Tighten ASTM A307 bolts to a "snug tight" fit. "Snug tight" is the tightness that exists when plies in a joint are in firm contact. If firm contact of joint plies cannot be obtained with a few impacts of an impact wrench, or the full effort of a man using a spud wrench, contact the Contracting Officer for further instructions.

3.2.2 High-Strength Bolts

Provide direct tension indicator washers in all ASTM F3125/F3125M, Grade A325 and Grade A490 bolted connections. Bolts must be installed in connection holes and initially brought to a snug tight fit. After the initial tightening procedure, fully tension bolts, progressing from the most rigid part of a connection to the free edges.

Fastener components shall be protected from dirt and moisture in closed containers at the site of the installation. Fastener components that are not incorporated into the work shall be returned to protected storage at the end of the work shift.

3.2.2.1 Installation of Direct Tension Indicator Washers (DTIW)

Where possible, install the DTIW under the bolt head and tighten the nut. If the DTIW is installed adjacent to the turned element, provide a flat washer between the DTIW and nut when the nut is turned for tightening, and between the DTIW and bolt head when the bolt head is turned for tightening. In addition to the DTIW, provide flat washers under both the bolt head and nut when ASTM F3125/F3125M, Grade A490 bolts are used.

3.2.3 Tension Control Bolts

Bolts must be installed in connection holes and initially brought to a snug tight fit. After the initial tightening procedure, fully tension bolts, progressing from the most rigid part of a connection to the free edges.

3.3 GAS CUTTING

Use of gas-cutting torch in the field for correcting fabrication errors is not permitted on any major member in the structural framing. Use of a gas cutting torch will be permitted on minor members not under stress only after approval has been obtained from the Contracting Officer.

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3.4 WELDING

Welding must be in accordance with AWS D1.1/D1.1M. Grind exposed welds smooth as indicated. Provide AWS D1.1/D1.1M qualified welders, welding operators, and tackers.

Develop and submit the Welding Procedure Specifications (WPS) for all welding, including welding done using prequalified procedures. Submit for approval all WPS, whether prequalified or qualified by testing.

3.4.1 Removal of Temporary Welds, Run-Off Plates, and Backing Strips

Remove only from finished areas.

3.5 SHOP PRIMER REPAIR

Repair shop primer in accordance with the paint manufacturer's recommendation for surfaces damaged by handling, transporting, cutting, welding, or bolting.

3.5.1 Field Priming

Field prime steel exposed to the weather, or located in building areas without HVAC for control of relative humidity. After erection, the field bolt heads and nuts, field welds, and any abrasions in the shop coat must be cleaned and primed with paint of the same quality as that used for the shop coat.

3.6 GALVANIZING REPAIR

Repair damage to galvanized coatings using ASTM A780/A780M zinc rich paint for galvanizing damaged by handling, transporting, cutting, welding, or bolting. Do not heat surfaces to which repair paint has been applied.

3.7 FIELD QUALITY CONTROL

Perform field tests, and provide labor, equipment, and incidentals required for testing. Notify the Contracting Officer in writing of defective welds, bolts, nuts, and washers within 7 working days of the date of the inspection.

3.7.1 Welds

3.7.1.1 Visual Inspection

AWS D1.1/D1.1M. Furnish the services of AWS-certified welding inspectors for fabrication and erection inspection and testing and verification inspections. A Certified Welding Inspector must perform visual inspection on 100 percent of all welds. Document this inspection in the Visual Weld Inspection Log. Submit certificates indicating that certified welding inspectors meet the requirements of AWS QC1.

3.7.1.2 Nondestructive Testing

Nondestructive testing must be in accordance with AWS D1.1/D1.1M. Ultrasonic testing must be performed in accordance with Table 6.2 of AWS D1.1/D1.1M. Test locations must be selected by the Contracting Officer. All personnel performing NDT must be certified in accordance with ANSI/ASNT CP-189 in the method of testing being performed. Submit

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certificates showing compliance with ANSI/ASNT CP-189 for all NDT technicians. If more than 20 percent of welds made by a welder contain defects identified by testing, then all groove welds made by that welder must be tested by ultrasonic testing, and all fillet welds made by that welder must be inspected by magnetic particle testing (MT) or dye penetrant testing (PT) as approved by the Contracting Officer. When groove welds made by an individual welder are required to be tested, magnetic particle or dye penetrant testing may be used only in areas inaccessible to ultrasonic testing. Retest all repaired areas. Submit weld inspection reports.

Testing frequency: Provide the following types and number of tests:

<u>Test Type</u>	<u>Number of Tests</u>
Ultrasonic	50 percent of CJP Welds

3.7.2 Direct Tension Indicator Washers

3.7.2.1 Direct Tension Indicator Washer Compression

Test direct tension indicator washers in place to verify that they have been compressed sufficiently to provide the 0.015 inch gap, as required by ASTM F959/F959M. Submit direct tension indicator washer inspection reports.

3.7.3 High-Strength Bolts

3.7.3.1 Testing Bolt, Nut, and Washer Assemblies

Test a minimum of 3 bolt, nut, and washer assemblies from each mill certificate batch in a tension measuring device at the job site prior to the beginning of bolting start-up. Demonstrate that the bolts and nuts, when used together, can develop tension not less than the provisions specified in AISC 360, depending on bolt size and grade. The bolt tension must be developed by tightening the nut. A representative of the manufacturer or supplier must be present to ensure that the fasteners are properly used, and to demonstrate that the fastener assemblies supplied satisfy the specified requirements. Submit bolt testing reports.

3.7.3.2 Inspection

Inspection procedures must be in accordance with AISC 360. Confirm and report to the Contracting Officer that the materials meet the project specification and that they are properly stored. Confirm that the faying surfaces have been properly prepared before the connections are assembled. Observe the specified job site testing and calibration, and confirm that the procedure to be used provides the required tension. Monitor the work to ensure the testing procedures are routinely followed on joints that are specified to be fully tensioned.

3.7.3.3 Testing

The Government has the option to perform nondestructive tests on 5 percent

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of the installed bolts to verify compliance with pre-load bolt tension requirements. Provide the required access for the Government to perform the tests. The nondestructive testing will be done in-place using an ultrasonic measuring device or any other device capable of determining in-place pre-load bolt tension. The test locations must be selected by the Contracting Officer. If more than 10 percent of the bolts tested contain defects identified by testing, then all bolts used from the batch from which the tested bolts were taken, must be tested at the Contractor's expense. Retest new bolts after installation at the Contractor's expense.

3.7.4 Inspection and Testing of Steel Stud Welding

Perform verification inspection and testing of steel stud welding conforming to the requirements of AWS D1.1/D1.1M, Stud Welding Clause. The Contracting Officer will serve as the verification inspector. Bend test studs that do not show a full 360 degree weld flash or have been repaired by welding as required by AWS D1.1/D1.1M, Stud Welding Clause. Studs that crack under testing in the weld, base metal or shank will be rejected and replaced by the Contractor at no additional cost.

-- End of Section --

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SECTION 05 50 13

MISCELLANEOUS METAL FABRICATIONS

05/17, CHG 1: 08/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ALUMINUM ASSOCIATION (AA)

AA DAF45 (2003; Reaffirmed 2009) Designation System
for Aluminum Finishes

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 318 (2019; R 2022) Building Code Requirements
for Structural Concrete (ACI 318-19) and
Commentary (ACI 318R-19)

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 303 (2016) Code of Standard Practice for Steel
Buildings and Bridges

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.2.1 (2012; R 2021) Square, Hex, Heavy Hex, and
Askew Head Bolts and Hex, Heavy Hex, Hex
Flange, Lobed Head, and Lag Screws (Inch
Series)

ASME B18.2.2 (2022) Nuts for General Applications:
Machine Screw Nuts, and Hex, Square, Hex
Flange, and Coupling Nuts (Inch Series)

ASME B18.6.2 (2020) Square Head Set Screws and Slotted
Headless Set Screws (Inch Series)

ASME B18.6.3 (2024) Machine Screws, Tapping Screws, and
Metallic Drive Screws (Inch Series)

ASME B18.21.1 (2009; R 2016) Washers: Helical
Spring-Lock, Tooth Lock, and Plain Washers
(Inch Series)

ASME B18.21.2M (1999; R 2014) Lock Washers (Metric Series)

ASME B18.22M (1981; R 2017) Metric Plain Washers

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.3 (2020) Safety Requirements for
Powder-Actuated Fastening Systems American

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National Standard for Construction and
 Demolition Operations

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding
 Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon
 Structural Steel

ASTM A47/A47M (1999; R 2022; E 2022) Standard
 Specification for Ferritic Malleable Iron
 Castings

ASTM A53/A53M (2024) Standard Specification for Pipe,
 Steel, Black and Hot-Dipped, Zinc-Coated,
 Welded and Seamless

ASTM A123/A123M (2017) Standard Specification for Zinc
 (Hot-Dip Galvanized) Coatings on Iron and
 Steel Products

ASTM A153/A153M (2023) Standard Specification for Zinc
 Coating (Hot-Dip) on Iron and Steel
 Hardware

ASTM A307 (2021) Standard Specification for Carbon
 Steel Bolts, Studs, and Threaded Rod 60
 000 PSI Tensile Strength

ASTM A500/A500M (2023) Standard Specification for
 Cold-Formed Welded and Seamless Carbon
 Steel Structural Tubing in Rounds and
 Shapes

ASTM A653/A653M (2023) Standard Specification for Steel
 Sheet, Zinc-Coated (Galvanized) or
 Zinc-Iron Alloy-Coated (Galvannealed) by
 the Hot-Dip Process

ASTM A786/A786M (2015; R 2021) Standard Specification for
 Hot-Rolled Carbon, Low-Alloy,
 High-Strength Low-Alloy, and Alloy Steel
 Floor Plates

ASTM A924/A924M (2022a) Standard Specification for General
 Requirements for Steel Sheet,
 Metallic-Coated by the Hot-Dip Process

ASTM B26/B26M (2018; E 2018) Standard Specification for
 Aluminum-Alloy Sand Castings

ASTM B108/B108M (2019) Standard Specification for
 Aluminum-Alloy Permanent Mold Castings

ASTM B209 (2014) Standard Specification for Aluminum

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and Aluminum-Alloy Sheet and Plate

ASTM B209M	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate (Metric)
ASTM B221	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM B221M	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes (Metric)
ASTM C1513	(2018) Standard Specification for Steel Tapping Screws for Cold-Formed Steel Framing Connections
ASTM D1187/D1187M	(1997; R 2018) Standard Specification for Asphalt-Base Emulsions for Use as Protective Coatings for Metal
ASTM E488/E488M	(2022) Standard Test Methods for Strength of Anchors in Concrete Elements
ASTM F1554	(2020) Standard Specification for Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength

MASTER PAINTERS INSTITUTE (MPI)

MPI 79	(2016) Primer, Alkyd, Anti-Corrosive for Metal
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NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM MBG 531	(2017) Metal Bar Grating Manual
NAAMM MBG 532	(2019) Heavy Duty Metal Bar Grating Manual

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC SP 3	(2018) Power Tool Cleaning
SSPC SP 6/NACE No.3	(2007) Commercial Blast Cleaning

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Occupational Health (SOH) Requirements
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1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

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Bollards/Pipe Guards; G

Embedded Angles and Plates, Installation Drawings; G

Roof Hatches, Installation Drawings; G

SD-03 Product Data

Corner Guards

Floor Gratings; G

Roof Hatches; G

1.3 QUALIFICATION OF WELDERS

Qualify welders in accordance with AWS D1.1/D1.1M. Use procedures, materials, and equipment of the type required for the work.

1.4 DELIVERY, STORAGE, AND PROTECTION

Protect from corrosion, deformation, and other types of damage. Store items in an enclosed area free from contact with soil and weather. Remove and replace damaged items with new items.

1.5 MISCELLANEOUS REQUIREMENTS

1.5.1 Fabrication Drawings

Submit fabrication drawings showing layout(s), connections to structural system, and anchoring details as specified in AISC 303.

1.5.2 Installation Drawings

Submit templates, erection, and installation drawings indicating thickness, type, grade, class of metal, and dimensions. Show construction details, reinforcement, anchorage, and installation in relation to the building construction.

PART 2 PRODUCTS

2.1 RECYCLED CONTENT

Provide products with recycled content.

2.2 MATERIALS

Provide exposed fastenings of compatible materials (avoid contact of dissimilar metals). Coordinate color and finish with the material to which fastenings are applied.

2.2.1 Structural Carbon Steel

Provide in accordance with ASTM A36/A36M.

2.2.2 Structural Tubing

Provide in accordance with ASTM A500/A500M.

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2.2.3 Steel Pipe

Provide in accordance with ASTM A53/A53M, Type E or S, Grade B.

2.2.4 Fittings for Steel Pipe

Provide standard malleable iron fittings in accordance with ASTM A47/A47M.

2.2.5 Gratings

- a. Provide metal bar type grating in accordance with NAAMM MBG 531 and NAAMM MBG 532.

2.2.6 Floor Plates, Patterned

Provide floor plate in accordance with ASTM A786/A786M. Provide steel plate not less than 14 gage.

2.2.7 Anchor Bolts

Provide in accordance with ASTM F1554. Where exposed, provide anchor bolts of the same material, color, and finish as the metal to which they are applied.

2.2.7.1 Expansion Anchors Sleeve Anchors Adhesive Anchors

Provide bolt diameter and embedment as indicated. Design values listed are as tested in accordance with ASTM E488/E488M.

- a. Provide minimum ultimate pullout value as indicated by the bolt manufacturer. Calculate pullout capacity according to ACI 318.
- b. Provide minimum ultimate shear value as indicated by the bolt manufacturer. Calculate shear capacity according to ACI 318.

2.2.7.2 Lag Screws and Bolts

Provide in accordance with ASME B18.2.1, type and grade best suited for the purpose.

2.2.7.3 Toggle Bolts

Provide in accordance with ASME B18.2.1.

2.2.7.4 Bolts, Nuts, Studs and Rivets

Provide in accordance with ASME B18.2.2 or ASTM A307.

2.2.7.5 Powder Actuated Fasteners

Follow safety provisions in accordance with ASSP A10.3.

2.2.7.6 Screws

Provide in accordance with ASME B18.2.1, ASME B18.6.2, ASME B18.6.3 and ASTM C1513.

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2.2.7.7 Washers

Provide plain washers in accordance with ASME B18.22M, ASME B18.21.1. Provide beveled washers for American Standard beams and channels, square or rectangular, tapered in thickness, and smooth. Provide lock washers in accordance with ASME B18.21.2M, ASME B18.21.1.

2.2.8 Aluminum Alloy Products

Provide in accordance with ASTM B209M, ASTM B209 for sheet plate, ASTM B221M, ASTM B221M, ASTM B221 for extrusions and ASTM B26/B26M or ASTM B108/B108M for castings. Provide aluminum extrusions at least 1/8 inch thick and aluminum plate or sheet at least 0.050 inch thick.

2.3 FABRICATION FINISHES

2.3.1 Galvanizing

Hot-dip galvanize items specified to be zinc-coated, after fabrication where practicable. Provide galvanizing in accordance with ASTM A123/A123M, ASTM A153/A153M, ASTM A653/A653M or ASTM A924/A924M, Z275 G90.

2.3.2 Galvanize

Anchor bolts, grating fasteners, washers, and parts or devices necessary for proper installation, unless indicated otherwise.

2.3.3 Shop Cleaning and Painting

2.3.3.1 Surface Preparation

Blast clean surfaces in accordance with SSPC SP 6/NACE No.3. Surfaces that will be exposed in spaces above ceiling or in attic spaces, crawl spaces, furred spaces, and chases may be cleaned in accordance with SSPC SP 3 in lieu of being blast cleaned. Wash cleaned surfaces which become contaminated with rust, dirt, oil, grease, or other contaminants with solvents until thoroughly clean. Steel to be embedded in concrete must be free of dirt and grease prior to embed. Do not paint or galvanize bearing surfaces, including contact surfaces within slip critical joints. Shop coat these surfaces with rust prevention.

2.3.3.2 Pretreatment, Priming and Painting

Apply pre-treatment, primer, and paint in accordance with manufacturer's printed instructions.

2.3.4 Nonferrous Metal Surfaces

Protect by plating, anodic, or organic coatings.

2.3.5 Aluminum Surfaces

2.3.5.1 Surface Condition

Before finishes are applied, remove roll marks, scratches, rolled-in scratches, kinks, stains, pits, orange peel, die marks, structural streaks, and other defects which will affect uniform appearance of finished surfaces.

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2.3.5.2 Aluminum Finishes

Unexposed sheet, plate and extrusions may have mill finish as fabricated. Sandblast castings' finish, medium, AA DAF45. Unless otherwise specified, provide all other aluminum items with a standard mill finish. Provide a coating thickness not less than that specified for protective and decorative type finishes for items used in interior locations or architectural Class I type finish for items used in exterior locations. Provide in accordance with AA DAF45. Provide a polished satin finish on items to be anodized.

2.4 CORNER GUARDS

For jambs and sills of openings and edges of platforms provide steel shapes and plates anchored in masonry or concrete with welded steel straps or end-weld stud anchors. Form corner guards for use with glazed or ceramic tile finish on walls with 0.0625 inch thick corrosion-resisting steel with polished or satin finish, extend 5 feet above the top of cove base or to the top of the wainscot, whichever is less, and securely anchor to the supporting wall. Provide galvanized corner guards on exterior.

2.5 COVER PLATES AND FRAMES

Fabricate cover plates of 1/4 inch thick rolled steel weighing not more than 100 pounds per plate with a selected raised pattern nonslip top surface. Provide galvanized plate. Reinforce to sustain a live load of 100 pounds per square foot. Provide structural steel shapes and plates for frames, with bent steel bars or headed anchors welded to frame for anchoring to concrete. Miter and weld all corners. Butt joint straight runs. Allow for expansion on straight runs over 15 feet. Provide holes for lifting tools. Remove sharp edges and burrs from cover plates and exposed edges of frames. Weld all connections and grind top surface smooth. Weld bar stops every six inches. Provide 1/8 inch clearance at edges and between cover plates.

2.6 FLOOR GRATINGS

Design aluminum grating in accordance with NAAMM MBG 531 for bar type gratings, or in accordance with manufacturer's charts for plank grating.

- a. Design floor gratings to support a stress live load of 100 pounds per square foot for the spans indicated, with maximum deflection of $L/240$.
- b. In accordance with NAAMM MBG 531, NAAMM MBG 532, band edges of grating with bars of the same size as the bearing bars. Weld banding in accordance with the manufacturer's standard for trim unless otherwise indicated. Design tops of bearing bars, cross or intermediate bars to be in the same plane and to match grating finish.
- c. NAAMM MBG 531, NAAMM MBG 532, band ends of gratings with bars of the same or greater thickness than the metal used for grating. Weld banding bars to bearing bars or channels at least every fourth bar or channel and in every corner. Tack weld intervening bars or channels. Band diagonal or round cuts by welding bars of the same or greater thickness as the grating and in accordance with the manufacturer's standard for trim unless otherwise indicated.
- d. Provide slip resistant surface finishes.

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2.7 BOLLARDS/PIPE GUARDS

Provide galvanized prime coated extra strong weight steel pipe in accordance with ASTM A53/A53M. Anchor posts in concrete as indicated and fill solidly with concrete with minimum compressive strength of 2500 psi.

2.8 MISCELLANEOUS PLATES AND SHAPES

Provide items that do not form a part of the structural steel framework, such as lintels, sill angles, miscellaneous mountings and frames. Provide lintels fabricated from structural steel shapes over openings in masonry walls and partitions as indicated and as required to support wall loads over openings. Provide with connections and fasteners. Construct to have at least 8 in bearing on masonry at each end.

Provide angles and plates in accordance with ASTM A36/A36M, for embedment as indicated. Galvanize embedded items exposed to the elements in accordance with ASTM A123/A123M.

2.9 ROOF HATCHES (SCUTTLES)

Provide aluminum sheets not less than 14 gauge with 3 inch beaded flange, welded and ground at corners. Provide a minimum clear opening of 30 by 36 inches. Insulate cover and curb with one inch thick rigid fiberboard insulation, covered and protected by aluminum sheet of not less than 26 gage. Provide with 12 inches high curb, formed with 3 inch mounting flanges with holes for securing to the roof deck.

PART 3 EXECUTION

3.1 GENERAL INSTALLATION REQUIREMENTS

Install items at locations indicated in accordance with manufacturer's instructions. Verify all field dimensions prior to fabrication. Include materials and parts necessary to complete each assembly, whether indicated or not. Miss-alignment and miss-sizing of holes for fasteners is cause for rejection. Conceal fastenings where practicable. Joints exposed to weather must be watertight.

3.2 WORKMANSHIP

Provide miscellaneous metalwork that is true and accurate in shape, size, and profile. Make angles and lines continuous and straight. Make curves consistent, smooth and unfaceted. Provide continuous welding along the entire area of contact except where tack welding is permitted. Do not tack weld exposed connections. Unless otherwise indicated and approved, provide a smooth finish on exposed surfaces. Provide countersunk rivets where exposed. Provide coped and mitered corner joints aligned flush and without gaps.

3.3 ANCHORAGE, FASTENINGS, AND CONNECTIONS

Provide anchorage as necessary, whether indicated or not, for fastening miscellaneous metal items securely in place. Include slotted inserts, expansion shields, powder-driven fasteners, toggle bolts (when approved for concrete), through bolts for masonry, headed shear studs, machine and carriage bolts for steel, through bolts, lag bolts, and screws for wood. Do not use wood plugs. Provide non-ferrous attachments for non-ferrous metal. Provide exposed fastenings of compatible materials (avoid contact

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of dissimilar metals), that generally match in color and finish the surfaces to which they are applied. Conceal fastenings where practicable. Provide all fasteners flush with the surfaces they fasten, unless indicated otherwise.

3.4 BUILT-IN WORK

Where necessary and not otherwise indicated, form built-in metal work for anchorage with concrete or masonry. Provide built-in metal work in ample time for securing in place as the work progresses.

3.5 WELDING

Perform welding, welding inspection, and corrective welding in accordance with AWS D1.1/D1.1M. Use continuous welds on all exposed connections. Grind visible welds smooth in the finished installation. Provide welded headed shear studs in accordance with AWS D1.1/D1.1M, Clause 7, except as otherwise specified. Provide in accordance with the safety requirements of EM 385-1-1.

3.6 DISSIMILAR METALS

Where dissimilar metals are in contact, protect surfaces with a coating in accordance with MPI 79 to prevent galvanic or corrosive action. Where aluminum is in contact with concrete, plaster, mortar, masonry, wood, or absorptive materials subject to wetting, protect in accordance with ASTM D1187/D1187M, asphalt-base emulsion. Clean surfaces with metal shavings from installation at the end of each work day.

3.7 PREPARATION

3.7.1 Material Coatings and Surfaces

Remove rust preventive coating just prior to field erection, using a remover approved by the metal manufacturer. Surfaces, when assembled, must be free of rust, grease, dirt and other foreign matter.

3.7.2 Environmental Conditions

Do not clean or paint surfaces when damp or exposed to foggy or rainy weather, when metallic surface temperature is less than minus 5 degrees F above the dew point of the surrounding air, or when surface temperature is below 45 degrees F or over 95 degrees F, unless approved by the Contracting Officer. Metal surfaces to be painted must be dry for a minimum of 48 hours prior to the application of primer or paint.

3.8 ROOF HATCH (SCUTTLES)

Construction and accessories as follows:

- a. Provide insulated cover and curb with mounting flanges for securing to roof deck. Provide curbs with integral metal cap flashing of the same gage and metal as the curb, fully welded and ground at corners for weather tightness.
- b. Provide hatches completely assembled, with pintle hinges, compression spring operators enclosed in telescopic tubes, positive snap latches with turn handles on inside and outside, and neoprene draft seals. Provide fasteners for padlocking from the inside. Provide covers with

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automatic hold-open arms complete with grip handle to permit one hand release. Cover action must be smooth through its entire range of motion with an operating pressure of approximately 30 pounds.

3.9 INSTALLATION OF BOLLARDS/PIPE GUARDS

Set bollards/pipe guards vertically in concrete piers. Fill hollow cores with concrete having a compressive strength of 3000 psi.

3.10 INSTALLATION MISCELLANEOUS PLATES AND SHAPES

Provide lintels fabricated from structural steel shapes over openings in masonry walls and partitions as indicated and as required to support wall loads over openings. Provide with connections and fasteners. Construct to have at least 8 inches bearing on masonry at each end.

-- End of Section --

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SECTION 05 52 00

METAL RAILINGS

08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 325	(2017) Steel Construction Manual
AISC 326	(2009) Detailing for Steel Construction
AISC 360	(2016) Specification for Structural Steel Buildings

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-22	(2022; Supp 1 2023; Supp 2 2023) Minimum Design Loads and Associated Criteria for Buildings and Other Structures
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AMERICAN WELDING SOCIETY (AWS)

AWS A2.4	(2012) Standard Symbols for Welding, Brazing and Nondestructive Examination
AWS D1.1/D1.1M	(2020; Errata 1 2021) Structural Welding Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A53/A53M	(2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A123/A123M	(2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A153/A153M	(2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM A307	(2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
ASTM A500/A500M	(2023) Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes

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NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM AMP 521 (2001; R 2012) Pipe Railing Systems Manual

1.2 ADMINISTRATIVE REQUIREMENTS

1.2.1 Preinstallation Meetings

Within 30 days of contract award, submit fabrication drawings for the following items:

- a. Steel railings and handrails
- b. Anchorage and fastening systems

Submit manufacturer's catalog data, including two copies of manufacturers specifications, load tables, dimension diagrams, and anchor details for the following items:

- c. Steel railings and handrails
- d. Anchorage and fastening systems

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Fabrication Drawings; G,

SD-03 Product Data

Concrete Inserts; G,
Masonry Anchorage Devices; G,
Protective Coating; G,
Steel Railings and Handrails; G,
Anchorage and Fastening Systems; G,

SD-05 Design Data

Delegated Design Submittal; G

SD-08 Manufacturer's Instructions

Installation Instructions

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1.4 QUALITY CONTROL

1.5 QUALITY ASSURANCE

1.5.1 Fabrication Drawing Requirements

Submit fabrication drawings for approval prior to fabrication. Prepare in accordance with AISC 325, AISC 326, ASCE 7-22, and State Building Code where the project is located. Fabrication drawings will not be reproductions of contract drawings. Sign and seal fabrication drawings by a registered professional engineer. Include complete information for the fabrication and erection of the structure's components, including the location, type, and size of bolts, welds, member sizes and lengths, connection details, blocks, copes, and cuts. Use AWS A2.4 standard welding symbols. Clearly highlight any deviations from the details shown on the contract drawings highlighted on the fabrication drawings. Explain the reasons for any deviations from the contract drawings.

1.5.2 Delegated Connection Design

Design structural steel connection indicated in the contract documents per AISC 360. Submit Delegated Design Submittal signed and sealed by a registered professional engineer.

PART 2 PRODUCTS

2.1 FABRICATION

Preassemble items in the shop to the greatest extent possible. Disassemble units only to the extent necessary for shipping and handling. Clearly mark units for reassembly and coordinated installation.

For the fabrication of work exposed to view, use only materials that are smooth and free of surface blemishes, including pitting, seam marks, roller marks, rolled trade names, and roughness. Remove blemishes by grinding, or by welding and grinding, before cleaning, treating, and applying surface finishes, including zinc coatings.

Provide railing and handrail detail plans and elevations at not less than 1 inch to 1 foot. Provide details of sections and connections at not less than 3 inches to 1 foot. Also detail setting drawings, diagrams, templates for installation of anchorages, including concrete inserts, anchor bolts, and miscellaneous metal items having integral anchors.

Use materials of size and thicknesses indicated or, if not indicated, of the size and thickness necessary to produce adequate strength and durability in the finished product for its intended use. Work the materials to the dimensions indicated on approved detail drawings, using proven details of fabrication and support. Use the type of materials indicated or specified for the various components of work.

Form exposed work true to line and level, with accurate angles and surfaces and straight sharp edges. Ensure that all exposed edges are eased to a radius of approximately 1/32 inch. Bend metal corners to the smallest radius possible without causing grain separation or otherwise impairing the work.

Weld corners and seams continuously and in accordance with the recommendations of AWS D1.1/D1.1M. Grind exposed welds smooth and flush to

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match and blend with adjoining surfaces.

Form the exposed connections with hairline joints that are flush and smooth, using concealed fasteners wherever possible. Use exposed fasteners of the type indicated or, if not indicated, use countersunk Phillips flathead screws or bolts.

Provide anchorage of the type indicated and coordinated with the supporting structure. Fabricate anchoring devices and space as indicated and as required to provide adequate support for the intended use of the work.

Use hot-rolled steel bars for work fabricated from bar stock unless work is indicated or specified to be fabricated from cold-finished or cold-rolled stock.

2.1.1 Steel Handrails

Fabricate joint posts, rail, and corners by one of the following methods:

- a. Flush-type rail fittings of commercial standard, welded and ground smooth, with railing splice locks secured with 3/8 inch hexagonal-recessed-head setscrews.
- b. Mitered and welded joints made by fitting post to top rail and intermediate rail to post, mitering corners, groove-welding joints, and grinding smooth. Butt railing splices and reinforce them by a tight-fitting interior sleeve not less than 6 inches long.
- c. Railings may be bent at corners in lieu of jointing, provided that bends are made in suitable jigs and the pipe is not crushed.

2.1.2 Protective Coating

Provide hot-dipped galvanized steelwork as indicated in accordance with ASTM A123/A123M. Touch up abraded surfaces and cut ends of galvanized members with zinc-dust, zinc-oxide primer, or an approved galvanizing repair compound.

2.2 COMPONENTS

2.2.1 Performance Requirement

Delegated Design Submittal:

Engage a qualified professional engineer registered in the state where the project is located to design railings, including attachment to building construction. Provide signed and sealed drawings and calculations.

Structural Performance: Railing, including attachment to building construction, shall withstand the effects of gravity loads and the following loads and stresses within limits and under conditions indicated:

- a. Handrail and top rails of guards:
 1. Service deflection not to exceed $L/200$ for railing and $H/100$ for posts.
- b. Infill of Guards:
 1. Infill load and other loads need not be assumed to act

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concurrently.

- c. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes.

- 1. Temperature Change: 120 deg F.

2.2.2 Structural-Steel Tubing

Provide structural-steel tubing, hot-formed, welded or seamless, conforming to ASTM A500/A500M, Grade B, unless otherwise noted.

2.2.3 Fasteners

Provide galvanized zinc-coated fasteners in accordance with ASTM A153/A153M used for exterior applications or where built into exterior walls or floor systems. Select fasteners for the type, grade, and class required for the installation of steel stair items.

Provide standard hexagon-head bolts, conforming to ASTM A307, Grade A.

2.2.4 Steel Railings And Handrails

Design handrails to resist a concentrated load of 200 lb in any direction at any point of the top of the rail or 50 lb per foot applied horizontally to the top of the rail, whichever is more severe. NAAMM AMP 521, provide the same size rail and post. Provide pipe collars of the same material and finish as the handrail and posts.

2.2.4.1 Steel Handrails

Provide steel handrails, including inserts in concrete, steel pipe conforming to ASTM A53/A53M . Provide steel railings of 1 1/2 inch nominal size, hot-dip galvanized and shop-painted.

Provide kickplates between railing posts where indicated, and consisting of 1/8 inch steel flat bars not less than 6 inches high. Secure kickplates as indicated.

Provide galvanized railings, including pipe, fittings, brackets, fasteners, and other ferrous metal components.

PART 3 EXECUTION

3.1 PREPARATION

Adjust stair railings and handrails before securing in place in order to ensure proper matching at butting joints and correct alignment throughout their length. Space posts not more than 6 feet on center. Plumb posts in each direction. Secure posts and rail ends to building construction as follows:

Secure handrails to walls by means of wall brackets and wall return fitting at handrail ends. Provide brackets of malleable iron castings, with not less than 3 inch projection from the finished wall surface to the center of the pipe, drilled to receive one 3/8 inch bolt. Locate brackets not more than 60 inches on center. Provide wall return fittings of cast iron castings, flush type, with the same projection as that specified for wall brackets. Secure wall brackets and wall return fittings to building

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construction as follows:

- a. For hollow masonry and stud partition anchorage, use toggle bolts having square heads.

Install toe boards and brackets where indicated. Make splices, where required, at expansion joints. Install removable sections as indicated.

3.2 INSTALLATION

Submit manufacturer's installation instructions for the following products to be used in the fabrication of stair railing and hand rail work:

- a. Protective coating
- b. Masonry anchorage devices
- c. Steel railings and handrails
- d. Anchorage and fastening systems

Provide complete, detailed fabrication and installation drawings for all iron and steel hardware, and for all steel shapes, plates, bars, and strips used in accordance with the design specifications cited in this section.

3.2.1 Steel Handrail

Install handrail in pipe sleeves embedded in concrete and filled with nonshrink grout or quick-setting anchoring cement with anchorage covered with standard pipe collar pinned to post. Secure rail ends by steel pipe flanges

3.2.2 Touchup Painting

Immediately after installation, clean field welds, bolted connections, abraded areas of the shop paint, and exposed areas painted with the paint used for shop painting. Apply paint by brush or spray to provide a minimum dry-film thickness of 2 mils.

3.3 FIELD QUALITY CONTROL

3.3.1 Field Welding

Ensure that procedures of manual shielded metal arc welding, appearance and quality of welds made, and methods used in correcting welding work comply with AWS D1.1/D1.1M.

-- End of Section --

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SECTION 07 60 00

FLASHING AND SHEET METAL

08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A480/A480M	(2024) Standard Specification for General Requirements for Flat-Rolled Stainless and Heat-Resisting Steel Plate, Sheet, and Strip
ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A792/A792M	(2022) Standard Specification for Steel Sheet, 55% Aluminum-Zinc Alloy-Coated by the Hot-Dip Process
ASTM B32	(2020) Standard Specification for Solder Metal
ASTM B69	(2021) Standard Specification for Rolled Zinc
ASTM B209/B209M	(2021a) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM B221	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM D41/D41M	(2011; R 2016) Standard Specification for Asphalt Primer Used in Roofing, Dampproofing, and Waterproofing
ASTM D226/D226M	(2017) Standard Specification for Asphalt-Saturated Organic Felt Used in Roofing and Waterproofing
ASTM D1784	(2020) Standard Specification for Rigid Poly(Vinyl Chloride) (PVC) Compounds and Chlorinated Poly(Vinyl Chloride) (CPVC) Compounds
ASTM D2244	(2021) Standard Practice for Calculation of Color Tolerances and Color Differences from Instrumentally Measured Color

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Coordinates

ASTM D4214	(2007; R 2015) Standard Test Method for Evaluating the Degree of Chalking of Exterior Paint Films
ASTM D4586/D4586M	(2007; R 2018) Asphalt Roof Cement, Asbestos-Free
ASTM E2112	(2023) Standard Practice for Installation of Exterior Windows, Doors and Skylights

NATIONAL ROOFING CONTRACTORS ASSOCIATION (NRCA)

NRCA 0429	(2022) The NRCA Roofing Manual: Architectural Metal Flashing, Condensation and Air Leakage Control and Reroofing
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SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

SMACNA 1793	(2012) Architectural Sheet Metal Manual, 7th Edition
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SINGLE PLY ROOFING INDUSTRY (SPRI)

ANSI/SPRI RD-1	(2019) Performance Standard for Retrofit Drains
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1.2 GENERAL REQUIREMENTS

Finished sheet metal assemblies must form a weathertight enclosure without waves, warps, buckles, fastening stresses or distortion, while allowing for expansion and contraction without damage to the system. The sheet metal installer is responsible for cutting, fitting, drilling, and other operations in connection with sheet metal modifications required to accommodate the work of other trades. Coordinate installation of sheet metal items used in conjunction with roofing with roofing work to permit continuous, uninterrupted roofing operations.

1.2.1 General Material Requirements

All materials specified in this Section installed in conjunction with the roofing system must be provided by the roofing system manufacturer, or by a manufacturer approved by the roofing system manufacturer for use in the roofing system, and must form a part of the Warranty as required by the applicable roofing system Section.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Exposed Sheet Metal Coverings; G

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Expansion Joints; G

Flashing for Roof Drains; G

Base Flashing; G

Counterflashing; G

Flashing at Roof Penetrations and Equipment Supports; G

Reglets; G

Copings; G

SD-04 Samples

Finish Samples; G

SD-07 Certificates

Warranty on Finishes; G

SD-08 Manufacturer's Instructions

Instructions for Installation;

SD-10 Operation and Maintenance Data

Cleaning and Maintenance; G

1.4 MISCELLANEOUS REQUIREMENTS

1.4.1 Product Data

Indicate thicknesses, dimensions, fastenings, anchoring methods, expansion joints, and other provisions necessary for thermal expansion and contraction. Scaled manufacturer's catalog data may be submitted for factory fabricated items.

1.4.2 Finish Samples

Submit two color charts and two finish sample chips from manufacturer's standard color and finish options for each type of finish indicated.

1.4.3 Operation and Maintenance Data

Submit detailed instructions for installation and quality control during installation, cleaning and maintenance, for each type of assembly indicated.

1.5 DELIVERY, HANDLING, AND STORAGE

Package and protect materials during shipment. Uncrate and inspect materials for damage, dampness, and wet-storage stains upon delivery to the job site. Remove from the site and replace damaged materials that cannot be restored to like-new condition. Handle sheet metal items to avoid damage to surfaces, edges, and ends. Store materials in dry, weather-tight, ventilated areas until installation.

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PART 2 PRODUCTS

2.1 MATERIALS

Use any metal listed by NRCA 0429 or SMACNA 1793 for a particular item, unless otherwise indicated. Provide materials and configurations in accordance with NRCA 0429 or SMACNA 1793 for each material, while also meeting the minimum thickness requirements specified in this Section. Different items need not be of the same metal, except that contact between dissimilar metals must be avoided.

Provide sheet metal items in 8 to 10 foot lengths. Single pieces less than 8 feet long may be used to connect to factory-fabricated inside and outside corners, and at ends of runs. Factory fabricate corner pieces with minimum 12 inch legs. Provide accessories and other items essential to complete the sheet metal installation. Provide accessories made of the same or compatible materials as the items to which they are applied. Provide sheet metal items with mill finish unless specified otherwise.

2.1.1 Exposed Sheet Metal Items

Provide exposed sheet metal items of the same material. Consider the following as exposed sheet metal: cap, valley, steeped, base, and eave flashings and related accessories.

2.1.2 Steel Sheet, Zinc-Coated (Galvanized)

Provide in accordance with ASTM A653/A653M, a minimum of 24 gauge, and a minimum zinc coat weighting of 90 ounces per square feet.

2.1.3 Zinc Sheet and Strip

Provide in accordance with ASTM B69, Type I, a minimum of 0.024 inch thick.

2.1.4 Steel Sheet, Aluminum Zinc-Coated

Provide in accordance with ASTM A792/A792M, a minimum of 24 gauge.

2.1.5 Stainless Steel

Provide in accordance with ASTM A480/A480M, Type 302 or 304, 2D Finish, fully annealed, dead-soft temper, a minimum of 24 gauge.

2.1.6 Aluminum Alloy Sheet and Plate

Provide in accordance with ASTM B209/B209M anodized clear form alloy, and temper appropriate for use. Provide material not less than 0.032-in in thickness.

2.1.6.1 Alclad

When fabricated of aluminum, fabricate the following items with Alclad 3003, Alclad 3004, or Alclad 3005, clad on one side unless otherwise indicated.

a. Flashing

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2.1.7 Finishes

Provide exposed exterior sheet metal and aluminum with a baked on, factory applied coil fluoropolymer color coating or approved equal siliconized polyester coating. Dry film thickness of coatings must be 0.8 to 1.3 mils. Color to be selected from manufacturer's standard range of color choices. Field applications of color coatings are prohibited and will be rejected.

2.1.7.1 Warranty on Finishes

Provide a manufacturer's warranty to repair, or replace, sheet metal flashing and trim that shows evidence of deterioration of factory-applied finishes within 20 years from date of project substantial completion. Deterioration includes the following:

- a. Color fading more than 5 Delta E units when tested in accordance with ASTM D2244.
- b. Chalking in excess of a No. 8 rating when tested in accordance with ASTM D4214.
- c. Cracking, checking, peeling, or failure of paint to adhere to bare metal.

2.1.8 Aluminum Alloy, Extruded Bars, Rods, Shapes, and Tubes

ASTM B221.

2.1.9 Solder

Provide in accordance with ASTM B32, 95-5 tin-antimony.

2.1.10 Reglets

2.1.10.1 Polyvinyl Chloride Reglets

Provide in accordance with ASTM D1784, Type II, Grade 1, Class 14333-D, 0.075 inch minimum thickness.

2.1.10.2 Metal Reglets

Provide factory fabricated caulked type or friction type reglets with a minimum opening of 1/4 inch and a depth of 1-1/4 inch, as approved.

2.1.10.2.1 Caulked Reglets

Provide with rounded edges, temporary reinforcing cores, and accessories as required for securing to adjacent construction. Provide built-up mitered corner pieces for inside and outside corners.

2.1.10.2.2 Friction Reglets

Provide with flashing receiving slots not less than 5/8 inch deep, one inch jointing tongues, and upper and lower anchoring flanges installed at 24 inch maximum snap-lock type receiver.

2.1.11 Splash Pans

Provide splash pans where downspouts discharge onto roof surfaces and at

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locations indicated. Unless otherwise indicated, provide pans not less than 24 inches long by 18 inches wide with metal ribs across bottoms of pans. Provide sides of pans with vertical baffles not less than one inch high in the front, and 4 inches high in the back. Ground splash pans to be precast concrete.

2.1.12 Copings

Unless otherwise indicated, provide copings in copper sheets, 8 or 10 feet long, joined by a 3/4 inch locked and soldered seam.

2.1.13 Bituminous Plastic Cement

Provide in accordance with ASTM D4586/D4586M, Type I.

2.1.14 Roofing Felt

Provide in accordance with ASTM D226/D226M Type I.

2.1.15 Asphalt Primer

Provide in accordance with ASTM D41/D41M.

2.1.16 Fasteners

Use the same metal as, or a metal compatible with the item fastened. Confirm compatibility of fasteners and items to be fastened to avoid galvanic corrosion due to dissimilar materials.

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 Workmanship

Make lines and angles sharp and true. Free exposed surfaces from visible wave, warp, buckle, and tool marks. Fold back exposed edges neatly to form a 1/2 inch hem on the concealed side. Make sheet metal exposed to the weather watertight with provisions for expansion and contraction.

Make surfaces to receive sheet metal plumb and true, clean, even, smooth, dry, and free of defects and projections. For installation of items not shown in detail or not covered by specifications conform to the applicable requirements of SMACNA 1793, Architectural Sheet Metal Manual. Provide sheet metal flashing in the angles formed where roof decks abut walls, curbs, ventilators, pipes, or other vertical surfaces and wherever indicated and necessary to make the work watertight.

3.1.2 Cleats

Provide cleats for sheet metal 18 inches and over in width. Space cleats evenly not over 12 inches on center unless otherwise specified or indicated. Unless otherwise specified, provide cleats of 2 inches wide by 3 inches long and of the same material and thickness as the sheet metal being installed. Secure one end of the cleat with two nails and the cleat folded back over the nailheads. Lock the other end into the seam. Pre-tin cleats for soldered seams.

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3.1.3 Bolts, Rivets, and Screws

Install bolts, rivets, and screws where indicated or required. Provide compatible washers where required to protect surface of sheet metal and to provide a watertight connection. Provide mechanically formed joints in aluminum sheets 0.040 inches or less in thickness.

3.1.4 Seams

Straight and uniform in width and height with no solder showing on the face.

3.1.5 Mechanical Fastening

3.1.5.1 Mechanical Fastening of Aluminum

Use No. 12, aluminum alloy, sheet metal screws or other suitable aluminum alloy or stainless steel fasteners. Drive fasteners in holes made with a No. 26 drill in securing side laps, end laps, and flashings. Space fasteners 12 inches maximum on center. Where end lap fasteners are required to improve closure, locate the end lap fasteners not more than 2 inches from the end of the overlapping sheet.

3.1.6 Protection from Contact with Dissimilar Materials

3.1.6.1 Copper or Copper-bearing Alloys

Paint with heavy-bodied bituminous paint surfaces in contact with dissimilar metal, or separate the surfaces by means of moistureproof building felts.

3.1.6.2 Aluminum

Do not allow aluminum surfaces in direct contact with other metals except stainless steel, zinc, or zinc coating. Where aluminum contacts another metal, paint the dissimilar metal with a primer followed by two coats of aluminum paint. Where drainage from a dissimilar metal passes over aluminum, paint the dissimilar metal with a non-lead pigmented paint.

3.1.6.3 Metal Surfaces

Paint surfaces in contact with mortar, concrete, or other masonry materials with alkali-resistant coatings such as heavy-bodied bituminous paint.

3.1.6.4 Wood or Other Absorptive Materials

Paint surfaces that may become repeatedly wet and in contact with metal with two coats of aluminum paint or a coat of heavy-bodied bituminous paint.

3.1.7 Expansion and Contraction

Provide expansion and contraction joints at not more than 32 foot intervals for aluminum and at not more than 40 foot intervals for other metals. Provide an additional joint where the distance between the last expansion joint and the end of the continuous run is more than half the required interval. Space joints evenly. Join extruded aluminum gravel stops and fascia by expansion and contraction joints spaced not more than 12 feet apart.

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3.1.8 Base Flashing

Extend up vertical surfaces of the flashing not less than 8 inches and not less than 4 inches under the roof covering. Where finish wall coverings form a counterflashing, extend the vertical leg of the flashing up behind the applied wall covering not less than 6 inches. Overlap the flashing strips with the previously laid flashing not less than 3 inches. Fasten the strips at their upper edge to the deck. Horizontal flashing at vertical surfaces must extend vertically above the roof surface and fastened at their upper edge to the deck a minimum of 6 inches on center with a minimum of 2 inch lap of any surface. Solder end laps and provide for expansion and contraction. Extend the metal flashing over crickets at the up-slope side of curbs, and similar vertical surfaces extending through sloping roofs, the metal flashings. Extend the metal flashings onto the roof covering not less than 4.5 inches at the lower side of vertical surfaces extending through the roof decks. Install and fit the flashings so as to be completely weathertight. Provide factory-fabricated base flashing for interior and exterior corners. Do not use metal base flashing on built-up roofing.

3.1.9 Counterflashing

Except where indicated or specified otherwise, insert counterflashing in reglets located from 9 to 10 inches above roof decks, extend down vertical surfaces over upturned vertical leg of base flashings not less than 3 inches. Fold the exposed edges of counterflashings 1/2 inch. Where stepped counterflashings are required, they may be installed in short lengths a minimum 8 inches by 8 inches or may be of the preformed single piece type. Provide end laps in counterflashings not less than 3 inches and make it weathertight with plastic cement. Do not make lengths of metal counterflashings exceed 10 feet. Form flashings to the required shapes before installation. Factory form corners not less than 12 inches from the angle. Secure the flashings in the reglets with lead wedges and space not more than 18 inches apart; on chimneys and short runs, place wedges closer together. Fill caulked-type reglets or raked joints which receive counterflashing with caulking compound. Turn up the concealed edge of counterflashings built into masonry or concrete walls not less than 1/4 inch and extend not less than 2 inches into the walls. Install counterflashing to provide a spring action against base flashing. Utilize two piece counterflashing components to minimize damage of counterflashing and reglets during future reroofing activities.

3.1.10 Metal Reglets

Keep temporary cores in place during installation. Ensure factory fabricated caulked type or friction type, reglets have a minimum opening of 1/4 inch and a minimum depth of 1-1/4 inch, when installed.

3.1.10.1 Caulked Reglets

Wedge flashing in reglets with lead wedges every 18 inches, caulked full and solid with an approved compound.

3.1.10.2 Friction Reglets

Install flashing snap lock receivers at 24 inches on center maximum. When flashing has been inserted the full depth of the slot, caulk the slot, lock, and fill with sealant.

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3.1.11 Polyvinyl Chloride Reglets for Temporary Construction

Rigid polyvinyl chloride reglets may be provided in lieu of metal reglets for temporary construction.

3.1.12 Gravel Stops and fascia

Prefabricate in the shapes and sizes indicated and in lengths not less than 8 feet. Extend flange at least 4 inches onto roofing. Provide prefabricated, mitered corners internal and external corners. Install gravel stops and fascia after all plies of the roofing membrane have been applied, but before the flood coat of bitumen is applied. Prime roof flange of gravel stops and fascia on both sides with an asphalt primer. After primer has dried, set flange on roofing membrane and strip-in. Nail flange securely to wood nailer with large-head, barbed-shank roofing nails 1.5 inch long spaced not more than 3 inches on center, in two staggered rows.

3.1.12.1 Edge Strip

Hook the lower edge of fascia at least 3/4 inch over a continuous strip of the same material bent outward at an angle not more than 45 degrees to form a drip. Nail hook strip to a wood nailer at 6 inches maximum on center. Where fastening is made to concrete or masonry, use screws spaced 12 inches on center driven in expansion shields set in the concrete or masonry. Where horizontal wood nailers are slotted to provide for insulation venting, install strips to prevent obstruction of vent slots. Where necessary, install strips over 1/16 inch thick compatible spacer or washers.

3.1.12.2 Joints

Leave open the section ends of gravel stops and fascia 1/4 inch and backed with a formed flashing plate, mechanically fastened in place and lapping each section end a minimum of 4 inches set laps in plastic cement. Face nailing is not permitted. Install prefabricated aluminum gravel stops and fascia in accordance with the manufacturer's printed instructions and details.

3.1.13 Flashing for Roof Drains

Provide a 30 inches square sheet indicated. Taper insulation to drain from 24 inches out. Set flashing on finished felts in a full bed of asphalt roof cement, ASTM D4586/D4586M. Heavily coat the drain flashing ring with asphalt roof cement. Clamp the roof membrane, flashing sheet, and stripping felt in the drain clamping ring. Secure clamps so that felts and drain flashing are free of wrinkles and folds. Retrofit roof drains must conform to ANSI/SPRI RD-1.

3.1.14 Flashing at Wall Openings

Install pan flashing in the rough opening sill at all penetrations in the exterior wall assemblies, such as windows, louvers, storefronts and curtain walls. Pan sill flashing must have end dams at both jambs a minimum of 2 in high and a rear dam of 2 in high. Flashing must comply with ASTM E2112 and SMACNA 1793.

3.1.15 Expansion Joints

Provide expansion joints for roofs, walls, and floors as indicated. Provide.

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3.1.16 Flashing at Roof Penetrations and Equipment Supports

Provide metal flashing for all pipes, ducts, and conduits projecting through the roof surface and for equipment supports, guy wire anchors, and similar items supported by or attached to the roof deck.

3.1.17 Single Pipe Vents

Bend the top of sleeve over and extend down into the vent pipe a minimum of 2 inches. For long runs or long rises above the deck, where it is impractical to cover the vent pipe with lead, use a two-piece formed metal housing. Set metal housing with a metal sleeve having a 4 inches roof flange in bituminous plastic cement and nailed 3 inches on center. Extend sleeve a minimum of 8 inches above the roof deck and lapped a minimum of 3 inches by a metal hood secured to the vent pipe by a draw band. Seal the area of hood in contact with vent pipe with an approved sealant.

3.1.18 Copings

Provide copings as indicated in the Drawings.

3.1.19 Through Wall Flashing

Provide through wall flashing as required in the applicable wall system Sections.

3.2 PAINTING

Touch ups in the field may be applied only after metal substrates have been cleaned and pretreated in accordance with manufacturer's written instructions and products.

Field-paint dissimilar sheet metals in contact to separate and deter galvanic interactions.

3.2.1 Aluminum Surfaces

Clean with solvent and apply one coat of zinc-molybdate primer and one coat of aluminum paint.

3.3 CLEANING

Clean exposed sheet metal work at completion of installation. Remove grease and oil films, handling marks, contamination from steel wool, fittings and drilling debris, and scrub-clean. Free the exposed metal surfaces of dents, creases, waves, scratch marks, and solder or weld marks.

3.4 REPAIRS TO FINISH

Scratches, abrasions, and minor surface defects of finish may be repaired in accordance with the manufacturer's printed instructions and as approved. Repair damaged surfaces caused by scratches, blemishes, and variations of color and surface texture. Replace items which cannot be repaired.

-- End of Section --

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SECTION 07 84 00

FIRESTOPPING

05/10, CHG 1: 08/13

PART 1 GENERAL

1.1 SUMMARY

Furnish and install tested and listed firestopping systems, combination of materials, or devices to form an effective barrier against the spread of flame, smoke and gases, and maintain the integrity of fire resistance rated walls, partitions, floors, and ceiling-floor assemblies, including through-penetrations and construction joints and gaps.

- a. Through-penetrations include the annular space around pipes, tubes, conduit, wires, cables and vents.
- b. Construction joints include those used to accommodate expansion, contraction, wind, or seismic movement; do not allow firestopping material to interfere with the required movement of the joint.

Gaps requiring firestopping include gaps between the curtain wall and the floor slab and between the top of the fire-rated walls and the roof or floor deck above and at the intersection of shaft assemblies and adjoining fire resistance rated assemblies.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E84	(2023) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E119	(2022) Standard Test Methods for Fire Tests of Building Construction and Materials
ASTM E814	(2013a; R 2017) Standard Test Method for Fire Tests of Penetration Firestop Systems
ASTM E1399/E1399M	(1997; R 2022) Standard Test Method for Cyclic Movement and Measuring the Minimum and Maximum Joint Widths of Architectural Joint Systems
ASTM E1966	(2015; R 2019) Standard Test Method for Fire-Resistive Joint Systems
ASTM E2307	(2020) Standard Test Method for Determining Fire Resistance of Perimeter Fire Barrier Systems Using Intermediate-Scale, Multi-story Test

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Apparatus

FM GLOBAL (FM)

FM 4991 (2013) Approval of Firestop Contractors

FM APP GUIDE (updated on-line) Approval Guide
<http://www.approvalguide.com/>

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC (2021) International Building Code

UNDERWRITERS LABORATORIES (UL)

UL 723 (2020) UL Standard for Safety Test for
 Surface Burning Characteristics of
 Building Materials

UL 1479 (2015; Reprint May 2021) Fire Tests of
 Through-Penetration Firestops

UL 2079 (2015; Reprint Jul 2020) Tests for Fire
 Resistance of Building Joint Systems

UL Fire Resistance (2014) Fire Resistance Directory

1.3 SEQUENCING

Coordinate the specified work with other trades. Apply firestopping materials, at penetrations of pipes and ducts, prior to insulating, unless insulation meets requirements specified for firestopping. Apply firestopping materials. at building joints and construction gaps, prior to completion of enclosing walls or assemblies. Locate cast-in-place firestop devices and install in place before concrete placement. Install pipe, conduit or cable bundles through cast-in-place device after concrete placement but before area is concealed or made inaccessible. Firestop material must be inspected and approved prior to final completion and enclosing of any assemblies that may conceal installed firestop.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Firestopping System

SD-03 Product Data

Firestopping Materials

SD-06 Test Reports

Inspection

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SD-07 Certificates

Inspector Qualifications

Firestopping Materials

Installer Qualifications;

1.5 QUALITY ASSURANCE

1.5.1 Installer

Engage an experienced Installer who is:

- a. FM Research approved in accordance with FM 4991, operating as a UL Certified Firestop Contractor, or
- b. Certified, licensed, or otherwise qualified by the firestopping manufacturer as having the necessary staff, training, and a minimum of 3 years experience in the installation of manufacturer's products in accordance with specified requirements. Submit documentation of this experience. A manufacturer's willingness to sell its firestopping products to the Contractor or to an installer engaged by the Contractor does not in itself confer installer qualifications on the buyer. The Installer must be a trained representative of the manufacturer (not distributor or agent) in the proper selection and installation procedures. Obtain and submit installer's written certification of training, and retain proof of certification for duration of firestop installation.

1.5.2 Inspector Qualifications

The inspector must have a minimum of two years experience in construction field inspections of firestopping systems, products, and assemblies. The inspector must be completely independent of, and divested from, the installer, the manufacturer, and the supplier of any material or item being inspected. The inspector must not be a competitor of the installer, the contractor, the manufacturer, or supplier of any material or item being inspected. Include in the qualifications submittal a notarized statement assuring compliance with the requirements stated herein.

1.6 DELIVERY, STORAGE, AND HANDLING

Deliver materials in the original unopened packages or containers showing name of the manufacturer and the brand name. Store materials off the ground, protected from damage and exposure to elements and temperatures in accordance with manufacturer requirements. Remove damaged or deteriorated materials from the site. Use materials within their indicated shelf life.

PART 2 PRODUCTS

2.1 FIRESTOPPING SYSTEM

Submit detail drawings including manufacturer's descriptive data, typical details conforming to UL Fire Resistance or other details certified by another nationally recognized testing laboratory, installation instructions or UL listing details for a firestopping assembly in lieu of fire-test data or report. For those firestop applications for which no UL tested system

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is available through a manufacturer, submit a manufacturer's engineering judgment, derived from similar UL system designs or other tests for review and approval prior to installation. Submittal must indicate the firestopping material to be provided for each type of application. When more than a total of 5 penetrations and/or construction joints are to receive firestopping, provide drawings that indicate location, "F" "T" and "L" ratings, and type of application.

Also, submit a written report indicating locations of and types of penetrations and types of firestopping used at each location; record type by UL list printed numbers.

2.2 FIRESTOPPING MATERIALS

Provide firestopping materials, supplied from a single domestic manufacturer, consisting of commercially manufactured, asbestos-free, nontoxic products FM APP GUIDE approved, or UL listed, for use with applicable construction and penetrating items, complying with the following minimum requirements:

2.2.1 Fire Hazard Classification

Provide material that has a flame spread of 25 or less, and a smoke developed rating of 50 or less, when tested in accordance with ASTM E84 or UL 723. Provide an approved firestopping material as listed in UL Fire Resistance or by a nationally recognized testing laboratory.

2.2.2 Toxicity

Provide material that is nontoxic and carcinogen free to humans at all stages of application or during fire conditions and does not contain hazardous chemicals or require harmful chemicals to clean material or equipment.

2.2.3 Fire Resistance Rating

Firestop systems must be UL Fire Resistance listed or FM APP GUIDE approved with "F" rating at least equal to fire-rating of fire wall or floor in which penetrated openings are to be protected. Where required, firestop systems must also have "T" rating at least equal to the fire-rated floor in which the openings are to be protected.

2.2.3.1 Through-Penetrations

Firestopping materials for through-penetrations, as described in paragraph SUMMARY, must provide "F", "T" and "L" fire resistance ratings in accordance with ASTM E814 or UL 1479. Provide fire resistance ratings as follows:

2.2.3.1.1 Penetrations of Fire Resistance Rated Walls and Partitions

F Rating = Rating of wall or partition being penetrated.

2.2.3.1.2 Penetrations of Fire Resistance Rated Floors, Floor-Ceiling Assemblies and the Ceiling Membrane of Roof-Ceiling Assemblies

F Rating Where the penetrating item is outside of a wall cavity the F rating must be equal to the fire resistance rating of the floor penetrated, and the T rating must be in accordance with the requirements of ICC IBC.

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2.2.3.2 Construction Joints and Gaps

Fire resistance ratings of construction joints, as described in paragraph SUMMARY, and gaps such as those between floor slabs and curtain walls must be Provide construction joints and gaps with firestopping materials and systems that have been tested in accordance with ASTM E119, ASTM E1966 or UL 2079 to meet the required fire resistance rating. Provide curtain wall joints with firestopping materials and systems that have been tested in accordance with ASTM E2307 to meet the required fire resistance rating. Systems installed at construction joints must meet the cycling requirements of ASTM E1399/E1399M or UL 2079. Provide a minimum class II movement capability for all joints at the intersection of the top of a fire resistance rated wall and the underside of a fire-rated floor, floor ceiling, or roof ceiling assembly.

2.2.4 Material Certification

Submit certificates attesting that firestopping material complies with the specified requirements. Provide certification of compliance with UL 1479 for all intumescent firestop materials used in through penetration systems.

PART 3 EXECUTION

3.1 PREPARATION

Areas to receive firestopping must be free of dirt, grease, oil, or loose materials which may affect the fitting or fire resistance of the firestopping system. For cast-in-place firestop devices, formwork or metal deck to receive device prior to concrete placement must be sound and capable of supporting device. Prepare surfaces as recommended by the manufacturer.

3.2 INSTALLATION

Completely fill void spaces with firestopping material regardless of geometric configuration, subject to tolerance established by the manufacturer. Firestopping systems for filling floor voids 4 inches or more in any direction must be capable of supporting the same load as the floor is designed to support or be protected by a permanent barrier to prevent loading or traffic in the firestopped area. Install firestopping in accordance with manufacturer's written instructions. Provide tested and listed firestop systems in the following locations, except in floor slabs on grade:

- a. Penetrations of duct, conduit, tubing, cable and pipe through floors and through fire-resistance rated walls, partitions, and ceiling-floor assemblies.
- b. Penetrations of vertical shafts such as pipe chases, elevator shafts, and utility chutes.
- c. Gaps at the intersection of floor slabs and curtain walls, including inside of hollow curtain walls at the floor slab.
- d. Gaps at perimeter of fire-resistance rated walls and partitions, such as between the top of the walls and the bottom of roof decks.
- e. Construction joints in floors and fire rated walls and partitions.

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- f. Other locations where required to maintain fire resistance rating of the construction.

3.2.1 Insulated Pipes and Ducts

Cut and remove thermal insulation where pipes or ducts pass through firestopping, unless insulation meets requirements specified for firestopping. Replace thermal insulation with a material having equal thermal insulating and firestopping characteristics.

3.2.2 Fire Dampers

Install and firestop fire dampers in accordance with Division 23. Firestop installed with fire damper must be tested and approved for use in fire damper system. Firestop installed with fire damper must be tested and approved for use in fire damper system.

3.2.3 Data and Communication Cabling

Seal cabling for data and communication applications with re-enterable firestopping products and devices.

3.2.3.1 Re-Enterable Devices

Provide firestopping devices that are pre-manufactured modular devices, containing built-in self-sealing intumescent inserts. Allow for cable moves, additions or changes without the need to remove or replace any firestop materials. Devices must be capable of maintaining the fire resistance rating of the penetrated membrane at 0 percent to 100 percent visual fill of penetrants; while maintaining "L" rating of <10 cfm/sf at 0 percent to 100 percent visual fill.

3.2.3.2 Re-Sealable Products

Provide firestopping pre-manufactured modular products, containing self-sealing intumescent inserts. Allow for cable moves, additions or changes. Provide devices capable of maintaining the fire resistance rating of the penetrated membrane at 0 percent to 100 percent visual fill of penetrants.

3.3 INSPECTION

For all projects, do not cover or enclose the firestopped areas until inspection is complete and approved by the Contracting Officer. The inspector must inspect the applications initially to ensure adequate preparations (clean surfaces suitable for application, etc.) and periodically during the work to assure that the completed work has been accomplished according to the manufacturer's written instructions and the specified requirements. Submit written reports indicating locations of and types of penetrations and types of firestopping used at each location; record type by UL listed printed numbers.

Install firestopping manufacturer's label(s) at each firestopped penetration. Labels shall be filled out and include, at the minimum, the UL listed number of the firestopping system, date installed, installer's name (the person doing the installation), and the company's name for whom the installer works.

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3.3.1 Inspection Standards

Inspect all firestopping and document inspection results to be submitted.

3.3.2 Inspection Reports

Submit inspection report stating that firestopping work has been inspected and found to be applied according to the manufacturer's recommendations and the specified requirements.

-- End of Section --

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SECTION 07 92 00

JOINT SEALANTS

08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C509	(2006; R 2021) Standard Specification for Elastomeric Cellular Preformed Gasket and Sealing Material
ASTM C734	(2015; R 2019) Standard Test Method for Low-Temperature Flexibility of Latex Sealants After Artificial Weathering
ASTM C834	(2017; R 2023) Standard Specification for Latex Sealants
ASTM C919	(2022) Standard Practice for Use of Sealants in Acoustical Applications
ASTM C920	(2018) Standard Specification for Elastomeric Joint Sealants
ASTM C1193	(2016) Standard Guide for Use of Joint Sealants
ASTM C1311	(2022) Standard Specification for Solvent Release Sealants
ASTM C1521	(2019; R 2020) Standard Practice for Evaluating Adhesion of Installed Weatherproofing Sealant Joints
ASTM D217	(2021a) Standard Test Methods for Cone Penetration of Lubricating Grease
ASTM D1056	(2020) Standard Specification for Flexible Cellular Materials - Sponge or Expanded Rubber
ASTM D1667	(2022) Standard Specification for Flexible Cellular Materials - Poly (Vinyl Chloride) Foam (Closed-Cell)
ASTM D2452	(2015; R 2019) Standard Test Method for Extrudability of Oil- and Resin-Base Caulking Compounds
ASTM D2453	(2015; R 2020; E 2020) Standard Test

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Method for Shrinkage and Tenacity of Oil-
 and Resin-Base Caulking Compounds

ASTM E84

(2023) Standard Test Method for Surface
 Burning Characteristics of Building
 Materials

CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

CDPH SECTION 01350

(2017; Version 1.2) Standard Method for
 the Testing and Evaluation of Volatile
 Organic Chemical Emissions from Indoor
 Sources using Environmental Chambers

SCIENTIFIC CERTIFICATION SYSTEMS (SCS)

SCS

SCS Global Services (SCS) Indoor Advantage

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168

(2017) Adhesive and Sealant Applications

UNDERWRITERS LABORATORIES (UL)

UL 2818

(2022) GREENGUARD Certification Program
 For Chemical Emissions For Building
 Materials, Finishes And Furnishings

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Sealants;

Primers;

Bond Breakers;

Backstops;

Caulking;

Cleaning Solvents;

SD-06 Test Reports

Field Adhesion;

SD-07 Certificates

Indoor Air Quality For Interior Sealants

Indoor Air Quality For Interior Floor Joint Sealants

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Indoor Air Quality For Interior Acoustical Sealants

Indoor Air Quality For Interior Caulking

1.3 PRODUCT DATA

Include storage requirements, shelf life, curing time, instructions for mixing and application, and accessories. Provide manufacturer's Safety Data Sheets (SDS) for each solvent, primer and sealant material proposed.

1.4 CERTIFICATIONS

1.4.1 Indoor Air Quality Certifications

Submit required indoor air quality certifications in one submittal package.

1.4.1.1 Adhesives and Sealants

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party program that products meet the requirements of this Section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein.

1.5 ENVIRONMENTAL CONDITIONS

Apply sealant when the ambient temperature is between 40 and 90 degrees F.

1.6 DELIVERY AND STORAGE

Deliver materials to the jobsite in unopened manufacturers' sealed shipping containers, with brand name, date of manufacture, and material designation clearly marked thereon. Label elastomeric sealant containers to identify type, class, grade, and use. Handle and store materials in accordance with manufacturer's printed instructions. Prevent exposure to foreign materials or subjection to sustained temperatures exceeding 90 degrees F or lower than 0 degrees F. Keep materials and containers closed and separated from absorptive materials such as wood and insulation.

1.7 QUALITY ASSURANCE

1.7.1 Compatibility with Substrate

Verify that each sealant is compatible for use with each joint substrate in accordance with sealant manufacturer's printed recommendations for each application.

1.7.2 Joint Tolerance

Provide joint tolerances in accordance with manufacturer's printed instructions.

1.7.3 Mock-Up

Provide a mock-up of each type of sealant using materials, colors, and techniques approved for use on the project. Approved mock-ups may be

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incorporated into the Work.

1.7.4 Adhesion

Provide in accordance with ASTM C1193 or ASTM C1521.

PART 2 PRODUCTS

2.1 SEALANTS

Provide sealant products that have been tested, found suitable, and documented as such by the manufacturer for the particular substrates to which they will be applied.

In areas with ambient temperatures that exceed 110 degrees F, do not use polybutene, bituminous, acrylic-latex, polyvinyl acetate latex sealants, polychloroprene (neoprene), polyvinyl chloride (PVC), and polyurethane foams, and neoprene, PVC, and styrene butadiene rubber extruded seals and closure strips due to these materials having maximum recommended surface temperature ranges from 130 to 180 degrees F.

2.1.1 Interior Sealants

Provide ASTM C834 ASTM C920, Type S or M, Grade NS, Class 12.5, Use NT. Provide sealant products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide certification or validation of indoor air quality for interior sealants. Location(s) and color(s) of sealant for the following (Note, color "as selected" refers to manufacturer's full range of color options):

LOCATION	COLOR
a. Small voids between walls or partitions and adjacent lockers, casework, shelving, door frames, built-in or surface mounted equipment and fixtures, and similar items.	Match adjacent surface color.
b. Perimeter of frames at doors, windows, and access panels which adjoin exposed interior concrete and masonry surfaces.	Match adjacent surface color.
c. Joints of interior masonry walls and partitions which adjoin columns, pilasters, concrete walls, and exterior walls unless otherwise detailed.	N.A.
d. Joints between edge members for acoustical tile and adjoining vertical surfaces.	N.A.
e. Interior locations, not otherwise indicated or specified, where small voids exist between materials specified to be painted.	White - paintable.

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LOCATION	COLOR
f. Joints between bathtubs and ceramic tile; joints between shower receptors and ceramic tile; joints formed where non-planar tile surfaces meet.	Match grout color.
g. Joints formed between tile floors and tile base cove; joints between tile and dissimilar materials; joints occurring where substrates change.	Match grout color.
h. Behind escutcheon plates at valve pipe penetrations and showerheads in showers.	White - paintable.

2.1.2 Exterior Sealants

For joints in vertical surfaces, provide ASTM C920, Type S or M, Grade NS, Class 25, Use NT. For joints in horizontal surfaces, provide ASTM C920, Type S or M, Grade P, Class 25, Use T. Provide location(s) and color(s) of sealant as follows (Note, color "as selected" refers to manufacturer's full range of color options):

LOCATION	COLOR
a. Joints and recesses formed where frames and subsills of windows, doors, louvers, and vents adjoin masonry, concrete, or metal frames. Use sealant at both exterior and interior surfaces of exterior wall penetrations.	Match adjacent surface color
b. Joints between new and existing exterior masonry walls.	N.A.
c. Masonry joints where shelf angles occur.	Match adjacent surface color.
d. Joints in wash surfaces of stonework.	N.A.
e. Expansion and control joints.	Match grout color in masonry. Match adjacent surface color.
f. Interior face of expansion joints in exterior concrete or masonry walls where metal expansion joint covers are not required.	Match adjacent surface color.
g. Voids where items pass through exterior walls.	Match adjacent surface color.

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LOCATION	COLOR
h. Metal reglets, where flashing is inserted into masonry joints, and where flashing is penetrated by coping dowels.	Match grout color.
i. Metal-to-metal joints where sealant is indicated or specified.	Match metal color.
j. Joints between ends of gravel stops, fascia, copings, and adjacent walls.	Match adjacent wall surface color.

2.1.1.3 Floor Joint Sealants

ASTM C920, Type S or M, Grade P, Class 25, Use T. Provide sealant products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide certification or validation of indoor air quality for interior floor joint sealants. Provide location(s) and color(s) of sealant as follows (Note, color "as selected" refers to manufacturer's full range of color options):

LOCATION	COLOR
a. Seats of metal thresholds for exterior doors.	Gray
b. Control and expansion joints in floors, slabs, ceramic tile, and walkways.	Match adjacent surface color.

2.1.1.4 Acoustical Sealants

Rubber or polymer based acoustical sealant in accordance with ASTM C919 to have a flame spread of 25 or less and a smoke developed rating of 50 or less when tested in accordance with ASTM E84. Provide non-staining acoustical sealant with a consistency of 250 to 310 when tested in accordance with ASTM D217. Acoustical sealant must remain flexible and adhesive after 500 hours of accelerated weathering as specified in ASTM C734. Provide sealant products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide certification or validation of indoor air quality for interior acoustical sealants.

2.1.1.5 Preformed Sealants

Provide preformed sealants of polybutylene or isoprene-butylene based pressure sensitive weather resistant tape or bead sealants capable of sealing out moisture, air and dust when installed as recommended by the manufacturer. At temperatures from minus 30 to plus 160 degrees F, sealants must be non-bleeding and have no loss of adhesion.

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2.1.5.1 Tape

Tape sealant: Provide cross section dimensions of required width.

2.1.5.2 Bead

Bead sealant: Provide cross section dimensions of required width.

2.1.5.3 Foam Strip

Provide foam strip of polyurethane foam with cross section dimensions of required width. Provide foam strip capable of sealing out moisture, air, and dust when installed and compressed in accordance with manufacturer's printed instructions. Service temperature must be minus 40 to plus 275 degrees F. Install untreated strips with adhesive to hold them in place. Do not allow adhesive to stain or bleed onto adjacent finishes. Saturate treated strips with butylene waterproofing or impregnate with asphalt.

2.2 PRIMERS

Non-staining, quick drying type and consistency as recommended by the sealant manufacturer for the particular application. Provide primers for interior applications that meet the indoor air quality requirements of the paragraph INTERIOR SEALANTS above.

2.3 BOND BREAKERS

Type and consistency as recommended by the sealant manufacturer to prevent adhesion of the sealant to the backing or to the bottom of the joint. Provide bond breakers for interior applications that meet the indoor air quality requirements of the paragraph INTERIOR SEALANTS above.

2.4 BACKSTOPS

Provide glass fiber roving, neoprene, butyl, polyurethane, or polyethylene foams free from oil or other staining elements as recommended by sealant manufacturer. Provide 25 to 33 percent oversized backing for closed cell and 40 to 50 percent oversized backing for open cell material, unless otherwise indicated. Provide backstop material that is compatible with sealant. Do not use oakum or other types of absorptive materials as backstops.

2.4.1 Rubber

Provide in accordance with ASTM D1056, Type 1, open cell, or Type 2, closed cell, Class A typical and Class B where petroleum oil or fuel resistance is required, Grade (as recommended by Manufacturer for intended use), round cross section for cellular rubber sponge backing.

2.4.2 PVC

Provide in accordance with ASTM D1667, Grade VO 12, open-cell foam, round cross section for polyvinyl chloride (PVC) backing.

2.4.3 Synthetic Rubber

Provide in accordance with ASTM C509, Option I, Type I preformed rods or tubes for synthetic rubber backing.

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2.4.4 Neoprene

Provide in accordance with ASTM D1056, closed cell expanded neoprene cord Type 2, Class C, Grade 2C2 open cell neoprene sponge Type 1, Class C, Grade 1C3 for neoprene backing.

2.4.5 Butyl Rubber Based

Provide in accordance with ASTM C1311, from a single component, with solvent release. color as selected from manufacturer's full range of color choices.

2.4.6 Silicone Rubber Base

Provide in accordance with ASTM C920, from a single component, with solvent release, Non-sag, Type 'S', Grade 'NS', Class 25. Color as selected from manufacturer's full range of color choices.

2.5 CAULKING

For interior use and only where there is little or no anticipated joint movement. Provide in accordance with ASTM D2452 and ASTM D2453, Type 'S', for oil and resin-based caulking. Provide products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide certification or validation of indoor air quality for interior caulking.

2.6 CLEANING SOLVENTS

Provide type(s) recommended by the sealant manufacturer and in accordance with environmental requirements herein. Protect adjacent aluminum and bronze surfaces from solvents. Provide solvents for interior applications that meet the indoor air quality requirements of the paragraph INTERIOR SEALANTS above.

PART 3 EXECUTION

3.1 FIELD QUALITY CONTROL

Perform a field adhesion test in accordance with manufacturer's instructions and ASTM C1193, Method A or ASTM C1521, Method A, Tail Procedure. Remove sealants that fail adhesion testing; clean substrates, reapply sealants, and re-test. Test sealants adjacent to failed sealants. Submit field adhesion test report indicating tests, locations, dates, results, and remedial actions taken.

3.2 SURFACE PREPARATION

Prepare surfaces according to manufacturer's printed installation instructions. Clean surfaces from dirt, frost, moisture, grease, oil, wax, lacquer, paint, or other foreign matter that would destroy or impair adhesion. Remove oil and grease with solvent; thoroughly remove solvents prior to sealant installation. Wipe surfaces dry with clean cloths. When resealing an existing joint, remove existing caulk or sealant prior to applying new sealant. For surface types not listed below, provide in accordance with sealant manufacturer's printed instructions for each specific surface.

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3.2.1 Steel Surfaces

Remove loose mill scale by sandblasting or, if sandblasting is impractical or would damage finished work, scraping and wire brushing. Remove protective coatings by sandblasting or using a residue free solvent. Remove resulting debris and solvent residue prior to sealant installation.

3.2.2 Aluminum or Bronze Surfaces

Remove temporary protective coatings from surfaces that will be in contact with sealant. When masking tape is used as a protective coating, remove tape and residual adhesive prior to sealant application. For removing protective coatings and final cleaning, use non-staining solvents recommended by the manufacturer of the item(s) containing aluminum or bronze surfaces.

3.2.3 Concrete and Masonry Surfaces

Where surfaces have been treated with curing compounds, oil, or other such materials, remove materials by sandblasting or wire brushing. Remove laitance, efflorescence and loose mortar from the joint cavity. Remove resulting debris prior to sealant installation.

3.2.4 Wood Surfaces

Ensure wood surfaces that will be in contact with sealants are free of splinters, sawdust and other loose particles.

3.2.5 Removing Existing Hazardous Sealants

For sealants applied prior to 1979, or that have been tested and found to contain polychlorinated biphenyls (PCBs), remove and dispose of these sealants in accordance with Division 02 Existing Conditions.

3.3 SEALANT PREPARATION

Do not add liquids, solvents, or powders to sealants. Mix multicomponent elastomeric sealants in accordance with manufacturer's printed instructions.

3.4 APPLICATION

3.4.1 Joint Width-To-Depth Ratios

Acceptable Ratios:

<u>JOINT WIDTH</u>	<u>JOINT DEPTH</u>	
	Minimum	Maximum
For metal, glass, or other nonporous surfaces:		
1/4 inch (minimum)	1/4 inch	1/4 inch
over 1/4 inch	1/2 of width	Equal to width

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<u>JOINT WIDTH</u>	<u>JOINT DEPTH</u>	
	Minimum	Maximum
For wood, concrete, masonry, or stone:		
1/4 inch (minimum)	1/4 inch	1/4 inch
over 1/4 inch to 1/2 inch	1/4 inch	Equal to width
over 1/2 inch to 1 inch	1/2 inch	5/8 inch
Over 1 inch	prohibited	

Unacceptable Ratios: Where joints of acceptable width-to-depth ratios have not been provided, clean out joints to acceptable depths and grind or cut to acceptable widths without damage to the adjoining work. Grinding is prohibited at metal surfaces.

3.4.2 Unacceptable Sealant Use

Do not install sealants in lieu of other required building enclosure weatherproofing components such as flashing, drainage components, and joint closure accessories, or to close gaps between walls, floors, roofs, windows, and doors, that exceed acceptable installation tolerances. Remove sealants that have been used in an unacceptable manner and correct building enclosure deficiencies to comply with contract documents requirements.

3.4.3 Masking Tape

Place masking tape on the finished surface on one or both sides of joint cavities to protect adjacent finished surfaces from primer or sealant smears. Remove masking tape within 10 minutes of joint filling and tooling.

3.4.4 Backstops

Provide backstops dry and free of tears or holes. Tightly pack the back or bottom of joint cavities with backstop material to provide joints in specified depths. Provide backstops where indicated and where backstops are not indicated but joint cavities exceed the acceptable maximum depths specified in JOINT WIDTH-TO-DEPTH RATIOS Table.

3.4.5 Primer

Clean out loose particles from joints immediately prior to application of. Apply primer to joints in concrete masonry units, wood, and other porous surfaces in accordance with sealant manufacturer's printed instructions. Do not apply primer to exposed finished surfaces.

3.4.6 Bond Breaker

Provide bond breakers to surfaces not intended to bond in accordance with, sealant manufacturer's printed instructions for each type of surface and sealant combination specified.

3.4.7 Sealants

Provide sealants compatible with the material(s) to which they are applied.

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Do not use a sealant that has exceeded its shelf life or has jelled and cannot be discharged in a continuous flow from the sealant gun. Apply sealants in accordance with the manufacturer's printed instructions with a gun having a nozzle that fits the joint width. Work sealant into joints so as to fill the joints solidly without air pockets. Tool sealant after application to ensure adhesion. Apply sealant uniformly smooth and free of wrinkles. Upon completion of sealant application, roughen partially filled or unfilled joints, apply additional sealant, and tool smooth as specified. Apply sealer over sealants in accordance with the sealant manufacturer's printed instructions.

3.5 PROTECTION AND CLEANING

3.5.1 Protection

Protect areas adjacent to joints from sealant smears. Masking tape may be used for this purpose if removed 5 to 10 minutes after the joint is filled and no residual tape marks remain.

3.5.2 Final Cleaning

Upon completion of sealant application, remove remaining smears and stains and leave the work in a clean and neat condition.

- a. Masonry and Other Porous Surfaces: Immediately remove fresh sealant that has been smeared on adjacent masonry, rub clean with a solvent, and remove solvent residue, in accordance with sealant manufacturer's printed instructions. Allow excess sealant to cure for 24 hours then remove by wire brushing or sanding. Remove resulting debris.
- b. Metal and Other Non-Porous Surfaces: Remove excess sealant with a solvent moistened cloth. Remove solvent residue in accordance with solvent manufacturer's printed instructions.

-- End of Section --

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SECTION 08 91 00

METAL WALL LOUVERS

08/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

- | | |
|------------|--|
| AMCA 500-L | (2015) Laboratory Methods of Testing Louvers for Rating |
| AMCA 511 | (2010; R 2016) Certified Ratings Program for Air Control Devices |

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

- | | |
|-----------|---|
| AAMA 2603 | (2020) Voluntary Specification, Performance Requirements and Test Procedures for Pigmented Organic Coatings on Aluminum Extrusions and Panels |
|-----------|---|

ASTM INTERNATIONAL (ASTM)

- | | |
|-------------------|--|
| ASTM A123/A123M | (2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products |
| ASTM A653/A653M | (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process |
| ASTM A780/A780M | (2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings |
| ASTM A1008/A1008M | (2021a) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable |
| ASTM B209 | (2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate |
| ASTM B221 | (2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes |

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1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Wall Louvers

SD-03 Product Data

Metal Wall Louvers

SD-04 Samples

Wall Louver Samples; G

1.3 DELIVERY, STORAGE, AND PROTECTION

Deliver materials to the site in an undamaged condition. Carefully store materials off the ground to provide proper ventilation, drainage, and protection against dampness. Louvers must be free from nicks, scratches, and blemishes. Replace defective or damaged materials with new.

1.4 DETAIL DRAWINGS

Show all information necessary for fabrication and installation of wall louvers. Indicate materials, sizes, thicknesses, fastenings, and profiles.

1.5 COLOR SAMPLES

Colors of finishes for wall louver samples must closely approximate colors indicated. Where color is not indicated, submit the manufacturer's standard colors to the Contracting Officer for selection.

PART 2 PRODUCTS

2.1 MATERIALS

2.1.1 Galvanized Steel Sheet

ASTM A653/A653M, coating designation G90.

2.1.2 Aluminum Sheet

ASTM B209, alloy 3003 or 5005 with temper as required for forming.

2.1.3 Extruded Aluminum

ASTM B221, alloy 6063-T5 or -T52.

2.1.4 Cold Rolled Steel Sheet

ASTM A1008/A1008M, Class 1, with matte finish. Use for interior louvers only.

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2.2 METAL WALL LOUVERS

Wind driven rain resistant type, with bird screens and made to withstand a wind load of not less than 30 pounds per square foot. Wall louvers must bear the AMCA certified ratings program seal for air performance and water penetration in accordance with AMCA 500-L and AMCA 511. The rating must show a water penetration of 0.20 or less ounce per square foot of free area at a free velocity of 800 feet per minute.

2.2.1 Extruded Aluminum Louvers

Fabricated of extruded 6063-T5 or -T52 aluminum with a wall thickness of not less than 0.081 inch.

2.2.2 Formed Metal Louvers

Formed of zinc-coated steel sheet not thinner than 16 U.S. gage, or aluminum sheet not less than 0.08 inch thick.

2.2.3 Mullions and Mullion Covers

Same material and finish as louvers. Provide mullions where indicated and for all louvers more than 5 feet in width at not more than 5 feet on centers. Provide mullion covers on both faces of joints between louvers.

2.2.4 Screens and Frames

For aluminum louvers, provide 1/2 inch square mesh, 14 or 16 gage aluminum or 1/4 inch square mesh, 16 gage aluminum bird screening. For steel louvers, provide 1/2 inch square mesh, 12 or 16 gage zinc-coated steel; 1/2 inch square mesh, 16 gage copper; or 1/4 inch square mesh, 16 gage zinc-coated steel or copper bird screening. Mount screens in removable, rewirable frames of same material and finish as the louvers.

2.3 FASTENERS AND ACCESSORIES

Provide stainless steel screws and fasteners for aluminum louvers and zinc-coated or stainless steel screws and fasteners for steel louvers. Provide other accessories as required for complete and proper installation.

2.4 FINISHES

2.4.1 Aluminum

Exposed aluminum surfaces must be factory finished with an organic coating. Color must be as indicated. All louvers must have the same finish.

2.4.1.1 Organic Coating

Clean and prime exposed aluminum surfaces. Provide a baked enamel finish conforming to AAMA 2603, with total dry film thickness not less than 0.8 mil, color as indicated.

2.4.2 Steel

Surfaces specified must have a zinc coating, a phosphate treatment, and a shop prime coat of rust-inhibitive paint. The galvanized coating must conform to ASTM A653/A653M, coating designation Z275 (G90), except that louvers located in conditioned spaces on interior of the building may be

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Z180 (G60). The weight of zinc coatings must be as designated in Table I of ASTM A123/A123M for the thickness of base metal to be coated. The prime coat must be a type especially developed for materials treated by phosphates and adapted to application by dipping or spraying. Repair damaged zinc-coated surfaces by the materials and methods conforming to ASTM A780/A780M and spot prime. At the option of the Contractor, a two-part system including bonderizing, baked-on epoxy primer, and baked-on enamel top coat may be applied before forming, in lieu of prime coat specified.

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 Wall Louvers

Install using stops or moldings, flanges, strap anchors, or jamb fasteners as appropriate for the wall construction and in accordance with manufacturer's recommendations.

3.1.2 Screens and Frames

Attach frames to louvers with screws or bolts.

3.2 PROTECTION FROM CONTACT OF DISSIMILAR MATERIALS

3.2.1 Copper or Copper-Bearing Alloys

Paint copper or copper-bearing alloys in contact with dissimilar metal with heavy-bodied bituminous paint or separate with inert membrane.

3.2.2 Aluminum

Where aluminum contacts metal other than zinc, paint the dissimilar metal with a primer and two coats of aluminum paint.

3.2.3 Metal

Paint metal in contact with mortar, concrete, or other masonry materials with alkali-resistant coatings such as heavy-bodied bituminous paint.

3.2.4 Wood

Paint wood or other absorptive materials that may become repeatedly wet and in contact with metal with two coats of aluminum paint or a coat of heavy-bodied bituminous paint.

-- End of Section --

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SECTION 09 29 00

GYPSUM BOARD

08/16, CHG 4: 02/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C475/C475M	(2017; R 2022) Standard Specification for Joint Compound and Joint Tape for Finishing Gypsum Board
ASTM C840	(2020) Standard Specification for Application and Finishing of Gypsum Board
ASTM C954	(2022) Standard Specification for Steel Drill Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Steel Studs from 0.033 in. (0.84 mm) to 0.112 in. (2.84 mm) in Thickness
ASTM C1002	(2022) Standard Specification for Steel Self-Piercing Tapping Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Wood Studs or Steel Studs
ASTM C1047	(2019) Standard Specification for Accessories for Gypsum Wallboard and Gypsum Veneer Base
ASTM C1178/C1178M	(2013) Standard Specification for Glass Mat Water-Resistant Gypsum Backing Panel
ASTM C1396/C1396M	(2017) Standard Specification for Gypsum Board
ASTM C1766	(2015; R2019) Standard Specification for Factory-Laminated Gypsum Panel Products
ASTM D226/D226M	(2017) Standard Specification for Asphalt-Saturated Organic Felt Used in Roofing and Waterproofing
ASTM D3273	(2021) Standard Test Method for Resistance to Growth of Mold on the Surface of Interior Coatings in an Environmental Chamber
ASTM E336	(2023) Standard Test Method for Measurement of Airborne Sound Attenuation

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between Rooms in Buildings

CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

CDPH SECTION 01350 (2017; Version 1.2) Standard Method for
 the Testing and Evaluation of Volatile
 Organic Chemical Emissions from Indoor
 Sources using Environmental Chambers

FM GLOBAL (FM)

FM APP GUIDE (updated on-line) Approval Guide
<http://www.approvalguide.com/>

GYPSUM ASSOCIATION (GA)

GA 214 (2010) Recommended Levels of Gypsum Board
 Finish

GA 216 (2016) Application and Finishing of Gypsum
 Panel Products

GA 600 (2015) Fire Resistance Design Manual Sound
 Control

SCIENTIFIC CERTIFICATION SYSTEMS (SCS)

SCS SCS Global Services (SCS) Indoor Advantage

UNDERWRITERS LABORATORIES (UL)

UL 2818 (2022) GREENGUARD Certification Program
 For Chemical Emissions For Building
 Materials, Finishes And Furnishings

UL Fire Resistance (2014) Fire Resistance Directory

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Cementitious Backer Units

Glass Mat Water-Resistant Gypsum Tile Backing Board

Water-Resistant Gypsum Backing Board

Accessories

Submit for each type of gypsum board and for cementitious backer units.

Gypsum Board

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Recycled Content for Gypsum Board

Recycled Content for Paper Facing and Gypsum Cores

VOC Content of Joint Compound

SD-06 Test Reports

ASTM E90 Factory Test Report;

ASTM E336 Field Test Report;

SD-07 Certificates

Asbestos Free Materials;

Certify that gypsum board types, gypsum backing board types, cementitious backer units, and joint treating materials do not contain asbestos.

Indoor Air Quality for Gypsum Board

SD-08 Manufacturer's Instructions

Safety Data Sheets

SD-10 Operation and Maintenance Data

Manufacturer Maintenance Instructions

1.3 CERTIFICATIONS

1.3.1 Indoor Air Quality Certifications

Submit required indoor air quality certifications in one submittal package.

1.3.1.1 Ceiling and Wall Systems

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party program that products meet the requirements of this Section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein.

1.3.1.2 Adhesives and Sealants

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party program that products meet the requirements of this Section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein.

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1.4 DELIVERY, STORAGE, AND HANDLING

1.4.1 Delivery

Deliver materials in the original packages, containers, or bundles with each bearing the brand name, applicable standard designation, and name of manufacturer, or supplier.

1.4.2 Storage

Keep materials dry by storing inside a sheltered building. Where necessary to store gypsum board and cementitious backer units outside, store off the ground, properly supported on a level platform, and protected from direct exposure to rain, snow, sunlight, and other extreme weather conditions. Provide adequate ventilation to prevent condensation. Store per manufacturer's recommendations for allowable temperature and humidity range.

Do not store gypsum wallboard with materials which have high emissions of volatile organic compounds (VOCs) or other contaminants, including. Do not store panels near materials that may offgas or emit harmful fumes, such as kerosene heaters, fresh paint, or adhesives. Do not use materials that have visible moisture or biological growth.

1.4.3 Handling

Neatly stack gypsum board and cementitious backer units flat to prevent sagging or damage to the edges, ends, and surfaces.

1.5 QUALIFICATIONS

Furnish type of gypsum board work specialized by the installer with a minimum of 3 years of documented successful experience.

1.6 SCHEDULING

The gypsum wallboard must be taped, finished and primed before the installation of the highly-emitting materials, or installed after the installation and ventilation period of the highly-emitting materials.

Commence application only after the area scheduled for gypsum board work is completely weathertight. The heating, ventilating, and air-conditioning systems must be complete and in operation prior to application of the gypsum board. If the mechanical system cannot be activated before gypsum board is begun, the gypsum board work may proceed in accordance with an approved plan to maintain the environmental conditions specified below. Apply gypsum board prior to the installation of finish flooring and acoustic ceiling.

1.7 ENVIRONMENTAL REQUIREMENTS

Do not expose the gypsum board to excessive sunlight prior to gypsum board application. Maintain a continuous uniform temperature of not less than 50 degrees F and not more than 80 degrees F for at least one week prior to the application of gypsum board work, while the gypsum board application is being done, and for at least one week after the gypsum board is set. Shield air supply and distribution devices to prevent any uneven flow of air across the plastered surfaces. Provide ventilation to exhaust moist air to the outside during gypsum board application, set, and until gypsum board jointing is dry. In glazed areas, keep windows open top and bottom

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or side to side 3 to 4 inches. Reduce openings in cold weather to prevent freezing of joint compound when applied. For enclosed areas lacking natural ventilation, provide temporary mechanical means for ventilation. In unglazed areas subjected to hot, dry winds or temperature differentials from day to night of 20 degrees F or more, screen openings with cheesecloth or similar materials. Avoid rapid drying. During periods of low indoor humidity, provide minimum air circulation following gypsum boarding and until gypsum board jointing complete and is dry.

1.8 FIRE RESISTIVE CONSTRUCTION

Comply with specified fire-rated assemblies for design numbers indicated per UL Fire Resistance or FM APP GUIDE.

PART 2 PRODUCTS

2.1 MATERIALS

Conform to specifications, standards and requirements specified. Provide gypsum board types, gypsum backing board types, cementitious backing units, and joint treating materials manufactured from asbestos free materials only.

Submit Safety Data Sheets and manufacturer maintenance instructions for gypsum materials including adhesives.

2.1.1 Gypsum Board

ASTM C1396/C1396M. Gypsum board must contain a minimum of 5 percent post-consumer recycled content, or a minimum of 20 percent post-industrial recycled content. Provide data identifying percentage of recycled content for gypsum board. Paper facings must contain a minimum of 100 percent recycled paper content. Gypsum cores must contain a minimum of 95 percent post-industrial recycled gypsum content. Provide data identifying percentage of recycled content for paper facing and gypsum cores. Provide gypsum wall board and panels meeting the emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type). Provide certification or validation of indoor air quality for gypsum board.

2.1.1.1 Regular

48 inch wide, 1/2, 5/8 inch thick, tapered edges.

2.1.1.2 Type X (Special Fire-Resistant)

48 inch wide, 1/2, 5/8 inch thick, tapered edges.

2.1.1.3 Mold Resistant / Anti-Microbial Gypsum

ASTM D3273. 48 inch wide, 1/2, 5/8 inch thick, tapered edges. This product must be used on the interior face of all exterior metal stud walls.

2.1.2 Gypsum Backing Board

ASTM C1396/C1396M, gypsum backing board must be used as a base in a multilayer system.

2.1.2.1 Regular

48 inch wide, 1/2, 5/8 inch thick, square edges.

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2.1.2.2 Foil-Backed

48 inch wide, 1/2, 5/8 inch thick, square edges.

2.1.2.3 Type X (Special Fire-Resistant)

48 inch wide, 1/2, 5/8 inch thick, square edges.

2.1.3 Regular Water-Resistant Gypsum Backing Board

ASTM C1396/C1396M

2.1.3.1 Regular

48 inch wide, 1/2, 5/8 inch thick, tapered edges.

2.1.3.2 Type X (Special Fire-Resistant)

48 inch wide, 1/2, 5/8 inch thick, tapered edges.

2.1.4 Glass Mat Water-Resistant Gypsum Tile Backing Board

ASTM C1178/C1178M

2.1.4.1 Regular

48 inch wide, 1/2, 5/8 inch thick, square edges.

2.1.4.2 Type X (Special Fire-Resistant)

48 inch wide, 1/2, 5/8 inch thick, square edges.

2.1.5 Factory-Laminated Gypsum Board

ASTM C1766, regular or Type X, 48 inch wide 1/2, 5/8 inch thick, sound dampening gypsum panel products composed of factory-laminated gypsum panels laminated into a composite panel.

2.1.6 Joint Treatment Materials

ASTM C475/C475M. Product must be low emitting VOC types with VOC limits not exceeding 50 g/L. Provide data identifying VOC content of joint compound. Use all purpose joint and texturing compound containing inert fillers and natural binders, including lime compound. Pre-mixed compounds must be free of antifreeze, vinyl adhesives, preservatives, biocides and other slow releasing compounds.

2.1.6.1 Embedding Compound

Specifically formulated and manufactured for use in embedding tape at gypsum board joints and compatible with tape, substrate and fasteners.

2.1.6.2 Finishing or Topping Compound

Specifically formulated and manufactured for use as a finishing compound.

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2.1.6.3 All-Purpose Compound

Specifically formulated and manufactured to serve as both a taping and a finishing compound and compatible with tape, substrate and fasteners.

2.1.6.4 Setting or Hardening Type Compound

Specifically formulated and manufactured for use with fiber glass mesh tape.

2.1.6.5 Joint Tape

Use cross-laminated, tapered edge, reinforced paper, or fiber glass mesh tape recommended by the manufacturer.

2.1.7 Fasteners

2.1.7.1 Screws

ASTM C1002, Type "G", Type "S" or Type "W" steel drill screws for fastening gypsum board to gypsum board, wood framing members and steel framing members less than 0.033 inch thick. ASTM C954 steel drill screws for fastening gypsum board to steel framing members 0.033 to 0.112 inch thick. Provide cementitious backer unit screws with a polymer coating.

2.1.8 Accessories

ASTM C1047. Fabricate from corrosion protected steel or plastic designed for intended use. Accessories manufactured with paper flanges are not acceptable. Flanges must be free of dirt, grease, and other materials that may adversely affect bond of joint treatment. Provide prefinished or job decorated materials.

2.1.9 Asphalt Impregnated Building Felt

Provide a 15 lb asphalt moisture barrier over glass mat covered or reinforced gypsum sheathing. Conforming to ASTM D226/D226M Type 1 (No. 15) for asphalt impregnated building felt.

2.1.10 Water

Provide clean, fresh, and potable water.

PART 3 EXECUTION

3.1 EXAMINATION

3.1.1 Framing and Furring

Verify that framing and furring are securely attached and of sizes and spacing to provide a suitable substrate to receive gypsum board and cementitious backer units. Verify that all blocking, headers and supports are in place to support plumbing fixtures and to receive soap dishes, grab bars, towel racks, and similar items. Do not proceed with work until framing and furring are acceptable for application of gypsum board and cementitious backer units.

3.1.2 Building Construction Materials

Do not install building construction materials that show visual evidence of

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biological growth.

3.2 APPLICATION OF GYPSUM BOARD

Apply gypsum board to framing and furring members in accordance with ASTM C840 or GA 216 and the requirements specified. Apply gypsum board with separate panels in moderate contact; do not force in place. Stagger end joints of adjoining panels. Neatly fit abutting end and edge joints. Use gypsum board of maximum practical length; select panel sizes to minimize waste. Cut out gypsum board to make neat, close, and tight joints around openings. In vertical application of gypsum board, provide panels in lengths required to reach full height of vertical surfaces in one continuous piece. Lay out panels to minimize waste; reuse cutoffs whenever feasible. Surfaces of gypsum board and substrate members may be bonded together with an adhesive, except where prohibited by fire rating(s). Treat edges of cutouts for plumbing pipes, screwheads, and joints with water-resistant compound as recommended by the gypsum board manufacturer. Minimize framing by floating corners with single studs and drywall clips. Install 5/8 inch gypsum or 1/2 inch ceiling board over framing at 24 inch on center. Provide type of gypsum board for use in each system specified herein as indicated.

3.2.1 Application of Gypsum Board to Steel Framing and Furring

Apply in accordance with ASTM C840, System VIII or GA 216.

3.2.2 Arches and Bending Radii

Apply gypsum board in accordance with ASTM C840, System IX or GA 216.

3.2.3 Gypsum Board for Wall Tile or Tile Base Applied with Adhesive

In dry areas (areas other than tubs, shower enclosures, saunas, steam rooms, gang shower rooms), apply glass mat water-resistant gypsum tile backing board or water-resistant gypsum backing board in accordance with ASTM C840, System X or GA 216.

3.2.4 Exterior Application

Apply exterior gypsum board (such as at soffits) in accordance with ASTM C840, System XI or GA 216.

3.2.5 Floating Interior Angles

Minimize framing by floating corners with single studs and drywall clips. Locate the attachment fasteners adjacent to ceiling and wall intersections in accordance with ASTM C840, System XII or GA 216, for single-ply and two-ply applications of gypsum board to wood framing.

3.2.6 Control Joints

Install expansion and contraction joints in ceilings and walls in accordance with ASTM C840, System XIII or GA 216. Fill control joints between studs in fire-rated construction with firesafing insulation to match the fire-rating of construction.

3.3 FINISHING OF GYPSUM BOARD

Tape and finish gypsum board in accordance with ASTM C840, GA 214 and GA 216.

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Finish plenum areas above ceilings to Level 1 in accordance with GA 214. Finish water resistant gypsum backing board, ASTM C1396/C1396M, to receive ceramic tile to Level 2 in accordance with GA 214. Finish walls and ceilings to receive a heavy-grade wall covering or heave textured finish before painting to Level 3 in accordance with GA 214. Finish walls and ceilings without critical lighting to receive flat paints, light textures, or wall coverings to Level 4 in accordance with GA 214. Unless otherwise specified, finish all gypsum board walls, partitions and ceilings to Level 5 in accordance with GA 214. Provide joint, fastener depression, and corner treatment. Tool joints as smoothly as possible to minimize sanding and dust. Do not use self-adhering fiber glass mesh tape with conventional drying type joint compounds; use setting or hardening type compounds only. Provide treatment for water-resistant gypsum board as recommended by the gypsum board manufacturer. Protect workers, building occupants, and HVAC systems from gypsum dust.

3.3.1 Uniform Surface

Wherever gypsum board is to receive eggshell, semigloss or gloss paint finish, or where severe, up or down lighting conditions occur, finish gypsum wall surface in accordance to GA 214 Level 5. In accordance with GA 214 Level 5, apply a thin skim coat of joint compound to the entire gypsum board surface, after the two-coat joint and fastener treatment is complete and dry.

3.4 SEALING

Seal openings around pipes, fixtures, and other items projecting through gypsum board and cementitious backer units as specified in Section 07 92 00 JOINT SEALANTS. Apply material with exposed surface flush with gypsum board or cementitious backer units.

3.4.1 Sealing for Glass Mat or Reinforced Gypsum Board Sheathing

Apply silicone sealant in a 3/8 inch bead to all joints and trowel flat. Apply enough of the same sealant to all fasteners penetrating through the glass mat gypsum board surface to completely cover the penetration when troweled flat.

3.5 FIRE-RESISTANT ASSEMBLIES

Wherever fire-rated construction is indicated, provide materials and application methods, including types and spacing of fasteners, wall and ceiling framing in accordance with the specifications contained in UL Fire Resistance for the Design Number(s) indicated, or GA 600 for the File Number(s) indicated. Joints of fire-rated gypsum board enclosures must be closed and sealed in accordance with UL test requirements or GA requirements. Seal penetrations through rated partitions and ceilings tight in accordance with tested systems.

3.6 SOUND RATED ASSEMBLIES

When sound rated assemblies are required, provide materials and application methods, including panels, insulation, types and spacing of fasteners, wall and ceiling framing in accordance with the contract document and the description of the assembly in the ASTM E90 Factory Test Report. Seal partitions continuously with acoustical foam or sealant (both sides) and finished to match wall wherever it abuts another element such as the floor, ceiling, wall, column, mullion, or another system or assembly.

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3.7 PATCHING

Patch surface defects in gypsum board to a smooth, uniform appearance, ready to receive finishes.

3.8 SHAFTWALL FRAMING

Install the shaftwall system in accordance with the system manufacturer's published instructions. Coordinate bucks, anchors, blocking and other items placed in or behind shaftwall framing with electrical and mechanical work. Patch or replace fireproofing materials which are damaged or removed during shaftwall construction.

3.9 SOUND RATED ASSEMBLY FIELD TESTING

Provide third party testing of sound rated assemblies tested in accordance with ASTM E336. Provide the ASTM E336 Field Test Report verifying that the installed assemblies perform no less than five ASTC rating points below the ASTM E90 Factory Test Report. Examine, modify adjust, and retest any installation not meeting the STC Rating until compliance is obtained.

-- End of Section --

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SECTION 09 51 00

ACOUSTICAL CEILINGS

08/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A641/A641M	(2019) Standard Specification for Zinc-Coated (Galvanized) Carbon Steel Wire
ASTM C423	(2023) Sound Absorption and Sound Absorption Coefficients by the Reverberation Room Method
ASTM C635/C635M	(2022) Standard Specification for Manufacture, Performance, and Testing of Metal Suspension Systems for Acoustical Tile and Lay-In Panel Ceilings
ASTM C636/C636M	(2013) Standard Practice for Installation of Metal Ceiling Suspension Systems for Acoustical Tile and Lay-In Panels
ASTM C834	(2017; R 2023) Standard Specification for Latex Sealants
ASTM E413	(2022) Classification for Rating Sound Insulation
ASTM E580/E580M	(2022) Standard Practice for Installation of Ceiling Suspension Systems for Acoustical Tile and Lay-in Panels in Areas Subject to Earthquake Ground Motions
ASTM E795	(2023) Standard Practices for Mounting Test Specimens During Sound Absorption Tests
ASTM E1111/E1111M	(2014; R 2022) Standard Test Method for Measuring the Interzone Attenuation of Open Office Components
ASTM E1264	(2023) Standard Classification for Acoustical Ceiling Products
ASTM E1414/E1414M	(2021a) Standard Test Method for Airborne Sound Attenuation Between Rooms Sharing a Common Ceiling Plenum
ASTM E1477	(1998; R 2022a) Standard Test Method for

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Luminous Reflectance Factor of Acoustical
Materials by Use of Integrating-Sphere
Reflectometers

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-301-01

(2023) Structural Engineering

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Drawings

SD-03 Product Data

Acoustical Ceiling Systems

Acoustical Performance

SD-04 Samples

Acoustical Units; G

Acoustical Ceiling Tiles; G

1.3 DELIVERY, STORAGE. AND HANDLING

Deliver materials to the site in the manufacturer's original unopened containers with brand name and type clearly marked. Carefully handle and store materials in dry, watertight enclosures. Immediately before installation, store acoustical units for not less than 24 hours at the same temperature and relative humidity as the space where they will be installed in order to assure proper temperature and moisture acclimation.

1.4 ENVIRONMENTAL REQUIREMENTS

Maintain a uniform temperature of not less than 60 degrees F nor more than 85 degrees F and a relative humidity of not more than 70 percent for 24 hours before, during, and 24 hours after installation of acoustical units.

1.5 SCHEDULING

Complete and dry interior finish work such as plastering, concrete and terrazzo work before ceiling installation. Complete mechanical, electrical, and other work above the ceiling line; install and start operating heating, ventilating, and air conditioning systems in order to maintain temperature and humidity requirements.

1.6 WARRANTY

Provide manufacturer's warranty to repair or replace defective materials

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and workmanship including but not limited to, sagging and warping of panels and rusting and of grid systems, for a period of one (1) year from date of final acceptance of the work.

1.7 EXTRA MATERIALS

Furnish spare tiles, from the same lot as those installed, of each color at the rate of 5 tiles for each 1000 tiles installed.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Provide sound controlling units mechanically mounted on a ceiling suspension system for acoustical treatment. Provide the unit size, texture, finish, and color as specified. The Contractor has the option to substitute inch-pound (I-P) Recessed Light Fixtures (RLF) for metric RLF. If the Contractor opts to provide I-P RLF, then provide I-P products for other ceiling elements like acoustical ceiling tiles, air diffusers, air registers and grills. Coordinate the entire ceiling system with other details, like the location of access panels and ceiling penetrations, for instance, shown on the drawings. Submit drawings showing the the location, extent and details of acoustical treatment including suspension system, method of anchoring and fastening, and reflected ceiling plan. Coordinate with paragraph RECLAMATION PROCEDURES for reclamation of mineral fiber acoustical ceiling panels to be removed from the job site.

2.1.1 Acoustical Performance

2.1.1.1 Ceiling Sound Transmission

Provide ceiling systems with the specified Ceiling Attenuation Class (CAC) ratings as determined in accordance with ASTM E1414/E1414M and ASTM E413. Provide sound attenuators over light fixtures, air terminals and other ceiling penetrations, provide acoustical blanket insulation on top of the ceiling or adjacent to partitions to provide lightweight acoustical plenum barriers above partitions as required to achieve the specified CAC ratings. Provide test ceiling continuous at the partition and assembled in the suspension system in the same manner that the ceiling will be installed on the project.

2.1.1.2 Ceiling Sound Absorption

Determine the Noise Reduction Coefficient (NRC) in accordance with ASTM C423. Determine Articulation Class (AC) in accordance with ASTM E1111/E1111M.

2.1.2 Light Reflectance

Determine light reflectance factor in accordance with ASTM E1477 test method.

2.2 ACOUSTICAL UNITS

Submit samples of each type of acoustical unit and each type of suspension grid tee section showing texture, finish, and color. Conform acoustical units to ASTM E1264, Class A, and the following requirements:

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2.2.1 Humidity Resistant Composition Units

2.2.1.1 Type I

Non-asbestos mineral or glass fibers bonded with ceramic, moisture resistant thermo-setting resin, or other moisture resistant material and having a factory applied white paint finish. Provide panels that do not sag or warp under conditions of heat, high humidity or chemical fumes.

2.2.1.2 Flame Spread

Class: A, 25 or less

2.2.1.3 Pattern

As indicated on Drawings.

2.2.1.4 Minimum NRC

0.75 when tested on Mounting Type E-400 of ASTM E795.

2.2.1.5 Minimum Light Reflectance Coefficient

LR-1, 0.88 or greater

2.2.1.6 Nominal Size

24 by 24 inch

2.2.1.7 Edge Detail

Square

2.2.1.8 Finish

Factory-applied standard finish. See paragraph COLORS AND PATTERNS.

2.2.2 Type II

Scrubbable, washable, water repellent, anti-mold/mildew with bioblock performance to resist growth of mold and mildew on the tile surface.

Include gasketed ceiling tile edge option.

2.2.2.1 Flame Spread / Smoke Spread Index

Class A, 25 or less. Smoke Developed Index of 50 or less.

2.2.2.2 Pattern

As indicated on Drawings.

2.2.2.3 Minimum Light Reflectance Coefficient

LR-1, 0.89 or greater

2.2.2.4 Nominal Size

24 x 24 inch

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2.2.2.5 Edge Detail

Square

2.3 SUSPENSION SYSTEM

Provide as indicated suspension system conforming to ASTM C635/C635M for intermediate-duty systems. Provide surfaces exposed to view of aluminum or steel with a factory-applied white baked-enamel finish. Provide wall molding having a flange of not less than 15/16 inch. Provide mitered corners. Provide a suspension system with a maximum deflection of 1/360 of the span length capable of supporting the finished ceiling, light fixtures, air diffusers, and accessories, as shown. Conform seismic details to the guidance in UFC 3-301-01, ASTM E580/E580M and as indicated Drawings.

2.4 HANGERS

Provide hangers and attachment capable of supporting a minimum 300 pound ultimate vertical load without failure of supporting material or attachment.

2.4.1 Wires

Conform wires to ASTM A641/A641M, Class 1, 0.11 in diameter.

2.5 FINISHES

Use manufacturer's standard textures, patterns and finishes as specified for acoustical units and suspension system members. Treat ceiling suspension system components to inhibit corrosion.

2.6 COLORS AND PATTERNS

Use colors and patterns for acoustical units and suspension system components as indicated; colors listed are not intended to limit the selection of equal colors from other manufacturers.

2.7 ACOUSTICAL SEALANT

Conform acoustical sealant to ASTM C834, nonstaining.

PART 3 EXECUTION

3.1 INSTALLATION

Do not install building construction materials that show visual evidence of biological growth.

Examine surfaces to receive directly attached acoustical units for unevenness, irregularities, and dampness that would affect quality and execution of the work. Rid areas, where acoustical units will be cemented, of oils, form residue, or other materials that reduce bonding capabilities of the adhesive. Complete and dry interior finish work such as plastering, concrete, and terrazzo work before installation. Complete and approve mechanical, electrical, and other work above the ceiling line prior to the start of acoustical ceiling systems installation. Provide acoustical work complete with necessary fastenings, clips, and other accessories required for a complete installation. Do not expose mechanical fastenings in the finished work. Lay out hangers for each individual room or space. Provide

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hangers to support framing around beams, ducts, columns, grilles, and other penetrations through ceilings. Keep main runners and carrying channels clear of abutting walls and partitions. Provide at least two main runners for each ceiling span. Wherever required to bypass an object with the hanger wires, install a subsuspension system so that all hanger wires will be plumb.

3.1.1 Suspension System

Install suspension system in accordance with ASTM C636/C636M and as specified herein. Do not suspend hanger wires or other loads from underside of steel decking.

3.1.1.1 Plumb Hangers

Install hangers plumb and not pressing against insulation covering ducts and pipes. Where lighting fixtures are supported from the suspended ceiling system, provide hangers at a minimum of four hangers per fixture and located not more than 6 inch from each corner of each fixture.

3.1.1.2 Splayed Hangers

Splay (slope or slant) hangers around obstructions, offsetting the resulting horizontal force by bracing, countersplaying, or other acceptable means.

3.1.2 Wall Molding

Provide wall molding where ceilings abut vertical surfaces. Miter corners where wall moldings intersect or install corner caps. Secure wall molding not more than 3 inch from ends of each length and not more than 16 inch on centers between end fastenings. Provide wall molding springs at each acoustical unit in semi-exposed or concealed systems.

3.1.3 Acoustical Units

Install acoustical units in accordance with the approved installation instructions of the manufacturer. Ensure that edges of acoustical units are in close contact with metal supports, with each other, and in true alignment. Arrange acoustical units so that units less than one-half width are minimized. Hold units in exposed-grid system in place with manufacturer's standard hold-down clips, if units weigh less than 1 psf or if required for fire resistance rating.

3.1.4 Acoustical Sealant

Seal all joints around pipes, ducts or electrical outlets penetrating the ceiling. Apply a continuous ribbon of acoustical sealant on vertical web of wall or edge moldings.

3.2 CEILING ACCESS PANELS

Locate ceiling access panels directly under the items which require access.

3.3 CLEANING

Following installation, clean dirty or discolored surfaces of acoustical units and leave them free from defects. Remove units that are damaged or improperly installed and provide new units as directed.

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-- End of Section --

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SECTION 09 90 00

PAINTS AND COATINGS

02/21

PART 1 GENERAL

1.1 RELATED REQUIREMENTS

1.1.1 Painting Included

Where a space or surface is indicated to be painted, include the following unless indicated otherwise.

- a. Surfaces behind portable objects and surface mounted articles readily detachable by removal of fasteners, such as screws and bolts.
- b. New factory finished surfaces that require identification or color coding and factory finished surfaces that are damaged during performance of the work.
- c. Existing coated surfaces that are damaged during performance of the work.

1.1.1.1 Exterior Painting

Includes new surfaces of the building and appurtenances.

1.1.1.2 Interior Painting

Includes new surfaces of the building and appurtenances as indicated. Where a space or surface is indicated to be painted, include the following items, unless indicated otherwise.

- a. Exposed columns, girders, beams, joists, and metal deck; and
- b. Other contiguous surfaces.

1.1.2 Painting Excluded

Do not paint the following unless indicated otherwise.

- a. Surfaces concealed and made inaccessible by panelboards, fixed ductwork, machinery, and equipment fixed in place.
- b. Surfaces in concealed spaces. Concealed spaces are defined as enclosed spaces above suspended ceilings, furred spaces, attic spaces, crawl spaces, elevator shafts and chases.
- c. Steel to be embedded in concrete.
- d. Copper, stainless steel, aluminum, anodized aluminum, brass, and lead except existing coated surfaces.
- e. Hardware, fittings, and other factory finished items.

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1.1.3 Mechanical and Electrical Painting

Includes field coating of interior new surfaces.

- a. Where a space or surface is indicated to be painted, include the following items unless indicated otherwise.

- (1) Exposed piping, conduit, and ductwork;
- (2) Supports, hangers, air grilles, and registers;
- (3) Miscellaneous metalwork and insulation coverings.

- b. Do not paint the following, unless indicated otherwise:

- (1) New zinc-coated, aluminum, and copper surfaces under insulation
- (2) New aluminum jacket on piping
- (3) New interior ferrous piping under insulation.

1.1.3.1 Fire Extinguishing Sprinkler Systems

Clean, pretreat, prime, and paint new fire extinguishing sprinkler systems including valves, piping, conduit, hangers, supports, miscellaneous metalwork, and accessories. Apply coatings to clean, dry surfaces, using clean brushes.

1.1.4 Exterior Painting of Site Work Items

Field coat the following items:

	New Surfaces	Existing Surfaces
a.	Steel Doors and Frames	N.A.
b.	Exposed Structural Steel	N.A.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS (ACGIH)

ACGIH 0100 (2017; Suppl 2020) Documentation of the Threshold Limit Values and Biological Exposure Indices

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A13.1 (2020) Scheme for the Identification of Piping Systems

ASTM INTERNATIONAL (ASTM)

ASTM D523 (2014; R 2018) Standard Test Method for Specular Gloss

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ASTM D4263	(1983; R 2018) Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method
ASTM D4444	(2013; R 2018) Standard Test Method for Laboratory Standardization and Calibration of Hand-Held Moisture Meters
ASTM D6386	(2016a) Standard Practice for Preparation of Zinc (Hot-Dip Galvanized) Coated Iron and Steel Product and Hardware Surfaces for Painting
ASTM F1869	(2023) Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride

CENTERS FOR DISEASE CONTROL AND PREVENTION (CDC)

Intelligence Bulletin 65	(2013) Occupational Exposure to Carbon Nanotubes and Nanofibers
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MASTER PAINTERS INSTITUTE (MPI)

MPI 3	(2016) Primer, Alkali Resistant, Water Based
MPI 4	(2016) Interior/Exterior Latex Block Filler
MPI 9	(2016) Alkyd, Exterior Gloss (MPI Gloss Level 6)
MPI 10	(2016) Latex, Exterior Flat (MPI Gloss Level 1)
MPI 11	(2016) Latex, Exterior Semi-Gloss, MPI Gloss Level 5
MPI 44	(2016) Latex, Interior, (MPI Gloss Level 2)
MPI 50	(2015) Primer Sealer, Latex, Interior
MPI 52	(2016) Latex, Interior, (MPI Gloss Level 3)
MPI 54	(2016) Latex, Interior, Semi-Gloss (MPI Gloss Level 5)
MPI 72	(2016) Polyurethane, Two-Component, Pigmented, Gloss (MPI Gloss Level 6-7)
MPI 77	(2015) Epoxy, Gloss
MPI 79	(2016) Primer, Alkyd, Anti-Corrosive for Metal
MPI 94	(2016) Alkyd, Exterior, Semi-Gloss (MPI Gloss Level 5)

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MPI 95	(2015) Primer, Quick Dry, for Aluminum
MPI 101	(2016) Primer, Epoxy, Anti-Corrosive, for Metal
MPI 107	(2016) Primer, Rust-Inhibitive, Water Based
MPI 108	(2015) Epoxy, High Build, Low Gloss
MPI 119	(2016) Latex, Exterior, Gloss (MPI Gloss Level 6)
MPI 134	(2015) Primer, Galvanized, Water Based
MPI 138	(2016) Latex, Interior, High Performance Architectural, (MPI Gloss Level 2)
MPI 139	(2016) Latex, Interior, High Performance Architectural, (MPI Gloss Level 3)
MPI 141	(2016) Latex, Interior, High Performance Architectural, Semi-Gloss (MPI Gloss Level 5)
MPI 144	(2016) Latex, Interior, Institutional Low Odor/VOC, (MPI Gloss Level 2)
MPI 145	(2016) Latex, Interior, Institutional Low Odor/VOC, (MPI Gloss Level 3)
MPI 146	(2016) Latex, Interior, Institutional Low Odor/VOC, (MPI Gloss Level 4)
MPI 147	(May 2016) Latex, Interior, Institutional Low Odor/VOC, Semi-Gloss (MPI Gloss Level 5)
MPI 149	(2016) Primer Sealer, Interior, Institutional Low Odor/VOC
MPI 161	(2016) Light Industrial Coating, Exterior, Water Based (MPI Gloss Level 3)
MPI 163	(2016) Light Industrial Coating, Exterior, Water Based, Semi-Gloss (MPI Gloss Level 5)
MPI 164	(2016) Light Industrial Coating, Exterior, Water Based, Gloss (MPI Gloss Level 6)
MPI 214	(2016) Latex, Exterior (MPI Gloss Level 2)
MPI ASM	(2019) Architectural Painting Specification Manual
MPI GPS-1-14	(2014) Green Performance Standard GPS-1-14
MPI GPS-2-14	(2014) Green Performance Standard GPS-2-14
MPI MRM	(2015) Maintenance Repainting Manual

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SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC 7/NACE No.4	(2007) Brush-Off Blast Cleaning
SSPC Glossary	(2011) SSPC Protective Coatings Glossary
SSPC Guide 6	(2021) Guide for Containing Surface Preparation Debris Generated During Paint Removal Operations
SSPC Guide 7	(2015) Guide to the Disposal of Lead-Contaminated Surface Preparation Debris
SSPC PA 1	(2016) Shop, Field, and Maintenance Coating of Metals
SSPC QP 1	(2019) Standard Procedure for Evaluating the Qualifications of Industrial/Marine Painting Contractors (Field Application to Complex Industrial Steel Structures and Other Metal Components)
SSPC SP 1	(2015) Solvent Cleaning
SSPC SP 2	(2018) Hand Tool Cleaning
SSPC SP 3	(2018) Power Tool Cleaning
SSPC SP 6/NACE No.3	(2007) Commercial Blast Cleaning
SSPC SP 10/NACE No. 2	(2015) Near-White Blast Cleaning
SSPC VIS 1	(2002; E 2004) Guide and Reference Photographs for Steel Surfaces Prepared by Dry Abrasive Blast Cleaning
SSPC VIS 3	(2004) Guide and Reference Photographs for Steel Surfaces Prepared by Hand and Power Tool Cleaning
SSPC VIS 4/NACE VIS 7	(1998; E 2000; E 2004) Guide and Reference Photographs for Steel Surfaces Prepared by Waterjetting
SSPC-SP WJ-1/NACE WJ-1	(2012) Clean to Bare Substrate, Waterjet Cleaning of Metals
SSPC-SP WJ-2/NACE WJ-2	(2012) Very Thorough Cleaning, Waterjet Cleaning of Metals
SSPC-SP WJ-3/NACE WJ-3	(2012) Thorough Cleaning, Waterjet Cleaning of Metals
SSPC-SP WJ-4/NACE WJ-4	(2012) Light Cleaning, Waterjet Cleaning of Metals

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U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-STD-101 (2014; Rev C) Color Code for Pipelines and
for Compressed Gas Cylinders

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA Method 24 (2000) Determination of Volatile Matter
Content, Water Content, Density, Volume
Solids, and Weight Solids of Surface
Coatings

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

FED-STD-313 (2018) Material Safety Data,
Transportation Data and Disposal Data for
Hazardous Materials Furnished to
Government Activities

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.1000 Air Contaminants
29 CFR 1910.1025 Lead
29 CFR 1926.62 Lead

1.3 DEFINITIONS

1.3.1 Qualification Testing

Qualification testing is the performance of all test requirements listed in the product specification. This testing is accomplished by MPI to qualify each product for the MPI Approved Product List, and may also be accomplished by Contractor's third-party testing lab if an alternative to Batch Quality Conformance Testing by MPI is desired.

1.3.2 Batch Quality Conformance Testing

Batch quality conformance testing determines that the product provided is the same as the product qualified to the appropriate product specification. This testing must be accomplished by an MPI testing lab.

1.3.3 Coating

SSPC Glossary; (1) A liquid, liquefiable, or mastic composition that is converted to a solid protective, decorative, or functional adherent film after application as a thin layer; (2) Generic term for paint, lacquer, enamel.

1.3.4 DFT or dft

Dry film thickness, the film thickness of the fully cured, dry paint or coating.

1.3.5 DSD

Degree of Surface Degradation, the MPI system of defining degree of surface degradation. Five levels are generically defined under the Assessment

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sections in the MPI MRM, MPI Maintenance Repainting Manual.

1.3.6 EXT

MPI short term designation for an exterior coating system.

1.3.7 INT

MPI short term designation for an interior coating system.

1.3.8 Loose Paint

Paint or coating that can be removed with a dull putty knife.

1.3.9 mil / mils

The English measurement for 0.001 in or one one-thousandth of an inch.

1.3.10 MPI Gloss Levels

MPI system of defining gloss. Seven gloss levels (G1 to G7) are generically defined under the Evaluation sections of the MPI Manuals. Traditionally, Flat refers to G1/G2, Eggshell refers to G3, Semigloss refers to G5, and Gloss refers to G6.

Gloss levels are defined by MPI as follows:

Gloss Level	Description	Units at 60 degree angle	Units at 80 degree angle
G1	Matte or Flat	0 to 5	10 max
G2	Velvet	0 to 10	10 to 35
G3	Eggshell	10 to 25	10 to 35
G4	Satin	20 to 35	35 min
G5	Semi-Gloss	35 to 70	
G6	Gloss	70 to 85	
G7	High Gloss		

Gloss is tested in accordance with ASTM D523. Historically, the Government has used Flat (G1 / G2), Eggshell (G3), Semi-Gloss (G5), and Gloss (G6).

1.3.11 MPI System Number

The MPI coating system number in each MPI Division found in either the MPI Architectural Painting Specification Manual or the Maintenance Repainting Manual and defined as an exterior (EXT/REX) or interior system (INT/RIN).

1.3.12 Paint

SSPC Glossary; (1) Any pigmented liquid, liquefiable, or mastic composition designed for application to a substrate in a thin layer that is converted to an opaque solid film after application. Used for protection, decoration, identification, or to serve some other functional purposes; (2)

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Application of a coating material.

1.3.13 REX

MPI short term designation for an exterior coating system used in repainting projects or over existing coating systems.

1.3.14 RIN

MPI short term designation for an interior coating system used in repainting projects or over existing coating systems.

1.4 SCHEDULING

Allow paint, polyurethane, varnish, and wood stain installations to cure prior to the installation of materials that adsorb VOCs, including carpets, textiles, unprimed gypsum wallboard, acoustical ceiling panels,.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

Samples of specified materials may be taken and tested for compliance with specification requirements.

SD-02 Shop Drawings

Piping Identification

SD-03 Product Data

Coating

Product Data Sheets

SD-04 Samples

Color; G

SD-07 Certificates

Qualification Testing laboratory for coatings

Indoor Air Quality for Paints and Primers

Indoor Air Quality for Consolidated Latex Paints

SD-08 Manufacturer's Instructions

Application Instructions

Mixing

Manufacturer's Safety Data Sheets

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SD-10 Operation and Maintenance Data

Coatings, Data Package 1; G

1.6 QUALITY ASSURANCE

1.6.1 Regulatory Requirements

1.6.1.1 Environmental Protection

In addition to requirements specified elsewhere for environmental protection, provide coating materials that conform to the restrictions of the local Air Pollution Control District and regional jurisdiction. Notify Contracting Officer of any paint specified herein which fails to conform.

1.6.1.2 Lead Content

Do not use coatings having a lead content over 0.06 percent by weight of nonvolatile content.

1.6.1.3 Chromate Content

Do not use coatings containing zinc-chromate or strontium-chromate.

1.6.1.4 Asbestos Content

Provide asbestos-free materials.

1.6.1.5 Mercury Content

Provide materials free of mercury or mercury compounds.

1.6.1.6 Silica

Provide abrasive blast media containing no free crystalline silica.

1.6.1.7 Human Carcinogens

Provide materials that do not contain ACGIH 0100 confirmed human carcinogens (A1) or suspected human carcinogens (A2).

1.6.1.8 Carbon Based Fibers / Tubes

Materials must not contain carbon based fibers such as carbon nanotubes or carbon nanofibers. Intelligence Bulletin 65 ranks toxicity of carbon nanotubes on a par with asbestos.

1.6.2 Coating Contractor's Qualification

Submit the name, address, telephone number, and e-mail address of the Contractor that will be performing all surface preparation and coating application. Submit evidence that key personnel have successfully performed surface preparation and application of coatings on a minimum of three similar projects within the past three years. List information by individual and include the following:

- a. Name of individual and proposed position for this work.

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b. Information about each previous assignment including:

Position or responsibility

Employer (if other than the Contractor)

Name of facility owner

Mailing address and telephone number of facility owner

Name of individual in facility owner's organization who can be contacted as a reference

Location, size and description of structure

Dates work was carried out

Description of work carried out on structure

1.6.3 SSPC QP 1 Certification

Contractors that perform surface preparation or coating application on steel substrates must be certified by the Society for Protective Coatings (formerly Steel Structures Painting Council) (SSPC) to the requirements of SSPC QP 1 prior to Contract award, and must remain certified while accomplishing any surface preparation or coating application. If a Contractor's certification expires, the firm will not be allowed to perform any work until the certification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered. Notify the Contracting Officer of any change in Contractor certification status. Notify the Contracting Officer of all scheduled and unannounced on-site audits from SSPC and furnish a copy of all audit reports.

1.6.4 Approved Products List

The current MPI, "Approved Product List" which lists paint by brand, label, product name and product code as of the date of Contract award, will be used to determine compliance with the submittal requirements of this specification. The Contractor may choose to use a subsequent MPI "Approved Product List", however, only one list may be used for the entire Contract and each coating system is to be from a single manufacturer. Provide all coats on a particular substrate from a single manufacturer. No variation from the MPI Approved Products List is acceptable.

1.6.5 Paints and Coatings Indoor Air Quality Certifications

Provide paint and coating products certified to meet indoor air quality requirements by MPI GPS-1-14, MPI GPS-2-14 or provide certification by other third-party programs. Provide current product certification documentation from certification body.

Provide certification of Indoor Air Quality for Paints and Primers.
Provide certification of Indoor Air Quality for Consolidated Latex Paints.
Submit required indoor air quality certifications in one submittal package.

1.6.6 Field Samples and Tests

The Contracting Officer may choose up to two coatings that have been

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delivered to the site to be tested at no cost to the Government. Take samples of each chosen product as specified in the paragraph SAMPLING PROCEDURE. Test each chosen product as specified in the paragraph TESTING PROCEDURE. Remove products from the job site which do not conform, and replace with new products that conform to the referenced specification. Test replacement products that failed initial testing as specified in the paragraph TESTING PROCEDURE at no cost to the Government.

1.6.6.1 Sampling Procedure

Select paint at random from the products that have been delivered to the job site for sample testing. The Contractor must provide one quart samples of the selected paint materials. Take samples in the presence of the Contracting Officer, and label, and identify each sample. Provide labels in accordance with the paragraph PACKAGING, LABELING, AND STORAGE.

1.6.6.2 Testing Procedure

Provide Batch Quality Conformance Testing for specified products, as defined by and performed by MPI. As an alternative to Batch Quality Conformance Testing, the Contractor may provide Qualification Testing for specified products above to the appropriate MPI product specification, using the third-party laboratory approved under the paragraph QUALIFICATION TESTING laboratory for coatings. Include the backup data and summary of the test results within the qualification testing lab report. Provide a summary listing of all the reference specification requirements and the result of each test. Clearly indicate in the summary whether the tested paint meets each test requirement. Note that Qualification Testing may take 4 to 6 weeks to perform, due to the extent of testing required.

Submit name, address, telephone number, FAX number, and e-mail address of the independent third party laboratory selected to perform testing of coating samples for compliance with specification requirements. Submit documentation that laboratory is regularly engaged in testing of paint samples for conformance with specifications, and that employees performing testing are qualified. If MPI is chosen to perform the Batch Quality Conformance testing, the above submittal information is not required, only a letter is required from the Contractor stating that MPI will perform the testing.

1.7 PACKAGING, LABELING, AND STORAGE

Provide paints in sealed containers that legibly show the Contract specification number, designation name, formula or specification number, batch number, color, quantity, date of manufacture, manufacturer's formulation number, manufacturer's directions including any warnings and special precautions, and name and address of manufacturer. Furnish pigmented paints in containers not larger than 5 gallons. Store paints and thinners in accordance with the manufacturer's written directions, and as a minimum, stored off the ground, under cover, with sufficient ventilation to prevent the buildup of flammable vapors, and at temperatures between 40 to 95 degrees F. Do not store paint, polyurethane, varnish, or wood stain products with materials that have a high capacity to absorb VOC emissions. Do not store paint, polyurethane, varnish, or wood stain products in occupied spaces.

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1.8 SAFETY AND HEALTH

Comply with applicable Federal, State, and local laws and regulations. Include in the Activity Hazard Analysis the potential impact of painting operations on painting personnel and on others involved in and adjacent to the work zone.

1.8.1 Toxic Materials

To protect personnel from overexposure to toxic materials, conform to the most stringent guidance of:

- a. The applicable manufacturer's Safety Data Sheets (SDS) or local regulation.
- b. 29 CFR 1910.1000.
- c. ACGIH 0100, threshold limit values.
- d. The appropriate OSHA standard in 29 CFR 1910.1025 and 29 CFR 1926.62 for surface preparation on painted surfaces containing lead. Additional guidance is given in SSPC Guide 6 and SSPC Guide 7. Refer to drawings for list of hazardous materials located on this project. Coordinate paint preparation activities with this specification section.

Submit manufacturer's Safety Data Sheets for coatings, solvents, and other potentially hazardous materials, as defined in FED-STD-313.

1.9 ENVIRONMENTAL REQUIREMENTS

Comply, at minimum, with manufacturer recommendations for space ventilation during and after installation. Isolate area of application from rest of building when applying high-emission paints or coatings.

1.9.1 Coatings

Do not apply coating when air or substrate conditions are:

- a. Less than 5 degrees F above dew point;
- b. Below 50 degrees F or over 95 degrees F, unless specifically pre-approved by the Contracting Officer and the product manufacturer. Do not, under any circumstances, violate the manufacturer's application recommendations.

1.9.2 Post-Application

Vacate space for as long as possible after application. Wait a minimum of 48 hours before occupying freshly painted rooms. Maintain one of the following ventilation conditions during the curing period, or for 72 hours after application:

- a. Supply 100 percent outside air 24 hours a day.
- b. Supply airflow at a rate of 6 air changes per hour, when outside temperatures are between 55 degrees F and 85 degrees F and humidity is between 30 percent and 60 percent.
- c. Supply airflow at a rate of 1.5 air changes per hour, when outside air

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conditions are not within the range stipulated above.

PART 2 PRODUCTS

2.1 MATERIALS

Conform to the coating specifications and standards referenced in PART 3. Submit Product Data Sheets for specified coatings and solvents. Provide preprinted cleaning and maintenance instructions for all coating systems. Submit Manufacturer's Instructions on Mixing: Detailed mixing instructions, minimum and maximum application temperature and humidity, pot life, and curing and drying times between coats.

2.2 COLOR SELECTION OF FINISH COATS

Provide colors of finish coats as indicated or specified. Allow Contracting Officer to select colors not indicated or specified. Manufacturers' names and color identification are used for the purpose of color identification only. Named products are acceptable for use only if they conform to specified requirements. Products of other manufacturers are acceptable if the colors are approximately the colors indicated and the product conforms to specified requirements.

Provide color, texture, and pattern of wall coating systems as indicated. Submit manufacturer's samples of paint colors. Cross reference color samples to color scheme as indicated. Submit color stencil codes. Tint each coat progressively darker to enable confirmation of the number of coats.

2.3 CONCRETE FLOOR SURFACE SEALER

Clear concrete water based, wet look, floor sealer system in conformance with MPI ASM, No. 99.

PART 3 EXECUTION

3.1 PROTECTION OF AREAS AND SPACES NOT TO BE PAINTED

Prior to surface preparation and coating applications, remove, mask, or otherwise protect hardware, hardware accessories, machined surfaces, radiator covers, plates, lighting fixtures, public and private property, and other such items not to be coated that are in contact with surfaces to be coated. Following completion of painting, reinstall removed items by workmen skilled in the trades. Restore surfaces contaminated by coating materials, to original condition and repair damaged items.

3.2 SURFACE PREPARATION

Remove dirt, splinters, loose particles, grease, oil, disintegrated coatings, and other foreign matter and substances deleterious to coating performance as specified for each substrate before application of paint or surface treatments. Remove oil and grease prior to mechanical cleaning. Schedule cleaning so that dust and other contaminants will not fall on wet, newly painted surfaces. Spot-prime exposed ferrous metals such as nail heads on or in contact with surfaces to be painted with water-thinned paints, with a suitable corrosion-inhibitive primer capable of preventing flash rusting and compatible with the coating specified for the adjacent areas. Refer to MPI ASM and MPI MRM for additional more specific substrate preparation requirements.

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3.2.1 Substrate Repair

- a. Repair substrate surface damaged during coating removal;
- b. Sand edges of adjacent soundly-adhered existing coatings so they are tapered as smooth as practical to areas involved with coating removal; and
- c. Clean and prime the substrate as specified.

3.3 PREPARATION OF METAL SURFACES

3.3.1 New Ferrous Surfaces

- a. Ferrous Surfaces including Shop-coated Surfaces and Small Areas That Contain Rust, Mill Scale and Other Foreign Substances: Solvent clean or detergent wash in accordance with SSPC SP 1 to remove oil and grease. Where shop coat is missing or damaged, clean according to SSPC SP 2, SSPC SP 3, SSPC SP 6/NACE No.3, or SSPC SP 10/NACE No. 2. Brush-off blast remaining surface in accordance with SSPC 7/NACE No.4; Water jetting to SSPC-SP WJ-4/NACE WJ-4 may be used to remove loose coating and other loose materials. Use inhibitor as recommended by coating manufacturer to prevent premature rusting. Protect shop-coated ferrous surfaces from corrosion by treating and touching up corroded areas immediately upon detection.
- b. Surfaces With More Than 20 Percent Rust, Mill Scale, and Other Foreign Substances: Clean entire surface in accordance with SSPC SP 6/NACE No.3 / SSPC-SP WJ-3/NACE WJ-3SSPC SP 10/NACE No. 2 / SSPC-SP WJ-2/NACE WJ-2.

3.3.2 Final Ferrous Surface Condition:

3.3.2.1 Tool Cleaned Surfaces

Comply with SSPC SP 2 and SSPC SP 3. Use as a visual reference, photographs in SSPC VIS 3 for the appearance of cleaned surfaces.

3.3.2.2 Abrasive Blast Cleaned Surfaces

Comply with SSPC 7/NACE No.4, SSPC SP 6/NACE No.3, and SSPC SP 10/NACE No. 2. Use as a visual reference, photographs in SSPC VIS 1 for the appearance of cleaned surfaces.

3.3.2.3 Waterjet Cleaned Surfaces

Comply with SSPC-SP WJ-1/NACE WJ-1, SSPC-SP WJ-2/NACE WJ-2, SSPC-SP WJ-3/NACE WJ-3 or SSPC-SP WJ-4/NACE WJ-4. Use as a visual reference, photographs in SSPC VIS 4/NACE VIS 7 for the appearance of cleaned surfaces.

3.3.3 Galvanized Surfaces

- a. New or Existing Galvanized Surfaces With Only Dirt and Zinc Oxidation Products: Clean with solvent, or non-alkaline detergent solution in accordance with SSPC SP 1. Completely remove coating by brush-off abrasive blast if the galvanized metal has been passivated or stabilized. Do not "passivate" or "stabilize" new galvanized steel to be coated. If the absence of hexavalent stain inhibitors is not

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documented, test as described in ASTM D6386, Appendix X2, and remove by one of the methods described therein.

- b. Galvanized with Slight Coating Deterioration or with Little or No Rusting: Water jetting to SSPC-SP WJ-3/NACE WJ-3 to remove loose coating from surfaces with less than 20 percent coating deterioration and no blistering, peeling, or cracking. Use inhibitor as recommended by the coating manufacturer to prevent rusting.
- c. Galvanized With Severe Deteriorated Coating or Severe Rusting: Water jet to SSPC-SP WJ-3/NACE WJ-3 degree of cleanliness. Spot abrasive blast rusted areas as described for steel in SSPC SP 6/NACE No.3, and waterjet to SSPC-SP WJ-3/NACE WJ-3 to remove existing coating.

3.3.4 Non-Ferrous Metallic Surfaces

Aluminum and aluminum-alloy, lead, copper, and other nonferrous metal surfaces.

Surface Cleaning: Solvent clean in accordance with SSPC SP 1 and wash with mild non-alkaline detergent to remove dirt and water soluble contaminants.

3.4 PREPARATION OF CONCRETE AND CEMENTITIOUS SURFACE

3.4.1 Concrete and Masonry

- a. Curing: Allow concrete, stucco and masonry surfaces to cure at least 30 days before painting, and concrete slab on grade to cure at least 90 days before painting.
- b. Surface Cleaning: Remove the following deleterious substances.
 - (1) Dirt, Chalking, Grease, and Oil: Wash new surfaces with a solution composed of 1/2 cup trisodium phosphate, 1/4 cup household detergent, and 4 quarts of warm water. Then rinse thoroughly with fresh water. For large areas, water blasting may be used.
 - (2) Fungus and Mold: Wash new surfaces with a solution composed of 1/2 cup trisodium phosphate, 1/4 cup household detergent, one quart 5 percent sodium hypochlorite solution and 3 quarts of warm water. Rinse thoroughly with fresh water.
 - (3) Paint and Loose Particles: Remove by wire brushing.
 - (4) Efflorescence: Remove by scraping or wire brushing followed by washing with a 5 to 10 percent by weight aqueous solution of hydrochloric (muriatic) acid. Do not allow acid to remain on the surface for more than five minutes before rinsing with fresh water. Do not acid clean more than 4 square feet of surface, per workman, at one time.
- c. Cosmetic Repair of Minor Defects: Repair or fill mortar joints and minor defects, including but not limited to spalls, in accordance with manufacturer's recommendations and prior to coating application.
- d. Allowable Moisture Content: Latex coatings may be applied to damp surfaces, but not to surfaces with droplets of water. Do not apply epoxies to damp vertical surfaces as determined by ASTM D4263 or

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horizontal surfaces that exceed 3 lbs of moisture per 1000 square feet in 24 hours as determined by ASTM F1869. In all cases follow manufacturer's recommendations. Allow surfaces to cure a minimum of 30 days before painting.

3.4.2 Gypsum Board, Plaster, and Stucco

3.4.2.1 Surface Cleaning

Verify that plaster and stucco surfaces are free from loose matter and that gypsum board is dry. Remove loose dirt and dust by brushing with a soft brush, rubbing with a dry cloth, or vacuum-cleaning prior to application of the first coat material. A damp cloth or sponge may be used if paint is water-based.

3.4.2.2 Repair of Minor Defects

Prior to painting, repair joints, cracks, holes, surface irregularities, and other minor defects with patching plaster or spackling compound and sand smooth.

3.4.2.3 Allowable Moisture Content

Latex coatings may be applied to damp surfaces, but not surfaces with droplets of water. Do not apply epoxies to damp surfaces as determined by ASTM D4263. Verify that new plaster to be coated has a maximum moisture content of 8 percent, when measured in accordance with ASTM D4444, Method A, unless otherwise authorized. In addition to moisture content requirements, allow new plaster to age a minimum of 30 days before preparation for painting.

3.5 PREPARATION OF WOOD AND PLYWOOD SURFACES

3.5.1 New Plywood and Wood Surfaces, Except Floors:

- a. Surface Cleaning: Clean wood surfaces of foreign matter. Verify that surfaces are free from dust and other deleterious substances and in a condition approved by the Contracting Officer prior to receiving paint or other finish. Do not use water to clean uncoated wood. Scrape to remove loose coatings. Lightly sand to roughen the entire area of previously enamel-coated wood surfaces.

3.6 APPLICATION

3.6.1 Coating Application

- a. Comply with applicable federal, state and local laws enacted to ensure compliance with Federal Clean Air Standards. Apply coating materials in accordance with SSPC PA 1. SSPC PA 1 methods are applicable to all substrates, except as modified herein.
- b. At the time of application, paint must show no signs of deterioration. Maintain uniform suspension of pigments during application.
- c. Unless otherwise specified or recommended by the paint manufacturer, paint may be applied by brush, roller, or spray. Use trigger operated spray nozzles for water hoses. Use rollers for applying paints and enamels of a type designed for the coating to be applied and the surface to be coated. Wear protective clothing and respirators when applying oil-based paints or using spray equipment with any paints.

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- d. Only apply paints, except water-thinned types, to surfaces that are completely free of moisture as determined by sight or touch.
- e. Thoroughly work coating materials into joints, crevices, and open spaces. Pay special attention to ensure that all edges, corners, crevices, welds, and rivets receive a film thickness equal to that of adjacent painted surfaces.
- f. Apply each coat of paint so that dry film is of uniform thickness and free from runs, drops, ridges, waves, pinholes or other voids, laps, brush marks, and variations in color, texture, and finish. Completely hide all blemishes.
- g. Touch up damaged coatings before applying subsequent coats. Broom clean and clear dust from interior areas before and during the application of coating material.
- g. Piping in Unfinished Areas: Provide primed surfaces with one coat of red alkyd gloss enamel (MPI 9) applied to a minimum dry film thickness of 1.0 mil in attic spaces, spaces above suspended ceilings, crawl spaces, pipe chases, mechanical equipment room, and spaces where walls or ceiling are not painted or not constructed of a prefinished material.
- h. Piping in Finished Areas: Provide primed surfaces with two coats of paint to match adjacent surfaces, except provide valves and operating accessories with one coat of red alkyd gloss enamel (MPI 9) applied to a minimum dry film thickness of 1.0 mil or two component gloss polyurethane (MPI 72) in exterior applications.
- i. Provide labeling on the surfaces of all feed and cross mains to show the pipe function such as "Sprinkler System", "Fire Department Connection", "Standpipe". For pipe sizes 4-inch and larger provide white painted stenciled letters and arrows, a minimum of 2 in in height and visible from at least two sides when viewed from the floor. For pipe sizes less than 4-inch, provide white painted stenciled letters and arrows, a minimum of 0.75 in in height and visible from the floor.
- j. All fire suppression system valves must be marked with permanent tags indicating normally open or normally closed.
- k. Drying Time: Allow time between coats, as recommended by the coating manufacturer, to permit thorough drying, but not to present topcoat adhesion problems. Provide each coat in specified condition to receive next coat.
- l. Primers, and Intermediate Coats: Do not allow primers or intermediate coats to dry more than 30 days, or longer than recommended by manufacturer, before applying subsequent coats. Follow manufacturer's recommendations for surface preparation if primers or intermediate coats are allowed to dry longer than recommended by manufacturers of subsequent coatings. Cover each preceding coat or surface completely by ensuring visually perceptible difference in shades of successive coats.
- m. Finished Surfaces: Provide finished surfaces free from runs, drops, ridges, waves, laps, brush marks, and variations in colors.
- n. Thermosetting Paints: Apply topcoats over thermosetting paints (epoxies and urethanes) within the overcoat window recommended by the

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manufacturer.

- o. Floors: For nonslip surfacing on level floors, as the intermediate coat is applied, cover wet surface completely with almandite garnet, Grit No. 36, with maximum passing U.S. Standard Sieve No. 40 less than 0.5 percent. When the coating is dry, use a soft bristle broom to sweep up excess grit, which may be reused, and vacuum up remaining residue before application of the topcoat.

3.6.2 Mixing and Thinning of Paints

Reduce paints to proper consistency by adding fresh paint, except when thinning is mandatory to suit surface, temperature, weather conditions, application methods, or for the type of paint being used. Obtain written permission from the Contracting Officer to use thinners. Verify that the written permission includes quantities and types of thinners to use.

When thinning is allowed, thin paints immediately prior to application with not more than one pint of suitable thinner per gallon. The use of thinner does not relieve the Contractor from obtaining complete hiding, full film thickness, or required gloss. Thinning cannot cause the paint to exceed limits on volatile organic compounds. Do not mix paints of different manufacturers.

3.6.3 Two-Component Systems

Mix two-component systems in accordance with manufacturer's instructions. Follow recommendation by the manufacturer for any thinning of the first coat to ensure proper penetration and sealing for each type of substrate.

3.6.4 Coating Systems

- a. Systems by Substrates: Apply coatings that conform to the respective specifications listed in the following Tables:

Table for Exterior Applications	
MPI Division	Substrate Application
MPI Division 3	Refer to Section 07 19 00 Water Repellents
MPI Division 4	Refer to Section 07 19 00 Water Repellents
MPI Division 5	Exterior Metal, Ferrous and Non-Ferrous Paint Table
MPI Division 6	Exterior Wood; Dressed Lumber, Paneling, Decking, Shingles Paint Table
MPI Division 9	Exterior Stucco Paint Table
MPI Division 10	Exterior Cloth Coverings and Bituminous Coated Surfaces Paint Table

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Table for Interior Applications	
MPI Division	Substrate Application
MPI Division 3	Interior Concrete Paint Table
MPI Division 5	Interior Metal, Ferrous and Non-Ferrous Paint Table
MPI Division 6	Interior Wood Paint Table
MPI Division 9	Gypsum Board, Paint Table

- b. Minimum Dry Film Thickness (DFT): Apply paints, primers, varnishes, enamels, undercoats, and other coatings to a minimum dry film thickness of 1.5 mil each coat unless specified otherwise in the Tables. Coating thickness, where specified, refers to the minimum dry film thickness.
- c. Coatings for Surfaces Not Specified Otherwise: Coat unspecified surfaces the same as surfaces having similar conditions of exposure.

3.7 COATING SYSTEMS FOR METAL

Apply coatings of Tables in MPI Division 5 for Exterior and Interior.

- a. Apply specified ferrous metal primer to steel surfaces on the same day that surface is cleaned, to surfaces that meet all specified surface preparation requirements at time of application.
- b. Inaccessible Surfaces: Prior to erection, use one coat of specified primer on metal surfaces that will be inaccessible after erection.
- c. Shop-primed Surfaces: Touch up exposed substrates and damaged coatings to protect from rusting prior to applying field primer.
- d. Pipes and Tubing: The semitransparent film applied to some pipes and tubing at the mill is not to be considered a shop coat. Overcoat these items with the specified ferrous-metal primer prior to application of finish coats.
- e. Exposed Nails, Screws, Fasteners, and Miscellaneous Ferrous Surfaces. On surfaces to be coated with water thinned coatings, spot prime exposed nails and other ferrous metal with latex primer MPI 107.

3.8 COATING SYSTEMS FOR CONCRETE AND CEMENTITIOUS SUBSTRATES

Apply coatings of Tables in MPI Division 3, 4 and 9 for Exterior and Interior.

3.9 COATING SYSTEMS FOR WOOD AND PLYWOOD

- a. Apply coatings of Tables in MPI Division 6 for Exterior and Interior.
- b. Prior to erection, apply two coats of specified primer to treat and prime wood and plywood surfaces which will be inaccessible after

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erection.

- c. Apply stains in accordance with manufacturer's printed instructions.

3.10 PIPING IDENTIFICATION

Piping Identification, Including Surfaces In Concealed Spaces: Provide in accordance with MIL-STD-101ASME A13.1. Place stenciling in clearly visible locations. On piping not covered by MIL-STD-101ASME A13.1, stencil approved names or code letters, in letters a minimum of 1/2 inch high for piping and a minimum of 2 inches high elsewhere. Stencil arrow-shaped markings on piping to indicate direction of flow using black stencil paint.

3.11 FIRE WALL MARKING AND IDENTIFICATION

For other than existing assemblies where there is an accessible concealed floor/ceiling or attic space, fire walls, fire barriers, smoke barriers, and smoke partitions must be permanently identified with an ink roller system within 15 feet of end of each wall, and at intervals not exceeding 30 feet measured horizontally along the wall. Include lettering not less than 3 inches in height with a minimum of 3/8 inch stroke in contrasting color. Identify the fire resistive rating as applicable.

3.12 INSPECTION AND ACCEPTANCE

In addition to meeting previously specified requirements, demonstrate mobility of moving components, including swinging and sliding doors, cabinets, and windows with operable sash, for inspection by the Contracting Officer. Perform this demonstration after appropriate curing and drying times of coatings have elapsed and prior to invoicing for final payment.

3.13 WASTE MANAGEMENT

As specified in the Waste Management Plan and as follows. Do not use kerosene or any such organic solvents to clean up water based paints. Properly dispose of paints or solvents in designated containers. Close and seal partially used containers of paint to maintain quality as necessary for reuse. Store in protected, well-ventilated, fire-safe area at moderate temperature. Place materials defined as hazardous or toxic waste in designated containers. Coordinate with manufacturer for take-back program. Set aside scrap to be returned to manufacturer for recycling into new product. When such a service is not available, contact local recyclers to reclaim the materials. Set aside extra paint for future color matches or reuse by the Government. Where local options exist for leftover paint recycling, collect all waste paint by type and provide for delivery to recycling or collection facility for reuse by local organizations.

3.14 PAINT TABLES

All DFT's are minimum values. Use only materials with a MPI GPS-1-14 green check mark having a minimum MPI "Environmentally Friendly" E2 or E3 rating based on VOC (EPA Method 24) content levels. Acceptable products are listed in the MPI Green Approved Products List, available at <http://www.specifygreen.com/APL/ProductIdxByMPInum.asp>.

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3.14.1 Exterior Paint Tables

3.14.1.1 MPI Division 3: Exterior Concrete Paint Table

A. Concrete; No Table Refer to Section 07 19 00 Water Repellents.

(1) New concrete; vertical surfaces, including undersides of balconies and soffits but excluding tops of slabs

Latex					
New and uncoated existing	Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI EXT 3.1A-G1 (Flat)	MPI REX 3.1A-G1 (Flat)	MPI 3	MPI 10	MPI 10	3.5 mils
MPI EXT 3.1A-G2 (Velvet)	MPI REX 3.1A-G2 (Velvet)	MPI 3	MPI 214	MPI 214	3.5 mils
MPI EXT 3.1A-G5 (Semigloss)	MPI REX 3.1A-G5 (Semigloss)	MPI 3	MPI 11	MPI 11	3.5 mils
MPI EXT 3.1A-G6 (Gloss)	MPI REX 3.1A-G6 (Gloss)	MPI 3	MPI 119	MPI 119	3.5 mils
Primer as recommended by manufacturer. Topcoat: Coating to match adjacent surfaces.					

3.14.1.2 MPI Division 5: Exterior Metal, Ferrous and Non-Ferrous Paint Table

A. Steel / Ferrous Surfaces

(1) New Steel that has been blast-cleaned to SSPC SP 6/NACE No.3

Option 1: Alkyd					
New	Existing, uncoated	Primer	Intermediate	Topcoat	System DFT
MPI EXT 5.1D-G5 (Semigloss)	MPI REX 5.1D-G5 (Semigloss)	MPI 79	MPI 94	MPI 94	5.25 mils
MPI EXT 5.1D-G6 (Gloss)	MPI REX 5.1D-G6 (Gloss)	MPI 79	MPI 9	MPI 9	5.25 mils
Topcoat: Coating to match adjacent surfaces.					

Option 2: Pigmented Polyurethane

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Existing, previously coated with epoxy	Primer	Intermediate	Topcoat	System DFT
MPI REX 5.1H-G6 (Gloss)	MPI 101	MPI 108	MPI 72	8.5 mils
Topcoat: Coating to match adjacent surfaces.				

B. Exterior Galvanized Surfaces (only where designated as "painted" on Drawings.

(1) New Galvanized surfaces

Waterborne Primer / Latex				
New Galvanized Surfaces	Primer	Intermediate	Topcoat	System DFT
MPI EXT 5.3H-G1 (Flat)	MPI 134	MPI 10	MPI 10	4.5 mils
EXT 5.3H-G5 (Semigloss)	MPI 134	MPI 11	MPI 11	4.5 mils
MPI EXT 5.3H-G6 (Gloss)	MPI 134	MPI 119	MPI 119	4.5 mils
Topcoat: Coating to match adjacent surfaces.				

Waterborne Primer / Waterborne Light Industrial Coating				
New Galvanized Surfaces	Primer	Intermediate	Topcoat	System DFT
MPI EXT 5.3J-G5 (Semigloss)	MPI 134	MPI 163	MPI 163	4.5 mils
MPI EXT 5.3J-G6 (Gloss)	MPI 134	MPI 164	MPI 164	4.5 mils
Topcoat: Coating to match adjacent surfaces.				

Epoxy Primer / Waterborne Light Industrial Coating				
New Galvanized Surfaces	Primer	Intermediate	Topcoat	System DFT
MPI EXT 5.3K-G5 (Semigloss)	MPI 101	MPI 163	MPI 163	5 mils

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MPI EXT 5.3K-G6 (Gloss)	MPI 101	MPI 164	MPI 164	5 mils
Topcoat: Coating to match adjacent surfaces.				

Option 1: Pigmented Polyurethane				
New Galvanized Surfaces	Primer	Intermediate	Topcoat	System DFT
MPI EXT 5.3L-G6 (Gloss)	MPI 101	N/A	MPI 72	5 mils
Topcoat: Coating to match adjacent surfaces.				

C. Exterior Surfaces, Other Metals (Non-Ferrous) - Only where designated as "Painted" on Drawings.

(1) Aluminum, aluminum alloy and other miscellaneous non-ferrous metal items not otherwise specified except hot metal surfaces, roof surfaces, and new prefinished equipment

Waterborne Light Industrial Coating				
New Galvanized Surfaces	Primer	Intermediate	Topcoat	System DFT
MPI EXT 5.4F-G5 (Semigloss)	MPI 95	MPI 163	MPI 163	5 mils
MPI EXT 5.4F-G6 (Gloss)	MPI 95	MPI 164	MPI 164	5 mils
Topcoat: Coating to match adjacent surfaces.				

(2) Surfaces adjacent to painted surfaces; Mechanical, Electrical, Fire extinguishing sprinkler systems including valves, conduit, hangers, supports, exposed copper piping, and miscellaneous metal items not otherwise specified except floors, hot metal surfaces, and new prefinished equipment

Waterborne Light Industrial Coating				
New	Primer	Intermediate	Topcoat	System DFT
MPI EXT 5.1C-G3 (Eggshell)	MPI 79	MPI 161	MPI 161	5 mils

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MPI EXT 5.1C-G5 (Semigloss)	MPI 79	MPI 163	MPI 163	5 mils
MPI EXT 5.1C-G6 (Gloss)	MPI 79	MPI 164	MPI 164	5 mils
Primer as recommended by manufacturer. Topcoat: Coating to match adjacent surfaces.				

3.14.2 Interior Paint Tables

3.14.2.1 MPI Division 3: Interior Concrete Paint Table

A. New Concrete, vertical surfaces, not specified otherwise

Institutional Low Odor / Low VOC Latex					
New, uncoated Existing	Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI INT 3.1M-G2 (Flat)	MPI RIN 3.1L-G2 (Flat)	MPI 149	MPI 144	MPI 144	4 mils
MPI INT 3.1M-G3 (Eggshell)	MPI RIN 3.1L-G3 (Eggshell)	MPI 149	MPI 145	MPI 145	4 mils
MPI INT 3.1M-G4 (satin)	MPI RIN 3.1L-G4	MPI 149	MPI 146	MPI 146	4 mils
MPI INT 3.1M-G5 (Semigloss)	MPI RIN 3.1L-G5 (Semigloss)	MPI 149	MPI 147	MPI 147	4 mils
Topcoat: Coating to match adjacent surfaces.					

E. New concrete floors:

Epoxy					
New, uncoated Existing	Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI INT 3.2C-G6 (Gloss)	MPI RIN 3.2C-G6 (Gloss)	MPI 77	MPI 77	MPI 77	5 mils
Note: Primer may be reduced for penetration per manufacturer's instructions.					

3.14.2.2 MPI Division 4: Interior Concrete Masonry Units Paint Table

A. New Concrete Masonry

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Institutional Low Odor / Low VOC Latex					
New, uncoated Existing	Filler	Primer	Intermediate	Topcoat	System DFT
MPI INT 4.2E-G2 (Flat)	MPI 4	N/A	MPI 144	MPI 144	4 mils
MPI INT 4.2E-G3 (Eggshell)	MPI 4	N/A	MPI 145	MPI 145	4 mils
MPI INT 4.2E-G4 (Satin)	MPI 4	N/A	MPI 146	MPI 146	4 mils
MPI INT 4.2E-G5 (Semigloss)	MPI 4	N/A	MPI 147	MPI 147	4 mils
Fill all holes in masonry surface					

3.14.2.3 MPI Division 9: Interior Plaster, Gypsum Board, Textured Surfaces
Paint Table - Only where designated as "Painted" on Drawings.

A. Interior New and Existing, previously painted not otherwise specified

Latex					
New	Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI INT 9.2A-G2 (Flat)	RIN 9.2A-G2 (Flat)	MPI 50	MPI 44	MPI 44	4 mils
MPI INT 9.2A-G3 (Eggshell)	RIN 9.2A-G3 (Eggshell)	MPI 50	MPI 52	MPI 52	4 mils
MPI INT 9.2A-G5 (Semigloss)	RIN 9.2A-G5 (Semigloss)	MPI 50	MPI 54	MPI 54	4 mils
Topcoat: Coating to match adjacent surfaces.					

High Performance Architectural Latex - High Traffic Areas					
New	Existing, previously painted	Primer	Intermediate	Topcoat	System DFT
MPI INT 9.2B-G2 (Flat)	MPI RIN 9.2B-G2 (Flat)	MPI 50	MPI 138	MPI 138	4 mils
MPI INT 9.2B-G3 (Eggshell)	MPI RIN 9.2B-G3 (Eggshell)	MPI 50	MPI 139	MPI 139	4 mils

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MPI INT 9.2B-G5 (Semigloss)	MPI RIN 9.2B-G5 (Semigloss)	MPI 50	MPI 141	MPI 141	4 mils
Topcoat: Coating to match adjacent surfaces.					

-- End of Section --

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SECTION 10 44 16

FIRE EXTINGUISHERS

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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 1	(2021) Fire Code
NFPA 10	(2022; ERTA 1 2021) Standard for Portable Fire Extinguishers
NFPA 101	(2021; TIA 21-1) Life Safety Code
NFPA 505	(2018) Fire Safety Standard for Powered Industrial Trucks Including Type Designations, Areas of Use, Conversions, Maintenance, and Operations

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.157	(2003) Portable Fire Extinguishers
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UNDERWRITERS LABORATORIES (UL)

UL 299	(2012; May 2021) Dry Chemical Fire Extinguishers
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1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Fire Extinguishers

Accessories

Cabinets

Wall Brackets

Schedule

SD-03 Product Data

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Fire Extinguishers

Accessories

Cabinets

Wall Brackets

Replacement Parts List

SD-07 Certificates

Fire Extinguishers Certifications

Manufacturer's Warranty with Inspection Tag

1.3 DELIVERY, STORAGE, AND HANDLING

Protect materials from weather, soil, and damage during delivery, storage, and construction.

Deliver materials in their original packages, containers, or bundles bearing the brand name and the name and type of the material.

Provide portable fire extinguishers in compliance with NFPA 505 for all ancillary vehicles where Fire Safety Standard for Powered Industrial Trucks, including type designations, special conditions relating to areas of use, conversions, maintenance, or specific operations apply.

1.4 WARRANTY

Guarantee that Fire Extinguishers are free of defects in materials, fabrication, finish, and installation and that they will remain so for a period of not less than five (5) years after completion.

Submit the manufacturer's warranty with inspection tag.

1.5 PROJECT SCHEDULE

For fire extinguishers. Coordinate final fire extinguisher schedule with fire protection cabinet schedule to ensure proper fit and function. Use same designations indicated on Drawings.

PART 2 PRODUCTS

Submit fabrication drawings consisting of fabrication and assembly details performed in the factory and product data for the following items: Fire Extinguishers; Accessories, cabinets, and Wall Brackets.

2.1 SYSTEM DESCRIPTION

2.1.1 Types

Submit fire extinguishers certifications showing compliance with local codes and regulations.

Provide fire extinguishers conforming to NFPA 10. Provide quantity and placement in compliance with the applicable sections of NFPA 1, NFPA 101, and 29 CFR 1910.157.

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Provide dry chemical type fire extinguishers compliant with UL 299.

2.1.2 Material

Provide corrosion-resistant steel extinguisher shell.

2.1.3 Size

Dimension cabinets to accommodate the specified fire extinguisher.

2.1.4 Accessories

2.2 EQUIPMENT

2.2.1 Cabinets

2.2.1.1 Material

Provide corrosion-resistant steel cabinets.

2.2.1.2 Type

Provided cabinet type for walls as scheduled on Drawings.

2.2.1.3 Size

Dimension cabinets to accommodate the specified fire extinguishers.

2.2.2 Wall Brackets

Provide spring-clip fire extinguisher wall brackets.

Provide wall bracket and accessories as approved.

2.2.2.1 Identification

Provide lettering complying with authorities having jurisdiction for letter style, size, spacing, and location. Locate as indicated by the drawings.

Identify bracket-mounted fire extinguishers with the words "FIRE EXTINGUISHER" in red letter decals applied to mounting surface.

Orientation: Vertical.

PART 3 EXECUTION

3.1 INSTALLATION

Install Fire Extinguishers where indicated on the drawings. Verify exact locations prior to installation.

Provide extinguishers which are fully charged and ready for operation upon installation. Provide extinguishers complete with Manufacturer's Warranty with Inspection Tag attached.

Install fire extinguishers in locations indicated and in compliance with requirements of authorities having jurisdiction.

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Comply with the manufacturer's recommendations for all installations.

3.2 PROTECTION

3.2.1 Repairing

Remove and replace damaged and unacceptable portions of completed work with new work at no additional cost to the Government.

Submit replacement parts list indicating specified items replacement part, replacement cost, and name, address and contact for replacement parts distributor.

3.2.2 Cleaning

Clean all surfaces of the work, and adjacent surfaces which are soiled as a result of the work. Remove from the site all construction equipment, tools, surplus materials and rubbish resulting from the work.

-- End of Section --